MA'ADEN معادن

Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**

Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **REPORT**
3000-OVERALL
GEOTECHNICAL REPORT

Document Number: **MD-216-3000-EG-CV-RPT-1008**
Revision: **D01**
Revision Date: **12.10.2023**
Status: **AS BUILT**

Outotec Document Number:	900707	GAU01	ZZ01	CED01	30001	OU100663477	3
L&T Document Number:	O19045	001	T	K	24006	001	3

REVISION HISTORY

Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer
A01	05.10.2019	FIRST ISSUE	MF	AW	HUR	
A02	26.11.2019	REVISED AS PER EMPLOYER/PMC COMMENTS DATED 23/10/2019	MF	AW/MLI	HUR	
C01	10.03.2020	ISSUED FOR CONSTRUCTION	MF	AW/MLI	HUR	
D01	11.10.2023	AS BUILT	MF	AW/MLI	HUR	

TABLE OF CONTENTS

Definitions and abbreviations 6

EXECUTIVE SUMMARY 7

1. Intent 9

2. Scope 9

3. SITE DESCRIPTION, GEOLOGY AND SEISMICITY 10

3.1. Site Description 10

3.2. Geology 10

3.3. Seismicity 11

4. FIELD INVESTIGATIONS 12

4.1. General 12

4.2. Execution of Boreholes 12

4.3. Excavation of Testpits 12

4.4. Installation of Standpipe Piezometers 12

4.5. In-Situ Sampling 13

4.5.1. Soil Sampling 13

4.5.2. Bulk Samples 13

4.5.3. Undisturbed Sampling 13

4.5.4. Rock Coring 13

4.6. In-Situ Testing 14

4.6.1. Standard Penetration Tests 14

4.6.2. Downhole Seismic Test 15

4.6.3. Field Electrical Resistivity Tests 16

4.6.4. Field Thermal Resistivity Tests 17

4.6.5. Field California Bearing Ratio (CBR) Tests 17

4.6.6. Plate Load Test 18

5. LABORATORY TESTING 18

5.1. Soil Testing 18

6. RESULTS OF INVESTIGATIONS 19

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This documents the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

6.1. General 19
6.2. Subsurface Strata 19
6.3. Groundwater Table 20
7. GEOTECHNICAL DESIGN PARAMETERS 20
7.1. General 20
7.2. Unit Weight 21
7.3. Shear Strength Parameters 21
7.4. Modulus Of Elasticity 21
7.5. Coefficient of Friction Between Soil and Concrete/Steel 22
7.6. Coefficient Of Earth Pressures 22
7.7. Modulus of Subgrade Reaction 23
7.8. Rock Design Parameters 24
8. RECOMMENDATIONS FOR DESIGN OF FOUNDATIONS 31
8.1. General 31
8.2. Geotechnical Evaluation 31
8.2.1. Subsurface Conditions 31
8.3. Shallow Foundation Recommendations 31
8.3.1. Geotechnical Design Criteria 31
8.3.2. Bearing Capacity Analysis 32
8.3.3. Tolerable Settlements 32
8.3.4. Allowable Bearing Pressure 32
8.4. Adjacent Footings 33
8.5. Foundations Subjected To Vibrations 34
8.5.1. Vibrating Foundation Systems 34
8.5.2. Dynamic Parameters 35
9. RECOMMENDATIONS FOR CONSTRUCTION OF FOUNDATIONS 38
9.1. The Earthwork 38
9.1.1. Fill Material 38
9.1.2. Grading and Compaction 40

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This documents the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

9.2. Soil Corrosivity 41
9.3. Drainage Provisions 41
9.4. Inspection And Monitoring 41
10. CONCLUSIONS AND RECOMMENDATIONS 42
11. LIMITATIONS 43
LIST OF REFERENCES 43

ATTACHMENTS

TABLES

Table 1 List of Coordinates and Elevations
Table 2 Summary of SPT N-Values
Table 3 Summary of SPT N-Values
Table 4 to 12 Bearing Capacity Analysis Results

FIGURES

Figure 1 Vicinity Map
Figure 2 Geotechnical Investigations Plan
Figure 3 to 9 Design SPT / RQD Profiles
Figure 10 RocLab Results

APPENDICES

APPENDIX "A"

Borehole Logs

Testpit Logs

Downhole Test Results

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This documents the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Electrical Resistivity Test Results

Thermal Resistivity Test Results

Field CBR Results

Plate Load Test Results

APPENDIX "B"

Laboratory Test Results

APPENDIX "C"

Design Aids

APPENDIX "D"

Bearing Capacity Calculation Sheets & Settle 3D Outputs

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This documents the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

DEFINITIONS AND ABBREVIATIONS

Employer	MA'ADEN GOLD AND BASE METALS or their representatives
EPC Contractor	Consortium of Outotec (EPC-OT) and Larsen & Toubro (EPC-L&T)
Project management Company (PMC)	BECHTEL appointed by EMPLOYER
Supplier	An entity(supplier, consultant, contractor) under the direct project management of EPC Contractor
Sub- supplier	A sub entity (supplier, consultant, contractor) under the direct project management of EPC Contractor
Shall	Is to be understood as mandatory in relation to the requirements of this Document
Should	Is to be understood as a strong recommendation to comply with the requirements of this Document
Client	Consortium of Outotec and Larsen & Tourbo

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This documents the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

EXECUTIVE SUMMARY

GENERAL

The Mansourah and Massarah Gold Project is part of the East Central Arabian Gold Region (ECAGR) within the Kingdom of Saudi Arabia, which consists of the Ar Rjum, Mansourah and Massarah deposits.

M/s L & T Construction, EPC Contractor of the project, contracted Riyadh GeoTechnique and Foundations (RGF), Al-Khobar for carrying out the geotechnical investigations for the proposed project site.

The investigations layout was planned by client and field work was carried out in August - September 2019by RGME. Subsequently, the laboratory testing was conducted at RGF Geotechnical and Materials Testing Laboratories, Al Khobar.

This report presents details of geotechnical investigations carried out for subject project, provide an evaluation of the subsurface conditions encountered at the investigated boreholes and recommendations for design of foundations for the proposed structures. The recommendations given in this report are based on the interpretation of field and laboratory test results, and on the state-of-the-art knowledge in geotechnical engineering, design and construction.

SUBSURFACE STRATA

The granular soil deposits consisting of SAND with varying amount of Silt, Clay and Gravel (SM / SC) and GRAVEL with Silt and Sand (GM / GW-GM) are present below existing grade level up to a depth varying from 0.75 to 7.5 m below existing grade level. The compactness of the sand layer is generally loose to medium dense. Highly to completely weathered rock recovered as Sand and Gravel are present below sand layers extending to the top of competent rock layers. The completely weathered rock are underlain by slightly to moderately weathered ANDESITE / SCHIST rock up to the maximum investigated depth of 30.0 m below exiting grade level.

GROUNDWATER TABLE

Observations were made for stabilized ground water table, measured after 24 hours or more after completion of drilling of boreholes. The groundwater table was encountered in three boreholes (BH-02, BH-15 & BH-19) at a depth of 23 and 24.0m below existing grade level during current investigations. A list of groundwater measurements at borehole locations is provided in **Table 3**.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This documents the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

GEOTECHNICAL DESIGN PARAMETERS

Engineering analyses for the determination of net allowable pressures and settlements for the various subsurface conditions have to be based on carefully selected representative soil and rock parameters. Different soil profiles have been developed for each area and are presented in attached **Figure 3 to Figure 9.** Design geotechnical parameters are provided in **Table 7-2 to Table 7-8** of Section 7.0.

SHALLOW FOUNDATION RECOMMENDATIONS

The net allowable bearing capacity analysis has been carried out for foundation sizes ranging from 1.0 x 1.0m to 5.0 x 5.0m at depth of 1.2m and 1.5m and specific foundations for the proposed structures provided by the client. The net allowable bearing capacity (NABC) values have been shown in attached **Table 4 to Table 12**. Bearing capacity calculation sheets are provided in **Appendix D**.

SOIL CORROSIVITY

The chemical tests performed on soil samples indicate that site have been contaminated with chemically aggressive salts. Therefore, It is recommended that foundations and other concrete structures coming in direct contact with the soil be constructed using **Type I+Silica** Fumes cement. Cement type should also comply with the project specification. The chemical tests results are provided in **Appendix B.**

EARTHWORK

The earthwork, selection of fill, its placement and compaction requirements shall be carried out as per section 9.0 of this report. Moreover, project specifications should also be followed.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This documents the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

1. INTENT

The Mansourah and Massarah Gold Project is part of the East Central Arabian Gold Region (ECAGR) within the Kingdom of Saudi Arabia, which consists of the Ar Rjum, Mansourah and Massarah deposits.

M/s L & T Construction, EPC Contractor of the project, contracted Riyadh GeoTechnique and Foundations (RGF), Al-Khobar for carrying out the geotechnical investigations for the proposed project site.

The investigations layout was planned by client and field work was carried out in August - September 2019by RGME. Subsequently, the laboratory testing was conducted at RGF Geotechnical and Materials Testing Laboratories, Al Khobar.

This report presents details of geotechnical investigations carried out for subject project, provide an evaluation of the subsurface conditions encountered at the investigated boreholes and recommendations for design of foundations for the proposed structures. The recommendations given in this report are based on the interpretation of field and laboratory test results, and on the state-of-the-art knowledge in geotechnical engineering, design and construction.

2. SCOPE

- A geotechnical investigations program was prepared, keeping in view the site conditions, the project components and to fulfill the objective of investigation in line with the scope of work, as described below.
- Surveying and Staking out of the borehole locations at the site in accordance with the locations / coordinates provided by client.
- Drilling and sampling of twenty nine (29) boreholes up to a depth of 8.0 to 30.0 m below existing ground level. It is to be noted that BH-24 is on-hold by the client.
- Excavation of fifty (50) test pits in soil downs to 1.50m depth or refusal whichever comes first. It is to be noted that TP-37 to TP-40 are on-hold by the client.
- Performance of three (03) Downhole Seismic Tests down to maximum 30.0 m depth.
- Performance of sixteen (16) Electrical Resistivity Tests at specified location using an electrode spacing of 1.0 m, 2.0 m, 5.0 m and 10.0 m. It is to be noted that ERT-01, ERT, 03 & ERT-08 are on-hold by the client.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This documents the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

- Performance of one (01) Thermal Resistivity Test
- Performance of seven (07) plate load tests. It is to be noted that PLT-07 is on-hold by the client.
- Performance of eighteen (18) field California Bearing Ratio (CBR) Tests. It is to be noted that CBR-18 is on-hold by the client.
- Performing laboratory tests to determine pertinent engineering and index properties as well as the chemical properties of the strata encountered at the project site.
- Analyzing the field and laboratory tests data and submitting an engineering report, furnishing recommendations for the design and construction of foundations and earthwork for the proposed structures at the site.

3. SITE DESCRIPTION, GEOLOGY AND SEISMICITY

3.1. Site Description

The Mansourah and Massarah Gold Project is part of the East Central Arabian Gold Region (ECAGR) within the Kingdom of Saudi Arabia, which consists of the Ar Rjum, Mansourah and Massarah deposits. Mansourah and Massarah are located about 55 kilometers East of Dhalim on the Taif-Riyadh highway within the Central Arabian Shield Gold Region. A Site Location Plan is provided as **Figure 1** whereas geotechnical investigation plan is provided as **Figure 2**.

3.2. Geology

The Mujayrib formation crops out in a north-trending belt in the center of the quadrangle. The formation was named by Agar and Hazmi (in preparation) after Jabal al Mujayrib in the north of the belt. The belt extends northward into the 'Afif quadrangle where its equivalent rocks, the Numan formation, were considered to be older than the Siham group (Letalenet, 1979). However, the base of the Mujayrib is exposed at Jabal Bani Ghayy and shows a clear unconformity between it and a red, alkali-feldspar granite that is intrusive into the Siham group. Furthermore, north of Jabal al Mujayrib and west of Jabal Qarnayn, the Bani Ghayy overlies Andesite of the Tandub formation of the Siham group, and, although the unconformity is not exposed, the cleavage and bedding of the Andesite is truncated beneath the Mujayrib formation.

In the north, the Mujayrib formation contains a coarse basal conglomerate (bje) that becomes finer upwards and is transitional into tuffaceous graywacke sandstone and siltstone (bjg) in a positive transgressional Lombard (1972)

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This documents the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

sequence. A thin agglomerate near the top of the basal conglomerate is a good marker horizon (Agar and Hazmi, in preparation) and shows that a thick sequence of volcanic rocks, intercalated with the conglomerate in the west near Jabal Bani Ghayy, is absent in the east near Jabal Ghuthaylib. The basal conglomerate in the east is generally finer grained and contains more interbedded graywacke (bjg) and siltstone (bjs) than in the west, and an east-west facies change is clearly demonstrated. However, in the south near Abraq al Bara'im, the same basal conglomerate and its intercalated volcanic rocks are exposed in the middle of the succession and are underlain by an older, positive transgressional Lomabrd (1972) sequence, which suggests that transgression took place from south to north (Agar, 1985). East of Jabal Hisan in the south of the quadrangle, a thin clastic limestone (bjl) is exposed within a thick graywacke sequence. The limestone is not seen anywhere else within the Mujayrib formation.

The volcanic rocks of the Mujayrib formation are geochemically bimodal, comprising basaltic Andesites and rhyolites (bjv, bjr) and are typical of continental-extensional environments (Agar,1985c).

3.3. Seismicity

As per Saudi Building Code Load and Forces Requirements SBC 301(2007), the following seismic design parameters for the subject project should be considered and seismic loading should be in accordance with following.

Process Plant Area

Spectral Response Acceleration (1 second) – S1 = 0.02g

Spectral Response Acceleration (0.2 second) – Ss = 0.10g

Non-Process Plant Area

Spectral Response Acceleration (1 second) – S1 = 0.02g

Spectral Response Acceleration (0.2 second) – Ss = 0.10g

Site Class

Based on the downhole seismic test results, it is recommended to consider Seismic Site **Class B** for the project site.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This documents the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

4. FIELD INVESTIGATIONS

4.1. General

The proposed field work consisted of exploratory drilling and sampling of the subsurface soil samples and performance of field tests at designated locations. The borehole locations were decided by the client and marked in the field by RGF. A list of coordinates and elevations of the geotechnical investigation locations is given in **Table 1**.

4.2. Execution of Boreholes

Boreholes were executed up to a depth of 8 to 30m below existing grades using straight rotary drilling method. The drilling of boreholes was carried out using ACKER ADII drilling rig. The subsurface stratum was penetrated by rotary drilling method using wash boring technique. Upon encountering rock or harder strata, the strata were penetrated through continuous coring using 101mm diamond core bit. Bentonite slurry of suitable consistency was circulated as a drilling fluid to provide stability to borehole walls. After completion of the testing, boreholes were backfilled with a mixture of cement, bentonite, sand and water. The grout backfill materials were proportioned to a groutable / pourable consistency and the proportions varied as per the specific requirements.

The borehole logs were prepared in the field and a description of soil samples recovered has been shown in the respective borehole logs. Field borehole logs were developed on the basis of the materials encountered at the site and were later confirmed / modified by the laboratory test results. The borehole logs are appended in **Appendix A** of this report.

4.3. Excavation of Test pits

Test pits were excavated at the specified locations up to a depth of 1.5m below existing grades using mechanical excavator. The test pit was carefully logged during excavation and field logs were developed, which were finalized after laboratory testing. The test pit logs are appended to this report as **Appendix A.**

4.4. Installation of Standpipe Piezometers

Piezometers were installed in pre-drilled boreholes (BH-01, BH-08 & BH-18), down to the maximum investigated depth of boreholes (15m) at the time of investigations.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This documents the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Prior to piezometers installation, each borehole was drilled as per the methodology described in Section 4.2. The soil boring was advanced to the specified depth by circulating the drilling fluid.

After completion of drilling, the borehole was converted to a temporary groundwater monitoring well by installing a 50mm diameter well in a 100mm diameter drilled borehole. The margin was allowed for sufficient annular space for placement of a filter pack around the well screen. All monitoring wells were constructed with 50mm inner diameter, flush threaded PVC and flush threaded riser pipe. The well-screened interval was tremie packed with filter pack sand to approximately 0.5m above the top of the screen. Fine sand was then placed above the filter pack sand upto the ground level. Groundwater was not observed in any of the piezometer during the time of investigation.

4.5. In-Situ Sampling

4.5.1. Soil Sampling

The samples recovered from hand excavation pits / split spoon sampler were visually inspected and classified as per ASTM D-2488. All the SPT were carried out through automatic SPT hammer and samples obtained from the boreholes were properly labeled, preserved in polythene bags and placed in plastic jars. All these samples were carefully transported to our laboratory for further evaluation and testing.

4.5.2. Bulk Samples

The Bulk soil samples from each test pits were collected to perform compaction and CBR tests. These samples were preserved in air tight plastic bags and were transported to our laboratory for the above mentioned testing.

4.5.3. Undisturbed Sampling

Un-disturbed Samples (UDS) were taken in cohesive soil layers using suitable sampler like Shellby tube and Denison sampler for soft to medium stiff layers and hard layers respectively. All UD samples were properly wrapped to preserve the moisture, labeled and transported to our geotechnical laboratory for required testing.

4.5.4. Rock Coring

Upon encountering rock or harder strata, the strata were penetrated through continuous coring using 101mm diamond core bit. The rock cores recovered were visually inspected and their percentage recovery (REC) and percentage

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This documents the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Rock Quality Designation (RQD) values were determined. All the samples recovered from rock coring were carefully preserved in rock core boxes and sent to our laboratory at Al-Khobar for further evaluation and testing. A summary of percentage recovery (REC) and percentage Rock Quality Designation (RQD) values for each borehole is given in **Table 2** of this report.

4.6. In-Situ Testing

4.6.1. Standard Penetration Tests

Standard Penetration Tests (SPTs) were performed using automatic hammers at 0.75m depth intervals down to 4.50m and at 1.5m depth intervals thereafter. These tests were performed generally in accordance with ASTM D-1586 using a split spoon sampler of 35mm inner dia. and 50mm outer dia. The data obtained from these tests were utilized to assess the in-situ compactness/denseness of the subsurface materials. The SPT blow counts were recorded for a total penetration of 45 cm of SPT sampler. The number of blows required to drive the sampler through the last 30 cm viz. 'N' values have been shown on the respective borehole logs. A summary of SPT-N values is given in Table 4.

Normalization of SPT Data

Corrected N values may be required for the design purposes. These corrections or adjustments to N values include:

- Correction for the actual energy delivered to the drill rod.
- Correction to account for the length of the drill rod.
- Correction to account for the absence or presence of a liner inside the split-spoon sampler.
- Correction to account for the influence of the diameter of the borehole.

It is necessary to normalize the N-values measured by any particular method to some standard rod energy ratio. Researchers show that the blow count in a given soil is inversely proportional to rod energy ratio (ER_r). N-values measured with a known or estimated rod energy ratio (ER_r) value can be normalized to a standard value by the following conversion:

$N_{60} = N\ (ER_r\ /60)$

Where

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This documents the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

$$ER_r = \frac{\text{actual hammer energy to the sampler}}{\text{theoretical input energy}} \times 100$$

Hammer energy calibration has been made and actual hammer energy transferred during the SPT was measured and shown below in Table 3-1.

Table 3-1: Hammer Energy Ratio for the Hammers

Hammer #	Energy Ratio (ER_r)	Hammer energy factor for N_{60} ($\eta_H = ER_r/60$)
AH-082	89.5	1.49
AH-600	90.2	1.50

These corrections were applied to the SPT blows as follows **(Ref. 2 & 3)**:

$(N)_{60} = N_{field} \times (\eta_H \times \eta_R \times \eta_S \times \eta_B)$

Where

$(N)_{60}$ = SPT blows normalized for 60% energy level

η_H = Hammer energy factor

η_R = Rod length factor

= 1.0 for rod length > 10m

= 0.95 for rod length of 6 to 10m

= 0.85 for rod length of 4 to 6m

= 0.75 for rod length of 0 to 4m

η_S= Sampler correction = 1.0 for sampler without liner

η_B= Bore diameter correction = 1.0 for borehole diameter between 60 to 120 mm

4.6.2. Downhole Seismic Test

Downhole seismic tests were conducted using a twelve-channel acquisition system in the designated borehole at each 1.0-meter depth interval down to the depth of 30m. These tests were performed generally in accordance with ASTM D-7400-08.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This documents the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Three channels were used for recording P and S wave signals. One channel recorded the P wave and the other two shall record S_v and S_h. Each reading is measured at 1-meter vertical interval. The evaluation and interpretation of the data is done with utmost care.

For down-hole seismic survey, PVC pipe of 3.0" diameter was lowered in the borehole of 4.0". The annular space between the pipe and borehole wall was grouted with cement and bentonite grout in an appropriate ratio. After giving sufficient time for setting and curing of grout, the down-hole seismic measurements were taken in the PVC pipe. For this purpose a tri-axial geophone was lowered into the borehole and clamped at the respective depth of measurements. The tri-axial geophone was connected to the 12-channel seismograph. The direct distance between the energy source at the surface and the recording geophone was calculated on the basis of known X, Y distances at the surface and within the bore hole. The arrival time measurements were recorded on the seismograph.

Downhole seismic data was processed using specific software. The velocity and density data for each depth increment is used to calculate Shear modulus, Poisson's ratio, Elastic modulus and Bulk modulus for the subsurface layers. Downhole tests results are provided in **Appendix A** of the report.

4.6.3. Field Electrical Resistivity Tests

Electrical Resistivity Tests were carried out at the specified locations of the site as per Standard ASTM G 57. The measurement of the electrical resistivity of the ground were conducted using a 4-Pin Soil Resistance Terrameter SAS 4000. The readings in the field were taken by using Wenner's four electrode configuration.

In Wenner electrode configuration, four electrodes are inserted into the ground at equal distance apart in a straight line. The outer electrodes are used as current electrodes while the inner ones act as potential electrodes. The ground is energized with the help of a battery pack. The current passing through the two current electrodes and the resulting difference between the two potential electrodes are recorded from the instrument. The natural or stray currents in the ground are nullified by the instrument. When the null detector is balanced, using the range switch and the dial, the resistance in ohms between potential electrodes is measured. The resistivity of the ground is calculated as follows:

$$R = \frac{V}{I} x K$$

Where

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This documents the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

R = Apparent Resistivity (Ohm-m)

a = Electrode spacing (meter)

V/ I = Ratio of Voltage to the current

K = 2 Π a (Wenner's configuration shape factor)

were performed on As the depth of measurement is dependent upon electrode spacing, the readings were taken starting from smaller electrode spacing and increased symmetrically with respect to center of electrode configuration. Electrode spacing of 1, 2, 5, and 10m were used. For each electrode spacing, apparent electrical resistivity values were then calculated using the above-mentioned formula.

Electrical resistivity test data and results have been provided in **Appendix A** of this report.

4.6.4. Field Thermal Resistivity Tests

Thermal resistivity tests were conducted as per related ASTM standard for the project at the specified locations. Thermal resistivity tests were performed as per test procedure outlined in ASTM D 5334-14. The test was performed using KD2 Pro, Thermal Properties Analyzer.

Thermal resistivity test data and results have been provided in **Appendix A** of this report.

4.6.5. Field California Bearing Ratio (CBR) Tests

Field CBR Tests were performed at the project site. This test is used for the determination of the California Bearing Ratio (CBR) of soil tested in place by the penetration load of the soil. This test was performed as per ASTM D 4429. The surface area to be tested was prepared by removing loose and dried material from the surface. The loading assembly (truck, excavator etc.) was located so that the center of the bearing attachment is directly over the surface to be tested. After the test arrangements, the load was applied to the penetration piston so that the rate of penetration is approximately 0.05 in. (1.3 mm)/min. The deflection of the proving ring or load cell reading was recorded at each 0.025-in. (0.64-mm) increment of penetration.

Tables and curves of the field CBR test results have been provided in **Appendix A** of this report.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This documents the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

4.6.6. Plate Load Test

Plate Load Tests were performed to evaluate the load-settlement behavior of the existing strata in accordance with ASTM D-1196. These tests were performed up to maximum pressure of 80T/m^2 (twice of maximum anticipated pressure). Bearing plate of 30 cm diameter was used to perform the tests. The maximum load test pressure was applied in four equal increments (each increment equal to 25% of maximum pressure) and was held for a minimum time of fifteen minutes. Load and deflection readings for each load increment were recorded. Load deflection curves are provided in **Appendix A**.

5. LABORATORY TESTING

5.1. Soil Testing

Samples retrieved from boreholes were examined in the field and then transported to our laboratories for relevant laboratory testing. Laboratory testing programs were prepared by the geotechnical engineers and were submitted for the execution at our Al-Khobar laboratory. The laboratory tests were conducted on the samples to determine the engineering characteristics of the subsurface strata. Samples collected from the boreholes were subjected to the following tests as per latest ASTM / BS or equivalent Standards.

- Particle Size Distribution of Soil using Sieve Analysis (ASTM D6913)
- Particle Size Distribution of Fine Grained Soil using Hydrometer (ASTM D7928-17)
- Atterberg Limits (ASTM D – 4318)
- Natural Moisture Content (ASTM D – 2216)
- Direct Shear Tests (ASTM D 3080)
- Unconfined Compression Strength (ASTM D2938)
- Laboratory Compaction of Soil using Modified Effort (ASTM D 1557)
- 3 Point Soaked CBR (ASTM D 1883)
- Sulphates Content (AASHTO T 290-94)
- Chloride Content (AASHTO T 291-94)
- pH (BS 1377 Part 3)
- Total Carbonate as Calcium Carbonate (ASTM D 4373)

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This documents the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

The laboratory tests were conducted on selected representative soil samples to determine the engineering characteristics of the subsurface strata. The summary of laboratory soil characterization tests along with the test result sheets are appended to this report as **Appendix B**. The laboratory chemical test result sheets are also provided in **Appendix B**.

6. RESULTS OF INVESTIGATIONS

6.1. General

Exploratory boreholes were drilled in the field and soil samples retrieved were tested in our laboratory. Results of the investigations discussed in this report are based on the data obtained at the investigated points, tests results and on the engineering interpretation of the data collected. In case of any significant variation revealed during execution of the work, the matter may be referred back to RGF for further analysis.

6.2. Subsurface Strata

The field investigations performed at the site through drilling of borehole have shown the following lithological units at site:

- The granular soil deposits consisting of Silty SAND with Gravel (SM), Clayey SAND with Gravel (SC), Silty GRAVEL with Sand (GM) and occasional Well Graded SAND with Silt and Gravel layers are present below existing grade level up to a depth varying from 0.75 to 7.5 m below existing grade level. The compactness of the sand layer is generally loose to medium dense.

- Highly to completely weathered rock recovered as Silty SAND with Gravel (SM), Clayey SAND with Gravel (SC), Silty GRAVEL with Sand (GM), Clayey GRAVEL with Sand (GC) Poorly to Well Graded GRAVEL with Silt and Sand (GP-GM / GW-GM / GW), Poorly to Well Graded SAND with Silt and Gravel (SP-SM / SW-SM) and occasional Silt and Clay with Sand and Gravel (CL/ML) are present below sand layers extending to the top of competent rock layers.

- The completely weathered rock are underlain by slightly to moderately weathered ANDESITE / SCHIST rock up to the maximum investigated depth of 30.0 m below exiting grade level.

- There has been no evidence of occurrence of any “cavities” in the rocky strata through borehole data down to the maximum investigated depth.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This documents the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Based on the thickness of overburden soil, proposed site has been divided into different zones as discussed in Section 7.0.

6.3. Groundwater Table

Observations were made for stabilized ground water table, measured after 24 hours or more after completion of drilling of boreholes. The groundwater table was encountered in three boreholes (BH-02, BH-15 & BH-19) at a depth of 23 and 24.0m below existing grade level during current investigations. A list of groundwater measurements at borehole locations is provided in **Table 3**.

7. GEOTECHNICAL DESIGN PARAMETERS

7.1. General

The geotechnical design parameters described in this report are based on the engineering evaluation of the subsurface conditions determined through drilling records, in-situ and laboratory testing of the representative samples from the investigation campaign.

Engineering analyses for the determination of net allowable pressures and settlements for the various subsurface conditions have to be based on carefully selected representative soil and rock parameters. Different soil profiles have been developed for each area and are presented in attached **Figure 3 to Figure 9.** Detail of the boreholes for each area is shown below

Figure No.	Area	Boreholes	Thickness of soil (m)	Thickness of rock (m)
Figure-3	Crushing Plant Area	BH-17, 18, 19, 25 & 29	0.0 – 15.0	-
Figure-4(a)	Process Plant-1 (Zone-1)	BH-04, 06 to 11, 16, 26 & 27	0.0 – 15.0	-
Figure-4(b)	Process Plant-1 (Zone-2)	BH-02, 12 to 15 & 20	0.0 – 3.0	3.0 - 30
Figure-4(c)	Process Plant-1 (BH-28)	BH-28	0.0 – 15.0	-
Figure-5	Process Plant-2	BH-01	0.0 – 10.5	10.5 – 15.0
Figure-6	Solar Plant & Workshop Area	BH-05	0.0 – 15.0	-
Figure-7	Village Area	BH-21	0.0 – 9.0	9.0 – 15.0
Figure-8	Explosive Magazine	BH-22	0.0 – 15.0	-
Figure-9	TMF	BH-03 & 23	0.0 – 15.0	15.0 – 30.0

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This documents the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

For the design of foundations of structures, the soil parameters for each zone have been adopted on the basis of the geotechnical data of the site (SPT N-values / RQD), laboratory test results (Direct Shear, UCS etc.), recent literature (Ref. 2 & 3) and engineering judgment.

The geotechnical design parameters are provided in the following sections.

7.2. Unit Weight

The unit weights of subsurface strata have been estimated through interpretation of the SPT / core data retrieved from boreholes. Design unit weight values are shown in Table 6-2 to Table 6.8 of this report.

7.3. Shear Strength Parameters

Shear strength of granular soil materials is usually expressed in the form of angle of internal friction "φ" and cohesion "c". These parameters for various subsurface layers are estimated on the basis of field test results (SPT N-values for soil) and laboratory test results (Direct Shear etc.).

Following SPT N-value correlations have also been used in addition to laboratory test results to estimate the angle of internal friction (φ).

Angle of Internal Friction – Cohesionless Soils

$\varphi = 1.89 + 20.56 \times D_{10} + 2.35 \times \gamma_d - 24.10 \times R_s$ **(Ref. 6)**

Where

D_{10} = Particle diameter corresponding to 10% Finer (mm)

γ_d= Dry Density (kN/m^3)

R_s = Particle Roundness

The recommended design values for shear strength parameters are provided in **Table 6-2 to Table 6-8** of this report.

7.4. Modulus Of Elasticity

The immediate settlement of foundation depends upon the elastic properties of the load bearing strata, characterized by Modulus of Elasticity (Es). The Modulus of Elasticity of soils is directly proportional to N-values recorded in SPTs. The constant of proportionality depends upon the type of soils in which the tests are

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This documents the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

carried out. For the strata encountered in this particular investigation, the following relationships between Es and N-values can be used (Ref. 2).

$E_s = 1000N_{60}$ *Normally Consolidated Sands*

$E_s = 1300N_{60} + 7200$ *Gravelly Sand*

$$E_s' = E_s \frac{1-\mu}{(1+\mu)(1-2\mu)}$$

Where:

E_s = Modulus of Elasticity (kPa) & N = SPT N-Value

E_s' = Constrained Modulus of Elasticity (kPa)

Considering the nature of overburden soil, modulus of elasticity has been calculated using correlation ($Es = 1000N_{60}$) for this project

Constrained Modulus of Elasticity (E_s') is used in the settlement analysis using Settle 3D software discussed in Section 8.0 of this report.

7.5. Coefficient of Friction Between Soil and Concrete/Steel

The coefficients of friction are defined as the tangent of the angle of friction and following values could be used for soil and concrete / steel (Ref 3, 4):.

(a) Coefficient of friction between soil and concrete= tan 30° (For Non Cohesive strata)
= 0.58

(b) Coefficient of friction between soil and steel = tan 20° (For Non Cohesive strata)
= 0.36

7.6. Coefficient Of Earth Pressures

The coefficients of earth pressures for granular soils are essentially functions of the angle of internal friction (ϕ), which can be estimated using the following relationships (Ref 3, 4).

(a) Coefficient of Active Earth Pressure (Ka)

$$= \frac{1 - Sin\ \phi}{1 + Sin\ \phi}$$

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This documents the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

(b) Coefficient of Passive Earth Pressure (Kp)

$$= \frac{1 + Sin\ \phi}{1 - Sin\ \phi}$$

(c) Coefficient of "At rest" Earth Pressure (Ko)

= 1 - Sinϕ

The static lateral earth pressure coefficients for active (Ka), at rest (Ko) and passive (Kp) conditions, using φ = 30˚ are recommended as follows:

K_a = 0.33

K_p = 3.00

K_o = 0.50

7.7. Modulus of Subgrade Reaction

Vertical Modulus of Subgrade Reaction

The value of vertical modulus of subgrade reaction is a function of the allowable bearing pressure and the following equations could be used to estimate its value. (Ref. 2).

K_s = 120 q_a (kN/m^3) – Spread Footings

K_s = 90 q_a (kN/m^3) – Mat Foundation

Where:

K_s = Modulus of subgrade reaction

q_a= Net Allowable bearing pressure

Horizontal Modulus of Subgrade Reaction

The value of horizontal modulus of subgrade reaction is calculated using following equations. (Ref. 2)

$k_s = A_s + B_s Z^n$

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This documents the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Where

A_s = Constant = $C(cN_cS_c + 05\gamma BN_\gamma S_\gamma)$

B_s = Coefficient for depth variation

Z = Depth of interest below ground

n = Exponent to give k_s the best fit (n = 1.0 has been considered)

$A_s = C(cN_cS_c + 05\gamma BN_\gamma S_\gamma)$

$B_sZ^1 = C(\gamma N_qS_q)Z^1$ (C = 40 for SI unit)

Modulus of vertical and horizontal Subgrade Reactions against each specific foundation case are evaluated and results are provided in **Table 4 to Table 12.** Moreover, modulus of subgrade reaction for testpit locations could be taken from adjacent borehole for similar soil conditions.Rock Design Parameters

RocLab software has been used to come up with the design parameters by inputting intact rock mass properties. RocLab procedure is detailed in **Appendix C.**

Hoek-Brown failure envelope in principal and / or shear-normal stress is plotted by using the following parameters:

- Unconfined compressive strength of intact rock, **sigci**
- Intact rock parameter, **mi**
- Geological strength index, **GSI**
- Disturbance factor, **D**

Determination of rock mass deformation modulus is carried out based on the following input data (Generalized Hoek-Diederichs equation):

- Intact rock deformation modulus, **Ei**
- Ei can be estimated using a modulus ratio, **MR**

Each of the above 5 input parameters (sigci, mi, GSI, D and Ei or MR), can be conveniently estimated from built-in charts and tables, based on rock type, geological conditions, etc. These built-in charts and tables used by the software are provided in **Appendix C**.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This documents the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

The RocLab parameters used for this project as input for the Hoek-Brown criterion (sigci, mi, GSI, D & MR) for determination of rock mass modulus and strength parameters are provided below:

Table 6-1: RocLab Input Parameters

Parameter	Value	Remarks
sigci	17 MPa	From Unconfined Compressive Strength Tests
mi	20	For Andesite Rock, mi = 25±5
GSI	30	Surface Condition=Poor to Fair, Structure = blocky/Disturbed
D	0	Applicable for Tunnels and slopes only
MR	300	For Andesite Rock, MR = 400± 100

The RocLab output sheets are attached as **Figure 10**.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This documents the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Table 6-2: Geotechnical Design Parameters – Crushing Plant

Soil Type	Depth (m)		Design SPT Value	Design RQD Value	Bulk Density "γ_b"	Undrained Cohesion "c_u"	Drained Cohesion "c"	Angle of Internal Friction "φ"	Modulus of Elasticity "E_s"
	From	To	N_{60}	(%)	(kN/m^3)	(kPa)	(kPa)	(degree)	(kPa)
SAND with Silt and Gravel / GRAVEL with Silt and Sand	0.00	0.75	9	-	17	-	0	30	9,000
	0.75	1.50	23	-	18	-	0	33	23,000
	1.50	4.50	74	-	20	-	0	35	74,000
	4.50	15.00	100	-	20	-	0	37	100,000

Table 6-3(a): Geotechnical Design Parameters – Process Plant-1 (Zone-1)

Soil Type	Depth (m)		Design SPT Value	Design RQD Value	Bulk Density "γ_b"	Undrained Cohesion "c_u"	Drained Cohesion "c"	Angle of Internal Friction "φ"	Modulus of Elasticity "E_s"
	From	To	N_{60}	(%)	(kN/m^3)	(kPa)	(kPa)	(degree)	(kPa)
SAND with Silt and Gravel / GRAVEL with Silt and Sand	0.00	1.50	16	-	18	-	0	32	16,000
	1.50	6.00	43	-	19	-	0	34	43,000
	6.00	9.00	85	-	20	-	0	37	85,000
	9.00	15.00	100	-	20	-	0	37	100,000

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This documents the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Table 6-3(b): Geotechnical Design Parameters – Process Plant-1 (Zone-2)

Soil Type	Depth (m)		Design SPT Value	Design RQD Value	Bulk Density "γ_b"	Undrained Cohesion "c_u"	Drained Cohesion "c"	Angle of Internal Friction "φ"	Modulus of Elasticity "E_s"
	From	To	N_{60}	(%)	(kN/m^3)	(kPa)	(kPa)	(degree)	(kPa)
SAND with Silt and Gravel / GRAVEL with Silt and Sand	0.00	1.50	19	0	18	-	0	32	19,000
	1.50	3.00	75	0	20	-	0	35	75,000
ADESITE Rock	3.00	30.00	-	74	24	-	174	47	415,000

Table 6-3(c): Geotechnical Design Parameters – Process Plant-1 (BH-28)

Soil Type	Depth (m)		Design SPT Value	Design RQD Value	Bulk Density "γ_b"	Undrained Cohesion "c_u"	Drained Cohesion "c"	Angle of Internal Friction "φ"	Modulus of Elasticity "E_s"
	From	To	N_{60}	(%)	(kN/m^3)	(kPa)	(kPa)	(degree)	(kPa)
SAND with Silt and Gravel / GRAVEL with Silt and Sand	0.00	0.75	10	-	17	-	0	30	10,000
	0.75	1.50	25	-	18	-	0	33	25,000
	1.50	6.00	41	-	19	-	0	34	41,000
	6.00	15.00	84	-	20	-	0	37	84,000

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This documents the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Table 6-4: Geotechnical Design Parameters – Process Plant-2 (BH-01)

Soil Type	Depth (m)		Design SPT Value	Design RQD Value	Bulk Density "γ_b"	Undrained Cohesion "c_u"	Drained Cohesion "c"	Angle of Internal Friction "φ"	Modulus of Elasticity "E_s"
	From	To	N_{60}	(%)	(kN/m^3)	(kPa)	(kPa)	(degree)	(kPa)
SAND with Silt and Gravel / GRAVEL with Silt and Sand	0.00	0.75	10	0	17	-	0	30	10,000
	0.75	1.50	26	0	18	-	0	33	26,000
	1.50	6.00	100	0	21	-	0	37	100,000
	6.00	9.00	51	0	20	-	0	35	51,000
	9.00	10.50	100	0	21	-	0	37	100,000
ADESITE Rock	10.50	15.00	-	63	24	-	174	47	415,000

Table 6-5: Geotechnical Design Parameters – Solar Plant & Workshop Area (BH-05)

Soil Type	Depth (m)		Design SPT Value	Design RQD Value	Bulk Density "γ_b"	Undrained Cohesion "c_u"	Drained Cohesion "c"	Angle of Internal Friction "φ"	Modulus of Elasticity "E_s"
	From	To	N_{60}	(%)	(kN/m^3)	(kPa)	(kPa)	(degree)	(kPa)
SAND with Silt and Gravel / GRAVEL with Silt and Sand	0.00	0.75	30	0	18	-	0	33	30,000
	0.75	15.0	100	0	20	-	0	37	100,000

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This documents the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Table 6-6: Geotechnical Design Parameters – Village Area (BH-21)

Soil Type	Depth (m)		Design SPT Value	Design RQD Value	Bulk Density "γ_b"	Undrained Cohesion "c_u"	Drained Cohesion "c"	Angle of Internal Friction "φ"	Modulus of Elasticity "E_s"
	From	To	N_{60}	(%)	(kN/m^3)	(kPa)	(kPa)	(degree)	(kPa)
SAND with Silt and Gravel / GRAVEL with Silt and Sand	0.00	2.25	21	-	18	-	0	33	21,000
	2.25	9.00	100	-	20	-	0	37	100,000
ADESITE Rock	9.00	15.0	-	55	23	-	150	45	375,000

Table 6-7: Geotechnical Design Parameters – Explosive Magazine (BH-22)

Soil Type	Depth (m)		Design SPT Value	Design RQD Value	Bulk Density "γ_b"	Undrained Cohesion "c_u"	Drained Cohesion "c"	Angle of Internal Friction "φ"	Modulus of Elasticity "E_s"
	From	To	N_{60}	(%)	(kN/m^3)	(kPa)	(kPa)	(degree)	(kPa)
SAND with Silt and Gravel / GRAVEL with Silt and Sand	0.00	1.50	25	-	18	-	0	33	25,000
	1.50	4.50	55	-	20	-	0	35	55,000
	4.50	15.00	100	-	20	-	0	37	100,000

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This documents the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Table 6-8: Geotechnical Design Parameters – TMF (BH-03 & BH-23)

Soil Type	Depth (m)		Design SPT Value	Design RQD Value	Bulk Density "γ_b"	Undrained Cohesion "c_u"	Drained Cohesion "c"	Angle of Internal Friction "φ"	Modulus of Elasticity "E_s"
	From	To	N_{60}	(%)	(kN/m^3)	(kPa)	(kPa)	(degree)	(kPa)
SAND with Silt and Gravel / GRAVEL with Silt and Sand	0.00	1.50	30	-	18	-	0	33	30,000
	1.50	3.00	84	-	20	-	0	38	84,000
	3.00	15.00	100	-	20	-	0	38	100,000
ADESITE Rock	15.00	30.0	-	74	24	-	170	45	400,000

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This documents the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

8. RECOMMENDATIONS FOR DESIGN OF FOUNDATIONS

8.1. General

The foundation design recommendations presented in this report are based on the data obtained at the investigated points, tests results and on the engineering interpretation of the data.

It is emphasized that all the required geotechnical recommendations for the foundations as noted above and as described in the Scope of Work do not depend only on the type, nature and engineering characteristics of the load bearing subsurface strata but to a larger extent on the type and nature of the proposed structures.

8.2. Geotechnical Evaluation

8.2.1. Subsurface Conditions

The borehole drilled at the site revealed the presence of medium dense to very dense overburden soil underlain by rocky strata. Considering the condition of the strata, as evaluated from the investigation results and the type of structures to be constructed, the subsurface strata are considered as competent to support the proposed structures using shallow foundations.

8.3. Shallow Foundation Recommendations

8.3.1. Geotechnical Design Criteria

The foundations of all the structures should meet the following design criteria: (Ref 2, 3)

i. These should be safe against shear failure of the supporting ground. A factor of safety of 3.0 has been adopted for isolated/strip footings to estimate allowable bearing capacity.

ii. These should not settle excessively under the service loads. A limit of 25mm has been put on the total settlement for isolated / strip footings. For mat foundations, the maximum tolerable total settlement is 50 mm.

iii. Differential settlement / angle of distortion with respect to horizontal should be within ratio of 1:500 for isolated footing and 1:300 for tank / raft foundation.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This documents the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

8.3.2. Bearing Capacity Analysis

In order to be safe, the load carrying strata must be competent to sustain the imposed loading without undergoing shear failure, and at the same time settlements of the foundations must not exceed the allowable limits. Therefore, the load carrying characteristics of the strata should be evaluated for the shear failure and potential volume variation considerations. In general, the net bearing pressure exerted by a footing on the supporting strata will depend upon the loading configuration and the contact areas of the footing. The contact pressure will have to be limited for a particular footing from the settlement considerations.

The allowable bearing capacity against shear failure and settlement criteria for shallow foundations of various widths was calculated using the procedures detailed in **Appendix C** and results are provided in section 8.3.4 of this report. Vesic equation has been used to estimate the bearing capacity against shear failure whereas Settle3D software has been used to estimate bearing capacity against allowable settlement. Methodology involved in Settle 3D software is provided in Appendix C.As settle3D software is used for settlement calculations, following points have been considered in the analysis:

- 1D modulus (constrained modulus) values are used for analysis. The modulus values used in the analysis are shown on the properties sheets in **Appendix D**.
- The settlement calculations are based on flexible and rigid foundation as per information provided by the client.

8.3.3. Tolerable Settlements

Tolerance for settlement is a function of type and size of foundation, and type and methodology of construction. The degree of settlement allowed for a structure depends on many factors, mainly on serviceability and sensitivity of the structures. Therefore, the amount of allowable settlement may vary as per the design tolerance of the structures. However, the limits set should not lead to the occurrence of an Ultimate Limit State (ULS) or Serviceability Limit State (SLS). Generally, 25mm and 50mm are considered maximum limits of the settlement tolerable for isolated/strip and mat type of foundations respectively. Accordingly, the total settlements associated with the recommended allowable bearing capacity values indicated above should be 25mm or less.

8.3.4. Allowable Bearing Pressure

The load carrying characteristics of the strata have been evaluated keeping in view both the shear failure and maximum allowable volume movement

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This documents the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

(settlement / uplift) criteria. The results of analysis show that for the subsurface strata and the envisaged loadings, both shear failure and maximum allowable settlement criteria govern the maximum allowable bearing pressures. The contact pressure will have to be limited for a particular foundation / structure from the allowable settlement considerations as presented in Section 8.3.3.

The net allowable bearing capacity analysis has been carried out for foundation sizes ranging from 1.0 x 1.0m to 5.0 x 5.0m at depth of 1.2m and 1.5m and specific foundations for the proposed structures provided by the client. The net allowable bearing capacity (NABC) values have been shown in attached **Table 4 to Table 12**. Bearing capacity calculation sheets are provided in **Appendix D**. The soil parameters and recommendation of bearing capacity for the design of pole like structures and fence foundations shall be taken from the nearest borehole data. The soil parameters for the design of foundations for boundary fence shall be taken from boreholes BH-3, BH-21, BH-22 & BH-23 which is indicated in table 10, 11 & 12.

8.4. Adjacent Footings

The settlement calculations for the footings are based on the consideration that the interaction of the adjacent footings will be negligible. An effect of interaction must be accounted for in case the footings are too close. The effect of adjacent footings at same grade will be negligible if 2:1 (vertical to horizontal) stress distribution lines from the edges of footings intersect at a depth greater than 2B. According to this concept, the interaction between 1.0, 2.0 and 3.0m wide footings will be negligible if the edge to edge distance between the footings is equal to or greater than 2.0, 4.0 and 6.0m respectively. In case where the lateral spacing between edges of footings is less than 2B, the stress distribution between the adjacent footings should be analyzed and reduction on the recommended net allowable bearing capacities (NABC) should be applied. The reduction values shown in Table 7-1 should be used to estimate the reduced net allowable bearing capacities in case of foundation interaction. The analysis presented in the report is based on non-interaction of adjacent footings. Suitable adjustments will be required to account for the interaction of adjacent footings wherever applicable.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This documents the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Table 7-1: Reduction in Net Allowable Bearing Capacities

Edge to Edge Distance between Footings, "d"	Reduction in Net Allowable Bearing Capacities (%)
d ≥ 2B	Nil
1B ≤ d < 2B	10 %
0.5B ≤ d < 1B	20 %
0.25B < d < 0.5B	30 %
d ≤ 0.25B	NABC equivalent to footing of width 2B

Note on Table 7-1:

These factors have been developed in-house by modeling foundations under different soil conditions and geometry using stress distribution by Westergaard method in settle3D software. Moreover, similar factors are also provided in book by Srinivasan and Ghosh, 2013.

8.5. Foundations Subjected To Vibrations

Foundations are subjected to vibrations from a number of natural and artificial sources. Mechanical equipment such as motors and rotary pumps produce continuous high-frequency vibrations. The design of such foundation must be based on studies of the response of foundations to dynamic loadings and the transmission of wave energy through soils. The amount of motion developed by the foundation in response to the dynamic loadings must be limited by design procedures, to values compatible with satisfactory operation of the facility supported by the foundation. In addition, the wave energy transmitted through the underlying soil from the moving foundation must not cause harmful effects to machines, structures, or people in the immediate vicinity. Thus the local soil conditions and the foundation-soil interaction must be evaluated and included as important factors in the design procedures.

8.5.1. Vibrating Foundation Systems

Vibrating foundation systems are comprised of the source of vibration (machine) and its foundation acting as a mass resting on soil that acts as a spring. The state of the art procedure for design of such foundation system involves modeling of the system characterized by springs, dash pots and connected masses. The

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This documents the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

design criteria for design of dynamically loaded foundations is developed by considering the foundations as a rigid block which may vibrate in anyone of the six degrees of freedom, described as follows:

i. Translatory motion along X-axis
ii. Translatory motion along Y-axis
iii. Translatory motion along Z-axis
iv. Pitching
v. Rocking
vi. Yawing

The condition of failure of vibrating foundations is reached when the motion exceeds a limiting value which is usually expressed in terms of velocity or acceleration of the foundation. For steady-state vibrations, these may also be expressed in terms of an allowable displacement at specified frequencies.

8.5.2. Dynamic Parameters

The dynamic parameters used for design of dynamically loaded foundations are all deduced from shear wave and compression wave velocities (Vs and Vp).

Natural Frequency

The natural frequency of a foundation-soil system is highly complex. It depends on both the modulus of elasticity of the soil and the size of the foundation. The weight of the vibrating body is the sum of the weight of the foundation and the portion of the soil mass below the foundation that is subjected to vibrations. Therefore, the natural frequency of the soil is not a property of the soil alone but also depends on the weight and size of the foundation and the load it carries. The natural frequency is somewhat less if energy is lost due to damping effect.

The frequency with which a body vibrates naturally when subjected to the application of an external force is called the “natural frequency”, fn. It is the frequency of the free vibration of a system.

$$fn = \frac{1}{2\Pi}\sqrt{\frac{k}{m}} \qquad (1)$$

Where,

k = spring constant

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This documents the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

m = mass density = $\frac{weight}{g}$

Spring constant K, depends on "Shear Modulus G", "equivalent radius of base ro", and "Poisson's Ratio "υ" as well as on the Mode of Vibration. The values of "K" are noted as follows:

$$K_z = \frac{4Gr_o}{1-\upsilon}$$ For vertical translatory movement denoted as Kz. (2)

$$K_x = \frac{32\,(1-\upsilon)}{7-8\upsilon} Gr_o$$ For sliding translatory movement denoted as Kx (3)

$$K_\psi = \frac{8\,Gr_o^{\,3}}{3\,(1-\upsilon)}$$ For rocking motion, denoted as Kψ (4)

$$K_\theta = \frac{16}{3} Gr_o^{\,3}$$ For Torsional motion, denoted as Kθ (5)

For determining "K" as noted above, we need:

G = Shear Modulus, determined from cross-hole test

υ = Poisson's ratio, determined from cross-hole test

ro determined by converting the rectangular base of dimensions "2c x 2d" into an equivalent circular base having a radius r_o, determined as follows:

$$r_o = \sqrt{\frac{4cd}{\Pi}}$$ For translation (vertical & sliding) (6)

$$r_o = \sqrt[4]{\frac{16cd^3}{3\Pi}}$$ For rocking (7)

$$r_o = \sqrt[4]{\frac{16cd(c^2+d^2)}{6\Pi}}$$ For torsion (8)

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This documents the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Wherein "2c" is the width of foundation (along axis of rotation in case of rocking) and 2d is the length of the foundation (in plane of rotation for rocking. The "shear modulus", G is determined from cross-hole or downhole test.

For the particular case of the proposed structures:

A. Determine r_o, using above equations 6, 7 & 8 for base (foundation) dimension "2c x 2d".

B. Determine k_Z, k_X, k_ψ or k_θ as the case may be for different applicable modes, using 2, 3, 4 & 5.

C. Take G for substitution in above equations 2, 3, 4 & 5.

D. Take υ for substitution in above equations 2, 3, 4 & 5.

E. Find natural frequency using equation 1, above, substituting for the relevant "K", and taking m = mass density, from m=$\frac{w}{g}$, where w=weight.

F. Be consistent with units.

Poisson's Ratio (υ)

Poisson's ratio is determined using the following relationship:

$$\upsilon = \frac{0.5\,(V_p / V_s)^2 - 1}{(V_p / V_s)^2 - 1}$$

Where Vp = Compression Wave Velocity (m/s)

Vs= Shear Wave Velocity (m/s)

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This documents the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Dynamic Shear Modulus (G)

This modulus is determined using the following equation:

$$G = \rho V_s^2$$

Where ρ = Mass Density of Soil = $\frac{\text{Unit Weight of soil}}{\text{Acceleration due to Gravity}}$

V_s = Shear Wave Velocity (m/s)

Dynamic Young's Modulus (E)

The dynamic Young's Modulus is a function to Dynamic Shear Modulus (G) and Poisson's Ratio (u) and is determined as follows;

$$E = 2 (1 + \upsilon) G$$

9. RECOMMENDATIONS FOR CONSTRUCTION OF FOUNDATIONS

9.1. The Earthwork

The earthwork, selection of fill, its placement and compaction requirements shall be carried out as per applicable Ma'aden Standards and project specifications.

9.1.1. Fill Material

The type of backfill/ fill, its placement and compaction requirements should be done as discussed below and in compliance with project specifications.

General Fill

General fill should have the following properties:

- A maximum particle size equal to one half of the layer thickness
- Not contain organic matter, chunks of highly plastic clay, contamination or other unsuitable material
- Not be gap graded or degradable

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This documents the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

- Fill materials shall be placed in lifts not exceeding 20cm (8 inches) in loose depths. For hand operated compaction equipment, however, fill material shall be placed in lifts not exceeding 10cm (4 inches) in loose depth.
- Compaction equipment's should be suitable for the soil being compacted

The Structural or Select Fill

Structural or select fill material should consist only of inorganic material and shall have 0 - 20% passing the sieve No. 200. Fill materials should have no particles greater than the 50 mm sieve size. Additionally, that portion of the material passing sieve No. 40, should not have liquid limit more than 35 and plasticity index of not more than 12. Atterberg limits are not required for select fill material with less than 15% passing sieve No. 200.

Fill materials shall be placed in lifts not exceeding 20cm (8 inches) in loose depths. For hand operated compaction equipment, however, fill material shall be placed in lifts not exceeding 10cm (4 inches) in loose depth. Compaction equipments should be suitable for the soil being compacted.

Backfill for Utility Trenches & around Sub-Structures

For proper control of backfilling in utility trenches all requirements for excavation, placement of sand bedding, selection of backfill materials, compaction, shoring, etc. must comply with project specifications. Minimum compaction requirements for non-traffic, medium traffic as well as heavy traffic areas must comply with project Specification.

Backfill materials shall be placed in lifts not exceeding 20cm (8 inches) in loose depths. For hand operated compaction equipment, however, fill material shall be placed in lifts not exceeding 10cm (4 inches) in loose depth. Compaction equipment's should be suitable for the soil being compacted.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This documents the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Floor Slabs

All the loose materials, organic and any debris etc. should be removed from the area of floor slabs. Any earth filling required to achieve the required grade for construction of floor slab should be composed of select fill material, referred as structural fill in above section of this report.

The floor slabs, designed as slab on-grade must not be structurally tied to the columns/walls. A flexible joint between the floor slab and column/walls should be provided to make the floor slab "float" independent of the settlements in column/wall, if any.

9.1.2. Grading and Compaction

The following section must be followed for the grading works. If the rough grading has not been included in a previous contract, the sub-grade shall be restored by removing all vegetation, filling all depressions, and leveling the surface. If the contract includes rough grading and surfacing, the rough grading shall be completed as far in advance of the construction of the surfacing as feasible. Soft and unstable material that will not compact when rolled or tamped shall be removed and replaced with material approved by client representative. Unless otherwise specified, the entire sub-grade shall be compacted to not less than 95 percent of the maximum dry density per (ASTM D 1557, or AASHTO T180) or to 85% of relative density determined by ASTM D4253 and D4254. All holes, ruts, soft places, and other defects shall be corrected. In no case shall the sub-base or base course be placed on soft or unstable material, or over areas that are not drained. If the sub-grade is dusty or muddy, operations shall be delayed until it is in a condition satisfactory to the client representative.

Where rolling of the sub-grade is required, any areas which are inaccessible to a roller shall be compacted either by a mechanical or hand tamper meeting the approval of the client representative.

The sub-grade shall be checked and approved by the client representative prior to starting construction of the sub-base or base course on any portion of the work.

The finished sub-grade shall be maintained in a smooth and compacted condition until the sub-base, base course and surface course are placed.

All other requirements regarding base & sub-base and wearing course materials / gradation placement & compaction must be followed as per the project specifications.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This documents the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

9.2. Soil Corrosivity

Protection of Concrete against Corrosion

The chemical tests performed on on-site soil samples indicated the various degrees of sulphate and chloride contents.

It is recommended that foundations and other concrete structures coming in direct contact with the soil be constructed using **Type I+Silica** Fumes cement. Cement type should also comply with the project specification. The chemical tests results are provided in **Appendix B**.

Protection of Utilities

It is necessary that all the utility lines (water, sewage, oil pipe lines and cables, etc.) be protected against corrosion due to presence of deleterious salts in the soil materials used for backfilling of utility trenches. Such protection can be provided by coating all the utility lines with bitumen/epoxy paint. For protection specifically required for certain underground utilities the specifications of the manufacturer of such utility should be followed.

9.3. Drainage Provisions

A well-planned drainage network should be designed for the site to take care of (i) rainfall; (ii) runoff water; (iii) irrigation water and (iv) sewage etc.

While preparing the site for the proposed construction, a suitable drainage network should be planned so as to remove the rainwater and other sources of water effectively away from the main structures. The drainage system should be capable of disposing, safely, waters from all sources without causing any adverse effect to structures and/or foundation soils. This should be continuously and effectively taken to a pre-planned zone so as to discharge into the designated outlets in the area.

9.4. Inspection And Monitoring

It is necessary that all the geotechnical aspects of foundation construction and earthwork be monitored by qualified and experienced geotechnical engineer as per applicable project standards and specifications. For this purpose a comprehensive inspection and monitoring program should be prepared. The program must specifically address the following aspects:

1. Inspection of all stripping work to ensure that the undesirable top soil containing refuse and organic materials has been adequately removed.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This documents the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

2. Making sure that all the stripped-off material and other unwanted materials from the site are disposed of in designated disposal area without any environmental hazard.
3. Identification of in-situ conditions of the sub-surface during foundation excavation and preparation of sub-grades, so that over excavation is avoided.
4. Inspection of all foundation excavations prior to placement of foundation concrete.
5. Preparation and implementation of a comprehensive quality control program including selection of earth materials and aggregates at the source and field compaction control.
6. The proof-rolling of soil surface to receive fill should be observed by a geotechnical engineer. If excessive pumping occurs, proof-rolling should be curtailed and the geotechnical engineer should be consulted, and other remedial actions employed as directed.

10. CONCLUSIONS AND RECOMMENDATIONS

Based on the geotechnical investigations carried out for the subject project, the following conclusions and recommendations have been arrived at:

a) The granular soil deposits consisting of Silty SAND with Gravel are present below existing grade level up to a depth varying from 0.75m to 7.5m below existing grade level underlain by completely to slightly weathered rock up to the maximum investigated depth of 30.0 m below exiting grade level.

b) Observations were made for stabilized ground water table, measured after 24 hours or more after completion of drilling of boreholes. The groundwater table was encountered in three boreholes (BH-02, BH-15 & BH-19) at a depth of about 24.0m below existing grade level during current investigations.

c) The net allowable bearing capacity analysis has been carried out for foundation sizes ranging from 1.0 x 1.0m to 5.0 x 5.0m at depth of 1.2m and 1.5m and specific foundations for the proposed structures provided by the client. The net allowable bearing capacity (NABC) values have been shown in attached **Table 4 to Table 12**. Bearing capacity calculation sheets are provided in **Appendix D**.

d) It is recommended that foundations and other concrete structures coming in direct contact with the soil be constructed using Type I+Silica Fumes cement. Cement type should also comply with the project specification.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This documents the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

e) Before placing foundations, make sure that all excavated surfaces are firm and free of any loose or undesirable soils. This should be checked by an experienced geotechnical engineer. In case loose/undesirable soils are encountered, the same must be replaced with engineered fill or crushed rock fill or lean concrete up to the desired foundation level.

11. LIMITATIONS

Results and recommendations given in this report are based on the analyses of the data collected at the investigation points, and best engineering judgment. All foundation excavations should be carried out under supervision of an experienced geotechnical engineer, to ensure that no deviation is observed from the recommendations discussed in this report. RGF bears no responsibility for the variations encountered from what has been described in this report. However, if matter is referred to RGF, an advice can be provided in resolving the issue.

LIST OF REFERENCES

1. United States Geological Survey (USGS), "Geologic Map GM-210A".

2. Bowles J.E 1996, "Foundation Analysis and Design" 5th Edition, McGraw-Hill Book Company, USA. Hunt, R.E, "Geotechnical Engineering Investigation Manual" McGraw-Hill Book Company

3. Das, B., "Principles of Foundation Engineering", 3rd Edition, PWS Publishing, Boston, MA, USA.

4. Schmertmann et al (1978), "Improved Strain Influence Factor Diagrams", Journal of Geotechnical Engineering Division, ASCE Vol. 104, No. GT 8, 1978.

5. USA. Department of Navy, USA (1982), "Design Manuals NAVFAC DM-7.1, DM-7.2"

6. Christopher A. Bareither; Tuncer B. Edil; Craig H. Benson and David M. Mickelson, "Geological and Physical Factors Affecting the Friction Angle of Compacted Sands" Journal of Geotechnical and Geoenvironmental Engineering, ASCE 2008.

7. Sorensen K.K. and Okkels N., "Relationship between Drained Shear Strength and Plasticity Index of Undisturbed Overconsolidated Clays" Proceedings of

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This documents the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

the 18th International Conference on Soil Mechanics and Geotechnical Engineering, Paris 2013.

8. Vesic A. S. (1967), “A study of bearing capacity on deep foundation.” Final Report, Project B-189, Georgia Inst. Tech., Atlanta, Ga., 231-236.

9. J. B. Burland& M. C. Burbidge, “Settlement of Foundation on Sand and Gravel”, Pro. Instn Civ, Part 1, 1985, Dec., 1325-1381

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This documents the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

TABLES

TABLE 1 : LIST OF COORDINATES & ELEVATION

CLIENT: **L & T CONSTRUCTION**

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**
GEOTECHNICAL INVESTIGATION

PROJ. NO.: **19-K-3054**

Description	UTM Coordinates (m)		Elevation (m)
	Northing	Easting	
Boreholes			
BH-01	2506230.93	263768.94	915.20
BH-02	2506179.071	264210.66	912.93
BH-03	2503834.172	265464.64	905.40
BH-04	2506042.964	264344.10	911.98
BH-05	2506162.43	263969.23	913.20
BH-06	2506071.234	263996.99	913.04
BH-07	2506130.933	264068.72	912.94
BH-08	2505913.117	264072.55	912.87
BH-09	2506010.372	264138.91	912.86
BH-10	2506142.028	264155.85	913.35
BH-11	2506094.386	264208.64	912.70
BH-12	2505967.37	264193.58	912.66

TABLE 1 : LIST OF COORDINATES & ELEVATION

CLIENT: **L & T CONSTRUCTION**

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**
GEOTECHNICAL INVESTIGATION

PROJ. NO.: **19-K-3054**

Description	UTM Coordinates (m)		Elevation (m)
	Northing	Easting	
BH-13	2505915.358	264216.03	912.52
BH-14	2505878.5	264355.99	911.87
BH-15	2506027.885	264280.82	912.50
BH-16	2506135.791	264290.48	912.07
BH-17	2506276.168	264362.88	912.15
BH-18	2506423.387	264270.89	912.46
BH-19	2506510.835	264424.58	912.34
BH-20	2506027.189	264219.07	912.52
BH-21	2508267.29	263178.80	921.35
BH-22	2509287.33	265495.28	911.50
BH-23	2505158.98	264306.20	911.15
BH-25	2506297.89	264397.65	912.25
BH-26	2506130.16	264334.91	911.85

TABLE 1 : LIST OF COORDINATES & ELEVATION

CLIENT: **L & T CONSTRUCTION**

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**
GEOTECHNICAL INVESTIGATION

PROJ. NO.: **19-K-3054**

Description	UTM Coordinates (m)		Elevation (m)
	Northing	Easting	
BH-27	2506067.53	264289.07	912.25
BH-28	2506033.19	263964.39	913.15
BH-29	2506526.60	264367.17	912.50

TABLE 2: SUMMARY OF SPT N-VALUE, REC & RQD

CLIENT: L & T CONSTRUCTION

PROJECT: MANSOURAH MASSARAH GOLD PROJECT
GEOTECHNICAL INVESTIGATION

PROJ. NO.: 19-K-3054

DEPTH	ηH	ηH	η_R	BH-01				BH-02			
(m)	RGE-AH-082	RGE-AH-600		N_{field}	N_{60}	REC	RQD	N_{field}	N_{60}	REC	RQD
0.00	1.49	1.50	0.75	9	10	-	-	40	45	-	-
0.75	1.49	1.50	0.75	23	26	-	-	R	R	-	-
1.50	1.49	1.50	0.75	R	R	-	-	R	R	-	-
1.60	1.49	1.50	0.75	-	-	-	-	-	-	95	80
2.25	1.49	1.50	0.75	R	R	-	-	-	-	-	-
2.60	1.49	1.50	0.75	-	-	-	-	-	-	-	-
3.00	1.49	1.50	0.75	R	R	-	-	-	-	95	75
3.20	1.49	1.50	0.75	-	-	-	-	-	-	-	-
3.75	1.49	1.50	0.75	R	R	-	-	-	-	-	-
3.80	1.49	1.50	0.75	-	-	50	10	-	-	-	-
4.50	1.49	1.50	0.85	R	R	-	-	-	-	90	70
4.60	1.49	1.50	0.85	-	-	-	-	-	-	-	-
5.00	1.49	1.50	0.85	-	-	-	-	-	-	-	-
6.00	1.49	1.50	0.95	47	42	-	-	-	-	95	80
6.10	1.49	1.50	0.95	-	-	-	-	-	-	-	-
7.50	1.49	1.50	0.95	67	60	-	-	-	-	97	90
7.60	1.49	1.50	0.95	-	-	-	-	-	-	-	-
8.00	1.49	1.50	0.95	-	-	17	0	-	-	-	-
8.50	1.49	1.50	0.95	-	-	-	-	-	-	-	-
9.00	1.49	1.50	0.95	R	R	-	-	-	-	90	80
9.10	1.49	1.50	0.95	-	-	-	-	-	-	-	-
10.50	1.49	1.50	1.00	-	-	80	50	-	-	90	85
11.00	1.49	1.50	1.00	-	-	-	-	-	-	-	-
12.00	1.49	1.50	1.00	-	-	90	60	-	-	90	80
12.10	1.49	1.50	1.00	-	-	-	-	-	-	-	-
12.50	1.49	1.50	1.00	-	-	-	-	-	-	-	-
13.50	1.49	1.50	1.00	-	-	95	80	-	-	95	90
13.60	1.49	1.50	1.00	-	-	-	-	-	-	-	-
14.00	1.49	1.50	1.00	-	-	-	-	-	-	-	-
15.00	1.49	1.50	1.00	-	-	-	-	-	-	97	90
16.50	1.49	1.50	1.00	-	-	-	-	-	-	98	85
16.60	1.49	1.50	1.00	-	-	-	-	-	-	-	-
18.00	1.49	1.50	1.00	-	-	-	-	-	-	90	80
19.50	1.49	1.50	1.00	-	-	-	-	-	-	90	85
21.00	1.49	1.50	1.00	-	-	-	-	-	-	97	90
22.50	1.49	1.50	1.00	-	-	-	-	-	-	90	80
24.00	1.49	1.50	1.00	-	-	-	-	-	-	90	75
25.50	1.49	1.50	1.00	-	-	-	-	-	-	98	90
27.00	1.49	1.50	1.00	-	-	-	-	-	-	85	70
28.50	1.49	1.34	1.00	-	-	-	-	-	-	90	80
30.00	1.49	1.34	1.00	-	-	-	-	-	-	-	-

REMARKS:

1. R: Refusal

2. REC : Core Recovery.

3. RQD : Rock Quality Designation.

TABLE 2: SUMMARY OF SPT N-VALUE, REC & RQD

CLIENT: L & T CONSTRUCTION

PROJECT: MANSOURAH MASSARAH GOLD PROJECT
GEOTECHNICAL INVESTIGATION

PROJ. NO.: 19-K-3054

DEPTH	BH-03				BH-04				BH-05			
(m)	N_{field}	N_{60}	REC	RQD	N_{field}	N_{60}	REC	RQD	N_{field}	N_{60}	REC	RQD
0.00	14	16	-	-	12	14	-	-	27	30	-	-
0.75	36	40	-	-	46	52	-	-	R	R	-	-
1.50	74	83	-	-	R	R	-	-	R	R	-	-
1.60	-	-	-	-	-	-	-	-	-	-	-	-
2.25	77	86	-	-	R	R	-	-	R	R	5	0
2.60	-	-	-	-	-	-	-	-	-	-	-	-
3.00	R	R	-	-	R	R	-	-	R	R	-	-
3.20	-	-	-	-	-	-	-	-	-	-	-	-
3.75	R	R	-	-	R	R	-	-	R	R	-	-
3.80	-	-	-	-	-	-	-	-	-	-	-	-
4.50	R	R	-	-	R	R	-	-	R	R	-	-
4.60	-	-	-	-	-	-	-	-	-	-	-	-
5.00	-	-	18	0	-	-	-	-	-	-	11	0
6.00	R	R	-	-	R	R	-	-	R	R	-	-
6.10	-	-	-	-	-	-	-	-	-	-	-	-
7.50	R	R	-	-	R	R	-	-	R	R	-	-
7.60	-	-	-	-	-	-	-	-	-	-	-	-
8.00	R	R	-	-	-	-	-	-	-	-	-	-
8.50	-	-	-	-	-	-	22	0	-	-	-	-
9.00	-	-	-	-	R	R	-	-	R	R	-	-
9.10	-	-	-	-	-	-	-	-	-	-	-	-
10.50	-	-	-	-	R	R	-	-	R	R	-	-
11.00	-	-	-	-	-	-	-	-	-	-	-	-
12.00	-	-	-	-	R	R	-	-	R	R	-	-
12.10	-	-	-	-	-	-	-	-	-	-	-	-
12.50	-	-	-	-	-	-	-	-	-	-	-	-
13.50	-	-	-	-	R	R	-	-	R	R	-	-
13.60	-	-	-	-	-	-	-	-	-	-	-	-
14.00	-	-	-	-	-	-	-	-	-	-	22	0
15.00	-	-	-	-	R	R	-	-	R	R	-	-
16.50	-	-	-	-	-	-	-	-	-	-	-	-
16.60	-	-	-	-	-	-	-	-	-	-	-	-
18.00	-	-	-	-	-	-	-	-	-	-	-	-
19.50	-	-	-	-	-	-	-	-	-	-	-	-
21.00	-	-	-	-	-	-	-	-	-	-	-	-
22.50	-	-	-	-	-	-	-	-	-	-	-	-
24.00	-	-	-	-	-	-	-	-	-	-	-	-
25.50	-	-	-	-	-	-	-	-	-	-	-	-
27.00	-	-	-	-	-	-	-	-	-	-	-	-
28.50	-	-	-	-	-	-	-	-	-	-	-	-
30.00	-	-	-	-	-	-	-	-	-	-	-	-

REMARKS:

1. R: Refusal

TABLE 2: SUMMARY OF SPT N-VALUE, REC & RQD

CLIENT: L & T CONSTRUCTION

PROJECT: MANSOURAH MASSARAH GOLD PROJECT
GEOTECHNICAL INVESTIGATION

PROJ. NO.: 19-K-3054

DEPTH	BH-06				BH-07		BH-08			
(m)	N_{field}	N_{60}	REC	RQD	N_{field}	N_{60}	N_{field}	N_{60}	REC	RQD
0.00	20	22	-	-	12	13	14	16	-	-
0.75	25	28	-	-	16	18	24	27	-	-
1.50	66	74	-	-	32	36	39	44	-	-
1.60	-	-	-	-	-	-	-	-	-	-
2.25	70	78	-	-	38	43	41	46	-	-
2.60	-	-	-	-	-	-	-	-	-	-
3.00	R	R	-	-	56	63	41	46	-	-
3.20	-	-	-	-	-	-	-	-	-	-
3.75	R	R	-	-	52	58	40	45	-	-
3.80	-	-	-	-	-	-	-	-	-	-
4.50	R	R	-	-	48	61	36	46	-	-
4.60	-	-	-	-	-	-	-	-	-	-
5.00	-	-	-	-	-	-	-	-	-	-
6.00	R	R	-	-	62	88	63	89	-	-
6.10	-	-	-	-	-	-	-	-	-	-
7.50	R	R	-	-	58	82	81	115	-	-
7.60	-	-	-	-	-	-	-	-	-	-
8.00	-	-	-	-	-	-	-	-	-	-
8.50	-	-	-	-	-	-	-	-	-	-
9.00	R	R	-	-	R	R	85	120	-	-
9.10	-	-	22	0	-	-	-	-	-	-
10.50	R	R	-	-	R	R	R	R	-	-
11.00	-	-	-	-	-	-	-	-	-	-
12.00	R	R	-	-	R	R	R	R	-	-
12.10	-	-	18	0	-	-	-	-	-	-
12.50	-	-	-	-	-	-	-	-	29	0
13.50	R	R	-	-	R	R	R	R	-	-
13.60	-	-	-	-	-	-	-	-	-	-
14.00	-	-	-	-	-	-	-	-	-	-
15.00	R	R	-	-	R	R	R	R	-	-
16.50	-	-	-	-	-	-	-	-	-	-
16.60	-	-	-	-	-	-	-	-	-	-
18.00	-	-	-	-	-	-	-	-	-	-
19.50	-	-	-	-	-	-	-	-	-	-
21.00	-	-	-	-	-	-	-	-	-	-
22.50	-	-	-	-	-	-	-	-	-	-
24.00	-	-	-	-	-	-	-	-	-	-
25.50	-	-	-	-	-	-	-	-	-	-
27.00	-	-	-	-	-	-	-	-	-	-
28.50	-	-	-	-	-	-	-	-	-	-
30.00	-	-	-	-	-	-	-	-	-	-

REMARKS:

1. R: Refusal

TABLE 2: SUMMARY OF SPT N-VALUE, REC & RQD

CLIENT: L & T CONSTRUCTION

PROJECT: MANSOURAH MASSARAH GOLD PROJECT
GEOTECHNICAL INVESTIGATION

PROJ. NO.: 19-K-3054

DEPTH	BH-09				BH-10				BH-11			
(m)	N_{field}	N_{60}	REC	RQD	N_{field}	N_{60}	REC	RQD	N_{field}	N_{60}	REC	RQD
0.00	23	26	-	-	32	36	-	-	15	17	-	-
0.75	56	63	-	-	R	R	-	-	22	25	-	-
1.50	66	74	-	-	R	R	-	-	R	R	-	-
1.60	-	-	-	-	-	-	-	-	-	-	-	-
2.25	72	81	-	-	R	R	-	-	R	R	-	-
2.60	-	-	-	-	-	-	-	-	-	-	-	-
3.00	R	R	-	-	R	R	-	-	R	R	-	-
3.20	-	-	-	-	-	-	-	-	-	-	-	-
3.75	R	R	-	-	R	R	-	-	R	R	-	-
3.80	-	-	-	-	-	-	-	-	-	-	-	-
4.50	R	R	-	-	R	R	-	-	R	R	-	-
4.60	-	-	-	-	-	-	-	-	-	-	-	-
5.00	-	-	-	-	-	-	80	60	-	-	-	-
6.00	R	R	-	-	R	R	-	-	R	R	-	-
6.10	-	-	-	-	-	-	-	-	-	-	-	-
7.50	R	R	-	-	R	R	-	-	R	R	-	-
7.60	-	-	-	-	-	-	-	-	-	-	28	0
8.00	-	-	-	-	-	-	-	-	-	-	-	-
8.50	-	-	-	-	-	-	-	-	-	-	-	-
9.00	R	R	-	-	-	-	-	-	R	R	-	-
9.10	-	-	-	-	-	-	-	-	-	-	-	-
10.50	R	R	-	-	-	-	-	-	R	R	-	-
11.00	-	-	-	-	-	-	-	-	-	-	-	-
12.00	R	R	-	-	-	-	-	-	R	R	-	-
12.10	-	-	-	-	-	-	-	-	-	-	-	-
12.50	-	-	-	-	-	-	-	-	-	-	-	-
13.50	R	R	-	-	-	-	-	-	R	R	-	-
13.60	-	-	-	-	-	-	-	-	-	-	-	-
14.00	-	-	-	-	-	-	-	-	-	-	-	-
15.00	R	R	-	-	-	-	-	-	R	R	-	-
16.50	-	-	-	-	-	-	-	-	-	-	-	-
16.60	-	-	-	-	-	-	-	-	-	-	-	-
18.00	-	-	-	-	-	-	-	-	-	-	-	-
19.50	-	-	-	-	-	-	-	-	-	-	-	-
21.00	-	-	-	-	-	-	-	-	-	-	-	-
22.50	-	-	-	-	-	-	-	-	-	-	-	-
24.00	-	-	-	-	-	-	-	-	-	-	-	-
25.50	-	-	-	-	-	-	-	-	-	-	-	-
27.00	-	-	-	-	-	-	-	-	-	-	-	-
28.50	-	-	-	-	-	-	-	-	-	-	-	-
30.00	-	-	-	-	-	-	-	-	-	-	-	-

REMARKS:

1. R: Refusal

TABLE 2: SUMMARY OF SPT N-VALUE, REC & RQD

CLIENT: L & T CONSTRUCTION

PROJECT: MANSOURAH MASSARAH GOLD PROJECT
GEOTECHNICAL INVESTIGATION

PROJ. NO.: 19-K-3054

DEPTH	BH-12				BH-13				BH-14			
(m)	N_{field}	N_{60}	REC	RQD	N_{field}	N_{60}	REC	RQD	N_{field}	N_{60}	REC	RQD
0.00	18	20	-	-	44	49	-	-	26	29	-	-
0.75	42	47	-	-	R	R	-	-	43	48	-	-
1.50	R	R	-	-	R	R	-	-	65	73	-	-
1.60	-	-	-	-	-	-	95	80	-	-	-	-
2.25	R	R	-	-	-	-	-	-	69	77	-	-
2.60	-	-	-	-	-	-	-	-	-	-	-	-
3.00	-	-	70	40	-	-	95	85	R	R	-	-
3.20	-	-	-	-	-	-	-	-	-	-	95	60
3.75	-	-	-	-	-	-	-	-	-	-	-	-
3.80	-	-	-	-	-	-	-	-	-	-	-	-
4.50	-	-	70	30	-	-	80	55	-	-	95	70
4.60	-	-	-	-	-	-	-	-	-	-	-	-
5.00	-	-	-	-	-	-	-	-	-	-	-	-
6.00	-	-	80	30	-	-	90	70	-	-	98	85
6.10	-	-	-	-	-	-	-	-	-	-	-	-
7.50	-	-	70	20	-	-	90	70	-	-	95	80
7.60	-	-	-	-	-	-	-	-	-	-	-	-
8.00	-	-	-	-	-	-	-	-	-	-	-	-
8.50	-	-	-	-	-	-	-	-	-	-	-	-
9.00	-	-	100	70	-	-	90	70	-	-	97	90
9.10	-	-	-	-	-	-	-	-	-	-	-	-
10.50	-	-	95	75	-	-	90	75	-	-	100	95
11.00	-	-	-	-	-	-	-	-	-	-	-	-
12.00	-	-	90	65	-	-	95	85	-	-	98	95
12.10	-	-	-	-	-	-	-	-	-	-	-	-
12.50	-	-	-	-	-	-	-	-	-	-	-	-
13.50	-	-	90	80	-	-	95	80	-	-	100	95
13.60	-	-	-	-	-	-	-	-	-	-	-	-
14.00	-	-	-	-	-	-	-	-	-	-	-	-
15.00	-	-	-	-	-	-	-	-	-	-	-	-
16.50	-	-	-	-	-	-	-	-	-	-	-	-
16.60	-	-	-	-	-	-	-	-	-	-	-	-
18.00	-	-	-	-	-	-	-	-	-	-	-	-
19.50	-	-	-	-	-	-	-	-	-	-	-	-
21.00	-	-	-	-	-	-	-	-	-	-	-	-
22.50	-	-	-	-	-	-	-	-	-	-	-	-
24.00	-	-	-	-	-	-	-	-	-	-	-	-
25.50	-	-	-	-	-	-	-	-	-	-	-	-
27.00	-	-	-	-	-	-	-	-	-	-	-	-
28.50	-	-	-	-	-	-	-	-	-	-	-	-
30.00	-	-	-	-	-	-	-	-	-	-	-	-

REMARKS:

1. R: Refusal

TABLE 2: SUMMARY OF SPT N-VALUE, REC & RQD

CLIENT: L & T CONSTRUCTION

PROJECT: MANSOURAH MASSARAH GOLD PROJECT
GEOTECHNICAL INVESTIGATION

PROJ. NO.: 19-K-3054

DEPTH	BH-15				BH-16				BH-17			
(m)	N_{field}	N_{60}	REC	RQD	N_{field}	N_{60}	REC	RQD	N_{field}	N_{60}	REC	RQD
0.00	9	10	-	-	15	17	-	-	14	16	-	-
0.75	25	28	-	-	31	35	-	-	20	23	-	-
1.50	R	R	-	-	48	54	-	-	R	R	-	-
1.60	-	-	-	-	-	-	-	-	-	-	-	-
2.25	R	R	-	-	62	69	-	-	R	R	-	-
2.60	-	-	90	80	-	-	-	-	-	-	-	-
3.00	-	-	97	80	R	R	-	-	R	R	-	-
3.20	-	-	-	-	-	-	-	-	-	-	-	-
3.75	-	-	-	-	73	82	-	-	R	R	-	-
3.80	-	-	-	-	-	-	-	-	-	-	-	-
4.50	-	-	98	90	R	R	-	-	R	R	-	-
4.60	-	-	-	-	-	-	-	-	-	-	-	-
5.00	-	-	-	-	-	-	-	-	-	-	-	-
6.00	-	-	100	95	R	R	-	-	R	R	-	-
6.10	-	-	-	-	-	-	-	-	-	-	-	-
7.50	-	-	95	90	R	R	-	-	R	R	-	-
7.60	-	-	-	-	-	-	-	-	-	-	-	-
8.00	-	-	-	-	-	-	-	-	-	-	-	-
8.50	-	-	-	-	-	-	-	-	-	-	-	-
9.00	-	-	95	90	R	R	-	-	R	R	-	-
9.10	-	-	-	-	-	-	-	-	-	-	100	70
10.50	-	-	90	70	R	R	-	-	-	-	95	60
11.00	-	-	-	-	-	-	95	80	-	-	-	-
12.00	-	-	95	75	-	-	90	75	-	-	90	60
12.10	-	-	-	-	-	-	-	-	-	-	-	-
12.50	-	-	-	-	-	-	-	-	-	-	-	-
13.50	-	-	95	80	-	-	90	80	-	-	80	60
13.60	-	-	-	-	-	-	-	-	-	-	-	-
14.00	-	-	-	-	-	-	-	-	-	-	-	-
15.00	-	-	98	90	-	-	-	-	-	-	-	-
16.50	-	-	95	80	-	-	-	-	-	-	-	-
16.60	-	-	-	-	-	-	-	-	-	-	-	-
18.00	-	-	90	70	-	-	-	-	-	-	-	-
19.50	-	-	90	65	-	-	-	-	-	-	-	-
21.00	-	-	90	70	-	-	-	-	-	-	-	-
22.50	-	-	85	60	-	-	-	-	-	-	-	-
24.00	-	-	80	55	-	-	-	-	-	-	-	-
25.50	-	-	85	65	-	-	-	-	-	-	-	-
27.00	-	-	90	70	-	-	-	-	-	-	-	-
28.50	-	-	90	60	-	-	-	-	-	-	-	-
30.00	-	-	-	-	-	-	-	-	-	-	-	-

REMARKS:

1. R: Refusal

3. RQD : Rock Quality Designation.

TABLE 2: SUMMARY OF SPT N-VALUE, REC & RQD

CLIENT: L & T CONSTRUCTION

PROJECT: MANSOURAH MASSARAH GOLD PROJECT
GEOTECHNICAL INVESTIGATION

PROJ. NO.: 19-K-3054

DEPTH	BH-18				BH-19				BH-20			
(m)	N_{field}	N_{60}	REC	RQD	N_{field}	N_{60}	REC	RQD	N_{field}	N_{60}	REC	RQD
0.00	18	20	-	-	8	9	-	-	15	17	-	-
0.75	61	68	-	-	42	47	-	-	36	40	-	-
1.50	R	R	-	-	81	91	-	-	71	79	-	-
1.60	-	-	-	-	-	-	-	-	-	-	-	-
2.25	R	R	-	-	R	R	-	-	R	R	-	-
2.60	-	-	-	-	-	-	-	-	-	-	-	-
3.00	R	R	-	-	R	R	-	-	R	R	-	-
3.20	-	-	-	-	-	-	-	-	-	-	90	75
3.75	R	R	-	-	R	R	-	-	-	-	-	-
3.80	-	-	-	-	-	-	-	-	-	-	-	-
4.50	R	R	-	-	R	R	-	-	-	-	90	65
4.60	-	-	-	-	-	-	-	-	-	-	-	-
5.00	-	-	-	-	-	-	-	-	-	-	-	-
6.00	R	R	-	-	R	R	-	-	-	-	95	75
6.10	-	-	-	-	-	-	-	-	-	-	-	-
7.50	R	R	-	-	R	R	-	-	-	-	100	80
7.60	-	-	-	-	-	-	95	80	-	-	-	-
8.00	-	-	-	-	-	-	-	-	-	-	-	-
8.50	-	-	-	-	-	-	-	-	-	-	-	-
9.00	R	R	-	-	-	-	90	80	-	-	98	90
9.10	-	-	-	-	-	-	-	-	-	-	-	-
10.50	R	R	-	-	-	-	95	80	-	-	100	95
11.00	-	-	-	-	-	-	-	-	-	-	-	-
12.00	R	R	-	-	-	-	95	90	-	-	100	95
12.10	-	-	-	-	-	-	-	-	-	-	-	-
12.50	-	-	-	-	-	-	-	-	-	-	-	-
13.50	R	R	-	-	-	-	-	-	-	-	100	98
13.60	-	-	90	70	-	-	-	-	-	-	-	-
14.00	-	-	-	-	-	-	-	-	-	-	-	-
15.00	-	-	-	-	R	R	-	-	-	-	-	-
16.50	-	-	-	-	R	R	-	-	-	-	-	-
16.60	-	-	-	-	-	-	90	75	-	-	-	-
18.00	-	-	-	-	-	-	90	80	-	-	-	-
19.50	-	-	-	-	-	-	95	85	-	-	-	-
21.00	-	-	-	-	-	-	98	80	-	-	-	-
22.50	-	-	-	-	-	-	98	90	-	-	-	-
24.00	-	-	-	-	-	-	95	70	-	-	-	-
25.50	-	-	-	-	-	-	90	65	-	-	-	-
27.00	-	-	-	-	-	-	90	60	-	-	-	-
28.50	-	-	-	-	-	-	90	65	-	-	-	-
30.00	-	-	-	-	-	-	-	-	-	-	-	-

REMARKS:

1. R: Refusal

TABLE 2: SUMMARY OF SPT N-VALUE, REC & RQD

CLIENT: L & T CONSTRUCTION

PROJECT: MANSOURAH MASSARAH GOLD PROJECT
GEOTECHNICAL INVESTIGATION

PROJ. NO.: 19-K-3054

DEPTH	BH-21				BH-22				BH-23			
(m)	N_{field}	N_{60}	REC	RQD	N_{field}	N_{60}	REC	RQD	N_{field}	N_{60}	REC	RQD
0.00	11	12	-	-	20	22	-	-	33	37	-	-
0.75	19	21	-	-	25	28	-	-	R	R	-	-
1.50	25	28	-	-	44	49	-	-	R	R	-	-
1.60	-	-	-	-	-	-	-	-	-	-	-	-
2.25	R	R	-	-	46	51	-	-	R	R	-	-
2.60	-	-	-	-	-	-	-	-	-	-	-	-
3.00	R	R	-	-	48	54	-	-	R	R	-	-
3.20	-	-	-	-	-	-	-	-	-	-	-	-
3.75	R	R	-	-	60	67	-	-	R	R	-	-
3.80	-	-	-	-	-	-	-	-	-	-	-	-
4.50	R	R	-	-	78	99	-	-	R	R	-	-
4.60	-	-	-	-	-	-	-	-	-	-	-	-
5.00	-	-	-	-	-	-	-	-	-	-	-	-
6.00	R	R	-	-	80	113	-	-	R	R	-	-
6.10	-	-	-	-	-	-	-	-	-	-	95	75
7.50	R	R	-	-	R	R	-	-	-	-	90	75
7.60	-	-	-	-	-	-	-	-	-	-	-	-
8.00	-	-	-	-	-	-	30	0	-	-	-	-
8.50	-	-	-	-	-	-	-	-	-	-	-	-
9.00	-	-	90	55	R	R	-	-	-	-	90	80
9.10	-	-	-	-	-	-	-	-	-	-	-	-
10.50	-	-	95	65	R	R	-	-	-	-	95	90
11.00	-	-	-	-	-	-	45	0	-	-	-	-
12.00	-	-	95	75	R	R	-	-	-	-	97	90
12.10	-	-	-	-	-	-	-	-	-	-	-	-
12.50	-	-	-	-	-	-	-	-	-	-	-	-
13.50	-	-	90	70	R	R	-	-	-	-	95	90
13.60	-	-	-	-	-	-	-	-	-	-	-	-
14.00	-	-	-	-	-	-	-	-	-	-	-	-
15.00	-	-	-	-	R	R	-	-	-	-	-	-
16.50	-	-	-	-	-	-	-	-	-	-	-	-
16.60	-	-	-	-	-	-	-	-	-	-	-	-
18.00	-	-	-	-	-	-	-	-	-	-	-	-
19.50	-	-	-	-	-	-	-	-	-	-	-	-
21.00	-	-	-	-	-	-	-	-	-	-	-	-
22.50	-	-	-	-	-	-	-	-	-	-	-	-
24.00	-	-	-	-	-	-	-	-	-	-	-	-
25.50	-	-	-	-	-	-	-	-	-	-	-	-
27.00	-	-	-	-	-	-	-	-	-	-	-	-
28.50	-	-	-	-	-	-	-	-	-	-	-	-
30.00	-	-	-	-	-	-	-	-	-	-	-	-

REMARKS:

1. R: Refusal

TABLE 2: SUMMARY OF SPT N-VALUE, REC & RQD

CLIENT: L & T CONSTRUCTION

PROJECT: MANSOURAH MASSARAH GOLD PROJECT
GEOTECHNICAL INVESTIGATION

PROJ. NO.: 19-K-3054

DEPTH	BH-25				BH-26				BH-27		BH-28		BH-29	
(m)	N_{field}	N_{60}	REC	RQD	N_{field}	N_{60}	REC	RQD	N_{field}	N_{60}	N_{field}	N_{60}	N_{field}	N_{60}
0.00	9	10	-	-	18	20	-	-	13	15	9	10	15	17
0.75	26	29	-	-	76	85	-	-	31	35	22	25	31	35
1.50	R	R	-	-	R	R	-	-	48	54	33	37	62	69
1.60	-	-	-	-	-	-	-	-	-	-	-	-	-	-
2.25	R	R	-	-	R	R	-	-	R	R	36	40	64	72
2.60	-	-	-	-	-	-	-	-	-	-	-	-	-	-
3.00	R	R	-	-	R	R	-	-	R	R	36	40	62	69
3.20	-	-	-	-	-	-	-	-	-	-	-	-	-	-
3.75	R	R	-	-	R	R	-	-	R	R	35	39	78	87
3.80	-	-	-	-	-	-	-	-	-	-	-	-	-	-
4.50	R	R	-	-	R	R	-	-	R	R	38	48	R	R
4.60	-	-	-	-	-	-	-	-	-	-	-	-	-	-
5.00	-	-	31	0	-	-	-	-	-	-	-	-	-	-
6.00	R	R	-	-	R	R	-	-	R	R	43	61	R	R
6.10	-	-	-	-	-	-	-	-	-	-	-	-	-	-
7.50	R	R	-	-	R	R	-	-	R	R	72	102	R	R
7.60	-	-	-	-	-	-	-	-	-	-	-	-	-	-
8.00	-	-	-	-	-	-	-	-	R	R	-	-	-	-
8.50	-	-	-	-	-	-	-	-	-	-	-	-	-	-
9.00	R	R	-	-	R	R	-	-	-	-	62	88	R	R
9.10	-	-	70	40	-	-	-	-	-	-	-	-	-	-
10.50	-	-	80	60	R	R	-	-	-	-	49	73	R	R
11.00	-	-	-	-	-	-	-	-	-	-	-	-	-	-
12.00	-	-	90	65	R	R	-	-	-	-	57	85	R	R
12.10	-	-	-	-	-	-	-	-	-	-	-	-	-	-
12.50	-	-	-	-	-	-	90	70	-	-	-	-	-	-
13.50	-	-	90	70	-	-	95	85	-	-	60	90	R	R
13.60	-	-	-	-	-	-	-	-	-	-	-	-	-	-
14.00	-	-	-	-	-	-	-	-	-	-	-	-	-	-
15.00	-	-	-	-	-	-	-	-	-	-	63	94	R	R
16.50	-	-	-	-	-	-	-	-	-	-	-	-	-	-
16.60	-	-	-	-	-	-	-	-	-	-	-	-	-	-
18.00	-	-	-	-	-	-	-	-	-	-	-	-	-	-
19.50	-	-	-	-	-	-	-	-	-	-	-	-	-	-
21.00	-	-	-	-	-	-	-	-	-	-	-	-	-	-
22.50	-	-	-	-	-	-	-	-	-	-	-	-	-	-
24.00	-	-	-	-	-	-	-	-	-	-	-	-	-	-
25.50	-	-	-	-	-	-	-	-	-	-	-	-	-	-
27.00	-	-	-	-	-	-	-	-	-	-	-	-	-	-
28.50	-	-	-	-	-	-	-	-	-	-	-	-	-	-
30.00	-	-	-	-	-	-	-	-	-	-	-	-	-	-

REMARKS:

1. R: Refusal

TABLE 3 : BOREHOLE DEPTH & GROUND WATER MEASUREMENTS

CLIENT: L & T CONSTRUCTION

PROJECT: MANSOURAH MASSARAH GOLD PROJECT
GEOTECHNICAL INVESTIGATION

PROJ. NO.: 19-K-3054

Description	UTM Coordinates (m)		Borehole Depth (m)	Elevation (m)	GW Depth (m)	GW Elevation (m)
	Northing	Easting				
BH-01	2506230.93	263768.94	15.00	915.20	NE	-
BH-02	2506179.07	264210.66	30.00	912.93	23.00	889.93
BH-03	2503834.17	265464.64	8.00	905.40	NE	-
BH-04	2506042.96	264344.10	15.00	911.98	NE	-
BH-05	2506162.43	263969.23	15.00	913.20	NE	-
BH-06	2506071.23	263996.99	15.00	913.04	NE	-
BH-07	2506130.93	264068.72	15.00	912.94	NE	-
BH-08	2505913.12	264072.55	15.00	912.87	NE	-
BH-09	2506010.37	264138.91	15.00	912.86	NE	-
BH-10	2506142.03	264155.85	15.00	913.35	NE	-
BH-11	2506094.39	264208.64	15.00	912.70	NE	-
BH-12	2505967.37	264193.58	15.00	912.66	NE	-
BH-13	2505915.36	264216.03	15.00	912.52	NE	-
BH-14	2505878.50	264355.99	15.00	911.87	NE	-

TABLE 3 : BOREHOLE DEPTH & GROUND WATER MEASUREMENTS

CLIENT: L & T CONSTRUCTION

PROJECT: MANSOURAH MASSARAH GOLD PROJECT
GEOTECHNICAL INVESTIGATION

PROJ. NO.: 19-K-3054

Description	UTM Coordinates (m)		Borehole Depth (m)	Elevation (m)	GW Depth (m)	GW Elevation (m)
	Northing	Easting				
BH-15	2506027.89	264280.82	30.00	912.50	24.00	888.50
BH-16	2506135.79	264290.48	15.00	912.07	NE	-
BH-17	2506276.17	264362.88	15.00	912.15	NE	-
BH-18	2506423.39	264270.89	15.00	912.46	NE	-
BH-19	2506510.84	264424.58	30.00	912.34	24.00	888.34
BH-20	2506027.19	264219.07	15.00	912.52	NE	-
BH-21	2508267.29	263178.80	15.00	921.35	NE	-
BH-22	2509287.33	265495.28	15.00	911.50	NE	-
BH-23	2505158.98	264306.20	15.00	911.15	NE	-
BH-25	2506297.89	264397.65	15.00	912.25	NE	-
BH-26	2506130.16	264334.91	15.00	911.85	NE	-
BH-27	2506067.53	264289.07	8.00	912.25	NE	-
BH-28	2506033.19	263964.39	15.00	913.15	NE	-
BH-29	2506526.60	264367.17	15.00	912.50	NE	-

TABLE 4: RECOMMENDED NET ALLOWABLE BEARING CAPACITIES

PRPJECT: MANSOURAH - MANSSARAH GOLD PROJECT
CLIENT: L & T CONSTRUCTION

Area	Borehole No.	Proposed Structures	Foundation Type	Depth of Foundation below FGL	Footing Size (B x L)	Net Allowable Bearing Capacity Based on		Recommended Net Allowable Bearing Capacity	Estimated Total Settlement	Vertical Modulus of Subgrade Reaction (Ks)	Horizontal Modulus of Subgrade Reaction (Kh)
						Settlement	Shear Failure				
				(m)	(m)	(kN/m^2)	(kN/m^2)	(kN/m^2)	(mm)	($kN/m^2/m$)	(kN/m^2)
Crushing Plant Area	BH-17, 18, 19, 25 & 29	-	Isolated Square	1.20	1.0 x 1.0	950	570	570	25	68,400	32,200
		-		1.50		1000	700	700	25	84,000	38,900
		-		1.20	2.0 x 2.0	850	700	700	25	84,000	37,500
		-		1.50		700	980	700	25	84,000	44,200
		-		1.20	3.0 x 3.0	700	900	700	25	84,000	42,800
		-		1.50		700	1120	700	25	84,000	49,500
		-		1.20	4.0 x 4.0	650	1040	650	25	78,000	48,100
		-		1.50		600	1250	600	25	72,000	54,800
		-		1.20	5.0 x 5.0	550	1172	550	25	66,000	53,400
		-		1.50		500	1342	500	25	45,000	60,200
	BH-17	Crushed ore stockpile	Ring Foundation / Raft	1.20	Dia.= 28m	550	*	550	50	49,500	175,000
				1.50		600	*	600	50	54,000	180,000
	BH-19	Primary crushing	Mat	2.50	5.0 x 24.0	900	*	900	50	81,000	79,900
				2.50	8.0 x 24.0	750	*	750	50	67,500	103,300
				2.50	10.0 x 40.0	700	*	700	50	63,000	119,800
	BH-25	Crushed ore stockpile	Ring Foundation / Raft	1.20	Dia. 28m	550	*	550	50	49,500	175,000
				1.50		600	*	600	50	54,000	180,000

* Not governing

TABLE 5: RECOMMENDED NET ALLOWABLE BEARING CAPACITIES

PRPJECT: MANSOURAH - MANSSARAH GOLD PROJECT
CLIENT: L & T CONSTRUCTION

Area	Borehole No.	Proposed Structures	Foundation Type	Depth of Foundation below FGL	Footing Size (B x L)	Net Allowable Bearing Capacity Based on		Recommended Net Allowable Bearing Capacity	Estimated Total Settlement	Vertical Modulus of Subgrade Reaction (Ks)	Horizontal Modulus of Subgrade Reaction (Kh)
						Settlement	Shear Failure				
				(m)	(m)	(kN/m^2)	(kN/m^2)	(kN/m^2)	(mm)	(kN/m^2/m)	(kN/m^2)
Process Plant Area-1 (Zone-1)	BH-04, 06 to 11, 16, 26 & 27	-	Isolated Square	1.20	1.0 x 1.0	750	380	380	25	45,600	32,200
		-		1.50		1000	510	510	25	61,200	38,900
		-		1.20	2.0 x 2.0	500	480	480	25	57,600	37,500
		-		1.50		600	590	590	25	70,800	44,200
		-		1.20	3.0 x 3.0	400	560	400	25	48,000	42,800
		-		1.50		450	670	450	25	54,000	49,500
		-		1.20	4.0 x 4.0	350	640	350	25	42,000	48,100
		-		1.50		375	750	375	25	45,000	54,800
		-		1.20	5.0 x 5.0	300	740	300	25	36,000	53,400
		-		1.50		325	870	325	25	39,000	60,200
	BH-06	Power Plant	Rectangular	1.20	3.0 x 5.0	425	680	425	25	51,000	43,000
				1.50		450	800	450	25	54,000	48,500
	BH-07	Concentrate filtration	Rectangular	1.20	4.0 x 6.0	350	800	350	25	42,000	49,500
				1.50		400	920	400	25	48,000	55,300
	BH-08	Raw Water Tank	Ring Foundation /Raft	0.60	Dia. = 27.5m	375	*	375	50	33,750	155,000
				1.20		425	*	425	50	38,250	170,000
				1.50		450	*	450	50	40,500	175,000

TABLE 5: RECOMMENDED NET ALLOWABLE BEARING CAPACITIES

PRPJECT: MANSOURAH - MANSSARAH GOLD PROJECT
CLIENT: L & T CONSTRUCTION

Area	Borehole No.	Proposed Structures	Foundation Type	Depth of Foundation below FGL	Footing Size (B x L)	Net Allowable Bearing Capacity Based on		Recommended Net Allowable Bearing Capacity	Estimated Total Settlement	Vertical Modulus of Subgrade Reaction (Ks)	Horizontal Modulus of Subgrade Reaction (Kh)
						Settlement	Shear Failure				
				(m)	(m)	(kN/m^2)	(kN/m^2)	(kN/m^2)	(mm)	($kN/m^2/m$)	(kN/m^2)
Process Plant Area-1	BH-09	Pressure oxidation	Rectangular	3.00	4.0 x 6.0	475	1490	475	25	57,000	84,800
	BH-10	Flotation	Rectangular	3.00	4.0 x 4.0	525	1570	525	25	63,000	88,500
	BH-11	Pox leaching	combined raft	1.20	13.0 x 27.0	450	*	450	50	40,500	114,400
				1.50		500	*	500	50	45,000	119,800
	BH-16	Low grade thickening	Raft	1.20	30.0 x 80.0	450	*	450	50	40,500	245,000
				1.50		450	*	450	50	40,500	250,000
	BH-26	Low grade thickening	Rectangular	1.20	3.0 x 4.0	450	680	450	25	54,000	42,000
				1.50		500	810	500	25	60,000	49,000
			Square	1.20	6.0 x 6.0	325	1030	325	25	39,000	58,000
				1.50		350	1200	350	25	42,000	65,500
		Flotation & Process Water Storage Tank	Ring Foundation /Raft	0.60	Dia. = 30m	375	*	375	50	33,750	186,000
	BH-27	Low grade thickening	Raft	1.20	30.0 x 80.0	450	*	450	50	40,500	245,000
				1.50		475	*	475	50	42,750	250,000

* Not governing

TABLE 6: RECOMMENDED NET ALLOWABLE BEARING CAPACITIES

PRPJECT: MANSOURAH - MANSSARAH GOLD PROJECT
CLIENT: L & T CONSTRUCTION

Area	Borehole No.	Proposed Structures	Foundation Type	Depth of Foundation below FGL	Footing Size (B x L)	Net Allowable Bearing Capacity Based on		Recommended Net Allowable Bearing Capacity	Estimated Total Settlement	Vertical Modulus of Subgrade Reaction (Ks)	Horizontal Modulus of Subgrade Reaction (Kh)
						Settlement	Shear Failure				
				(m)	(m)	(kN/m^2)	(kN/m^2)	(kN/m^2)	(mm)	($kN/m^2/m$)	(kN/m^2)
Process Plant Area-1 (Zone-2)	BH-02, 12 to 15, & 20	-	Isolated Square	1.20	1.0 x 1.0	1600	570	570	25	68,400	32,200
		-		1.50		2700	850	850	25	102,000	38,900
		-		1.20	2.0 x 2.0	1250	750	750	25	90,000	37,500
		-		1.50		1800	980	980	25	117,600	44,200
		-		1.20	3.0 x 3.0	1100	900	900	25	108,000	42,800
		-		1.50		1500	1120	1120	25	134,400	49,500
		-		1.20	4.0 x 4.0	1000	1040	1000	25	120,000	48,100
		-		1.50		1300	1250	1250	25	150,000	54,800
		-		1.20	5.0 x 5.0	900	1172	900	25	108,000	53,400
		-		1.50		1200	1342	1200	25	144,000	60,200
	BH-02	Ball Mill	Raft	2.50	18.0 x 25.0	1300	*	1300	50	117,000	137,000
				3.00		1350	*	1350	50	121,500	143,000
	BH-12	CCD	Rectangular	1.20	3.0 x 4.0	1000	900	900	25	81,000	42,000
				1.50		1400	1120	1120	25	100,800	49,000
	BH-13	Neutralization and thickening	combined raft	1.20	17.0 x 23.0	1000	*	1000	50	90,000	130,000
				1.50		1200	*	1200	50	108,000	136,000
	BH-14	Tailing Paste Thickening	Ring wall	1.20	Dia. = 40m	800	*	800	50	72,000	235,000
				1.50		950	*	950	50	85,500	240,000
	BH-15	Primary cyanide destruction	combined raft	1.20	28.0 x 98.5.0	800	*	800	50	72,000	235,000
				1.50		900	*	900	50	81,000	240,000
	BH-20	Pox leaching	combined raft	1.20	13.0 x 27.0	1100	*	1100	50	99,000	114,000
				1.50		1250	*	1250	50	112,500	119,000

* Not governing

TABLE 7: RECOMMENDED NET ALLOWABLE BEARING CAPACITIES

PRPJECT: MANSOURAH - MANSSARAH GOLD PROJECT
CLIENT: L & T CONSTRUCTION

Area	Borehole No.	Proposed Structures	Foundation Type	Depth of Foundation below FGL	Footing Size (B x L)	Net Allowable Bearing Capacity Based on		Recommended Net Allowable Bearing Capacity	Estimated Total Settlement	Vertical Modulus of Subgrade Reaction (Ks)	Horizontal Modulus of Subgrade Reaction (Kh)
						Settlement	Shear Failure				
				(m)	(m)	(kN/m^2)	(kN/m^2)	(kN/m^2)	(mm)	($kN/m^2/m$)	(kN/m^2)
Process Plant Area-1 (BH-28)	BH-28	Power Plant	Isolated Square	1.20	1.0 x 1.0	900	400	400	25	48,000	32,200
				1.50		1000	510	510	25	61,200	38,900
				1.20	2.0 x 2.0	575	490	490	25	58,800	37,500
				1.50		600	590	590	25	70,800	44,200
				1.20	3.0 x 3.0	450	570	450	25	54,000	42,800
				1.50		475	670	475	25	57,000	49,500
				2.00		500	820	500	25	60,000	60,800
				1.20	4.0 x 4.0	375	650	375	25	45,000	48,100
				1.50		400	750	400	25	48,000	54,800
				1.20	5.0 x 5.0	325	740	325	25	39,000	53,400
				1.50		350	870	350	25	42,000	60,200
			Rectangular	1.20	3.0 x 5.0	650	525	525	25	63,000	42,800
				1.50		750	650	650	25	78,000	49,500

TABLE 8: RECOMMENDED NET ALLOWABLE BEARING CAPACITIES

PRPJECT: MANSOURAH - MANSSARAH GOLD PROJECT
CLIENT: L & T CONSTRUCTION

Area	Borehole No.	Proposed Structures	Foundation Type	Depth of Foundation below FGL	Footing Size (B x L)	Net Allowable Bearing Capacity Based on		Recommended Net Allowable Bearing Capacity	Estimated Total Settlement	Vertical Modulus of Subgrade Reaction (Ks)	Horizontal Modulus of Subgrade Reaction (Kh)
						Settlement	Shear Failure				
				(m)	(m)	(kN/m^2)	(kN/m^2)	(kN/m^2)	(mm)	($kN/m^2/m$)	(kN/m^2)
Processing Plant-2	BH-01	Admin Building	Isolated Square	1.20	1.0 x 1.0	1500	570	570	25	68,400	32,200
				1.50		2000	850	850	25	102,000	38,900
				1.20	2.0 x 2.0	1000	750	750	25	90,000	37,500
				1.50		1200	850	850	25	102,000	44,200
				1.20	3.0 x 3.0	775	900	775	25	93,000	42,800
				1.50		950	850	850	25	102,000	49,500
				1.20	4.0 x 4.0	625	1250	625	25	75,000	48,100
				1.50		750	1172	750	25	90,000	54,800
				1.20	5.0 x 5.0	525	1342	525	25	63,000	53,400
				1.50		600	870	600	25	72,000	60,200

TABLE 9: RECOMMENDED NET ALLOWABLE BEARING CAPACITIES

PRPJECT: MANSOURAH - MANSSARAH GOLD PROJECT
CLIENT: L & T CONSTRUCTION

Area	Borehole No.	Proposed Structures	Foundation Type	Depth of Foundation below FGL	Footing Size (B x L)	Net Allowable Bearing Capacity Based on		Recommended Net Allowable Bearing Capacity	Estimated Total Settlement	Vertical Modulus of Subgrade Reaction (Ks)	Horizontal Modulus of Subgrade Reaction (Kh)
						Settlement	Shear Failure				
				(m)	(m)	(kN/m^2)	(kN/m^2)	(kN/m^2)	(mm)	($kN/m^2/m$)	(kN/m^2)
Solar Plant & Workshop Area	BH-05	Workshop	Isolated Square	1.20	1.0 x 1.0	3200	410	410	25	49,200	32,200
				1.50		3250	750	750	25	90,000	38,900
				1.20	2.0 x 2.0	1700	720	720	25	86,400	37,500
				1.50		1700	880	880	25	105,600	44,200
				1.20	3.0 x 3.0	1200	870	870	25	104,400	42,800
				1.50		1200	920	920	25	110,400	49,500
				1.20	4.0 x 4.0	850	1100	850	25	102,000	48,100
				1.50		850	1250	850	25	102,000	54,800
				1.20	5.0 x 5.0	850	1240	850	25	102,000	53,400
				1.50		850	1380	850	25	102,000	60,200
			Rectangular	1.20	4.0 x 6.0	850	1130	850	26	102,000	48,100
				1.50		850	1260	850	27	102,000	54,800

TABLE 10: RECOMMENDED NET ALLOWABLE BEARING CAPACITIES

PRPJECT: MANSOURAH - MANSSARAH GOLD PROJECT
CLIENT: L & T CONSTRUCTION

Area	Borehole No.	Proposed Structures	Foundation Type	Depth of Foundation below FGL	Footing Size (B x L)	Net Allowable Bearing Capacity Based on		Recommended Net Allowable Bearing Capacity	Estimated Total Settlement	Vertical Modulus of Subgrade Reaction (Ks)	Horizontal Modulus of Subgrade Reaction (Kh)
						Settlement	Shear Failure				
				(m)	(m)	(kN/m^2)	(kN/m^2)	(kN/m^2)	(mm)	($kN/m^2/m$)	(kN/m^2)
Village Area	BH-21	Village	Isolated Square	0.85	0.35 x 0.35	1500	225	225	25	27,000	20,900
				1.20	1.0 x 1.0	1000	350	350	25	42,000	32,200
				1.50		1100	500	500	25	60,000	38,900
				1.20	2.0 x 2.0	700	580	580	25	69,600	37,500
				1.50		850	750	750	25	90,000	44,200
				1.20	3.0 x 3.0	600	750	600	25	72,000	42,800
				1.50		700	930	700	25	84,000	49,500
				1.20	4.0 x 4.0	550	910	550	25	66,000	48,100
				1.50		625	1090	625	25	75,000	54,800
				1.20	5.0 x 5.0	475	1060	475	25	57,000	53,400
				1.50		550	1240	550	25	66,000	60,200

TABLE 11: RECOMMENDED NET ALLOWABLE BEARING CAPACITIES

PRPJECT: MANSOURAH - MANSSARAH GOLD PROJECT
CLIENT: L & T CONSTRUCTION

Area	Borehole No.	Proposed Structures	Foundation Type	Depth of Foundation below FGL	Footing Size (B x L)	Net Allowable Bearing Capacity Based on		Recommended Net Allowable Bearing Capacity	Estimated Total Settlement	Vertical Modulus of Subgrade Reaction (Ks)	Horizontal Modulus of Subgrade Reaction (Kh)
						Settlement	Shear Failure				
				(m)	(m)	(kN/m^2)	(kN/m^2)	(kN/m^2)	(mm)	($kN/m^2/m$)	(kN/m^2)
Explosive Magazine Area	BH-22	Explosive Magazine	Isolated Square	0.85	0.35 x 0.35	2000	225	225	25	27,000	20,900
				1.20	1.0 x 1.0	1400	440	440	25	52,800	32,200
				1.50		1800	600	600	25	72,000	38,900
				1.20	2.0 x 2.0	900	550	550	25	66,000	37,500
				1.50		1050	690	690	25	82,800	44,200
				1.20	3.0 x 3.0	700	650	650	25	78,000	42,800
				1.50		750	780	750	25	90,000	49,500
				1.20	4.0 x 4.0	575	800	575	25	69,000	48,100
				1.50		625	950	625	25	75,000	54,800
				1.20	5.0 x 5.0	500	950	500	25	60,000	53,400
				1.50		550	1110	550	25	66,000	60,200

TABLE 12: RECOMMENDED NET ALLOWABLE BEARING CAPACITIES

PRPJECT: MANSOURAH - MANSSARAH GOLD PROJECT
CLIENT: L & T CONSTRUCTION

Area	Borehole No.	Proposed Structures	Foundation Type	Depth of Foundation below FGL	Footing Size (B x L)	Net Allowable Bearing Capacity Based on		Recommended Net Allowable Bearing Capacity	Estimated Total Settlement	Vertical Modulus of Subgrade Reaction (Ks)	Horizontal Modulus of Subgrade Reaction (Kh)
						Settlement	Shear Failure				
				(m)	(m)	(kN/m^2)	(kN/m^2)	(kN/m^2)	(mm)	($kN/m^2/m$)	(kN/m^2)
TMT Area	BH-03 & BH-23	TMT	Isolated Square	0.85	0.35 x 0.35	2200	235	235	25	28,200	20,900
				1.20	1.0 x 1.0	2000	580	580	25	69,600	32,200
				1.50		2500	850	850	25	102,000	38,900
				1.20	2.0 x 2.0	1250	760	760	25	91,200	37,500
				1.50		850	980	850	25	102,000	44,200
				1.20	3.0 x 3.0	850	910	850	25	102,000	42,800
				1.50		850	1120	850	25	102,000	49,500
				1.20	4.0 x 4.0	750	1050	750	25	90,000	48,100
				1.50		825	1250	825	25	99,000	54,800
				1.20	5.0 x 5.0	650	1190	650	25	78,000	53,400
				1.50		700	1380	700	25	84,000	60,200

FIGURES

FIGURE 1 : VICINITY MAP

الرياض للجيوتقنية والأساسات (ش ذ م م) RIYADH GEOTECHNIQUE & FOUNDATIONS (LLC) AL-KHOBAR	PROJECT : MANSOURAH - MASSARAH GOLD PROJECT CLIENT : L & T CONSTRUCTION PROJECT # : 19-K-3054

DRAWING NUMBER: MD-216-1000-EG-CV-LAD-1001 REV A03

LEGENDS

AREA	DESCRIPTION
1	WEIGHBRIDGE
2	NOT USED
3	WAREHOUSE OUTSIDE LAYDOWN
4	NOT USED
5	WORKSHOP
6	LABORATORY
7	ENGINEERING OFFICE
8	CONTROL ROOM
9	NOT USED
25	ACIDIC EVENT POND
26	CYANIDE EVENT POND
27	PASTE THICKENER EVENT POND
37	BUS STOP
38	SECURITY GATE 5
39	MOSQUE
43	ABLUTION
44	ADMIN BUILDING
45	EMERGENCY SERVICES BUILDING
46	NOT USED
47	COMMUNICATION TOWER

CRUSHING PLANT AREA

PROCESS PLANT AREA-1 (ZONE-1)

PROCESS PLANT AREA-1 (ZONE-2)

PROCESS PLANT AREA-1 (POWER PLANT)

SOLAR PLANT & WORKSHOP AREA

PROCESS PLANT AREA-2

ROM PAD

S16

PROCESS AREA (3000)
1:2000

LEGENDS

3000	PROCESS PLANT
4100	ACCOMODATION VILLAGE
50	TAILING POND
51	MINING OPERATION MAINTENANCE (BY OTHERS)
52	WATER TREATMENT AND STORAGE
53	WASTE DRYING BED
54	WASTE WATER TREATMENT
55	GARBAGE COLLECTION AREA
56	MINING VILLAGE (BY OTHERS)
57	MAADEN PMC TEMP ACCOMODATION (BY OTHERS)
58	CONSTRUCTION CAMP
59	CONSTRUCTION LAYDOWN YARD
60	HELICOPTER PAD
61	NOT USED
62	FACILITY SECURITY FORCE BASE CAMP
63	GOVERNMENT SECURITY BUILDING
64	OVERWATCH TOWER
65	ROM PAD
66	FACILITY BY OTHERS

SITE PLAN
1:30000

KEY PLAN

TAILING AREA
1:20000

PROPOSED TESTS TO BE CONDUCTED

BH	-	BORE HOLE
TP	-	TRIAL PIT
CBR	-	CALIFORNIA BEARING RATIO
PLT	-	PLATE LOAD TEST
EBH	-	ENVIRONMENTAL BORE HOLE
ERT	-	ELECTRICAL RESISTIVITY TEST
DH	-	DOWN HOLE SEISMIC TEST
TRT	-	THERMAL RESISTIVITY TEST

LIST OF UNITS :-

AREA	DESCRIPTION
3110	PRIMARY CRUSHING
3120	SECONDARY CRUSHING
3130	TERTIARY CRUSHING
3210	CRUSHED ORE STORAGE AND TRANSFER
3220	GRINDING AND CLASSIFICATION – BALL MILL
3230	GRINDING AND CLASSIFICATION – GRAVITY CONCENTRATOR
3310	FLOTATION ROUGHER/SCAVENGER/CLEANER
3320	LOW GRADE THICKENING
3340	CONCENTRATE THICKENING
3350	CONCENTRATE FILTRATION
3360	CONCENTRATE STORAGE AND REPULP
3370	FLOTATION PROCESS WATER STORAGE AND DISTRIBUTION
3380	ACIDULATION
3410	PRESSURE OXIDATION – AUTOCLAVE FEED STORAGE
3420	PRESSURE OXIDATION – AUTOCLAVE FLASH VESSEL & SCRUBBER
3440	PRESSURE OXIDATION – HOT ACID CURE
3450	PRESSURE OXIDATION – SERVICES
3510	CCD CIRCUIT & POX CONDITIONING
3520	NEUTRALIZATION AND THICKENING
3530	CCD & POX – PROCESS WATER COOLING
3540	CCD & POX – PROCESS WATER STORAGE
3610	POX LEACHING/ADSORPTION
3620	POX INTENSIVE CYANIDE LEACHING
3640	LOW GRADE CONDITIONING/LEACHING/ADSORPTION
3650	POX ACID WASH/ELUTION/CARBON REGENERATION
3660	LOW GRADE ACID WASH / ELUTION / CARBON REGEN
3670	PRIMARY CYANIDE DESTRUCTION
3680	CONDITIONING & LEACHING & CARBON CIRCUITS ELECTROWIN
3690	GOLD ROOM & ELECTROWINNING & CALCINING & SMELTING
3710	TAILINGS PASTE THICKENING
3720	TAILINGS DISPOSAL/TAILINGS MANAGEMENT FACILITY PIPELINE
3740	TAILINGS & CIL SECONDARY CYANIDE DESTRUCTION
3750	TAILINGS & CIL PROCESS WATER STORAGE
3810	REAGENTS – CAUSTIC AND CYANIDE
3820	REAGENTS – HYDROCHLORIC ACID/SULPHAMIC ACID
3830	REAGENTS – FLOTATION
3840	REAGENTS – FLOCCULANT
3850	REAGENTS – LIME SLAKING
3860	REAGENTS – LIMESTONE
3890	REAGENTS – SMBS & PEROXIDE
39WL	RAW WATER
39LA	FIRE WATER SYSTEM
39WG	GLAND WATER / MINE WATER
39WF	POTABLE WATER STORAGE AND DISTRIBUTION
39EJ	DEMINERALISATION SYSTEM
39AG	LOW PRESSURE AIR
39AB	PLANT AIR DISTRIBUTION
39IA	INSTRUMENT AIR
39MA	PIPE RACKS
3980	OXYGEN PLANT
4000	OTHER BUILDINGS
11BG	HEALTH & SAFETY & SECURITY AND RISK
49RD	INTERNAL ROADS AND DRAINAGE
4400	TAILINGS MANAGEMENT FACILITY CHANNEL/DRAINAGE/POND
49EE	WASTE DISPOSAL
49PR	POWER GENERATION
49PG	POWER DISTRIBUTION
49PS	SOLAR POWER PLANT
49FE	FUEL STORAGE
49FE	FUEL STORAGE(LPG)
49RA	ACCESS ROAD
39W1	BOOSTER PUMP STATION & PROCESS WATER SUPPLY
49WF	POTABLE WATER SUPPLY
49BL	LABORATORY
49CD	IT AND COMMUNICATIONS SYSTEM

VILLAGE AREA (4100)
1:2500

EXPLOSIVE MAGAZINE AREA (2500)
(FUTURE SCOPE)
1:4000

SECURITY BASE CAMP AREA
1:1000

TRIAL PITS

TP. MKD.	N - CORD.	E - CORD.	TP TERMINATION DEPTH
TP-1	2505871.024	263813.787	1.5 METERS
TP-2	2506290.709	263710.923	1.5 METERS
TP-3	2506407.793	263670.981	1.5 METERS
TP-4	2506095.352	263862.547	1.5 METERS
TP-5	2505773.757	264071.603	1.5 METERS
TP-6	2505856.456	264251.253	1.5 METERS
TP-7	2506207.693	264054.015	1.5 METERS
TP-8	2506296.885	264215.955	1.5 METERS
TP-9	2506108.527	264399.718	1.5 METERS
TP-10	2506147.896	264457.770	1.5 METERS
TP-11	2506472.313	264419.582	1.5 METERS
TP-12	2508492.393	262866.087	1.5 METERS
TP-13	2508703.913	263193.745	1.5 METERS
TP-14	2508295.601	263457.331	1.5 METERS
TP-15	2508084.082	263129.673	1.5 METERS
TP-16	2506818.334	263202.020	1.5 METERS
TP-17	2508357.653	263260.338	1.5 METERS
TP-18	2508310.025	263158.627	1.5 METERS
TP-19	2509426.370	264899.010	1.5 METERS
TP-20	2509357.805	264761.267	1.5 METERS
TP-21	2509648.592	265327.557	1.5 METERS
TP-22	2509648.592	265827.862	1.5 METERS
TP-23	2509031.305	265827.826	1.5 METERS
TP-24	2509031.300	265327.560	1.5 METERS
TP-25	2508286.441	262982.705	1.5 METERS
TP-26	2506202.226	263912.559	1.5 METERS
TP-27	2505941.986	264051.068	1.5 METERS
TP-28	2504886.341	262381.915	1.5 METERS
TP-29	2502543.025	263678.220	1.5 METERS
TP-30	2503431.597	268590.000	1.5 METERS
TP-31	2506475.534	268590.000	1.5 METERS
TP-32	2510479.237	266377.762	1.5 METERS
TP-33	2514933.124	263916.776	1.5 METERS
TP-34	2514886.696	262156.429	1.5 METERS
TP-35	2510985.789	262295.647	1.5 METERS
TP-36	2505881.869	263947.835	1.5 METERS
TP-41	2509194.335	261836.091	1.5 METERS
TP-42	2509152.968	261832.920	1.5 METERS
TP-43	2506673.817	263692.737	1.5 METERS
TP-44	2506177.327	263802.776	1.5 METERS
TP-45	2505965.004	263870.119	1.5 METERS
TP-46	2505805.123	264254.097	4 METERS
TP-47	2505926.765	264444.343	4 METERS
TP-48	2506447.348	264066.431	1.5 METERS
TP-49	2506322.996	264137.259	1.5 METERS
TP-50	2506325.231	263940.870	1.5 METERS

ELECTRICAL RESISTIVITY TEST

ERT MKD.	N - CORD.	E - CORD.
ERT-1	2506127.818	264163.775
ERT-2	2505905.541	264145.847
ERT-3	2506362.987	264034.715
ERT-4	2506481.090	264440.232
ERT-5	2506086.398	264003.793
ERT-6	2506024.040	263874.764
ERT-7	2509393.269	264858.439
ERT-9	2508495.158	263255.302
ERT-10	2508260.166	263217.165
ERT-11	2508419.379	263058.270
ERT-12	2509398.759	265649.455
ERT-13	2506800.380	263714.984
ERT-14	2509177.693	261827.469
ERT-15	2505924.444	264333.985
ERT-16	2506141.706	264233.672

CALIFORNIA BEARING RATIO

CBR. MKD.	N - CORD.	E - CORD.
CBR-1	2506293.900	263547.697
CBR-2	2505980.974	263787.176
CBR-3	2506131.287	263858.582
CBR-4	2505648.408	264049.895
CBR-5	2505810.594	264116.104
CBR-6	2506156.333	264018.629
CBR-7	2506040.884	264179.207
CBR-8	2505857.328	264393.021
CBR-9	2506165.392	264522.509
CBR-10	2506280.463	264309.592
CBR-11	2505969.066	264358.264
CBR-12	2506551.675	264125.477
CBR-13	2508722.706	263203.218
CBR-14	2508063.024	263118.687
CBR-15	2509004.189	265854.978
CBR-16	2509675.708	265300.441
CBR-17	2504718.247	264445.364
CBR-19	2509171.635	261859.561

BOREHOLES

BH MKD.	N - CORD.	E - CORD.	BH TERMINATION DEPTH★
BH-1	2506230.930	263768.938	15 METERS
BH-2	2506179.071	264210.656	30 METERS
BH-3	2503834.172	265464.639	8 METERS
BH-4	2506042.964	264344.101	15 METERS
BH-5	2506162.430	263969.229	15 METERS
BH-6	2506071.234	263996.986	15 METERS
BH-7	2506130.933	264068.724	15 METERS
BH-8	2505913.117	264072.548	15 METERS
BH-9	2506010.372	264138.908	15 METERS
BH-10	2506142.028	264155.853	15 METERS
BH-11	2506094.386	264208.644	15 METERS
BH-12	2505967.370	264193.575	15 METERS
BH-13	2505915.358	264216.032	15 METERS
BH-14	2505878.500	264355.994	15 METERS
BH-15	2506027.885	264280.822	30 METERS
BH-16	2506135.791	264290.480	15 METERS
BH-17	2506276.168	264362.880	15 METERS
BH-18	2506423.387	264270.887	15 METERS
BH-19	2506510.835	264424.583	30 METERS
BH-20	2506027.189	264219.066	15 METERS
BH-21	2505267.290	263178.795	15 METERS
BH-22	2509287.328	265495.284	15 METERS
BH-23	2505158.977	264306.200	15 METERS
BH-25	2506297.894	264397.650	15 METERS
BH-26	2506130.164	264334.912	15 METERS
BH-27	2506067.527	264289.071	15 METERS
BH-28	2506033.191	263964.386	15 METERS
BH-29	2506526.604	264367.168	15 METERS

★ NOTE:– DEPTH TO BE FINALISED AFTER DISCUSSION WITH TESTING AGENCY

DOWN HOLE SEISMIC TEST

DH. MKD.	N - CORD.	E - CORD.
DH-1(WITHIN BH-2)	2506179.071	264210.656
DH-2 (WITHIN BH-15)	2506027.885	264280.822
DH-3 (WITHIN BH-19)	2506510.835	264424.583

PIEZOMETER TEST

PZ. MKD.	N - CORD.	E - CORD.
PZ-1(WITHIN BH-1)	2506230.930	263768.938
PZ-2 (WITHIN BH-18)	2506423.387	264270.887
PZ-3 (WITHIN BH-8)	2505913.117	264072.548

THERMAL RESISTIVITY TEST

TRT. MKD.	N - CORD.	E - CORD.
TRT-1	2506046.160	263966.171

ENVIRONMENTAL BOREHOLES

EBH. MKD.	N - CORD.	E - CORD.
EBH-1	2505642.661	264066.961
EBH-2	2506254.243	264059.434
EBH-3	2509547.711	265768.778

PLATE LOAD TESTS

PLT MKD.	N - CORD.	E - CORD.
PLT-1	2505956.533	264115.134
PLT-2	2505962.600	264303.475
PLT-3	2508417.463	263018.466
PLT-6	2509176.016	261819.089

NOTES:–
1. ALL DIMENSIONS ARE IN M.
2. ALL CO ORDINATES ARE IN M.
3. THIS DWG SHALL BE READ IN CONJUNCTION WITH DWG NO. 019045_001_T_B_24754_002
5. THE GEOTECHNICAL TEST POINTS ARE INDICATED AS PER RECENT LAYOUT DATED 13.09.19.

DOCUMENT NUMBER	REFERENCE DOCUMENTS	DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
				05	16-09-2019	ADDITION OF TEST POINTS AT EXPLOSIVE MAGAZINE AREA & BH-28 & BH-29 ADDED DUE TO LAYOUT CHANGE ON 10.09.19	SYA	MLI	MLI	SJM	
				04	10-09-2019	BORE HOLE MARKED BH-26 & BH-27 ADDED, ERT16 SHIFTED DUE TO LAYOUT CHANGE ON 09.09.19 & TRT-1 ADDED (AS MKD.)	SYA	MLI	MLI	SJM	
				03	23-08-2019	HOLD RELEASED	SYA	MLI	MLI	SJM	
				02	09-08-2019	REVISED DUE TO LAYOUT CHANGE & DELETION OF TEST POINTS AT EXPLOSIVE MAGAZINE	SYA	MLI	MLI	SJM	
				01	02-08-2019	REVISED DUE TO LAYOUT CHANGE & FOLLOWING TEST POINTS ARE RELEASED FOR SITE EXECUTION:- BH-2, BH-9, BH10, BH-18, BH-19, DH-1, DH-3, PZ-2	SYA	MLI	MLI	SJM	
				00	08-07-2019	FIRST ISSUE	SYA	MLI	MLI	SJM	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

This drawing is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

The content of these documents are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without prior written permission of Outotec is prohibited.

This document and the information contained herein is Ma'aden proprietary know-how. This document or any part thereof should not be reproduced, stored in a retrieval system or transmitted by any means, electronic or otherwise, or divulged without prior written permission of Ma'aden.

LARSEN & TOUBRO DRAWING NUMBER	019045_001_T_B_24754_001
REVISION	5
OUTOTEC DRAWING NUMBER	900707_GAU01_ZZ01_CLD01_00003
DOCUMENT ID	OU100656268
REVISION	2
DRAWN	SYA 08-07-2019
CHECKED	MLI 08-07-2019
ENGINEER	MLI 08-07-2019
APPROVED	SJM 08-07-2019
EMPLOYER	
SCALE	1:30000,20000,4500 1:3000,2500,2000
	A0

MANSOURAH – MASSARAH GOLD PROJECT

SITE PLAN

PLANT LAYOUT AND SOIL TEST LOCATIONS

DRAWING NUMBER	REV
MD-216-1000-EG-CV-LAD-1001	A03

FIGURE-2

FIGURE 3: VARIATION OF FIELD SPT BLOWS (N_{60}) & RQD (%) WITH DEPTH

MANSOURAH - MASSARAH GOLD PROJECT

CRUSHING & SCREENING AREA

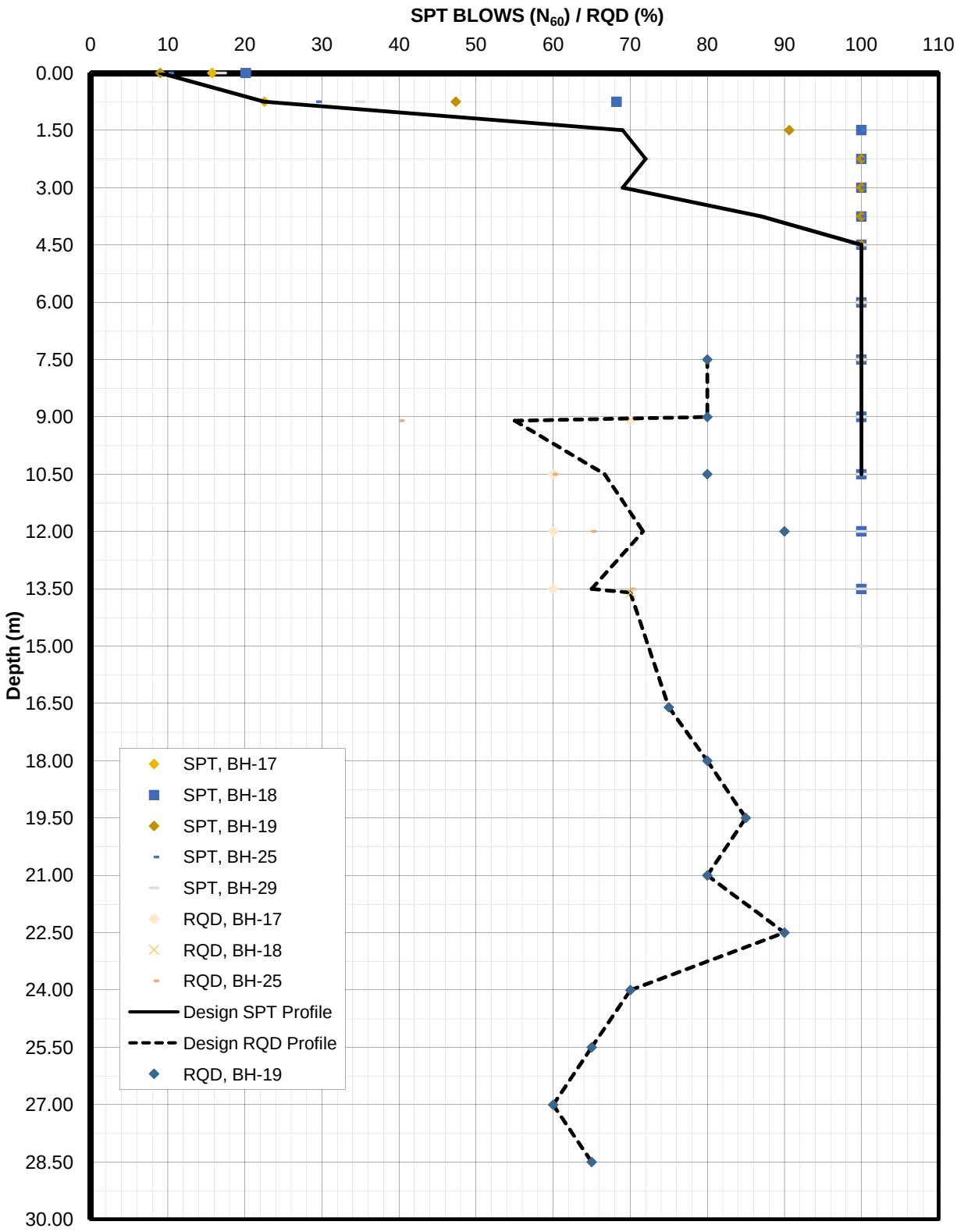


FIGURE 4(a): VARIATION OF FIELD SPT BLOWS (N_{60}) & RQD (%) WITH DEPTH

MANSOURAH - MASSARAH GOLD PROJECT

PROCESS PLANT AREA-1 (ZONE-1)

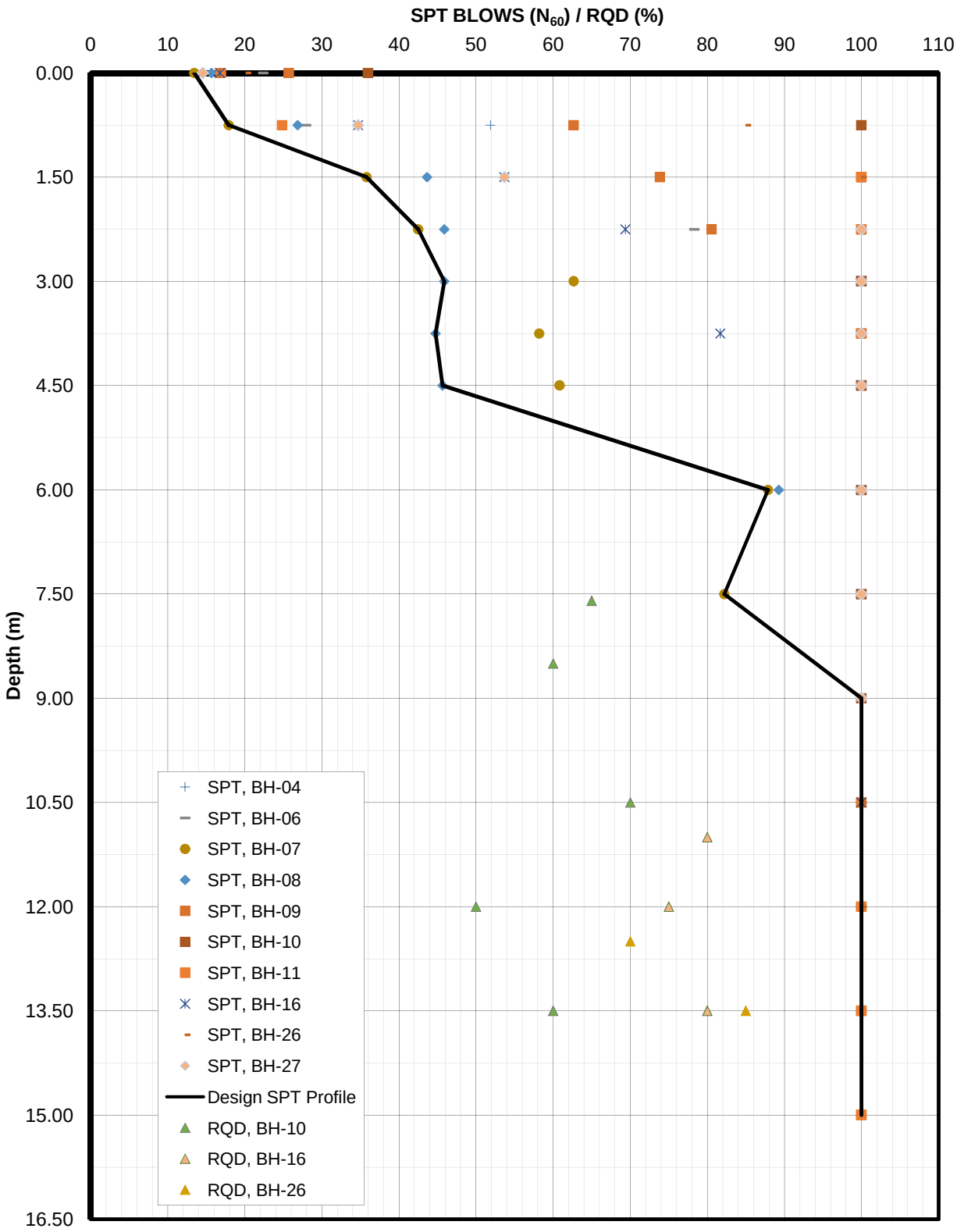


FIGURE 4(b): VARIATION OF FIELD SPT BLOWS (N_{60}) & RQD (%) WITH DEPTH

MANSOURAH - MASSARAH GOLD PROJECT
PROCESS PLANT AREA-1 (ZONE-2)

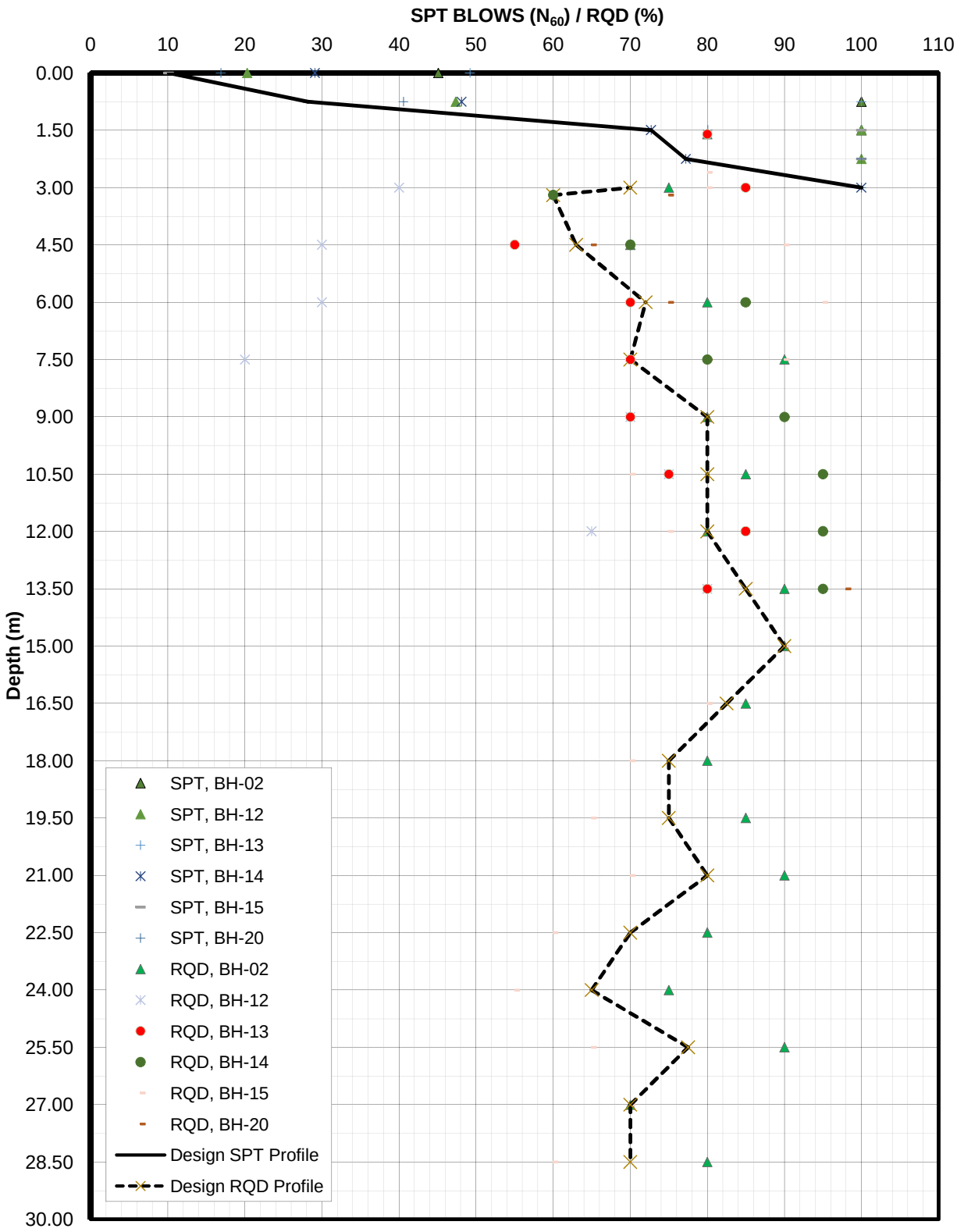


FIGURE 4(c): VARIATION OF FIELD SPT BLOWS (N_{60}) & RQD (%) WITH DEPTH

MANSOURAH - MASSARAH GOLD PROJECT

PROCESS PLANT-1 (BH-28)

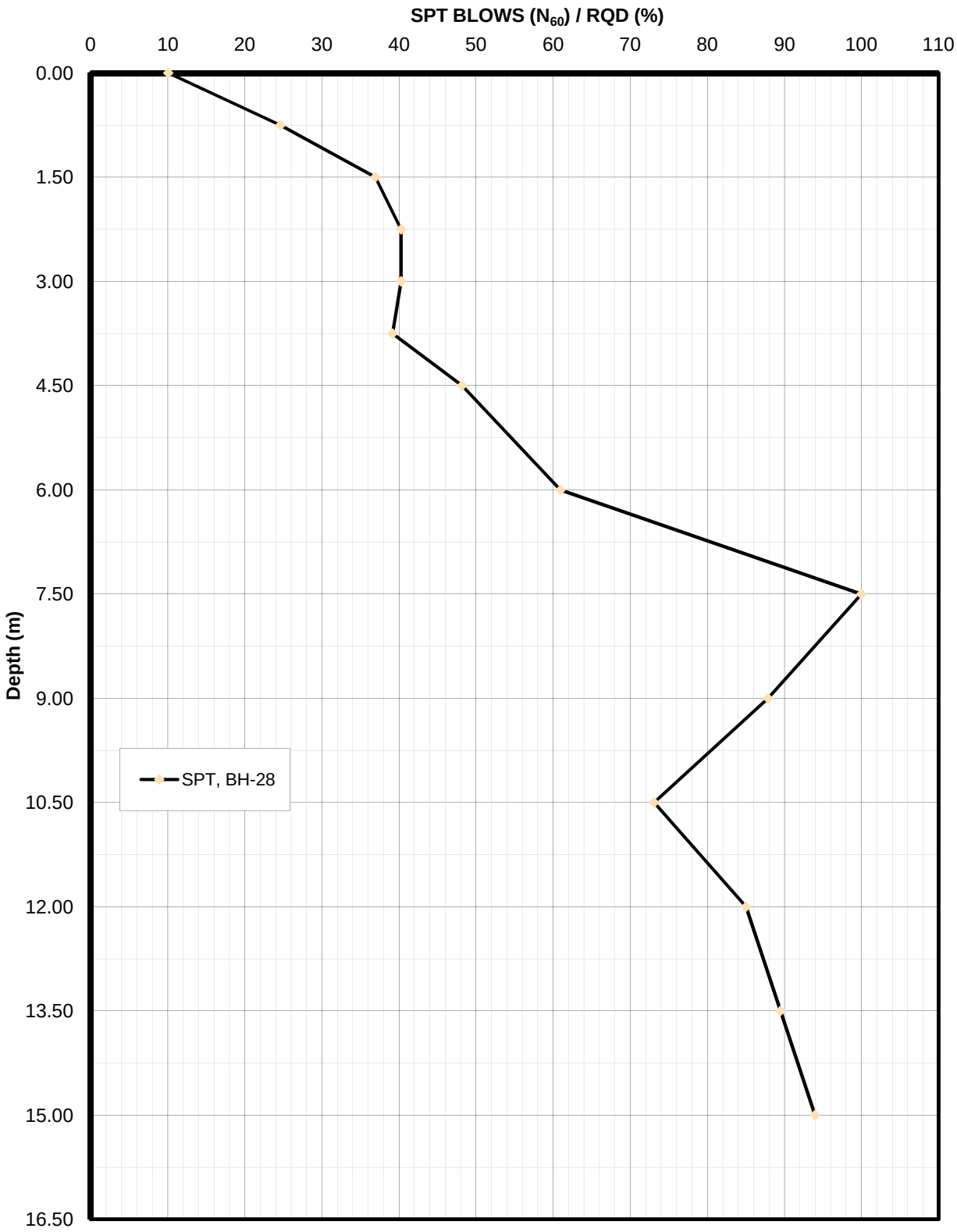


FIGURE 5: VARIATION OF FIELD SPT BLOWS (N_{60}) & RQD (%) WITH DEPTH

MANSOURAH - MASSARAH GOLD PROJECT

PROCESS PLANT-2

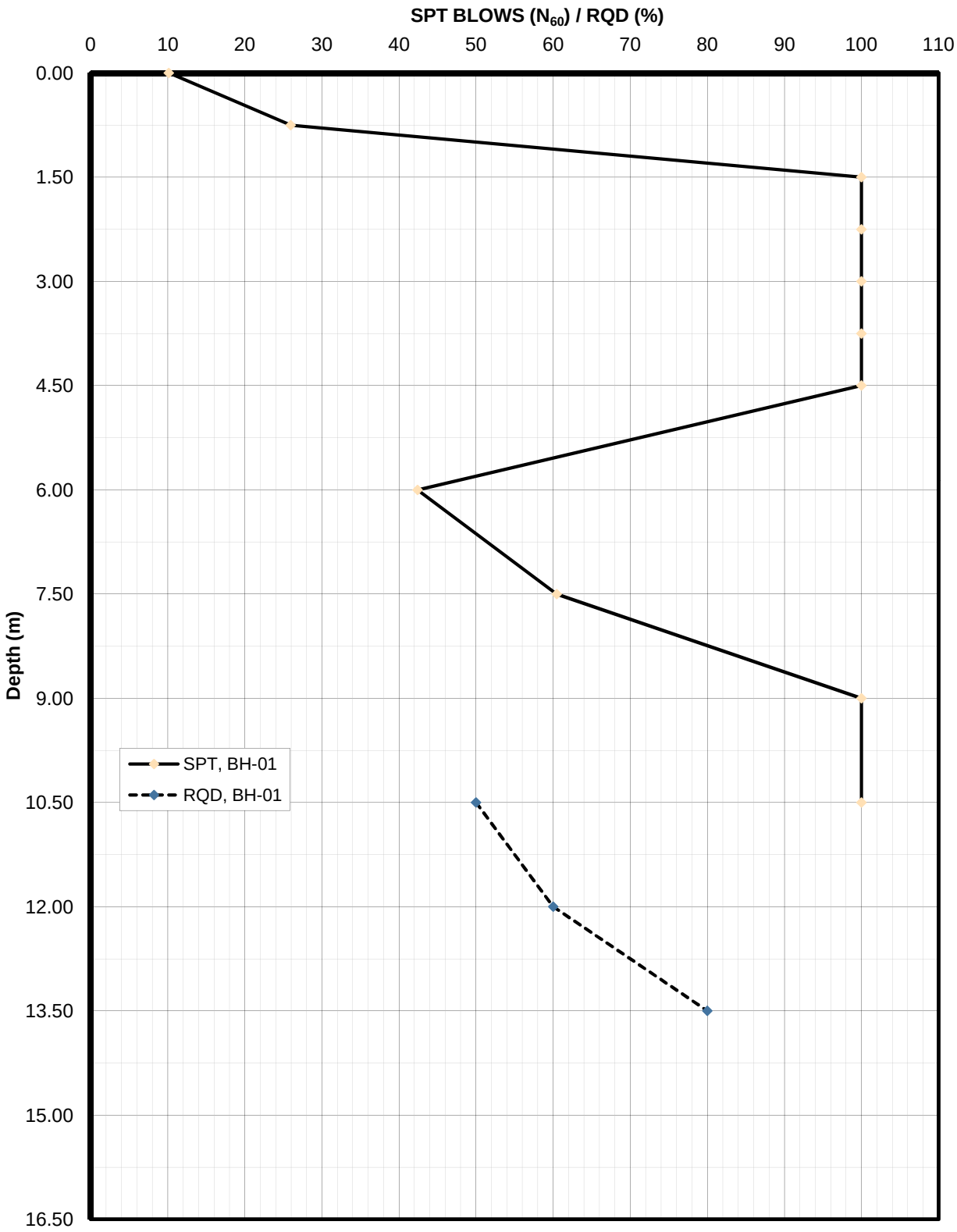


FIGURE 6: VARIATION OF FIELD SPT BLOWS (N_{60}) & RQD (%) WITH DEPTH

MANSOURAH - MASSARAH GOLD PROJECT
SOLAR PLANT & WORKSHOP AREA

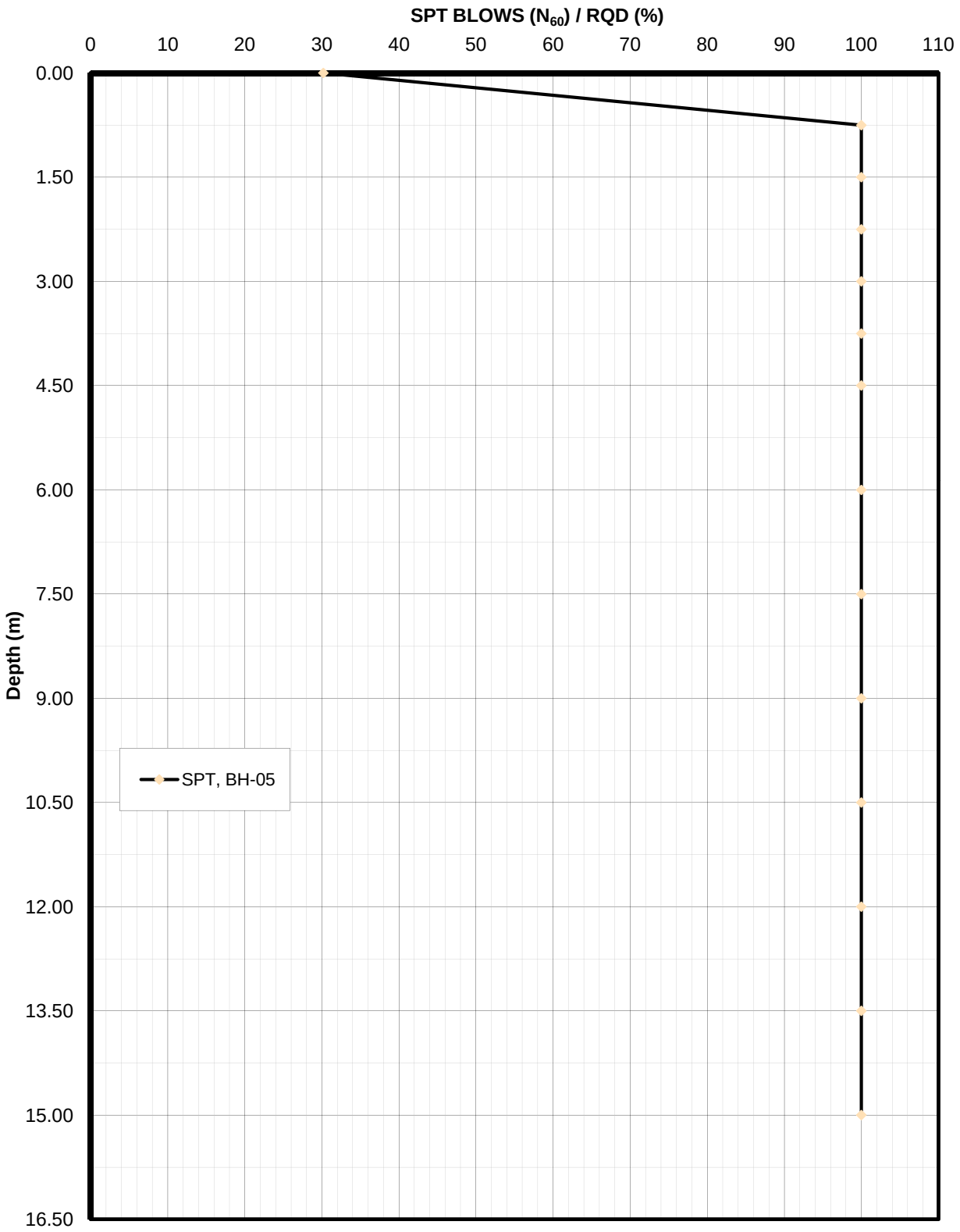


FIGURE 7: VARIATION OF FIELD SPT BLOWS (N_{60}) & RQD (%) WITH DEPTH

MANSOURAH - MASSARAH GOLD PROJECT

VILLAGE AREA

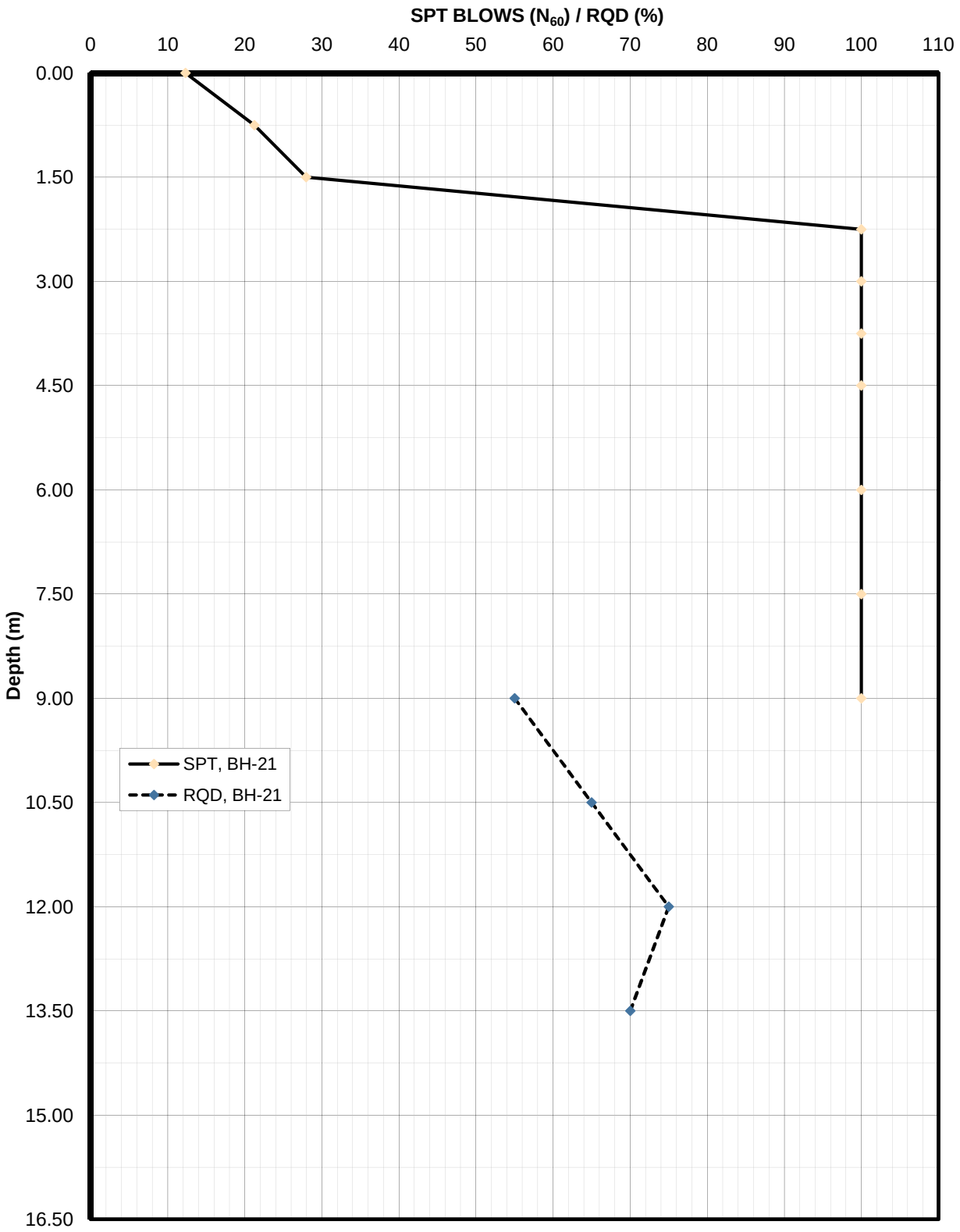


FIGURE 8: VARIATION OF FIELD SPT BLOWS (N_{60}) & RQD (%) WITH DEPTH

MANSOURAH - MASSARAH GOLD PROJECT

SOLAR PLANT & WORKSHOP AREA

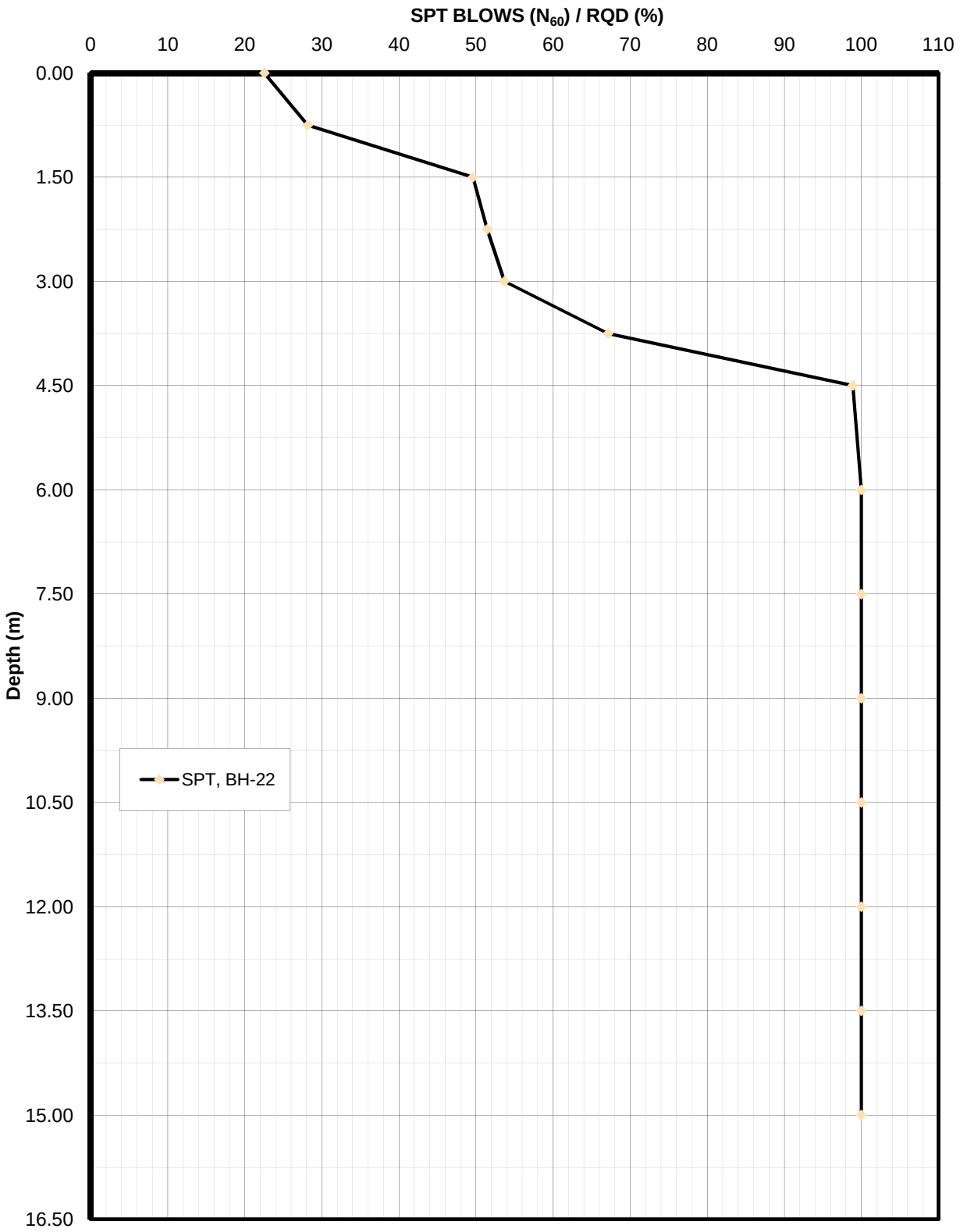


FIGURE 9: VARIATION OF FIELD SPT BLOWS (N_{60}) & RQD (%) WITH DEPTH

MANSOURAH - MASSARAH GOLD PROJECT

TMT

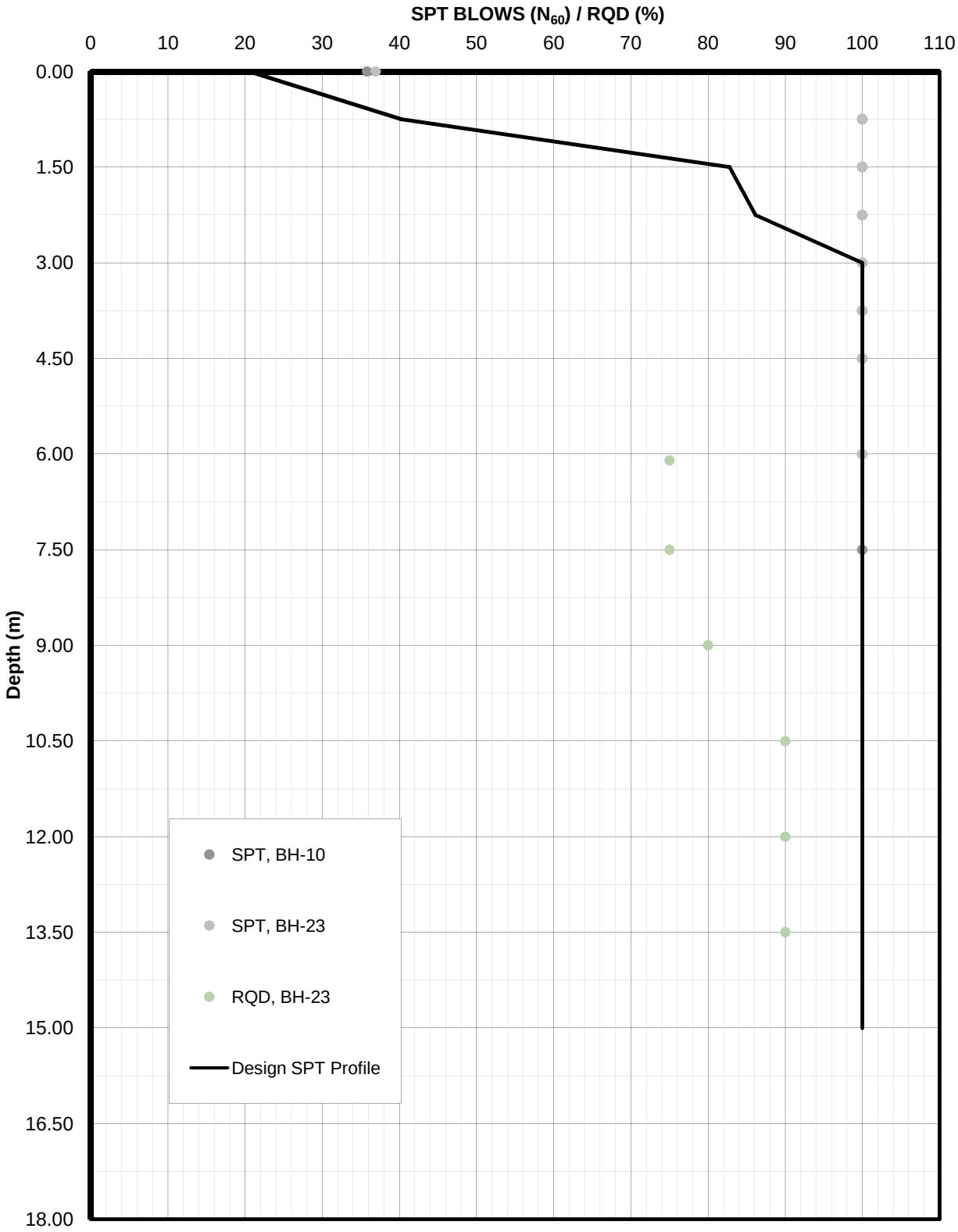
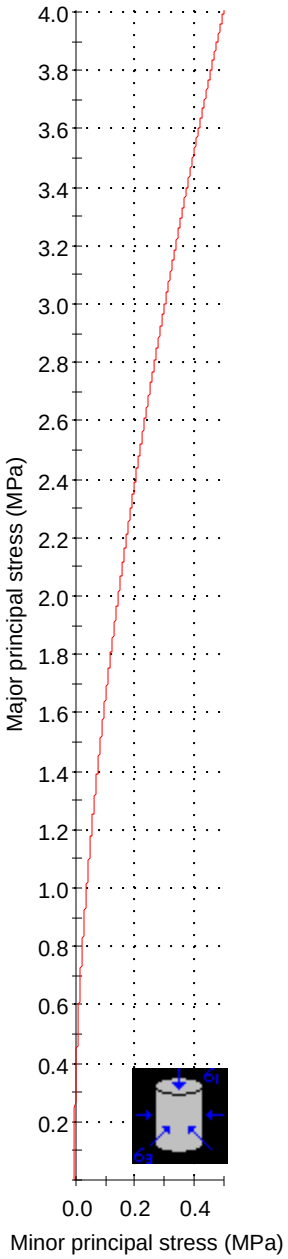


Figure 10: Analysis of Rock Strength using RocLab



Hoek-Brown Classification

intact uniaxial comp. strength (sigci) = 17 MPa
GSI = 30 mi = 20 Disturbance factor (D) = 0
intact modulus (Ei) = 5100 MPa
modulus ratio (MR) = 300

Hoek-Brown Criterion

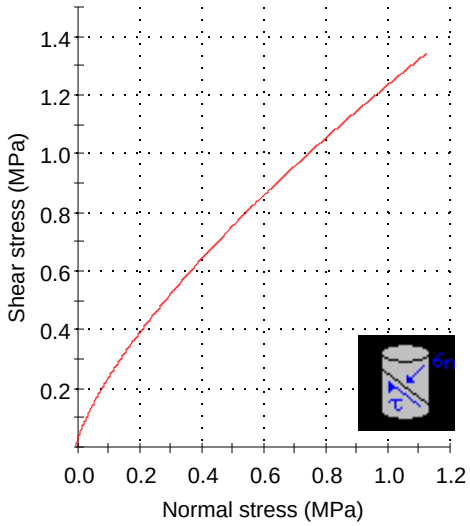
mb = 1.642 s = 0.0004 a = 0.522

Mohr-Coulomb Fit

cohesion = 0.174 MPa friction angle = 47.68 deg

Rock Mass Parameters

tensile strength = -0.004 MPa
uniaxial compressive strength = 0.292 MPa
global strength = 2.666 MPa
deformation modulus = 415.05 MPa



APPENDIX "A"

FIELD DATA

BOREHOLE LOGS

RIYADH GEOTECHNIQUE & FOUNDATIONS

Test Boring Log

BH NO. **BH-01**

SHEET **1** OF **2**

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**		ELEVATION: **915.20**	N: **2506230.93** E: **263768.94**
PROJECT NO.: 19-K-3054	CLIENT: **L & T CONSTRUCTION**	DATE STARTED **9-1-19**	DATE COMPLETED **9-1-19**
DRILLING EQUIPMENT: **Acker AD II**		GROUND WATER	
SIZE & TYPE OF CASING: **100mm diameter pipe casing**		DATE / TIME / WATER DEPTH (m) / CASING DEPTH	
DRILLING METHOD: **Straight Rotary**			
CORE BAR.: **-**	LENGTH: **-** BIT: **-**	DRILLER: **Sanaullah**	
SAMPLER: **S.P.T.**	WEIGHT: **63.50 Kg** DROP: **76 cm**	INSPECTOR: **Azam S.**	

DEPTH (m)	DESCRIPTION	PROFILE	SAMPLES SPT-N	SAMPLES PENET BL/15cm	ROCK REC. %	ROCK RQD. %	REMARKS
	Light brown, loose, silty SAND with gravel.		9	2 3 6			SPT at 0.15m depth.
1	Changing to very stiff.		23	4 8 15			SPT at 0.75m depth.
2	Light brown, very dense, completely weathered rock (recovered as silty SAND with gravel).		R	50/ 14			SPT at 1.50m depth.
	Grey, very dense, completely weathered rock (recovered as poorly graded GRAVEL with sand).		R	50/ 11			SPT at 2.25m depth.
3	-ditto-, with sand.		R	50/ 12			SPT at 3.0m depth.
	-ditto-		R	50/ 10	50	10	SPT at 3.75m depth. Coring at 3.80m to 4.50m depth.
4	Light grey, moderately weathered, moderately fractured, ANDESITE.						
	Brown, very dense, completely weathered rock (recovered as clayey SAND).		R	50/ 12			SPT at 4.50m depth.
5							
6	-ditto-		47	35 30 17			SPT at 6.0m depth.
	UDS-1						Undisturbed Sample @ 6.5m to 7.5m depth.
7							
	-ditto-		67	19 19 48			SPT at 7.50m depth.
8					17	0	Coring at 8.00m to 9.0m depth.
9	-ditto-		R	50/ 10			SPT at 9.0m depth.
10							

RIYADH GEOTECHNIQUE & FOUNDATIONS

Test Boring Log

BH NO. **BH-01**

SHEET **2** OF **2**

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**

CLIENT: **L & T CONSTRUCTION**

PROJECT NO.: **19-K-3054**

DEPTH (m)	DESCRIPTION	PROFILE	SAMPLES		ROCK		REMARKS
			SPT-N	PENET BL/15cm	REC. %	RQD. %	
	Brown, very dense, completely weathered rock (recovered as clayey SAND). *(continued)*						
11	Light grey, moderately weathered, slightly fractured, ANDESITE.				80	50	Coring at 10.50m to 12.0m depth.
12	-ditto-				90	60	Coring at 12.00m to 13.50m depth.
13							
14	-ditto-				95	80	Coring at 13.50m to 15.0m depth.
15	***End of borehole at 15.0m depth.***						
16							
17							
18							
19							
20							
21							

RIYADH GEOTECHNIQUE & FOUNDATIONS

Test Boring Log

BH NO. **BH-02**

SHEET **1** OF **3**

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**		ELEVATION: **912.93**	N: **2506179.07** E: **264210.66**
PROJECT NO.: 19-K-3054	CLIENT: **L & T CONSTRUCTION**	DATE STARTED **8-15-19**	DATE COMPLETED **8-22-19**
DRILLING EQUIPMENT: **Acker AD II**		GROUND WATER	
SIZE & TYPE OF CASING: **100mm diameter pipe casing**		DATE	TIME / WATER DEPTH (m) / CASING DEPTH
DRILLING METHOD: **Straight Rotary**		**23-Aug-19**	**08:30** / **23**
CORE BAR.: **-**	LENGTH: **-** BIT: **-**	DRILLER: **Sanaullah**	
SAMPLER: **S.P.T.**	WEIGHT: **63.50 Kg** DROP: **76 cm**	INSPECTOR: **Azam S.**	

DEPTH (m)	DESCRIPTION	PROFILE	SAMPLES SPT-N	SAMPLES PENET BL/15cm	ROCK REC. %	ROCK RQD. %	REMARKS
	Light brown, dense, silty SAND.		40	7 13 27			SPT at 0.0m depth.
1	Brown, very dense, completely weathered rock (recovered as clayey SAND with gravel).		R	33 50/ 6			SPT at 0.75m depth.
	-ditto-		R	50/ 3	95	80	SPT at 1.5.0m depth. Coring at 1.60m to 3.00m depth.
2	Light grey, slightly weathered, slightly fractured, ANDESITE.						
3	-ditto-				95	75	Coring at 3.00m to 4.50m depth.
4							
5	-ditto-, quartz and feldspar intruded as with thin layer.				90	70	Coring at 4.50m to 6.00m depth.
6	-ditto-, water loss 100% @ 6.0m to 15.0m depth.				95	80	Coring at 6.00m to 7.50m depth.
7							
8	-ditto-				97	90	Coring at 7.50m to 9.00m depth.
9	Light grey, slightly weathered, slightly fractured, ANDESITE. with layers of quarzatic veins.				90	80	Coring at 9.00m to 10.50m depth.
10							

RIYADH GEOTECHNIQUE & FOUNDATIONS

Test Boring Log

BH NO. **BH-02**

SHEET **2** OF **3**

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**

CLIENT: **L & T CONSTRUCTION**

PROJECT NO.: **19-K-3054**

DEPTH (m)	DESCRIPTION	PROFILE	SAMPLES		ROCK		REMARKS
			SPT-N	PENET BL/15cm	REC. %	RQD. %	
11	Light grey, slightly weathered, slightly fractured, ANDESITE. with layers of quarzatic veins. *(continued)* -ditto-				90	85	Coring at 10.50m to 12.00m depth.
12	-ditto-				90	80	Coring at 12.00m to 13.50m depth.
13	-ditto-				95	90	Coring at 13.50m to 15.00m depth.
14							
15	-ditto-				97	90	Coring at 15.00m to 16.50m depth.
16	-ditto-				98	85	Coring at 16.50m to 18.00m depth.
17							
18	-ditto-				90	80	Coring at 18.00m to 19.50m depth.
19	-ditto-				90	85	Coring at 19.50m to 21.00m depth.
20							
21	-ditto-				97	90	Coring at 21.00m to 22.50m depth.

RIYADH GEOTECHNIQUE & FOUNDATIONS

Test Boring Log

BH NO. **BH-02**

SHEET **3** OF **3**

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**

CLIENT: **L & T CONSTRUCTION**

PROJECT NO.: **19-K-3054**

DEPTH (m)	DESCRIPTION	PROFILE	SAMPLES SPT-N	SAMPLES PENET BL/15cm	ROCK REC. %	ROCK RQD. %	REMARKS
22	Light grey, slightly weathered, slightly fractured, ANDESITE. with layers of quarzatic veins. *(continued)*						
	-ditto-				90	80	Coring at 22.50m to 24.00m depth.
23							
24	Light grey, slightly weathered, slightly fractured, ANDESITE. with layers of quarzatic veins.				90	75	Coring at 24.00m to 25.50m depth.
25							
	-ditto-				98	90	Coring at 25.50m to 27.00m depth.
26							
27	-ditto-				85	70	Coring at 27.00m to 28.50m depth.
28							
	-ditto-				90	80	Coring at 28.50m to 30.00m depth.
29							
30	***End of borehole at 30.0m depth.***						
31							
32							

RIYADH GEOTECHNIQUE & FOUNDATIONS

Test Boring Log

BH NO. **BH-03**

SHEET **1** OF **1**

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**		ELEVATION: **905.40**	N: **2503834.17** E: **265464.64**
PROJECT NO.: 19-K-3054	CLIENT: **L & T CONSTRUCTION**	DATE STARTED **9-1-19**	DATE COMPLETED **9-1-19**
DRILLING EQUIPMENT: **Acker AD II**		GROUND WATER	
SIZE & TYPE OF CASING: **100mm diameter pipe casing**		DATE / TIME / WATER DEPTH (m) / CASING DEPTH	
DRILLING METHOD: **Straight Rotary**			
CORE BAR.: **-**	LENGTH: **-** BIT: **-**	DRILLER: **Azim**	
SAMPLER: **S.P.T.**	WEIGHT: **63.50 Kg** DROP: **76 cm**	INSPECTOR: **Azam S.**	

DEPTH (m)	DESCRIPTION	PROFILE	SAMPLES SPT-N	SAMPLES PENET BL/15cm	ROCK REC. %	ROCK RQD. %	REMARKS
	Light brown, medium dense, silty SAND.		14	5 5 9			SPT at 0.15m depth.
1	-ditto-, trace gravel.		36	10 17 19			SPT at 0.75m depth.
	Grey, very dense, completely weathered rock (recovered as silty GRAVEL with sand).		74	16 34 40			SPT at 1.50m depth.
2							
	Grey, very dense, completely weathered rock (recovered as silty SAND with gravel).		77	19 36 41			SPT at 2.25m depth.
3	-ditto- UDS-1		R	34 50/ 13			SPT at 3.0m depth. Undisturbed Sample @ 3.15m to 3.75m depth.
4	-ditto-		R	40 50/ 14			SPT at 3.75m depth.
	-ditto-		R	50/ 10			SPT at 4.50m depth.
5					18	0	Coring at 5.00m to 6.00m depth.
6	-ditto-		R	50/ 6			SPT at 6.0m depth.
7							
	-ditto-		R	50/ 5			SPT at 7.50m depth.
8	-ditto-		R	50/ 3			SPT at 8.0m depth.
	End of borehole at 8.0m depth.						
9							
10							

RIYADH GEOTECHNIQUE & FOUNDATIONS

Test Boring Log

BH NO. **BH-04**

SHEET **1** OF **2**

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**		ELEVATION: **911.98**	N: **2506042.96** E: **264344.10**
PROJECT NO.: 19-K-3054	CLIENT: **L & T CONSTRUCTION**	DATE STARTED **8-31-19**	DATE COMPLETED **8-31-19**
DRILLING EQUIPMENT: **Acker AD II**		GROUND WATER	
SIZE & TYPE OF CASING: **100mm diameter pipe casing**		DATE / TIME / WATER DEPTH (m) / CASING DEPTH	
DRILLING METHOD: **Straight Rotary**			
CORE BAR.: **-**	LENGTH: **-** BIT: **-**	DRILLER: **Azim**	
SAMPLER: **S.P.T.**	WEIGHT: **63.50 Kg** DROP: **76 cm**	INSPECTOR: **Azam S.**	

DEPTH (m)	DESCRIPTION	PROFILE	SAMPLES SPT-N	SAMPLES PENET BL/15cm	ROCK REC. %	ROCK RQD. %	REMARKS
	Light brown, medium dense, silty SAND.		12	3 5 7			SPT at 0.15m depth.
1	Brown, hard, sandy lean CLAY.		46	13 21 25			SPT at 0.75m depth.
2	Brown to light grey, very dense, completely weathered rock (recovered as silty SAND with gravel).		R	28 50/ 10			SPT at 1.50m depth.
	-ditto-		R	50/ 13			SPT at 2.25m depth.
3	Brown, very dense, completely weathered rock (recovered as silty SAND with gravel).		R	50/ 11			SPT at 3.0m depth.
4	-ditto-		R	50/ 9			SPT at 3.75m depth.
5	-ditto-		R	50/ 12			SPT at 4.50m depth.
6	-ditto-		R	50/ 10			SPT at 6.0m depth.
7							
8	Grey, hard, completely weathered rock (recovered as sandy CLAY).		R	50/ 11			SPT at 7.50m depth.
	UDS-1						Undisturbed Sample @ 8.0m to 9.0m depth.
					22	0	Coring at 8.00m to 9.00m depth.
9	Grey, very dense, completely weathered rock (recovered as clayey SAND).		R	50/ 10			SPT at 9.0m depth.
10							

RIYADH GEOTECHNIQUE & FOUNDATIONS

Test Boring Log

BH NO. **BH-04**

SHEET **2** OF **2**

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**

CLIENT: **L & T CONSTRUCTION**

PROJECT NO.: **19-K-3054**

DEPTH (m)	DESCRIPTION	PROFILE	SAMPLES SPT-N	SAMPLES PENET BL/15cm	ROCK REC. %	ROCK RQD. %	REMARKS
11	Grey, very dense, completely weathered rock (recovered as clayey SAND). *(continued)*						
	-ditto-		R	50/9			SPT at 10.50m depth.
12	-ditto-		R	50/7			SPT at 12.0m depth.
13							
14	Grey, very dense, completely weathered rock (recovered as silty SAND with gravel).		R	50/3			SPT at 13.50m depth.
15	Grey, very dense, completely weathered rock (recovered as silty GRAVEL with sand).		R	50/3			SPT at 15.0m depth.
	End of borehole at 15.0m depth.						
16							
17							
18							
19							
20							
21							

RIYADH GEOTECHNIQUE & FOUNDATIONS

Test Boring Log

BH NO. **BH-05**

SHEET **1** OF **2**

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**		ELEVATION: **913.20**	N: **2506162.43** E: **263969.23**
PROJECT NO.: 19-K-3054	CLIENT: **L & T CONSTRUCTION**	DATE STARTED **9-1-19**	DATE COMPLETED **9-1-19**
DRILLING EQUIPMENT: **Acker AD II**		GROUND WATER	
SIZE & TYPE OF CASING: **100mm diameter pipe casing**		DATE / TIME / WATER DEPTH (m) / CASING DEPTH	
DRILLING METHOD: **Straight Rotary**			
CORE BAR.: **-**	LENGTH: **-** BIT: **-**	DRILLER: **Azim**	
SAMPLER: **S.P.T.**	WEIGHT: **63.50 Kg** DROP: **76 cm**	INSPECTOR: **Azam S.**	

DEPTH (m)	DESCRIPTION	PROFILE	SAMPLES SPT-N	SAMPLES PENET BL/15cm	ROCK REC. %	ROCK RQD. %	REMARKS
1	Light brown, medium dense, completely weathered rock (recovered as silty GRAVEL with sand).		27	7 10 17			SPT at 0.15m depth.
	Changing to hard.		R	30 50/ 14			SPT at 0.75m depth.
2	Brown, very dense, completely weathered rock (recovered as silty SAND with gravel).		R	50/ 13			SPT at 1.50m depth.
	-ditto-				5	0	Coring 2.25m to 3.00m depth.
3	-ditto-		R	50/ 9			SPT at 3.0m depth.
4	Brown, very dense, completely weathered rock (recovered as silty SAND with gravel).		R	50/ 10			SPT at 3.75m depth.
5	Grey, very dense, completely weathered rock (recovered as silty GRAVEL with sand).		R	50/ 9			SPT at 4.50m depth.
					11	0	Coring at 5.00m to 6.00m depth.
6	-ditto-		R	50/ 14			SPT at 6.0m depth.
7							
8	Grey, very dense, completely weathered rock (recovered as silty SAND).		R	50/ 6			SPT at 7.50m depth.
9	-ditto-, trace clay.		R	50/ 10			SPT at 9.0m depth.
10							

RIYADH GEOTECHNIQUE & FOUNDATIONS

Test Boring Log

BH NO. **BH-05**

SHEET **2** OF **2**

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**

CLIENT: **L & T CONSTRUCTION**

PROJECT NO.: **19-K-3054**

DEPTH (m)	DESCRIPTION	PROFILE	SAMPLES SPT-N	SAMPLES PENET BL/15cm	ROCK REC. %	ROCK RQD. %	REMARKS
11	Grey, very dense, completely weathered rock (recovered as silty SAND). *(continued)*						
	-ditto-		R	50/ 8			SPT at 10.50m depth.
12	Brown, very dense, completely weathered rock (recovered as clayey SAND with gravel).		R	50/ 14			SPT at 12.0m depth.
13							
	-ditto-, trace sand.		R	28 50/ 9			SPT at 13.50m depth.
14					22	0	Coring at 14.00m to 15.00m depth.
15	Brown, very dense, completely weathered rock (recovered as silty SAND with gravel).		R	50/ 12			SPT at 15.0m depth.
	End of borehole at 15.0m depth.						
16							
17							
18							
19							
20							
21							

RIYADH GEOTECHNIQUE & FOUNDATIONS

Test Boring Log

BH NO. **BH-06**

SHEET **1** OF **2**

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**		ELEVATION: **913.04**	N: **2506071.23** E: **263996.99**
PROJECT NO.: 19-K-3054	CLIENT: **L & T CONSTRUCTION**	DATE STARTED **8-28-19**	DATE COMPLETED **8-28-19**
DRILLING EQUIPMENT: **Acker AD II**		GROUND WATER	
SIZE & TYPE OF CASING: **100mm diameter pipe casing**		DATE / TIME / WATER DEPTH (m) / CASING DEPTH	
DRILLING METHOD: **Straight Rotary**			
CORE BAR.: **-**	LENGTH: **-** BIT: **-**	DRILLER: **Azim**	
SAMPLER: **S.P.T.**	WEIGHT: **63.50 Kg** DROP: **76 cm**	INSPECTOR: **Azam S.**	

DEPTH (m)	DESCRIPTION	PROFILE	SAMPLES SPT-N	SAMPLES PENET BL/15cm	ROCK REC. %	ROCK RQD. %	REMARKS
	Brown, stiff, sandy lean CLAY.		20	6 9 11			SPT at 0.15m depth.
1	Brown to grey, medium dense, clayey SAND.		25	10 14 11			SPT at 0.75m depth.
	Light grey, very dense, clayey SAND with gravel.		66	24 32 34			SPT at 1.50m depth.
2	-ditto-		70	25 34 36			SPT at 2.25m depth.
3	Light grey, very dense, completely weathered rock (recovered as clayey SAND with gravel).		R	44 50/ 6			SPT at 3.0m depth.
	Grey, very dense, completely weathered rock (recovered as silty SAND with gravel).		R	50/ 14			SPT at 3.75m depth.
4	-ditto-		R	50/ 12			SPT at 4.50m depth.
5							
6	-ditto-		R	50/ 10			SPT at 6.0m depth.
7							
	-ditto-		R	50/ 13			SPT at 7.50m depth.
8							
9	-ditto-		R	50/ 14	22	0	SPT at 9.0m depth. Coring at 9.15m to 10.50m depth.
10							

RIYADH GEOTECHNIQUE & FOUNDATIONS

Test Boring Log

BH NO. **BH-06**

SHEET **2** OF **2**

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**

CLIENT: **L & T CONSTRUCTION**

PROJECT NO.: **19-K-3054**

DEPTH (m)	DESCRIPTION	PROFILE	SAMPLES SPT-N	SAMPLES PENET BL/15cm	ROCK REC. %	ROCK RQD. %	REMARKS
11	Grey, very dense, completely weathered rock (recovered as silty SAND with gravel). *(continued)*						
	-ditto-		R	50/10			SPT at 10.50m depth.
12	-ditto-		R	50/9	18	0	SPT at 12.0m depth. Coring at 12.10m to 13.50m depth.
13							
	-ditto-		R	50/7			SPT at 13.50m depth.
14							
15	Grey, very dense, completely weathered rock (recovered as silty SAND with gravel).		R	50/5			SPT at 15.0m depth.
	End of borehole at 15.0m depth.						
16							
17							
18							
19							
20							
21							

RIYADH GEOTECHNIQUE & FOUNDATIONS

Test Boring Log

BH NO. **BH-07**

SHEET **1** OF **2**

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**		ELEVATION: **912.94**	N: **2506130.93** E: **264068.72**
PROJECT NO.: 19-K-3054	CLIENT: **L & T CONSTRUCTION**	DATE STARTED **8-27-19**	DATE COMPLETED **8-28-19**
DRILLING EQUIPMENT: **Acker AD II**		GROUND WATER	
SIZE & TYPE OF CASING: **100mm diameter pipe casing**		DATE / TIME / WATER DEPTH (m) / CASING DEPTH	
DRILLING METHOD: **Straight Rotary**			
CORE BAR.: **-**	LENGTH: **-** BIT: **-**	DRILLER: **Azim**	
SAMPLER: **S.P.T.**	WEIGHT: **63.50 Kg** DROP: **76 cm**	INSPECTOR: **Azam S.**	

DEPTH (m)	DESCRIPTION	PROFILE	SAMPLES SPT-N	SAMPLES PENET BL/15cm	ROCK REC. %	ROCK RQD. %	REMARKS
	Light brown, stiff, sandy silty CLAY.		12	4 5 7			SPT at 0.15m depth.
1	Brown, medium dense, clayey SAND.		16	7 7 9			SPT at 0.75m depth.
	Changing to dense.		32	10 14 18			SPT at 1.50m depth.
2	-ditto-		38	12 17 21			SPT at 2.25m depth.
3	Brown to light grey, very dense, clayey SAND with gravel.		56	22 28 28			SPT at 3.0m depth.
	-ditto-, with gravel.		52	22 25 27			SPT at 3.75m depth.
4	-ditto-		48	19 22 26			SPT at 4.50m depth.
5	UDS-1						Undisturbed Sample @ 5.0m depth.
6	-ditto-		62	24 28 34			SPT at 6.0m depth.
7							
8	Brown, very dense, completely weathered rock (recovered as silty SAND with gravel).		58	29 21 37			SPT at 7.50m depth.
9	-ditto-		R	30 50/ 14			SPT at 9.0m depth.
10							

RIYADH GEOTECHNIQUE & FOUNDATIONS

Test Boring Log

BH NO. **BH-07**

SHEET **2** OF **2**

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**

CLIENT: **L & T CONSTRUCTION**

PROJECT NO.: **19-K-3054**

DEPTH (m)	DESCRIPTION	PROFILE	SAMPLES SPT-N	SAMPLES PENET BL/15cm	ROCK REC. %	ROCK RQD. %	REMARKS
11	Brown, very dense, completely weathered rock (recovered as silty SAND with gravel). *(continued)* -ditto-		R	34 50/ 12			SPT at 10.50m depth.
12	-ditto-		R	50/ 14			SPT at 12.0m depth.
13	-ditto-		R	50/ 9			SPT at 13.50m depth.
14	-ditto-		R	50/ 8			SPT at 15.0m depth.
15	***End of borehole at 15.0m depth.***						
16							
17							
18							
19							
20							
21							

RIYADH GEOTECHNIQUE & FOUNDATIONS

Test Boring Log

BH NO. **BH-08**

SHEET **1** OF **2**

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**		ELEVATION: **912.87**	N: **2505913.12** E: **264072.55**
PROJECT NO.: 19-K-3054	CLIENT: **L & T CONSTRUCTION**	DATE STARTED **8-29-19**	DATE COMPLETED **8-29-19**
DRILLING EQUIPMENT: **Acker AD II**		GROUND WATER	
SIZE & TYPE OF CASING: **100mm diameter pipe casing**		DATE / TIME / WATER DEPTH (m) / CASING DEPTH	
DRILLING METHOD: **Straight Rotary**			
CORE BAR.: **-**	LENGTH: **-** BIT: **-**	DRILLER: **Azim**	
SAMPLER: **S.P.T.**	WEIGHT: **63.50 Kg** DROP: **76 cm**	INSPECTOR: **Azam S.**	

DEPTH (m)	DESCRIPTION	PROFILE	SAMPLES SPT-N	SAMPLES PENET BL/15cm	ROCK REC. %	ROCK RQD. %	REMARKS
	Brown, stiff, sandy lean CLAY.		14	6 7 7			SPT at 0.15m depth.
1	Changing to very stiff.		24	9 11 14			SPT at 0.75m depth.
	Brown, dense, clayey SAND with gravel.		39	14 18 21			SPT at 1.50m depth.
2	-ditto-		41	15 20 21			SPT at 2.25m depth.
3	-ditto-		41	15 19 22			SPT at 3.0m depth.
4	-ditto-		40	14 19 21			SPT at 3.75m depth.
	Brown, dense, completely weathered rock (recovered as silty GRAVEL with sand, water loss 20% @ 4.50m to 15.0m depth).		36	14 18 18			SPT at 4.50m depth.
5							
6	-ditto-		63	22 30 33			SPT at 6.0m depth.
	UDS-1						Undisturbed sample at 6.50m to 7.50m depth.
7							
	Changing to very dense.		81	26 36 45			SPT at 7.50m depth.
8							
9	Brown, very dense, clayey SAND with gravel.		85	27 38 47			SPT at 9.0m depth.
10							

RIYADH GEOTECHNIQUE & FOUNDATIONS

Test Boring Log

BH NO. **BH-08**

SHEET **2** OF **2**

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**

CLIENT: **L & T CONSTRUCTION**

PROJECT NO.: **19-K-3054**

DEPTH (m)	DESCRIPTION	PROFILE	SAMPLES SPT-N	SAMPLES PENET BL/15cm	ROCK REC. %	ROCK RQD. %	REMARKS
11	Brown, very dense, clayey SAND with gravel. *(continued)*						
	-ditto-		R	38 50/ 13			SPT at 10.50m depth.
12	-ditto-		R	40 50/ 12			SPT at 12.0m depth.
13					29	0	Coring at 12.50m to 13.50m depth.
	Brown, very dense, clayey SAND.		R	42 50/ 8			SPT at 13.50m depth.
14							
15	-ditto-		R	50/ 5			SPT at 15.0m depth.
	End of borehole at 15.0m depth.						
16							
17							
18							
19							
20							
21							

RIYADH GEOTECHNIQUE & FOUNDATIONS

Test Boring Log

BH NO. **BH-09**

SHEET **1** OF **2**

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**		ELEVATION: **912.75**	N: **2506010.38** E: **264138.91**
PROJECT NO.: 19-K-3054	CLIENT: **L & T CONSTRUCTION**	DATE STARTED **8-29-19**	DATE COMPLETED **8-29-19**
DRILLING EQUIPMENT: **Acker AD II**		GROUND WATER	
SIZE & TYPE OF CASING: **100mm diameter pipe casing**		DATE / TIME / WATER DEPTH (m) / CASING DEPTH	
DRILLING METHOD: **Straight Rotary**			
CORE BAR.: **-**	LENGTH: **-** BIT: **-**	DRILLER: **Azim**	
SAMPLER: **S.P.T.**	WEIGHT: **63.50 Kg** DROP: **76 cm**	INSPECTOR: **Azam S.**	

DEPTH (m)	DESCRIPTION	PROFILE	SAMPLES SPT-N	SAMPLES PENET BL/15cm	ROCK REC. %	ROCK RQD. %	REMARKS
	Brown, medium dense, clayey SAND.		23	5 10 13			SPT at 0.15m depth.
1	Brown, very dense, sandy lean CLAY.		56	18 26 30			SPT at 0.75m depth.
	-ditto-		66	19 28 38			SPT at 1.50m depth.
2	Brown, very dense, clayey GRAVEL with sand.		72	21 31 41			SPT at 2.25m depth.
3	-ditto-		R	21 42 50/ 9			SPT at 3.0m depth.
	-ditto-		R	34 50/ 11			SPT at 3.75m depth.
4	Brown, very dense, completely weathered rock (recovered as clayey SAND with gravel).		R	42 50/ 9			SPT at 4.50m depth.
5							
6	-ditto-		R	50/ 13			SPT at 6.0m depth.
7	UDS-1						Undisturbed sample at 7.0m to 7.50m depth.
	-ditto-		R	50/ 13			SPT at 7.50m depth.
8							
9	-ditto-		R	50/ 11			SPT at 9.0m depth.
10							

RIYADH GEOTECHNIQUE & FOUNDATIONS

Test Boring Log

BH NO. **BH-09**

SHEET **2** OF **2**

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**

CLIENT: **L & T CONSTRUCTION**

PROJECT NO.: **19-K-3054**

DEPTH (m)	DESCRIPTION	PROFILE	SAMPLES SPT-N	SAMPLES PENET BL/15cm	ROCK REC. %	ROCK RQD. %	REMARKS
11	Brown, very dense, completely weathered rock (recovered as clayey SAND with gravel). *(continued)* -ditto-		R	50/ 10			SPT at 10.50m depth.
12	-ditto-		R	50/ 9			SPT at 12.0m depth.
13							
14	-ditto-		R	50/ 9			SPT at 13.50m depth.
15	-ditto-		R	50/ 7			SPT at 15.0m depth.
	End of borehole at 15.0m depth.						
16							
17							
18							
19							
20							
21							

RIYADH GEOTECHNIQUE & FOUNDATIONS

Test Boring Log

BH NO. **BH-10**

SHEET **1** OF **2**

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**		ELEVATION: **913.35**	N: **2506142.03** E: **264155.85**
PROJECT NO.: 19-K-3054	CLIENT: **L & T CONSTRUCTION**	DATE STARTED **8-7-19**	DATE COMPLETED **8-8-19**
DRILLING EQUIPMENT: **Acker AD II**		GROUND WATER	
SIZE & TYPE OF CASING: **100mm diameter pipe casing**		DATE / TIME / WATER DEPTH (m) / CASING DEPTH	
DRILLING METHOD: **Straight Rotary**			
CORE BAR.: **-**	LENGTH: **-** BIT: **-**	DRILLER: **Sanaullah**	
SAMPLER: **S.P.T.**	WEIGHT: **63.50 Kg** DROP: **76 cm**	INSPECTOR: **Azam S.**	

DEPTH (m)	DESCRIPTION	PROFILE	SAMPLES SPT-N	SAMPLES PENET BL/15cm	ROCK REC. %	ROCK RQD. %	REMARKS
	Light brown, dense, silty SAND with gravel.		32	6 13 19			SPT at 0.15m depth.
1	Light brown, very dense, completely weathered rock (recovered as silty SAND with gravel)		R	31 50/ 11			SPT at 0.75m depth.
	-ditto-		R	50/ 12			SPT at 1.50m depth.
2	-ditto-		R	50/ 9			SPT at 2.25m depth.
3	-ditto-		R	50/ 6			SPT at 3.0m depth.
4	Brownish grey, very dense, completely weathered rock (recovered as poorly graded GRAVEL with silt).		R	50/ 6			SPT at 3.75m depth.
	Brown, very dense, completely weathered rock (recovered as poorly graded SAND with silt).		R	50/ 5			SPT at 4.50m depth.
5	Grey, medium weathered, medium fractured, ANDESITE.				80	60	Coring at 5.00m to 6.00m depth.
6	Grey, very dense, completely weathered rock (recovered as poorly graded SAND with silt and gravel).		R	50/ 10			SPT at 6.00m depth.
7							
	-ditto-		R	50/ 7	85	65	SPT at 7.50m depth.
8	Grey, medium fractured, medium weathered, ANDESITE.						Coring at 7.60m to 9.0m depth.
9	-ditto-				80	60	Coring at 9.00m to 10.50m depth.
10							

RIYADH GEOTECHNIQUE & FOUNDATIONS

Test Boring Log

BH NO. **BH-10**

SHEET **2** OF **2**

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**

CLIENT: **L & T CONSTRUCTION**

PROJECT NO.: **19-K-3054**

DEPTH (m)	DESCRIPTION	PROFILE	SAMPLES		ROCK		REMARKS
			SPT-N	PENET BL/15cm	REC. %	RQD. %	
11	Grey, medium fractured, medium weathered, ANDESITE. *(continued)* -ditto-, water loss 20-30% @ 10.00m to 15.0m depth. Changing to slightly weathered.				90	70	Coring at 10.50m to 12.00m depth.
12	-ditto-				80	50	Coring at 12.00m to 13.50m depth.
13							
14	-ditto-				85	60	Coring at 13.50m to 15.00m depth.
15	***End of Borehole at 15.0m depth.***						
16							
17							
18							
19							
20							
21							

RIYADH GEOTECHNIQUE & FOUNDATIONS

Test Boring Log

BH NO. **BH-11**

SHEET **1** OF **2**

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**		ELEVATION: **912.70**	N: **2506094.39** E: **264208.64**
PROJECT NO.: 19-K-3054	CLIENT: **L & T CONSTRUCTION**	DATE STARTED **8-24-19**	DATE COMPLETED **8-25-19**
DRILLING EQUIPMENT: **Acker AD II**		GROUND WATER	
SIZE & TYPE OF CASING: **100mm diameter pipe casing**		DATE / TIME / WATER DEPTH (m) / CASING DEPTH	
DRILLING METHOD: **Straight Rotary**			
CORE BAR.: **-**	LENGTH: **-** BIT: **-**	DRILLER: **Sanaullah**	
SAMPLER: **S.P.T.**	WEIGHT: **63.50 Kg** DROP: **76 cm**	INSPECTOR: **Azam S.**	

DEPTH (m)	DESCRIPTION	PROFILE	SAMPLES SPT-N	SAMPLES PENET BL/15cm	ROCK REC. %	ROCK RQD. %	REMARKS
	Light brown, medium dense, clayey SAND.		15	3 5 10			SPT at 0.15m depth.
1	Brown, medium dense, silty SAND.		22	6 9 13			SPT at 0.75m depth.
2	Brown, hard, completely weathered rock (recovered as sandy SILT with gravel).		R	50/ 14			SPT at 1.50m depth.
	Brown, very dense, clayey SAND.		R	50/ 13			SPT at 2.25m depth.
3	-ditto-		R	42 50/ 9			SPT at 3.0m depth.
4	-ditto-		R	50/ 10			SPT at 3.75m depth.
	Grey, very dense, completely weathered rock (recovered as silty GRAVEL with sand).		R	50/ 5			SPT at 4.50m depth.
6	Brown, very dense, completely weathered rock (recovered as clayey SAND).		R	40 50/ 8			SPT at 6.0m depth.
7							
8	-ditto-, water loss 50%-70% @ 7.50m to 15.0m depth.		R	50/ 8	28	0	SPT at 7.50m depth. Coring at 7.60m to 9.0m depth.
9	-ditto-		R	50/ 9			SPT at 9.0m depth.
10							

RIYADH GEOTECHNIQUE & FOUNDATIONS

Test Boring Log

BH NO. **BH-11**

SHEET **2** OF **2**

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**

CLIENT: **L & T CONSTRUCTION**

PROJECT NO.: **19-K-3054**

DEPTH (m)	DESCRIPTION	PROFILE	SAMPLES SPT-N	SAMPLES PENET BL/15cm	ROCK REC. %	ROCK RQD. %	REMARKS
11	Brown, very dense, completely weathered rock (recovered as clayey SAND). *(continued)*						
	Grey, very dense, completely weathered rock (recovered as silty SAND with gravel).		R	50/4			SPT at 10.50m depth.
12	-ditto-		R	50/5			SPT at 12.0m depth.
13	-ditto-		R	50/6			SPT at 13.50m depth.
14							
15	Grey, very dense, completely weathered rock (recovered as silty GRAVEL with sand).		R	50/4			SPT at 15.0m depth.
	End of borehole at 15.0m depth.						
16							
17							
18							
19							
20							
21							

RIYADH GEOTECHNIQUE & FOUNDATIONS

Test Boring Log

BH NO. **BH-12**

SHEET **1** OF **2**

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**		ELEVATION: **912.66**	N: **2505967.37** E: **264193.58**
PROJECT NO.: 19-K-3054	CLIENT: **L & T CONSTRUCTION**	DATE STARTED **8-25-19**	DATE COMPLETED **8-26-19**
DRILLING EQUIPMENT: **Acker AD II**		GROUND WATER	
SIZE & TYPE OF CASING: **100mm diameter pipe casing**		DATE / TIME / WATER DEPTH (m) / CASING DEPTH	
DRILLING METHOD: **Straight Rotary**			
CORE BAR.: **-**	LENGTH: **-** BIT: **-**	DRILLER: **Sanaullah**	
SAMPLER: **S.P.T.**	WEIGHT: **63.50 Kg** DROP: **76 cm**	INSPECTOR: **Azam S.**	

DEPTH (m)	DESCRIPTION	PROFILE	SAMPLES SPT-N	SAMPLES PENET BL/15cm	ROCK REC. %	ROCK RQD. %	REMARKS
	Light brown, medium dense, silty GRAVEL with sand.		18	5 7 11			SPT at 0.15m depth.
1	Brown, dense, sandy SILT with gravel.		42	13 17 25			SPT at 0.75m depth.
2	Grey, very dense, completely weathered rock (recovered as clayey SAND with gravel).		R	50/ 12			SPT at 1.50m depth.
	Grey, very dense, completely weathered rock (recovered as silty GRAVEL with sand).		R	50/ 7			SPT at 2.25m depth.
3	Brown, moderately weathered, slightly fractured, green ANDESITE., quartz and feldspar intruded as thin layer.				70	40	Coring at 3.00m to 4.50m depth.
4							
5	-ditto-, with vugs.				70	30	Coring at 4.50m to 6.00m depth.
6	Changing to slightly weathered. -ditto-, water loss 100% @ 6.30m to 15.0m depth.				80	30	Coring at 6.00m to 7.50m depth.
7							
8	-ditto-				70	20	Coring at 7.50m to 9.00m depth.
9	Changing to moderately weathered.				100	70	Coring at 9.00m to 10.50m depth.
10							

RIYADH GEOTECHNIQUE & FOUNDATIONS

Test Boring Log

BH NO. **BH-12**

SHEET **2** OF **2**

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**

CLIENT: **L & T CONSTRUCTION**

PROJECT NO.: **19-K-3054**

DEPTH (m)	DESCRIPTION	PROFILE	SAMPLES		ROCK		REMARKS
			SPT-N	PENET BL/15cm	REC. %	RQD. %	
11	Brown, moderately weathered, slightly fractured, green ANDESITE., quartz and feldspar intruded as thin layer. *(continued)* -ditto-				95	75	Coring at 10.50m to 12.00m depth.
12	Changing to slightly weathered.				90	65	Coring at 12.00m to 13.50m depth.
13							
14	-ditto-				90	80	Coring at 13.50m to 15.00m depth.
15	***End of borehole at 15.0m depth.***						
16							
17							
18							
19							
20							
21							

RIYADH GEOTECHNIQUE & FOUNDATIONS

Test Boring Log

BH NO. **BH-13**

SHEET **1** OF **2**

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**		ELEVATION: **912.52**	N: **2505915.36** E: **264216.03**
PROJECT NO.: 19-K-3054	CLIENT: **L & T CONSTRUCTION**	DATE STARTED **8-26-19**	DATE COMPLETED **8-27-19**
DRILLING EQUIPMENT: **Acker AD II**		GROUND WATER	
SIZE & TYPE OF CASING: **100mm diameter pipe casing**		DATE / TIME / WATER DEPTH (m) / CASING DEPTH	
DRILLING METHOD: **Straight Rotary**			
CORE BAR.: **-**	LENGTH: **-** BIT: **-**	DRILLER: **Azim**	
SAMPLER: **S.P.T.**	WEIGHT: **63.50 Kg** DROP: **76 cm**	INSPECTOR: **Azam S.**	

DEPTH (m)	DESCRIPTION	PROFILE	SAMPLES SPT-N	SAMPLES PENET BL/15cm	ROCK REC. %	ROCK RQD. %	REMARKS
	Light brown, dense, silty SAND with gravel.		44	10 14 30			SPT at 0.0m depth.
1	Changing to very dense.		R	50/ 10			SPT at 0.75m depth.
	Grey, very dense, silty GRAVEL with sand.		R	50/ 5	95	80	SPT at 1.5.0m depth.
2	Grey, slightly weathered, slightly fractured, ANDESITE. (green SCHIST), medium to low straight (Quartz and feldspar intruded.						Coring at 1.60m to 3.00m depth.
3	Changing to moderately weathered.				95	85	Coring at 3.00m to 4.50m depth.
4							
	-ditto-, water loss 100% @ 4.50m to 15.0m depth.				80	55	Coring at 4.50m to 6.00m depth.
5							
6	-ditto-				95	90	Coring at 6.00m to 7.50m depth.
7							
	Changing to slightly weathered.				90	70	Coring at 7.50m to 9.00m depth.
8							
9	-ditto-				90	70	Coring at 9.00m to 10.50m depth.
10							

RIYADH GEOTECHNIQUE & FOUNDATIONS

Test Boring Log

BH NO. **BH-13**

SHEET **2** OF **2**

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**

CLIENT: **L & T CONSTRUCTION**

PROJECT NO.: **19-K-3054**

DEPTH (m)	DESCRIPTION	PROFILE	SAMPLES		ROCK		REMARKS
			SPT-N	PENET BL/15cm	REC. %	RQD. %	
11	Grey, slightly weathered, slightly fractured, ANDESITE. (green SCHIST), medium to low straight (Quartz and feldspar intruded. *(continued)* -ditto-				90	75	Coring at 10.50m to 12.00m depth.
12	-ditto-				95	85	Coring at 12.00m to 13.50m depth.
13							
14	-ditto-				95	80	Coring at 13.50m to 15.00m depth.
15	***End of borehole at 15.0m depth.***						
16							
17							
18							
19							
20							
21							

RIYADH GEOTECHNIQUE & FOUNDATIONS

Test Boring Log

BH NO. **BH-14**

SHEET **1** OF **2**

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**		ELEVATION: **911.87**	N: **2505878.50** E: **264355.99**
PROJECT NO.: 19-K-3054	CLIENT: **L & T CONSTRUCTION**	DATE STARTED **9-3-19**	DATE COMPLETED **9-3-19**
DRILLING EQUIPMENT: **Acker AD II**		GROUND WATER	
SIZE & TYPE OF CASING: **100mm diameter pipe casing**		DATE / TIME / WATER DEPTH (m) / CASING DEPTH	
DRILLING METHOD: **Straight Rotary**			
CORE BAR.: **-**	LENGTH: **-** BIT: **-**	DRILLER: **Azim**	
SAMPLER: **S.P.T.**	WEIGHT: **63.50 Kg** DROP: **76 cm**	INSPECTOR: **Azam S.**	

DEPTH (m)	DESCRIPTION	PROFILE	SAMPLES SPT-N	SAMPLES PENET BL/15cm	ROCK REC. %	ROCK RQD. %	REMARKS
	Light brown, medium dense, silty SAND with gravel.		26	7 10 16			SPT at 0.15m depth.
1	Brown, grey, dense, silty GRAVEL with sand.		43	10 19 24			SPT at 0.75m depth.
	Changing to very dense.		65	20 30 35			SPT at 1.50m depth.
2	-ditto-		69	24 32 37			SPT at 2.25m depth.
3	Grey, very dense, well graded GRAVEL with silt.		R	50/ 5			SPT at 3.0m depth.
	Grey, green, slightly weathered, slightly fractured, ANDESITE.. (medium to high strength).				95	60	Coring at 3.20m to 4.50m depth.
4							
	-ditto-				95	70	Coring at 4.50m to 6.00m depth.
5							
6	Changing to high strength.				98	85	Coring at 6.00m to 7.50m depth.
7							
	Grey, slightly weathered, slightly fractured, ANDESITE. (very high strength).				95	80	Coring at 7.50m to 9.00m depth.
8							
9	-ditto- Water loss 10% to 20% at 9.0m to 15.0m depth				97	90	Coring at 9.00m to 10.50m depth.
10							

RIYADH GEOTECHNIQUE & FOUNDATIONS

Test Boring Log

BH NO. **BH-14**

SHEET **2** OF **2**

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**

CLIENT: **L & T CONSTRUCTION**

PROJECT NO.: **19-K-3054**

DEPTH (m)	DESCRIPTION	PROFILE	SAMPLES SPT-N	SAMPLES PENET BL/15cm	ROCK REC. %	ROCK RQD. %	REMARKS
11	Grey, slightly weathered, slightly fractured, ANDESITE. (very high strength). *(continued)* -ditto-				100	95	Coring at 10.50m to 12.00m depth.
12	-ditto-				98	95	Coring at 12.00m to 13.50m depth.
13							
14	-ditto-				100	95	Coring at 13.50m to 15.00m depth.
15	***End of Borehole at 15.0m depth.***						
16							
17							
18							
19							
20							
21							

RIYADH GEOTECHNIQUE & FOUNDATIONS

Test Boring Log

BH NO. BH-15

SHEET 1 OF 3

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**		ELEVATION: **912.50**	N: **2506027.89** E: **264280.82**
PROJECT NO.: 19-K-3054	CLIENT: **L & T CONSTRUCTION**	DATE STARTED **9-2-19**	DATE COMPLETED **9-2-19**
DRILLING EQUIPMENT: **Acker AD II**		GROUND WATER	
SIZE & TYPE OF CASING: **100mm diameter pipe casing**		DATE **03-Sep-19**	TIME **08:30** / WATER DEPTH (m) **24** / CASING DEPTH
DRILLING METHOD: **Straight Rotary**			
CORE BAR.: **-**	LENGTH: **-** BIT: **-**	DRILLER: **Sanaullah**	
SAMPLER: **S.P.T.**	WEIGHT: **63.50 Kg** DROP: **76 cm**	INSPECTOR: **Azam S.**	

DEPTH (m)	DESCRIPTION	PROFILE	SAMPLES SPT-N	SAMPLES PENET BL/15cm	ROCK REC. %	ROCK RQD. %	REMARKS
1	Light brown, loose, silty SAND with gravel.		9	3 3 6			SPT at 0.15m depth.
	Changing to medium dense.		25	7 11 14			SPT at 0.75m depth.
2	Grey, very dense, completely weathered rock (recovered as well graded GRAVEL with silt).		R	36 50/ 16			SPT at 1.50m depth.
	-ditto-		R	50/ 11			SPT at 2.25m depth.
3	Green to dark green, moderately weathered, slightly fractured, ANDESITE. (medium to high strength).				90	80	Coring at 2.60m to 3.0m depth.
	-ditto-				97	80	Coring at 3.0m to 4.50m depth.
4							
5	-ditto-				98	90	Coring at 4.50m to 6.00m depth.
6	-ditto-				100	95	Coring at 6.00m to 7.50m depth.
7							
8	-ditto-				95	90	Coring at 7.50m to 9.00m depth.
9	Dark green to grayish, moderately weathered, slightly fractured, ANDESITE. (medium to high strength).				95	90	Coring at 9.00m to 10.50m depth.
10							

RIYADH GEOTECHNIQUE & FOUNDATIONS

Test Boring Log

BH NO. **BH-15**

SHEET **2** OF **3**

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**

CLIENT: **L & T CONSTRUCTION**

PROJECT NO.: **19-K-3054**

DEPTH (m)	DESCRIPTION	PROFILE	SAMPLES SPT-N	SAMPLES PENET BL/15cm	ROCK REC. %	ROCK RQD. %	REMARKS
11	Dark green to grayish, moderately weathered, slightly fractured, ANDESITE. (medium to high strength). *(continued)* -ditto-				90	70	Coring at 10.50m to 12.00m depth.
12	Grey to dark grey, moderately weathered, slightly fractured, ANDESITE..				95	75	Coring at 12.00m to 13.50m depth.
13							
14	Dark grey, slightly weathered, slightly weathered, ANDESITE.. Water loss 100% at 13.50m to 17.0m depth.				95	80	Coring at 13.50m to 15.00m depth.
15	-ditto-				98	90	Coring at 15.00m to 16.50m depth.
16							
17	-ditto- Water loss 100% at 17.20m to 21.0m depth.				95	80	Coring at 16.50m to 18.00m depth.
18	Dark grey, moderately weathered, slightly fractured, ANDESITE..				90	70	Coring at 18.00m to 19.50m depth.
19							
20	-ditto-				90	65	Coring at 19.50m to 21.00m depth.
21	Changing to dark grey to black color.				90	70	Coring at 21.00m to 22.50m depth.

RIYADH GEOTECHNIQUE & FOUNDATIONS

Test Boring Log

BH NO. **BH-15**

SHEET **3** OF **3**

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**

CLIENT: **L & T CONSTRUCTION**

PROJECT NO.: **19-K-3054**

DEPTH (m)	DESCRIPTION	PROFILE	SAMPLES SPT-N	SAMPLES PENET BL/15cm	ROCK REC. %	ROCK RQD. %	REMARKS
22	Dark grey, moderately weathered, slightly fractured, ANDESITE.. *(continued)* Water loss 100% at 22.0m depth.						
	-ditto-				85	60	Coring at 22.50m to 24.00m depth.
23							
24	-ditto-				80	55	Coring at 24.00m to 25.50m depth.
25							
	-ditto-				85	65	Coring at 25.50m to 27.00m depth.
26							
27	-ditto-				90	70	Coring at 27.00m to 28.50m depth.
28							
	-ditto-				90	60	Coring at 28.50m to 30.00m depth.
29							
30	***End of Borehole at 30.0m depth.***						
31							
32							

RIYADH GEOTECHNIQUE & FOUNDATIONS

Test Boring Log

BH NO. **BH-16**

SHEET **1** OF **2**

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**		ELEVATION: **912.07**	N: **2506135.79** E: **264290.48**
PROJECT NO.: 19-K-3054	CLIENT: **L & T CONSTRUCTION**	DATE STARTED **9-4-19**	DATE COMPLETED **9-4-19**
DRILLING EQUIPMENT: **Acker AD II**		GROUND WATER	
SIZE & TYPE OF CASING: **100mm diameter pipe casing**		DATE / TIME / WATER DEPTH (m) / CASING DEPTH	
DRILLING METHOD: **Straight Rotary**			
CORE BAR.: **-**	LENGTH: **-** BIT: **-**	DRILLER: **Azim**	
SAMPLER: **S.P.T.**	WEIGHT: **63.50 Kg** DROP: **76 cm**	INSPECTOR: **Azam S.**	

DEPTH (m)	DESCRIPTION	PROFILE	SAMPLES SPT-N	SAMPLES PENET BL/15cm	ROCK REC. %	ROCK RQD. %	REMARKS
	Light brown, medium dense, clayey SAND with gravel.		15	4 6 9			SPT at 0.15m depth.
1	Brown, dense, clayey SAND.		31	10 14 17			SPT at 0.75m depth.
	-ditto-		48	14 20 28			SPT at 1.50m depth.
2	Changing to very dense.		62	14 23 39			SPT at 2.25m depth.
3	Brown, very dense, completely weathered rock (recovered as calyey SAND).		R	16 35 50/ 14			SPT at 3.0m depth.
4	Brown, very dense, clayey SAND.		73	21 29 44			SPT at 3.75m depth.
	-ditto-		R	25 36 50/ 13			SPT at 4.50m depth.
5	UDS - 1						Undisturbed Sample @ 5.0m to 6.0m depth.
6	Brown, very dense, completely weathered rock (recovered as clayey SAND with gravel).		R	50/ 14			SPT at 6.0m depth.
7							
8	Brown, very dense, completely weathered rock (recovered as clayey SAND).		R	50/ 13			SPT at 7.50m depth.
9	-ditto-		R	50/ 12			SPT at 9.0m depth.
10							

RIYADH GEOTECHNIQUE & FOUNDATIONS

Test Boring Log

BH NO. **BH-16**

SHEET **2** OF **2**

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**

CLIENT: **L & T CONSTRUCTION**

PROJECT NO.: **19-K-3054**

DEPTH (m)	DESCRIPTION	PROFILE	SAMPLES SPT-N	SAMPLES PENET BL/15cm	ROCK REC. %	ROCK RQD. %	REMARKS
11	Brown, very dense, completely weathered rock (recovered as clayey SAND). *(continued)*						
	-ditto-		R	50/9			SPT at 10.50m depth.
	Grey to greenish, slightly weathered, slightly fractured, ANDESITE., interbedded Diorite and Gneiss.				95	80	Coring at 11.0m to 12.0m depth.
12	-ditto-				90	75	Coring at 12.0m to 13.50m depth.
13							
	-ditto-				90	80	Coring at 13.50m to 13.50m depth.
14							
15	***End of Borehole at 15.0m depth.***						
16							
17							
18							
19							
20							
21							

RIYADH GEOTECHNIQUE & FOUNDATIONS

Test Boring Log

BH NO. **BH-17**

SHEET **1** OF **2**

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**		ELEVATION: **912.15**	N: **2506276.17** E: **264362.88**
PROJECT NO.: 19-K-3054	CLIENT: **L & T CONSTRUCTION**	DATE STARTED **8-27-19**	DATE COMPLETED **8-28-19**
DRILLING EQUIPMENT: **Acker AD II**		GROUND WATER	
SIZE & TYPE OF CASING: **100mm diameter pipe casing**		DATE / TIME / WATER DEPTH (m) / CASING DEPTH	
DRILLING METHOD: **Straight Rotary**			
CORE BAR.: **-**	LENGTH: **-** BIT: **-**	DRILLER: **Sanaullah**	
SAMPLER: **S.P.T.**	WEIGHT: **63.50 Kg** DROP: **76 cm**	INSPECTOR: **Azam S.**	

DEPTH (m)	DESCRIPTION	PROFILE	SAMPLES SPT-N	SAMPLES PENET BL/15cm	ROCK REC. %	ROCK RQD. %	REMARKS
	Light brown, medium dense, clayey SAND.		14	4 6 8			SPT at 0.15m depth.
1	Brown, medium dense, clayey SAND with gravel.		20	7 9 11			SPT at 0.75m depth.
2	Brown to grey, very dense, completely weathered rock (recovered as silty GRAVEL with sand).		R	50/ 11			SPT at 1.50m depth.
	-ditto-		R	50/ 9			SPT at 2.25m depth.
3	-ditto-		R	50/ 5			SPT at 3.0m depth.
4	-ditto-		R	50/ 4			SPT at 3.75m depth.
5	Grey, very dense, completely weathered rock (recovered as poorly graded SAND with silt).		R	50/ 3			SPT at 4.50m depth.
6	Grey, very dense, completely weathered rock (recovered as well graded GRAVEL with sand).		R	50/ 5			SPT at 6.0m depth.
7							
8	Grey, very dense, completely weathered rock (recovered as silty GRAVEL with sand).		R	50/ 4			SPT at 7.50m depth.
9	-ditto-		R	50/ 4	100	70	SPT at 9.0m depth. Coring at 9.10m to 10.50m depth.
10	Light grey, moderately weathered, slightly fractured, quartzite ANDESITE..						

RIYADH GEOTECHNIQUE & FOUNDATIONS

Test Boring Log

BH NO. **BH-17**

SHEET **2** OF **2**

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**

CLIENT: **L & T CONSTRUCTION**

PROJECT NO.: **19-K-3054**

DEPTH (m)	DESCRIPTION	PROFILE	SAMPLES		ROCK		REMARKS
			SPT-N	PENET BL/15cm	REC. %	RQD. %	
11	Light grey, moderately weathered, slightly fractured, quartzite ANDESITE.. *(continued)* Changing to slightly weathered.				95	60	Coring at 10.50m to 12.00m depth.
12	-ditto-				90	60	Coring at 12.00m to 13.50m depth.
13							
14	Changing to moderately weathered.				80	60	Coring at 13.50m to 15.00m depth.
15	***End of borehole at 15.0m depth.***						
16							
17							
18							
19							
20							
21							

RIYADH GEOTECHNIQUE & FOUNDATIONS

Test Boring Log

BH NO. **BH-18**

SHEET **1** OF **2**

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**		ELEVATION: **912.46**	N: **2506423.39** E: **264270.89**
PROJECT NO.: 19-K-3054	CLIENT: **L & T CONSTRUCTION**	DATE STARTED **8-14-19**	DATE COMPLETED **8-15-19**
DRILLING EQUIPMENT: **Acker AD II**		GROUND WATER	
SIZE & TYPE OF CASING: **100mm diameter pipe casing**		DATE / TIME / WATER DEPTH (m) / CASING DEPTH	
DRILLING METHOD: **Straight Rotary**			
CORE BAR.: **-**	LENGTH: **-** BIT: **-**	DRILLER: **Sanaullah**	
SAMPLER: **S.P.T.**	WEIGHT: **63.50 Kg** DROP: **76 cm**	INSPECTOR: **Azam S.**	

DEPTH (m)	DESCRIPTION	PROFILE	SAMPLES SPT-N	SAMPLES PENET BL/15cm	ROCK REC. %	ROCK RQD. %	REMARKS
	Light brown, clayey SAND with gravel.		18	4 6 12			SPT at 0.15m depth.
1	Changing to very dense.		61	19 23 38			SPT at 0.75m depth.
	-ditto-		R	30 50/ 8			SPT at 1.50m depth.
2	Light brown, very dense, completely weathered rock (recovered as clayey SAND)		R	36 50/ 8			SPT at 2.25m depth.
3	Grey, hard, sandy SILT completely weathered rock (recovered as sandy SILT).		R	37 50/ 12			SPT at 3.0m depth.
	Grey, hard, SILT with sand.		R	40 50/ 6			SPT at 3.75m depth.
4	-ditto-		R	40 50/ 7			SPT at 4.50m depth.
5							
6	Grey, very dense, completely weathered rock (recovered as silty SAND with gravel).		R	42,50/ 5			SPT at 6.0m depth.
7							
	-ditto-		R	50/ 10			SPT at 7.50m depth.
8							
9	-ditto-		R	50/ 6			SPT at 9.0m depth.
10							

RIYADH GEOTECHNIQUE & FOUNDATIONS

Test Boring Log

BH NO. **BH-18**

SHEET **2** OF **2**

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**

CLIENT: **L & T CONSTRUCTION**

PROJECT NO.: **19-K-3054**

DEPTH (m)	DESCRIPTION	PROFILE	SAMPLES SPT-N	SAMPLES PENET BL/15cm	ROCK REC. %	ROCK RQD. %	REMARKS
11	Grey, very dense, completely weathered rock (recovered as silty SAND with gravel). *(continued)*						
	Grey, hard, completely weathered rock (recovered as sandy SILT).		R	50/12			SPT at 10.50m depth.
12	-ditto-		R	50/10			SPT at 12.0m depth.
13							
	-ditto-		R	50/2	90	70	SPT at 13.50m depth. Coring at 13.60m to 15.00m depth.
14	Grey, slightly weathered, slightly fractured, ANDESITE.						
15	***End of Borehole at 15.0m depth.***						
16							
17							
18							
19							
20							
21							

RIYADH GEOTECHNIQUE & FOUNDATIONS

Test Boring Log

BH NO. BH-19

SHEET 1 OF 3

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**		ELEVATION: **912.34**	N: **2506510.84** E: **264424.58**
PROJECT NO.: 19-K-3054	CLIENT: **L & T CONSTRUCTION**	DATE STARTED **9-5-19**	DATE COMPLETED **9-8-19**
DRILLING EQUIPMENT: **Acker AD II**		GROUND WATER	
SIZE & TYPE OF CASING: **100mm diameter pipe casing**		DATE **09-Sep-19**	TIME **08:30** / WATER DEPTH (m) **24** / CASING DEPTH
DRILLING METHOD: **Straight Rotary**			
CORE BAR.: **-**	LENGTH: **-** / BIT: **-**	DRILLER: **Sanaullah**	
SAMPLER: **S.P.T.**	WEIGHT: **63.50 Kg** / DROP: **76 cm**	INSPECTOR: **Azam S.**	

DEPTH (m)	DESCRIPTION	PROFILE	SAMPLES SPT-N	SAMPLES PENET BL/15cm	ROCK REC. %	ROCK RQD. %	REMARKS
	Light brown, loose, silty SAND with gravel.		8	2 3 5			SPT at 0.15m depth.
1	Changing to brown color and dense.		42	11 19 23			SPT at 0.75m depth.
	Changing to very dense.		81	24 37 44			SPT at 1.50m depth.
2	Brown, very dense, completely weathered rock (recovered as clayey SAND with gravel).		R	29 50/ 11			SPT at 2.25m depth.
3	-ditto-		R	23 43 50/ 8			SPT at 3.0m depth.
4	-ditto-		R	37 50/ 13			SPT at 3.75m depth.
	-ditto-		R	50/ 5			SPT at 4.50m depth.
5	UDS - 1						Undisturbed Sample @ 5.0m to 6.0m depth.
6	Grey, very dense, completely weathered rock (recovered as silty GRAVEL with sand).		R	50/ 6			SPT at 6.0m depth.
7							
	No sample recovered.		R	50/ 3	95	80	SPT at 7.50m depth.
8	Grey to blackish, slightly weathered, slightly fractured, ANDESITE.						Coring at 7.60m to 9.00m depth.
9	-ditto-				90	80	Coring at 9.00m to 10.50m depth.
10							

RIYADH GEOTECHNIQUE & FOUNDATIONS

Test Boring Log

BH NO. **BH-19**

SHEET **2** OF **3**

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**

CLIENT: **L & T CONSTRUCTION**

PROJECT NO.: **19-K-3054**

DEPTH (m)	DESCRIPTION	PROFILE	SAMPLES SPT-N	SAMPLES PENET BL/15cm	ROCK REC. %	ROCK RQD. %	REMARKS
11	Grey to blackish, slightly weathered, slightly fractured, ANDESITE. *(continued)*						
	-ditto-				95	80	Coring at 10.50m to 12.00m depth.
12	-ditto-				95	90	Coring at 12.00m to 13.50m depth.
13							
							Coring at 13.50m to 15.00m depth.
14							
15	Grey, very dense, completely weathered rock (recovered as silty GRAVEL).		R	50/ 8			SPT at 15.0m depth.
16							
	No sample recovered.		R	50/ 11	90	75	SPT at 16.5m depth. Coring at 16.60m to 18.0m depth.
17	Grey, slightly weathered, slightly fractured, ANDESITE.						
18	-ditto-				90	80	Coring at 18.00m to 19.50m depth.
19							
	-ditto-				95	85	Coring at 19.50m to 21.00m depth.
20							
21	-ditto-				98	80	Coring at 21.00m to 22.50m depth.

RIYADH GEOTECHNIQUE & FOUNDATIONS

Test Boring Log

BH NO. **BH-19**

SHEET **3** OF **3**

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**

CLIENT: **L & T CONSTRUCTION**

PROJECT NO.: **19-K-3054**

DEPTH (m)	DESCRIPTION	PROFILE	SAMPLES SPT-N	SAMPLES PENET BL/15cm	ROCK REC. %	ROCK RQD. %	REMARKS
22	Grey, slightly weathered, slightly fractured, ANDESITE. *(continued)*						
23	-ditto-				98	90	Coring at 22.50m to 24.00m depth.
24	Grey, moderately weathered, moderately fractured, quartzite ANDESITE.				95	70	Coring at 24.00m to 25.50m depth.
25	-ditto-				90	65	Coring at 25.50m to 27.00m depth.
26							
27	-ditto-				90	60	Coring at 27.00m to 28.50m depth.
28							
29	-ditto-				90	65	Coring at 28.50m to 30.00m depth.
30	***End of Borehole at 30.0m depth.***						
31							
32							

RIYADH GEOTECHNIQUE & FOUNDATIONS

Test Boring Log

BH NO. **BH-20**

SHEET **1** OF **2**

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**		ELEVATION: **912.52**	N: **2506027.19** E: **264219.07**
PROJECT NO.: 19-K-3054	CLIENT: **L & T CONSTRUCTION**	DATE STARTED **9-4-19**	DATE COMPLETED **9-5-19**
DRILLING EQUIPMENT: **Acker AD II**		GROUND WATER	
SIZE & TYPE OF CASING: **100mm diameter pipe casing**		DATE / TIME / WATER DEPTH (m) / CASING DEPTH	
DRILLING METHOD: **Straight Rotary**			
CORE BAR.: **-**	LENGTH: **-** BIT: **-**	DRILLER: **Azim**	
SAMPLER: **S.P.T.**	WEIGHT: **63.50 Kg** DROP: **76 cm**	INSPECTOR: **Azam S.**	

DEPTH (m)	DESCRIPTION	PROFILE	SAMPLES SPT-N	SAMPLES PENET BL/15cm	ROCK REC. %	ROCK RQD. %	REMARKS
	Greenish grey, medium dense, well graded SAND with silt and gravel.		15	4 7 8			SPT at 0.15m depth.
1	Brown, dense, silty GRAVEL with sand.		36	10 15 21			SPT at 0.75m depth.
	Changing to very dense.		71	35 43 28			SPT at 1.50m depth.
2	Brown, very dense, completely weathered rock (recovered as silty GRAVEL).		R	37 50/ 13			SPT at 2.25m depth.
3	-ditto-, water loss 100% @ 3.0m to 15.0m depth.		R	50/ 4			SPT at 3.0m depth.
	Light grey, slightly weathered, slightly fractured, ANDESITE.				90	75	Coring at 3.30m to 4.50m depth.
4							
	-ditto-				90	65	Coring at 4.50m to 6.00m depth.
5							
6	-ditto-				95	75	Coring at 6.00m to 7.50m depth.
7							
	Dark grey, slightly weathered, slightly fractured, ANDESITE.				100	80	Coring at 7.50m to 9.00m depth.
8							
9	-ditto-				98	90	Coring at 9.00m to 10.50m depth.
10							

RIYADH GEOTECHNIQUE & FOUNDATIONS

Test Boring Log

BH NO. **BH-20**

SHEET **2** OF **2**

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**

CLIENT: **L & T CONSTRUCTION**

PROJECT NO.: **19-K-3054**

DEPTH (m)	DESCRIPTION	PROFILE	SAMPLES		ROCK		REMARKS
			SPT-N	PENET BL/15cm	REC. %	RQD. %	
11	Dark grey, slightly weathered, slightly fractured, ANDESITE. *(continued)* -ditto-				100	95	Coring at 10.50m to 12.00m depth.
12	-ditto-				100	95	Coring at 12.00m to 13.50m depth.
13							
14	-ditto-				100	98	Coring at 13.50m to 15.00m depth.
15	End of borehole at 15.0m depth.						
16							
17							
18							
19							
20							
21							

RIYADH GEOTECHNIQUE & FOUNDATIONS

Test Boring Log

BH NO. **BH-21**

SHEET **1** OF **2**

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**		ELEVATION: **921.35**	N: **2508267.29** E: **263178.80**
PROJECT NO.: 19-K-3054	CLIENT: **L & T CONSTRUCTION**	DATE STARTED **9-5-19**	DATE COMPLETED **9-5-19**
DRILLING EQUIPMENT: **Acker AD II**		GROUND WATER	
SIZE & TYPE OF CASING: **100mm diameter pipe casing**		DATE / TIME / WATER DEPTH (m) / CASING DEPTH	
DRILLING METHOD: **Straight Rotary**			
CORE BAR.: **-**	LENGTH: **-** BIT: **-**	DRILLER: **Azim**	
SAMPLER: **S.P.T.**	WEIGHT: **63.50 Kg** DROP: **76 cm**	INSPECTOR: **Azam S.**	

DEPTH (m)	DESCRIPTION	PROFILE	SAMPLES SPT-N	SAMPLES PENET BL/15cm	ROCK REC. %	ROCK RQD. %	REMARKS
	Light brown, medium dense, poorly graded SAND with silt.		11	3 5 6			SPT at 0.15m depth.
1	Brown, medium dense, clayey SAND.		19	5 9 10			SPT at 0.75m depth.
	-ditto-		25	10 12 13			SPT at 1.50m depth.
2	Brown, very dense, completely weathered rock (recovered as silty SAND with gravel).		R	50/ 10			SPT at 2.25m depth.
3	-ditto-		R	35 50/ 9			SPT at 3.0m depth.
	Brown, very dense, completely weathered rock (recovered as poorly graded SAND with silt and gravel).		R	20 50/ 10			SPT at 3.75m depth.
4	-ditto-		R	25 50/ 8			SPT at 4.50m depth.
5							
6	-ditto-		R	50/ 5			SPT at 6.0m depth.
7							
	No sample recovered.		R	50/ 10			SPT at 7.50m depth.
8							
9	Light grey, slightly weathered, slightly fractured, ANDESITE.				90	55	Coring at 9.0m to 10.50m depth.
10							

RIYADH GEOTECHNIQUE & FOUNDATIONS

Test Boring Log

BH NO. **BH-21**

SHEET **2** OF **2**

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**

CLIENT: **L & T CONSTRUCTION**

PROJECT NO.: **19-K-3054**

DEPTH (m)	DESCRIPTION	PROFILE	SAMPLES SPT-N	SAMPLES PENET BL/15cm	ROCK REC. %	ROCK RQD. %	REMARKS
11	Light grey, slightly weathered, slightly fractured, ANDESITE. *(continued)*						
	Light grey, slightly weathered, slightly fractured, ANDESITE.				95	65	Coring at 10.50m to 12.00m depth.
12	-ditto-				95	75	Coring at 12.00m to 13.50m depth.
13							
	-ditto-				90	70	Coring at 13.50m to 15.00m depth.
14							
15	***End of Borehole at 15.0m depth.***						
16							
17							
18							
19							
20							
21							

RIYADH GEOTECHNIQUE & FOUNDATIONS

Test Boring Log

BH NO. **BH-22**

SHEET **1** OF **2**

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**		ELEVATION: **911.50**	N: **2509287.33** E: **265495.28**
PROJECT NO.: 19-K-3054	CLIENT: **L & T CONSTRUCTION**	DATE STARTED **9-19-19**	DATE COMPLETED **9-19-19**
DRILLING EQUIPMENT: **Acker AD II**		GROUND WATER	
SIZE & TYPE OF CASING: **100mm diameter pipe casing**		DATE / TIME / WATER DEPTH (m) / CASING DEPTH	
DRILLING METHOD: **Straight Rotary**			
CORE BAR.:	LENGTH: / BIT:	DRILLER: **Azim**	
SAMPLER: **S.P.T.**	WEIGHT: **63.50 Kg** / DROP: **76 cm**	INSPECTOR: **Azam S.**	

DEPTH (m)	DESCRIPTION	PROFILE	SAMPLES SPT-N	SAMPLES PENET BL/15cm	ROCK REC. %	ROCK RQD. %	REMARKS
0	Light brown, medium dense, poorly graded SAND with silt.		20	7 9 11			SPT at 0.15m depth.
1	Changing to brown color.		25	7 11 14			SPT at 0.75m depth.
	Brown to grey, dense, silty SAND.		44	19 21 23			SPT at 1.50m depth.
2	-ditto-, with gravel.		46	19 22 24			SPT at 2.25m depth.
3	-ditto-		48	20 23 25			SPT at 3.0m depth.
	Grey, hard, clayey SILT with gravel trace sand.		60	19 24 36			SPT at 3.75m depth.
4	-ditto-		78	17 33 45			SPT at 4.50m depth.
5							
6	-ditto-		80	20 34 46			SPT at 6.0m depth.
7							
	Dark grey, hard, completely weathered rock (recovered as elastic SILT).		R	50/ 8			SPT at 7.50m depth.
8					30	0	Coring at 8.0m to 9.0m depth.
9	-ditto-		R	50/ 5			SPT at 9.0m depth.
	UDS-1						Undisturbed Sample @ 9.50 to 10.50m depth.
10							

RIYADH GEOTECHNIQUE & FOUNDATIONS

Test Boring Log

BH NO. **BH-22**

SHEET **2** OF **2**

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**

CLIENT: **L & T CONSTRUCTION**

PROJECT NO.: **19-K-3054**

DEPTH (m)	DESCRIPTION	PROFILE	SAMPLES SPT-N	SAMPLES PENET BL/15cm	ROCK REC. %	ROCK RQD. %	REMARKS
	Dark grey, hard, completely weathered rock (recovered as elastic SILT). *(continued)*						
	Grey, hard, completely weathered rock (recovered as elastic SILT with gravel).		R	50/5			SPT at 10.50m depth.
11					45	0	Coring at 11.0m to 12.0m depth.
12	-ditto-		R	50/4			SPT at 12.0m depth.
13							
	Grey, hard, completely weathered rock (recovered as clayey SILT with gravel).		R	50/4			SPT at 13.50m depth.
14							
15	-ditto-		R	50/3			SPT at 15.0m depth.
	End of borehole at 15.0m depth.						
16							
17							
18							
19							
20							
21							

RIYADH GEOTECHNIQUE & FOUNDATIONS

Test Boring Log

BH NO. BH-23

SHEET 1 OF 2

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**		ELEVATION: **911.15**	N: **2505158.98** E: **264306.20**
PROJECT NO.: 19-K-3054	CLIENT: **L & T CONSTRUCTION**	DATE STARTED **2-9-19**	DATE COMPLETED **9-2-19**
DRILLING EQUIPMENT: **Acker AD II**		GROUND WATER	
SIZE & TYPE OF CASING: **100mm diameter pipe casing**		DATE / TIME / WATER DEPTH (m) / CASING DEPTH	
DRILLING METHOD: **Straight Rotary**			
CORE BAR.: **-**	LENGTH: **-** BIT: **-**	DRILLER: **Azim**	
SAMPLER: **S.P.T.**	WEIGHT: **63.50 Kg** DROP: **76 cm**	INSPECTOR: **Azam S.**	

DEPTH (m)	DESCRIPTION	PROFILE	SAMPLES SPT-N	SAMPLES PENET BL/15cm	ROCK REC. %	ROCK RQD. %	REMARKS
	Light brown, dense, silty GRAVEL with sand.		33	10 14 19			SPT at 0.15m depth.
1	Grey, very dense, completely weathered rock (recovered as silty SAND with gravel).		R	50/ 3			SPT at 0.75m depth.
	-ditto- and trace clay.		R	50/ 6			SPT at 1.50m depth.
2	-ditto-		R	27 50/ 10			SPT at 2.25m depth.
3	-ditto-		R	30 50/ 7			SPT at 3.0m depth.
	Grey, very dense, completely weathered rock (recovered as silty GRAVEL with sand). UDS - 1		R	39 50/ 6			SPT at 3.75m depth. Undisturbed Sample @ 4.0m to 4.50m depth.
4	-ditto-		R	45 50/ 4			SPT at 4.50m depth.
5							
6	Grey, very dense, completely weathered rock (recovered as silty SAND with gravel).		R	50/ 5	95	75	SPT at 6.0m depth. Coring at 6.10m to 7.50m depth.
7	Dark brown, moderately weathered, slightly fractured, ANDESITE. Water loss 20% at 6.0m to 7.50m depth.						
8	Dark green, slightly weathered, slightly fractured, ANDESITE, (medium to high strength).				90	75	Coring at 7.50m to 9.00m depth.
9	Grey, slightly weathered, slightly fractured, ANDESITE, (high to very high strength).				90	80	Coring at 9.00m to 10.50m depth.
10							

RIYADH GEOTECHNIQUE & FOUNDATIONS

Test Boring Log

BH NO. **BH-23**

SHEET **2** OF **2**

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**

CLIENT: **L & T CONSTRUCTION**

PROJECT NO.: **19-K-3054**

DEPTH (m)	DESCRIPTION	PROFILE	SAMPLES SPT-N	SAMPLES PENET BL/15cm	ROCK REC. %	ROCK RQD. %	REMARKS
11	Grey, slightly weathered, slightly fractured, ANDESITE, (high to very high strength). *(continued)* -ditto-				95	90	Coring at 10.50m to 12.00m depth.
12	Grey, slightly weathered, slightly fractured, ANDESITE, (very high strength).				97	90	Coring at 12.00m to 13.50m depth.
13							
14	Dark brown, slightly weathered, slightly fractured, ANDESITE, (medium to high strength).				95	90	Coring at 13.50m to 15.00m depth.
15	***End of Borehole at 15.0m depth.***						
16							
17							
18							
19							
20							
21							

RIYADH GEOTECHNIQUE & FOUNDATIONS

Test Boring Log

BH NO. **BH-25**

SHEET **1** OF **2**

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**		ELEVATION: **912.25**	N: **2506297.89** E: **264397.65**
PROJECT NO.: 19-K-3054	CLIENT: **L & T CONSTRUCTION**	DATE STARTED **8-28-19**	DATE COMPLETED **8-28-19**
DRILLING EQUIPMENT: **Acker AD II**		GROUND WATER	
SIZE & TYPE OF CASING: **100mm diameter pipe casing**		DATE / TIME / WATER DEPTH (m) / CASING DEPTH	
DRILLING METHOD: **Straight Rotary**			
CORE BAR.: **-**	LENGTH: **-** BIT: **-**	DRILLER: **Sanaullah**	
SAMPLER: **S.P.T.**	WEIGHT: **63.50 Kg** DROP: **76 cm**	INSPECTOR: **Azam S.**	

DEPTH (m)	DESCRIPTION	PROFILE	SAMPLES SPT-N	SAMPLES PENET BL/15cm	ROCK REC. %	ROCK RQD. %	REMARKS
	Light brown, loose, silty SAND with gravel.		9	4 4 5			SPT at 0.15m depth.
1	Brown, medium dense, poorly graded SAND with silt and gravel.		26	7 11 15			SPT at 0.75m depth.
2	Grey, very dense, completely weathered rock (recovered as silty GRAVEL).		R	50/ 6			SPT at 1.50m depth.
	-ditto-		R	50/ 5			SPT at 2.25m depth.
3	Grey, very dense, completely weathered rock (recovered as poorly graded GRAVEL with silt).		R	50/ 6			SPT at 3.0m depth.
4	-ditto-		R	50/ 7			SPT at 3.75m depth.
5	-ditto-		R	50/ 10	31	0	SPT at 4.50m depth. Coring at 4.60m to 6.0m depth.
6	-ditto-		R	50/ 5			SPT at 6.0m depth.
7							
8	Grey, very dense, completely weathered rock (recovered as silty SAND with gravel).		R	50/ 6			SPT at 7.50m depth.
9	-ditto-		R	50/ 2	70	40	SPT at 9.0m depth. Coring at 9.10m to 10.50m depth.
	grey, moderately weathered, slightly fractured, ANDESITE, (green/Quartzite).						
10							

RIYADH GEOTECHNIQUE & FOUNDATIONS

Test Boring Log

BH NO. **BH-25**

SHEET **2** OF **2**

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**

CLIENT: **L & T CONSTRUCTION**

PROJECT NO.: **19-K-3054**

DEPTH (m)	DESCRIPTION	PROFILE	SAMPLES		ROCK		REMARKS
			SPT-N	PENET BL/15cm	REC. %	RQD. %	
	grey, moderately weathered, slightly fractured, ANDESITE, (green/Quartzite). *(continued)*						
	-ditto-				80	60	Coring at 10.50m to 12.00m depth.
11							
12	-ditto-				90	65	Coring at 12.00m to 13.50m depth.
13							
	-ditto-				90	70	Coring at 13.50m to 15.00m depth.
14							
15	***End of borehole at 15.0m depth.***						
16							
17							
18							
19							
20							
21							

RIYADH GEOTECHNIQUE & FOUNDATIONS

Test Boring Log

BH NO.	**BH-26**
SHEET	**1** OF **2**

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**		ELEVATION: **911.85**	N: **2506130.16** E: **264334.91**
PROJECT NO.: 19-K-3054	CLIENT: **L & T CONSTRUCTION**	DATE STARTED **8-9-19**	DATE COMPLETED **8-9-19**
DRILLING EQUIPMENT: **Acker AD II**		GROUND WATER	
SIZE & TYPE OF CASING: **100mm diameter pipe casing**		DATE / TIME / WATER DEPTH (m) / CASING DEPTH	
DRILLING METHOD: **Straight Rotary**			
CORE BAR.: **-**	LENGTH: **-** BIT: **-**	DRILLER: **Azim**	
SAMPLER: **S.P.T.**	WEIGHT: **63.50 Kg** DROP: **76 cm**	INSPECTOR: **Azam S.**	

DEPTH (m)	DESCRIPTION	PROFILE	SAMPLES SPT-N	SAMPLES PENET BL/15cm	ROCK REC. %	ROCK RQD. %	REMARKS
	Light brown, medium dense, clayey SAND with gravel.		18	7 8 10			SPT at 0.15m depth.
1	Grey, very dense, completely weathered rock (recovered as silty SAND).		76	30 35 41			SPT at 0.75m depth.
	-ditto-		R	40 50/ 10			SPT at 1.50m depth.
2							
	Grey, very dense, completely weathered rock (recovered as poorly graded GRAVEL with silt).		R	50/ 14			SPT at 2.25m depth.
3	-ditto-		R	50/ 13			SPT at 3.0m depth.
	-ditto-		R	50/ 11			SPT at 3.75m depth.
4							
	-ditto-		R	50/ 10			SPT at 4.50m depth.
5							
6	-ditto-		R	50/ 5			SPT at 6.0m depth.
7							
	Grey, very dense, completely weathered rock (recovered as silty SAND with gravel).		R	50/ 12			SPT at 7.50m depth.
8							
9	-ditto-		R	50/ 11			SPT at 9.0m depth.
10							

RIYADH GEOTECHNIQUE & FOUNDATIONS

Test Boring Log

BH NO. **BH-26**

SHEET **2** OF **2**

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**

CLIENT: **L & T CONSTRUCTION**

PROJECT NO.: **19-K-3054**

DEPTH (m)	DESCRIPTION	PROFILE	SAMPLES SPT-N	SAMPLES PENET BL/15cm	ROCK REC. %	ROCK RQD. %	REMARKS
11	Grey, very dense, completely weathered rock (recovered as silty SAND with gravel). *(continued)*						
	-ditto-		R	50/14			SPT at 10.50m depth.
12	-ditto-		R	50/2			SPT at 12.0m depth.
	Grey, slightly weathered, slightly fractured, ANDESITE, (interbedded some gnesis).				90	70	Coring at 12.50m to 13.50m depth.
13							
	-ditto-				95	85	Coring at 13.50m to 15.00m depth.
14							
15	***End of borehole at 15.0m depth.***						
16							
17							
18							
19							
20							
21							

RIYADH GEOTECHNIQUE & FOUNDATIONS

Test Boring Log

BH NO. **BH-27**

SHEET **1** OF **1**

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**		ELEVATION: **912.25**	N: **2506067.53** E: **264289.07**
PROJECT NO.: 19-K-3054	CLIENT: **L & T CONSTRUCTION**	DATE STARTED **8-9-19**	DATE COMPLETED **8-9-19**
DRILLING EQUIPMENT: **Acker AD II**		GROUND WATER	
SIZE & TYPE OF CASING: **100mm diameter pipe casing**		DATE / TIME / WATER DEPTH (m) / CASING DEPTH	
DRILLING METHOD: **Straight Rotary**			
CORE BAR.: **-**	LENGTH: **-** BIT: **-**	DRILLER: **Azim**	
SAMPLER: **S.P.T.**	WEIGHT: **63.50 Kg** DROP: **76 cm**	INSPECTOR: **Azam S.**	

DEPTH (m)	DESCRIPTION	PROFILE	SAMPLES SPT-N	SAMPLES PENET BL/15cm	ROCK REC. %	ROCK RQD. %	REMARKS
	Light brown, medium dense, silty SAND with gravel.		13	4 6 7			SPT at 0.15m depth.
1	Changing to brown color and hard.		31	1 014 17			SPT at 0.75m depth.
	Brown, dense, clayey SAND.		48	16 24 24			SPT at 1.50m depth.
2	Brown, very dense, completely weathered rock (recovered as clayey SAND). UDS - 1		R	50/ 14			SPT at 2.25m depth. Undisturbed Sample @ 2.50m to 3.0m depth.
3	Brown, very dense, completely weathered rock (recovered as silty SAND with gravel).		R	50/ 13			SPT at 3.0m depth.
4	-ditto-		R	50/ 10			SPT at 3.75m depth.
5	Brown, very dense, completely weathered rock (recovered as clayey SAND).		R	50/ 9			SPT at 4.50m depth.
6	Brown to grey, very dense, completely weathered rock (recovered as silty SAND with gravel).		R	50/ 6			SPT at 6.0m depth.
7	-ditto-		R	50/ 10			SPT at 7.50m depth.
8	-ditto-		R	50/ 7			SPT at 8.0m depth.
	End of Borehole at 8.0m depth.						
9							
10							

RIYADH GEOTECHNIQUE & FOUNDATIONS

Test Boring Log

BH NO. **BH-28**

SHEET **1** OF **2**

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**		ELEVATION: **913.15**	N: **2506033.19** E: **263964.39**
PROJECT NO.: 19-K-3054	CLIENT: **L & T CONSTRUCTION**	DATE STARTED **9-16-19**	DATE COMPLETED **9-16-19**
DRILLING EQUIPMENT: **Acker AD II**		GROUND WATER	
SIZE & TYPE OF CASING: **100mm diameter pipe casing**		DATE / TIME / WATER DEPTH (m) / CASING DEPTH	
DRILLING METHOD: **Straight Rotary**			
CORE BAR.:	LENGTH: / BIT:	DRILLER: **Azim**	
SAMPLER: **S.P.T.**	WEIGHT: **63.50 Kg** / DROP: **76 cm**	INSPECTOR: **Azam S.**	

DEPTH (m)	DESCRIPTION	PROFILE	SAMPLES SPT-N	PENET BL/15cm	ROCK REC. %	ROCK RQD. %	REMARKS
	Brown, loose, poorly graded SAND with silt.		9	4 4 5			SPT at 0.15m depth.
1	Brown, medium dense, silty SAND.		22	8 10 12			SPT at 0.75m depth.
	Brown to off white, hard, sandy SILT trace clay.		33	10 15 18			SPT at 1.50m depth.
2	-ditto-		36	10 17 19			SPT at 2.25m depth.
3	Brown to grey, hard, clayey SILT with gravel trace clay.		36	10 16 20			SPT at 3.0m depth.
4	Brown, dense, silty SAND (few gravel).		35	10 15 20			SPT at 3.75m depth.
5	Grey, dense, poorly graded SAND with silt and gravel.		38	10 16 22			SPT at 4.50m depth.
6	-ditto-		43	13 19 24			SPT at 6.0m depth.
7							
8	Brown to off white, very dense, silty SAND with gravel.		72	16 32 40			SPT at 7.50m depth.
9	-ditto-		62	16 25 37			SPT at 9.0m depth.
10							

RIYADH GEOTECHNIQUE & FOUNDATIONS

Test Boring Log

BH NO. **BH-28**

SHEET **2** OF **2**

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**

CLIENT: **L & T CONSTRUCTION**

PROJECT NO.: **19-K-3054**

DEPTH (m)	DESCRIPTION	PROFILE	SAMPLES SPT-N	SAMPLES PENET BL/15cm	ROCK REC. %	ROCK RQD. %	REMARKS
11	Brown to off white, very dense, silty SAND with gravel. *(continued)*						
	Brown to grey, hard, clayey SILT with gravel trace sand.		49	14 19 30			SPT at 10.50m depth.
12	-ditto-, with gravel.		57	24 26 31			SPT at 12.0m depth.
13							
14	Brown, very dense, poorly graded SAND with silt and gravel.		60	25 28 32			SPT at 13.50m depth.
15	-ditto-		63	25 30 33			SPT at 15.0m depth.
	End of borehole at 15.0m depth.						
16							
17							
18							
19							
20							
21							

RIYADH GEOTECHNIQUE & FOUNDATIONS

Test Boring Log

BH NO. **BH-29**

SHEET **1** OF **2**

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**		ELEVATION: **912.50**	N: **2506526.60** E: **264367.17**
PROJECT NO.: 19-K-3054	CLIENT: **L & T CONSTRUCTION**	DATE STARTED **9-21-19**	DATE COMPLETED **9-21-19**
DRILLING EQUIPMENT: **Acker AD II**		GROUND WATER	
SIZE & TYPE OF CASING: **100mm diameter pipe casing**		DATE / TIME / WATER DEPTH (m) / CASING DEPTH	
DRILLING METHOD: **Straight Rotary**			
CORE BAR.:	LENGTH: / BIT:	DRILLER: **Azim**	
SAMPLER: **S.P.T.**	WEIGHT: **63.50 Kg** / DROP: **76 cm**	INSPECTOR: **Azam S.**	

DEPTH (m)	DESCRIPTION	PROFILE	SAMPLES SPT-N	SAMPLES PENET BL/15cm	ROCK REC. %	ROCK RQD. %	REMARKS
0	Light brown, medium dense, silty SAND (few gravel).		15	5 7 8			SPT at 0.15m depth.
1	Changing to dense.		31	10 14 17			SPT at 0.75m depth.
2	Grey, hard, sandy SILT with gravel.		62	23 29 33			SPT at 1.50m depth.
	-ditto-		64	23 30 34			SPT at 2.25m depth.
3	Grey, very dense, gravely SAND with trave silt.		62	23 30 32			SPT at 3.0m depth.
4	Brown to grey, hard, clayey SILT with gravel trace sand.		78	29 36 42			SPT at 3.75m depth.
5	Brown, hard, completely weathered rock (recovered as clayey SILT).		R	36 45 50/ 14			SPT at 4.50m depth.
6	-ditto-		R	40 50/ 13			SPT at 6.0m depth.
7							
8	-ditto-		R	21 34 50/ 13			SPT at 7.50m depth.
9	Grey, very dense, poorly graded SAND with silt trace clay.		R	50/ 12			SPT at 9.0m depth.
10							

RIYADH GEOTECHNIQUE & FOUNDATIONS

Test Boring Log

BH NO. **BH-29**

SHEET **2** OF **2**

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**

CLIENT: **L & T CONSTRUCTION**

PROJECT NO.: **19-K-3054**

DEPTH (m)	DESCRIPTION	PROFILE	SAMPLES SPT-N	SAMPLES PENET BL/15cm	ROCK REC. %	ROCK RQD. %	REMARKS
11	Grey, very dense, poorly graded SAND with silt trace clay. *(continued)*						
	Grey, very dense, poorly graded SAND with silt.		R	50/10			SPT at 10.50m depth.
12	-ditto-		R	50/5			SPT at 12.0m depth.
							Coring at 12.50m to 13.00m depth.
13	-ditto-		R	50/4			SPT at 13.50m depth.
14							
15	-ditto-		R	50/3			SPT at 15.0m depth.
	End of borehole at 15.0m depth.						
16							
17							
18							
19							
20							
21							

TEST PIT LOGS

TEST PIT LOG

GEOTECHNICAL & CIVIL TESTING UNIT

PROJECT NAME:	MANSOURAH MASSARAH GOLD PROJECT	PROJECT NO:	19-K-3054
CLIENT:	L & T CONSTRUCTION	FIELD INVESTIGATION:	
COORDINATES:	NORTH: 2,505,871.02 meter	TYPE OF BORING:	
	EAST: 263,813.79 meter	DIA. OF BORING: CORE SIZE:	TESTPIT NO: TP-01
GROUND ELEVATION:	meter	DIA. OF UNDISTURBED SAMPLE:	DATE COMMENCED: 8-27-19
DEPTH OF GROUNDWATER TABLE:	meter	CASING HAMMER WT. & DROP:	DATE COMPLETED: 8-27-19
MEASURED AT:	HOURS ON:	HAMMER WT. & DROP:	SUPERVISOR: Azam S.

Depth (feet)	Dry Density (gm/cm3)	Natural Moisture Content (%)	Maximum Dry Density (gm/cm3)	Optimum Moisture Content (%)	In place % Compaction	Soil Symbol	Classi-fication (D-2487)	DESCRIPTION	Depth (meters)
1							SM	Light brown, medium dense, silty SAND with gravel.	0.3
2									0.6
3								End of testpit @ 0.60m depth.	0.9
4									1.2
5									1.5
6									1.8
7									2.1
8									2.4
9									2.7

L E G E N D:

SPT-1 : STANDARD PENETRATION TEST & NUMBER PER 12 INCHES
H-1 : HAND SAMPLE & NUMBER
R-1 : ROCK CORE RUN & NUMBER
RQD : ROCK QUALITY DESIGNATION
▼ : WATER TABLE

REC : ROCK CORE RECOVERY
HWR : HIGHLY WEATHERED ROCK
PWR : PARTIALLY WEATHERED ROCK
ROCK : UNWEATHERED ROCK
NE NOT ENCOUNTERED

TEST PIT LOG

GEOTECHNICAL & CIVIL TESTING UNIT

PROJECT NAME:	MANSOURAH MASSARAH GOLD PROJECT	PROJECT NO:	19-K-3054
CLIENT:	L & T CONSTRUCTION	FIELD INVESTIGATION:	
COORDINATES:	NORTH: 2,506,290.71 meter	TYPE OF BORING:	
	EAST: 263,710.92 meter	DIA. OF BORING: CORE SIZE:	TESTPIT NO: TP-02
GROUND ELEVATION:	meter	DIA. OF UNDISTURBED SAMPLE:	DATE COMMENCED: 8-27-19
DEPTH OF GROUNDWATER TABLE:	meter	CASING HAMMER WT. & DROP:	DATE COMPLETED: 8-27-19
MEASURED AT:	HOURS ON:	HAMMER WT. & DROP:	SUPERVISOR: Azam S.

Depth (feet)	Dry Density (gm/cm3)	Natural Moisture Content (%)	Maximum Dry Density (gm/cm3)	Optimum Moisture Content (%)	In place % Compaction	Soil Symbol	Classi-fication (D-2487)	DESCRIPTION	Depth (meters)
								Light brown, medium dense to dense, clayey SAND with gravel.	
1									0.3
							SC		
2									0.6
								Greenish grey, dense, silty GRAVEL with sand.	
3							GM		0.9
								End of testpit @ 1.0m depth.	
4									1.2
5									1.5
6									1.8
7									2.1
8									2.4
9									2.7

L E G E N D:

SPT-1 : STANDARD PENETRATION TEST & NUMBER PER 12 INCHES
H-1 : HAND SAMPLE & NUMBER
R-1 : ROCK CORE RUN & NUMBER
RQD : ROCK QUALITY DESIGNATION
▼ : WATER TABLE

REC : ROCK CORE RECOVERY
HWR : HIGHLY WEATHERED ROCK
PWR : PARTIALLY WEATHERED ROCK
ROCK : UNWEATHERED ROCK
NE NOT ENCOUNTERED

TEST PIT LOG

GEOTECHNICAL & CIVIL TESTING UNIT

PROJECT NAME:	MANSOURAH MASSARAH GOLD PROJECT	PROJECT NO:	19-K-3054
CLIENT:	L & T CONSTRUCTION	FIELD INVESTIGATION:	
COORDINATES:	NORTH: 2,506,407.79 meter	TYPE OF BORING:	
	EAST: 263,670.98 meter	DIA. OF BORING: CORE SIZE:	TESTPIT NO: TP-03
GROUND ELEVATION:	meter	DIA. OF UNDISTURBED SAMPLE:	DATE COMMENCED: 8-27-19
DEPTH OF GROUNDWATER TABLE:	meter	CASING HAMMER WT. & DROP:	DATE COMPLETED: 8-27-19
MEASURED AT:	HOURS ON:	HAMMER WT. & DROP:	SUPERVISOR: Azam S.

Depth (feet)	Dry Density (gm/cm3)	Natural Moisture Content (%)	Maximum Dry Density (gm/cm3)	Optimum Moisture Content (%)	In place % Compaction	Soil Symbol	Classi-fication (D-2487)	DESCRIPTION	Depth (meters)
								Brown, medium dense to dense, silty SAND with gravel.	
1									0.3
							SM		
2									0.6
3								End of testpit @ 0.8m depth.	0.9
4									1.2
5									1.5
6									1.8
7									2.1
8									2.4
9									2.7

L E G E N D:

SPT-1 : STANDARD PENETRATION TEST & NUMBER PER 12 INCHES
H-1 : HAND SAMPLE & NUMBER
R-1 : ROCK CORE RUN & NUMBER
RQD : ROCK QUALITY DESIGNATION
▼ : WATER TABLE

REC : ROCK CORE RECOVERY
HWR : HIGHLY WEATHERED ROCK
PWR : PARTIALLY WEATHERED ROCK
ROCK : UNWEATHERED ROCK
NE NOT ENCOUNTERED

TEST PIT LOG

GEOTECHNICAL & CIVIL TESTING UNIT

PROJECT NAME:	MANSOURAH MASSARAH GOLD PROJECT	PROJECT NO:	19-K-3054
CLIENT:	L & T CONSTRUCTION	FIELD INVESTIGATION:	
COORDINATES:	NORTH: 2,506,095.35 meter	TYPE OF BORING:	
	EAST: 263,862.55 meter	DIA. OF BORING: CORE SIZE:	TESTPIT NO: TP-04
GROUND ELEVATION:	meter	DIA. OF UNDISTURBED SAMPLE:	DATE COMMENCED: 8-27-19
DEPTH OF GROUNDWATER TABLE:	meter	CASING HAMMER WT. & DROP:	DATE COMPLETED: 8-27-19
MEASURED AT:	HOURS ON:	HAMMER WT. & DROP:	SUPERVISOR: Azam S.

Depth (feet)	Dry Density (gm/cm3)	Natural Moisture Content (%)	Maximum Dry Density (gm/cm3)	Optimum Moisture Content (%)	In place % Compaction	Soil Symbol	Classi-fication (D-2487)	DESCRIPTION	Depth (meters)
							SM	Light brown, medium dense to dense, silty SAND with gravel.	
1									0.3
							GP-GM	Grey, dense, poorly graded GRAVEL with silt.	
2									0.6
3									0.9
								End of testpit @ 0.9m depth.	
4									1.2
5									1.5
6									1.8
7									2.1
8									2.4
9									2.7

L E G E N D:

SPT-1 : STANDARD PENETRATION TEST & NUMBER PER 12 INCHES
H-1 : HAND SAMPLE & NUMBER
R-1 : ROCK CORE RUN & NUMBER
RQD : ROCK QUALITY DESIGNATION
▼ : WATER TABLE

REC : ROCK CORE RECOVERY
HWR : HIGHLY WEATHERED ROCK
PWR : PARTIALLY WEATHERED ROCK
ROCK : UNWEATHERED ROCK
NE NOT ENCOUNTERED

TEST PIT LOG

GEOTECHNICAL & CIVIL TESTING UNIT

PROJECT NAME:	MANSOURAH MASSARAH GOLD PROJECT	PROJECT NO:	19-K-3054
CLIENT:	L & T CONSTRUCTION	FIELD INVESTIGATION:	
COORDINATES:	NORTH: 2,505,773.76 meter	TYPE OF BORING:	
	EAST: 264,071.60 meter	DIA. OF BORING: CORE SIZE:	TESTPIT NO: TP-05
GROUND ELEVATION:	meter	DIA. OF UNDISTURBED SAMPLE:	DATE COMMENCED: 8-27-19
DEPTH OF GROUNDWATER TABLE:	meter	CASING HAMMER WT. & DROP:	DATE COMPLETED: 8-27-19
MEASURED AT:	HOURS ON:	HAMMER WT. & DROP:	SUPERVISOR: Azam S.

Depth (feet)	Dry Density (gm/cm3)	Natural Moisture Content (%)	Maximum Dry Density (gm/cm3)	Optimum Moisture Content (%)	In place % Compaction	Soil Symbol	Classification (D-2487)	DESCRIPTION	Depth (meters)
								Brown, medium dense to dense, silty SAND with gravel.	
							SM		
1									0.3
								Brown, dense, clayey SAND with gravel.	
2									0.6
							SC		
3									0.9
								End of testpit @ 1.0m depth.	
4									1.2
5									1.5
6									1.8
7									2.1
8									2.4
9									2.7

L E G E N D:

SPT-1 : STANDARD PENETRATION TEST & NUMBER PER 12 INCHES
H-1 : HAND SAMPLE & NUMBER
R-1 : ROCK CORE RUN & NUMBER
RQD : ROCK QUALITY DESIGNATION
▼ : WATER TABLE

REC : ROCK CORE RECOVERY
HWR : HIGHLY WEATHERED ROCK
PWR : PARTIALLY WEATHERED ROCK
ROCK : UNWEATHERED ROCK
NE NOT ENCOUNTERED

TEST PIT LOG

GEOTECHNICAL & CIVIL TESTING UNIT

PROJECT NAME: MANSOURAH MASSARAH GOLD PROJECT			PROJECT NO: 19-K-3054
CLIENT: L & T CONSTRUCTION		FIELD INVESTIGATION:	
COORDINATES:	NORTH: 2,505,856.46 meter	TYPE OF BORING:	
	EAST: 264,251.25 meter	DIA. OF BORING: CORE SIZE:	TESTPIT NO: TP-06
GROUND ELEVATION: meter		DIA. OF UNDISTURBED SAMPLE:	DATE COMMENCED: 8-27-19
DEPTH OF GROUNDWATER TABLE: meter		CASING HAMMER WT. & DROP:	DATE COMPLETED: 8-27-19
MEASURED AT:	HOURS ON:	HAMMER WT. & DROP:	SUPERVISOR: Azam S.

Depth (feet)	Dry Density (gm/cm3)	Natural Moisture Content (%)	Maximum Dry Density (gm/cm3)	Optimum Moisture Content (%)	In place % Compaction	Soil Symbol	Classi-fication (D-2487)	DESCRIPTION	Depth (meters)
1							SC	Light brown, medium dense to dense, clayey SAND with gravel.	0.3
2								End of testpit @ 0.55m depth.	0.6
3									0.9
4									1.2
5									1.5
6									1.8
7									2.1
8									2.4
9									2.7

L E G E N D:

SPT-1 : STANDARD PENETRATION TEST & NUMBER PER 12 INCHES
H-1 : HAND SAMPLE & NUMBER
R-1 : ROCK CORE RUN & NUMBER
RQD : ROCK QUALITY DESIGNATION
▼ : WATER TABLE

REC : ROCK CORE RECOVERY
HWR : HIGHLY WEATHERED ROCK
PWR : PARTIALLY WEATHERED ROCK
ROCK : UNWEATHERED ROCK
NE NOT ENCOUNTERED

TEST PIT LOG

GEOTECHNICAL & CIVIL TESTING UNIT

PROJECT NAME:	MANSOURAH MASSARAH GOLD PROJECT	PROJECT NO:	19-K-3054
CLIENT:	L & T CONSTRUCTION	FIELD INVESTIGATION:	
COORDINATES:	NORTH: 2,506,207.69 meter	TYPE OF BORING:	
	EAST: 264,054.02 meter	DIA. OF BORING: CORE SIZE:	TESTPIT NO: TP-07
GROUND ELEVATION:	meter	DIA. OF UNDISTURBED SAMPLE:	DATE COMMENCED: 8-27-19
DEPTH OF GROUNDWATER TABLE:	meter	CASING HAMMER WT. & DROP:	DATE COMPLETED: 8-27-19
MEASURED AT:	HOURS ON:	HAMMER WT. & DROP:	SUPERVISOR: Azam S.

Depth (feet)	Dry Density (gm/cm3)	Natural Moisture Content (%)	Maximum Dry Density (gm/cm3)	Optimum Moisture Content (%)	In place % Compaction	Soil Symbol	Classi-fication (D-2487)	DESCRIPTION	Depth (meters)
								Brown, medium dense to dense, silty SAND with gravel.	
1									0.3
2									0.6
3							SM		0.9
4									1.2
5								End of testpit @ 1.3m depth.	1.5
6									1.8
7									2.1
8									2.4
9									2.7

L E G E N D:

SPT-1 : STANDARD PENETRATION TEST & NUMBER PER 12 INCHES
H-1 : HAND SAMPLE & NUMBER
R-1 : ROCK CORE RUN & NUMBER
RQD : ROCK QUALITY DESIGNATION
▼ : WATER TABLE

REC : ROCK CORE RECOVERY
HWR : HIGHLY WEATHERED ROCK
PWR : PARTIALLY WEATHERED ROCK
ROCK : UNWEATHERED ROCK
NE NOT ENCOUNTERED

TEST PIT LOG

GEOTECHNICAL & CIVIL TESTING UNIT

PROJECT NAME: MANSOURAH MASSARAH GOLD PROJECT			PROJECT NO: 19-K-3054
CLIENT: L & T CONSTRUCTION		FIELD INVESTIGATION:	
COORDINATES:	NORTH: 2,506,296.89 meter	TYPE OF BORING:	
	EAST: 264,215.96 meter	DIA. OF BORING: CORE SIZE:	TESTPIT NO: TP-08
GROUND ELEVATION: meter		DIA. OF UNDISTURBED SAMPLE:	DATE COMMENCED: 8-27-19
DEPTH OF GROUNDWATER TABLE: meter		CASING HAMMER WT. & DROP:	DATE COMPLETED: 8-27-19
MEASURED AT:	HOURS ON:	HAMMER WT. & DROP:	SUPERVISOR: Azam S.

Depth (feet)	Dry Density (gm/cm3)	Natural Moisture Content (%)	Maximum Dry Density (gm/cm3)	Optimum Moisture Content (%)	In place % Compaction	Soil Symbol	Classi-fication (D-2487)	DESCRIPTION	Depth (meters)
1							SC	Brown, medium dense to dense, clayey SAND with gravel.	0.3
2								End of testpit @ 0.50m depth.	0.6
3									0.9
4									1.2
5									1.5
6									1.8
7									2.1
8									2.4
9									2.7

L E G E N D:

SPT-1 : STANDARD PENETRATION TEST & NUMBER PER 12 INCHES
H-1 : HAND SAMPLE & NUMBER
R-1 : ROCK CORE RUN & NUMBER
RQD : ROCK QUALITY DESIGNATION
▼ : WATER TABLE

REC : ROCK CORE RECOVERY
HWR : HIGHLY WEATHERED ROCK
PWR : PARTIALLY WEATHERED ROCK
ROCK : UNWEATHERED ROCK
NE NOT ENCOUNTERED

TEST PIT LOG

GEOTECHNICAL & CIVIL TESTING UNIT

PROJECT NAME:	MANSOURAH MASSARAH GOLD PROJECT	PROJECT NO:	19-K-3054
CLIENT:	L & T CONSTRUCTION	FIELD INVESTIGATION:	
COORDINATES:	NORTH: 2,506,108.53 meter	TYPE OF BORING:	
	EAST: 264,399.72 meter	DIA. OF BORING: CORE SIZE:	TESTPIT NO: TP-09
GROUND ELEVATION:	meter	DIA. OF UNDISTURBED SAMPLE:	DATE COMMENCED: 8-27-19
DEPTH OF GROUNDWATER TABLE:	meter	CASING HAMMER WT. & DROP:	DATE COMPLETED: 8-27-19
MEASURED AT:	HOURS ON:	HAMMER WT. & DROP:	SUPERVISOR: Azam S.

Depth (feet)	Dry Density (gm/cm3)	Natural Moisture Content (%)	Maximum Dry Density (gm/cm3)	Optimum Moisture Content (%)	In place % Compaction	Soil Symbol	Classi-fication (D-2487)	DESCRIPTION	Depth (meters)
1							GM	Brown, medium dense to dense, silty GRAVEL with sand.	0.3
2									0.6
3							SW-SM	Grey, dense, well graded SAND with silt and gravel.	0.9
4								End of testpit @ 1.0m depth.	1.2
5									1.5
6									1.8
7									2.1
8									2.4
9									2.7

L E G E N D:

SPT-1 : STANDARD PENETRATION TEST & NUMBER PER 12 INCHES
H-1 : HAND SAMPLE & NUMBER
R-1 : ROCK CORE RUN & NUMBER
RQD : ROCK QUALITY DESIGNATION
▼ : WATER TABLE

REC : ROCK CORE RECOVERY
HWR : HIGHLY WEATHERED ROCK
PWR : PARTIALLY WEATHERED ROCK
ROCK : UNWEATHERED ROCK
NE NOT ENCOUNTERED

TEST PIT LOG

GEOTECHNICAL & CIVIL TESTING UNIT

PROJECT NAME: MANSOURAH MASSARAH GOLD PROJECT			PROJECT NO: 19-K-3054
CLIENT: L & T CONSTRUCTION		FIELD INVESTIGATION:	
COORDINATES:	NORTH: 2,506,147.90 meter	TYPE OF BORING:	
	EAST: 264,457.77 meter	DIA. OF BORING: CORE SIZE:	TESTPIT NO: TP-10
GROUND ELEVATION: meter		DIA. OF UNDISTURBED SAMPLE:	DATE COMMENCED: 8-27-19
DEPTH OF GROUNDWATER TABLE: meter		CASING HAMMER WT. & DROP:	DATE COMPLETED: 8-27-19
MEASURED AT:	HOURS ON:	HAMMER WT. & DROP:	SUPERVISOR: Azam S.

Depth (feet)	Dry Density (gm/cm3)	Natural Moisture Content (%)	Maximum Dry Density (gm/cm3)	Optimum Moisture Content (%)	In place % Compaction	Soil Symbol	Classi-fication (D-2487)	DESCRIPTION	Depth (meters)
								brown, medium dense to dense, clayey SAND.	
1									0.3
2									0.6
3							SC		0.9
4									1.2
5								End of testpit @ 1.3m depth.	1.5
6									1.8
7									2.1
8									2.4
9									2.7

L E G E N D:

SPT-1 : STANDARD PENETRATION TEST & NUMBER PER 12 INCHES
H-1 : HAND SAMPLE & NUMBER
R-1 : ROCK CORE RUN & NUMBER
RQD : ROCK QUALITY DESIGNATION
▼ : WATER TABLE

REC : ROCK CORE RECOVERY
HWR : HIGHLY WEATHERED ROCK
PWR : PARTIALLY WEATHERED ROCK
ROCK : UNWEATHERED ROCK
NE NOT ENCOUNTERED

TEST PIT LOG

GEOTECHNICAL & CIVIL TESTING UNIT

PROJECT NAME:	MANSOURAH MASSARAH GOLD PROJECT	PROJECT NO:	19-K-3054
CLIENT:	L & T CONSTRUCTION	FIELD INVESTIGATION:	
COORDINATES:	NORTH: 2,506,472.31 meter	TYPE OF BORING:	
	EAST: 264,419.58 meter	DIA. OF BORING: CORE SIZE:	TESTPIT NO: TP-11
GROUND ELEVATION:	meter	DIA. OF UNDISTURBED SAMPLE:	DATE COMMENCED: 8-28-19
DEPTH OF GROUNDWATER TABLE:	meter	CASING HAMMER WT. & DROP:	DATE COMPLETED: 8-28-19
MEASURED AT:	HOURS ON:	HAMMER WT. & DROP:	SUPERVISOR: Azam S.

Depth (feet)	Dry Density (gm/cm3)	Natural Moisture Content (%)	Maximum Dry Density (gm/cm3)	Optimum Moisture Content (%)	In place % Compaction	Soil Symbol	Classification (D-2487)	DESCRIPTION	Depth (meters)
								Light brown to offwhite, medium dense to dense, silty SAND.	
1							SM		0.3
								Grey, dense, poorly graded GRAVEL with sand.	
2							GP		0.6
3									0.9
								End of testpit @ 0.9m depth.	
4									1.2
5									1.5
6									1.8
7									2.1
8									2.4
9									2.7

L E G E N D:

SPT-1 : STANDARD PENETRATION TEST & NUMBER PER 12 INCHES
H-1 : HAND SAMPLE & NUMBER
R-1 : ROCK CORE RUN & NUMBER
RQD : ROCK QUALITY DESIGNATION
▼ : WATER TABLE

REC : ROCK CORE RECOVERY
HWR : HIGHLY WEATHERED ROCK
PWR : PARTIALLY WEATHERED ROCK
ROCK : UNWEATHERED ROCK
NE NOT ENCOUNTERED

TEST PIT LOG

GEOTECHNICAL & CIVIL TESTING UNIT

PROJECT NAME:	MANSOURAH MASSARAH GOLD PROJECT	PROJECT NO:	19-K-3054
CLIENT:	L & T CONSTRUCTION	FIELD INVESTIGATION:	
COORDINATES:	NORTH: 2,508,492.39 meter	TYPE OF BORING:	
	EAST: 262,866.09 meter	DIA. OF BORING: CORE SIZE:	TESTPIT NO: TP-12
GROUND ELEVATION:	meter	DIA. OF UNDISTURBED SAMPLE:	DATE COMMENCED: 8-28-19
DEPTH OF GROUNDWATER TABLE:	meter	CASING HAMMER WT. & DROP:	DATE COMPLETED: 8-28-19
MEASURED AT:	HOURS ON:	HAMMER WT. & DROP:	SUPERVISOR: Azam S.

Depth (feet)	Dry Density (gm/cm3)	Natural Moisture Content (%)	Maximum Dry Density (gm/cm3)	Optimum Moisture Content (%)	In place % Compaction	Soil Symbol	Classi-fication (D-2487)	DESCRIPTION	Depth (meters)
								Brown, medium dense to dense, silty SAND with gravel.	
1									0.3
							SM		
2									0.6
3								End of testpit @ 0.75m depth.	0.9
4									1.2
5									1.5
6									1.8
7									2.1
8									2.4
9									2.7

L E G E N D:

SPT-1 : STANDARD PENETRATION TEST & NUMBER PER 12 INCHES
H-1 : HAND SAMPLE & NUMBER
R-1 : ROCK CORE RUN & NUMBER
RQD : ROCK QUALITY DESIGNATION
▼ : WATER TABLE

REC : ROCK CORE RECOVERY
HWR : HIGHLY WEATHERED ROCK
PWR : PARTIALLY WEATHERED ROCK
ROCK : UNWEATHERED ROCK
NE NOT ENCOUNTERED

TEST PIT LOG

GEOTECHNICAL & CIVIL TESTING UNIT

PROJECT NAME:	MANSOURAH MASSARAH GOLD PROJECT	PROJECT NO:	19-K-3054
CLIENT:	L & T CONSTRUCTION	FIELD INVESTIGATION:	
COORDINATES:	NORTH: 2,508,703.91 meter	TYPE OF BORING:	
	EAST: 263,193.75 meter	DIA. OF BORING: CORE SIZE:	TESTPIT NO: TP-13
GROUND ELEVATION:	meter	DIA. OF UNDISTURBED SAMPLE:	DATE COMMENCED: 8-28-19
DEPTH OF GROUNDWATER TABLE:	meter	CASING HAMMER WT. & DROP:	DATE COMPLETED: 8-28-19
MEASURED AT:	HOURS ON:	HAMMER WT. & DROP:	SUPERVISOR: Azam S.

Depth (feet)	Dry Density (gm/cm3)	Natural Moisture Content (%)	Maximum Dry Density (gm/cm3)	Optimum Moisture Content (%)	In place % Compaction	Soil Symbol	Classi-fication (D-2487)	DESCRIPTION	Depth (meters)
								brown, medium dense to dense, clayey SAND with gravel.	
1									0.3
							SC		
2									0.6
								Light grey, dense, poorly graded GRAVEL with sand.	
3							GP		0.9
								End of testpit @ 0.95m depth.	
4									1.2
5									1.5
6									1.8
7									2.1
8									2.4
9									2.7

L E G E N D:

SPT-1 : STANDARD PENETRATION TEST & NUMBER PER 12 INCHES
H-1 : HAND SAMPLE & NUMBER
R-1 : ROCK CORE RUN & NUMBER
RQD : ROCK QUALITY DESIGNATION
▼ : WATER TABLE

REC : ROCK CORE RECOVERY
HWR : HIGHLY WEATHERED ROCK
PWR : PARTIALLY WEATHERED ROCK
ROCK : UNWEATHERED ROCK
NE NOT ENCOUNTERED

TEST PIT LOG

GEOTECHNICAL & CIVIL TESTING UNIT

PROJECT NAME: MANSOURAH MASSARAH GOLD PROJECT			PROJECT NO: 19-K-3054
CLIENT: L & T CONSTRUCTION		FIELD INVESTIGATION:	
COORDINATES:	NORTH: 2,508,295.60 meter	TYPE OF BORING:	
	EAST: 263,457.33 meter	DIA. OF BORING: CORE SIZE:	TESTPIT NO: TP-14
GROUND ELEVATION: meter		DIA. OF UNDISTURBED SAMPLE:	DATE COMMENCED: 8-28-19
DEPTH OF GROUNDWATER TABLE: meter		CASING HAMMER WT. & DROP:	DATE COMPLETED: 8-28-19
MEASURED AT:	HOURS ON:	HAMMER WT. & DROP:	SUPERVISOR: Azam S.

Depth (feet)	Dry Density (gm/cm3)	Natural Moisture Content (%)	Maximum Dry Density (gm/cm3)	Optimum Moisture Content (%)	In place % Compaction	Soil Symbol	Classi-fication (D-2487)	DESCRIPTION	Depth (meters)
								brown, medium dense to dense, clayey SAND with gravel.	
1							SC		0.3
2							GP-GM	Brown, dense, poorly graded GRAVEL with silt.	0.6
3								End of testpit @ 0.75m depth.	0.9
4									1.2
5									1.5
6									1.8
7									2.1
8									2.4
9									2.7

L E G E N D:

SPT-1 : STANDARD PENETRATION TEST & NUMBER PER 12 INCHES
H-1 : HAND SAMPLE & NUMBER
R-1 : ROCK CORE RUN & NUMBER
RQD : ROCK QUALITY DESIGNATION
▼ : WATER TABLE

REC : ROCK CORE RECOVERY
HWR : HIGHLY WEATHERED ROCK
PWR : PARTIALLY WEATHERED ROCK
ROCK : UNWEATHERED ROCK
NE NOT ENCOUNTERED

TEST PIT LOG

GEOTECHNICAL & CIVIL TESTING UNIT

PROJECT NAME: MANSOURAH MASSARAH GOLD PROJECT			PROJECT NO: 19-K-3054
CLIENT: L & T CONSTRUCTION		FIELD INVESTIGATION:	
COORDINATES:	NORTH: 2,508,084.08 meter	TYPE OF BORING:	
	EAST: 263,129.67 meter	DIA. OF BORING: CORE SIZE:	TESTPIT NO: TP-15
GROUND ELEVATION: meter		DIA. OF UNDISTURBED SAMPLE:	DATE COMMENCED: 8-28-19
DEPTH OF GROUNDWATER TABLE: meter		CASING HAMMER WT. & DROP:	DATE COMPLETED: 8-28-19
MEASURED AT:	HOURS ON:	HAMMER WT. & DROP:	SUPERVISOR: Azam S.

Depth (feet)	Dry Density (gm/cm3)	Natural Moisture Content (%)	Maximum Dry Density (gm/cm3)	Optimum Moisture Content (%)	In place % Compaction	Soil Symbol	Classi-fication (D-2487)	DESCRIPTION	Depth (meters)
1							SC	Brown, medium dense to dense, clayey SAND with gravel.	0.3
2								End of testpit @ 0.50m depth.	0.6
3									0.9
4									1.2
5									1.5
6									1.8
7									2.1
8									2.4
9									2.7

L E G E N D:

SPT-1 : STANDARD PENETRATION TEST & NUMBER PER 12 INCHES
H-1 : HAND SAMPLE & NUMBER
R-1 : ROCK CORE RUN & NUMBER
RQD : ROCK QUALITY DESIGNATION
▼ : WATER TABLE

REC : ROCK CORE RECOVERY
HWR : HIGHLY WEATHERED ROCK
PWR : PARTIALLY WEATHERED ROCK
ROCK : UNWEATHERED ROCK
NE NOT ENCOUNTERED

TEST PIT LOG

GEOTECHNICAL & CIVIL TESTING UNIT

PROJECT NAME:	MANSOURAH MASSARAH GOLD PROJECT	PROJECT NO:	19-K-3054
CLIENT:	L & T CONSTRUCTION	FIELD INVESTIGATION:	
COORDINATES:	NORTH: 2,508,518.33 meter	TYPE OF BORING:	
	EAST: 263,202.02 meter	DIA. OF BORING: CORE SIZE:	TESTPIT NO: TP-16
GROUND ELEVATION:	meter	DIA. OF UNDISTURBED SAMPLE:	DATE COMMENCED: 8-28-19
DEPTH OF GROUNDWATER TABLE:	meter	CASING HAMMER WT. & DROP:	DATE COMPLETED: 8-28-19
MEASURED AT:	HOURS ON:	HAMMER WT. & DROP:	SUPERVISOR: Azam S.

Depth (feet)	Dry Density (gm/cm3)	Natural Moisture Content (%)	Maximum Dry Density (gm/cm3)	Optimum Moisture Content (%)	In place % Compaction	Soil Symbol	Classi-fication (D-2487)	DESCRIPTION	Depth (meters)
1							SC	Brown, medium dense to dense, clayey SAND with gravel.	0.3
2									0.6
3									0.9
4								End of testpit @ 1.0m depth.	1.2
5									1.5
6									1.8
7									2.1
8									2.4
9									2.7

L E G E N D:

SPT-1 : STANDARD PENETRATION TEST & NUMBER PER 12 INCHES
H-1 : HAND SAMPLE & NUMBER
R-1 : ROCK CORE RUN & NUMBER
RQD : ROCK QUALITY DESIGNATION
▼ : WATER TABLE

REC : ROCK CORE RECOVERY
HWR : HIGHLY WEATHERED ROCK
PWR : PARTIALLY WEATHERED ROCK
ROCK : UNWEATHERED ROCK
NE NOT ENCOUNTERED

TEST PIT LOG

GEOTECHNICAL & CIVIL TESTING UNIT

PROJECT NAME:	MANSOURAH MASSARAH GOLD PROJECT	PROJECT NO:	19-K-3054
CLIENT:	L & T CONSTRUCTION	FIELD INVESTIGATION:	
COORDINATES:	NORTH: 2,508,357.65 meter	TYPE OF BORING:	
	EAST: 263,260.34 meter	DIA. OF BORING: CORE SIZE:	TESTPIT NO: TP-17
GROUND ELEVATION:	meter	DIA. OF UNDISTURBED SAMPLE:	DATE COMMENCED: 8-28-19
DEPTH OF GROUNDWATER TABLE:	meter	CASING HAMMER WT. & DROP:	DATE COMPLETED: 8-28-19
MEASURED AT:	HOURS ON:	HAMMER WT. & DROP:	SUPERVISOR: Azam S.

Depth (feet)	Dry Density (gm/cm3)	Natural Moisture Content (%)	Maximum Dry Density (gm/cm3)	Optimum Moisture Content (%)	In place % Compaction	Soil Symbol	Classi-fication (D-2487)	DESCRIPTION	Depth (meters)
								Brown, medium dense to dense, silty SAND with gravel.	
1									0.3
2							SM		0.6
3									0.9
4								End of testpit @ 0.90m depth.	1.2
5									1.5
6									1.8
7									2.1
8									2.4
9									2.7

L E G E N D:

SPT-1 : STANDARD PENETRATION TEST & NUMBER PER 12 INCHES
H-1 : HAND SAMPLE & NUMBER
R-1 : ROCK CORE RUN & NUMBER
RQD : ROCK QUALITY DESIGNATION
▼ : WATER TABLE

REC : ROCK CORE RECOVERY
HWR : HIGHLY WEATHERED ROCK
PWR : PARTIALLY WEATHERED ROCK
ROCK : UNWEATHERED ROCK
NE NOT ENCOUNTERED

TEST PIT LOG

GEOTECHNICAL & CIVIL TESTING UNIT

PROJECT NAME: MANSOURAH MASSARAH GOLD PROJECT			PROJECT NO: 19-K-3054
CLIENT: L & T CONSTRUCTION		FIELD INVESTIGATION:	
COORDINATES:	NORTH: 2,508,310.03 meter	TYPE OF BORING:	
	EAST: 263,158.63 meter	DIA. OF BORING: CORE SIZE:	TESTPIT NO: TP-18
GROUND ELEVATION: meter		DIA. OF UNDISTURBED SAMPLE:	DATE COMMENCED: 8-28-19
DEPTH OF GROUNDWATER TABLE: meter		CASING HAMMER WT. & DROP:	DATE COMPLETED: 8-28-19
MEASURED AT:	HOURS ON:	HAMMER WT. & DROP:	SUPERVISOR: Azam S.

Depth (feet)	Dry Density (gm/cm3)	Natural Moisture Content (%)	Maximum Dry Density (gm/cm3)	Optimum Moisture Content (%)	In place % Compaction	Soil Symbol	Classi-fication (D-2487)	DESCRIPTION	Depth (meters)
1							SM	Brown, medium dense to dense, silty SAND with gravel.	0.3
2								End of testpit @ 0.50m depth.	0.6
3									0.9
4									1.2
5									1.5
6									1.8
7									2.1
8									2.4
9									2.7

L E G E N D:

SPT-1 : STANDARD PENETRATION TEST & NUMBER PER 12 INCHES
H-1 : HAND SAMPLE & NUMBER
R-1 : ROCK CORE RUN & NUMBER
RQD : ROCK QUALITY DESIGNATION
▼ : WATER TABLE

REC : ROCK CORE RECOVERY
HWR : HIGHLY WEATHERED ROCK
PWR : PARTIALLY WEATHERED ROCK
ROCK : UNWEATHERED ROCK
NE NOT ENCOUNTERED

TEST PIT LOG

GEOTECHNICAL & CIVIL TESTING UNIT

PROJECT NAME: MANSOURAH MASSARAH GOLD PROJECT			PROJECT NO: 19-K-3054
CLIENT: L & T CONSTRUCTION			FIELD INVESTIGATION:
COORDINATES:	NORTH: 2,509,426.37 meter	TYPE OF BORING:	
	EAST: 264,899.01 meter	DIA. OF BORING: CORE SIZE:	TESTPIT NO: TP-19
GROUND ELEVATION: meter		DIA. OF UNDISTURBED SAMPLE:	DATE COMMENCED: 9-22-19
DEPTH OF GROUNDWATER TABLE: meter		CASING HAMMER WT. & DROP:	DATE COMPLETED: 9-22-19
MEASURED AT:	HOURS ON:	HAMMER WT. & DROP:	SUPERVISOR: Azam S.

Depth (feet)	Dry Density (gm/cm3)	Natural Moisture Content (%)	Maximum Dry Density (gm/cm3)	Optimum Moisture Content (%)	In place % Compaction	Soil Symbol	Classi-fication (D-2487)	DESCRIPTION	Depth (meters)
							SM	Light brown, medium dense to dense, silty SAND.	
1									0.3
							SC	Brown, dense, clayey SAND with gravel..	
2								End of testpit at 0.50m depth.	0.6
3									0.9
4									1.2
5									1.5
6									1.8
7									2.1
8									2.4
9									2.7

L E G E N D:

SPT-1 : STANDARD PENETRATION TEST & NUMBER PER 12 INCHES
H-1 : HAND SAMPLE & NUMBER
R-1 : ROCK CORE RUN & NUMBER
RQD : ROCK QUALITY DESIGNATION
▼ : WATER TABLE

REC : ROCK CORE RECOVERY
HWR : HIGHLY WEATHERED ROCK
PWR : PARTIALLY WEATHERED ROCK
ROCK : UNWEATHERED ROCK
NE NOT ENCOUNTERED

TEST PIT LOG

GEOTECHNICAL & CIVIL TESTING UNIT

PROJECT NAME: MANSOURAH MASSARAH GOLD PROJECT			PROJECT NO: 19-K-3054
CLIENT: L & T CONSTRUCTION			FIELD INVESTIGATION:
COORDINATES:	NORTH: 2,509,357.81 meter	TYPE OF BORING:	
	EAST: 264,761.27 meter	DIA. OF BORING: CORE SIZE:	TESTPIT NO: TP-20
GROUND ELEVATION: meter		DIA. OF UNDISTURBED SAMPLE:	DATE COMMENCED: 9-22-19
DEPTH OF GROUNDWATER TABLE: meter		CASING HAMMER WT. & DROP:	DATE COMPLETED: 9-22-19
MEASURED AT:	HOURS ON:	HAMMER WT. & DROP:	SUPERVISOR: Azam S.

Depth (feet)	Dry Density (gm/cm3)	Natural Moisture Content (%)	Maximum Dry Density (gm/cm3)	Optimum Moisture Content (%)	In place % Compaction	Soil Symbol	Classi-fication (D-2487)	DESCRIPTION	Depth (meters)
1							SC	Light brown, medium dense to dense, clayey SAND with gravel.	0.3
2								End of testpit at 0.50m depth.	0.6
3									0.9
4									1.2
5									1.5
6									1.8
7									2.1
8									2.4
9									2.7

L E G E N D:

SPT-1 : STANDARD PENETRATION TEST & NUMBER PER 12 INCHES
H-1 : HAND SAMPLE & NUMBER
R-1 : ROCK CORE RUN & NUMBER
RQD : ROCK QUALITY DESIGNATION
▼ : WATER TABLE

REC : ROCK CORE RECOVERY
HWR : HIGHLY WEATHERED ROCK
PWR : PARTIALLY WEATHERED ROCK
ROCK : UNWEATHERED ROCK
NE NOT ENCOUNTERED

TEST PIT LOG

GEOTECHNICAL & CIVIL TESTING UNIT

PROJECT NAME:	MANSOURAH MASSARAH GOLD PROJECT	PROJECT NO:	19-K-3054
CLIENT:	L & T CONSTRUCTION	FIELD INVESTIGATION:	
COORDINATES:	NORTH: 2,509,648.59 meter	TYPE OF BORING:	
	EAST: 265,327.56 meter	DIA. OF BORING: CORE SIZE:	TESTPIT NO: TP-21
GROUND ELEVATION:	meter	DIA. OF UNDISTURBED SAMPLE:	DATE COMMENCED: 9-24-19
DEPTH OF GROUNDWATER TABLE:	meter	CASING HAMMER WT. & DROP:	DATE COMPLETED: 9-24-19
MEASURED AT:	HOURS ON:	HAMMER WT. & DROP:	SUPERVISOR: Azam S.

Depth (feet)	Dry Density (gm/cm3)	Natural Moisture Content (%)	Maximum Dry Density (gm/cm3)	Optimum Moisture Content (%)	In place % Compaction	Soil Symbol	Classi-fication (D-2487)	DESCRIPTION	Depth (meters)
								Light brown, medium dense to dense, silty SAND with gravel.	
1							SM		0.3
								brown, dense, clayey SAND with gravel.	
2									0.6
3							SC		0.9
4									1.2
								End of testpit at 1.30m depth.	
5									1.5
6									1.8
7									2.1
8									2.4
9									2.7

L E G E N D:

SPT-1 : STANDARD PENETRATION TEST & NUMBER PER 12 INCHES
H-1 : HAND SAMPLE & NUMBER
R-1 : ROCK CORE RUN & NUMBER
RQD : ROCK QUALITY DESIGNATION
▼ : WATER TABLE

REC : ROCK CORE RECOVERY
HWR : HIGHLY WEATHERED ROCK
PWR : PARTIALLY WEATHERED ROCK
ROCK : UNWEATHERED ROCK
NE NOT ENCOUNTERED

TEST PIT LOG

GEOTECHNICAL & CIVIL TESTING UNIT

PROJECT NAME: MANSOURAH MASSARAH GOLD PROJECT			PROJECT NO: 19-K-3054
CLIENT: L & T CONSTRUCTION		FIELD INVESTIGATION:	
COORDINATES:	NORTH: 2,509,648.59 meter	TYPE OF BORING:	
	EAST: 265,827.86 meter	DIA. OF BORING: CORE SIZE:	TESTPIT NO: TP-22
GROUND ELEVATION: meter		DIA. OF UNDISTURBED SAMPLE:	DATE COMMENCED: 9-24-19
DEPTH OF GROUNDWATER TABLE: meter		CASING HAMMER WT. & DROP:	DATE COMPLETED: 9-24-19
MEASURED AT:	HOURS ON:	HAMMER WT. & DROP:	SUPERVISOR: Azam S.

Depth (feet)	Dry Density (gm/cm3)	Natural Moisture Content (%)	Maximum Dry Density (gm/cm3)	Optimum Moisture Content (%)	In place % Compaction	Soil Symbol	Classi-fication (D-2487)	DESCRIPTION	Depth (meters)
								Light brown, medium dense to dense, silty SAND.	
1							SM		0.3
2								Brown, dense, silty SAND with gravel.	0.6
3							SM		0.9
4									1.2
								End of testpit at 1.20m depth.	
5									1.5
6									1.8
7									2.1
8									2.4
9									2.7

L E G E N D:

SPT-1	: STANDARD PENETRATION TEST & NUMBER PER 12 INCHES	REC	: ROCK CORE RECOVERY
H-1	: HAND SAMPLE & NUMBER	HWR	: HIGHLY WEATHERED ROCK
R-1	: ROCK CORE RUN & NUMBER	PWR	: PARTIALLY WEATHERED ROCK
RQD	: ROCK QUALITY DESIGNATION	ROCK	: UNWEATHERED ROCK
▼	: WATER TABLE	NE	NOT ENCOUNTERED

TEST PIT LOG

GEOTECHNICAL & CIVIL TESTING UNIT

PROJECT NAME: MANSOURAH MASSARAH GOLD PROJECT			PROJECT NO: 19-K-3054
CLIENT: L & T CONSTRUCTION		FIELD INVESTIGATION:	
COORDINATES:	NORTH: 2,509,031.31 meter	TYPE OF BORING:	
	EAST: 265,827.83 meter	DIA. OF BORING: CORE SIZE:	TESTPIT NO: TP-23
GROUND ELEVATION: meter		DIA. OF UNDISTURBED SAMPLE:	DATE COMMENCED: 9-22-19
DEPTH OF GROUNDWATER TABLE: meter		CASING HAMMER WT. & DROP:	DATE COMPLETED: 9-22-19
MEASURED AT:	HOURS ON:	HAMMER WT. & DROP:	SUPERVISOR: Azam S.

Depth (feet)	Dry Density (gm/cm3)	Natural Moisture Content (%)	Maximum Dry Density (gm/cm3)	Optimum Moisture Content (%)	In place % Compaction	Soil Symbol	Classi-fication (D-2487)	DESCRIPTION	Depth (meters)
							SM	Light brown, medium dense to dense, silty SAND with gravel.	
1									0.3
							GM	brown, dense, silty GRAVEL with sand.	
2								End of testpit at 0.55m depth.	0.6
3									0.9
4									1.2
5									1.5
6									1.8
7									2.1
8									2.4
9									2.7

L E G E N D:

SPT-1 : STANDARD PENETRATION TEST & NUMBER PER 12 INCHES
H-1 : HAND SAMPLE & NUMBER
R-1 : ROCK CORE RUN & NUMBER
RQD : ROCK QUALITY DESIGNATION
▼ : WATER TABLE

REC : ROCK CORE RECOVERY
HWR : HIGHLY WEATHERED ROCK
PWR : PARTIALLY WEATHERED ROCK
ROCK : UNWEATHERED ROCK
NE NOT ENCOUNTERED

TEST PIT LOG

GEOTECHNICAL & CIVIL TESTING UNIT

PROJECT NAME:	MANSOURAH MASSARAH GOLD PROJECT	PROJECT NO:	19-K-3054
CLIENT:	L & T CONSTRUCTION	FIELD INVESTIGATION:	
COORDINATES:	NORTH: 2,509,031.30 meter	TYPE OF BORING:	
	EAST: 265,327.56 meter	DIA. OF BORING: CORE SIZE:	TESTPIT NO: TP-24
GROUND ELEVATION:	meter	DIA. OF UNDISTURBED SAMPLE:	DATE COMMENCED: 9-22-19
DEPTH OF GROUNDWATER TABLE:	meter	CASING HAMMER WT. & DROP:	DATE COMPLETED: 9-22-19
MEASURED AT:	HOURS ON:	HAMMER WT. & DROP:	SUPERVISOR: Azam S.

Depth (feet)	Dry Density (gm/cm3)	Natural Moisture Content (%)	Maximum Dry Density (gm/cm3)	Optimum Moisture Content (%)	In place % Compaction	Soil Symbol	Classi-fication (D-2487)	DESCRIPTION	Depth (meters)
								Light brown, medium dense to dense, clayey SAND.	
1							SC		0.3
2								brown, dense, clayey SAND with gravel.	0.6
3							SC		0.9
4								End of testpit at 1.00m depth.	1.2
5									1.5
6									1.8
7									2.1
8									2.4
9									2.7

L E G E N D:

SPT-1 :	STANDARD PENETRATION TEST & NUMBER PER 12 INCHES	REC :	ROCK CORE RECOVERY
H-1 :	HAND SAMPLE & NUMBER	HWR :	HIGHLY WEATHERED ROCK
R-1 :	ROCK CORE RUN & NUMBER	PWR :	PARTIALLY WEATHERED ROCK
RQD :	ROCK QUALITY DESIGNATION	ROCK :	UNWEATHERED ROCK
▼ :	WATER TABLE	NE	NOT ENCOUNTERED

TEST PIT LOG

GEOTECHNICAL & CIVIL TESTING UNIT

PROJECT NAME: MANSOURAH MASSARAH GOLD PROJECT			PROJECT NO: 19-K-3054
CLIENT: L & T CONSTRUCTION		FIELD INVESTIGATION:	
COORDINATES:	NORTH: 2,508,286.44 meter	TYPE OF BORING:	
	EAST: 262,982.71 meter	DIA. OF BORING: CORE SIZE:	TESTPIT NO: TP-25
GROUND ELEVATION: meter		DIA. OF UNDISTURBED SAMPLE:	DATE COMMENCED: 8-28-19
DEPTH OF GROUNDWATER TABLE: meter		CASING HAMMER WT. & DROP:	DATE COMPLETED: 8-28-19
MEASURED AT:	HOURS ON:	HAMMER WT. & DROP:	SUPERVISOR: Azam S.

Depth (feet)	Dry Density (gm/cm3)	Natural Moisture Content (%)	Maximum Dry Density (gm/cm3)	Optimum Moisture Content (%)	In place % Compaction	Soil Symbol	Classi-fication (D-2487)	DESCRIPTION	Depth (meters)
1							SM	Brown, medium dense to dense, silty SAND with gravel.	0.3
2								End of testpit @ 0.50m depth.	0.6
3									0.9
4									1.2
5									1.5
6									1.8
7									2.1
8									2.4
9									2.7

L E G E N D:

SPT-1 : STANDARD PENETRATION TEST & NUMBER PER 12 INCHES
H-1 : HAND SAMPLE & NUMBER
R-1 : ROCK CORE RUN & NUMBER
RQD : ROCK QUALITY DESIGNATION
▼ : WATER TABLE

REC : ROCK CORE RECOVERY
HWR : HIGHLY WEATHERED ROCK
PWR : PARTIALLY WEATHERED ROCK
ROCK : UNWEATHERED ROCK
NE NOT ENCOUNTERED

TEST PIT LOG

GEOTECHNICAL & CIVIL TESTING UNIT

PROJECT NAME:	MANSOURAH MASSARAH GOLD PROJECT	PROJECT NO:	19-K-3054
CLIENT:	L & T CONSTRUCTION	FIELD INVESTIGATION:	
COORDINATES:	NORTH: 2,506,202.23 meter	TYPE OF BORING:	
	EAST: 263,912.56 meter	DIA. OF BORING: CORE SIZE:	TESTPIT NO: TP-26
GROUND ELEVATION:	meter	DIA. OF UNDISTURBED SAMPLE:	DATE COMMENCED: 8-28-19
DEPTH OF GROUNDWATER TABLE:	meter	CASING HAMMER WT. & DROP:	DATE COMPLETED: 8-28-19
MEASURED AT:	HOURS ON:	HAMMER WT. & DROP:	SUPERVISOR: Azam S.

Depth (feet)	Dry Density (gm/cm3)	Natural Moisture Content (%)	Maximum Dry Density (gm/cm3)	Optimum Moisture Content (%)	In place % Compaction	Soil Symbol	Classi-fication (D-2487)	DESCRIPTION	Depth (meters)
1							SM	Brown, medium dense to dense, silty SAND with gravel.	0.3
2									0.6
3								End of testpit @ 0.70m depth.	0.9
4									1.2
5									1.5
6									1.8
7									2.1
8									2.4
9									2.7

L E G E N D:

SPT-1 : STANDARD PENETRATION TEST & NUMBER PER 12 INCHES
H-1 : HAND SAMPLE & NUMBER
R-1 : ROCK CORE RUN & NUMBER
RQD : ROCK QUALITY DESIGNATION
▼ : WATER TABLE

REC : ROCK CORE RECOVERY
HWR : HIGHLY WEATHERED ROCK
PWR : PARTIALLY WEATHERED ROCK
ROCK : UNWEATHERED ROCK
NE NOT ENCOUNTERED

TEST PIT LOG

GEOTECHNICAL & CIVIL TESTING UNIT

PROJECT NAME:	MANSOURAH MASSARAH GOLD PROJECT	PROJECT NO:	19-K-3054
CLIENT:	L & T CONSTRUCTION	FIELD INVESTIGATION:	
COORDINATES:	NORTH: 2,505,941.99 meter	TYPE OF BORING:	
	EAST: 264,051.07 meter	DIA. OF BORING: CORE SIZE:	TESTPIT NO: TP-27
GROUND ELEVATION:	meter	DIA. OF UNDISTURBED SAMPLE:	DATE COMMENCED: 8-28-19
DEPTH OF GROUNDWATER TABLE:	meter	CASING HAMMER WT. & DROP:	DATE COMPLETED: 8-28-19
MEASURED AT:	HOURS ON:	HAMMER WT. & DROP:	SUPERVISOR: Azam S.

Depth (feet)	Dry Density (gm/cm3)	Natural Moisture Content (%)	Maximum Dry Density (gm/cm3)	Optimum Moisture Content (%)	In place % Compaction	Soil Symbol	Classi-fication (D-2487)	DESCRIPTION	Depth (meters)
1							SM	Brown, medium dense to dense, silty SAND with gravel.	0.3
2							SM	Grey, dense to dense, silty SAND with gravel.	0.6
3								End of testpit @ 0.80m depth.	0.9
4									1.2
5									1.5
6									1.8
7									2.1
8									2.4
9									2.7

L E G E N D:

SPT-1 : STANDARD PENETRATION TEST & NUMBER PER 12 INCHES
H-1 : HAND SAMPLE & NUMBER
R-1 : ROCK CORE RUN & NUMBER
RQD : ROCK QUALITY DESIGNATION
▼ : WATER TABLE

REC : ROCK CORE RECOVERY
HWR : HIGHLY WEATHERED ROCK
PWR : PARTIALLY WEATHERED ROCK
ROCK : UNWEATHERED ROCK
NE NOT ENCOUNTERED

TEST PIT LOG

GEOTECHNICAL & CIVIL TESTING UNIT

PROJECT NAME: MANSOURAH MASSARAH GOLD PROJECT			PROJECT NO: 19-K-3054
CLIENT: L & T CONSTRUCTION		FIELD INVESTIGATION:	
COORDINATES:	NORTH: 2,504,886.34 meter	TYPE OF BORING:	
	EAST: 262,381.92 meter	DIA. OF BORING: CORE SIZE:	TESTPIT NO: TP-28
GROUND ELEVATION: meter		DIA. OF UNDISTURBED SAMPLE:	DATE COMMENCED: 8-28-19
DEPTH OF GROUNDWATER TABLE: meter		CASING HAMMER WT. & DROP:	DATE COMPLETED: 8-28-19
MEASURED AT:	HOURS ON:	HAMMER WT. & DROP:	SUPERVISOR: Azam S.

Depth (feet)	Dry Density (gm/cm3)	Natural Moisture Content (%)	Maximum Dry Density (gm/cm3)	Optimum Moisture Content (%)	In place % Compaction	Soil Symbol	Classi-fication (D-2487)	DESCRIPTION	Depth (meters)
								Brown, medium dense to dense, clayey SAND.	
1									0.3
							SC		
2									0.6
							SM	Dark grey, dense, silty SAND with gravel.	
3								End of testpit @ 0.80m depth.	0.9
4									1.2
5									1.5
6									1.8
7									2.1
8									2.4
9									2.7

L E G E N D:

SPT-1 : STANDARD PENETRATION TEST & NUMBER PER 12 INCHES
H-1 : HAND SAMPLE & NUMBER
R-1 : ROCK CORE RUN & NUMBER
RQD : ROCK QUALITY DESIGNATION
▼ : WATER TABLE

REC : ROCK CORE RECOVERY
HWR : HIGHLY WEATHERED ROCK
PWR : PARTIALLY WEATHERED ROCK
ROCK : UNWEATHERED ROCK
NE NOT ENCOUNTERED

TEST PIT LOG

GEOTECHNICAL & CIVIL TESTING UNIT

PROJECT NAME:	MANSOURAH MASSARAH GOLD PROJECT	PROJECT NO:	19-K-3054
CLIENT:	L & T CONSTRUCTION	FIELD INVESTIGATION:	
COORDINATES:	NORTH: 2,502,543.03 meter	TYPE OF BORING:	
	EAST: 263,678.22 meter	DIA. OF BORING: CORE SIZE:	TESTPIT NO: TP-29
GROUND ELEVATION:	meter	DIA. OF UNDISTURBED SAMPLE:	DATE COMMENCED: 8-28-19
DEPTH OF GROUNDWATER TABLE:	meter	CASING HAMMER WT. & DROP:	DATE COMPLETED: 8-28-19
MEASURED AT:	HOURS ON:	HAMMER WT. & DROP:	SUPERVISOR: Azam S.

Depth (feet)	Dry Density (gm/cm3)	Natural Moisture Content (%)	Maximum Dry Density (gm/cm3)	Optimum Moisture Content (%)	In place % Compaction	Soil Symbol	Classi-fication (D-2487)	DESCRIPTION	Depth (meters)
1							SC	Brown, medium dense to dense, clayey SAND with gravel.	0.3
2									0.6
3									0.9
4								End of testpit @ 1.00m depth.	1.2
5									1.5
6									1.8
7									2.1
8									2.4
9									2.7

L E G E N D:

SPT-1 : STANDARD PENETRATION TEST & NUMBER PER 12 INCHES
H-1 : HAND SAMPLE & NUMBER
R-1 : ROCK CORE RUN & NUMBER
RQD : ROCK QUALITY DESIGNATION
▼ : WATER TABLE

REC : ROCK CORE RECOVERY
HWR : HIGHLY WEATHERED ROCK
PWR : PARTIALLY WEATHERED ROCK
ROCK : UNWEATHERED ROCK
NE NOT ENCOUNTERED

TEST PIT LOG

GEOTECHNICAL & CIVIL TESTING UNIT

PROJECT NAME: MANSOURAH MASSARAH GOLD PROJECT			PROJECT NO: 19-K-3054
CLIENT: L & T CONSTRUCTION		FIELD INVESTIGATION:	
COORDINATES:	NORTH: 2,503,431.60 meter	TYPE OF BORING:	
	EAST: 268,590.00 meter	DIA. OF BORING: CORE SIZE:	TESTPIT NO: TP-30
GROUND ELEVATION: meter		DIA. OF UNDISTURBED SAMPLE:	DATE COMMENCED: 8-28-19
DEPTH OF GROUNDWATER TABLE: meter		CASING HAMMER WT. & DROP:	DATE COMPLETED: 8-28-19
MEASURED AT:	HOURS ON:	HAMMER WT. & DROP:	SUPERVISOR: Azam S.

Depth (feet)	Dry Density (gm/cm3)	Natural Moisture Content (%)	Maximum Dry Density (gm/cm3)	Optimum Moisture Content (%)	In place % Compaction	Soil Symbol	Classi-fication (D-2487)	DESCRIPTION	Depth (meters)
1							SC	Light brown, medium dense to dense, clayey SAND.	0.3
2								Grey, dense, clayey SAND with gravel.	0.6
3							SC		0.9
4								End of testpit @ 1.05m depth.	1.2
5									1.5
6									1.8
7									2.1
8									2.4
9									2.7

L E G E N D:

SPT-1 : STANDARD PENETRATION TEST & NUMBER PER 12 INCHES
H-1 : HAND SAMPLE & NUMBER
R-1 : ROCK CORE RUN & NUMBER
RQD : ROCK QUALITY DESIGNATION
▼ : WATER TABLE

REC : ROCK CORE RECOVERY
HWR : HIGHLY WEATHERED ROCK
PWR : PARTIALLY WEATHERED ROCK
ROCK : UNWEATHERED ROCK
NE NOT ENCOUNTERED

TEST PIT LOG

GEOTECHNICAL & CIVIL TESTING UNIT

PROJECT NAME:	MANSOURAH MASSARAH GOLD PROJECT	PROJECT NO:	19-K-3054
CLIENT:	L & T CONSTRUCTION	FIELD INVESTIGATION:	
COORDINATES:	NORTH: 2,506,475.53 meter	TYPE OF BORING:	
	EAST: 268,590.00 meter	DIA. OF BORING: CORE SIZE:	TESTPIT NO: TP-31
GROUND ELEVATION:	meter	DIA. OF UNDISTURBED SAMPLE:	DATE COMMENCED: 8-28-19
DEPTH OF GROUNDWATER TABLE:	meter	CASING HAMMER WT. & DROP:	DATE COMPLETED: 8-28-19
MEASURED AT:	HOURS ON:	HAMMER WT. & DROP:	SUPERVISOR: Azam S.

Depth (feet)	Dry Density (gm/cm3)	Natural Moisture Content (%)	Maximum Dry Density (gm/cm3)	Optimum Moisture Content (%)	In place % Compaction	Soil Symbol	Classi-fication (D-2487)	DESCRIPTION	Depth (meters)
								Light brown, medium dense to dense, silty SAND.	
1							SM		0.3
								Brown, dense, clayey SAND.	
2									0.6
3							SC		0.9
4									1.2
								End of testpit @ 1.20m depth.	
5									1.5
6									1.8
7									2.1
8									2.4
9									2.7

L E G E N D:

SPT-1 : STANDARD PENETRATION TEST & NUMBER PER 12 INCHES
H-1 : HAND SAMPLE & NUMBER
R-1 : ROCK CORE RUN & NUMBER
RQD : ROCK QUALITY DESIGNATION
▼ : WATER TABLE

REC : ROCK CORE RECOVERY
HWR : HIGHLY WEATHERED ROCK
PWR : PARTIALLY WEATHERED ROCK
ROCK : UNWEATHERED ROCK
NE NOT ENCOUNTERED

TEST PIT LOG

GEOTECHNICAL & CIVIL TESTING UNIT

PROJECT NAME:	MANSOURAH MASSARAH GOLD PROJECT	PROJECT NO:	19-K-3054
CLIENT:	L & T CONSTRUCTION	FIELD INVESTIGATION:	
COORDINATES:	NORTH: 2,510,479.24 meter	TYPE OF BORING:	
	EAST: 266,377.76 meter	DIA. OF BORING: CORE SIZE:	TESTPIT NO: TP-32
GROUND ELEVATION:	meter	DIA. OF UNDISTURBED SAMPLE:	DATE COMMENCED: 8-29-19
DEPTH OF GROUNDWATER TABLE:	meter	CASING HAMMER WT. & DROP:	DATE COMPLETED: 8-29-19
MEASURED AT:	HOURS ON:	HAMMER WT. & DROP:	SUPERVISOR: Azam S.

Depth (feet)	Dry Density (gm/cm3)	Natural Moisture Content (%)	Maximum Dry Density (gm/cm3)	Optimum Moisture Content (%)	In place % Compaction	Soil Symbol	Classi-fication (D-2487)	DESCRIPTION	Depth (meters)
1							SM	Light brown, medium dense to dense, silty SAND.	0.3
2									0.6
3									0.9
4								End of testpit @ 1.00m depth.	1.2
5									1.5
6									1.8
7									2.1
8									2.4
9									2.7

L E G E N D:

SPT-1 : STANDARD PENETRATION TEST & NUMBER PER 12 INCHES
H-1 : HAND SAMPLE & NUMBER
R-1 : ROCK CORE RUN & NUMBER
RQD : ROCK QUALITY DESIGNATION
▼ : WATER TABLE

REC : ROCK CORE RECOVERY
HWR : HIGHLY WEATHERED ROCK
PWR : PARTIALLY WEATHERED ROCK
ROCK : UNWEATHERED ROCK
NE NOT ENCOUNTERED

TEST PIT LOG

GEOTECHNICAL & CIVIL TESTING UNIT

PROJECT NAME:	MANSOURAH MASSARAH GOLD PROJECT	PROJECT NO:	19-K-3054
CLIENT:	L & T CONSTRUCTION	FIELD INVESTIGATION:	
COORDINATES:	NORTH: 2,514,933.12 meter	TYPE OF BORING:	
	EAST: 263,916.78 meter	DIA. OF BORING: CORE SIZE:	TESTPIT NO: TP-33
GROUND ELEVATION:	meter	DIA. OF UNDISTURBED SAMPLE:	DATE COMMENCED: 8-29-19
DEPTH OF GROUNDWATER TABLE:	meter	CASING HAMMER WT. & DROP:	DATE COMPLETED: 8-29-19
MEASURED AT:	HOURS ON:	HAMMER WT. & DROP:	SUPERVISOR: Azam S.

Depth (feet)	Dry Density (gm/cm3)	Natural Moisture Content (%)	Maximum Dry Density (gm/cm3)	Optimum Moisture Content (%)	In place % Compaction	Soil Symbol	Classi-fication (D-2487)	DESCRIPTION	Depth (meters)
								Light brown to brown, medium dense to dense, silty SAND with gravel.	
1							SM		0.3
								Grey, dense, clayey SAND with gravel.	
2							SC		0.6
								End of testpit @ 0.70m depth.	
3									0.9
4									1.2
5									1.5
6									1.8
7									2.1
8									2.4
9									2.7

L E G E N D:

SPT-1 : STANDARD PENETRATION TEST & NUMBER PER 12 INCHES
H-1 : HAND SAMPLE & NUMBER
R-1 : ROCK CORE RUN & NUMBER
RQD : ROCK QUALITY DESIGNATION
▼ : WATER TABLE

REC : ROCK CORE RECOVERY
HWR : HIGHLY WEATHERED ROCK
PWR : PARTIALLY WEATHERED ROCK
ROCK : UNWEATHERED ROCK
NE NOT ENCOUNTERED

TEST PIT LOG

GEOTECHNICAL & CIVIL TESTING UNIT

PROJECT NAME:	MANSOURAH MASSARAH GOLD PROJECT	PROJECT NO:	19-K-3054
CLIENT:	L & T CONSTRUCTION	FIELD INVESTIGATION:	
COORDINATES:	NORTH: 2,514,686.70 meter	TYPE OF BORING:	
	EAST: 262,156.43 meter	DIA. OF BORING: CORE SIZE:	TESTPIT NO: TP-34
GROUND ELEVATION:	meter	DIA. OF UNDISTURBED SAMPLE:	DATE COMMENCED: 8-29-19
DEPTH OF GROUNDWATER TABLE:	meter	CASING HAMMER WT. & DROP:	DATE COMPLETED: 8-29-19
MEASURED AT:	HOURS ON:	HAMMER WT. & DROP:	SUPERVISOR: Azam S.

Depth (feet)	Dry Density (gm/cm3)	Natural Moisture Content (%)	Maximum Dry Density (gm/cm3)	Optimum Moisture Content (%)	In place % Compaction	Soil Symbol	Classi-fication (D-2487)	DESCRIPTION	Depth (meters)
1							SC	Light brown, medium dense to dense, clayey SAND with gravel.	0.3
2							SW-SM	Grey, dense, well graded SAND with silt.	0.6
3								End of testpit @ 0.70m depth.	0.9
4									1.2
5									1.5
6									1.8
7									2.1
8									2.4
9									2.7

L E G E N D:

SPT-1 : STANDARD PENETRATION TEST & NUMBER PER 12 INCHES
H-1 : HAND SAMPLE & NUMBER
R-1 : ROCK CORE RUN & NUMBER
RQD : ROCK QUALITY DESIGNATION
▼ : WATER TABLE

REC : ROCK CORE RECOVERY
HWR : HIGHLY WEATHERED ROCK
PWR : PARTIALLY WEATHERED ROCK
ROCK : UNWEATHERED ROCK
NE NOT ENCOUNTERED

TEST PIT LOG

GEOTECHNICAL & CIVIL TESTING UNIT

PROJECT NAME: MANSOURAH MASSARAH GOLD PROJECT			PROJECT NO: 19-K-3054
CLIENT: L & T CONSTRUCTION		FIELD INVESTIGATION:	
COORDINATES:	NORTH: 2,510,985.79 meter	TYPE OF BORING:	
	EAST: 262,295.65 meter	DIA. OF BORING: CORE SIZE:	TESTPIT NO: TP-35
GROUND ELEVATION: meter		DIA. OF UNDISTURBED SAMPLE:	DATE COMMENCED: 8-29-19
DEPTH OF GROUNDWATER TABLE: meter		CASING HAMMER WT. & DROP:	DATE COMPLETED: 8-29-19
MEASURED AT:	HOURS ON:	HAMMER WT. & DROP:	SUPERVISOR: Azam S.

Depth (feet)	Dry Density (gm/cm3)	Natural Moisture Content (%)	Maximum Dry Density (gm/cm3)	Optimum Moisture Content (%)	In place % Compaction	Soil Symbol	Classi-fication (D-2487)	DESCRIPTION	Depth (meters)
1							SM	Light brown, medium dense, silty SAND with gravel.	0.3
2									0.6
3								End of testpit @ 0.60m depth.	0.9
4									1.2
5									1.5
6									1.8
7									2.1
8									2.4
9									2.7

L E G E N D:

SPT-1 : STANDARD PENETRATION TEST & NUMBER PER 12 INCHES
H-1 : HAND SAMPLE & NUMBER
R-1 : ROCK CORE RUN & NUMBER
RQD : ROCK QUALITY DESIGNATION
▼ : WATER TABLE

REC : ROCK CORE RECOVERY
HWR : HIGHLY WEATHERED ROCK
PWR : PARTIALLY WEATHERED ROCK
ROCK : UNWEATHERED ROCK
NE NOT ENCOUNTERED

TEST PIT LOG

GEOTECHNICAL & CIVIL TESTING UNIT

PROJECT NAME:	MANSOURAH MASSARAH GOLD PROJECT	PROJECT NO:	19-K-3054
CLIENT:	L & T CONSTRUCTION	FIELD INVESTIGATION:	
COORDINATES:	NORTH: 2,505,881.87 meter	TYPE OF BORING:	
	EAST: 263,947.84 meter	DIA. OF BORING: CORE SIZE:	TESTPIT NO: TP-36
GROUND ELEVATION:	meter	DIA. OF UNDISTURBED SAMPLE:	DATE COMMENCED: 8-29-19
DEPTH OF GROUNDWATER TABLE:	meter	CASING HAMMER WT. & DROP:	DATE COMPLETED: 8-29-19
MEASURED AT:	HOURS ON:	HAMMER WT. & DROP:	SUPERVISOR: Azam S.

Depth (feet)	Dry Density (gm/cm3)	Natural Moisture Content (%)	Maximum Dry Density (gm/cm3)	Optimum Moisture Content (%)	In place % Compaction	Soil Symbol	Classi-fication (D-2487)	DESCRIPTION	Depth (meters)
1							SM	Light brown, medium dense to dense, silty SAND with gravel.	0.3
2								End of testpit @ 0.45m depth.	0.6
3									0.9
4									1.2
5									1.5
6									1.8
7									2.1
8									2.4
9									2.7

L E G E N D:

SPT-1 : STANDARD PENETRATION TEST & NUMBER PER 12 INCHES
H-1 : HAND SAMPLE & NUMBER
R-1 : ROCK CORE RUN & NUMBER
RQD : ROCK QUALITY DESIGNATION
▼ : WATER TABLE

REC : ROCK CORE RECOVERY
HWR : HIGHLY WEATHERED ROCK
PWR : PARTIALLY WEATHERED ROCK
ROCK : UNWEATHERED ROCK
NE NOT ENCOUNTERED

TEST PIT LOG

GEOTECHNICAL & CIVIL TESTING UNIT

PROJECT NAME:	MANSOURAH MASSARAH GOLD PROJECT	PROJECT NO:	19-K-3054
CLIENT:	L & T CONSTRUCTION	FIELD INVESTIGATION:	
COORDINATES:	NORTH: 2,509,194.34 meter	TYPE OF BORING:	
	EAST: 261,836.09 meter	DIA. OF BORING: CORE SIZE:	TESTPIT NO: TP-41
GROUND ELEVATION:	meter	DIA. OF UNDISTURBED SAMPLE:	DATE COMMENCED: 8-29-19
DEPTH OF GROUNDWATER TABLE:	meter	CASING HAMMER WT. & DROP:	DATE COMPLETED: 8-29-19
MEASURED AT:	HOURS ON:	HAMMER WT. & DROP:	SUPERVISOR: Azam S.

Depth (feet)	Dry Density (gm/cm3)	Natural Moisture Content (%)	Maximum Dry Density (gm/cm3)	Optimum Moisture Content (%)	In place % Compaction	Soil Symbol	Classi-fication (D-2487)	DESCRIPTION	Depth (meters)
1							SM	Light brown, medium dense to dense, silty SAND with gravel.	0.3
2							GP-GM	Grey, dense, poorly graded GRAVEL with silt and sand.	0.6
3								End of testpit @ 0.70m depth.	0.9
4									1.2
5									1.5
6									1.8
7									2.1
8									2.4
9									2.7

L E G E N D:

SPT-1 : STANDARD PENETRATION TEST & NUMBER PER 12 INCHES
H-1 : HAND SAMPLE & NUMBER
R-1 : ROCK CORE RUN & NUMBER
RQD : ROCK QUALITY DESIGNATION
▼ : WATER TABLE

REC : ROCK CORE RECOVERY
HWR : HIGHLY WEATHERED ROCK
PWR : PARTIALLY WEATHERED ROCK
ROCK : UNWEATHERED ROCK
NE NOT ENCOUNTERED

TEST PIT LOG

GEOTECHNICAL & CIVIL TESTING UNIT

PROJECT NAME:	MANSOURAH MASSARAH GOLD PROJECT	PROJECT NO:	19-K-3054
CLIENT:	L & T CONSTRUCTION	FIELD INVESTIGATION:	
COORDINATES:	NORTH: 2,509,152.97 meter	TYPE OF BORING:	
	EAST: 261,832.92 meter	DIA. OF BORING: CORE SIZE:	TESTPIT NO: TP-42
GROUND ELEVATION:	meter	DIA. OF UNDISTURBED SAMPLE:	DATE COMMENCED: 8-29-19
DEPTH OF GROUNDWATER TABLE:	meter	CASING HAMMER WT. & DROP:	DATE COMPLETED: 8-29-19
MEASURED AT:	HOURS ON:	HAMMER WT. & DROP:	SUPERVISOR: Azam S.

Depth (feet)	Dry Density (gm/cm3)	Natural Moisture Content (%)	Maximum Dry Density (gm/cm3)	Optimum Moisture Content (%)	In place % Compaction	Soil Symbol	Classi-fication (D-2487)	DESCRIPTION	Depth (meters)
1							SM	Light brown, medium dense to dense, silty SAND with gravel.	0.3
2									0.6
3							SP-SM	Grey, dense, poorly graded SAND with silt and gravel.	0.9
4								End of testpit @ 1.00m depth.	1.2
5									1.5
6									1.8
7									2.1
8									2.4
9									2.7

L E G E N D:

SPT-1 : STANDARD PENETRATION TEST & NUMBER PER 12 INCHES
H-1 : HAND SAMPLE & NUMBER
R-1 : ROCK CORE RUN & NUMBER
RQD : ROCK QUALITY DESIGNATION
▼ : WATER TABLE

REC : ROCK CORE RECOVERY
HWR : HIGHLY WEATHERED ROCK
PWR : PARTIALLY WEATHERED ROCK
ROCK : UNWEATHERED ROCK
NE NOT ENCOUNTERED

TEST PIT LOG

GEOTECHNICAL & CIVIL TESTING UNIT

PROJECT NAME: MANSOURAH MASSARAH GOLD PROJECT			PROJECT NO: 19-K-3054
CLIENT: L & T CONSTRUCTION		FIELD INVESTIGATION:	
COORDINATES:	NORTH: 2,506,673.82 meter	TYPE OF BORING:	
	EAST: 263,692.74 meter	DIA. OF BORING: CORE SIZE:	TESTPIT NO: TP-43
GROUND ELEVATION: meter		DIA. OF UNDISTURBED SAMPLE:	DATE COMMENCED: 8-29-19
DEPTH OF GROUNDWATER TABLE: meter		CASING HAMMER WT. & DROP:	DATE COMPLETED: 8-29-19
MEASURED AT:	HOURS ON:	HAMMER WT. & DROP:	SUPERVISOR: Azam S.

Depth (feet)	Dry Density (gm/cm3)	Natural Moisture Content (%)	Maximum Dry Density (gm/cm3)	Optimum Moisture Content (%)	In place % Compaction	Soil Symbol	Classi-fication (D-2487)	DESCRIPTION	Depth (meters)
							SC	Brown, medium dense to dense, clayey SAND.	
1								brown, dense, clayey SAND with gravel.	0.3
2							SC		0.6
3									0.9
								End of testpit @ 0.90m depth.	
4									1.2
5									1.5
6									1.8
7									2.1
8									2.4
9									2.7

L E G E N D:

SPT-1 : STANDARD PENETRATION TEST & NUMBER PER 12 INCHES
H-1 : HAND SAMPLE & NUMBER
R-1 : ROCK CORE RUN & NUMBER
RQD : ROCK QUALITY DESIGNATION
▼ : WATER TABLE

REC : ROCK CORE RECOVERY
HWR : HIGHLY WEATHERED ROCK
PWR : PARTIALLY WEATHERED ROCK
ROCK : UNWEATHERED ROCK
NE NOT ENCOUNTERED

TEST PIT LOG

GEOTECHNICAL & CIVIL TESTING UNIT

PROJECT NAME:	MANSOURAH MASSARAH GOLD PROJECT	PROJECT NO:	19-K-3054
CLIENT:	L & T CONSTRUCTION	FIELD INVESTIGATION:	
COORDINATES:	NORTH: 2,506,177.33 meter	TYPE OF BORING:	
	EAST: 263,802.78 meter	DIA. OF BORING: CORE SIZE:	TESTPIT NO: TP-44
GROUND ELEVATION:	meter	DIA. OF UNDISTURBED SAMPLE:	DATE COMMENCED: 8-29-19
DEPTH OF GROUNDWATER TABLE:	meter	CASING HAMMER WT. & DROP:	DATE COMPLETED: 8-29-19
MEASURED AT:	HOURS ON:	HAMMER WT. & DROP:	SUPERVISOR: Azam S.

Depth (feet)	Dry Density (gm/cm3)	Natural Moisture Content (%)	Maximum Dry Density (gm/cm3)	Optimum Moisture Content (%)	In place % Compaction	Soil Symbol	Classi-fication (D-2487)	DESCRIPTION	Depth (meters)
1								Light brown, medium dense to dense, silty SAND with gravel.	0.3
2							SM		0.6
3								End of testpit @ 0.85m depth.	0.9
4									1.2
5									1.5
6									1.8
7									2.1
8									2.4
9									2.7

L E G E N D:

SPT-1 : STANDARD PENETRATION TEST & NUMBER PER 12 INCHES
H-1 : HAND SAMPLE & NUMBER
R-1 : ROCK CORE RUN & NUMBER
RQD : ROCK QUALITY DESIGNATION
▼ : WATER TABLE

REC : ROCK CORE RECOVERY
HWR : HIGHLY WEATHERED ROCK
PWR : PARTIALLY WEATHERED ROCK
ROCK : UNWEATHERED ROCK
NE NOT ENCOUNTERED

TEST PIT LOG

GEOTECHNICAL & CIVIL TESTING UNIT

PROJECT NAME:	MANSOURAH MASSARAH GOLD PROJECT	PROJECT NO:	19-K-3054
CLIENT:	L & T CONSTRUCTION	FIELD INVESTIGATION:	
COORDINATES:	NORTH: 2,505,965.00 meter	TYPE OF BORING:	
	EAST: 263,870.12 meter	DIA. OF BORING: CORE SIZE:	TESTPIT NO: TP-45
GROUND ELEVATION:	meter	DIA. OF UNDISTURBED SAMPLE:	DATE COMMENCED: 8-29-19
DEPTH OF GROUNDWATER TABLE:	meter	CASING HAMMER WT. & DROP:	DATE COMPLETED: 8-29-19
MEASURED AT:	HOURS ON:	HAMMER WT. & DROP:	SUPERVISOR: Azam S.

Depth (feet)	Dry Density (gm/cm3)	Natural Moisture Content (%)	Maximum Dry Density (gm/cm3)	Optimum Moisture Content (%)	In place % Compaction	Soil Symbol	Classi-fication (D-2487)	DESCRIPTION	Depth (meters)
1							SM	Light brown, medium dense to dense, silty SAND with gravel.	0.3
2							GP	Grey, dense, poorly graded GRAVEL with sand.	0.6
3								End of testpit @ 0.65m depth.	0.9
4									1.2
5									1.5
6									1.8
7									2.1
8									2.4
9									2.7

L E G E N D:

SPT-1 : STANDARD PENETRATION TEST & NUMBER PER 12 INCHES
H-1 : HAND SAMPLE & NUMBER
R-1 : ROCK CORE RUN & NUMBER
RQD : ROCK QUALITY DESIGNATION
▼ : WATER TABLE

REC : ROCK CORE RECOVERY
HWR : HIGHLY WEATHERED ROCK
PWR : PARTIALLY WEATHERED ROCK
ROCK : UNWEATHERED ROCK
NE NOT ENCOUNTERED

TEST PIT LOG

GEOTECHNICAL & CIVIL TESTING UNIT

PROJECT NAME: MANSOURAH MASSARAH GOLD PROJECT			PROJECT NO: 19-K-3054
CLIENT: L & T CONSTRUCTION		FIELD INVESTIGATION:	
COORDINATES:	NORTH: 2,505,805.12 meter	TYPE OF BORING:	
	EAST: 264,254.10 meter	DIA. OF BORING: CORE SIZE:	TESTPIT NO: TP-46
GROUND ELEVATION: meter		DIA. OF UNDISTURBED SAMPLE:	DATE COMMENCED: 8-29-19
DEPTH OF GROUNDWATER TABLE: meter		CASING HAMMER WT. & DROP:	DATE COMPLETED: 8-29-19
MEASURED AT:	HOURS ON:	HAMMER WT. & DROP:	SUPERVISOR: Azam S.

Depth (feet)	Dry Density (gm/cm3)	Natural Moisture Content (%)	Maximum Dry Density (gm/cm3)	Optimum Moisture Content (%)	In place % Compaction	Soil Symbol	Classi-fication (D-2487)	DESCRIPTION	Depth (meters)
1							SC	Light brown, medium dense to dense, clayey SAND with gravel.	0.3
2									0.6
3								End of testpit @ 0.60m depth.	0.9
4									1.2
5									1.5
6									1.8
7									2.1
8									2.4
9									2.7

L E G E N D:

SPT-1 : STANDARD PENETRATION TEST & NUMBER PER 12 INCHES
H-1 : HAND SAMPLE & NUMBER
R-1 : ROCK CORE RUN & NUMBER
RQD : ROCK QUALITY DESIGNATION
▼ : WATER TABLE

REC : ROCK CORE RECOVERY
HWR : HIGHLY WEATHERED ROCK
PWR : PARTIALLY WEATHERED ROCK
ROCK : UNWEATHERED ROCK
NE NOT ENCOUNTERED

TEST PIT LOG

GEOTECHNICAL & CIVIL TESTING UNIT

PROJECT NAME:	MANSOURAH MASSARAH GOLD PROJECT	PROJECT NO:	19-K-3054
CLIENT:	L & T CONSTRUCTION	FIELD INVESTIGATION:	
COORDINATES:	NORTH: 2,505,926.77 meter	TYPE OF BORING:	
	EAST: 264,444.34 meter	DIA. OF BORING: CORE SIZE:	TESTPIT NO: TP-47
GROUND ELEVATION:	meter	DIA. OF UNDISTURBED SAMPLE:	DATE COMMENCED: 8-29-19
DEPTH OF GROUNDWATER TABLE:	meter	CASING HAMMER WT. & DROP:	DATE COMPLETED: 8-29-19
MEASURED AT:	HOURS ON:	HAMMER WT. & DROP:	SUPERVISOR: Azam S.

Depth (feet)	Dry Density (gm/cm3)	Natural Moisture Content (%)	Maximum Dry Density (gm/cm3)	Optimum Moisture Content (%)	In place % Compaction	Soil Symbol	Classi-fication (D-2487)	DESCRIPTION	Depth (meters)
1							SC	Light brown, medium dense to dense, clayey SAND with gravel.	0.3
2							GP	Grey, dense, poorly graded GRAVEL with sand.	0.6
3								End of testpit @ 0.75m depth.	0.9
4									1.2
5									1.5
6									1.8
7									2.1
8									2.4
9									2.7

L E G E N D:

SPT-1 : STANDARD PENETRATION TEST & NUMBER PER 12 INCHES
H-1 : HAND SAMPLE & NUMBER
R-1 : ROCK CORE RUN & NUMBER
RQD : ROCK QUALITY DESIGNATION
▼ : WATER TABLE

REC : ROCK CORE RECOVERY
HWR : HIGHLY WEATHERED ROCK
PWR : PARTIALLY WEATHERED ROCK
ROCK : UNWEATHERED ROCK
NE NOT ENCOUNTERED

TEST PIT LOG

GEOTECHNICAL & CIVIL TESTING UNIT

PROJECT NAME: MANSOURAH MASSARAH GOLD PROJECT			PROJECT NO: 19-K-3054
CLIENT: L & T CONSTRUCTION		FIELD INVESTIGATION:	
COORDINATES:	NORTH: 2,506,447.35 meter	TYPE OF BORING:	
	EAST: 264,066.43 meter	DIA. OF BORING: CORE SIZE:	TESTPIT NO: TP-48
GROUND ELEVATION: meter		DIA. OF UNDISTURBED SAMPLE:	DATE COMMENCED: 9-1-19
DEPTH OF GROUNDWATER TABLE: meter		CASING HAMMER WT. & DROP:	DATE COMPLETED: 9-1-19
MEASURED AT:	HOURS ON:	HAMMER WT. & DROP:	SUPERVISOR: Azam S.

Depth (feet)	Dry Density (gm/cm3)	Natural Moisture Content (%)	Maximum Dry Density (gm/cm3)	Optimum Moisture Content (%)	In place % Compaction	Soil Symbol	Classi-fication (D-2487)	DESCRIPTION	Depth (meters)
1							SC	Brown, medium dense to dense, clayey SAND.	0.3
2							SC	Grey, dense, clayey SAND with gravel.	0.6
3								End of testpit @ 0.7m depth.	0.9
4									1.2
5									1.5
6									1.8
7									2.1
8									2.4
9									2.7

L E G E N D:

SPT-1 : STANDARD PENETRATION TEST & NUMBER PER 12 INCHES
H-1 : HAND SAMPLE & NUMBER
R-1 : ROCK CORE RUN & NUMBER
RQD : ROCK QUALITY DESIGNATION
▼ : WATER TABLE

REC : ROCK CORE RECOVERY
HWR : HIGHLY WEATHERED ROCK
PWR : PARTIALLY WEATHERED ROCK
ROCK : UNWEATHERED ROCK
NE NOT ENCOUNTERED

TEST PIT LOG

GEOTECHNICAL & CIVIL TESTING UNIT

PROJECT NAME:	MANSOURAH MASSARAH GOLD PROJECT	PROJECT NO:	19-K-3054
CLIENT:	L & T CONSTRUCTION	FIELD INVESTIGATION:	
COORDINATES:	NORTH: 2,506,323.00 meter	TYPE OF BORING:	
	EAST: 264,137.26 meter	DIA. OF BORING: CORE SIZE:	TESTPIT NO: TP-49
GROUND ELEVATION:	meter	DIA. OF UNDISTURBED SAMPLE:	DATE COMMENCED: 9-1-19
DEPTH OF GROUNDWATER TABLE:	meter	CASING HAMMER WT. & DROP:	DATE COMPLETED: 9-1-19
MEASURED AT:	HOURS ON:	HAMMER WT. & DROP:	SUPERVISOR: Azam S.

Depth (feet)	Dry Density (gm/cm3)	Natural Moisture Content (%)	Maximum Dry Density (gm/cm3)	Optimum Moisture Content (%)	In place % Compaction	Soil Symbol	Classi-fication (D-2487)	DESCRIPTION	Depth (meters)
							SC	Light brown, medium dense to dense, clayey SAND with gravel.	
1								Grey, dense, poorly graded GRAVEL with sand.	0.3
2							GP		0.6
								End of testpit @ 0.6m depth.	
3									0.9
4									1.2
5									1.5
6									1.8
7									2.1
8									2.4
9									2.7

L E G E N D:

SPT-1 :	STANDARD PENETRATION TEST & NUMBER PER 12 INCHES	REC :	ROCK CORE RECOVERY
H-1 :	HAND SAMPLE & NUMBER	HWR :	HIGHLY WEATHERED ROCK
R-1 :	ROCK CORE RUN & NUMBER	PWR :	PARTIALLY WEATHERED ROCK
RQD :	ROCK QUALITY DESIGNATION	ROCK :	UNWEATHERED ROCK
▼ :	WATER TABLE	NE	NOT ENCOUNTERED

TEST PIT LOG

GEOTECHNICAL & CIVIL TESTING UNIT

PROJECT NAME: MANSOURAH MASSARAH GOLD PROJECT			PROJECT NO: 19-K-3054
CLIENT: L & T CONSTRUCTION		FIELD INVESTIGATION:	
COORDINATES:	NORTH: 2,506,325.23 meter	TYPE OF BORING:	
	EAST: 263,940.87 meter	DIA. OF BORING: CORE SIZE:	TESTPIT NO: TP-50
GROUND ELEVATION: meter		DIA. OF UNDISTURBED SAMPLE:	DATE COMMENCED: 9-1-19
DEPTH OF GROUNDWATER TABLE: meter		CASING HAMMER WT. & DROP:	DATE COMPLETED: 9-1-19
MEASURED AT:	HOURS ON:	HAMMER WT. & DROP:	SUPERVISOR: Azam S.

Depth (feet)	Dry Density (gm/cm3)	Natural Moisture Content (%)	Maximum Dry Density (gm/cm3)	Optimum Moisture Content (%)	In place % Compaction	Soil Symbol	Classi-fication (D-2487)	DESCRIPTION	Depth (meters)
1							SC	Light brown, medium dense to dense, clayey SAND with gravel.	0.3
2							GP	Grey, dense, poorly graded GRAVEL with sand.	0.6
3								End of testpit @ 0.7m depth.	0.9
4									1.2
5									1.5
6									1.8
7									2.1
8									2.4
9									2.7

L E G E N D:

SPT-1 : STANDARD PENETRATION TEST & NUMBER PER 12 INCHES
H-1 : HAND SAMPLE & NUMBER
R-1 : ROCK CORE RUN & NUMBER
RQD : ROCK QUALITY DESIGNATION
▼ : WATER TABLE

REC : ROCK CORE RECOVERY
HWR : HIGHLY WEATHERED ROCK
PWR : PARTIALLY WEATHERED ROCK
ROCK : UNWEATHERED ROCK
NE NOT ENCOUNTERED

DOWNHOLE TEST RESULTS

TEST REPORT - DOWNHOLE SEISMIC TEST RESULTS *(ASTM D7400 - 08)*

FORM # RGF-KHO-GEO-DH-1-00

Project: *MANSOURAH MASSARAH GOLD PROJECT*
Client: *L & T CONSTRUCTION*
Project No.: *19-K-3054* **Test No.** *DH-01*
Borehole No.: *BH-02* **Geotechnical File No.:** -
Source Type: *Sledge Hammer* **Source to BH Distance (m):** 3 m
Receiver Type: *Triaxial Geophones* **Coordinates (m):** ***N:** 2506179.07, **E:** 264210.66*

Sr. No.	Depth of Reciever (m)	Distance of Source to Receiver (m)	Average Time		Calculated Velocity		Assumed Density ρ (g/cm^3)	Poisson Ratio υ	Shear Modulus G (M Pa)	Young Modulus E (M Pa)	Bulk Modulus K (M Pa)
			P-wave (msec)	S-wave (msec)	P-wave (m/sec)	S-wave (m/sec)					
1	0.75	3.09	3.69	6.54	838	473	1.8	0.266	402	1019	728
2	1.50	3.35	3.79	6.72	2618	1454	2.1	0.277	4441	11342	8468
3	2.25	3.75	3.96	6.99	2329	1466	2.2	0.172	4730	11083	5625
4	3.00	4.24	4.17	7.33	2346	1449	2.2	0.192	4619	11008	5949
5	3.75	4.80	4.42	7.74	2239	1365	2.2	0.204	4100	9873	5561
6	4.50	5.41	4.70	8.18	2164	1377	2.2	0.160	4173	9679	4741
7	5.25	6.05	4.98	8.63	2280	1419	2.2	0.184	4427	10485	5532
8	6.00	6.71	5.26	9.09	2363	1438	2.2	0.206	4550	10971	6213
9	6.75	7.39	5.53	9.54	2513	1508	2.2	0.219	5001	12189	7223
10	7.50	8.08	5.80	9.98	2560	1571	2.2	0.198	5428	13005	7177
11	8.25	8.78	6.07	10.43	2595	1557	2.2	0.219	5335	13005	7707
12	9.00	9.49	6.37	10.92	2361	1446	2.2	0.200	4597	11035	6134
13	9.75	10.20	6.67	11.40	2381	1488	2.2	0.179	4872	11492	5976
14	10.50	10.92	6.97	11.90	2397	1438	2.2	0.219	4550	11091	6572
15	11.25	11.64	7.28	12.41	2332	1418	2.2	0.207	4421	10672	6071
16	12.00	12.37	7.58	12.91	2421	1452	2.2	0.219	4641	11312	6703
17	12.75	13.10	7.89	13.42	2351	1429	2.2	0.207	4493	10847	6171
18	13.50	13.83	8.20	13.90	2358	1523	2.2	0.142	5104	11660	5432
19	14.25	14.56	8.51	14.40	2365	1466	2.2	0.188	4729	11234	5997
20	15.00	15.30	8.82	14.91	2370	1441	2.2	0.207	4566	11021	6270
21	15.75	16.03	9.11	15.38	2538	1566	2.2	0.193	5397	12872	6979
22	16.50	16.77	9.42	15.89	2379	1446	2.2	0.207	4599	11101	6315
23	17.25	17.51	9.74	16.40	2308	1448	2.2	0.175	4612	10841	5565
24	18.00	18.25	10.06	16.94	2311	1369	2.2	0.229	4124	10141	6245
25	18.75	18.99	10.35	17.41	2552	1575	2.2	0.193	5457	13015	7057
26	19.50	19.73	10.66	17.93	2390	1425	2.2	0.224	4467	10937	6612
27	20.25	20.47	10.97	18.45	2392	1426	2.2	0.224	4475	10957	6624
28	21.00	21.21	11.29	18.95	2319	1484	2.2	0.153	4847	11179	5371
29	21.75	21.96	11.59	19.44	2476	1516	2.2	0.200	5055	12133	6745
30	22.50	22.70	11.93	19.98	2186	1376	2.2	0.172	4167	9764	4955
31	23.25	23.44	12.22	20.47	2564	1518	2.2	0.230	5067	12469	7710
32	24.00	24.19	12.52	20.96	2480	1518	2.2	0.200	5072	12176	6769

This sheet is underownership of RGF Geotechnical Department Al Khobar, Data on this sheet can not be reproduced / changed untill unless prior permission from section Incharge

Sr. No.	Depth of Reciever (m)	Distance of Source to Receiver (m)	Average Time		Calculated Velocity		Assumed Density	Poisson Ratio	Shear Modulus	Young Modulus	Bulk Modulus
			P-wave (msec)	S-wave (msec)	P-wave (m/sec)	S-wave (m/sec)	**ρ** (g/cm^3)	**υ**	**G** (M Pa)	**E** (M Pa)	**K** (M Pa)
33	24.75	24.93	12.82	21.47	2481	1460	2.2	0.235	4687	11581	7296
34	25.50	25.68	13.15	21.98	2257	1460	2.2	0.140	4691	10694	4949
35	26.25	26.42	13.48	22.51	2258	1406	2.2	0.183	4347	10289	5417
36	27.00	27.17	13.79	23.01	2404	1491	2.2	0.188	4888	11612	6199
37	27.75	27.91	14.13	23.54	2193	1407	2.2	0.150	4353	10015	4774
38	28.50	28.66	14.43	24.04	2486	1492	2.2	0.219	4894	11930	7070
39	29.25	29.40	14.74	24.54	2406	1492	2.2	0.188	4897	11633	6210
40	30.00	30.15	15.05	25.04	2407	1492	2.2	0.188	4900	11640	6213

$$\text{Poisson Ratio } (\upsilon) = \frac{[\,(Vp/Vs)^2 - 2\,]}{[\,2 \times (Vp/Vs)^2 - 2\,]}$$

$$\text{Shear Modulus } (G) = \frac{\rho \times (Vs)^2}{1000}$$

$$\text{Young Modulus } (E) = 2G \times (1 + \upsilon)$$

$$\text{Bulk Modulus } (K) = \frac{G \times E}{3\,(3G - E)}$$

M. Azam Khan
Tested By

Muhammad Asad Shafiq
Checked By

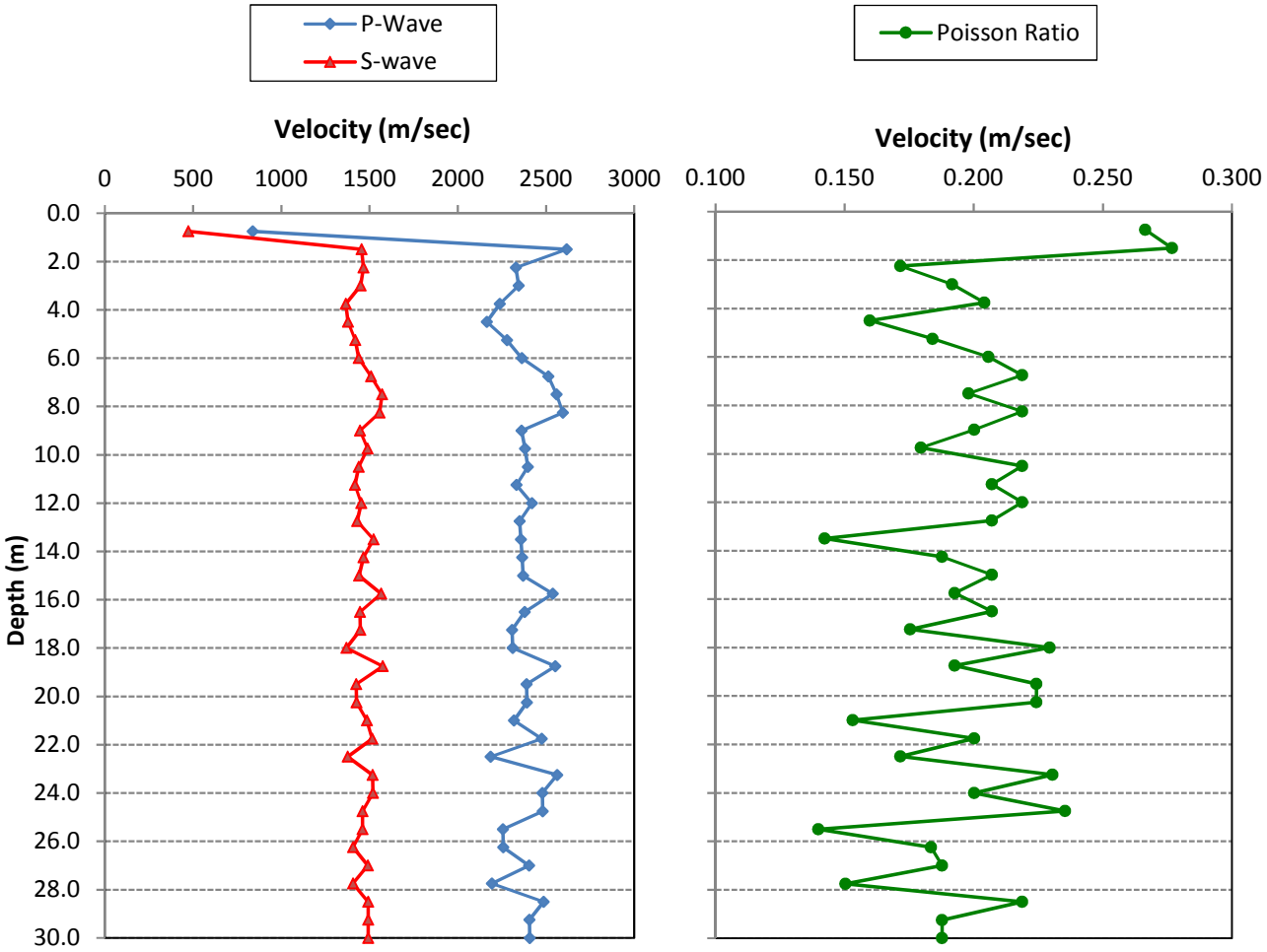
This sheet is underownership of RGF Geotechnical Department Al Khobar, Data on this sheet can not be reproduced / changed untill unless prior permission from section Incharge
RGF-KHO-GEO-DH-1-00

TEST REPORT - CROSSHOLE SEISMIC TEST RESULTS *(ASTM D4428/D4428M-07)*

FORM #	RGF-KHO-GEO-DH-1-00

Client:	*MANSOURAH MASSARAH GOLD PROJECT*	**Test No.**	*DH-01*
Project:	*L & T CONSTRUCTION*	**Geotechnical File No.:**	-
Project No.:	*19-K-3054*	**Coordinates (meters):**	*N: 2506179.07, E: 264210.66*
Borehole No.:	*BH-02*		

VELOCITY - DEPTH PLOTS



Subsurface Velocity Vs Depth profile

M. Azam Khan
Checked By

Muhammad Asad Shafiq
Approved By

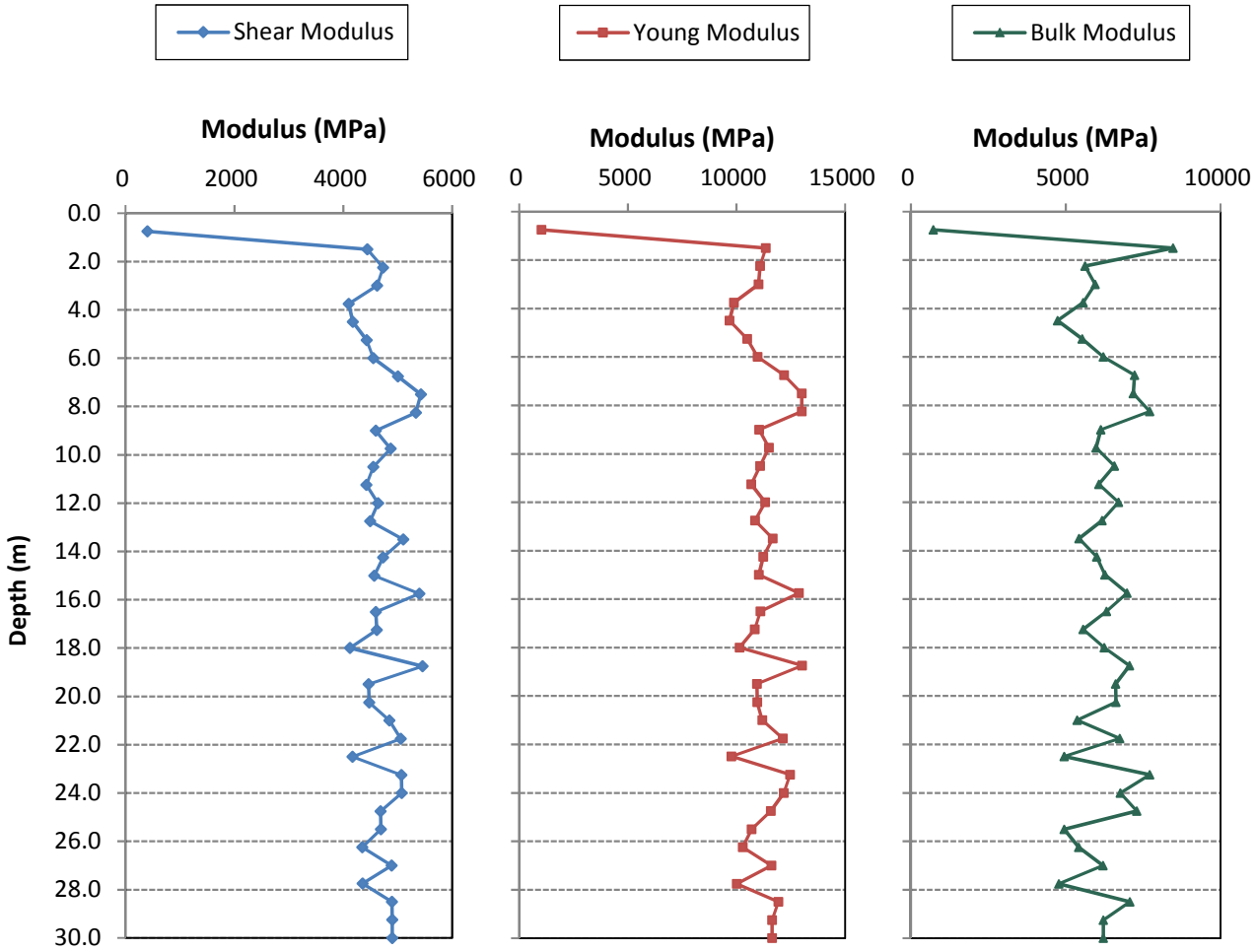
This sheet is underownership of RGF Geotechnical Department Al Khobar, Data on this sheet can not be reproduced / changed untill unless prior permission from section Incharge
RGF-KHO-GEO-DH-1-00

TEST REPORT - CROSSHOLE SEISMIC TEST RESULTS *(ASTM D4428/D4428M-07)*

FORM #	RGF-KHO-GEO-DH-1-00

Client:	*MANSOURAH MASSARAH GOLD PROJECT*	**Test No.**	*DH-01*
Project:	*L & T CONSTRUCTION*	**Geotechnical File No.:**	-
Project No.:	*19-K-3054*	**Coordinates (meters):**	*N: 2506179.07, E: 264210.66*
Borehole No.:	*BH-02*		

SUBSURFACE SOIL MODULUS - DEPTH PLOTS



Subsurface Soil Modulus Vs Depth profile

M. Azam Khan
Checked By

Muhammad Asad Shafiq
Approved By

This sheet is underownership of RGF Geotechnical Department Al Khobar, Data on this sheet can not be reproduced / changed untill unless prior permission from section Incharge
RGF-KHO-GEO-DH-1-00

TEST REPORT - DOWNHOLE SEISMIC TEST RESULTS *(ASTM D7400 - 08)*

FORM # RGF-KHO-GEO-DH-1-00

Project: *MANSOURAH MASSARAH GOLD PROJECT*
Client: *L & T CONSTRUCTION*
Project No.: *19-K-3054* **Test No.** *DH-02*
Borehole No.: *BH-15* **Geotechnical File No.:** -
Source Type: *Sledge Hammer* **Source to BH Distance (m):** 3 m
Receiver Type: *Triaxial Geophones* **Coordinates (m):** ***N:** 2506027.89, **E:** 264280.82*

Sr. No.	Depth of Reciever (m)	Distance of Source to Receiver (m)	Average Time		Calculated Velocity		Assumed Density ρ (g/cm^3)	Poisson Ratio υ	Shear Modulus G (M Pa)	Young Modulus E (M Pa)	Bulk Modulus K (M Pa)
			P-wave (msec)	S-wave (msec)	P-wave (m/sec)	S-wave (m/sec)					
1	0.75	3.09	8.24	15.41	375	201	1.7	0.300	68	178	148
2	1.50	3.35	8.51	15.92	970	513	1.8	0.305	474	1238	1060
3	2.25	3.75	8.91	16.65	990	542	2.0	0.285	588	1512	1175
4	3.00	4.24	9.11	16.99	2463	1449	2.1	0.235	4409	10894	6863
5	3.75	4.80	9.35	17.39	2332	1399	2.2	0.219	4307	10499	6222
6	4.50	5.41	9.60	17.80	2424	1478	2.2	0.204	4806	11573	6518
7	5.25	6.05	9.86	18.22	2455	1520	2.2	0.189	5082	12089	6486
8	6.00	6.71	10.15	18.69	2281	1407	2.2	0.193	4358	10395	5636
9	6.75	7.39	10.45	19.18	2261	1385	2.2	0.200	4217	10124	5628
10	7.50	8.08	10.74	19.65	2383	1470	2.2	0.193	4757	11346	6152
11	8.25	8.78	11.03	20.12	2416	1491	2.2	0.193	4891	11666	6325
12	9.00	9.49	11.33	20.61	2361	1446	2.2	0.200	4597	11035	6134
13	9.75	10.20	11.67	21.17	2101	1275	2.2	0.208	3579	8648	4937
14	10.50	10.92	12.02	21.73	2054	1284	2.2	0.179	3627	8557	4449
15	11.25	11.64	12.34	22.27	2259	1339	2.2	0.229	3943	9696	5972
16	12.00	12.37	12.70	22.84	2017	1274	2.2	0.168	3571	8343	4191
17	12.75	13.10	13.05	23.41	2082	1279	2.2	0.197	3597	8615	4744
18	13.50	13.83	13.37	23.93	2285	1406	2.2	0.195	4349	10397	5686
19	14.25	14.56	13.71	24.49	2156	1309	2.2	0.208	3770	9108	5200
20	15.00	15.30	14.06	25.06	2099	1289	2.2	0.197	3655	8753	4821
21	15.75	16.03	14.40	25.61	2165	1338	2.2	0.191	3941	9385	5058
22	16.50	16.77	14.75	26.17	2107	1317	2.2	0.179	3814	8997	4679
23	17.25	17.51	15.10	26.76	2110	1252	2.2	0.229	3446	8467	5198
24	18.00	18.25	15.48	27.38	1946	1193	2.2	0.199	3129	7503	4157
25	18.75	18.99	15.78	27.88	2467	1480	2.2	0.219	4821	11752	6964
26	19.50	19.73	16.09	28.40	2390	1425	2.2	0.224	4467	10937	6612
27	20.25	20.47	16.41	28.94	2317	1373	2.2	0.229	4149	10202	6283
28	21.00	21.21	16.75	29.48	2183	1374	2.2	0.172	4156	9738	4942
29	21.75	21.96	17.13	30.10	1955	1198	2.2	0.199	3157	7572	4195
30	22.50	22.70	17.52	30.74	1906	1161	2.2	0.205	2967	7148	4034
31	23.25	23.44	17.88	31.33	2066	1260	2.2	0.203	3495	8412	4727
32	24.00	24.19	18.27	31.95	1908	1200	2.2	0.173	3168	7430	3783

This sheet is underownership of RGF Geotechnical Department Al Khobar, Data on this sheet can not be reproduced / changed untill unless prior permission from section Incharge
RGF-KHO-GEO-DH-1-00

Sr. No.	Depth of Reciever (m)	Distance of Source to Receiver (m)	Average Time		Calculated Velocity		Assumed Density	Poisson Ratio	Shear Modulus	Young Modulus	Bulk Modulus
			P-wave (msec)	S-wave (msec)	P-wave (m/sec)	S-wave (m/sec)	**ρ** (g/cm^3)	**υ**	**G** (M Pa)	**E** (M Pa)	**K** (M Pa)
33	24.75	24.93	18.64	32.56	2012	1220	2.2	0.209	3276	7921	4536
34	25.50	25.68	19.00	33.16	2069	1241	2.2	0.219	3389	8261	4895
35	26.25	26.42	19.41	33.84	1817	1096	2.2	0.214	2641	6414	3743
36	27.00	27.17	19.80	34.47	1911	1183	2.2	0.189	3079	7324	3929
37	27.75	27.91	20.20	35.11	1864	1165	2.2	0.179	2985	7042	3662
38	28.50	28.66	20.59	35.74	1912	1184	2.2	0.189	3083	7333	3934
39	29.25	29.40	20.98	36.39	1913	1148	2.2	0.219	2898	7063	4186
40	30.00	30.15	21.39	37.03	1820	1166	2.2	0.152	2991	6890	3300

$$\text{Poisson Ratio } (\upsilon) = \frac{[\,(Vp/Vs)^2 - 2\,]}{[\,2 \times (Vp/Vs)^2 - 2\,]}$$

$$\text{Shear Modulus } (G) = \frac{\rho \times (Vs)^2}{1000}$$

$$\text{Young Modulus } (E) = 2G \times (1 + \upsilon)$$

$$\text{Bulk Modulus } (K) = \frac{G \times E}{3\,(3G - E)}$$

M. Azam Khan
Tested By

Muhammad Asad Shafiq
Checked By

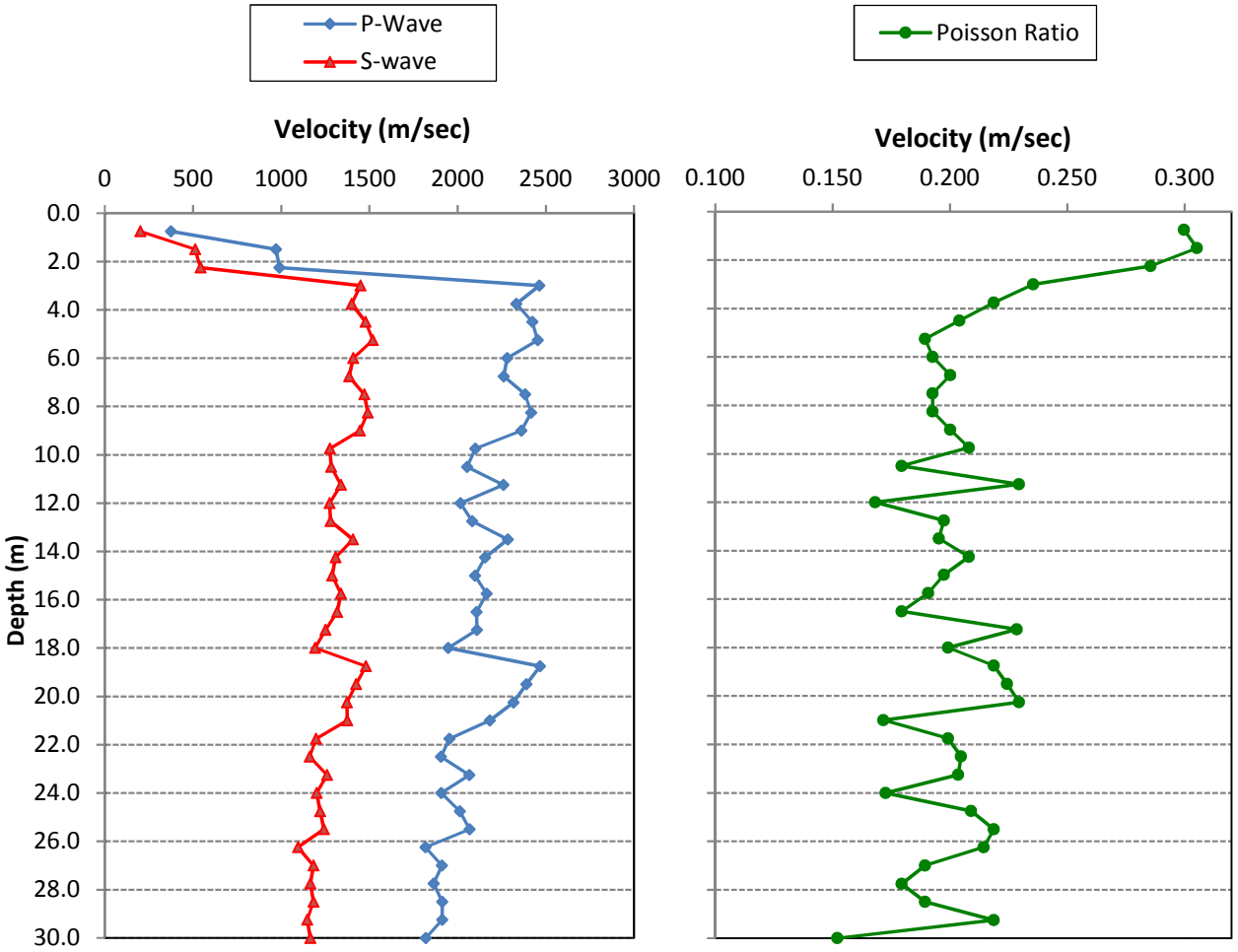
This sheet is underownership of RGF Geotechnical Department Al Khobar, Data on this sheet can not be reproduced / changed untill unless prior permission from section Incharge
RGF-KHO-GEO-DH-1-00

TEST REPORT - CROSSHOLE SEISMIC TEST RESULTS *(ASTM D4428/D4428M-07)*

FORM #	RGF-KHO-GEO-DH-1-00

Client:	*MANSOURAH MASSARAH GOLD PROJECT*	**Test No.**	*DH-02*
Project:	*L & T CONSTRUCTION*	**Geotechnical File No.:**	-
Project No.:	*19-K-3054*	**Coordinates (meters):**	*N: 2506027.89, E: 264280.82*
Borehole No.:	*BH-15*		

VELOCITY - DEPTH PLOTS



Subsurface Velocity Vs Depth profile

M. Azam Khan	***Muhammad Asad Shafiq***
Checked By	*Approved By*

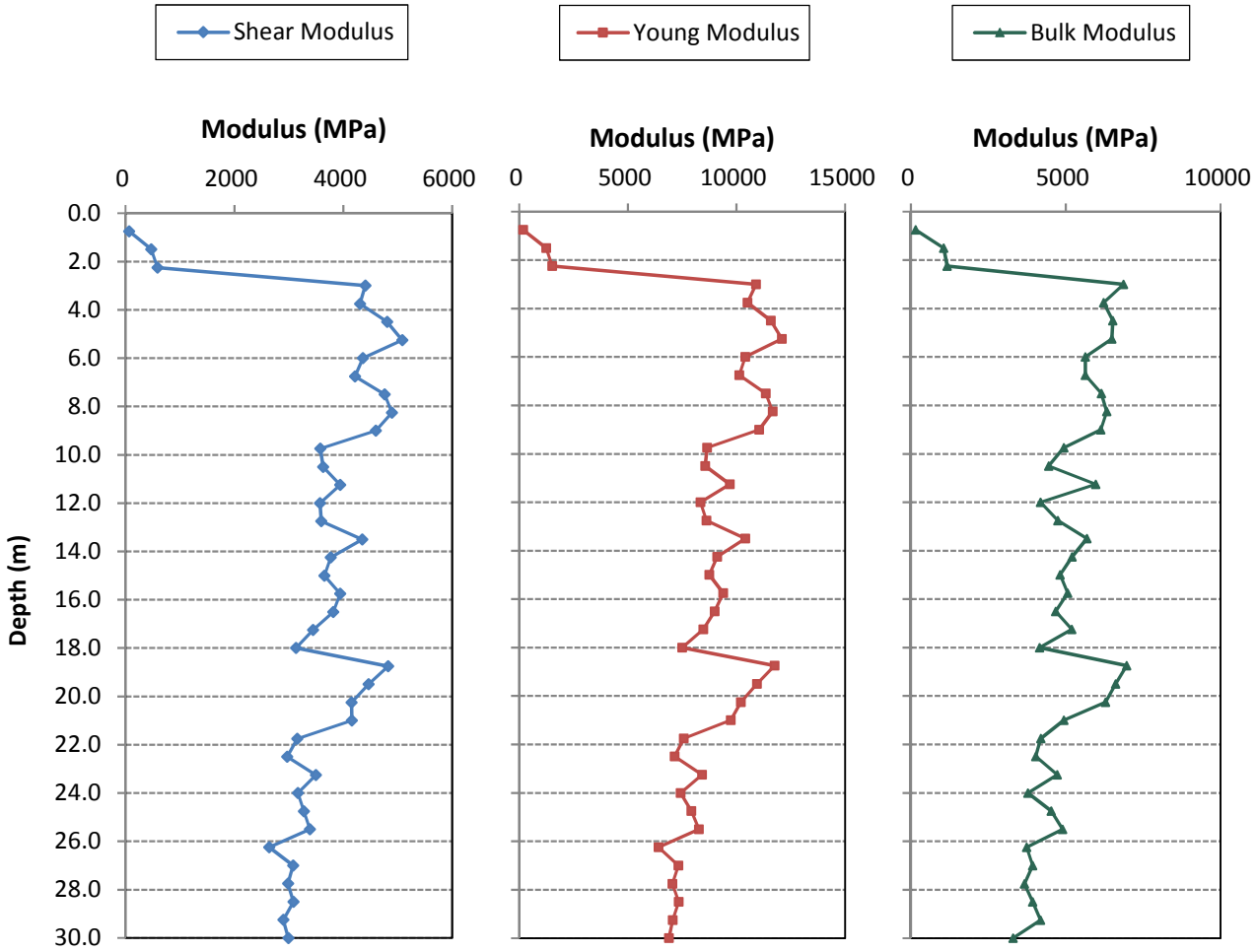
This sheet is underownership of RGF Geotechnical Department Al Khobar, Data on this sheet can not be reproduced / changed untill unless prior permission from section Incharge
RGF-KHO-GEO-DH-1-00

TEST REPORT - CROSSHOLE SEISMIC TEST RESULTS *(ASTM D4428/D4428M-07)*

FORM #	RGF-KHO-GEO-DH-1-00

Client:	*MANSOURAH MASSARAH GOLD PROJECT*	**Test No.**	*DH-02*
Project:	*L & T CONSTRUCTION*	**Geotechnical File No.:**	-
Project No.:	*19-K-3054*	**Coordinates (meters):**	*N: 2506027.89, E: 264280.82*
Borehole No.:	*BH-15*		

SUBSURFACE SOIL MODULUS - DEPTH PLOTS



Subsurface Soil Modulus Vs Depth profile

M. Azam Khan
Checked By

Muhammad Asad Shafiq
Approved By

This sheet is underownership of RGF Geotechnical Department Al Khobar, Data on this sheet can not be reproduced / changed untill unless prior permission from section Incharge
RGF-KHO-GEO-DH-1-00

TEST REPORT - DOWNHOLE SEISMIC TEST RESULTS *(ASTM D7400 - 08)*

FORM # RGF-KHO-GEO-DH-1-00

Project: *MANSOURAH MASSARAH GOLD PROJECT*
Client: *L & T CONSTRUCTION*
Project No.: *19-K-3054* **Test No.** *DH-03*
Borehole No.: *BH-19* **Geotechnical File No.:** -
Source Type: *Sledge Hammer* **Source to BH Distance (m):** 3 m
Receiver Type: *Triaxial Geophones* **Coordinates (m):** ***N:** 2506510.84, **E:** 264424.58*

Sr. No.	Depth of Reciever (m)	Distance of Source to Receiver (m)	Average Time		Calculated Velocity		Assumed Density ρ (g/cm^3)	Poisson Ratio υ	Shear Modulus G (M Pa)	Young Modulus E (M Pa)	Bulk Modulus K (M Pa)
			P-wave (msec)	S-wave (msec)	P-wave (m/sec)	S-wave (m/sec)					
1	0.75	3.09	5.10	9.09	606	340	1.7	0.27	197	500	363
2	1.50	3.35	5.38	9.56	935	557	1.7	0.22	527	1292	783
3	2.25	3.75	5.78	10.21	990	609	1.8	0.20	668	1596	873
4	3.00	4.24	6.28	11.01	985	616	1.8	0.18	683	1610	837
5	3.75	4.80	6.87	11.97	949	583	1.8	0.20	612	1464	804
6	4.50	5.41	7.48	12.93	993	631	1.8	0.16	717	1666	820
7	5.25	6.05	8.16	14.04	939	575	1.8	0.20	595	1428	793
8	6.00	6.71	8.81	15.09	1018	630	1.8	0.19	714	1699	912
9	6.75	7.39	9.44	16.15	1077	640	1.8	0.23	737	1809	1104
10	7.50	8.08	9.94	16.97	1382	843	1.9	0.20	1350	3250	1830
11	8.25	8.78	10.26	17.50	2190	1322	2.1	0.21	3671	8908	5176
12	9.00	9.49	10.59	18.04	2146	1312	2.2	0.20	3785	9099	5089
13	9.75	10.20	10.90	18.55	2304	1401	2.2	0.21	4315	10417	5926
14	10.50	10.92	11.22	19.08	2247	1357	2.2	0.21	4050	9825	5709
15	11.25	11.64	11.58	19.66	2008	1246	2.2	0.19	3418	8113	4315
16	12.00	12.37	11.90	20.18	2269	1397	2.2	0.20	4291	10256	5609
17	12.75	13.10	12.23	20.74	2209	1302	2.2	0.23	3727	9198	5763
18	13.50	13.83	12.56	21.26	2216	1406	2.2	0.16	4349	10115	5000
19	14.25	14.56	12.96	21.89	1833	1164	2.2	0.16	2979	6924	3417
20	15.00	15.30	13.42	22.63	1597	993	2.2	0.19	2169	5140	2721
21	15.75	16.03	13.87	23.39	1636	969	2.2	0.23	2064	5077	3135
22	16.50	16.77	14.26	24.02	1891	1170	2.2	0.19	3014	7168	3846
23	17.25	17.51	14.60	24.57	2172	1343	2.2	0.19	3966	9444	5090
24	18.00	18.25	14.93	25.11	2240	1369	2.2	0.20	4124	9915	5544
25	18.75	18.99	15.27	25.67	2177	1322	2.2	0.21	3844	9287	5302
26	19.50	19.73	15.60	26.23	2245	1323	2.2	0.23	3851	9505	5956
27	20.25	20.47	15.94	26.79	2181	1324	2.2	0.21	3858	9322	5322
28	21.00	21.21	16.28	27.33	2183	1374	2.2	0.17	4156	9738	4942
29	21.75	21.96	16.62	27.87	2184	1375	2.2	0.17	4162	9752	4949
30	22.50	22.70	16.97	28.44	2123	1304	2.2	0.20	3740	8957	4933
31	23.25	23.44	17.27	28.95	2479	1458	2.2	0.24	4677	11557	7281
32	24.00	24.19	17.60	29.49	2255	1378	2.2	0.20	4176	10040	5615

This sheet is underownership of RGF Geotechnical Department Al Khobar, Data on this sheet can not be reproduced / changed untill unless prior permission from section Incharge
RGF-KHO-GEO-DH-1-00

Sr. No.	Depth of Reciever (m)	Distance of Source to Receiver (m)	Average Time		Calculated Velocity		Assumed Density	Poisson Ratio	Shear Modulus	Young Modulus	Bulk Modulus
			P-wave (msec)	S-wave (msec)	P-wave (m/sec)	S-wave (m/sec)	ρ (g/cm^3)	υ	**G** (M Pa)	**E** (M Pa)	**K** (M Pa)
33	24.75	24.93	17.93	30.04	2256	1353	2.2	0.22	4030	9823	5821
34	25.50	25.68	18.35	30.71	1773	1112	2.2	0.18	2718	6394	3293
35	26.25	26.42	18.78	31.38	1733	1112	2.2	0.15	2720	6255	2977
36	27.00	27.17	19.18	32.03	1863	1147	2.2	0.20	2892	6914	3781
37	27.75	27.91	19.58	32.65	1864	1202	2.2	0.14	3181	7275	3401
38	28.50	28.66	19.98	33.31	1864	1130	2.2	0.21	2809	6796	3902
39	29.25	29.40	20.36	33.93	1963	1203	2.2	0.20	3185	7638	4232
40	30.00	30.15	20.76	34.57	1865	1166	2.2	0.18	2991	7055	3668

$$\text{Poisson Ratio } (\upsilon) = \frac{[\,(Vp/Vs)^2 - 2\,]}{[\,2 \times (Vp/Vs)^2 - 2\,]}$$

$$\text{Shear Modulus } (G) = \frac{\rho \times (Vs)^2}{1000}$$

$$\text{Young Modulus } (E) = 2G \times (1 + \upsilon)$$

$$\text{Bulk Modulus } (K) = \frac{G \times E}{3\,(3G - E)}$$

M. Azam Khan
Tested By

Muhammad Asad Shafiq
Checked By

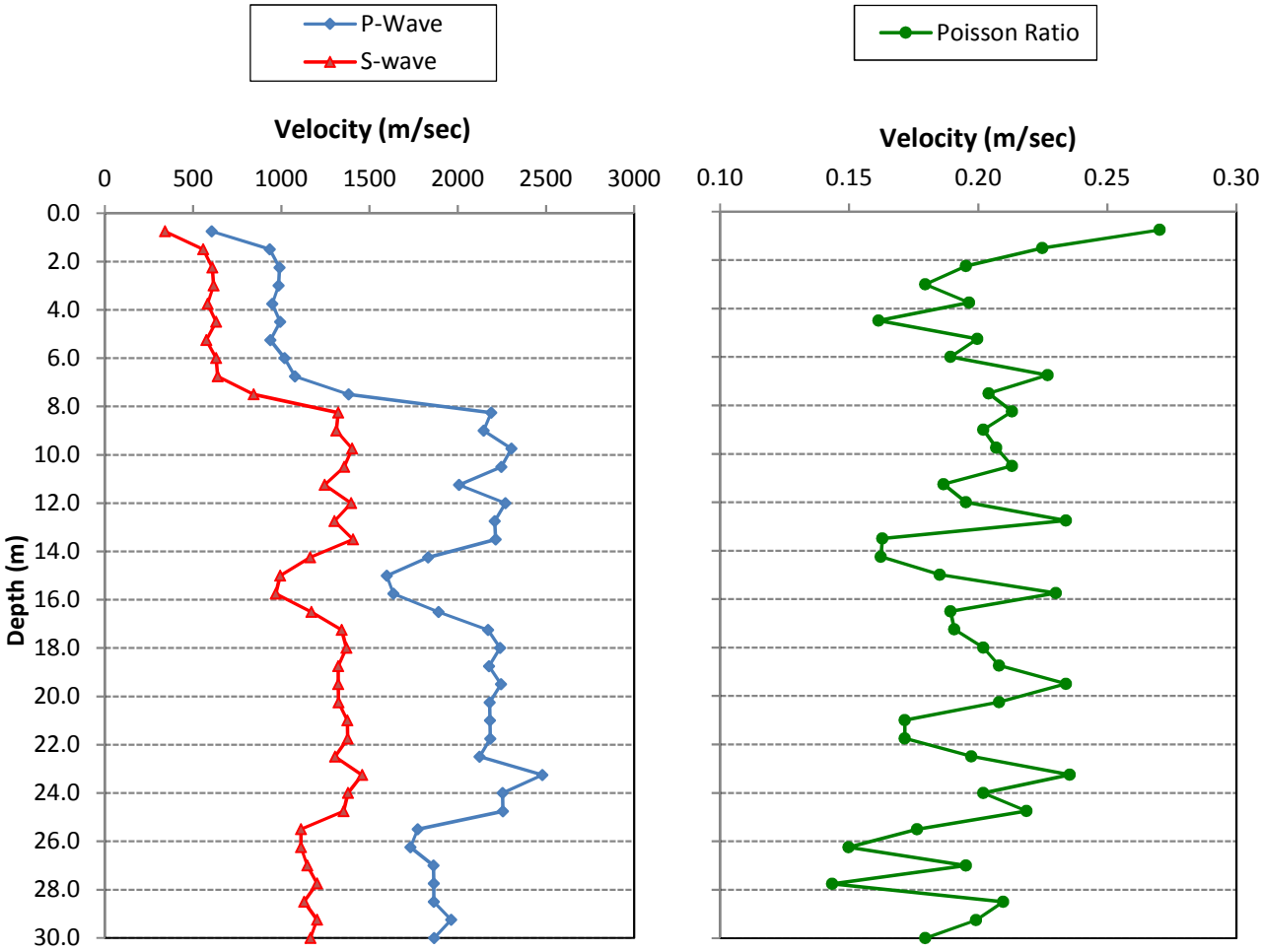
This sheet is underownership of RGF Geotechnical Department Al Khobar, Data on this sheet can not be reproduced / changed untill unless prior permission from section Incharge
RGF-KHO-GEO-DH-1-00

TEST REPORT - CROSSHOLE SEISMIC TEST RESULTS *(ASTM D4428/D4428M-07)*

FORM #	RGF-KHO-GEO-DH-1-00

Client:	*MANSOURAH MASSARAH GOLD PROJECT*	**Test No.**	*DH-03*
Project:	*L & T CONSTRUCTION*	**Geotechnical File No.:**	-
Project No.:	*19-K-3054*	**Coordinates (meters):**	*N: 2506510.84, E: 264424.58*
Borehole No.:	*BH-19*		

VELOCITY - DEPTH PLOTS



Subsurface Velocity Vs Depth profile

M. Azam Khan	***Muhammad Asad Shafiq***
Checked By	*Approved By*

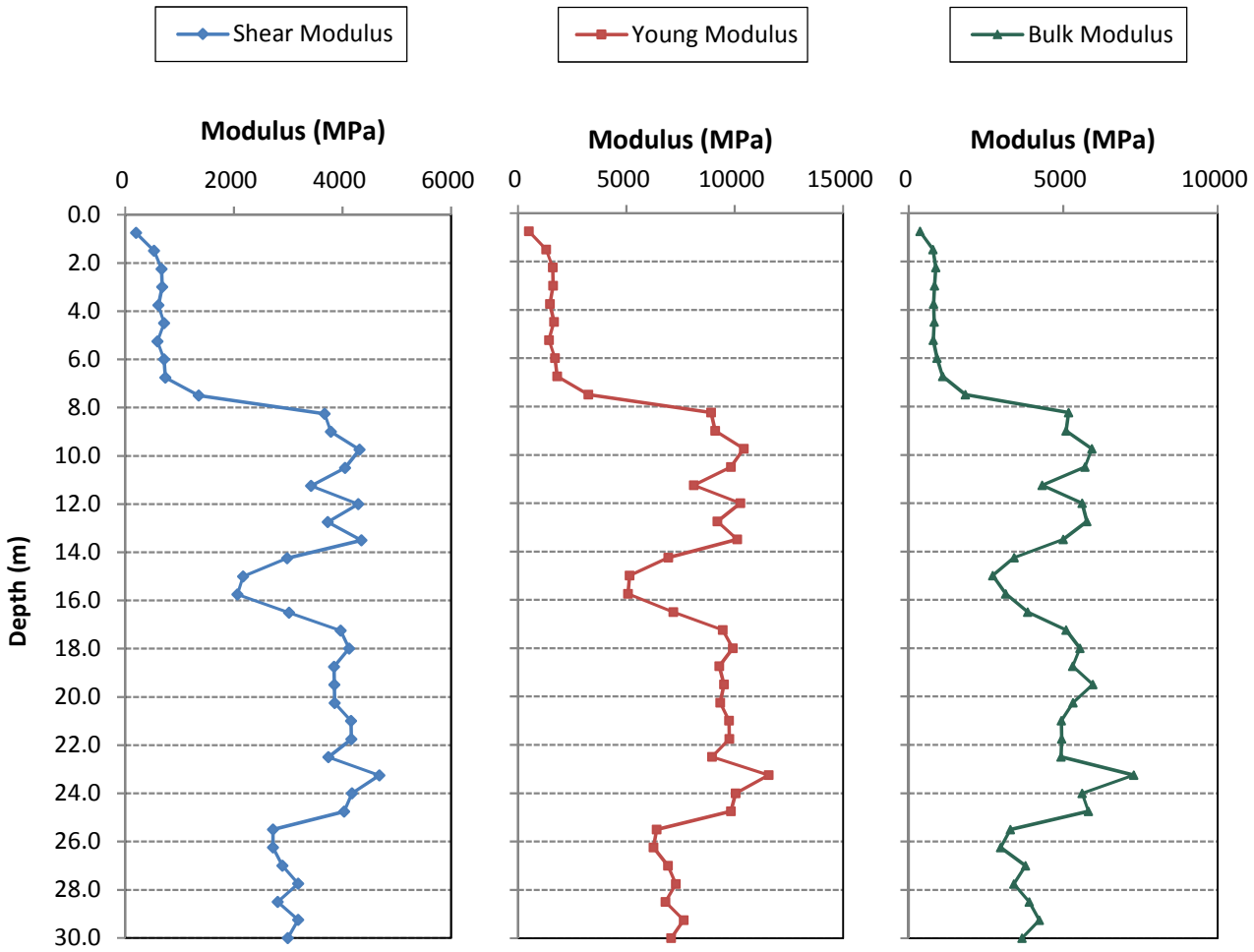
This sheet is underownership of RGF Geotechnical Department Al Khobar, Data on this sheet can not be reproduced / changed untill unless prior permission from section Incharge
RGF-KHO-GEO-DH-1-00

TEST REPORT - CROSSHOLE SEISMIC TEST RESULTS *(ASTM D4428/D4428M-07)*

FORM #	RGF-KHO-GEO-DH-1-00

Client:	*MANSOURAH MASSARAH GOLD PROJECT*	**Test No.**	*DH-03*
Project:	*L & T CONSTRUCTION*	**Geotechnical File No.:**	-
Project No.:	*19-K-3054*	**Coordinates (meters):**	*N: 2506510.84, E: 264424.58*
Borehole No.:	*BH-19*		

SUBSURFACE SOIL MODULUS - DEPTH PLOTS



Subsurface Soil Modulus Vs Depth profile

M. Azam Khan
Checked By

Muhammad Asad Shafiq
Approved By

This sheet is underownership of RGF Geotechnical Department Al Khobar, Data on this sheet can not be reproduced / changed untill unless prior permission from section Incharge
RGF-KHO-GEO-DH-1-00

ELECTRICAL RESISTIVITY TEST RESULTS

TEST REPORT FOR ELECTRICAL RESISTIVITY SURVEY

ASTM G57-06 (2012)

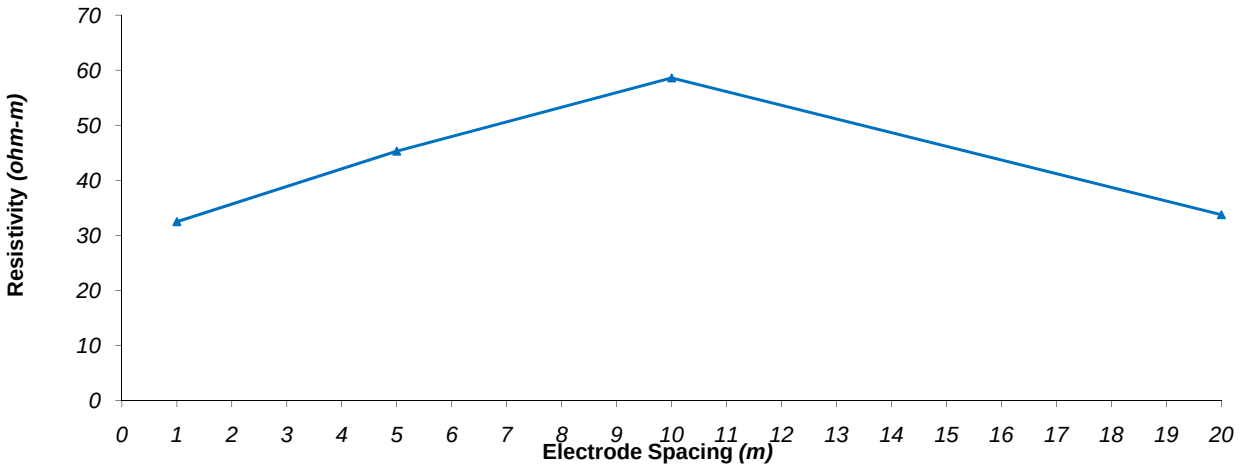
Project:	MANSOURAH MASSARAH GOLD PROJECT		
Client:	L & T CONSTRUCTION		
Configuration:	WENNER FOUR ELECTRODE METHOD	**Project No.:**	19-K-3054
Equipment:	TERRAMETER SAS 4000	**Geotech. File No.:**	-
Orientation:	EAST-WEST	**Date**	Sep 10, 2019

ELECTRICAL RESISTIVITY TEST # ERT-02

Coordinates (m): ***N:*** 2505905.54 ***E:*** 264145.85

Inter-Electrode Spacing a (m)	Geometric Factor K	Resistance R (Ω)	Apparent Resistivity ρ (Ω.m)
1	6.28	5.173	32.49
5	31.40	1.443	45.29
10	62.80	0.933	58.59
20	125.60	0.269	33.75

GRAPHICAL REPRESENTATION



Tested By:	*Checked By:*	*Approved By:*
Ghassan	*Amir yahya*	*M. Asad Shafiq*

TEST REPORT FOR ELECTRICAL RESISTIVITY SURVEY

ASTM G57-06 (2012)

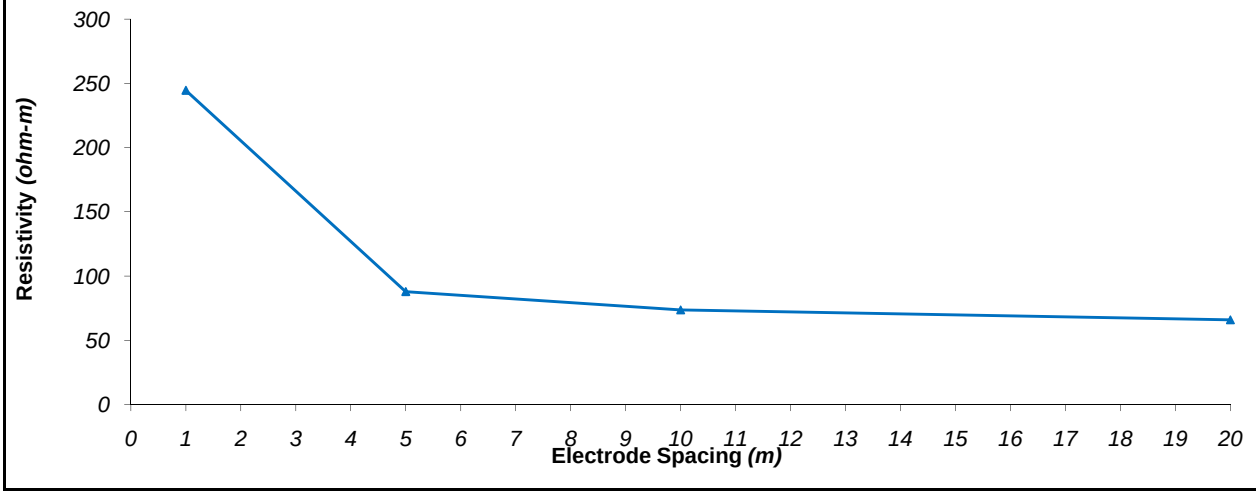
Project:	MANSOURAH MASSARAH GOLD PROJECT		
Client:	L & T CONSTRUCTION		
Configuration:	WENNER FOUR ELECTRODE METHOD	**Project No.:**	19-K-3054
Equipment:	TERRAMETER SAS 4000	**Geotech. File No.:**	-
Orientation:	NORTH-SOUTH	**Date**	Sep 10, 2019

ELECTRICAL RESISTIVITY TEST # ERT-04

Coordinates (m): ***N:*** 2506481.09 ***E:*** 264440.23

Inter-Electrode Spacing a (m)	Geometric Factor K	Resistance R (Ω)	Apparent Resistivity ρ (Ω.m)
1	6.28	38.943	244.56
5	31.40	2.798	87.87
10	62.80	1.172	73.61
20	125.60	0.525	65.89

GRAPHICAL REPRESENTATION



Tested By:	*Checked By:*	*Approved By:*
Ghassan	*Amir yahya*	*M. Asad Shafiq*

TEST REPORT FOR ELECTRICAL RESISTIVITY SURVEY

ASTM G57-06 (2012)

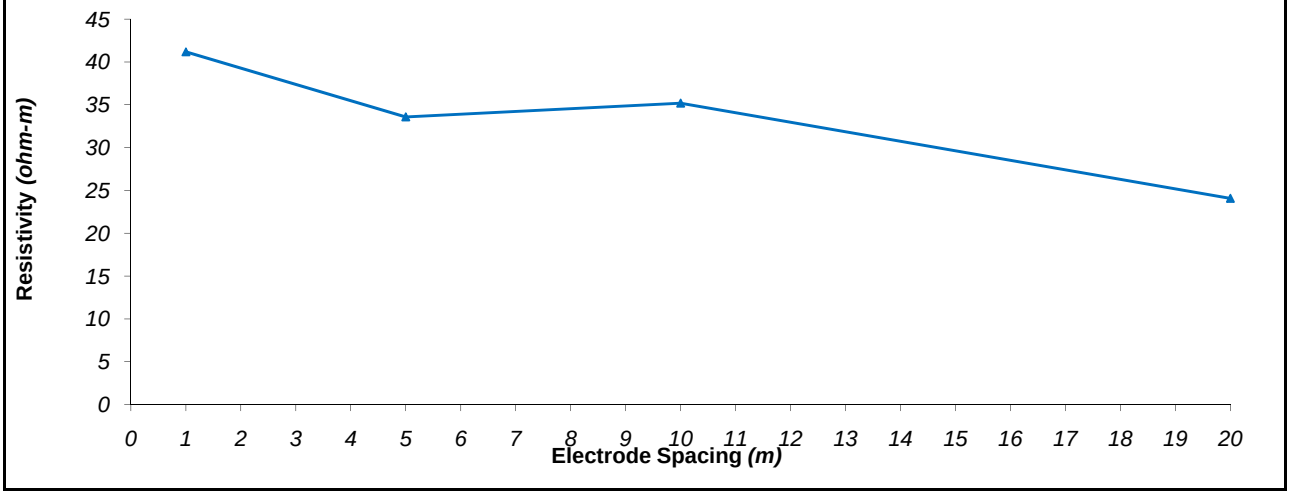
Project:	MANSOURAH MASSARAH GOLD PROJECT		
Client:	L & T CONSTRUCTION		
Configuration:	WENNER FOUR ELECTRODE METHOD	**Project No.:**	19-K-3054
Equipment:	TERRAMETER SAS 4000	**Geotech. File No.:**	-
Orientation:	EAST-WEST	**Date**	Sep 10, 2019

ELECTRICAL RESISTIVITY TEST # ERT-05

Coordinates (m): ***N:*** 2506086.40 ***E:*** 264003.79

Inter-Electrode Spacing a (m)	Geometric Factor K	Resistance R (Ω)	Apparent Resistivity ρ (Ω.m)
1	6.28	6.557	41.18
5	31.40	1.070	33.59
10	62.80	0.560	35.19
20	125.60	0.192	24.07

GRAPHICAL REPRESENTATION



Tested By:	*Checked By:*	*Approved By:*
Ghassan	*Amir yahya*	*M. Asad Shafiq*

TEST REPORT FOR ELECTRICAL RESISTIVITY SURVEY

ASTM G57-06 (2012)

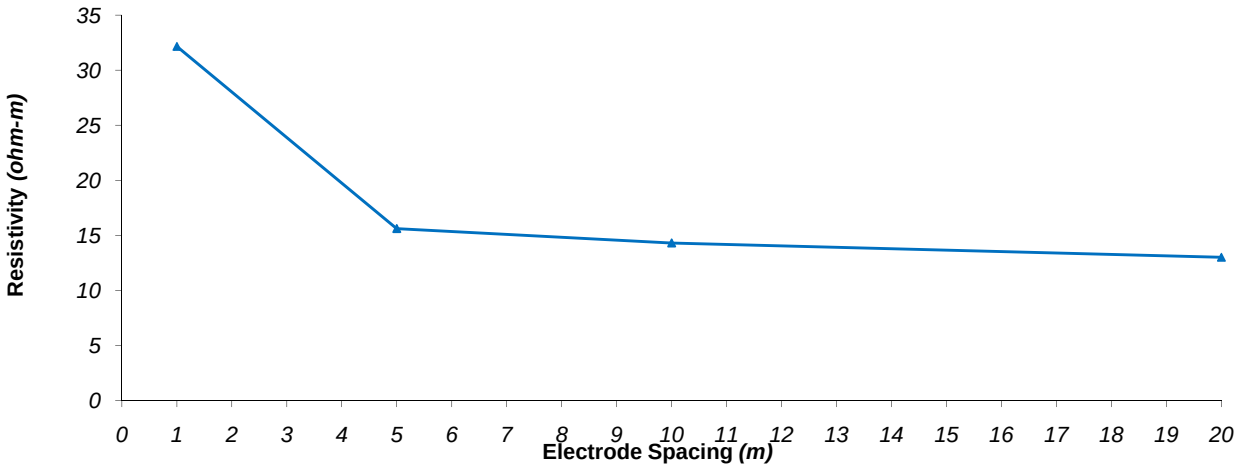
Project:	MANSOURAH MASSARAH GOLD PROJECT		
Client:	L & T CONSTRUCTION		
Configuration:	WENNER FOUR ELECTRODE METHOD	**Project No.:**	19-K-3054
Equipment:	TERRAMETER SAS 4000	**Geotech. File No.:**	-
Orientation:	NORTH-SOUTH	**Date**	Sep 10, 2019

ELECTRICAL RESISTIVITY TEST # ERT-06

Coordinates (m): ***N:*** 2506024.04 ***E:*** 263874.76

Inter-Electrode Spacing a (m)	Geometric Factor K	Resistance R (Ω)	Apparent Resistivity ρ (Ω.m)
1	6.28	5.121	32.16
5	31.40	0.497	15.61
10	62.80	0.228	14.30
20	125.60	0.104	13.01

GRAPHICAL REPRESENTATION



Tested By:	*Checked By:*	*Approved By:*
Ghassan	*Amir yahya*	*M. Asad Shafiq*

TEST REPORT FOR ELECTRICAL RESISTIVITY SURVEY

ASTM G57-06 (2012)

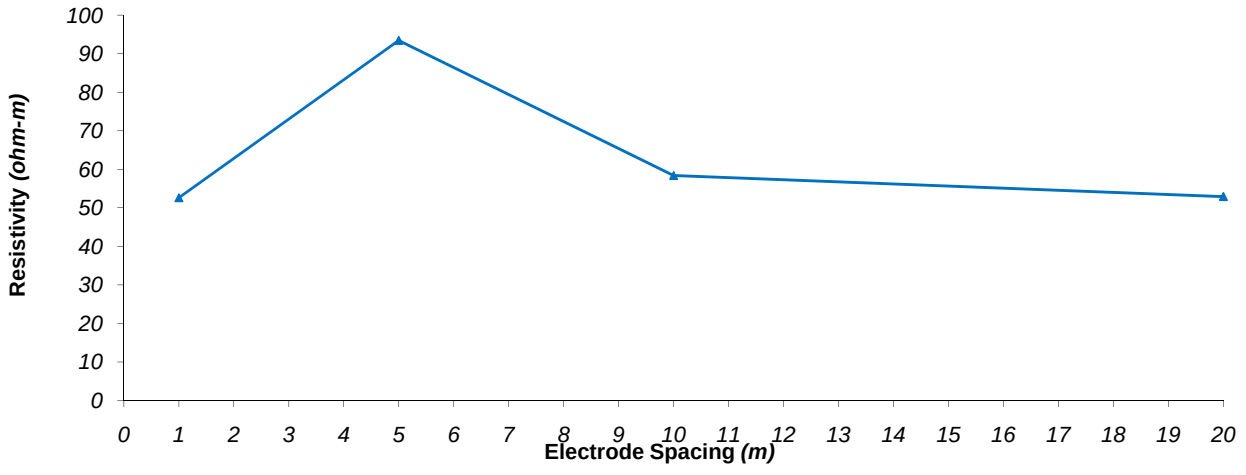
Project:	MANSOURAH MASSARAH GOLD PROJECT		
Client:	L & T CONSTRUCTION		
Configuration:	WENNER FOUR ELECTRODE METHOD	**Project No.:**	19-K-3054
Equipment:	TERRAMETER SAS 4000	**Geotech. File No.:**	-
Orientation:	NORTH-SOUTH	**Date**	Sep 18, 2019

ELECTRICAL RESISTIVITY TEST # ERT-07

Coordinates (m): ***N:*** 2509393.27 ***E:*** 264858.44

Inter-Electrode Spacing a (m)	Geometric Factor K	Resistance R (Ω)	Apparent Resistivity ρ (Ω.m)
1	6.28	8.377	52.60
5	31.40	2.976	93.45
10	62.80	0.930	58.38
20	125.60	0.421	52.92

GRAPHICAL REPRESENTATION



Tested By:	*Checked By:*	*Approved By:*
Ghassan	*Amir yahya*	*M. Asad Shafiq*

TEST REPORT FOR ELECTRICAL RESISTIVITY SURVEY

ASTM G57-06 (2012)

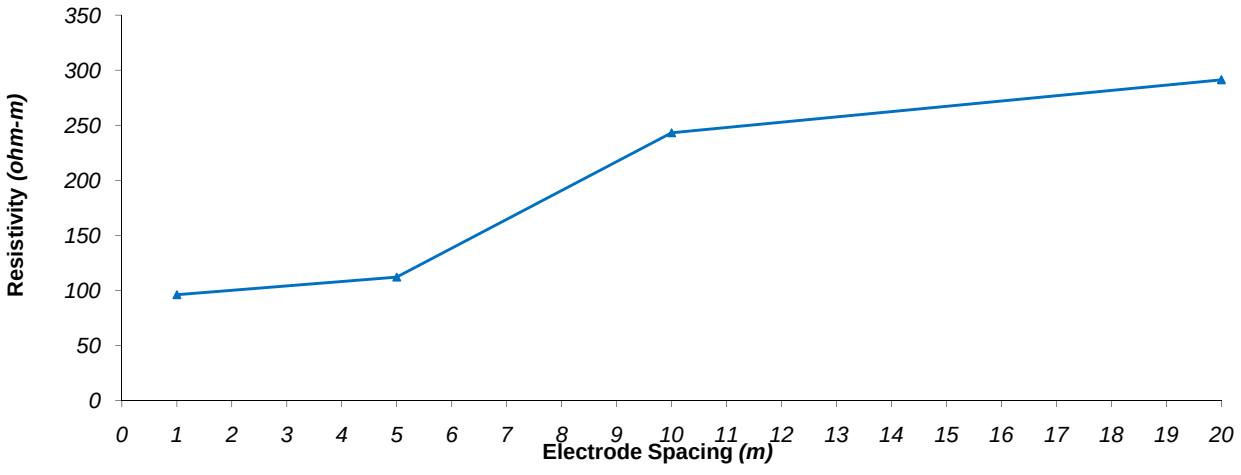
Project:	MANSOURAH MASSARAH GOLD PROJECT		
Client:	L & T CONSTRUCTION		
Configuration:	WENNER FOUR ELECTRODE METHOD	**Project No.:**	19-K-3054
Equipment:	TERRAMETER SAS 4000	**Geotech. File No.:**	-
Orientation:	NORTH-SOUTH	**Date**	Sep 11, 2019

ELECTRICAL RESISTIVITY TEST # ERT-09

Coordinates (m): ***N:*** 2508495.16 ***E:*** 263255.30

Inter-Electrode Spacing a (m)	Geometric Factor K	Resistance R (Ω)	Apparent Resistivity ρ (Ω.m)
1	6.28	15.291	96.03
5	31.40	3.569	112.05
10	62.80	3.871	243.09
20	125.60	2.319	291.29

GRAPHICAL REPRESENTATION



Tested By:	*Checked By:*	*Approved By:*
Ghassan	*Amir yahya*	*M. Asad Shafiq*

TEST REPORT FOR ELECTRICAL RESISTIVITY SURVEY

ASTM G57-06 (2012)

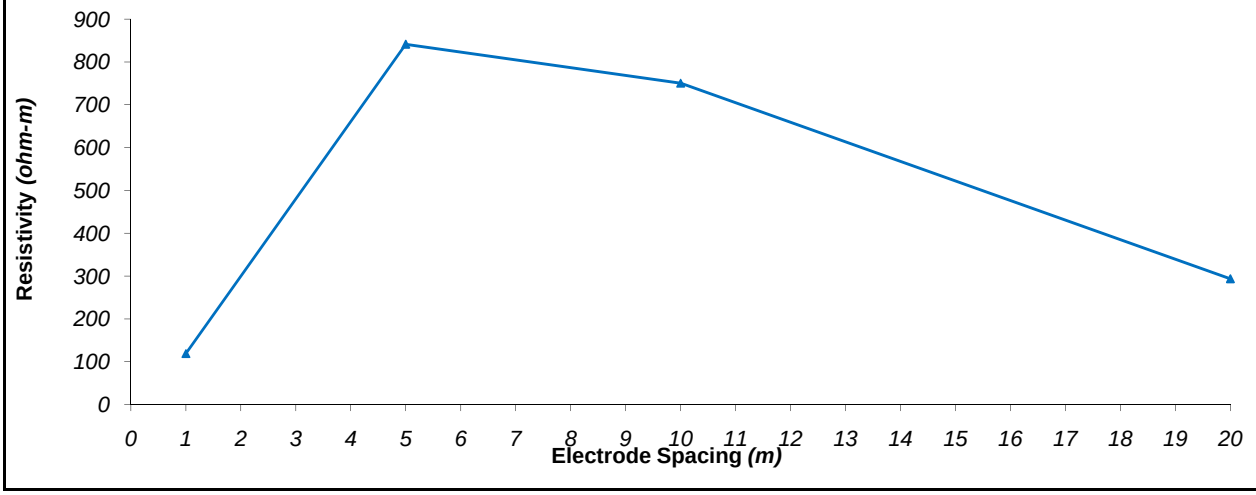
Project:	MANSOURAH MASSARAH GOLD PROJECT		
Client:	L & T CONSTRUCTION		
Configuration:	WENNER FOUR ELECTRODE METHOD	**Project No.:**	19-K-3054
Equipment:	TERRAMETER SAS 4000	**Geotech. File No.:**	-
Orientation:	EAST-WEST	**Date**	Sep 11, 2019

ELECTRICAL RESISTIVITY TEST # ERT-10

Coordinates (m): ***N:*** 2508260.17 ***E:*** 263217.17

Inter-Electrode Spacing a (m)	Geometric Factor K	Resistance R (Ω)	Apparent Resistivity ρ (Ω.m)
1	6.28	18.916	118.79
5	31.40	26.793	841.30
10	62.80	11.949	750.40
20	125.60	2.337	293.58

GRAPHICAL REPRESENTATION



Tested By:	***Checked By:***	***Approved By:***
Ghassan	*Amir yahya*	*M. Asad Shafiq*

TEST REPORT FOR ELECTRICAL RESISTIVITY SURVEY

ASTM G57-06 (2012)

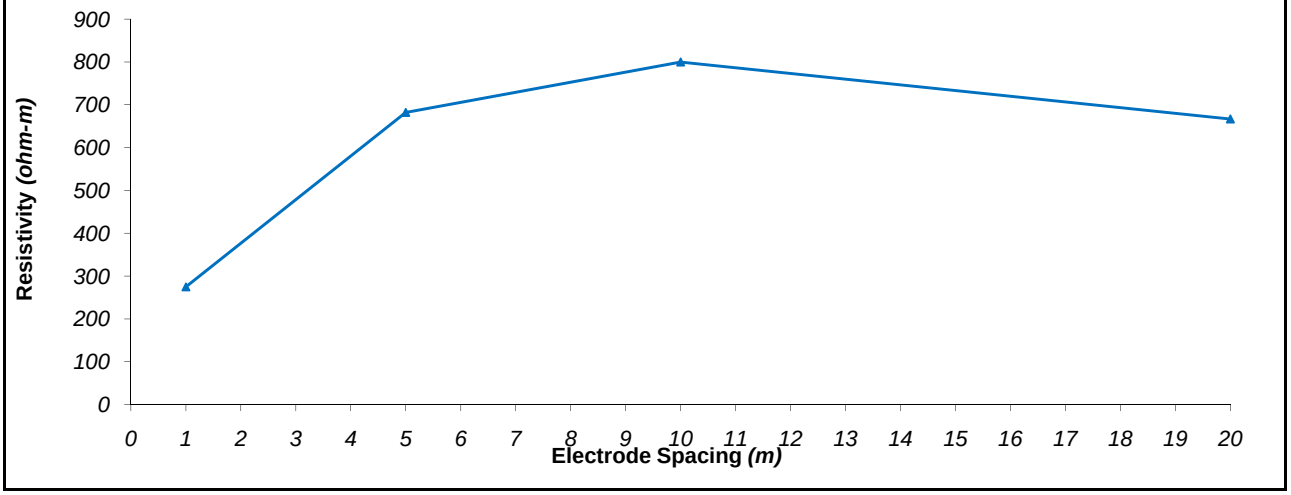
Project:	MANSOURAH MASSARAH GOLD PROJECT		
Client:	L & T CONSTRUCTION		
Configuration:	WENNER FOUR ELECTRODE METHOD	**Project No.:**	19-K-3054
Equipment:	TERRAMETER SAS 4000	**Geotech. File No.:**	-
Orientation:	EAST-WEST	**Date**	Sep 11, 2019

ELECTRICAL RESISTIVITY TEST # ERT-11

Coordinates (m): ***N:*** 2508419.38 ***E:*** 263058.27

Inter-Electrode Spacing a (m)	Geometric Factor K	Resistance R (Ω)	Apparent Resistivity ρ (Ω.m)
1	6.28	43.735	274.66
5	31.40	21.722	682.07
10	62.80	12.737	799.88
20	125.60	5.310	666.91

GRAPHICAL REPRESENTATION



Tested By:	***Checked By:***	***Approved By:***
Ghassan	*Amir yahya*	*M. Asad Shafiq*

TEST REPORT FOR ELECTRICAL RESISTIVITY SURVEY

ASTM G57-06 (2012)

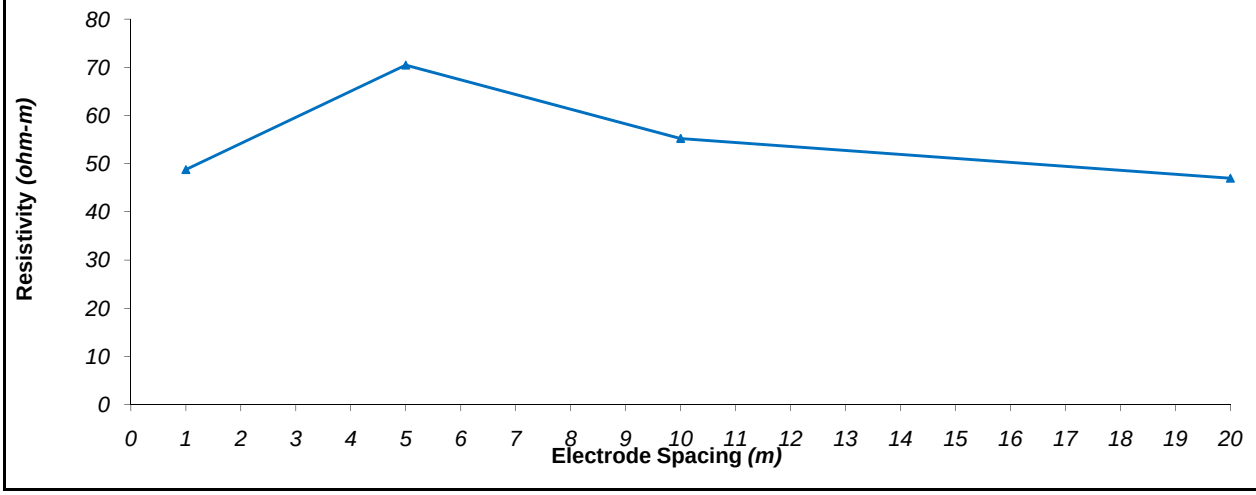
Project:	MANSOURAH MASSARAH GOLD PROJECT		
Client:	L & T CONSTRUCTION		
Configuration:	WENNER FOUR ELECTRODE METHOD	**Project No.:**	19-K-3054
Equipment:	TERRAMETER SAS 4000	**Geotech. File No.:**	-
Orientation:	EAST-WEST	**Date**	Sep 11, 2019

ELECTRICAL RESISTIVITY TEST # ERT-12

Coordinates (m): ***N:*** 2509398.76 ***E:*** 265649.46

Inter-Electrode Spacing a (m)	Geometric Factor K	Resistance R (Ω)	Apparent Resistivity ρ (Ω.m)
1	6.28	7.765	48.77
5	31.40	2.244	70.45
10	62.80	0.879	55.22
20	125.60	0.374	46.97

GRAPHICAL REPRESENTATION



Tested By:	*Checked By:*	*Approved By:*
Ghassan	*Amir yahya*	*M. Asad Shafiq*

TEST REPORT FOR ELECTRICAL RESISTIVITY SURVEY

ASTM G57-06 (2012)

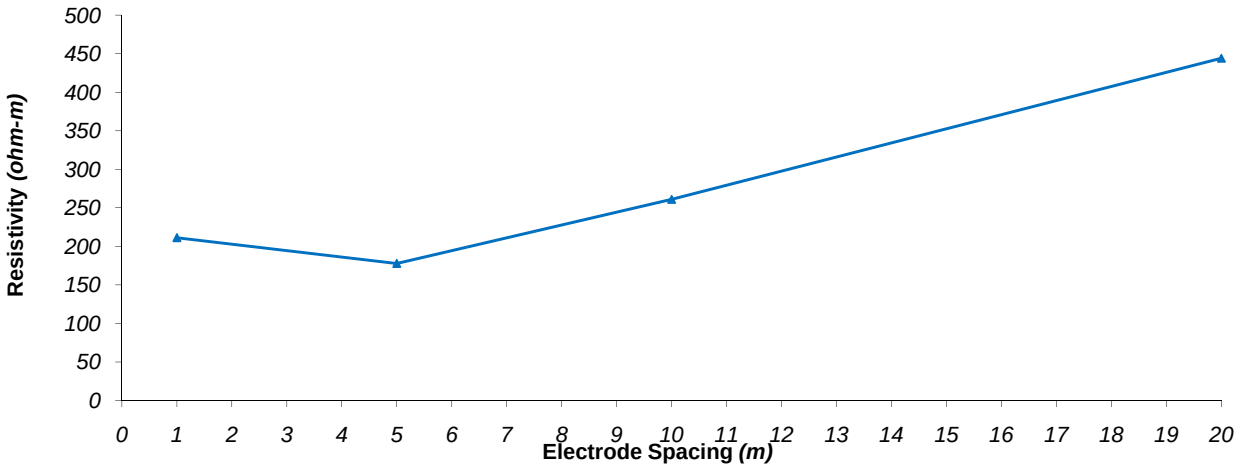
Project:	MANSOURAH MASSARAH GOLD PROJECT		
Client:	L & T CONSTRUCTION		
Configuration:	WENNER FOUR ELECTRODE METHOD	**Project No.:**	19-K-3054
Equipment:	TERRAMETER SAS 4000	**Geotech. File No.:**	-
Orientation:	EAST-WEST	**Date**	Sep 11, 2019

ELECTRICAL RESISTIVITY TEST # ERT-13

Coordinates (m): ***N:*** 2506800.38 ***E:*** 263714.98

Inter-Electrode Spacing a (m)	Geometric Factor K	Resistance R (Ω)	Apparent Resistivity ρ (Ω.m)
1	6.28	33.629	211.19
5	31.40	5.663	177.82
10	62.80	4.155	260.95
20	125.60	3.536	444.11

GRAPHICAL REPRESENTATION



Tested By:	***Checked By:***	***Approved By:***
Ghassan	*Amir yahya*	*M. Asad Shafiq*

TEST REPORT FOR ELECTRICAL RESISTIVITY SURVEY

ASTM G57-06 (2012)

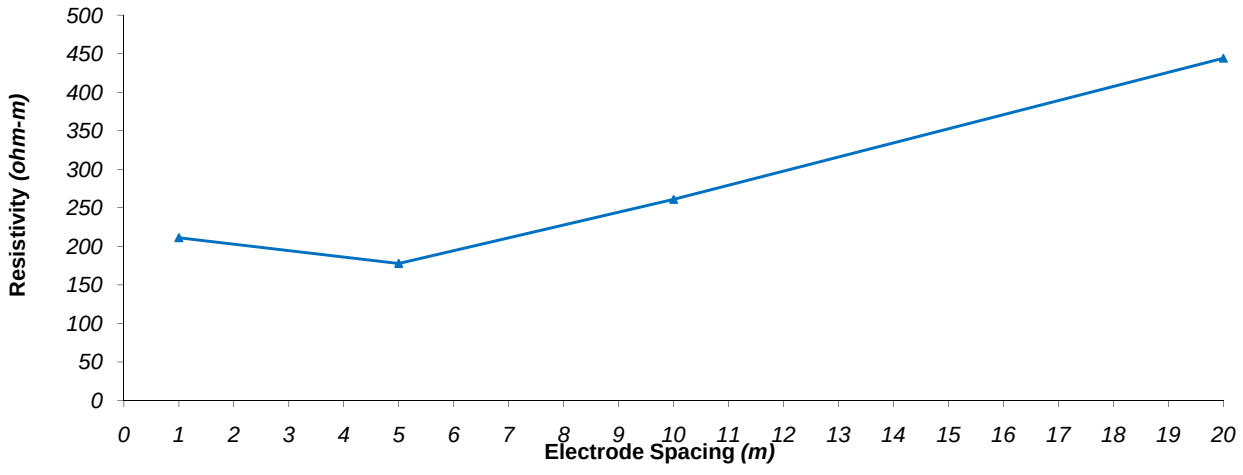
Project:	MANSOURAH MASSARAH GOLD PROJECT		
Client:	L & T CONSTRUCTION		
Configuration:	WENNER FOUR ELECTRODE METHOD	**Project No.:**	19-K-3054
Equipment:	TERRAMETER SAS 4000	**Geotech. File No.:**	-
Orientation:	EAST-WEST	**Date**	Sep 11, 2019

ELECTRICAL RESISTIVITY TEST # ERT-14

Coordinates (m): ***N:*** 2509177.69 ***E:*** 261827.47

Inter-Electrode Spacing a (m)	Geometric Factor K	Resistance R (Ω)	Apparent Resistivity ρ (Ω.m)
1	6.28	33.629	211.19
5	31.40	5.663	177.82
10	62.80	4.155	260.95
20	125.60	3.536	444.11

GRAPHICAL REPRESENTATION



Tested By:	***Checked By:***	***Approved By:***
Ghassan	*Amir yahya*	*M. Asad Shafiq*

TEST REPORT FOR ELECTRICAL RESISTIVITY SURVEY

ASTM G57-06 (2012)

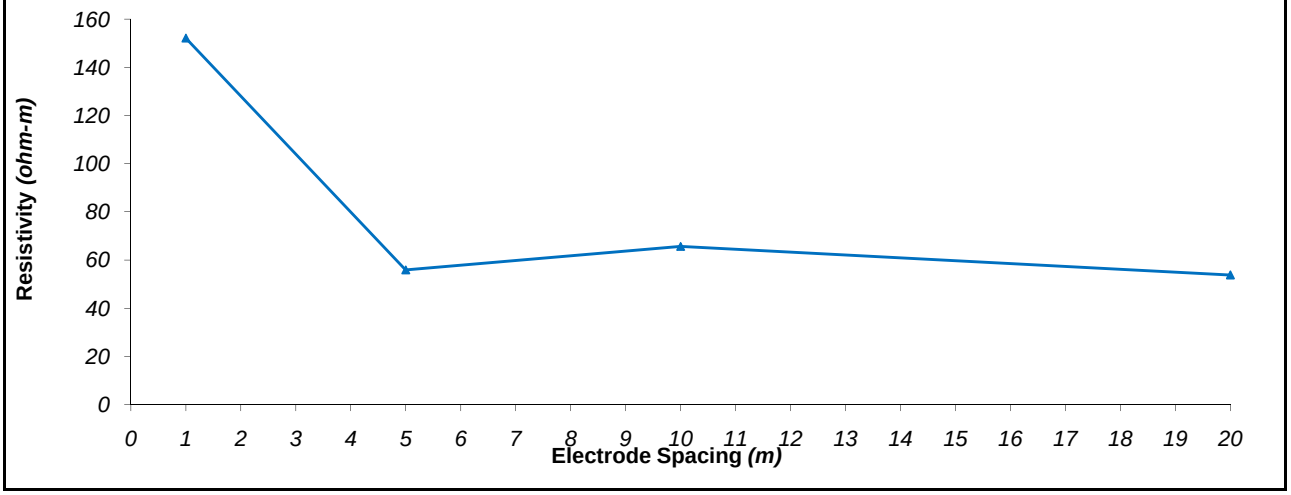
Project:	MANSOURAH MASSARAH GOLD PROJECT		
Client:	L & T CONSTRUCTION		
Configuration:	WENNER FOUR ELECTRODE METHOD	**Project No.:**	19-K-3054
Equipment:	TERRAMETER SAS 4000	**Geotech. File No.:**	-
Orientation:	EAST-WEST	**Date**	Sep 10, 2019

ELECTRICAL RESISTIVITY TEST # ERT-15

Coordinates (m): ***N:*** 2505924.44 ***E:*** 264333.99

Inter-Electrode Spacing a (m)	Geometric Factor K	Resistance R (Ω)	Apparent Resistivity ρ (Ω.m)
1	6.28	24.224	152.13
5	31.40	1.779	55.87
10	62.80	1.045	65.64
20	125.60	0.428	53.76

GRAPHICAL REPRESENTATION



Tested By:	***Checked By:***	***Approved By:***
Ghassan	*Amir yahya*	*M. Asad Shafiq*

TEST REPORT FOR ELECTRICAL RESISTIVITY SURVEY

ASTM G57-06 (2012)

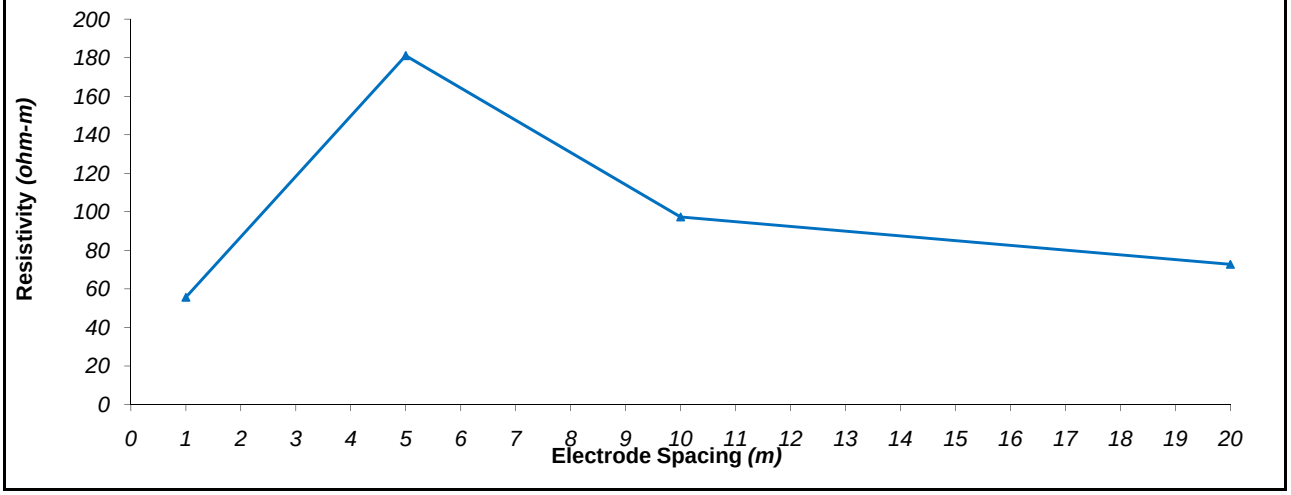
Project:	MANSOURAH MASSARAH GOLD PROJECT		
Client:	L & T CONSTRUCTION		
Configuration:	WENNER FOUR ELECTRODE METHOD	**Project No.:**	19-K-3054
Equipment:	TERRAMETER SAS 4000	**Geotech. File No.:**	-
Orientation:	NORTH-SOUTH	**Date**	Sep 10, 2019

ELECTRICAL RESISTIVITY TEST # ERT-16

Coordinates (m): ***N:*** 2506141.71 ***E:*** 264233.67

Inter-Electrode Spacing a (m)	Geometric Factor K	Resistance R (Ω)	Apparent Resistivity ρ (Ω.m)
1	6.28	8.858	55.63
5	31.40	5.767	181.08
10	62.80	1.550	97.35
20	125.60	0.579	72.76

GRAPHICAL REPRESENTATION



Tested By:	***Checked By:***	***Approved By:***
Ghassan	*Amir yahya*	*M. Asad Shafiq*

THERMAL RESISTIVITY TEST RESULTS

الرياض للجيوتقنية والأساسات (ش.م.م)
RIYADH GEOTECHNIQUE & FOUNDATIONS (LLC)

THERMAL RESISTIVITY TEST RESULTS

CLIENT: **L & T CONSTRUCTION**

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**
GEOTECHNICAL INVESTIGATION

PROJ. NO.: **19-K-3054**

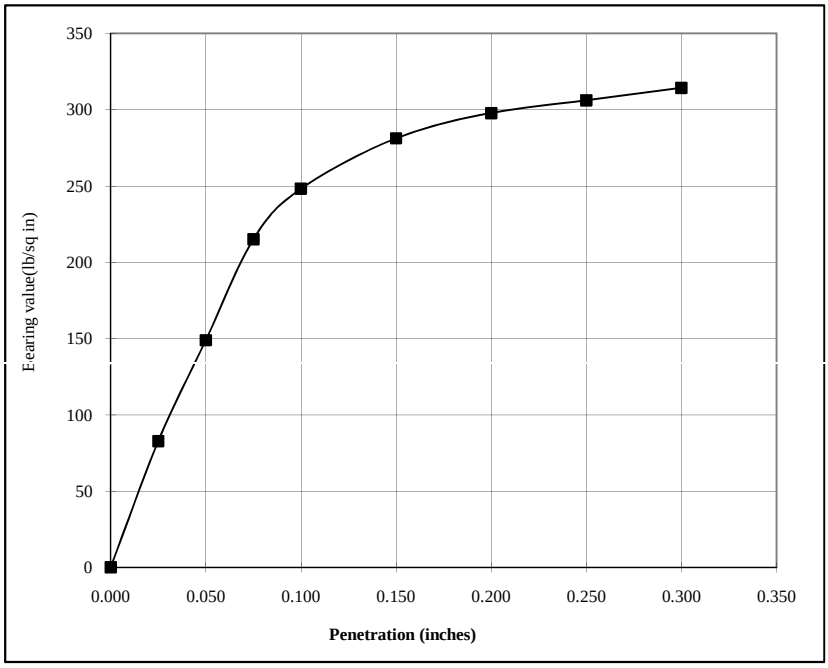
TEST NO.	COORDINATES (m)		THERMAL RESISTIVITY	THERMAL CONDUCTIVITY
	NORTHING	EASTING	(C°-cm/W)	(W/m-K)
TR-01	2506046.16	263966.17	185.00	0.54

FIELD CBR TEST RESULT

FIELD CALIFORNIA BEARING RATIO TEST(ASTM-D4429)

Project:	**Mansourah Massarah Gold Project**	**TEST DATE :**	**15-Sep-19**
Client:	**L & T Construction**		
Coordinate :	**N-2506293.9, E-263547.697**	**PROJECT # :**	**19-K-3054**
Location	**Ar Rjum**	**TEST PIT NO :**	**FCBR-1**

PENET. (IN INCHES)	DIAL READING	BEARING VALUE (Ib/sq in.)
0.000	0.00	0
0.025	5.00	83
0.050	9.00	149
0.075	13.00	215
0.100	15.00	248
0.150	17.00	281
0.200	18.00	298
0.250	18.50	306
0.300	19.00	314



Bearing Ratio (%) @ 0.1" Penetration: **24.8**
Bearing Ratio (%) @ 0.2" Penetration: **19.9**
CBR (%): At 0.2" **20**

REMARKS:

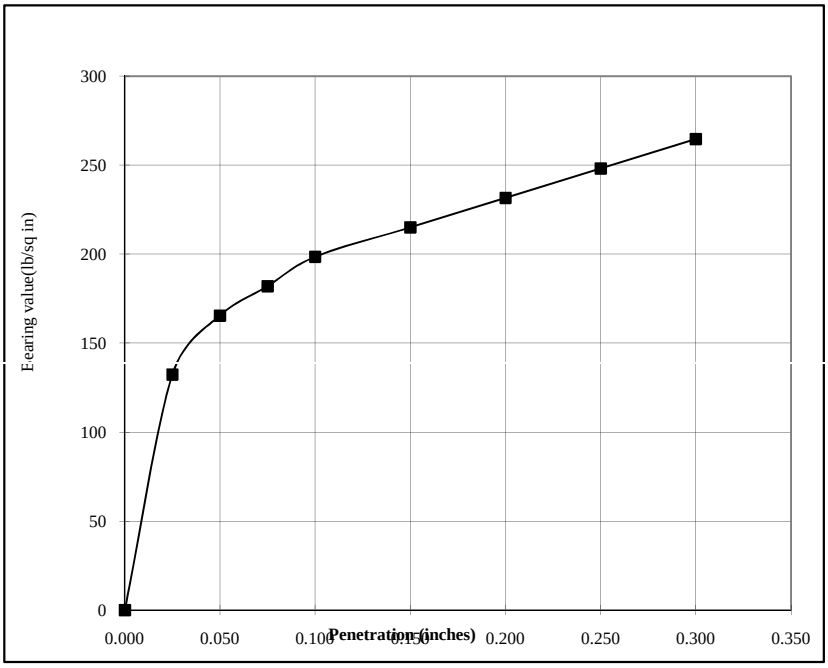
Test has been conducted after removal of 15cm material from surface level

TESTED BY **SUPERVISOR**
Mushtaq M.Tahir

FIELD CALIFORNIA BEARING RATIO TEST(ASTM-D4429)

Project:	**Mansourah Massarah Gold Project**	**TEST DATE :**	**15-Sep-19**
Client:	**L & T Construction**		
Coordinate :	**N-2505980.974, E-263787.176**	**PROJECT # :**	**19-K-3054**
Location	**Ar Rjum**	**TEST PIT NO :**	**FCBR-2**

PENET. (IN INCHES)	DIAL READING	BEARING VALUE (Ib/sq in.)
0.000	0.00	0
0.025	8.00	132
0.050	10.00	165
0.075	11.00	182
0.100	12.00	199
0.150	13.00	215
0.200	14.00	232
0.250	15.00	248
0.300	16.00	265



Bearing Ratio (%) @ 0.1" Penetration: **19.9**
Bearing Ratio (%) @ 0.2" Penetration: **15.4**
CBR (%): At 0.2" **15**

REMARKS:

Test has been conducted after removal of 15cm material from surface level

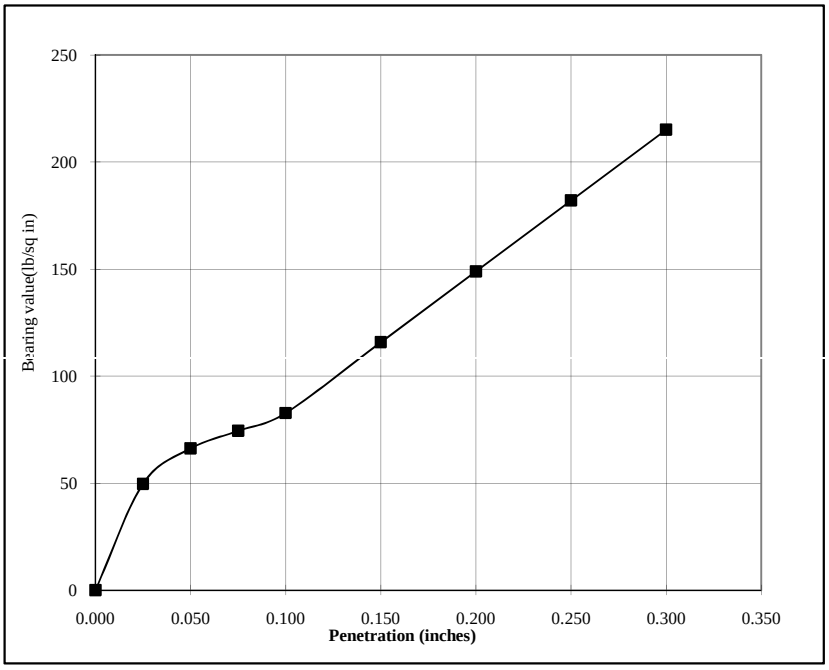
TESTED BY
Mushtaq

SUPERVISOR
M.Tahir

FIELD CALIFORNIA BEARING RATIO TEST(ASTM-D4429)

Project:	**Mansourah Massarah Gold Project**	**TEST DATE :**	**15-Sep-19**
Client:	**L & T Construction**		
Coordinate :	**N-2506131.287, E-263858.582**	**PROJECT # :**	**19-K-3054**
Location	**Ar Rjum**	**TEST PIT NO :**	**FCBR-3**

PENET. (IN INCHES)	DIAL READING	BEARING VALUE (Ib/sq in.)
0.000	0.00	0
0.025	3.00	50
0.050	4.00	66
0.075	4.50	74
0.100	5.00	83
0.150	7.00	116
0.200	9.00	149
0.250	11.00	182
0.300	13.00	215



Bearing Ratio (%) @ 0.1" Penetration: **8.3**
Bearing Ratio (%) @ 0.2" Penetration: **9.9**
CBR (%): At 0.2" **10**

REMARKS:

Test has been conducted after removal of 15cm material from surface level

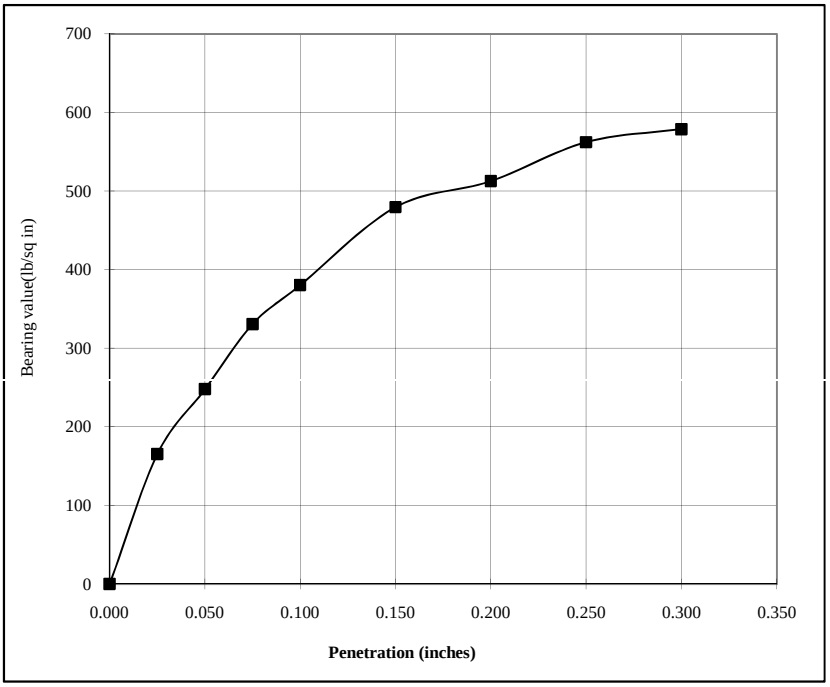
TESTED BY
Mushtaq

SUPERVISOR
M.Tahir

FIELD CALIFORNIA BEARING RATIO TEST(ASTM-D4429)

Project:	**Mansourah Massarah Gold Project**	**TEST DATE :**	**15-Sep-19**
Client:	**L & T Construction**		
Coordinate :	**N-2505648.408, E-264049.895**	**PROJECT # :**	**19-K-3054**
Location	**Ar Rjum**	**TEST PIT NO :**	**FCBR-04**

PENET. (IN INCHES)	DIAL READING	BEARING VALUE (Ib/sq in.)
0.000	0.00	0
0.025	10.00	165
0.050	15.00	248
0.075	20.00	331
0.100	23.00	381
0.150	29.00	480
0.200	31.00	513
0.250	34.00	563
0.300	35.00	579



Bearing Ratio (%) @ 0.1" Penetration: **38.1**
Bearing Ratio (%) @ 0.2" Penetration: **34.2**
CBR (%): At 0.2" **34**

REMARKS:

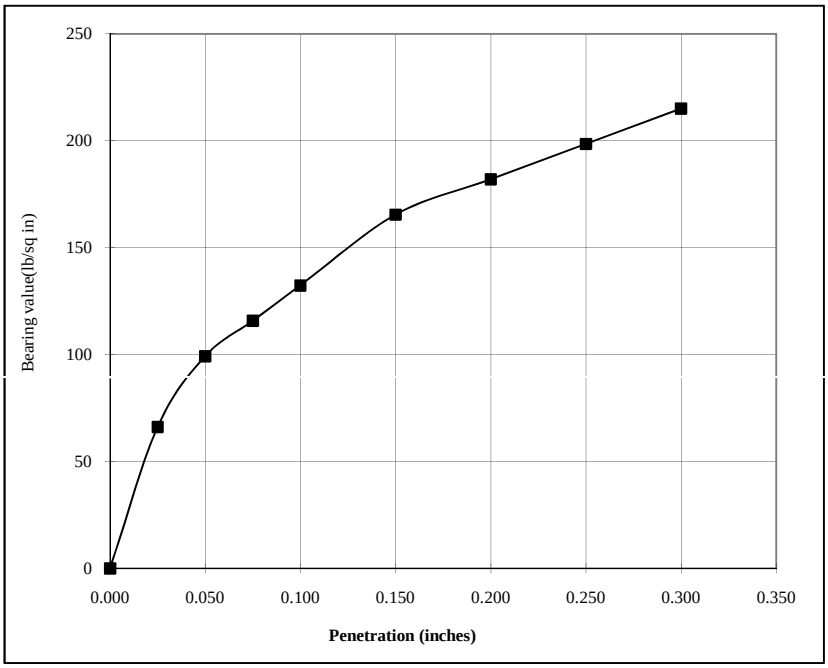
Test has been conducted after removal of 15cm material from surface level

TESTED BY **SUPERVISOR**
Mushtaq M.Tahir

FIELD CALIFORNIA BEARING RATIO TEST(ASTM-D4429)

Project:	**Mansourah Massarah Gold Project**	**TEST DATE :**	**15-Sep-19**
Client:	**L & T Construction**		
Coordinate :	**N-2505810.594, E-264116.104**	**PROJECT # :**	**19-K-3054**
Location	**Ar Rjum**	**TEST PIT NO :**	**FCBR-5**

PENET. (IN INCHES)	DIAL READING	BEARING VALUE (Ib/sq in.)
0.000	0.00	0
0.025	4.00	66
0.050	6.00	99
0.075	7.00	116
0.100	8.00	132
0.150	10.00	165
0.200	11.00	182
0.250	12.00	199
0.300	13.00	215



Bearing Ratio (%) @ 0.1" Penetration: **13.2**
Bearing Ratio (%) @ 0.2" Penetration: **12.1**
CBR (%): At 0.2" **12**

REMARKS:

Test has been conducted after removal of 15cm material from surface level

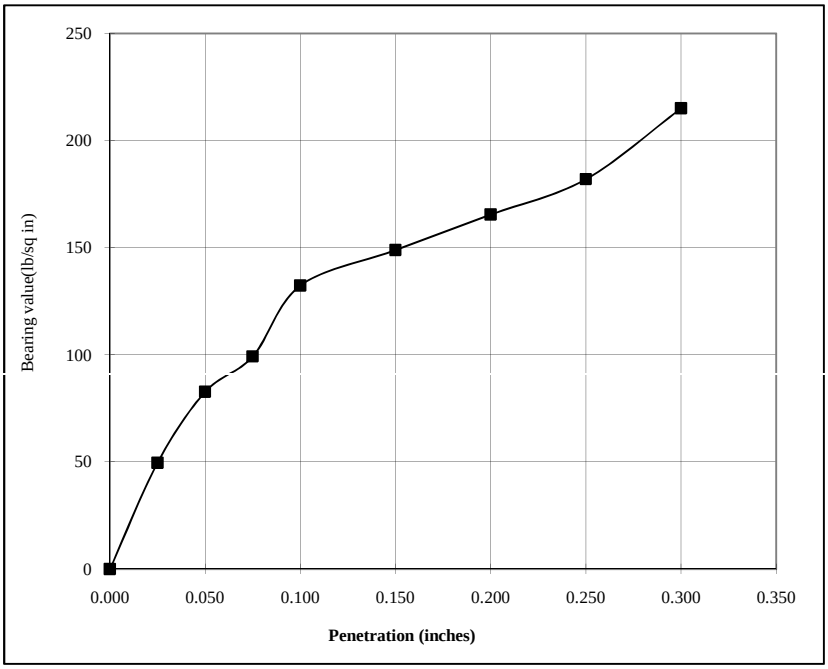
TESTED BY
Mushtaq

SUPERVISOR
M.Tahir

FIELD CALIFORNIA BEARING RATIO TEST(ASTM-D4429)

Project:	**Mansourah Massarah Gold Project**	**TEST DATE :**	**15-Sep-19**
Client:	**L & T Construction**		
Coordinate :	**N-2506156.333, E-264018.629**	**PROJECT # :**	**19-K-3054**
Location	**Ar Rjum**	**TEST PIT NO :**	**FCBR-06**

PENET. (IN INCHES)	DIAL READING	BEARING VALUE (Ib/sq in.)
0.000	0.00	0
0.025	3.00	50
0.050	5.00	83
0.075	6.00	99
0.100	8.00	132
0.150	9.00	149
0.200	10.00	165
0.250	11.00	182
0.300	13.00	215



Bearing Ratio (%) @ 0.1" Penetration: **13.2**
Bearing Ratio (%) @ 0.2" Penetration: **11.0**
CBR (%): At 0.2" **11**

REMARKS:

Test has been conducted after removal of 15cm material from surface level

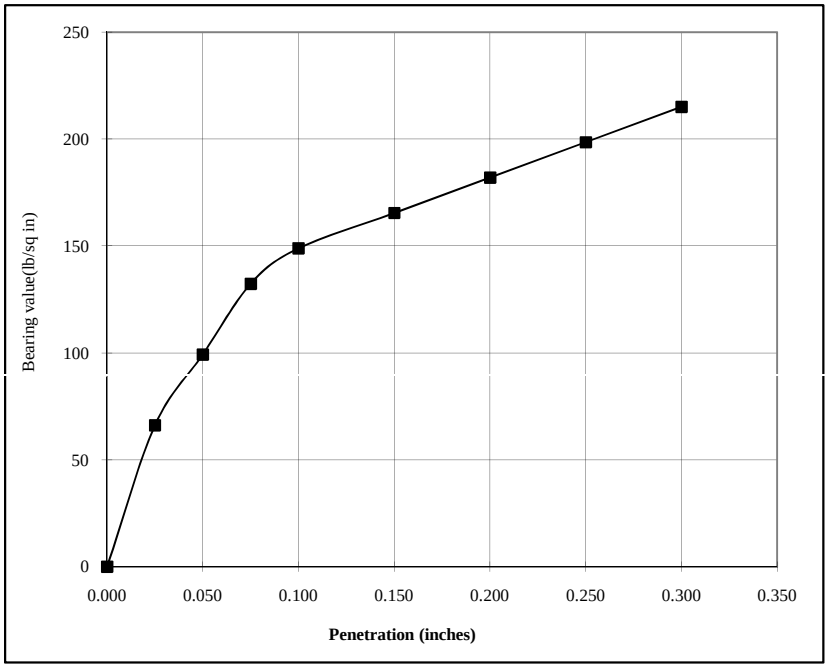
TESTED BY
Mushtaq

SUPERVISOR
M.Tahir

FIELD CALIFORNIA BEARING RATIO TEST(ASTM-D4429)

Project:	**Mansourah Massarah Gold Project**	**TEST DATE :**	**15-Sep-19**
Client:	**L & T Construction**		
Coordinate :	**N-2506040.884, E-264179.207**	**PROJECT # :**	**19-K-3054**
Location	**Ar Rjum**	**TEST PIT NO :**	**FCBR-07**

PENET. (IN INCHES)	DIAL READING	BEARING VALUE (Ib/sq in.)
0.000	0.00	0
0.025	4.00	66
0.050	6.00	99
0.075	8.00	132
0.100	9.00	149
0.150	10.00	165
0.200	11.00	182
0.250	12.00	199
0.300	13.00	215



Bearing Ratio (%) @ 0.1" Penetration: **14.9**
Bearing Ratio (%) @ 0.2" Penetration: **12.1**
CBR (%): At 0.2" **12**

REMARKS:

Test has been conducted after removal of 15cm material from surface level

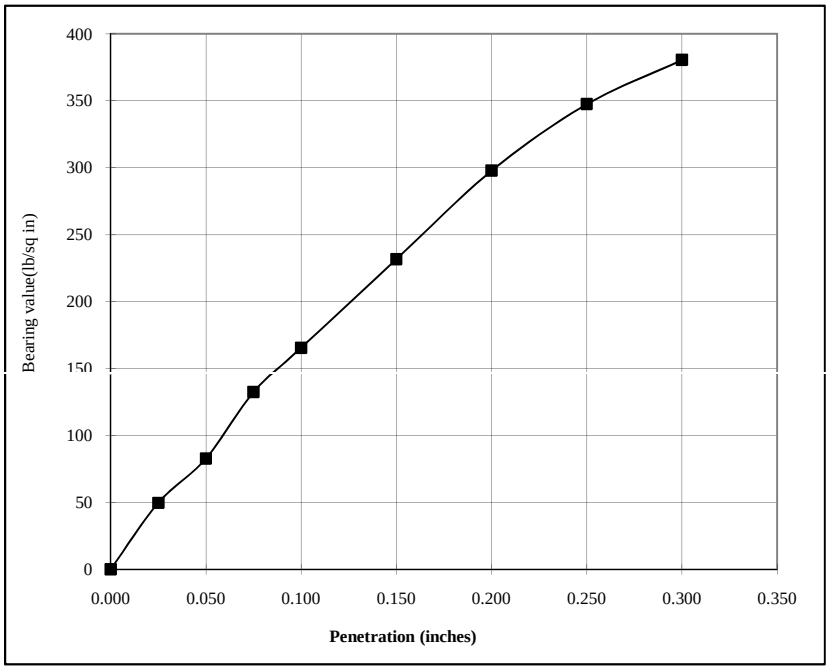
TESTED BY
Mushtaq

SUPERVISOR
M.Tahir

FIELD CALIFORNIA BEARING RATIO TEST(ASTM-D4429)

Project:	**Mansourah Massarah Gold Project**	**TEST DATE :**	**15-Sep-19**
Client:	**L & T Construction**		
Coordinate :	**N-2505857.328, E-264393.021**	**PROJECT # :**	**19-K-3054**
Location	**Ar Rjum**	**TEST PIT NO :**	**FCBR-08**

PENET. (IN INCHES)	DIAL READING	BEARING VALUE (Ib/sq in.)
0.000	0.00	0
0.025	3.00	50
0.050	5.00	83
0.075	8.00	132
0.100	10.00	165
0.150	14.00	232
0.200	18.00	298
0.250	21.00	347
0.300	23.00	381



Bearing Ratio (%) @ 0.1" Penetration: **16.5**
Bearing Ratio (%) @ 0.2" Penetration: **19.9**
CBR (%): At 0.2" **20**

REMARKS:

Test has been conducted after removal of 15cm material from surface level

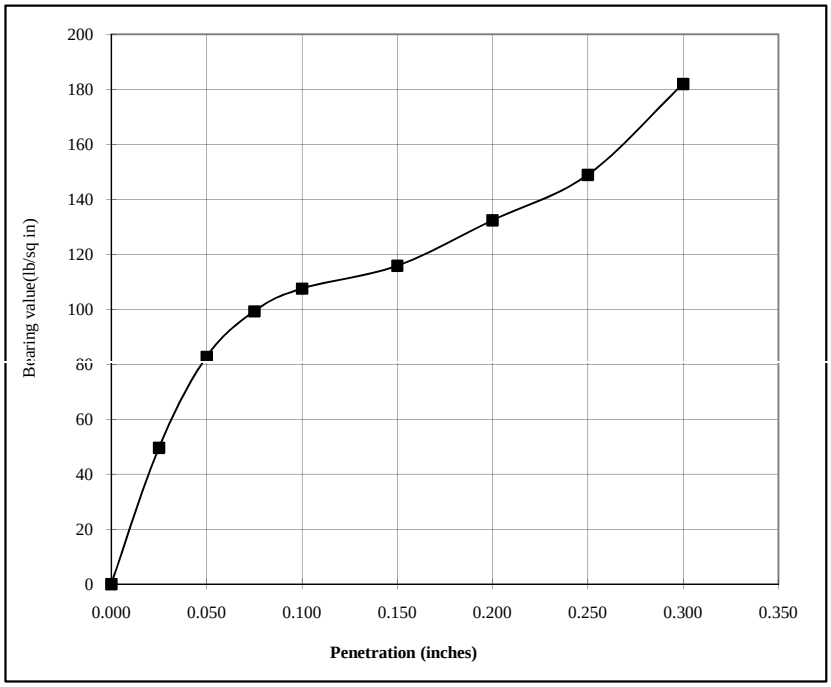
TESTED BY
Mushtaq

SUPERVISOR
M.Tahir

FIELD CALIFORNIA BEARING RATIO TEST(ASTM-D4429)

Project:	**Mansourah Massarah Gold Project**	**TEST DATE :**	**15-Sep-19**
Client:	**L & T Construction**		
Coordinate :	**N-2506165.392, E-264522.509**	**PROJECT # :**	**19-K-3054**
Location	**Ar Rjum**	**TEST PIT NO :**	**FCBR-9**

PENET. (IN INCHES)	DIAL READING	BEARING VALUE (Ib/sq in.)
0.000	0.00	0
0.025	3.00	50
0.050	5.00	83
0.075	6.00	99
0.100	6.50	108
0.150	7.00	116
0.200	8.00	132
0.250	9.00	149
0.300	11.00	182



Bearing Ratio (%) @ 0.1" Penetration: **10.8**
Bearing Ratio (%) @ 0.2" Penetration: **8.8**
CBR (%): At 0.2" **9**

REMARKS:

Test has been conducted after removal of 15cm material from surface level

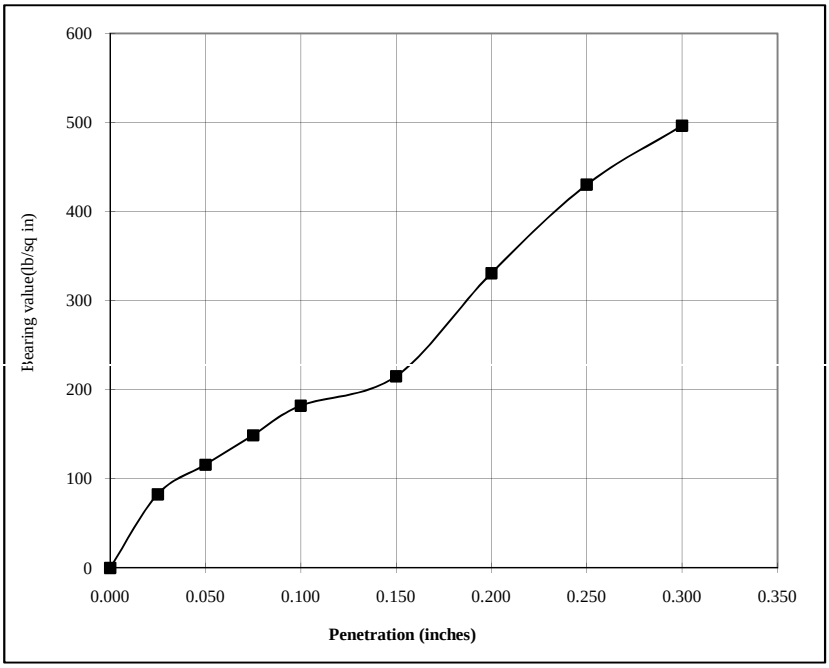
TESTED BY
Mushtaq

SUPERVISOR
M.Tahir

FIELD CALIFORNIA BEARING RATIO TEST(ASTM-D4429)

Project:	**Mansourah Massarah Gold Project**	**TEST DATE :**	**15-Sep-19**
Client:	**L & T Construction**		
Coordinate :	**N-2506280.463, E-264309.592**	**PROJECT # :**	**19-K-3054**
Location	**Ar Rjum**	**TEST PIT NO :**	**FCBR-10**

PENET. (IN INCHES)	DIAL READING	BEARING VALUE (Ib/sq in.)
0.000	0.00	0
0.025	5.00	83
0.050	7.00	116
0.075	9.00	149
0.100	11.00	182
0.150	13.00	215
0.200	20.00	331
0.250	26.00	430
0.300	30.00	496



Bearing Ratio (%) @ 0.1" Penetration: **18.2**
Bearing Ratio (%) @ 0.2" Penetration: **22.1**
CBR (%): At 0.2" **22**

REMARKS:

Test has been conducted after removal of 15cm material from surface level

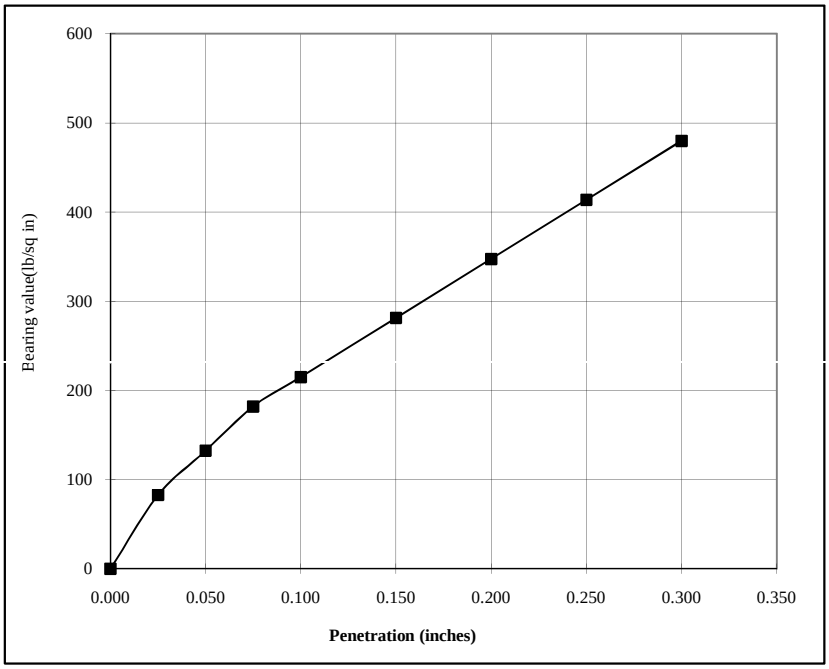
TESTED BY
Mushtaq

SUPERVISOR
M.Tahir

FIELD CALIFORNIA BEARING RATIO TEST(ASTM-D4429)

Project:	**Mansourah Massarah Gold Project**	**TEST DATE :**	**15-Sep-19**
Client:	**L & T Construction**		
Coordinate :	**N-2505969.066, E-264358.264**	**PROJECT # :**	**19-K-3054**
Location	**Ar Rjum**	**TEST PIT NO :**	**FCBR-11**

PENET. (IN INCHES)	DIAL READING	BEARING VALUE (Ib/sq in.)
0.000	0.00	0
0.025	5.00	83
0.050	8.00	132
0.075	11.00	182
0.100	13.00	215
0.150	17.00	281
0.200	21.00	347
0.250	25.00	414
0.300	29.00	480



Bearing Ratio (%) @ 0.1" Penetration: **21.5**
Bearing Ratio (%) @ 0.2" Penetration: **23.2**
CBR (%): At 0.2" **23**

REMARKS:

Test has been conducted after removal of 15cm material from surface level

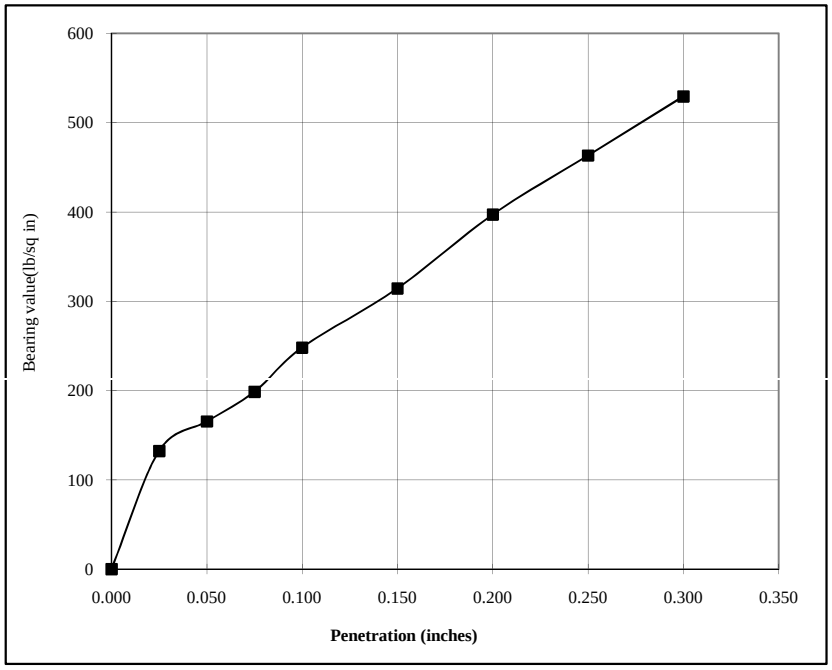
TESTED BY
Mushtaq

SUPERVISOR
M.Tahir

FIELD CALIFORNIA BEARING RATIO TEST(ASTM-D4429)

Project:	**Mansourah Massarah Gold Project**	**TEST DATE :**	**15-Sep-19**
Client:	**L & T Construction**		
Coordinate :	**N-2506551.675, E-264125.477**	**PROJECT # :**	**19-K-3054**
Location	**Ar Rjum**	**TEST PIT NO :**	**FCBR-12**

PENET. (IN INCHES)	DIAL READING	BEARING VALUE (Ib/sq in.)
0.000	0.00	0
0.025	8.00	132
0.050	10.00	165
0.075	12.00	199
0.100	15.00	248
0.150	19.00	314
0.200	24.00	397
0.250	28.00	463
0.300	32.00	529



Bearing Ratio (%) @ 0.1" Penetration: **24.8**
Bearing Ratio (%) @ 0.2" Penetration: **26.5**
CBR (%): At 0.2" **26**

REMARKS:

Test has been conducted after removal of 15cm material from surface level

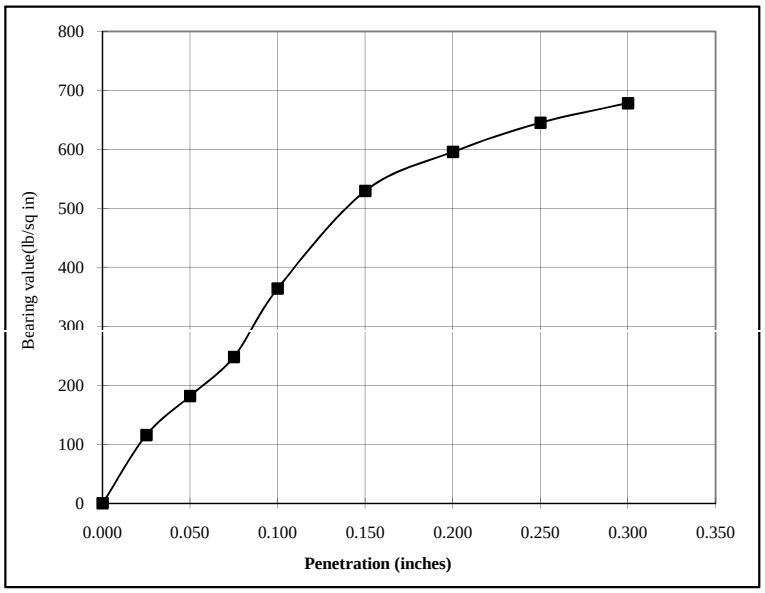
TESTED BY
Mushtaq

SUPERVISOR
M.Tahir

FIELD CALIFORNIA BEARING RATIO TEST(ASTM-D4429)

Project:	**Mansourah Massarah Gold Project**	**TEST DATE :**	**15-Sep-19**
Client:	**L & T Construction**		
Coordinate :	**N-2508722.706, E-263203.218**	**PROJECT # :**	**19-K-3054**
Location	**Ar Rjum**	**TEST PIT NO :**	**FCBR-13**

PENET. (IN INCHES)	DIAL READING	BEARING VALUE (Ib/sq in.)
0.000	0.00	0
0.025	7.00	116
0.050	11.00	182
0.075	15.00	248
0.100	22.00	364
0.150	32.00	529
0.200	36.00	596
0.250	39.00	645
0.300	41.00	678



Bearing Ratio (%) @ 0.1" Penetration: **36.4**
Bearing Ratio (%) @ 0.2" Penetration: **39.7**
CBR (%): At 0.2" **40**

REMARKS:

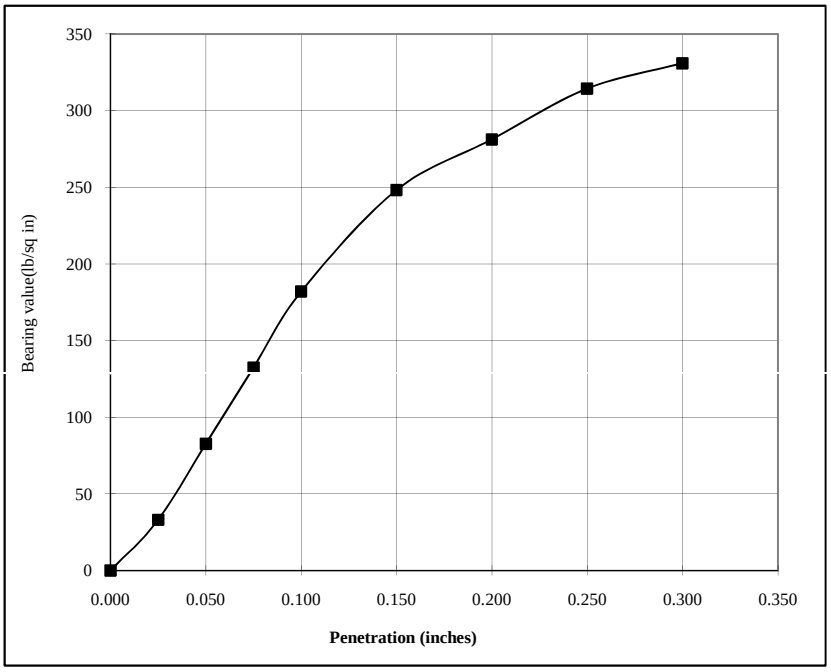
Test has been conducted after removal of 15cm material from surface level

TESTED BY **SUPERVISOR**
Mushtaq M.Tahir

FIELD CALIFORNIA BEARING RATIO TEST(ASTM-D4429)

Project:	**Mansourah Massarah Gold Project**	**TEST DATE :**	**15-Sep-19**
Client:	**L & T Construction**		
Coordinate :	**N-2508063.024, E-263118.687**	**PROJECT # :**	**19-K-3054**
Location	**Ar Rjum**	**TEST PIT NO :**	**FCBR-14**

PENET. (IN INCHES)	DIAL READING	BEARING VALUE (Ib/sq in.)
0.000	0.00	0
0.025	2.00	33
0.050	5.00	83
0.075	8.00	132
0.100	11.00	182
0.150	15.00	248
0.200	17.00	281
0.250	19.00	314
0.300	20.00	331



Bearing Ratio (%) @ 0.1" Penetration: **18.2**
Bearing Ratio (%) @ 0.2" Penetration: **18.8**
CBR (%): At 0.2" **19**

REMARKS:

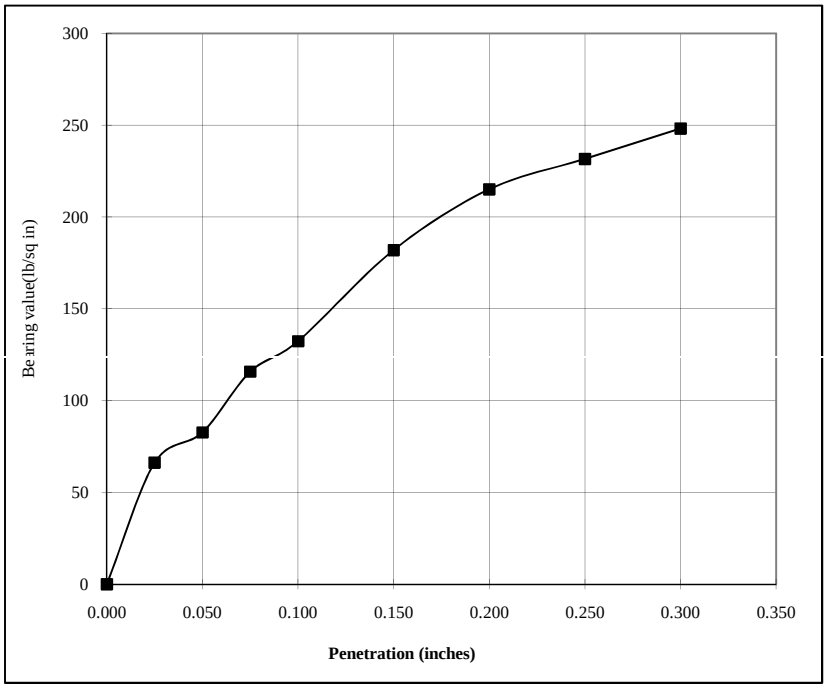
Test has been conducted after removal of 15cm material from surface level

TESTED BY **SUPERVISOR**
Mushtaq M.Tahir

FIELD CALIFORNIA BEARING RATIO TEST(ASTM-D4429)

Project:	**Mansourah Massarah Gold Project**	**TEST DATE :**	**15-Sep-19**
Client:	**L & T Construction**		
Coordinate :	**N-2509004.189, E-265854.978**	**PROJECT # :**	**19-K-3054**
Location	**Ar Rjum**	**TEST PIT NO :**	**FCBR-15**

PENET. (IN INCHES)	DIAL READING	BEARING VALUE (Ib/sq in.)
0.000	0.00	0
0.025	4.00	66
0.050	5.00	83
0.075	7.00	116
0.100	8.00	132
0.150	11.00	182
0.200	13.00	215
0.250	14.00	232
0.300	15.00	248



Bearing Ratio (%) @ 0.1" Penetration: **13.2**
Bearing Ratio (%) @ 0.2" Penetration: **14.3**
CBR (%): At 0.2" **14**

REMARKS:

Test has been conducted after removal of 15cm material from surface level

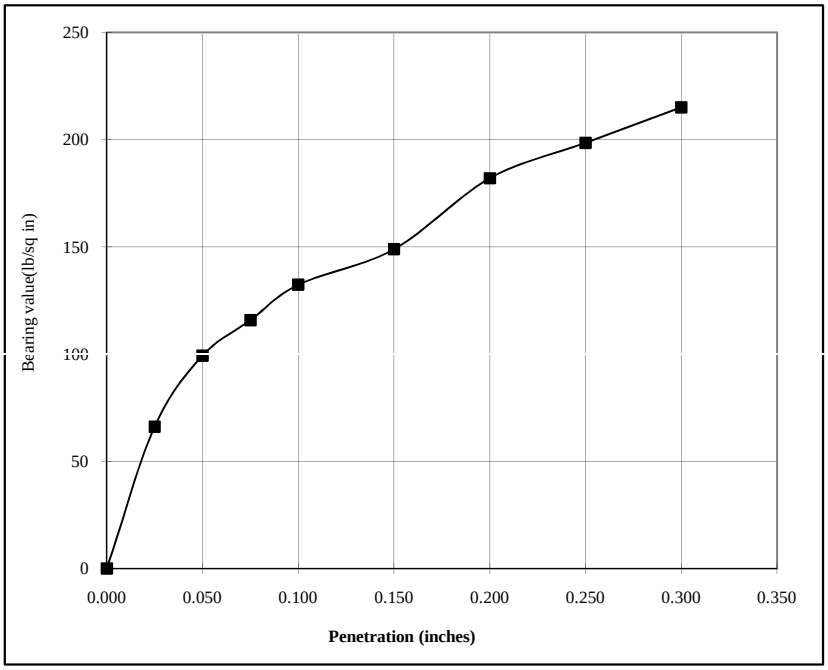
TESTED BY
Mushtaq

SUPERVISOR
M.Tahir

FIELD CALIFORNIA BEARING RATIO TEST(ASTM-D4429)

Project:	**Mansourah Massarah Gold Project**	**TEST DATE :**	**15-Sep-19**
Client:	**L & T Construction**		
Coordinate :	**N-2509675.708, E-265300.441**	**PROJECT # :**	**19-K-3054**
Location	**Ar Rjum**	**TEST PIT NO :**	**FCBR-16**

PENET. (IN INCHES)	DIAL READING	BEARING VALUE (Ib/sq in.)
0.000	0.00	0
0.025	4.00	66
0.050	6.00	99
0.075	7.00	116
0.100	8.00	132
0.150	9.00	149
0.200	11.00	182
0.250	12.00	199
0.300	13.00	215



Bearing Ratio (%) @ 0.1" Penetration: **13.2**
Bearing Ratio (%) @ 0.2" Penetration: **12.1**
CBR (%): At 0.2" **12**

REMARKS:

Test has been conducted after removal of 15cm material from surface level

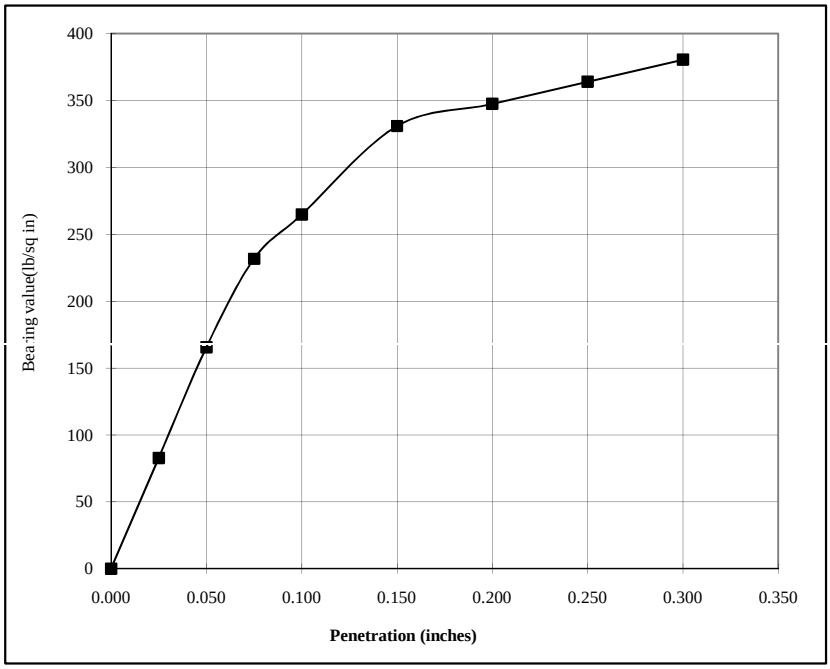
TESTED BY
Mushtaq

SUPERVISOR
M.Tahir

FIELD CALIFORNIA BEARING RATIO TEST(ASTM-D4429)

Project:	**Mansourah Massarah Gold Project**	**TEST DATE :**	**15-Sep-19**
Client:	**L & T Construction**		
Coordinate :	**N-2504718.247, E-264445.364**	**PROJECT # :**	**19-K-3054**
Location	**Ar Rjum**	**TEST PIT NO :**	**FCBR-17**

PENET. (IN INCHES)	DIAL READING	BEARING VALUE (Ib/sq in.)
0.000	0.00	0
0.025	5.00	83
0.050	10.00	165
0.075	14.00	232
0.100	16.00	265
0.150	20.00	331
0.200	21.00	347
0.250	22.00	364
0.300	23.00	381



Bearing Ratio (%) @ 0.1" Penetration: **26.5**
Bearing Ratio (%) @ 0.2" Penetration: **23.2**
CBR (%): At 0.2" **23**

REMARKS:

Test has been conducted after removal of 15cm material from surface level

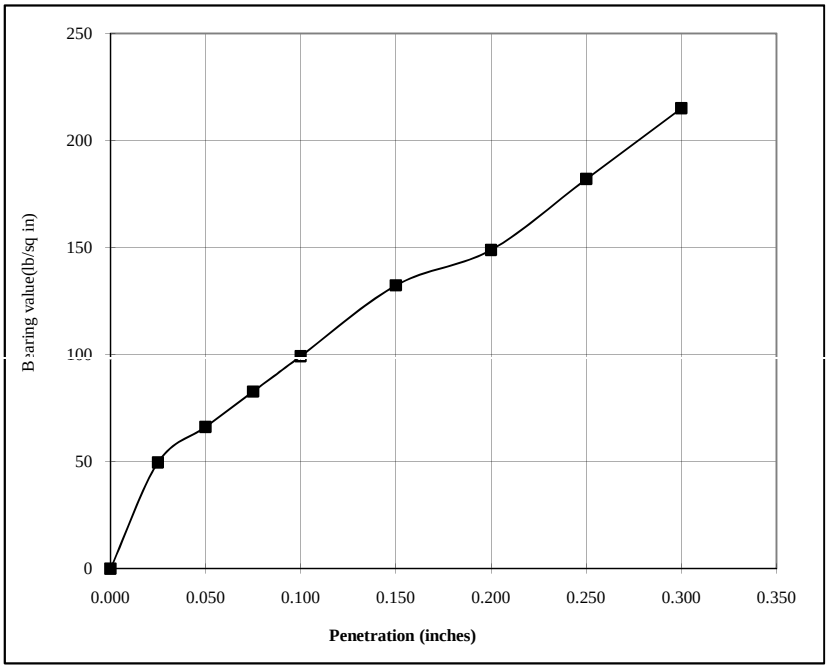
TESTED BY
Mushtaq

SUPERVISOR
M.Tahir

FIELD CALIFORNIA BEARING RATIO TEST(ASTM-D4429)

Project:	**Mansourah Massarah Gold Project**	**TEST DATE :**	**15-Sep-19**
Client:	**L & T Construction**		
Coordinate :	**N-2509171.635, E-261859.561**	**PROJECT # :**	**19-K-3054**
Location	**Ar Rjum**	**TEST PIT NO :**	**FCBR-19**

PENET. (IN INCHES)	DIAL READING	BEARING VALUE (Ib/sq in.)
0.000	0.00	0
0.025	3.00	50
0.050	4.00	66
0.075	5.00	83
0.100	6.00	99
0.150	8.00	132
0.200	9.00	149
0.250	11.00	182
0.300	13.00	215



Bearing Ratio (%) @ 0.1" Penetration: **9.9**
Bearing Ratio (%) @ 0.2" Penetration: **9.9**
CBR (%): At 0.2" **10**

REMARKS:

Test has been conducted after removal of 15cm material from surface level

TESTED BY
Mushtaq

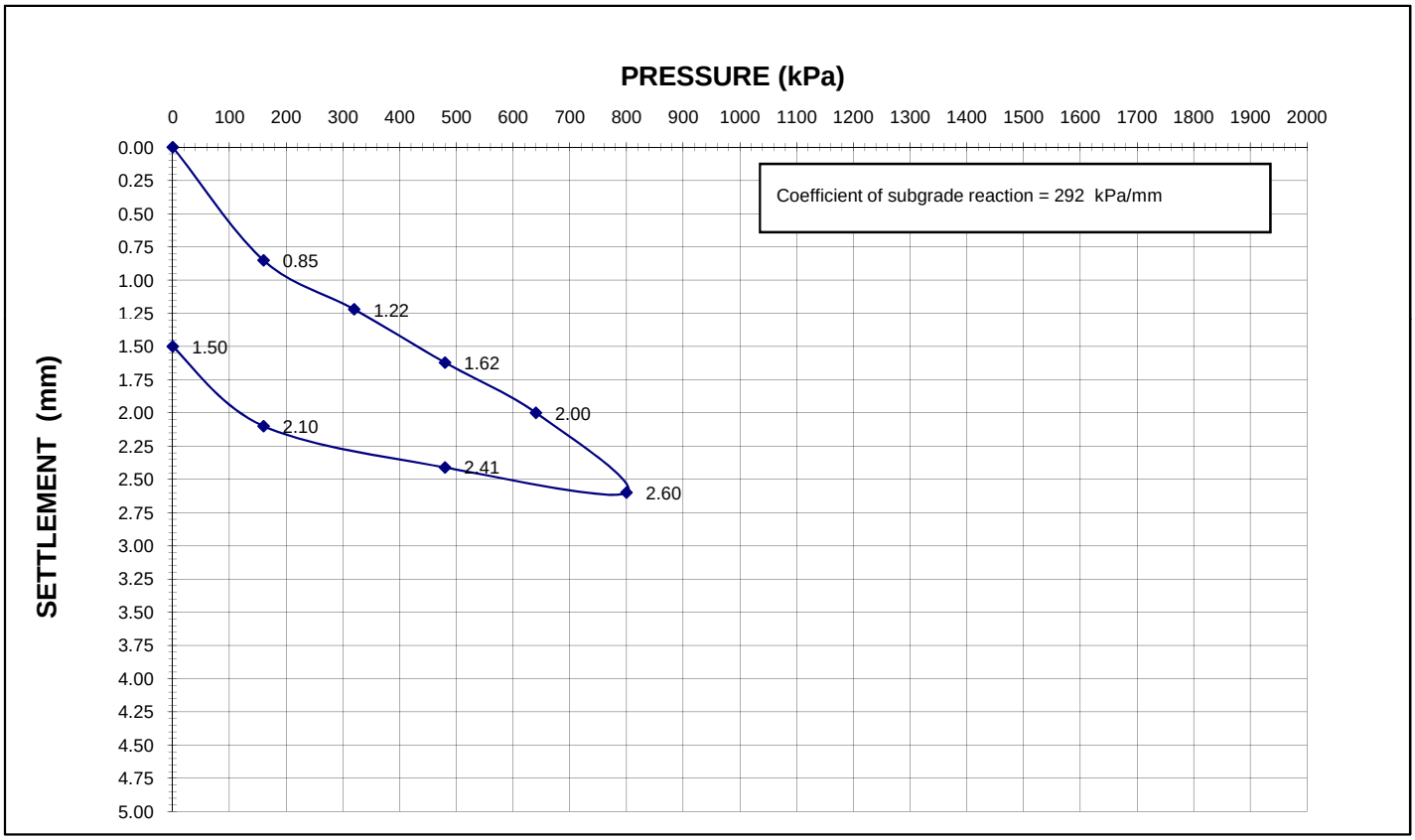
SUPERVISOR
M.Tahir

PLATE LOAD TEST

PLATE LOAD TEST (ASTM D 1196)

PROJECT:	**MMGP**	**LOAD TEST #**	**PLT-1**
CLIENT :	**L & T**	**TEST DATE:**	**12/9/2019**
		WEATHER:	**FAIR**
TEST PLATE DIA (cm	**30**	**TEST PLATE AREA (m^2) =**	**0.0707**
PROJECT :	**19-k-3054**		
Existing Grade E1	**Northing (m)**	**2505956.533**	**Easting (m)**
264115.134			

STAGE	**LOADING**							**UNLOADING**						
PRESSURE (KPa)	0	160	320	480	640	800	480	160	0					
SETTLE MENT (mm)	0	0.85	1.22	1.62	2.00	2.60	2.41	2.10	1.50					



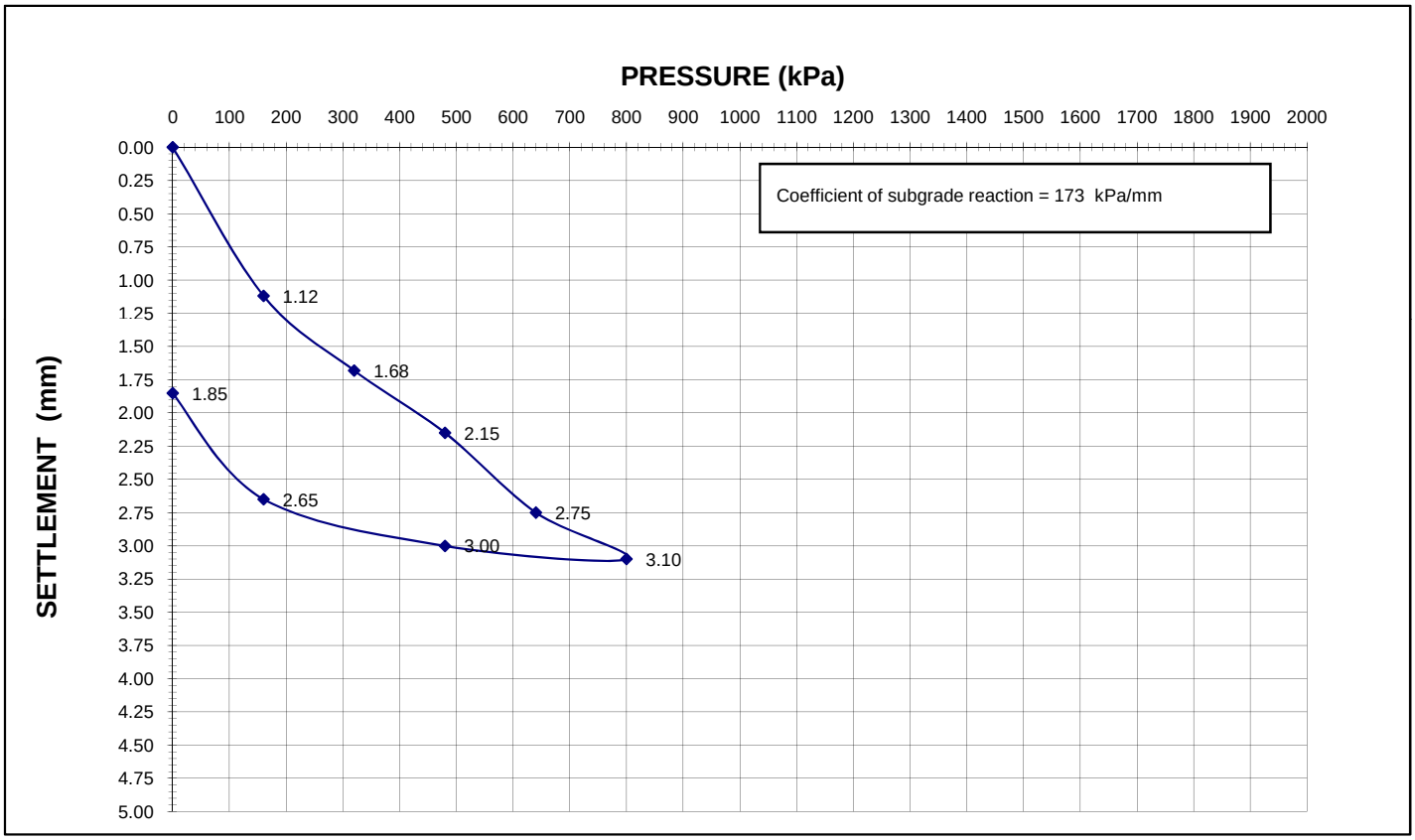
TESTED BY : Abdal Haque

CHECKED BY M.TAHIR

PLATE LOAD TEST (ASTM D 1196)

PROJECT:	**MMGP**	**LOAD TEST #**	**PLT-2**
CLIENT :	**L & T**	**TEST DATE:**	**12/9/2019**
		WEATHER:	**FAIR**
TEST PLATE DIA (cm	**30**	**TEST PLATE AREA (m^2) =**	**0.0707**
PROJECT :	**19-k-3054**		
Existing Grade E1		**Northing (m)**	**2505962.6**
Easting (m)	**264303.475**		

STAGE	**LOADING**							**UNLOADING**						
PRESSURE (KPa)	0	160	320	480	640	800	480	160	0					
SETTLE MENT (mm)	0	1.12	1.68	2.15	2.75	3.10	3.00	2.65	1.85					



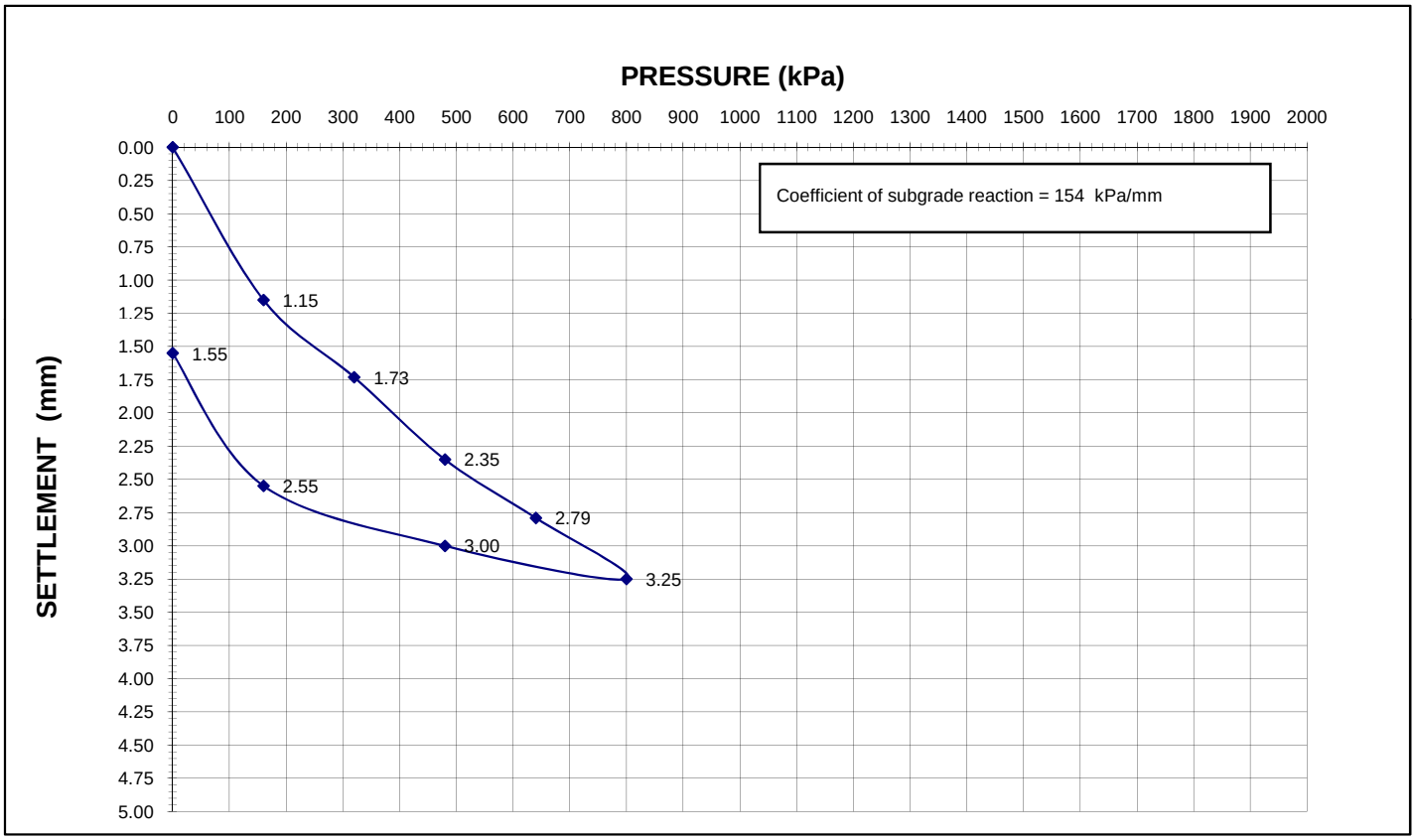
TESTED BY : Abdal Haque

CHECKED BY M.TAHIR

PLATE LOAD TEST (ASTM D 1196)

PROJECT:	**MMGP**	**LOAD TEST #**	**PLT-3**
CLIENT :	**L & T**	**TEST DATE:**	**12/9/2019**
		WEATHER:	**FAIR**
TEST PLATE DIA (cm	**30**	**TEST PLATE AREA (m^2) =**	**0.0707**
PROJECT :	**19-k-3054**		
Existing Grade E1		**Northing (m)**	**2508417.463**
Easting (m)	**263018.466**		

STAGE	**LOADING**							**UNLOADING**						
PRESSURE (KPa)	0	160	320	480	640	800	480	160	0					
SETTLE MENT (mm)	0	1.15	1.73	2.35	2.79	3.25	3.00	2.55	1.55					



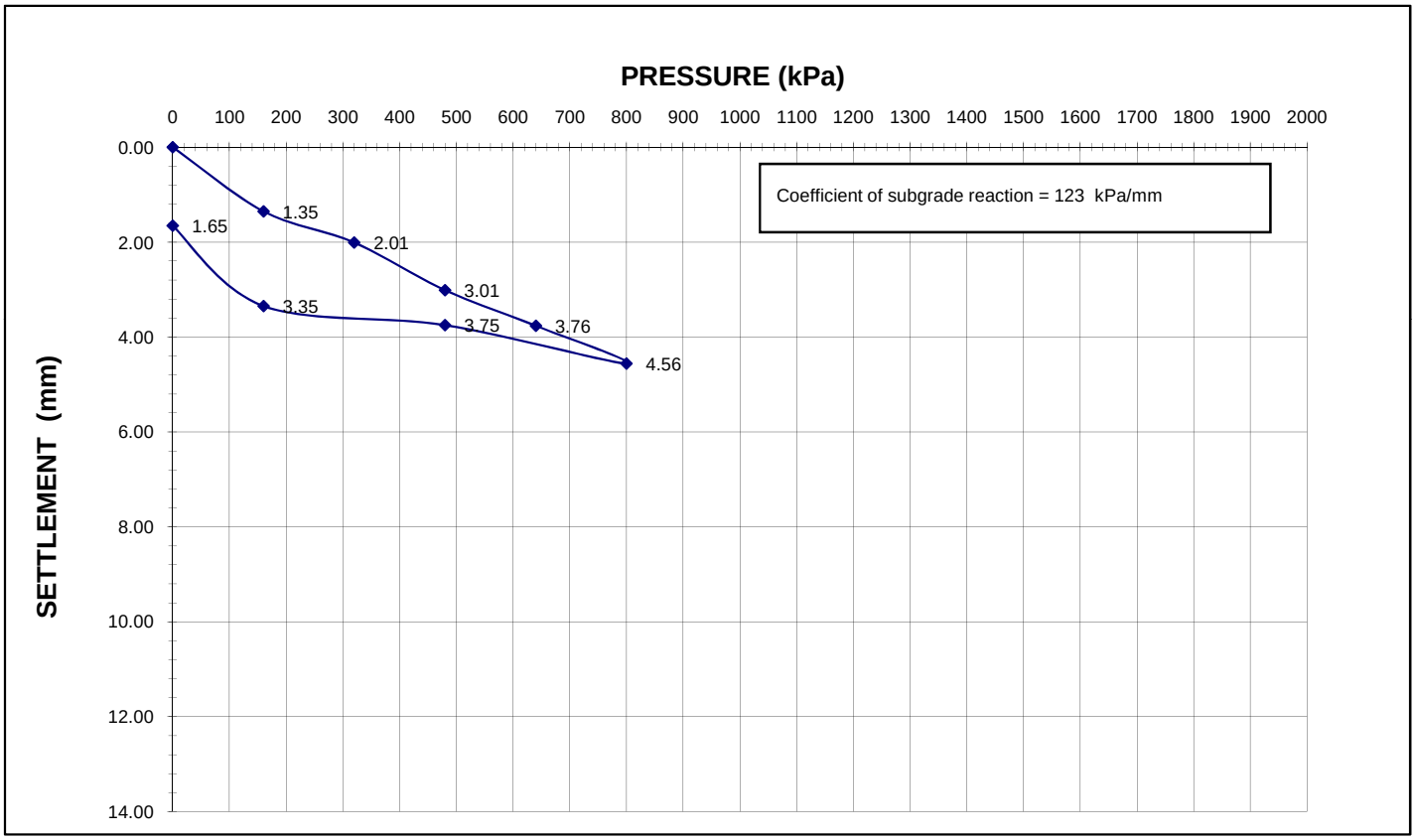
TESTED BY : Abdal Haque

CHECKED BY M.TAHIR

PLATE LOAD TEST (ASTM D 1196)

PROJECT:	**MMGP**	**LOAD TEST #**	**PLT-4**
CLIENT :	**L & T**	**TEST DATE:**	**17/9/2019**
		WEATHER:	**FAIR**
TEST PLATE DIA (cm	**30**	**TEST PLATE AREA (m^2) =**	**0.0707**
PROJECT :	**19-k-3054**		
Existing Grade E1		**Northing (m)**	**2504043.156**
Easting (m)	**248605.515**		

STAGE	**LOADING**							**UNLOADING**						
PRESSURE (KPa)	0	160	320	480	640	800	480	160	0					
SETTLE MENT (mm)	0	1.35	2.01	3.01	3.76	4.56	3.75	3.35	1.65					



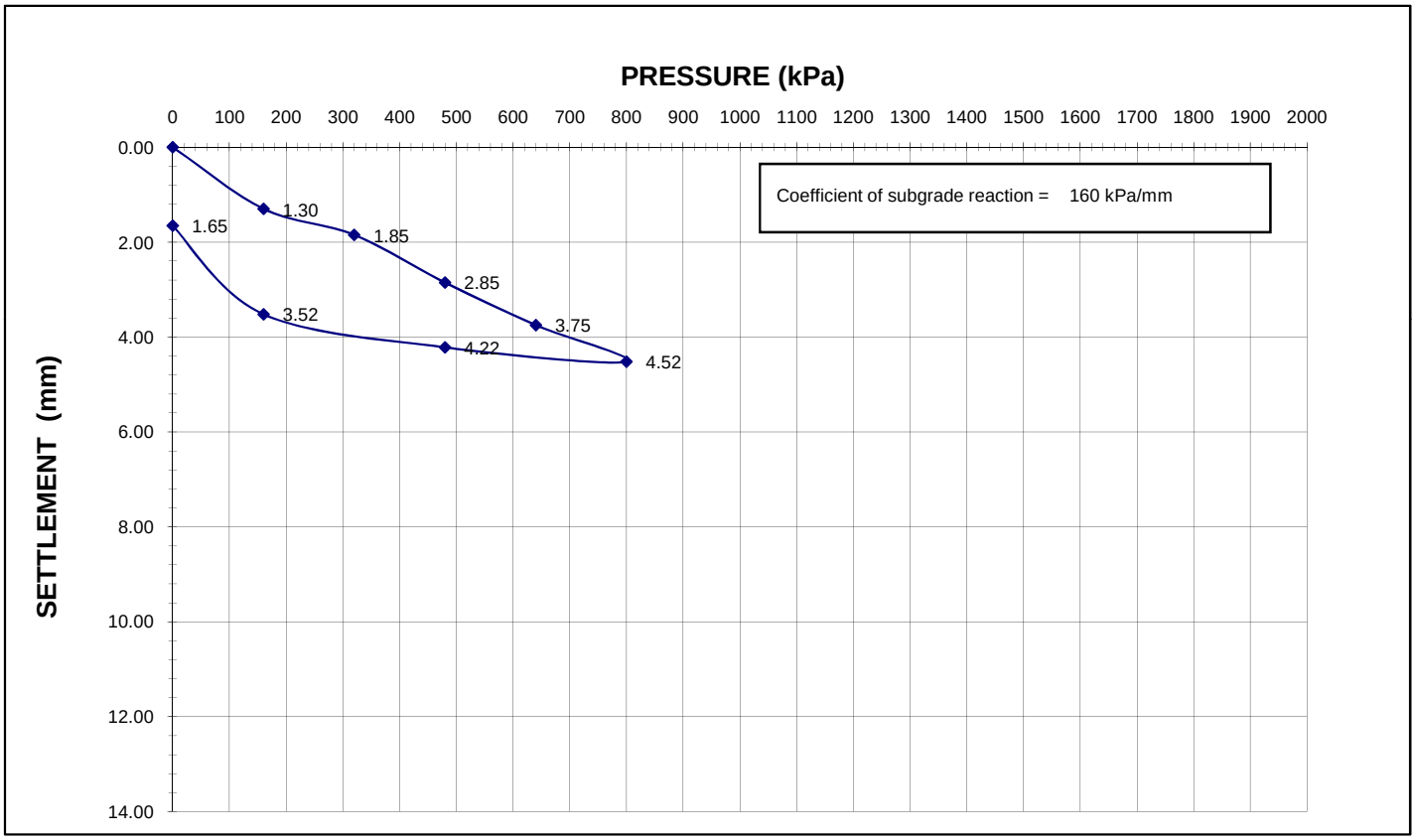
TESTED BY : Abdal Haque

CHECKED BY M.TAHIR

PLATE LOAD TEST (ASTM D 1196)

PROJECT:	**MMGP**	**LOAD TEST #**	**PLT-5**
CLIENT :	**L & T**	**TEST DATE:**	**12/9/2019**
		WEATHER:	**FAIR**
TEST PLATE DIA (cm	**30**	**TEST PLATE AREA (m^2) =**	**0.0707**
PROJECT :	**19-k-3054**		
Existing Grade E1		**Northing (m)**	**2519120.662**
Easting (m)	**264915.08**		

STAGE	**LOADING**							**UNLOADING**						
PRESSURE (KPa)	0	160	320	480	640	800	480	160	0					
SETTLE MENT (mm)	0	1.30	1.85	2.85	3.75	4.52	4.22	3.52	1.65					



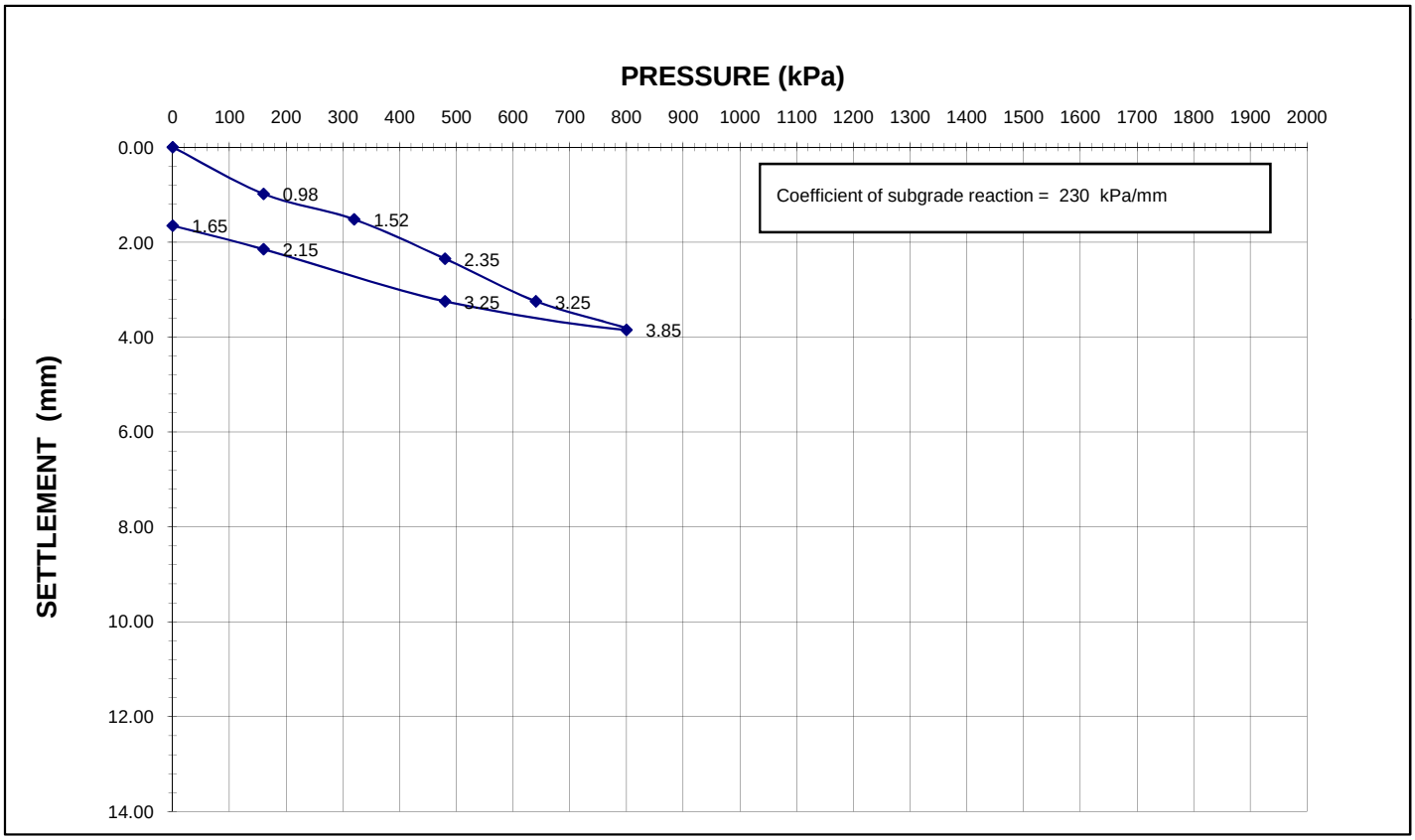
TESTED BY : Abdal Haque

CHECKED BY M.TAHIR

PLATE LOAD TEST (ASTM D 1196)

PROJECT:	**MMGP**	**LOAD TEST #**	**PLT-6**
CLIENT :	**L & T**	**TEST DATE:**	**12/9/2019**
		WEATHER:	**FAIR**
TEST PLATE DIA (cm	**30**	**TEST PLATE AREA (m^2) =**	**0.0707**
PROJECT :	**19-k-3054**		
Existing Grade E1		**Northing (m)**	**2509176.016**
Easting (m)	**261819.089**		

STAGE	**LOADING**							**UNLOADING**						
PRESSURE (KPa)	0	160	320	480	640	800	480	160	0					
SETTLE MENT (mm)	0	0.98	1.52	2.35	3.25	3.85	3.25	2.15	1.65					



TESTED BY : Abdal Haque

CHECKED BY M.TAHIR

APPENDIX "B"

LABORATORY TEST RESULTS

SIEVE ANALYSIS RESULTS

RIYADH GEOTECHNIQUE & FOUNDATIONS

AL-KHOBAR, SAUDI ARABIA

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**

CLIENT: **L & T CONSTRUCTION**

Project # : **19-K-3054**

SUMMARY OF LABORATORY TEST RESULTS

B.H No.	DEPTH (m)	DESCRIPTION	USCS ASTM D-2487	MC %	% PASSING #200	ATTERBERG LIMITS		
						LL	PL	PI
BH-01	0.15	SILTY SAND with GRAVEL	SM	2.8	32.9	NP	NP	NP
BH-01	1.50	SILTY SAND with GRAVEL	SM	8.5	13.0	NP	NP	NP
BH-01	3.00	POORLY GRADED GRAVEL with SILT and SAND	GP-GM	6.8	8.5	NP	NP	NP
BH-01	6.00	CLAYEY SAND	SC	11.4	45.9	30.0	20.0	10.0
BH-02	0.15	SILTY SAND with GRAVEL	SM	0.6	24.0	NP	NP	NP
BH-02	0.75	CLAYEY SAND with GRAVEL	SC	7.9	32.0	28.0	20.0	8.0
BH-03	0.15	SILTY SAND	SM	1.2	39.7	NP	NP	NP
BH-03	1.50	SILTY GRAVEL with SAND	GM	7.2	24.5	NP	NP	NP
BH-03	2.25	SILTY SAND with GRAVEL	SM	5.7	29.8	32.0	26.0	6.0
BH-03	6.00	SILTY SAND with GRAVEL	SM	13.8	18.7	NP	NP	NP
BH-04	0.75	SANDY LEAN CLAY	CL	17.7	52.1	36.0	24.0	12.0
BH-04	1.50	SILTY SAND with GRAVEL	SM	7.1	25.9	NP	NP	NP
BH-04	3.00	SILTY SAND with GRAVEL	SM	9.2	28.1	NP	NP	NP
BH-04	7.50	SANDY LEAN CLAY	CL	24.1	63.9	33.0	21.0	12.0
BH-04	9.00	CLAYEY SAND	SC	16.6	33.7	27.0	18.0	9.0
BH-04	10.50	CLAYEY SAND	SC	10.2	37.5	31.0	21.0	10.0
BH-04	13.50	SILTY SAND with GRAVEL	SM	11.6	25.1	NP	NP	NP
BH-04	15.00	SILTY GRAVEL with SAND	GM	7.5	16.9	NP	NP	NP
BH-05	0.15	SILTY GRAVEL with SAND	GM	1.6	23.2	NP	NP	NP
BH-05	1.50	SILTY SAND with GRAVEL	SM	9.2	22.8	NP	NP	NP

RIYADH GEOTECHNIQUE & FOUNDATIONS

AL-KHOBAR, SAUDI ARABIA

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**

CLIENT: **L & T CONSTRUCTION**

Project # : **19-K-3054**

SUMMARY OF LABORATORY TEST RESULTS

B.H No.	DEPTH (m)	DESCRIPTION	USCS ASTM D-2487	MC %	% PASSING #200	ATTERBERG LIMITS		
						LL	PL	PI
BH-05	3.75	SILTY SAND with GRAVEL	SM	13.2	17.7	NP	NP	NP
BH-05	7.50	SILTY SAND	SM	19.1	23.7	NP	NP	NP
BH-05	12.00	CLAYEY SAND with GRAVEL	SC	14.6	26.9	30.0	21.0	9.0
BH-05	15.00	SILTY SAND with GRAVEL	SM	13.7	23.5	NP	NP	NP
BH-06	0.15	SANDY LEAN CLAY	CL	6.3	55.4	32.0	20.0	12.0
BH-06	0.75	CLAYEY SAND	SC	11.6	27.7	30.0	22.0	8.0
BH-06	2.25	CLAYEY SAND with GRAVEL	SC	12.1	25.9	27.0	19.0	8.0
BH-06	3.75	SILTY SAND with GRAVEL	SM	12.1	21.4	NP	NP	NP
BH-06	7.50	SILTY SAND	SM	11.9	24.8	NP	NP	NP
BH-06	15.00	SILTY SAND with GRAVEL	SM	14.8	13.7	NP	NP	NP
BH-07	0.75	SANDY SILTY CLAY	CL-ML	9.1	50.4	27.0	21.0	6.0
BH-07	3.00	CLAYEY SAND with GRAVEL	SC	15.0	32.4	33.0	23.0	10.0
BH-07	7.50	SILTY SAND with GRAVEL	SM	12.7	19.1	NP	NP	NP
BH-07	10.50	SILTY SAND with GRAVEL	SM	13.7	22.8	NP	NP	NP
BH-07	13.50	SILTY SAND with GRAVEL	SM	14.9	21.9	NP	NP	NP
BH-07	15.00	SILTY SAND with GRAVEL	SM	15.4	15.4	NP	NP	NP
BH-08	0.15	SANDY LEAN CLAY	CL	3.7	66.0	33.0	22.0	11.0
BH-08	1.50	CLAYEY SAND with GRAVEL	SC	13.1	36.8	29.0	19.0	10.0
BH-08	4.50	SILTY GRAVEL with SAND	GM	17.2	18.9	NP	NP	NP
BH-08	13.50	CLAYEY SAND	SC	14.1	39.1	31.0	22.0	9.0

RIYADH GEOTECHNIQUE & FOUNDATIONS

AL-KHOBAR, SAUDI ARABIA

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**

CLIENT: **L & T CONSTRUCTION**

Project # : **19-K-3054**

SUMMARY OF LABORATORY TEST RESULTS

B.H No.	DEPTH (m)	DESCRIPTION	USCS ASTM D-2487	MC %	% PASSING #200	ATTERBERG LIMITS		
						LL	PL	PI
BH-09	0.00	CLAYEY SAND	SC	2.0	41.8	28.0	19.0	9.0
BH-09	0.75	SANDY LEAN CLAY	CL	10.0	56.4	32.0	20.0	12.0
BH-09	2.25	CLAYEY GRAVEL with SAND	GC	9.1	24.3	29.0	21.0	8.0
BH-09	6.00	CLAYEY SAND with GRAVEL	SC	9.6	30.8	31.0	22.0	9.0
BH-10	0.00	SILTY SAND with GRAVEL	SM	3.6	31.9	NP	NP	NP
BH-10	1.50	SILTY SAND with GRAVEL	SM	13.1	26.5	NP	NP	NP
BH-10	3.75	POORLY GRADED GRAVEL with SILT and SAND	GP-GM	3.8	11.0	NP	NP	NP
BH-10	4.50	POORLY GRADED SAND with SILT and GRAVEL	SP-SM	4.9	10.7	NP	NP	NP
BH-11	0.15	CLAYEY SAND	SC	1.6	46.5	29.0	21.0	8.0
BH-11	1.50	SANDY SILT with GRAVEL	ML	10.2	52.4	34.0	24.0	10.0
BH-11	3.00	CLAYEY SAND	SC	15.2	41.9	34.0	23.0	11.0
BH-11	6.00	CLAYEY SAND	SC	10.5	30.8	32.0	22.0	10.0
BH-11	9.00	CLAYEY SAND	SC	13.5	30.4	31.0	20.0	11.0
BH-11	10.50	SILTY SAND with GRAVEL	SM	12.6	17.5	NP	NP	NP
BH-11	15.00	SILTY GRAVEL with SAND	GM	13.1	14.3	NP	NP	NP
BH-12	0.15	SILTY GRAVEL with SAND	GM	0.9	19.9	NP	NP	NP
BH-12	0.75	SILTY SAND with GRAVEL	SM	11.0	13.6	NP	NP	NP
BH-12	2.25	SILTY GRAVEL with SAND	GM	5.0	23.9	NP	NP	NP
BH-13	0.15	SILTY SAND with GRAVEL	SM	1.0	26.4	NP	NP	NP
BH-13	0.75	SILTY SAND with GRAVEL	SM	1.4	29.7	NP	NP	NP

RIYADH GEOTECHNIQUE & FOUNDATIONS

AL-KHOBAR, SAUDI ARABIA

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**

CLIENT: **L & T CONSTRUCTION**

Project # : **19-K-3054**

SUMMARY OF LABORATORY TEST RESULTS

B.H No.	DEPTH (m)	DESCRIPTION	USCS ASTM D-2487	MC %	% PASSING #200	ATTERBERG LIMITS		
						LL	PL	PI
BH-13	1.50	SILTY GRAVEL with SAND	GM	5.9	12.8	NP	NP	NP
BH-14	0.15	SILTY SAND with GRAVEL	SM	1.4	27.8	NP	NP	NP
BH-14	0.75	SILTY GRAVEL with SAND	GM	4.1	23.8	NP	NP	NP
BH-14	3.00	WELL-GRADED GRAVEL with SILT and SAND	GW-GM	7.0	5.6	NP	NP	NP
BH-15	0.15	SILTY SAND with GRAVEL	SM	2.7	19.4	NP	NP	NP
BH-15	0.75	SILTY SAND with GRAVEL	SM	14.6	16.9	NP	NP	NP
BH-15	2.25	WELL-GRADED SAND with SILT and GRAVEL	SW-SM	14.2	6.5	NP	NP	NP
BH-16	0.75	CLAYEY SAND with GRAVEL	SC	11.3	27.3	28.0	20.0	8.0
BH-16	3.75	CLAYEY SAND	SC	13.6	49.9	31.0	22.0	9.0
BH-16	6.00	CLAYEY SAND with GRAVEL	SC	19.7	37.5	26.0	18.0	8.0
BH-16	9.00	CLAYEY SAND	SC	16.3	29.7	27.0	19.0	8.0
BH-17	0.15	CLAYEY SAND	SC	1.4	44.1	30.0	2.0	28.0
BH-17	0.75	CLAYEY SAND with GRAVEL	SC	10.6	38.7	29.0	19.0	10.0
BH-17	3.00	SILTY GRAVEL with SAND	GM	3.7	14.3	NP	NP	NP
BH-17	4.50	POORLY GRADED GRAVEL with SILT and SAND	GP-GM	3.8	10.2	NP	NP	NP
BH-17	6.00	WELL-GRADED GRAVEL with SAND	GW	5.1	4.2	NP	NP	NP
BH-17	7.50	SILTY GRAVEL with SAND	GM	4.1	14.4	NP	NP	NP
BH-18	0.15	CLAYEY SAND with GRAVEL	SC	1.4	35.3	25.0	17.0	8.0
BH-18	0.75	CLAYEY SAND with GRAVEL	SC	8.6	26.6	31.0	21.0	10.0
BH-18	3.00	SANDY SILT	ML	9.7	67.5	37.0	28.0	9.0

RIYADH GEOTECHNIQUE & FOUNDATIONS

AL-KHOBAR, SAUDI ARABIA

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**

CLIENT: **L & T CONSTRUCTION**

Project # : **19-K-3054**

SUMMARY OF LABORATORY TEST RESULTS

B.H No.	DEPTH (m)	DESCRIPTION	USCS ASTM D-2487	MC %	% PASSING #200	ATTERBERG LIMITS		
						LL	PL	PI
BH-18	3.75	SILT with SAND	ML	8.1	70.1	40.0	30.0	10.0
BH-18	7.50	SILTY SAND with GRAVEL	SM	12.3	43.0	37.0	29.0	8.0
BH-18	10.50	SANDY SILT	ML	19.0	67.6	36.0	27.0	9.0
BH-19	0.15	SILTY SAND with GRAVEL	SM	2.0	21.1	NP	NP	NP
BH-19	0.75	SILTY SAND with GRAVEL	SM	8.3	15.4	NP	NP	NP
BH-19	3.75	CLAYEY SAND with GRAVEL	SC	10.1	38.4	30.0	21.0	9.0
BH-19	15.00	POORLY GRADED GRAVEL with SILT and SAND	GP-GM	4.7	6.9	NP	NP	NP
BH-20	0.15	WELL-GRADED SAND with SILT and GRAVEL	SW-SM	1.7	6.3	NP	NP	NP
BH-20	1.50	SILTY GRAVEL with SAND	GM	9.4	15.2	NP	NP	NP
BH-20	2.25	SILTY GRAVEL with SAND	GM	6.2	20.3	NP	NP	NP
BH-21	0.15	POORLY GRADED SAND with SILT	SP-SM	2.3	9.1	NP	NP	NP
BH-21	0.75	CLAYEY SAND with GRAVEL	SC	8.4	31.5	29.0	20.0	9.0
BH-21	2.25	SILTY SAND with GRAVEL	SM	10.7	19.2	NP	NP	NP
BH-21	3.75	SILTY SAND with GRAVEL	SM	7.6	22.5	NP	NP	NP
BH-23	0.15	SILTY GRAVEL with SAND	GM	0.5	14.7	NP	NP	NP
BH-23	2.25	SILTY SAND with GRAVEL	SM	11.7	28.3	NP	NP	NP
BH-23	3.75	SILTY GRAVEL with SAND	GM	6.1	24.7	NP	NP	NP
BH-23	6.00	SILTY SAND with GRAVEL	SM	8.7	14.0	NP	NP	NP
BH-25	0.15	SILTY SAND with GRAVEL	SM	1.2	31.8	31.0	23.0	8.0
BH-25	1.50	SILTY GRAVEL with SAND	GM	4.7	16.9	NP	NP	NP

RIYADH GEOTECHNIQUE & FOUNDATIONS

AL-KHOBAR, SAUDI ARABIA

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**

CLIENT: **L & T CONSTRUCTION**

Project # : **19-K-3054**

SUMMARY OF LABORATORY TEST RESULTS

B.H No.	DEPTH (m)	DESCRIPTION	USCS ASTM D-2487	MC %	% PASSING #200	ATTERBERG LIMITS		
						LL	PL	PI
BH-25	3.00	POORLY GRADED GRAVEL with SILT and SAND	GP-GM	7.2	10.1	NP	NP	NP
BH-25	7.50	SILTY SAND with GRAVEL	SM	3.7	13.9	NP	NP	NP
BH-26	0.15	CLAYEY SAND with GRAVEL	SC	2.3	39.1	30.0	22.0	8.0
BH-26	0.75	SILTY SAND with GRAVEL	SM	8.8	26.2	NP	NP	NP
BH-26	2.25	POORLY GRADED GRAVEL with SILT and SAND	GP-GM	5.4	9.0	NP	NP	NP
BH-26	4.50	POORLY GRADED GRAVEL with SILT and SAND	GP-GM	9.1	10.7	NP	NP	NP
BH-26	7.50	SILTY SAND with GRAVEL	SM	16.2	20.7	NP	NP	NP
BH-27	0.15	SILTY SAND with GRAVEL	SM	2.5	22.3	NP	NP	NP
BH-27	1.50	CLAYEY SAND	SC	13.3	40.6	30.0	20.0	10.0
BH-27	3.00	SILTY SAND with GRAVEL	SM	11.0	21.8	NP	NP	NP
BH-27	4.50	CLAYEY SAND	SC	10.5	41.0	29.0	21.0	8.0
BH-27	7.50	SILTY SAND with GRAVEL	SM	12.4	21.7	NP	NP	NP
TP-01	0.00	SILTY SAND with GRAVEL	SM	0.7	18.3	NP	NP	NP
TP-02	0.00	CLAYEY SAND with GRAVEL	SC	1.4	30.1	28.0	20.0	8.0
TP-02	0.70	SILTY GRAVEL with SAND	GM	1.7	13.7	NP	NP	NP
TP-03	0.00	SILTY SAND with GRAVEL	SM	1.9	21.4	NP	NP	NP
TP-04	0.00	SILTY SAND with GRAVEL	SM	1.2	25.4	NP	NP	NP
TP-04	0.30	SILTY GRAVEL with SAND	GM	1.9	25.4	NP	NP	NP
TP-05	0.00	SILTY SAND with GRAVEL	SM	0.6	18.4	NP	NP	NP
TP-05	0.30	CLAYEY SAND with GRAVEL	SC	2.1	31.4	29.0	21.0	8.0

RIYADH GEOTECHNIQUE & FOUNDATIONS

AL-KHOBAR, SAUDI ARABIA

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**

CLIENT: **L & T CONSTRUCTION**

Project # : **19-K-3054**

SUMMARY OF LABORATORY TEST RESULTS

B.H No.	DEPTH (m)	DESCRIPTION	USCS ASTM D-2487	MC %	% PASSING #200	ATTERBERG LIMITS		
						LL	PL	PI
TP-06	0.00	CLAYEY SAND with GRAVEL	SC	2.7	33.4	30.0	21.0	9.0
TP-07	0.00	SILTY SAND with GRAVEL	SM	0.5	21.4	NP	NP	NP
TP-07	0.60	SILTY SAND with GRAVEL	SM	1.4	13.4	NP	NP	NP
TP-08	0.00	SILTY SAND with GRAVEL	SM	2.5	36.3	NP	NP	NP
TP-09	0.00	SILTY GRAVEL with SAND	GM	0.5	18.4	NP	NP	NP
TP-09	0.50	WELL-GRADED SAND with SILT and GRAVEL	SW-SM	1.3	7.4	NP	NP	NP
TP-10	0.00	CLAYEY SAND with GRAVEL	SC	2.7	34.0	30.0	20.0	10.0
TP-11	0.00	SILTY SAND	SM	0.2	18.4	NP	NP	NP
TP-11	0.30	WELL-GRADED GRAVEL with SILT and SAND	GW-GM	2.1	6.4	NP	NP	NP
TP-12	0.00	SILTY SAND with GRAVEL	SM	0.9	16.4	NP	NP	NP
TP-13	0.00	CLAYEY SAND with GRAVEL	SC	2.4	36.2	28.0	19.0	9.0
TP-13	0.70	SILTY GRAVEL with SAND	GM	1.7	21.4	NP	NP	NP
TP-14	0.00	CLAYEY SAND with GRAVEL	SC	2.1	34.3	29.0	21.0	8.0
TP-14	0.50	POORLY GRADED GRAVEL with SILT and SAND	GP-GM	1.3	8.4	NP	NP	NP
TP-15	0.00	CLAYEY SAND with GRAVEL	SC	2.6	29.4	31.0	22.0	9.0
TP-16	0.00	CLAYEY SAND with GRAVEL	SC	2.9	33.5	29.0	21.0	8.0
TP-17	0.00	SILTY SAND with GRAVEL	SM	1.2	22.6	NP	NP	NP
TP-18	0.00	SILTY SAND with GRAVEL	SM	1.6	18.2	NP	NP	NP
TP-19	0.00	SILTY SAND	SM	0.5	16.2	NP	NP	NP
TP-19	0.20	CLAYEY SAND with GRAVEL	SC	2.4	35.2	29.0	20.0	9.0

RIYADH GEOTECHNIQUE & FOUNDATIONS

AL-KHOBAR, SAUDI ARABIA

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**

CLIENT: **L & T CONSTRUCTION**

Project # : **19-K-3054**

SUMMARY OF LABORATORY TEST RESULTS

B.H No.	DEPTH (m)	DESCRIPTION	USCS ASTM D-2487	MC %	% PASSING #200	ATTERBERG LIMITS		
						LL	PL	PI
TP-20	0.00	CLAYEY SAND with GRAVEL	SC	2.0	31.3	28.0	20.0	8.0
TP-21	0.00	SILTY SAND with GRAVEL	SM	1.2	25.3	NP	NP	NP
TP-21	0.40	CLAYEY SAND with GRAVEL	SC	2.8	37.5	30.0	20.0	10.0
TP-22	0.00	SILTY SAND	SM	0.7	21.5	NP	NP	NP
TP-22	0.50	SILTY SAND with GRAVEL	SM	1.3	22.4	NP	NP	NP
TP-23	0.00	SILTY SAND with GRAVEL	SM	0.8	15.4	NP	NP	NP
TP-23	0.30	POORLY GRADED GRAVEL with SAND	GP	1.2	4.3	NP	NP	NP
TP-24	0.00	SILTY SAND	SM	1.7	41.5	NP	NP	NP
TP-24	0.55	CLAYEY SAND with GRAVEL	SC	2.9	32.3	29.0	21.0	8.0
TP-25	0.00	SILTY SAND with GRAVEL	SM	0.9	13.4	NP	NP	NP
TP-26	0.00	SILTY SAND with GRAVEL	SM	0.7	18.5	NP	NP	NP
TP-27	0.00	SILTY SAND with GRAVEL	SM	0.5	23.4	NP	NP	NP
TP-27	0.50	SILTY SAND with GRAVEL	SM	1.2	18.3	NP	NP	NP
TP-28	0.00	CLAYEY SAND	SC	1.9	41.4	29.0	20.0	9.0
TP-28	0.70	SILTY SAND with GRAVEL	SM	1.5	25.4	NP	NP	NP
TP-29	0.00	SILTY SAND with GRAVEL	SM	2.2	32.3	31.0	23.0	8.0
TP-30	0.00	CLAYEY SAND	SC	1.5	39.4	32.0	22.0	10.0
TP-30	0.40	CLAYEY SAND with GRAVEL	SC	2.7	34.1	28.0	19.0	9.0
TP-31	0.00	SILTY SAND	SM	0.6	23.5	NP	NP	NP
TP-31	0.40	CLAYEY SAND	SC	2.1	35.6	33.0	23.0	10.0

RIYADH GEOTECHNIQUE & FOUNDATIONS

AL-KHOBAR, SAUDI ARABIA

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**

CLIENT: **L & T CONSTRUCTION**

Project # : **19-K-3054**

SUMMARY OF LABORATORY TEST RESULTS

B.H No.	DEPTH (m)	DESCRIPTION	USCS ASTM D-2487	MC %	% PASSING #200	ATTERBERG LIMITS		
						LL	PL	PI
TP-32	0.00	SILTY SAND	SM	0.9	21.4	NP	NP	NP
TP-33	0.00	SILTY SAND with GRAVEL	SM	0.6	17.2	NP	NP	NP
TP-33	0.40	CLAYEY SAND with GRAVEL	SC	2.4	35.6	30.0	20.0	10.0
TP-34	0.00	CLAYEY SAND with GRAVEL	SC	2.0	32.4	28.0	20.0	8.0
TP-34	0.40	WELL-GRADED SAND with SILT	SW-SM	1.2	7.3	NP	NP	NP
TP-35	0.00	SILTY SAND with GRAVEL	SM	1.2	15.3	NP	NP	NP
TP-36	0.00	SILTY SAND with GRAVEL	SM	0.8	18.5	NP	NP	NP
TP-41	0.00	SILTY SAND with GRAVEL	SM	0.6	21.5	NP	NP	NP
TP-41	0.50	POORLY GRADED GRAVEL with SILT and SAND	GP-GM	1.5	10.2	NP	NP	NP
TP-42	0.00	SILTY SAND with GRAVEL	SM	0.9	19.5	NP	NP	NP
TP-42	0.60	POORLY GRADED SAND with SILT and GRAVEL	SP-SM	1.3	7.5	NP	NP	NP
TP-43	0.00	CLAYEY SAND	SC	1.6	41.5	29.0	19.0	10.0
TP-43	0.20	CLAYEY SAND with GRAVEL	SC	2.7	35.1	28.0	20.0	8.0
TP-44	0.00	SILTY SAND with GRAVEL	SM	0.9	22.4	NP	NP	NP
TP-45	0.00	SILTY SAND with GRAVEL	SM	1.0	17.5	NP	NP	NP
TP-45	0.30	SILTY GRAVEL with SAND	GM	1.3	15.2	NP	NP	NP
TP-46	0.00	SILTY SAND with GRAVEL	SM	2.2	34.2	NP	NP	NP
TP-47	0.00	CLAYEY SAND with GRAVEL	SC	2.4	30.3	29.0	20.0	9.0
TP-47	0.45	SILTY GRAVEL with SAND	GM	1.9	21.3	NP	NP	NP
TP-48	0.00	CLAYEY SAND	SC	1.3	45.3	31.0	21.0	10.0

RIYADH GEOTECHNIQUE & FOUNDATIONS

AL-KHOBAR, SAUDI ARABIA

PROJECT: **MANSOURAH MASSARAH GOLD PROJECT**

CLIENT: **L & T CONSTRUCTION**

Project # : **19-K-3054**

SUMMARY OF LABORATORY TEST RESULTS

B.H No.	DEPTH (m)	DESCRIPTION	USCS ASTM D-2487	MC %	% PASSING #200	ATTERBERG LIMITS		
						LL	PL	PI
TP-48	0.50	CLAYEY SAND with GRAVEL	SC	2.3	32.3	29.0	20.0	9.0
TP-49	0.00	CLAYEY SAND with GRAVEL	SC	1.9	29.4	26.0	18.0	8.0
TP-49	0.20	POORLY GRADED GRAVEL with SILT and SAND	GP-GM	2.3	7.3	NP	NP	NP
TP-50	0.00	CLAYEY SAND with GRAVEL	SC	1.6	35.4	28.0	20.0	8.0
TP-50	0.35	POORLY GRADED GRAVEL with SILT and SAND	GP-GM	1.3	5.3	NP	NP	NP

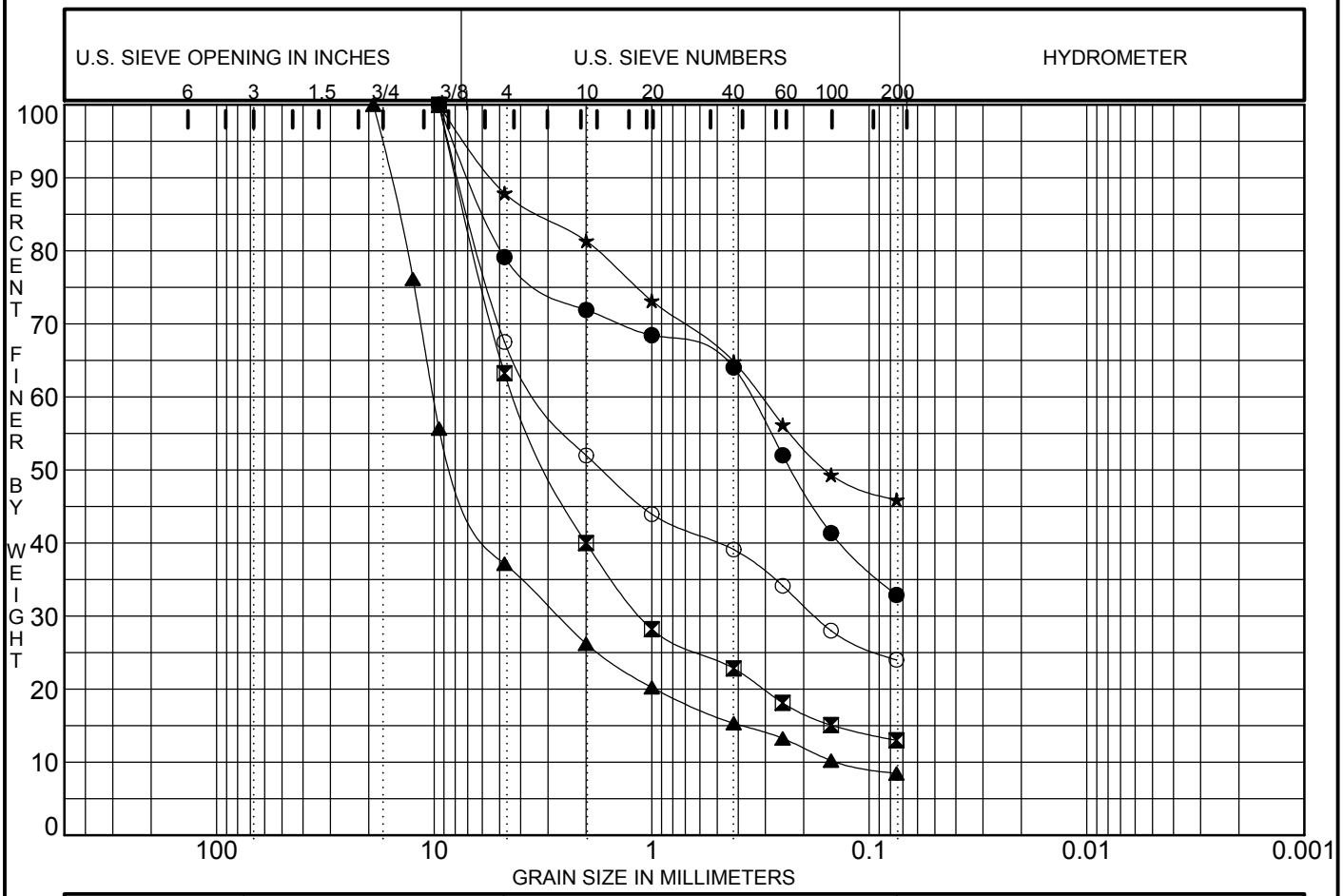
RIYADH GEOTECHNIQUE & FOUNDATIONS
AL-KHOBAR, SAUDI ARABIA

PROJECT # **MANSOURAH MASSARAH GOLD PROJECT**

CLIENT: **L & T CONSTRUCTION**

BOREHOLE NO. **BH-02**
LOCATION:

SIEVE ANALYSIS
ASTM D 422



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Sample No.	Depth(m)	Classification	MC%	LL	PL	PI	Cc	Cu
●	BH-01	0.15	SILTY SAND with GRAVEL (SM)	3	NP	NP	NP		
⊠	BH-01	1.50	SILTY SAND with GRAVEL (SM)	9	NP	NP	NP		
▲	BH-01	3.00	POORLY GRADED GRAVEL with SILT and SAND (GP-GM)	7	NP	NP	NP	5.26	73.4
★	BH-01	6.00	CLAYEY SAND (SC)	11	30	20	10		
⊙	BH-02	0.15	SILTY SAND with GRAVEL (SM)	1	NP	NP	NP		

	Sample No.	Depth(m)	D100	D50	D30	D10	%Gravel	%Sand	%Silt	%Clay
●	BH-01	0.15	9.50	0.23			20.8	46.3	32.9	
⊠	BH-01	1.50	9.50	2.90	1.111		36.8	50.2	13.0	
▲	BH-01	3.00	19.00	7.68	2.695	0.1371	62.8	28.7	8.5	
★	BH-01	6.00	9.50	0.16			12.1	42.0	45.9	
⊙	BH-02	0.15	9.50	1.68	0.177		32.5	43.5	24.0	

RIYADH GEOTECHNIQUE & FOUNDATIONS

AL-KHOBAR, SAUDI ARABIA

PROJECT # **MANSOURAH MASSARAH GOLD PROJECT**

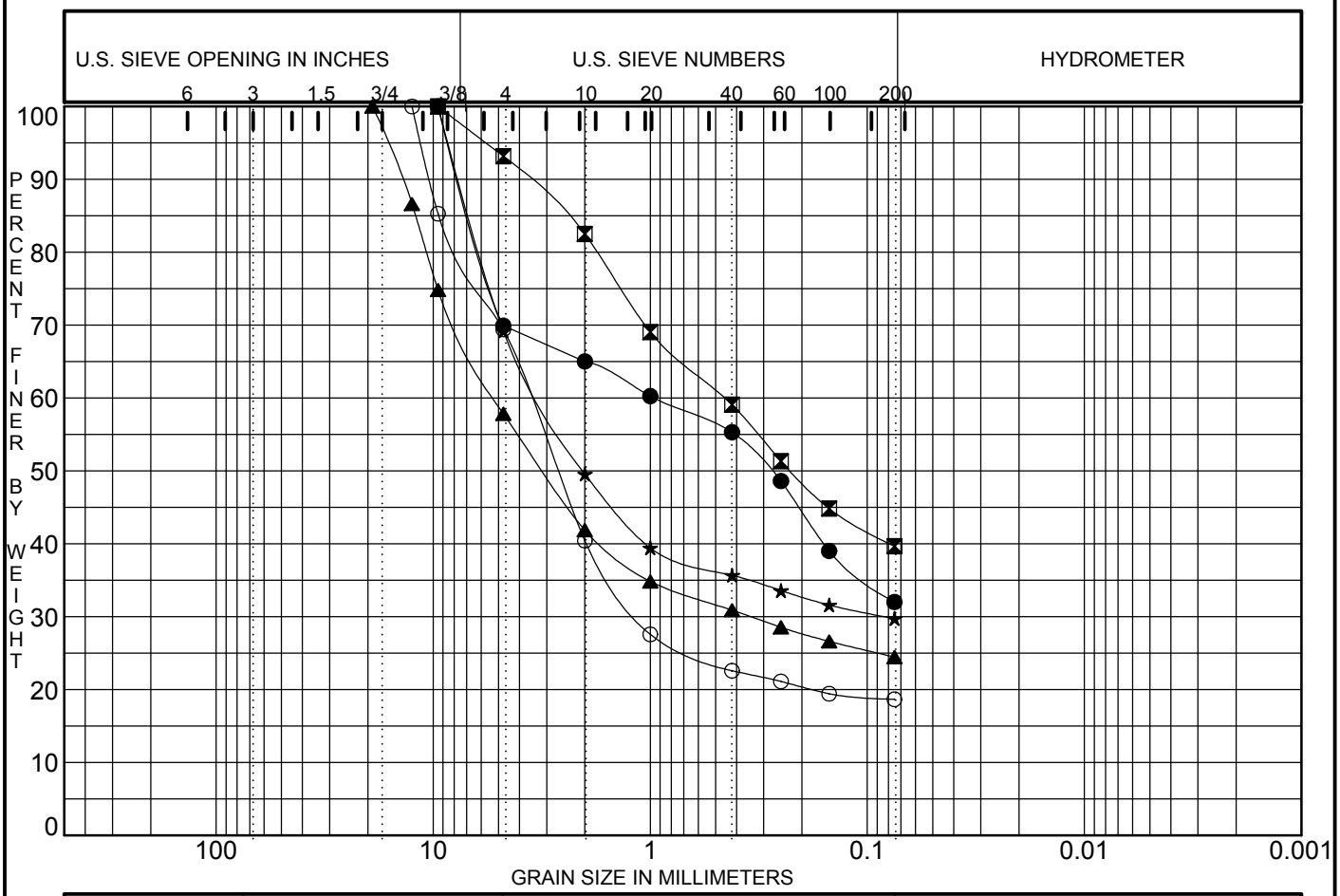
CLIENT: **L & T CONSTRUCTION**

BOREHOLE NO. **BH-03**

LOCATION:

SIEVE ANALYSIS

ASTM D 422



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Sample No.	Depth(m)	Classification	MC%	LL	PL	PI	Cc	Cu
●	BH-02	0.75	CLAYEY SAND with GRAVEL (SC)	8	28	20	8		
⊠	BH-03	0.15	SILTY SAND (SM)	1	NP	NP	NP		
▲	BH-03	1.50	SILTY GRAVEL with SAND (GM)	7	NP	NP	NP		
★	BH-03	2.25	SILTY SAND with GRAVEL (SM)	6	32	26	6		
⊙	BH-03	6.00	SILTY SAND with GRAVEL (SM)	14	NP	NP	NP		

	Sample No.	Depth(m)	D100	D50	D30	D10	%Gravel	%Sand	%Silt	%Clay
●	BH-02	0.75	9.50	0.28			30.1	37.9	32.0	
⊠	BH-03	0.15	9.50	0.23			6.8	53.5	39.7	
▲	BH-03	1.50	19.00	3.10	0.344		42.2	33.4	24.5	
★	BH-03	2.25	9.50	2.04	0.082		30.8	39.5	29.8	
⊙	BH-03	6.00	12.50	2.66	1.139		30.6	50.7	18.7	

RIYADH GEOTECHNIQUE & FOUNDATIONS

AL-KHOBAR, SAUDI ARABIA

PROJECT # **MANSOURAH MASSARAH GOLD PROJECT**

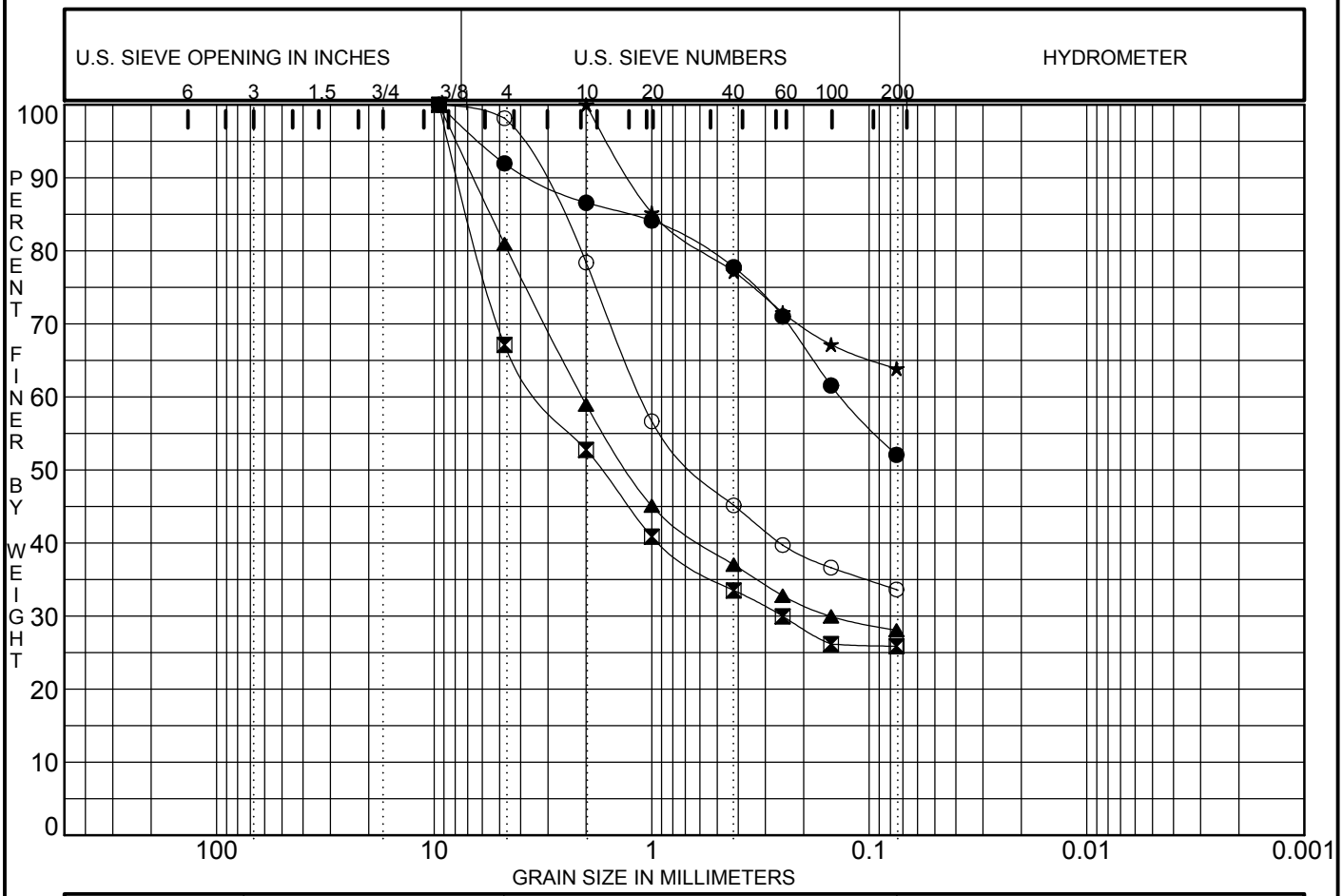
CLIENT: **L & T CONSTRUCTION**

BOREHOLE NO. **BH-04**

LOCATION:

SIEVE ANALYSIS

ASTM D 422



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Sample No.	Depth(m)	Classification	MC%	LL	PL	PI	Cc	Cu
●	BH-04	0.75	SANDY LEAN CLAY (CL)	18	36	24	12		
⊠	BH-04	1.50	SILTY SAND with GRAVEL (SM)	7	NP	NP	NP		
▲	BH-04	3.00	SILTY SAND with GRAVEL (SM)	9	NP	NP	NP		
★	BH-04	7.50	SANDY LEAN CLAY (CL)	24	33	21	12		
⊙	BH-04	9.00	CLAYEY SAND (SC)	17	27	18	9		

	Sample No.	Depth(m)	D100	D50	D30	D10	%Gravel	%Sand	%Silt	%Clay
●	BH-04	0.75	9.50				8.0	39.9	52.1	
⊠	BH-04	1.50	9.50	1.70	0.251		32.9	41.2	25.9	
▲	BH-04	3.00	9.50	1.28	0.151		19.0	52.9	28.1	
★	BH-04	7.50	2.00				0.0	36.1	63.9	
⊙	BH-04	9.00	9.50	0.60			1.8	64.5	33.7	

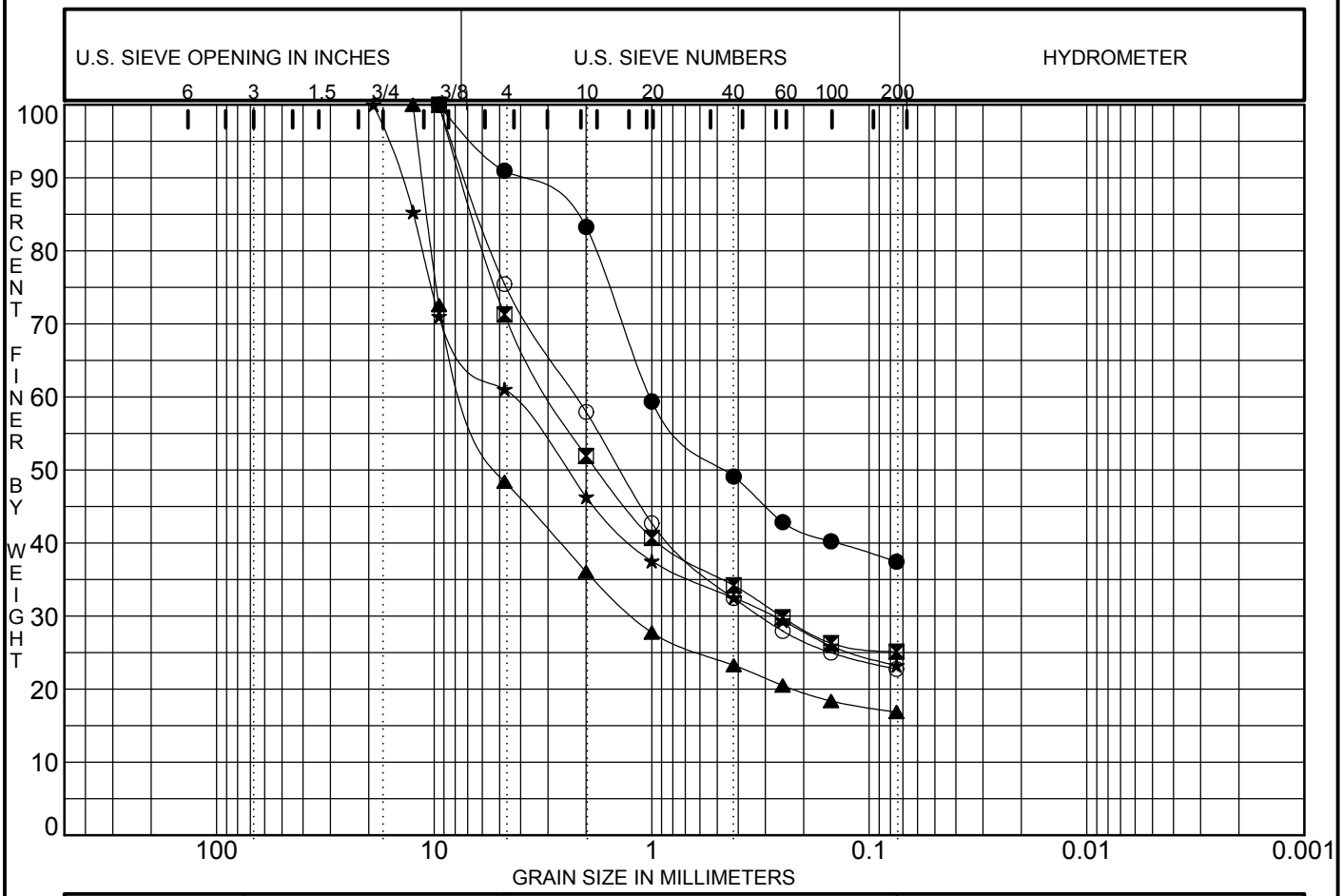
RIYADH GEOTECHNIQUE & FOUNDATIONS
AL-KHOBAR, SAUDI ARABIA

PROJECT # **MANSOURAH MASSARAH GOLD PROJECT**

CLIENT: **L & T CONSTRUCTION**

BOREHOLE NO. **BH-05**
LOCATION:

SIEVE ANALYSIS
ASTM D 422



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Sample No.	Depth(m)	Classification	MC%	LL	PL	PI	Cc	Cu
●	BH-04	10.50	CLAYEY SAND (SC)	10	31	21	10		
⊠	BH-04	13.50	SILTY SAND with GRAVEL (SM)	12	NP	NP	NP		
▲	BH-04	15.00	SILTY GRAVEL with SAND (GM)	7	NP	NP	NP		
★	BH-05	0.15	SILTY GRAVEL with SAND (GM)	2	NP	NP	NP		
⊙	BH-05	1.50	SILTY SAND with GRAVEL (SM)	9	NP	NP	NP		

	Sample No.	Depth(m)	D100	D50	D30	D10	%Gravel	%Sand	%Silt	%Clay
●	BH-04	10.50	9.50	0.45			9.0	53.5	37.5	
⊠	BH-04	13.50	9.50	1.77	0.255		28.7	46.1	25.1	
▲	BH-04	15.00	12.50	4.97	1.205		51.6	31.5	16.9	
★	BH-05	0.15	19.00	2.48	0.277		38.9	37.8	23.2	
⊙	BH-05	1.50	9.50	1.39	0.317		24.5	52.7	22.8	

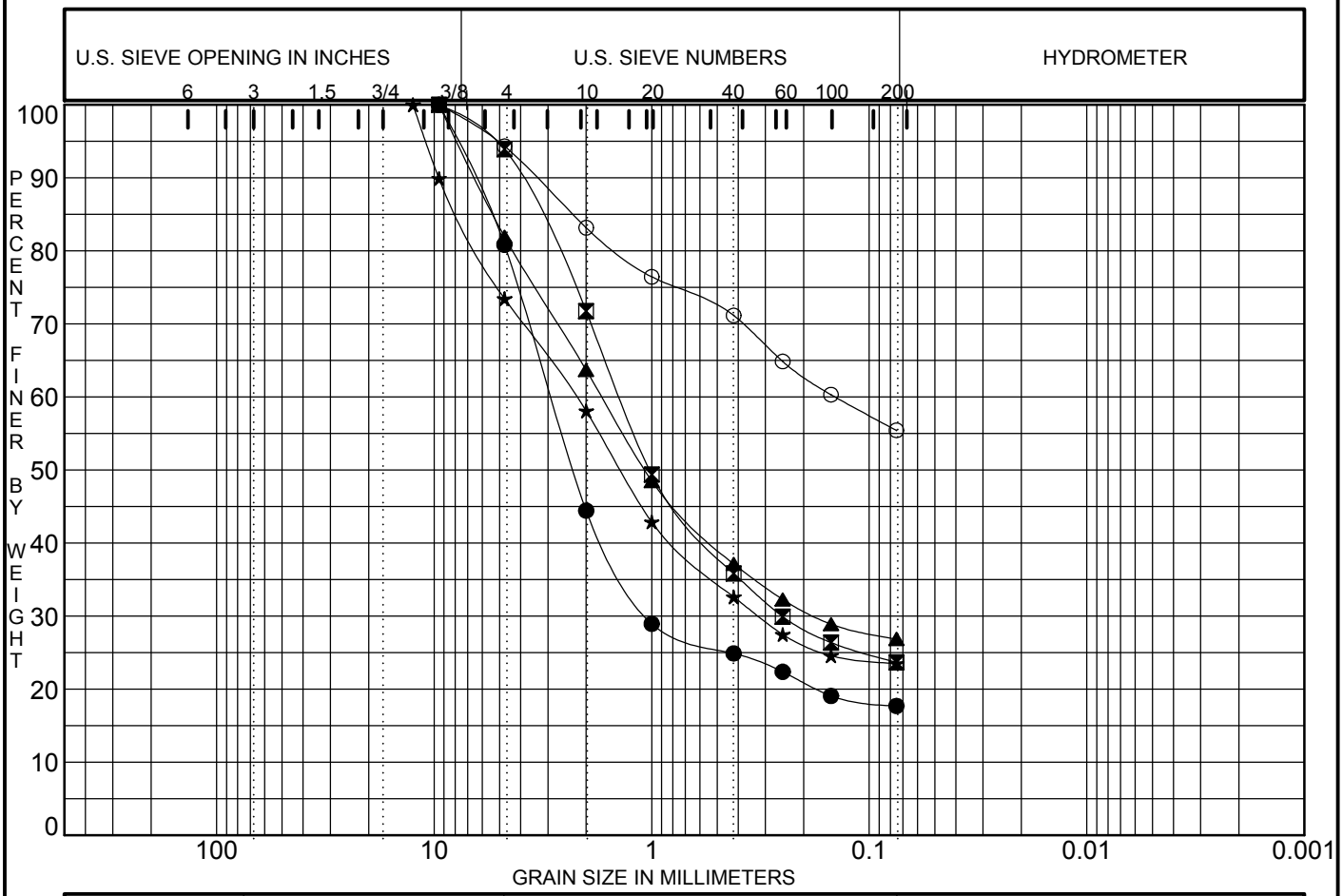
RIYADH GEOTECHNIQUE & FOUNDATIONS
AL-KHOBAR, SAUDI ARABIA

PROJECT # **MANSOURAH MASSARAH GOLD PROJECT**

CLIENT: **L & T CONSTRUCTION**

BOREHOLE NO. **BH-06**
LOCATION:

SIEVE ANALYSIS
ASTM D 422



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Sample No.	Depth(m)	Classification	MC%	LL	PL	PI	Cc	Cu
●	BH-05	3.75	SILTY SAND with GRAVEL (SM)	13	NP	NP	NP		
⊠	BH-05	7.50	SILTY SAND (SM)	19	NP	NP	NP		
▲	BH-05	12.00	CLAYEY SAND with GRAVEL (SC)	15	30	21	9		
★	BH-05	15.00	SILTY SAND with GRAVEL (SM)	14	NP	NP	NP		
⊙	BH-06	0.15	SANDY LEAN CLAY (CL)	6	32	20	12		

	Sample No.	Depth(m)	D100	D50	D30	D10	%Gravel	%Sand	%Silt	%Clay
●	BH-05	3.75	9.50	2.28	1.048		19.2	63.1	17.7	
⊠	BH-05	7.50	9.50	1.02	0.252		6.1	70.3	23.7	
▲	BH-05	12.00	9.50	1.07	0.177		18.1	55.1	26.9	
★	BH-05	15.00	12.50	1.38	0.322		26.5	49.9	23.5	
⊙	BH-06	0.15	9.50				5.7	38.9	55.4	

RIYADH GEOTECHNIQUE & FOUNDATIONS

AL-KHOBAR, SAUDI ARABIA

PROJECT # **MANSOURAH MASSARAH GOLD PROJECT**

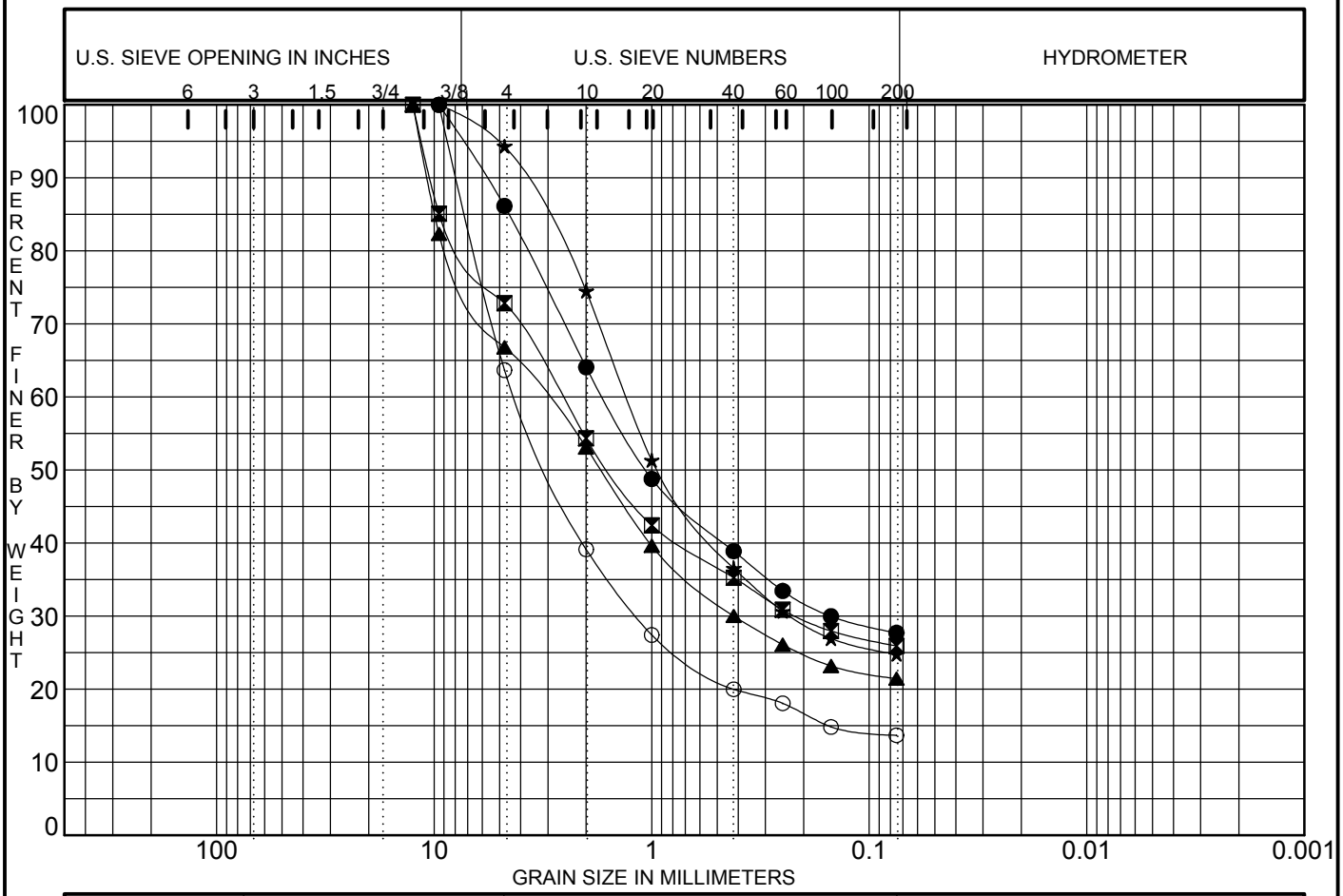
CLIENT: **L & T CONSTRUCTION**

BOREHOLE NO. **BH-06**

LOCATION:

SIEVE ANALYSIS

ASTM D 422



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Sample No.	Depth(m)	Classification	MC%	LL	PL	PI	Cc	Cu
●	BH-06	0.75	CLAYEY SAND (SC)	12	30	22	8		
⊠	BH-06	2.25	CLAYEY SAND with GRAVEL (SC)	12	27	19	8		
▲	BH-06	3.75	SILTY SAND with GRAVEL (SM)	12	NP	NP	NP		
★	BH-06	7.50	SILTY SAND (SM)	12	NP	NP	NP		
⊙	BH-06	15.00	SILTY SAND with GRAVEL (SM)	15	NP	NP	NP		

	Sample No.	Depth(m)	D100	D50	D30	D10	%Gravel	%Sand	%Silt	%Clay
●	BH-06	0.75	9.50	1.06	0.151		13.9	58.4	27.7	
⊠	BH-06	2.25	12.50	1.55	0.215		27.2	46.9	25.9	
▲	BH-06	3.75	12.50	1.71	0.419		33.2	45.4	21.4	
★	BH-06	7.50	9.50	0.92	0.229		5.7	69.5	24.8	
⊙	BH-06	15.00	9.50	2.93	1.165		36.3	50.0	13.7	

RIYADH GEOTECHNIQUE & FOUNDATIONS

AL-KHOBAR, SAUDI ARABIA

PROJECT # **MANSOURAH MASSARAH GOLD PROJECT**

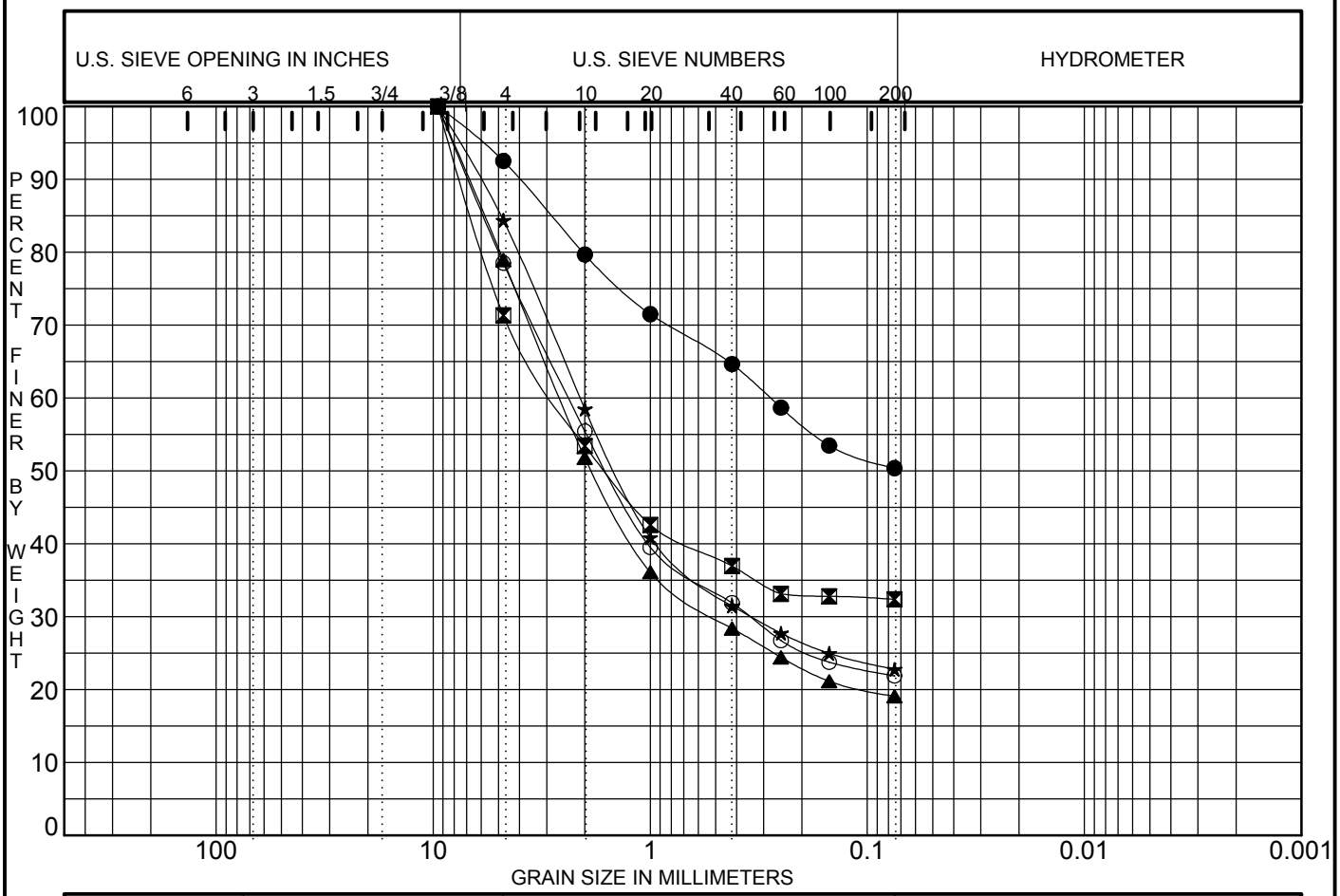
CLIENT: **L & T CONSTRUCTION**

BOREHOLE NO. **BH-07**

LOCATION:

SIEVE ANALYSIS

ASTM D 422



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Sample No.	Depth(m)	Classification	MC%	LL	PL	PI	Cc	Cu
●	BH-07	0.75	SANDY SILTY CLAY (CL-ML)	9	27	21	6		
⊠	BH-07	3.00	CLAYEY SAND with GRAVEL (SC)	15	33	23	10		
▲	BH-07	7.50	SILTY SAND with GRAVEL (SM)	13	NP	NP	NP		
★	BH-07	10.50	SILTY SAND with GRAVEL (SM)	14	NP	NP	NP		
⊙	BH-07	13.50	SILTY SAND with GRAVEL (SM)	15	NP	NP	NP		

	Sample No.	Depth(m)	D100	D50	D30	D10	%Gravel	%Sand	%Silt	%Clay
●	BH-07	0.75	9.50				7.5	42.1	50.4	
⊠	BH-07	3.00	9.50	1.61			28.7	39.0	32.4	
▲	BH-07	7.50	9.50	1.85	0.503		21.1	59.8	19.1	
★	BH-07	10.50	9.50	1.43	0.342		15.6	61.6	22.8	
⊙	BH-07	13.50	9.50	1.58	0.347		21.5	56.6	21.9	

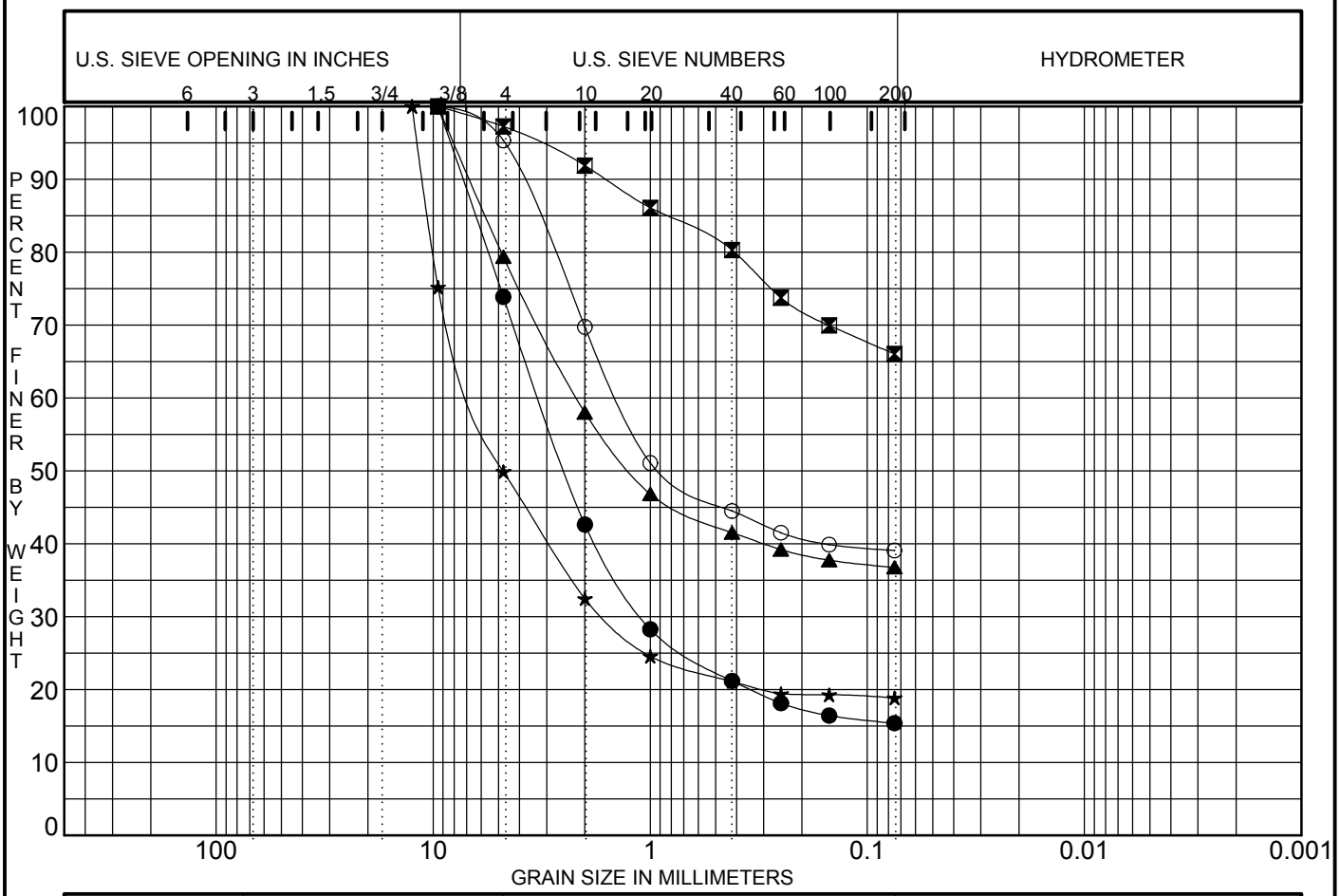
RIYADH GEOTECHNIQUE & FOUNDATIONS
AL-KHOBAR, SAUDI ARABIA

PROJECT # **MANSOURAH MASSARAH GOLD PROJECT**

CLIENT: **L & T CONSTRUCTION**

BOREHOLE NO. **BH-08**
LOCATION:

SIEVE ANALYSIS
ASTM D 422



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Sample No.	Depth(m)	Classification	MC%	LL	PL	PI	Cc	Cu
●	BH-07	15.00	SILTY SAND with GRAVEL (SM)	15	NP	NP	NP		
⊠	BH-08	0.15	SANDY LEAN CLAY (CL)	4	33	22	11		
▲	BH-08	1.50	CLAYEY SAND with GRAVEL (SC)	13	29	19	10		
★	BH-08	4.50	SILTY GRAVEL with SAND (GM)	17	NP	NP	NP		
⊙	BH-08	13.50	CLAYEY SAND (SC)	14	31	22	9		

	Sample No.	Depth(m)	D100	D50	D30	D10	%Gravel	%Sand	%Silt	%Clay
●	BH-07	15.00	9.50	2.45	1.087		26.1	58.5	15.4	
⊠	BH-08	0.15	9.50				2.7	31.2	66.0	
▲	BH-08	1.50	9.50	1.22			20.6	42.6	36.8	
★	BH-08	4.50	12.50	4.76	1.609		50.1	31.0	18.9	
⊙	BH-08	13.50	9.50	0.87			4.7	56.3	39.1	

RIYADH GEOTECHNIQUE & FOUNDATIONS

AL-KHOBAR, SAUDI ARABIA

PROJECT # **MANSOURAH MASSARAH GOLD PROJECT**

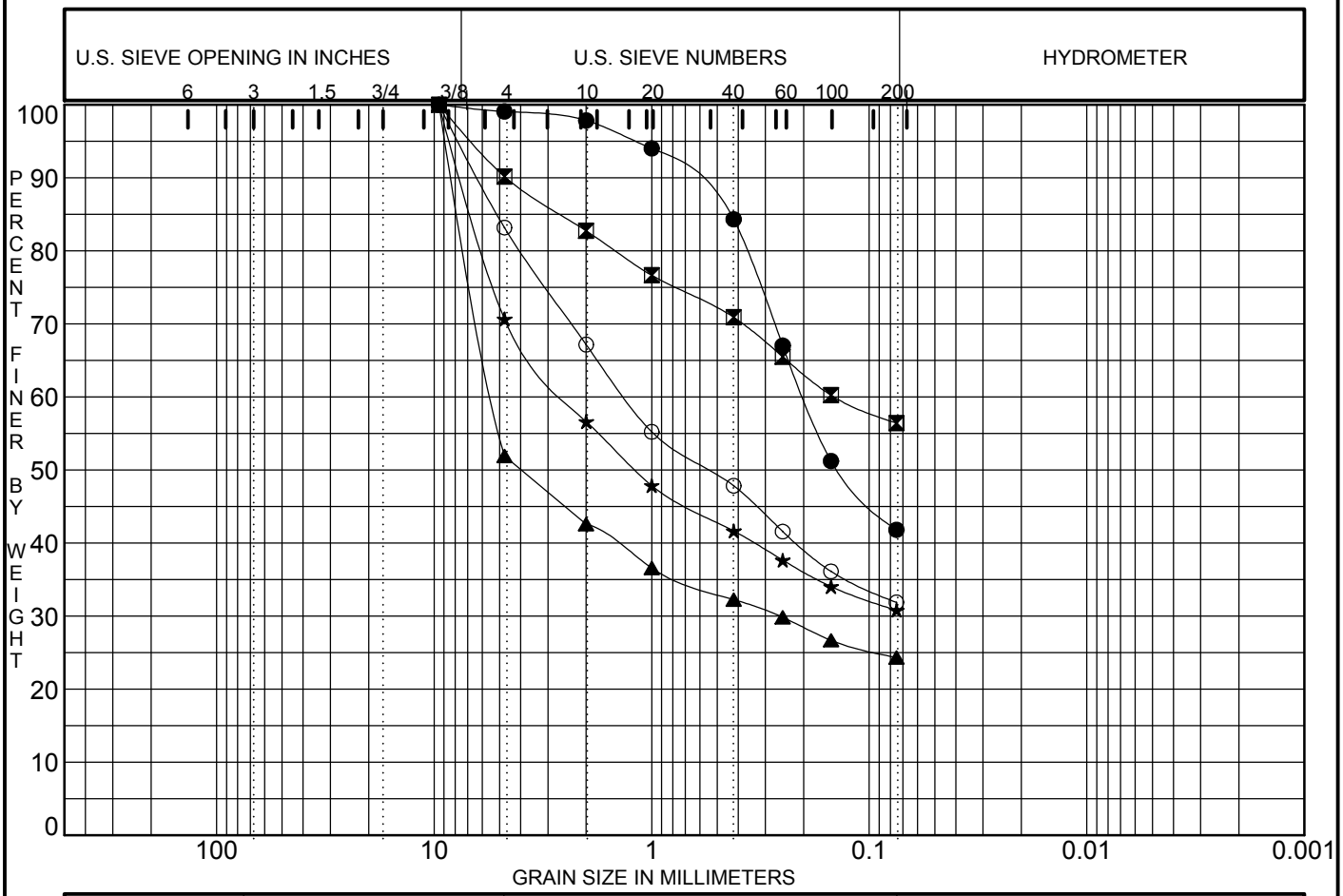
CLIENT: **L & T CONSTRUCTION**

BOREHOLE NO. **BH-10**

LOCATION:

SIEVE ANALYSIS

ASTM D 422



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Sample No.	Depth(m)	Classification	MC%	LL	PL	PI	Cc	Cu
●	BH-09	0.00	CLAYEY SAND (SC)	2	28	19	9		
⊠	BH-09	0.75	SANDY LEAN CLAY (CL)	10	32	20	12		
▲	BH-09	2.25	CLAYEY GRAVEL with SAND (GC)	9	29	21	8		
★	BH-09	6.00	CLAYEY SAND with GRAVEL (SC)	10	31	22	9		
⊙	BH-10	0.00	SILTY SAND with GRAVEL (SM)	4	NP	NP	NP		

	Sample No.	Depth(m)	D100	D50	D30	D10	%Gravel	%Sand	%Silt	%Clay
●	BH-09	0.00	9.50	0.14			0.9	57.3	41.8	
⊠	BH-09	0.75	9.50				9.8	33.8	56.4	
▲	BH-09	2.25	9.50	3.97	0.258		48.1	27.6	24.3	
★	BH-09	6.00	9.50	1.18			29.3	39.9	30.8	
⊙	BH-10	0.00	9.50	0.54			16.8	51.3	31.9	

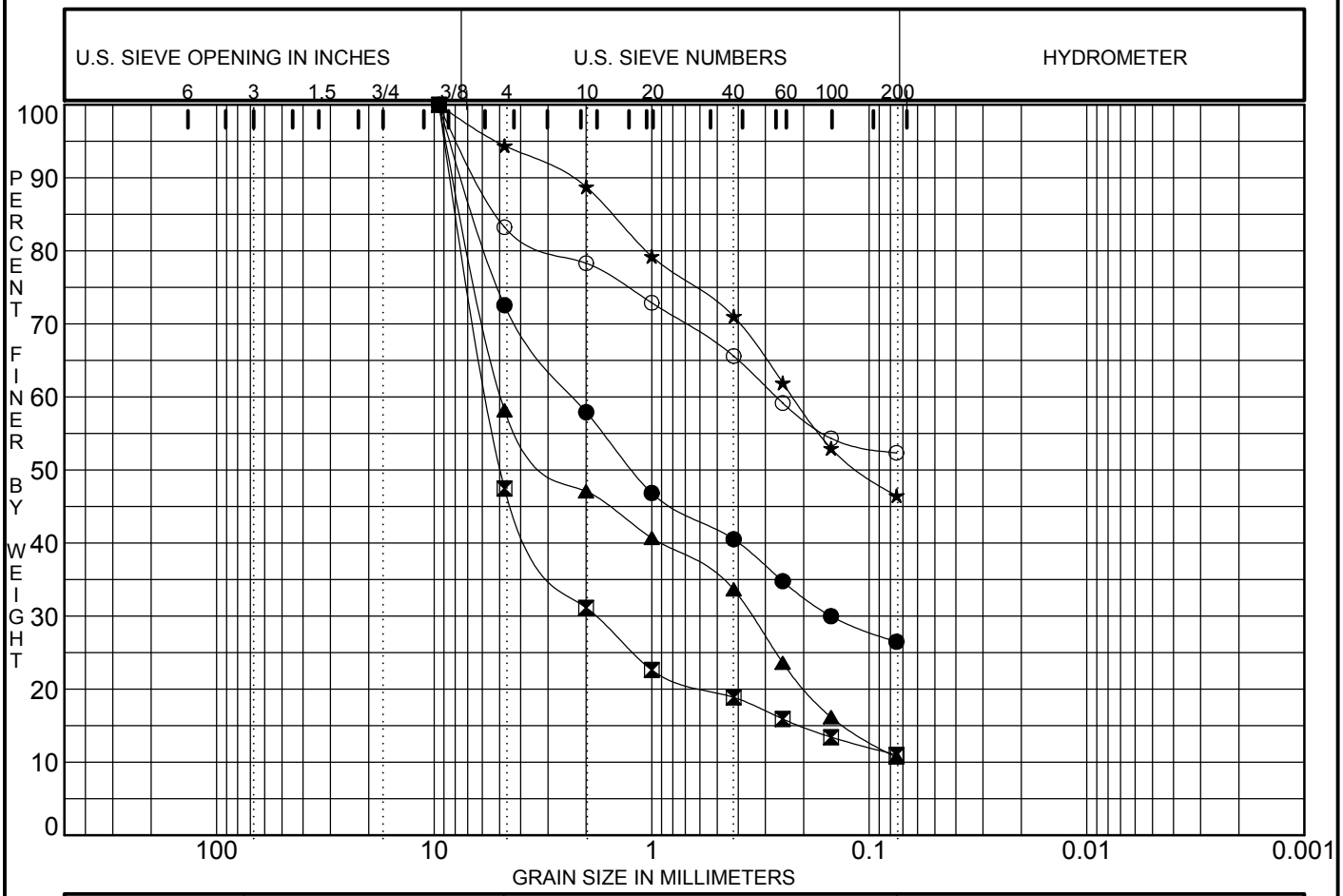
RIYADH GEOTECHNIQUE & FOUNDATIONS
AL-KHOBAR, SAUDI ARABIA

PROJECT # **MANSOURAH MASSARAH GOLD PROJECT**

CLIENT: **L & T CONSTRUCTION**

BOREHOLE NO. **BH-11**
LOCATION:

SIEVE ANALYSIS
ASTM D 422



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Sample No.	Depth(m)	Classification	MC%	LL	PL	PI	Cc	Cu
●	BH-10	1.50	SILTY SAND with GRAVEL (SM)	13	NP	NP	NP		
⊠	BH-10	3.75	POORLY GRADED GRAVEL with SILT and SAND (GP-GM)	4	NP	NP	NP	10.60	100.4
▲	BH-10	4.50	POORLY GRADED SAND with SILT and GRAVEL (SP-SM)	5	NP	NP	NP	0.36	71.0
★	BH-11	0.15	CLAYEY SAND (SC)	2	29	21	8		
⊙	BH-11	1.50	SANDY SILT with GRAVEL (ML)	10	34	24	10		

	Sample No.	Depth(m)	D100	D50	D30	D10	%Gravel	%Sand	%Silt	%Clay
●	BH-10	1.50	9.50	1.22	0.150		27.4	46.1	26.5	
⊠	BH-10	3.75	9.50	4.91	1.821		52.5	36.4	11.0	
▲	BH-10	4.50	9.50	2.51	0.348		41.8	47.5	10.7	
★	BH-11	0.15	9.50	0.11			5.6	47.9	46.5	
⊙	BH-11	1.50	9.50				16.8	30.9	52.4	

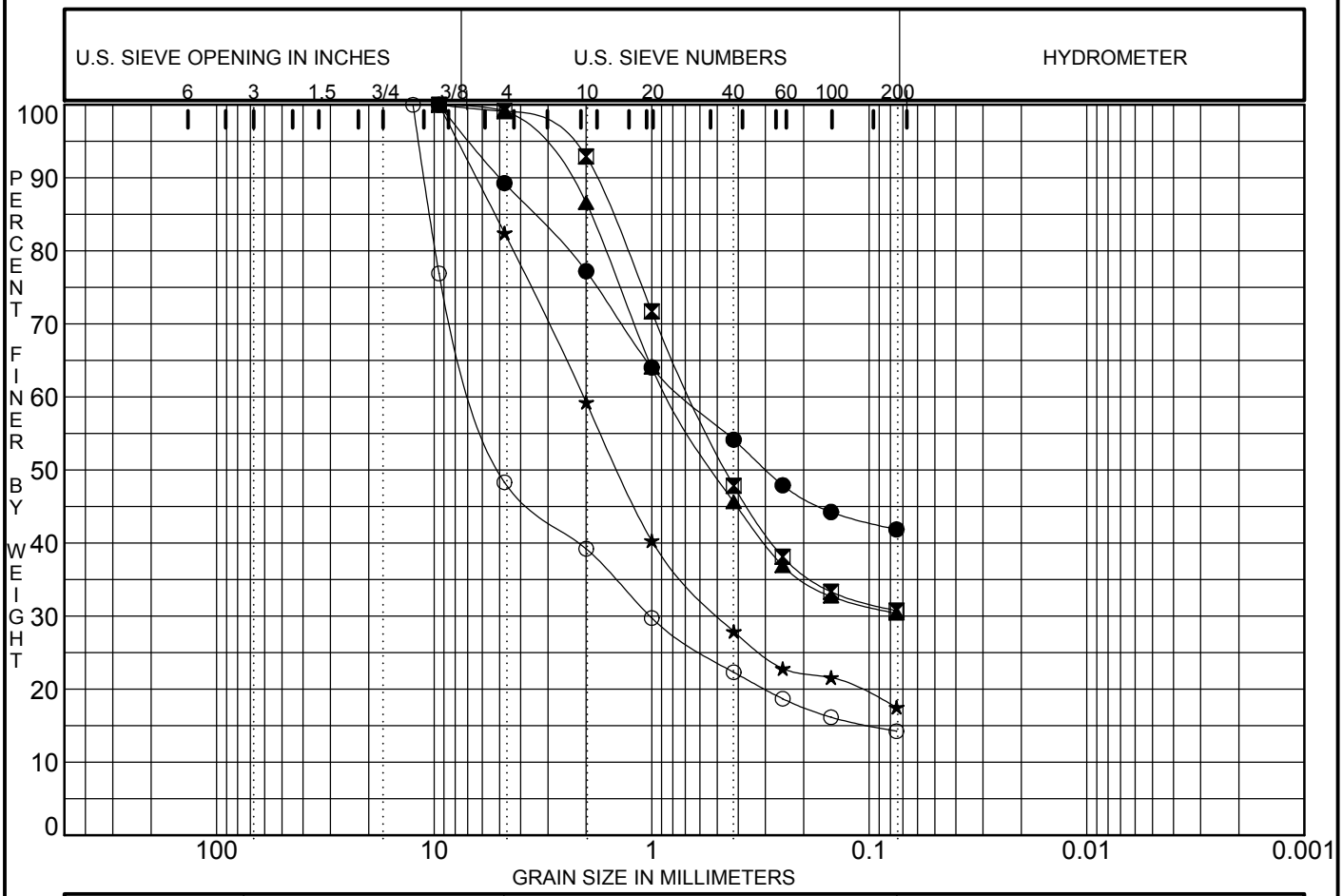
RIYADH GEOTECHNIQUE & FOUNDATIONS
AL-KHOBAR, SAUDI ARABIA

PROJECT # **MANSOURAH MASSARAH GOLD PROJECT**

CLIENT: **L & T CONSTRUCTION**

BOREHOLE NO. **BH-11**
LOCATION:

SIEVE ANALYSIS
ASTM D 422



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Sample No.	Depth(m)	Classification	MC%	LL	PL	PI	Cc	Cu
●	BH-11	3.00	CLAYEY SAND (SC)	15	34	23	11		
⊠	BH-11	6.00	CLAYEY SAND (SC)	10	32	22	10		
▲	BH-11	9.00	CLAYEY SAND (SC)	14	31	20	11		
★	BH-11	10.50	SILTY SAND with GRAVEL (SM)	13	NP	NP	NP		
⊙	BH-11	15.00	SILTY GRAVEL with SAND (GM)	13	NP	NP	NP		

	Sample No.	Depth(m)	D100	D50	D30	D10	%Gravel	%Sand	%Silt	%Clay
●	BH-11	3.00	9.50	0.30			10.7	47.4	41.9	
⊠	BH-11	6.00	9.50	0.45			0.9	68.4	30.8	
▲	BH-11	9.00	9.50	0.52			0.8	68.8	30.4	
★	BH-11	10.50	9.50	1.42	0.486		17.5	64.9	17.5	
⊙	BH-11	15.00	12.50	4.95	1.019		51.7	34.1	14.3	

RIYADH GEOTECHNIQUE & FOUNDATIONS

AL-KHOBAR, SAUDI ARABIA

PROJECT # **MANSOURAH MASSARAH GOLD PROJECT**

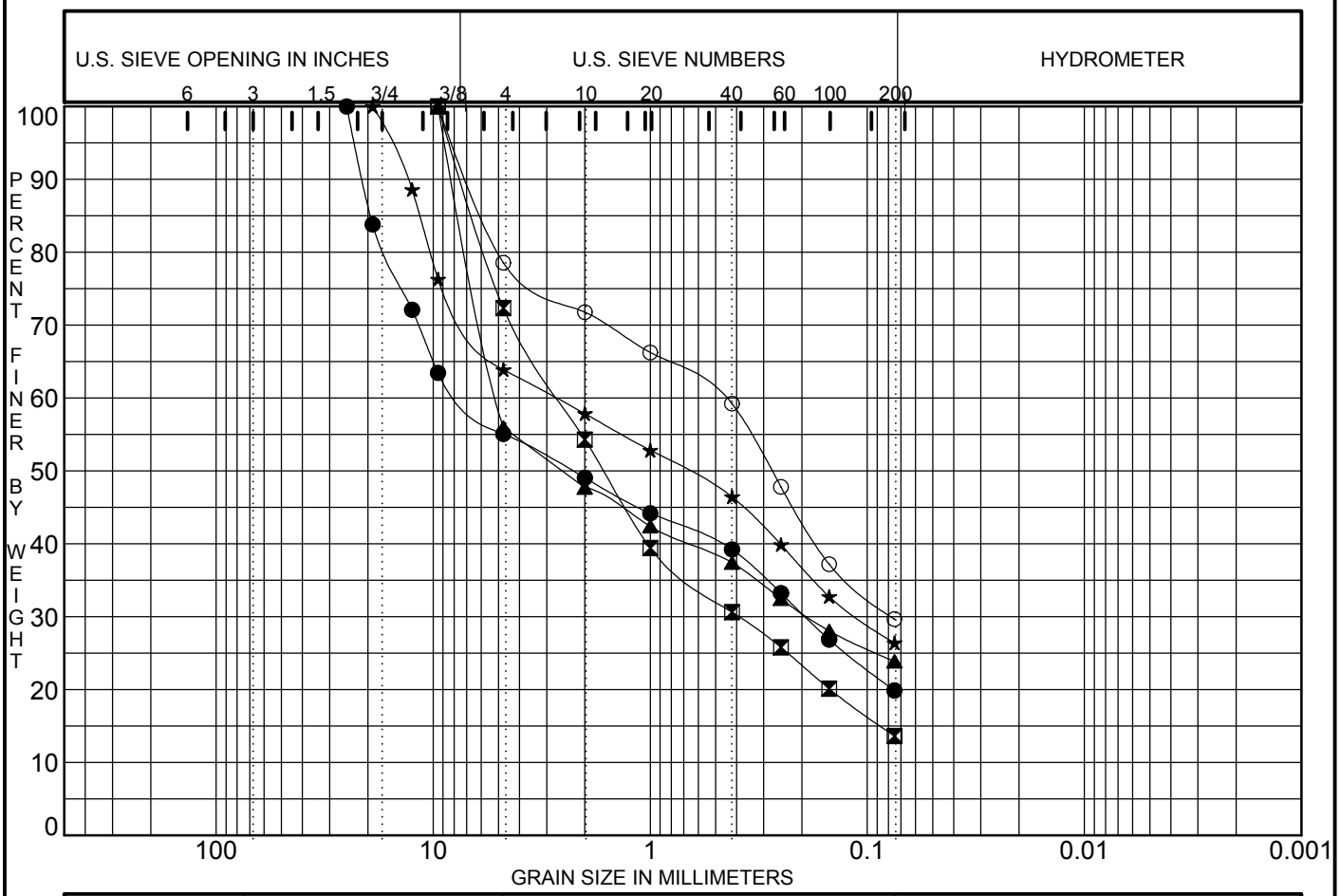
CLIENT: **L & T CONSTRUCTION**

BOREHOLE NO. **BH-13**

LOCATION:

SIEVE ANALYSIS

ASTM D 422



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Sample No.	Depth(m)	Classification	MC%	LL	PL	PI	Cc	Cu
●	BH-12	0.15	SILTY GRAVEL with SAND (GM)	1	NP	NP	NP		
⊠	BH-12	0.75	SILTY SAND with GRAVEL (SM)	11	NP	NP	NP		
▲	BH-12	2.25	SILTY GRAVEL with SAND (GM)	5	NP	NP	NP		
★	BH-13	0.15	SILTY SAND with GRAVEL (SM)	1	NP	NP	NP		
⊙	BH-13	0.75	SILTY SAND with GRAVEL (SM)	1	NP	NP	NP		

	Sample No.	Depth(m)	D100	D50	D30	D10	%Gravel	%Sand	%Silt	%Clay
●	BH-12	0.15	25.00	2.29	0.193		44.9	35.2	19.9	
⊠	BH-12	0.75	9.50	1.64	0.393		27.7	58.7	13.6	
▲	BH-12	2.25	9.50	2.54	0.188		44.0	32.1	23.9	
★	BH-13	0.15	19.00	0.68	0.111		36.1	37.5	26.4	
⊙	BH-13	0.75	9.50	0.28	0.077		21.4	48.9	29.7	

RIYADH GEOTECHNIQUE & FOUNDATIONS

AL-KHOBAR, SAUDI ARABIA

PROJECT # **MANSOURAH MASSARAH GOLD PROJECT**

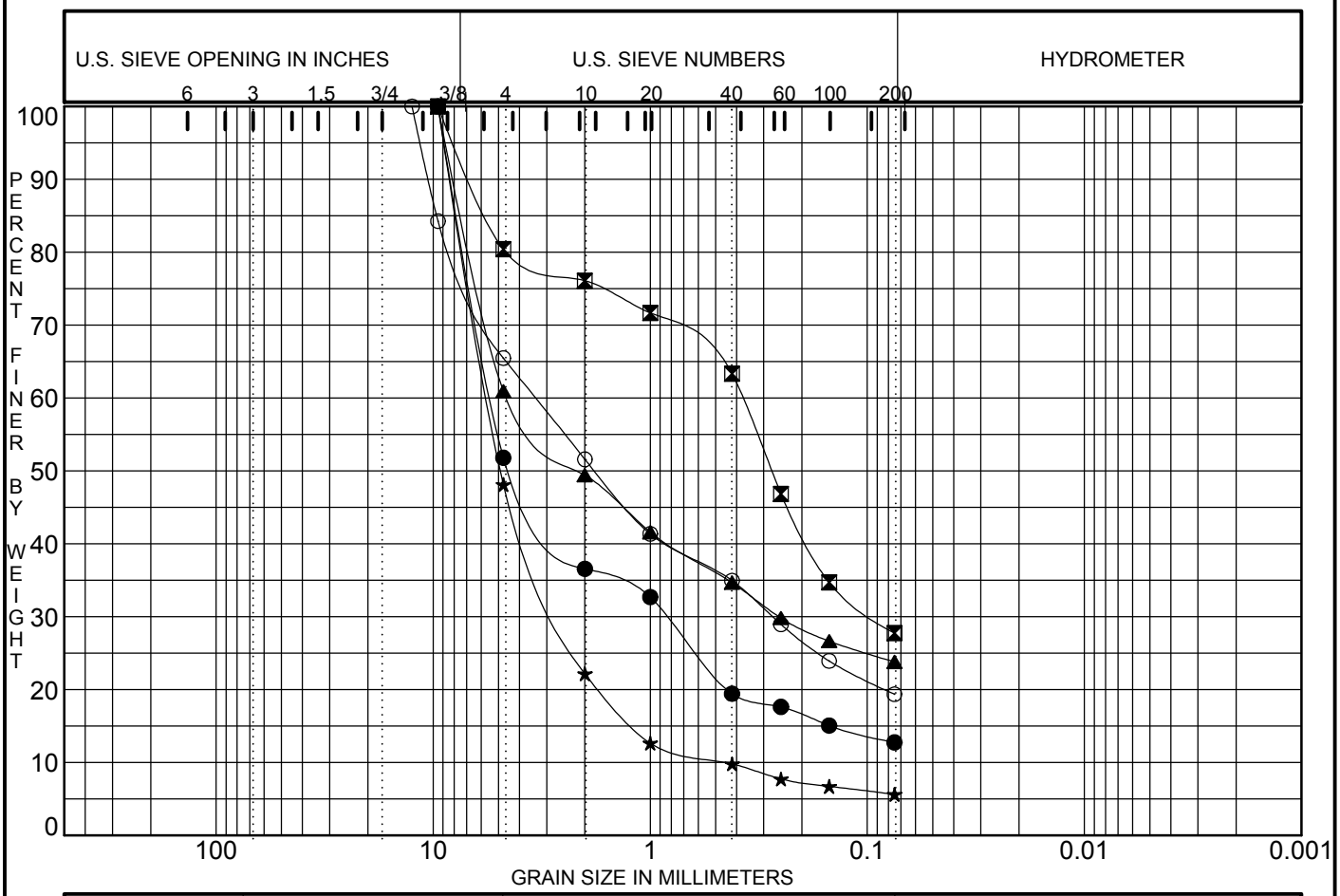
CLIENT: **L & T CONSTRUCTION**

BOREHOLE NO. **BH-15**

LOCATION:

SIEVE ANALYSIS

ASTM D 422



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Sample No.	Depth(m)	Classification	MC%	LL	PL	PI	Cc	Cu
●	BH-13	1.50	SILTY GRAVEL with SAND (GM)	6	NP	NP	NP		
⊠	BH-14	0.15	SILTY SAND with GRAVEL (SM)	1	NP	NP	NP		
▲	BH-14	0.75	SILTY GRAVEL with SAND (GM)	4	NP	NP	NP		
★	BH-14	3.00	WELL-GRADED GRAVEL with SILT and SAND (GW-GM)	7	NP	NP	NP	2.72	12.5
⊙	BH-15	0.15	SILTY SAND with GRAVEL (SM)	3	NP	NP	NP		

	Sample No.	Depth(m)	D100	D50	D30	D10	%Gravel	%Sand	%Silt	%Clay
●	BH-13	1.50	9.50	4.29	0.838		48.2	39.0	12.8	
⊠	BH-14	0.15	9.50	0.28	0.094		19.6	52.7	27.8	
▲	BH-14	0.75	9.50	2.09	0.255		39.1	37.1	23.8	
★	BH-14	3.00	9.50	4.87	2.596	0.4452	51.9	42.5	5.6	
⊙	BH-15	0.15	12.50	1.80	0.274		34.5	46.1	19.4	

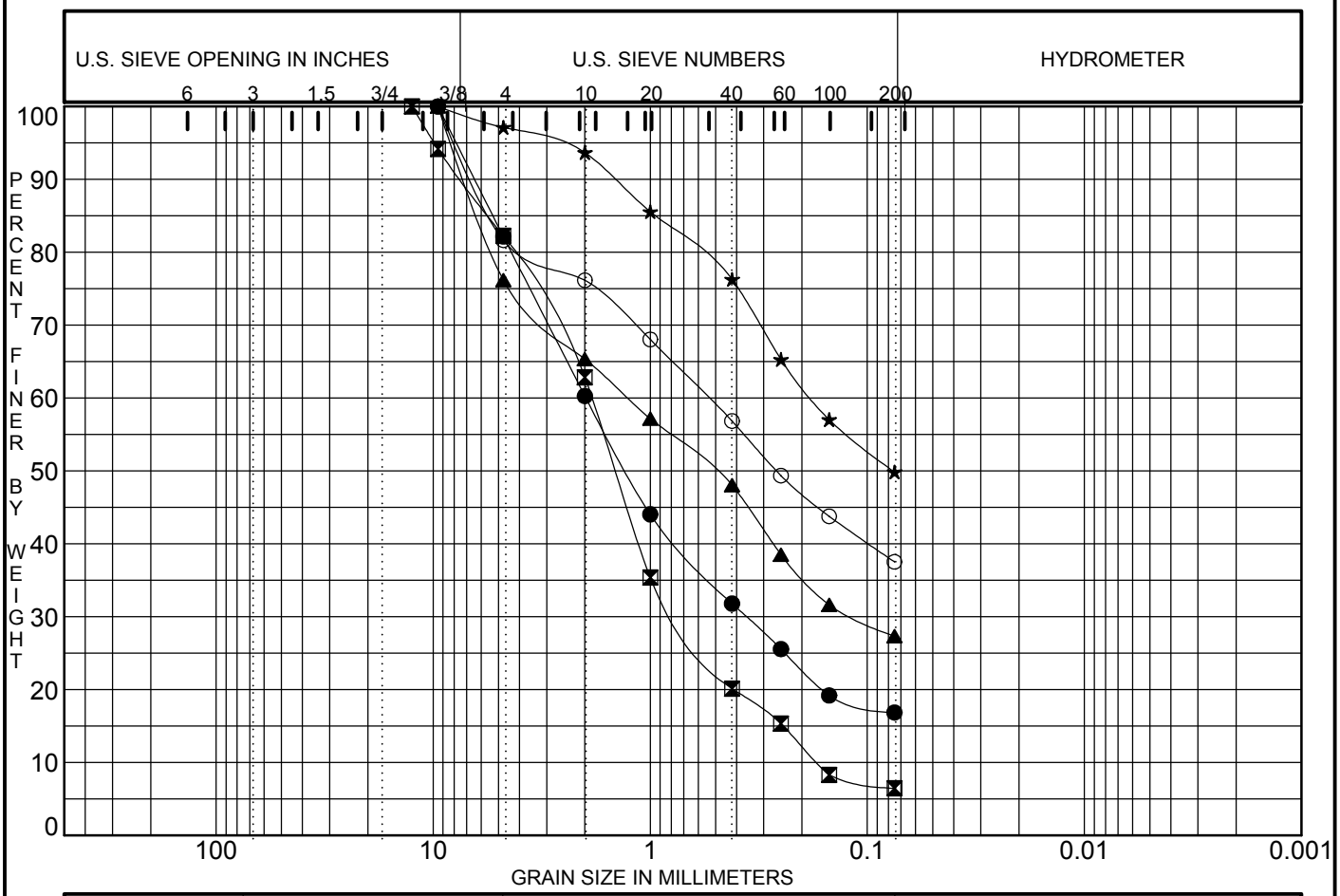
RIYADH GEOTECHNIQUE & FOUNDATIONS
AL-KHOBAR, SAUDI ARABIA

PROJECT # **MANSOURAH MASSARAH GOLD PROJECT**

CLIENT: **L & T CONSTRUCTION**

BOREHOLE NO. **BH-16**
LOCATION:

SIEVE ANALYSIS
ASTM D 422



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Sample No.	Depth(m)	Classification	MC%	LL	PL	PI	Cc	Cu
●	BH-15	0.75	SILTY SAND with GRAVEL (SM)	15	NP	NP	NP		
⊠	BH-15	2.25	WELL-GRADED SAND with SILT and GRAVEL (SW-SM)	14	NP	NP	NP	1.71	11.0
▲	BH-16	0.75	CLAYEY SAND with GRAVEL (SC)	11	28	20	8		
★	BH-16	3.75	CLAYEY SAND (SC)	14	31	22	9		
⊙	BH-16	6.00	CLAYEY SAND with GRAVEL (SC)	20	26	18	8		

	Sample No.	Depth(m)	D100	D50	D30	D10	%Gravel	%Sand	%Silt	%Clay
●	BH-15	0.75	9.50	1.29	0.362		17.8	65.3	16.9	
⊠	BH-15	2.25	12.50	1.45	0.735	0.1696	17.7	75.8	6.5	
▲	BH-16	0.75	9.50	0.51	0.115		23.8	48.8	27.3	
★	BH-16	3.75	9.50	0.08			2.9	47.3	49.9	
⊙	BH-16	6.00	9.50	0.26			18.4	44.1	37.5	

RIYADH GEOTECHNIQUE & FOUNDATIONS

AL-KHOBAR, SAUDI ARABIA

PROJECT # **MANSOURAH MASSARAH GOLD PROJECT**

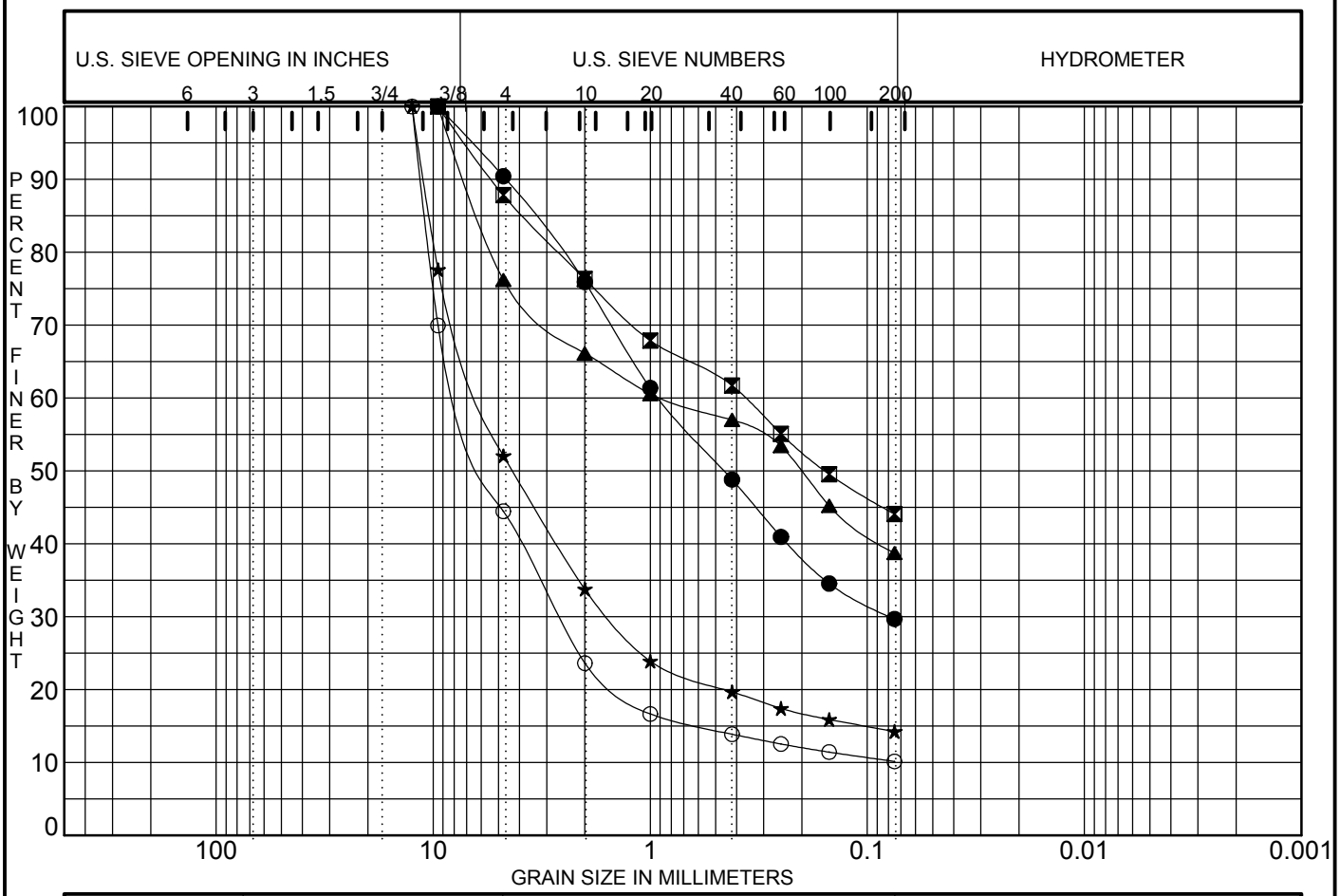
CLIENT: **L & T CONSTRUCTION**

BOREHOLE NO. **BH-17**

LOCATION:

SIEVE ANALYSIS

ASTM D 422



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Sample No.	Depth(m)	Classification	MC%	LL	PL	PI	Cc	Cu
●	BH-16	9.00	CLAYEY SAND (SC)	16	27	19	8		
⊠	BH-17	0.15	CLAYEY SAND (SC)	1	30	2	28		
▲	BH-17	0.75	CLAYEY SAND with GRAVEL (SC)	11	29	19	10		
★	BH-17	3.00	SILTY GRAVEL with SAND (GM)	4	NP	NP	NP		
⊙	BH-17	4.50	POORLY GRADED GRAVEL with SILT and SAND (GP-GM)	4	NP	NP	NP	13.57	105.0

	Sample No.	Depth(m)	D100	D50	D30	D10	%Gravel	%Sand	%Silt	%Clay
●	BH-16	9.00	9.50	0.46	0.078		9.6	60.7	29.7	
⊠	BH-17	0.15	9.50	0.16			12.2	43.8	44.1	
▲	BH-17	0.75	9.50	0.20			23.8	37.4	38.7	
★	BH-17	3.00	12.50	4.31	1.534		47.9	37.8	14.3	
⊙	BH-17	4.50	12.50	5.52	2.605		55.5	34.3	10.2	

RIYADH GEOTECHNIQUE & FOUNDATIONS

AL-KHOBAR, SAUDI ARABIA

PROJECT # **MANSOURAH MASSARAH GOLD PROJECT**

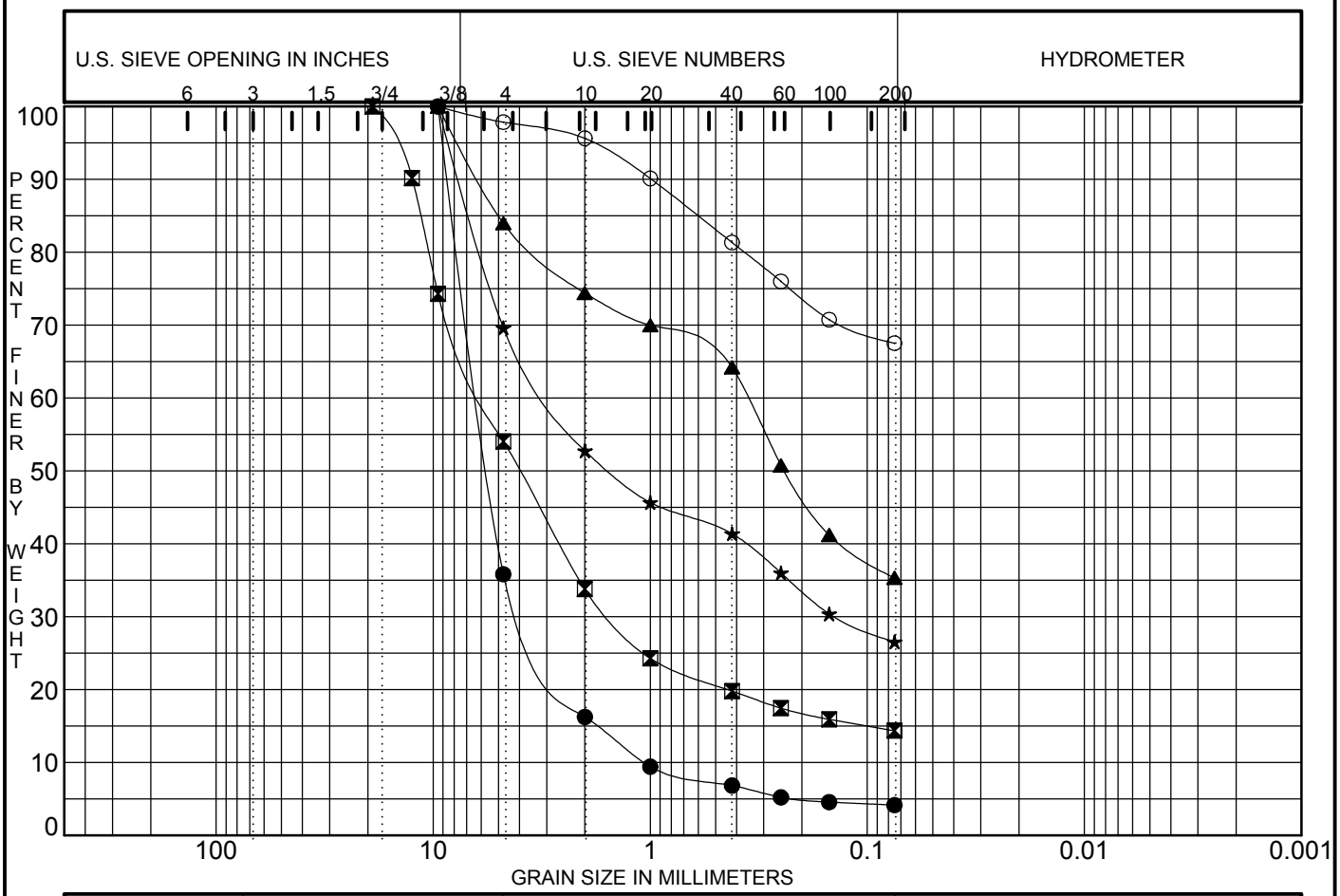
CLIENT: **L & T CONSTRUCTION**

BOREHOLE NO. **BH-18**

LOCATION:

SIEVE ANALYSIS

ASTM D 422



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Sample No.	Depth(m)	Classification	MC%	LL	PL	PI	Cc	Cu
●	BH-17	6.00	WELL-GRADED GRAVEL with SAND (GW)	5	NP	NP	NP	2.06	5.8
⊠	BH-17	7.50	SILTY GRAVEL with SAND (GM)	4	NP	NP	NP		
▲	BH-18	0.15	CLAYEY SAND with GRAVEL (SC)	1	25	17	8		
★	BH-18	0.75	CLAYEY SAND with GRAVEL (SC)	9	31	21	10		
⊙	BH-18	3.00	SANDY SILT (ML)	10	37	28	9		

	Sample No.	Depth(m)	D100	D50	D30	D10	%Gravel	%Sand	%Silt	%Clay
●	BH-17	6.00	9.50	5.54	3.672	1.0598	64.2	31.7	4.2	
⊠	BH-17	7.50	19.00	4.00	1.517		45.9	39.7	14.4	
▲	BH-18	0.15	9.50	0.24			16.0	48.6	35.3	
★	BH-18	0.75	9.50	1.53	0.140		30.4	43.1	26.6	
⊙	BH-18	3.00	9.50				2.2	30.3	67.5	

RIYADH GEOTECHNIQUE & FOUNDATIONS

AL-KHOBAR, SAUDI ARABIA

PROJECT # **MANSOURAH MASSARAH GOLD PROJECT**

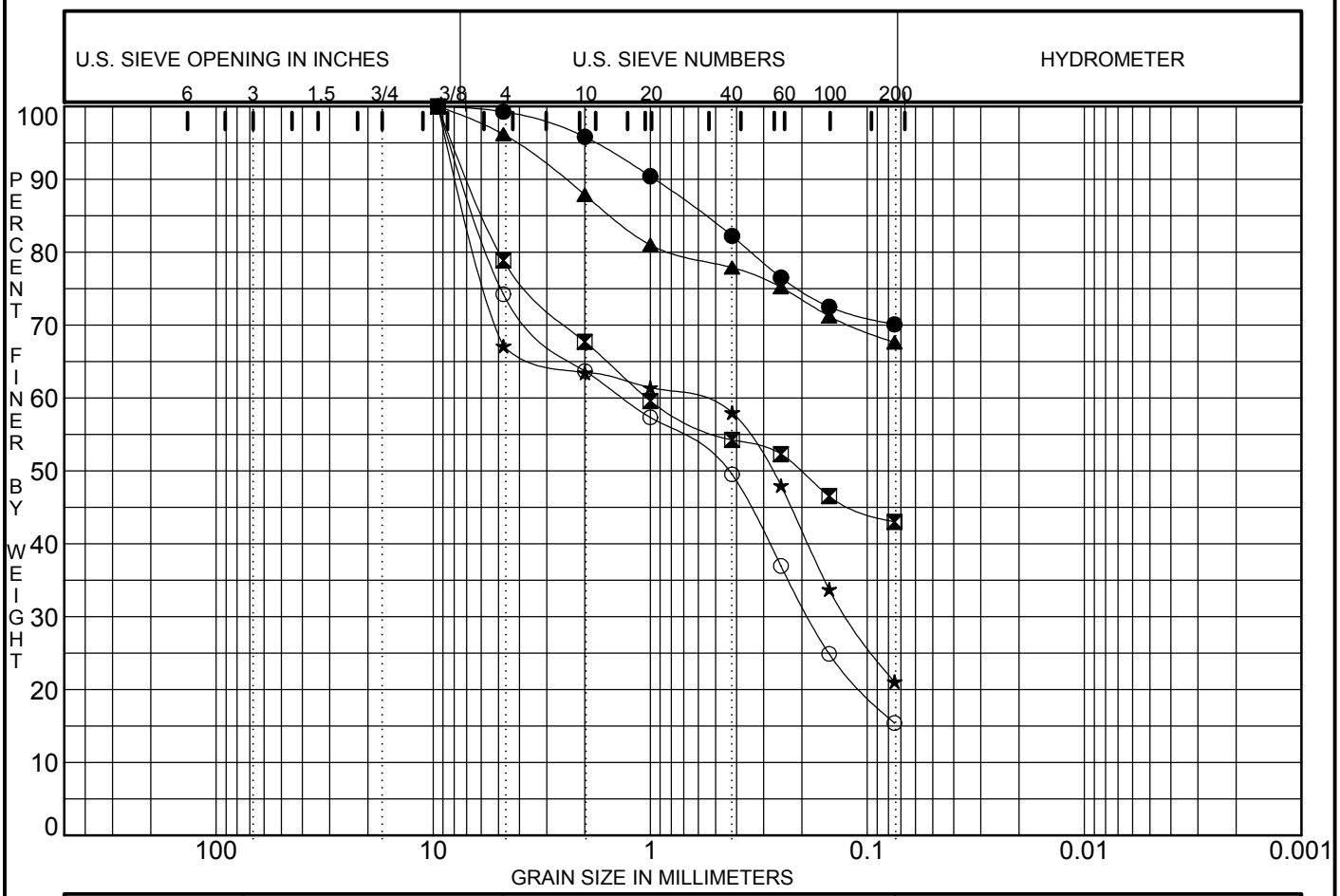
CLIENT: **L & T CONSTRUCTION**

BOREHOLE NO. **BH-19**

LOCATION:

SIEVE ANALYSIS

ASTM D 422



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Sample No.	Depth(m)	Classification	MC%	LL	PL	PI	Cc	Cu
●	BH-18	3.75	SILT with SAND (ML)	8	40	30	10		
⊠	BH-18	7.50	SILTY SAND with GRAVEL (SM)	12	37	29	8		
▲	BH-18	10.50	SANDY SILT (ML)	19	36	27	9		
★	BH-19	0.15	SILTY SAND with GRAVEL (SM)	2	NP	NP	NP		
⊙	BH-19	0.75	SILTY SAND with GRAVEL (SM)	8	NP	NP	NP		

	Sample No.	Depth(m)	D100	D50	D30	D10	%Gravel	%Sand	%Silt	%Clay
●	BH-18	3.75	9.50				0.7	29.2	70.1	
⊠	BH-18	7.50	9.50	0.20			21.1	35.9	43.0	
▲	BH-18	10.50	9.50				3.8	28.5	67.6	
★	BH-19	0.15	9.50	0.28	0.122		32.9	46.0	21.1	
⊙	BH-19	0.75	9.50	0.44	0.186		25.7	58.8	15.4	

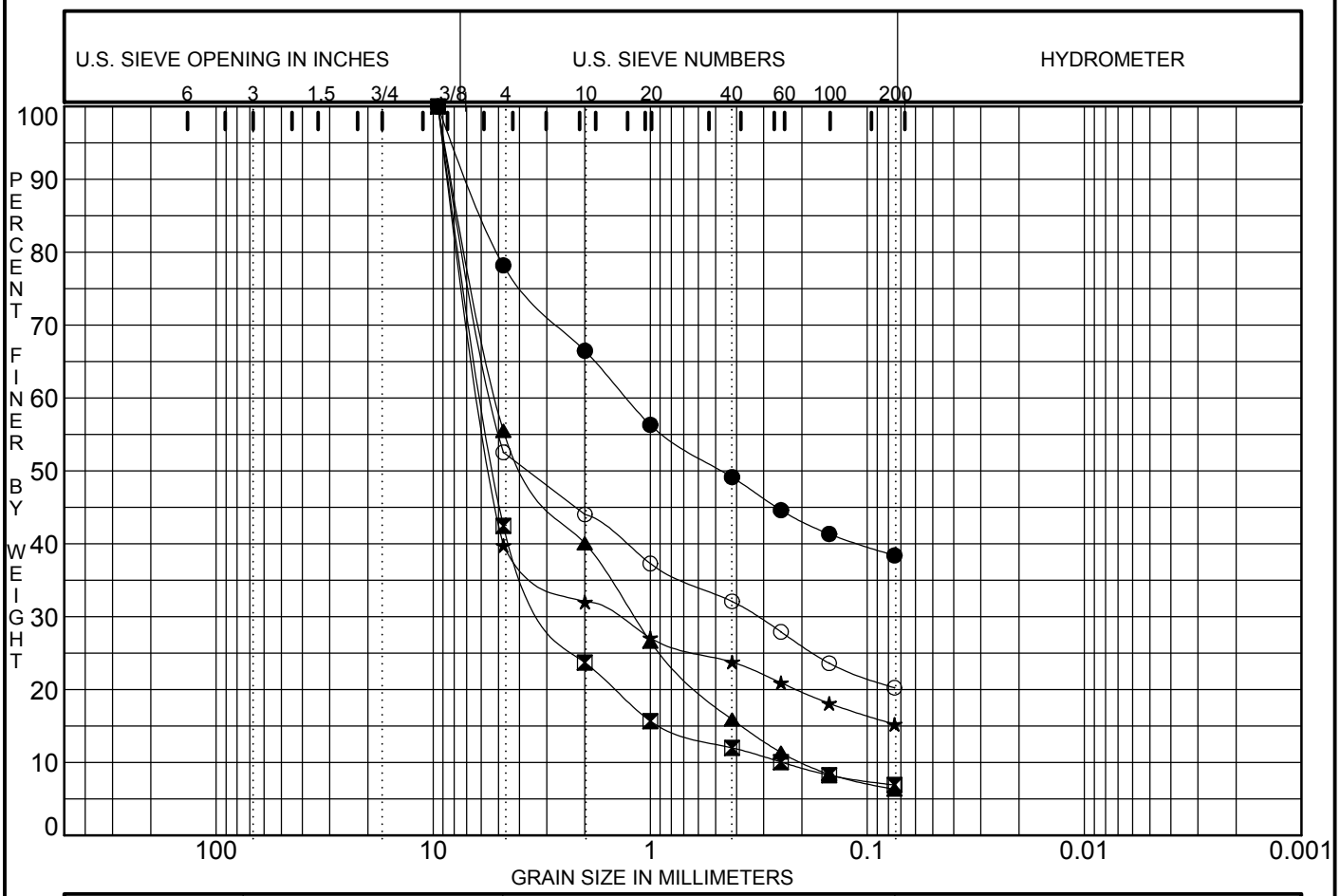
RIYADH GEOTECHNIQUE & FOUNDATIONS
AL-KHOBAR, SAUDI ARABIA

PROJECT # **MANSOURAH MASSARAH GOLD PROJECT**

CLIENT: **L & T CONSTRUCTION**

BOREHOLE NO. **BH-20**
LOCATION:

SIEVE ANALYSIS
ASTM D 422



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Sample No.	Depth(m)	Classification	MC%	LL	PL	PI	Cc	Cu
●	BH-19	3.75	CLAYEY SAND with GRAVEL (SC)	10	30	21	9		
⊠	BH-19	15.00	POORLY GRADED GRAVEL with SILT and SAND (GP-GM)	5	NP	NP	NP	4.96	23.9
▲	BH-20	0.15	WELL-GRADED SAND with SILT and GRAVEL (SW-SM)	2	NP	NP	NP	1.41	25.8
★	BH-20	1.50	SILTY GRAVEL with SAND (GM)	9	NP	NP	NP		
⊙	BH-20	2.25	SILTY GRAVEL with SAND (GM)	6	NP	NP	NP		

	Sample No.	Depth(m)	D100	D50	D30	D10	%Gravel	%Sand	%Silt	%Clay
●	BH-19	3.75	9.50	0.47			21.8	39.8	38.4	
⊠	BH-19	15.00	9.50	5.20	2.674	0.2459	57.5	35.5	6.9	
▲	BH-20	0.15	9.50	3.49	1.191	0.1975	44.5	49.2	6.3	
★	BH-20	1.50	9.50	5.34	1.506		60.2	24.5	15.2	
⊙	BH-20	2.25	9.50	3.67	0.323		47.4	32.3	20.3	

RIYADH GEOTECHNIQUE & FOUNDATIONS

AL-KHOBAR, SAUDI ARABIA

PROJECT # **MANSOURAH MASSARAH GOLD PROJECT**

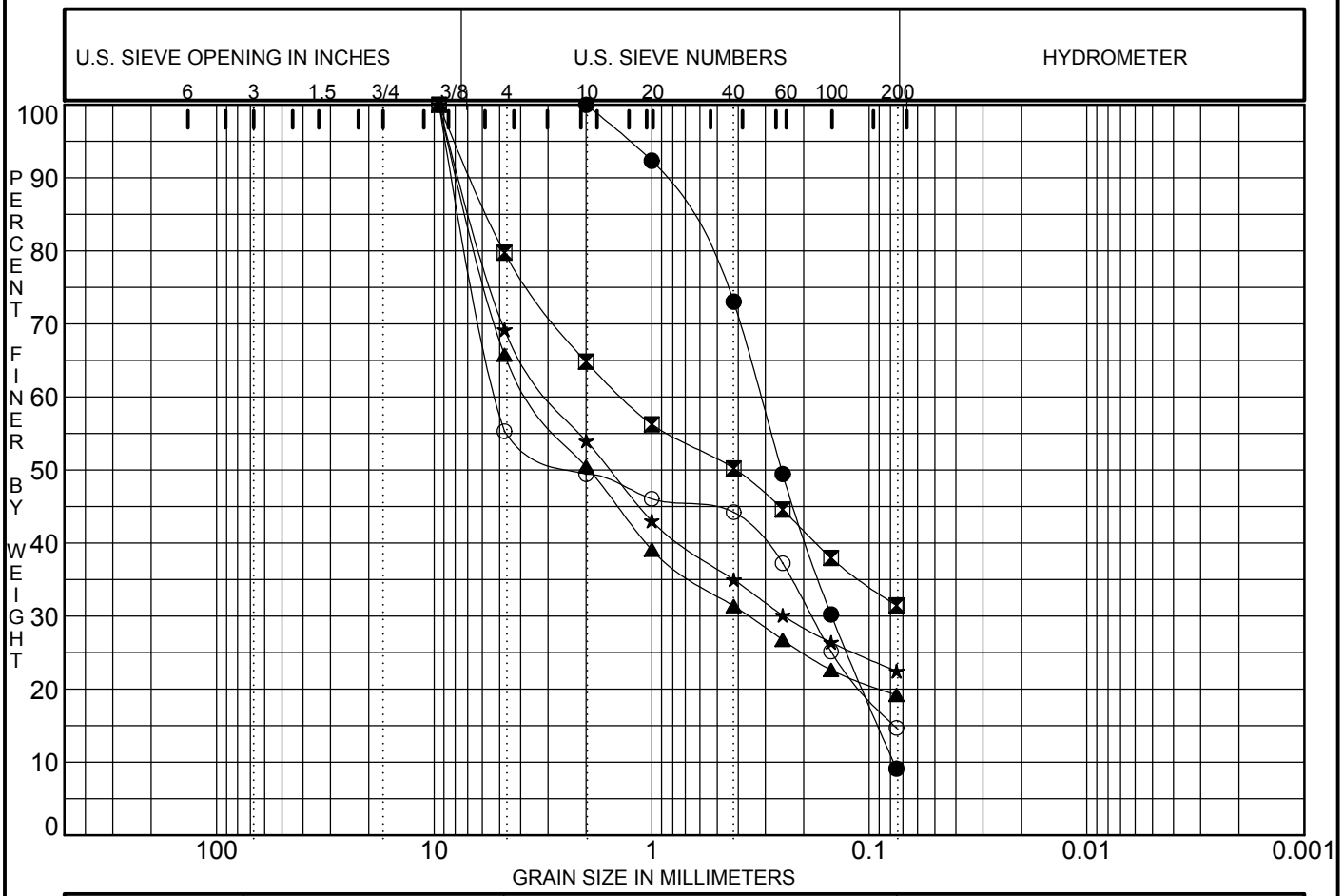
CLIENT: **L & T CONSTRUCTION**

BOREHOLE NO. **BH-23**

LOCATION:

SIEVE ANALYSIS

ASTM D 422



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Sample No.	Depth(m)	Classification	MC%	LL	PL	PI	Cc	Cu
●	BH-21	0.15	POORLY GRADED SAND with SILT (SP-SM)	2	NP	NP	NP	0.91	4.1
⊠	BH-21	0.75	CLAYEY SAND with GRAVEL (SC)	8	29	20	9		
▲	BH-21	2.25	SILTY SAND with GRAVEL (SM)	11	NP	NP	NP		
★	BH-21	3.75	SILTY SAND with GRAVEL (SM)	8	NP	NP	NP		
⊙	BH-23	0.15	SILTY GRAVEL with SAND (GM)	0	NP	NP	NP		

	Sample No.	Depth(m)	D100	D50	D30	D10	%Gravel	%Sand	%Silt	%Clay
●	BH-21	0.15	2.00	0.25	0.149	0.0772	0.0	90.9	9.1	
⊠	BH-21	0.75	9.50	0.41			20.2	48.3	31.5	
▲	BH-21	2.25	9.50	1.94	0.360		34.2	46.6	19.2	
★	BH-21	3.75	9.50	1.55	0.246		30.8	46.7	22.5	
⊙	BH-23	0.15	9.50	2.18	0.184		44.7	40.6	14.7	

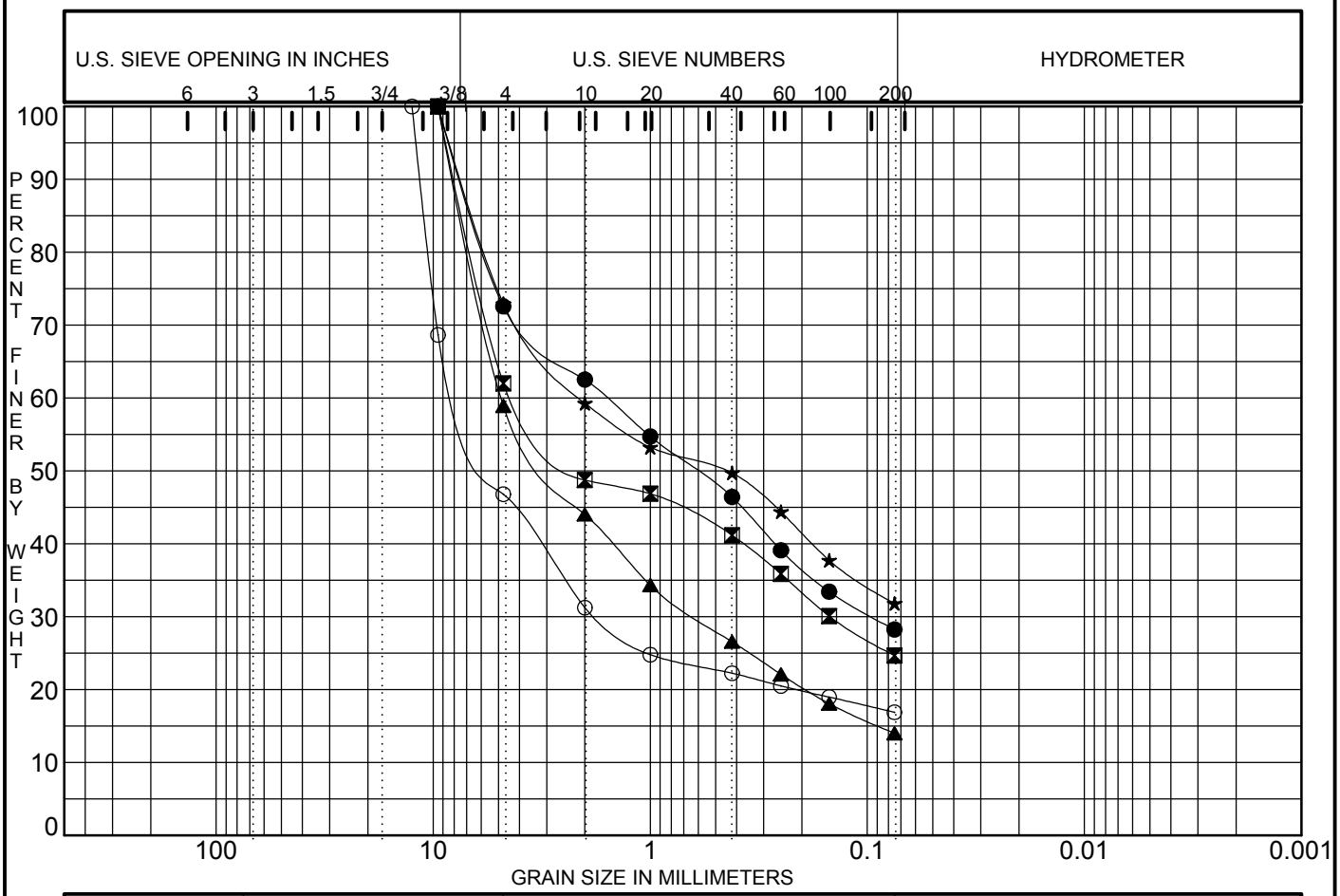
RIYADH GEOTECHNIQUE & FOUNDATIONS
AL-KHOBAR, SAUDI ARABIA

PROJECT # **MANSOURAH MASSARAH GOLD PROJECT**

CLIENT: **L & T CONSTRUCTION**

BOREHOLE NO. **BH-25**
LOCATION:

SIEVE ANALYSIS
ASTM D 422



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Sample No.	Depth(m)	Classification	MC%	LL	PL	PI	Cc	Cu
●	BH-23	2.25	SILTY SAND with GRAVEL (SM)	12	NP	NP	NP		
⊠	BH-23	3.75	SILTY GRAVEL with SAND (GM)	6	NP	NP	NP		
▲	BH-23	6.00	SILTY SAND with GRAVEL (SM)	9	NP	NP	NP		
★	BH-25	0.15	SILTY SAND with GRAVEL (SM)	1	31	23	8		
⊙	BH-25	1.50	SILTY GRAVEL with SAND (GM)	5	NP	NP	NP		

	Sample No.	Depth(m)	D100	D50	D30	D10	%Gravel	%Sand	%Silt	%Clay
●	BH-23	2.25	9.50	0.61	0.095		27.4	44.3	28.3	
⊠	BH-23	3.75	9.50	2.17	0.148		38.0	37.3	24.7	
▲	BH-23	6.00	9.50	2.83	0.616		41.1	44.9	14.0	
★	BH-25	0.15	9.50	0.45			27.1	41.1	31.8	
⊙	BH-25	1.50	12.50	5.26	1.750		53.2	29.9	16.9	

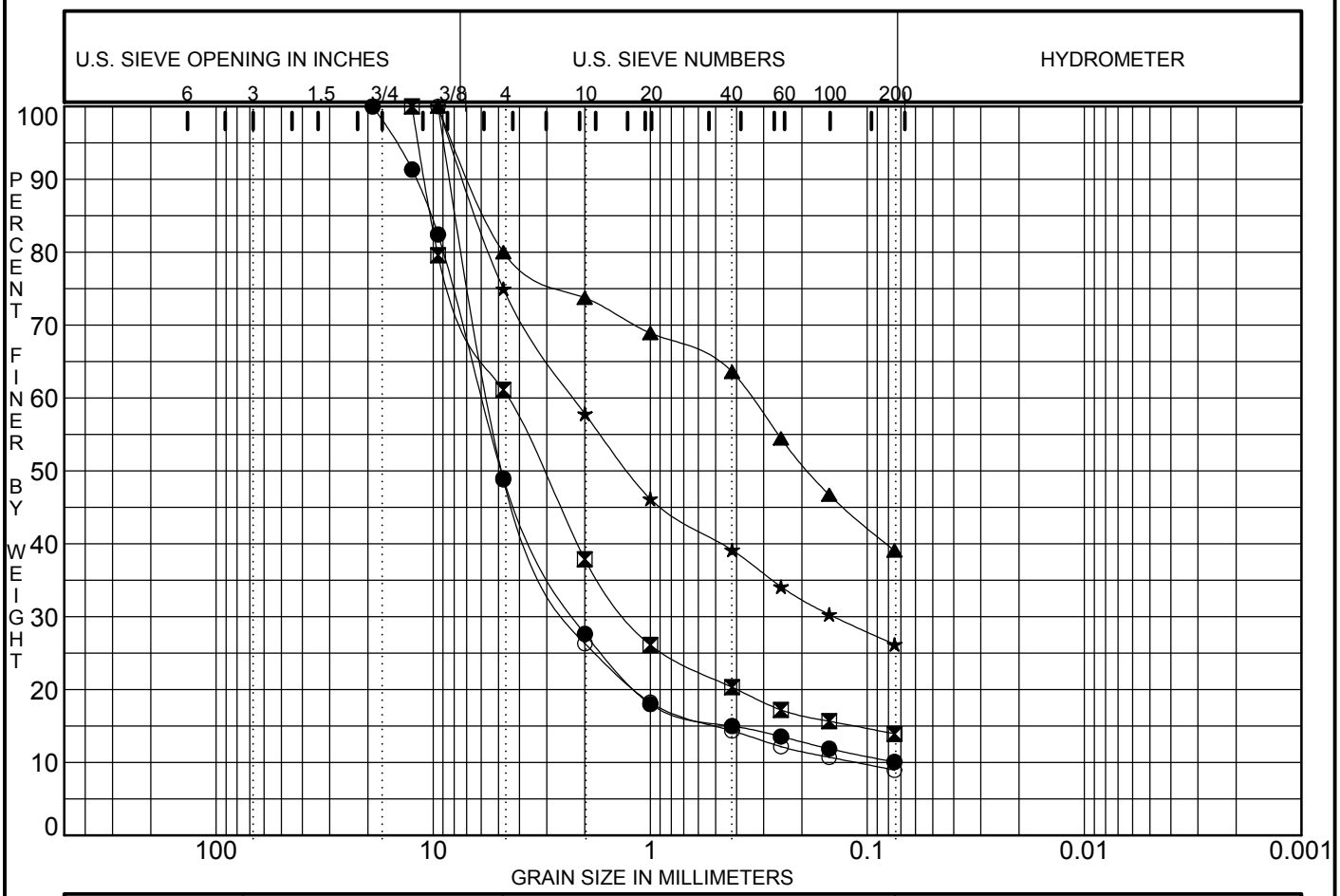
RIYADH GEOTECHNIQUE & FOUNDATIONS
AL-KHOBAR, SAUDI ARABIA

PROJECT # **MANSOURAH MASSARAH GOLD PROJECT**

CLIENT: **L & T CONSTRUCTION**

BOREHOLE NO. **BH-26**
LOCATION:

SIEVE ANALYSIS
ASTM D 422



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Sample No.	Depth(m)	Classification	MC%	LL	PL	PI	Cc	Cu
●	BH-25	3.00	POORLY GRADED GRAVEL with SILT and SAND (GP-GM)	7	NP	NP	NP	11.19	82.3
⊠	BH-25	7.50	SILTY SAND with GRAVEL (SM)	4	NP	NP	NP		
▲	BH-26	0.15	CLAYEY SAND with GRAVEL (SC)	2	30	22	8		
★	BH-26	0.75	SILTY SAND with GRAVEL (SM)	9	NP	NP	NP		
⊙	BH-26	2.25	POORLY GRADED GRAVEL with SILT and SAND (GP-GM)	5	NP	NP	NP	8.51	49.0

	Sample No.	Depth(m)	D100	D50	D30	D10	%Gravel	%Sand	%Silt	%Clay
●	BH-25	3.00	19.00	4.85	2.201		51.0	38.9	10.1	
⊠	BH-25	7.50	12.50	3.14	1.257		38.9	47.2	13.9	
▲	BH-26	0.15	9.50	0.19			20.0	40.9	39.1	
★	BH-26	0.75	9.50	1.26	0.142		25.0	48.8	26.2	
⊙	BH-26	2.25	9.50	4.83	2.304	0.1129	51.2	39.8	9.0	

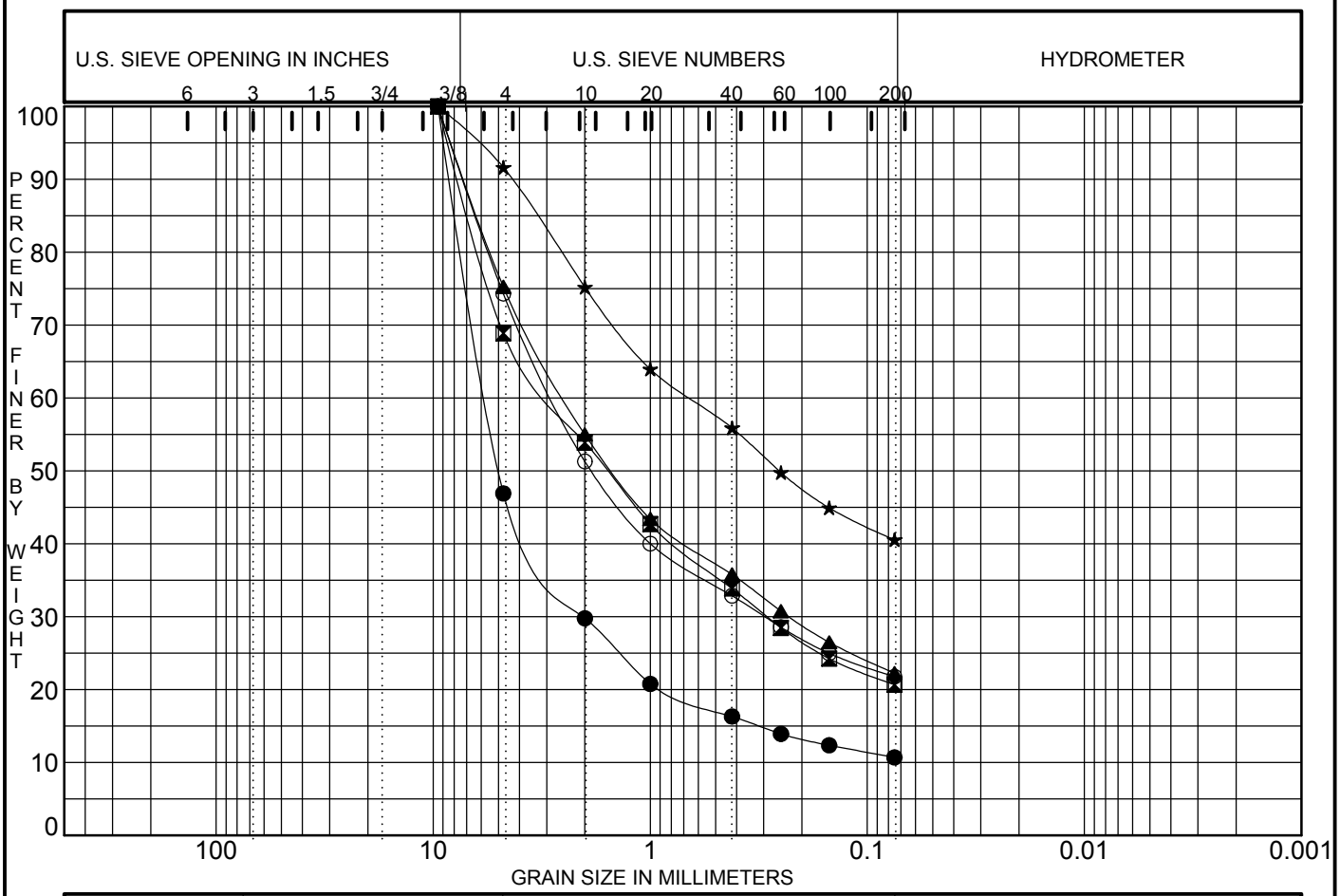
RIYADH GEOTECHNIQUE & FOUNDATIONS
AL-KHOBAR, SAUDI ARABIA

PROJECT # **MANSOURAH MASSARAH GOLD PROJECT**

CLIENT: **L & T CONSTRUCTION**

BOREHOLE NO. **BH-27**
LOCATION:

SIEVE ANALYSIS
ASTM D 422



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Sample No.	Depth(m)	Classification	MC%	LL	PL	PI	Cc	Cu
●	BH-26	4.50	POORLY GRADED GRAVEL with SILT and SAND (GP-GM)	9	NP	NP	NP	12.92	100.4
⊠	BH-26	7.50	SILTY SAND with GRAVEL (SM)	16	NP	NP	NP		
▲	BH-27	0.15	SILTY SAND with GRAVEL (SM)	3	NP	NP	NP		
★	BH-27	1.50	CLAYEY SAND (SC)	13	30	20	10		
⊙	BH-27	3.00	SILTY SAND with GRAVEL (SM)	11	NP	NP	NP		

	Sample No.	Depth(m)	D100	D50	D30	D10	%Gravel	%Sand	%Silt	%Clay
●	BH-26	4.50	9.50	4.95	2.022		53.1	36.2	10.7	
⊠	BH-26	7.50	9.50	1.57	0.290		31.1	48.2	20.7	
▲	BH-27	0.15	9.50	1.48	0.228		24.7	53.0	22.3	
★	BH-27	1.50	9.50	0.25			8.4	51.1	40.6	
⊙	BH-27	3.00	9.50	1.85	0.297		25.7	52.5	21.8	

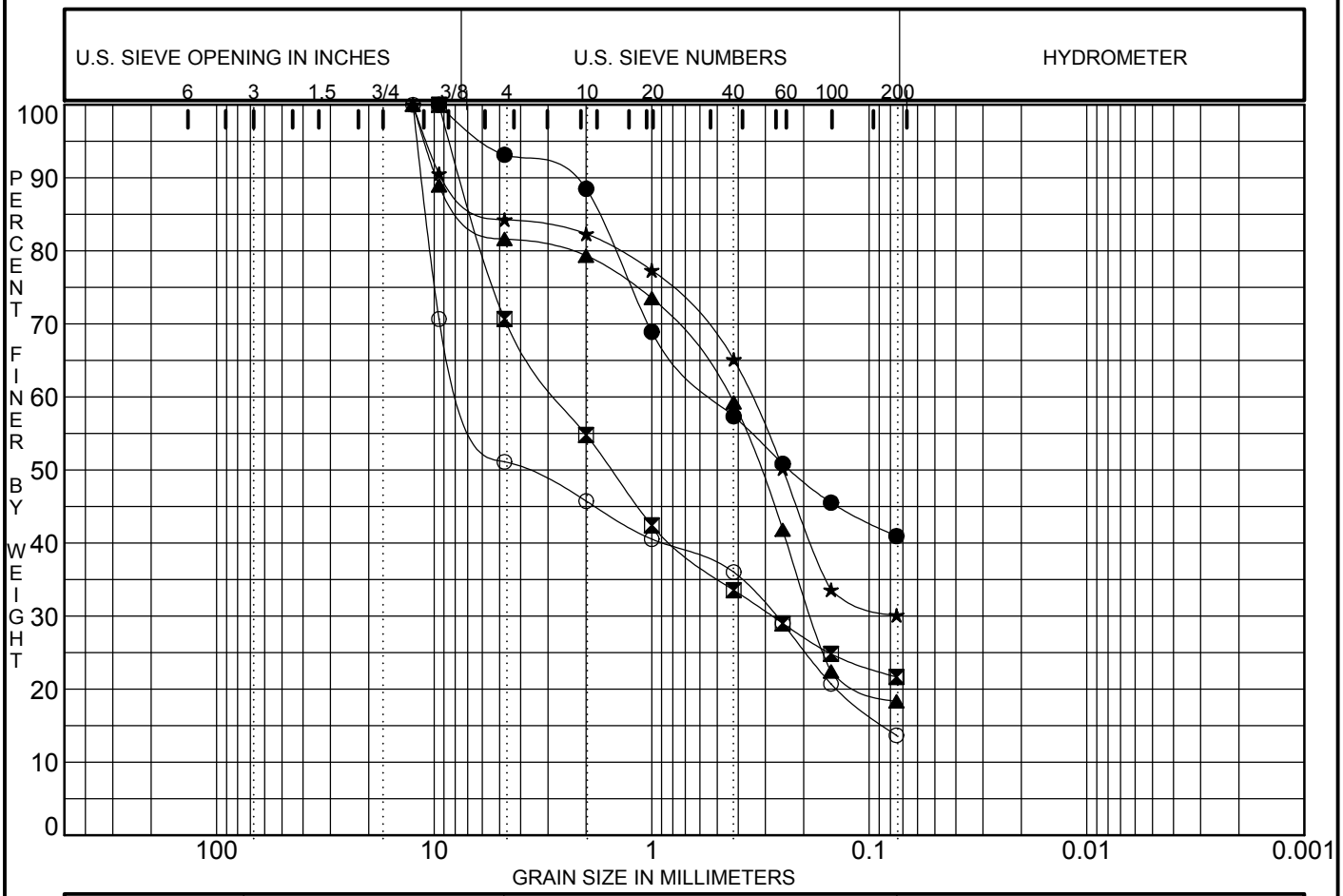
RIYADH GEOTECHNIQUE & FOUNDATIONS
AL-KHOBAR, SAUDI ARABIA

PROJECT # **MANSOURAH MASSARAH GOLD PROJECT**

CLIENT: **L & T CONSTRUCTION**

BOREHOLE NO. **TP-02**
LOCATION:

SIEVE ANALYSIS
ASTM D 422



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Sample No.	Depth(m)	Classification	MC%	LL	PL	PI	Cc	Cu
●	BH-27	4.50	CLAYEY SAND (SC)	11	29	21	8		
⊠	BH-27	7.50	SILTY SAND with GRAVEL (SM)	12	NP	NP	NP		
▲	TP-01	0.00	SILTY SAND with GRAVEL (SM)	1	NP	NP	NP		
★	TP-02	0.00	CLAYEY SAND with GRAVEL (SC)	1	28	20	8		
⊙	TP-02	0.70	SILTY GRAVEL with SAND (GM)	2	NP	NP	NP		

	Sample No.	Depth(m)	D100	D50	D30	D10	%Gravel	%Sand	%Silt	%Clay
●	BH-27	4.50	9.50	0.23			6.8	52.2	41.0	
⊠	BH-27	7.50	9.50	1.53	0.280		29.3	49.0	21.7	
▲	TP-01	0.00	12.50	0.32	0.183		18.4	63.3	18.3	
★	TP-02	0.00	12.50	0.25			15.7	54.2	30.1	
⊙	TP-02	0.70	12.50	3.96	0.269		48.9	37.4	13.7	

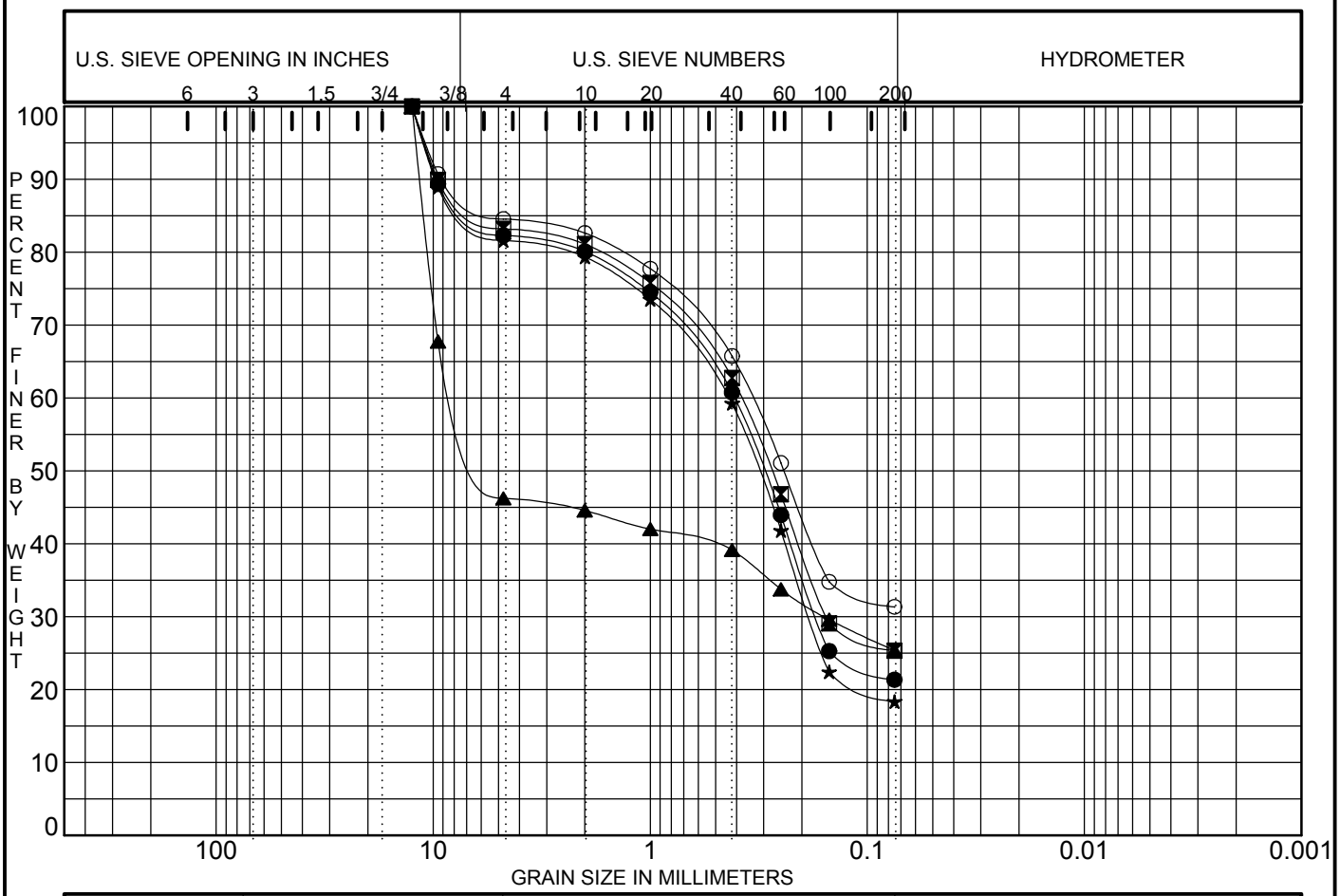
RIYADH GEOTECHNIQUE & FOUNDATIONS
AL-KHOBAR, SAUDI ARABIA

PROJECT # **MANSOURAH MASSARAH GOLD PROJECT**

CLIENT: **L & T CONSTRUCTION**

BOREHOLE NO. **TP-05**
LOCATION:

SIEVE ANALYSIS
ASTM D 422



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Sample No.	Depth(m)	Classification	MC%	LL	PL	PI	Cc	Cu
●	TP-03	0.00	SILTY SAND with GRAVEL (SM)	2	NP	NP	NP		
⊠	TP-04	0.00	SILTY SAND with GRAVEL (SM)	1	NP	NP	NP		
▲	TP-04	0.30	SILTY GRAVEL with SAND (GM)	2	NP	NP	NP		
★	TP-05	0.00	SILTY SAND with GRAVEL (SM)	1	NP	NP	NP		
⊙	TP-05	0.30	CLAYEY SAND with GRAVEL (SC)	2	29	21	8		

	Sample No.	Depth(m)	D100	D50	D30	D10	%Gravel	%Sand	%Silt	%Clay
●	TP-03	0.00	12.50	0.30	0.171		17.7	60.9	21.4	
⊠	TP-04	0.00	12.50	0.28	0.154		16.8	57.8	25.4	
▲	TP-04	0.30	12.50	5.36	0.158		53.8	20.9	25.4	
★	TP-05	0.00	12.50	0.32	0.183		18.4	63.3	18.4	
⊙	TP-05	0.30	12.50	0.24			15.4	53.2	31.4	

RIYADH GEOTECHNIQUE & FOUNDATIONS

AL-KHOBAR, SAUDI ARABIA

PROJECT # **MANSOURAH MASSARAH GOLD PROJECT**

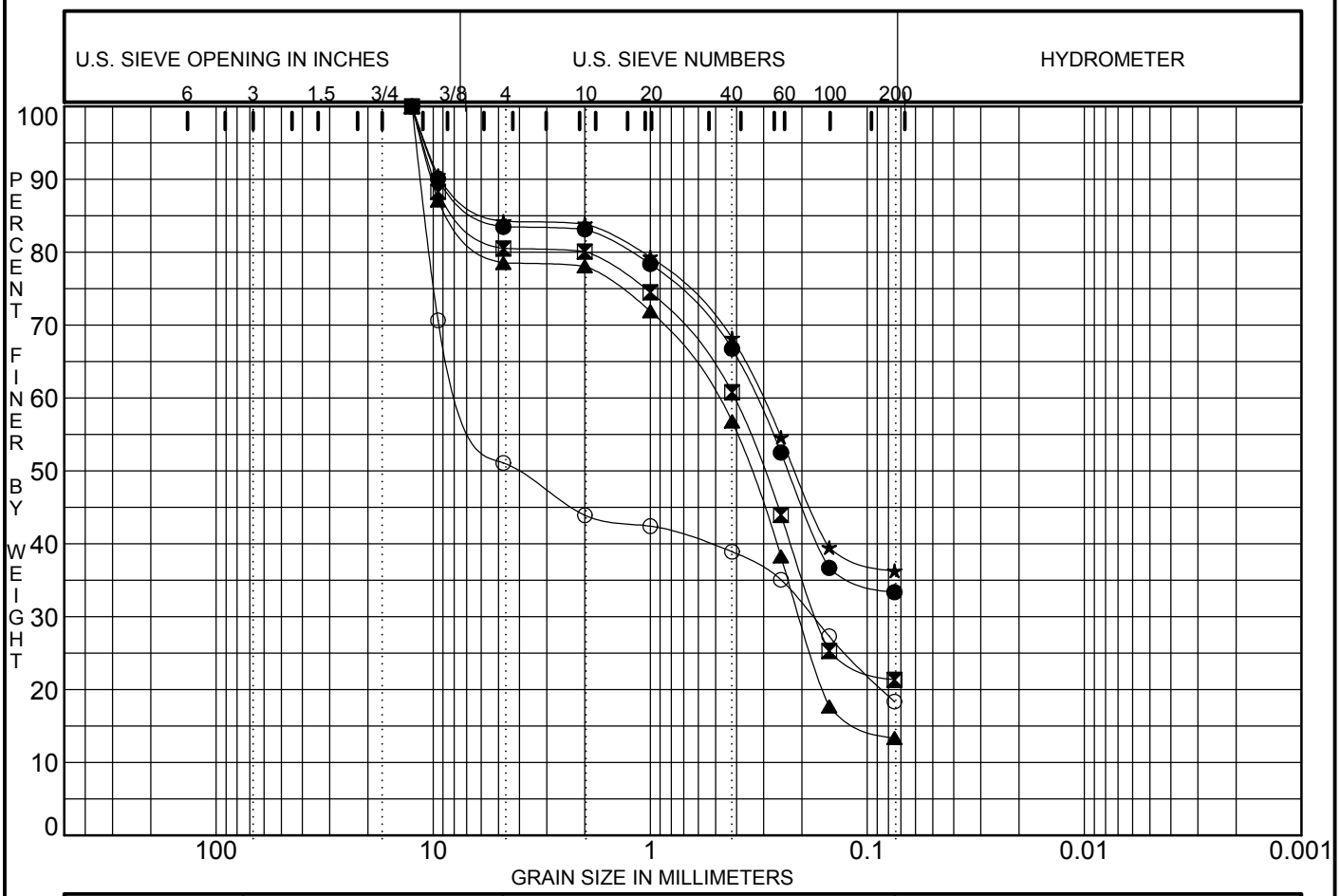
CLIENT: **L & T CONSTRUCTION**

BOREHOLE NO. **TP-09**

LOCATION:

SIEVE ANALYSIS

ASTM D 422



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Sample No.	Depth(m)	Classification	MC%	LL	PL	PI	Cc	Cu
●	TP-06	0.00	CLAYEY SAND with GRAVEL (SC)	3	30	21	9		
⊠	TP-07	0.00	SILTY SAND with GRAVEL (SM)	1	NP	NP	NP		
▲	TP-07	0.60	SILTY SAND with GRAVEL (SM)	1	NP	NP	NP		
★	TP-08	0.00	SILTY SAND with GRAVEL (SM)	3	NP	NP	NP		
⊙	TP-09	0.00	SILTY GRAVEL with SAND (GM)	0	NP	NP	NP		

	Sample No.	Depth(m)	D100	D50	D30	D10	%Gravel	%Sand	%Silt	%Clay
●	TP-06	0.00	12.50	0.23			16.5	50.1	33.4	
⊠	TP-07	0.00	12.50	0.30	0.171		19.5	59.1	21.4	
▲	TP-07	0.60	12.50	0.35	0.204		21.5	65.2	13.4	
★	TP-08	0.00	12.50	0.21			15.8	47.9	36.3	
⊙	TP-09	0.00	12.50	4.16	0.179		48.9	32.7	18.4	

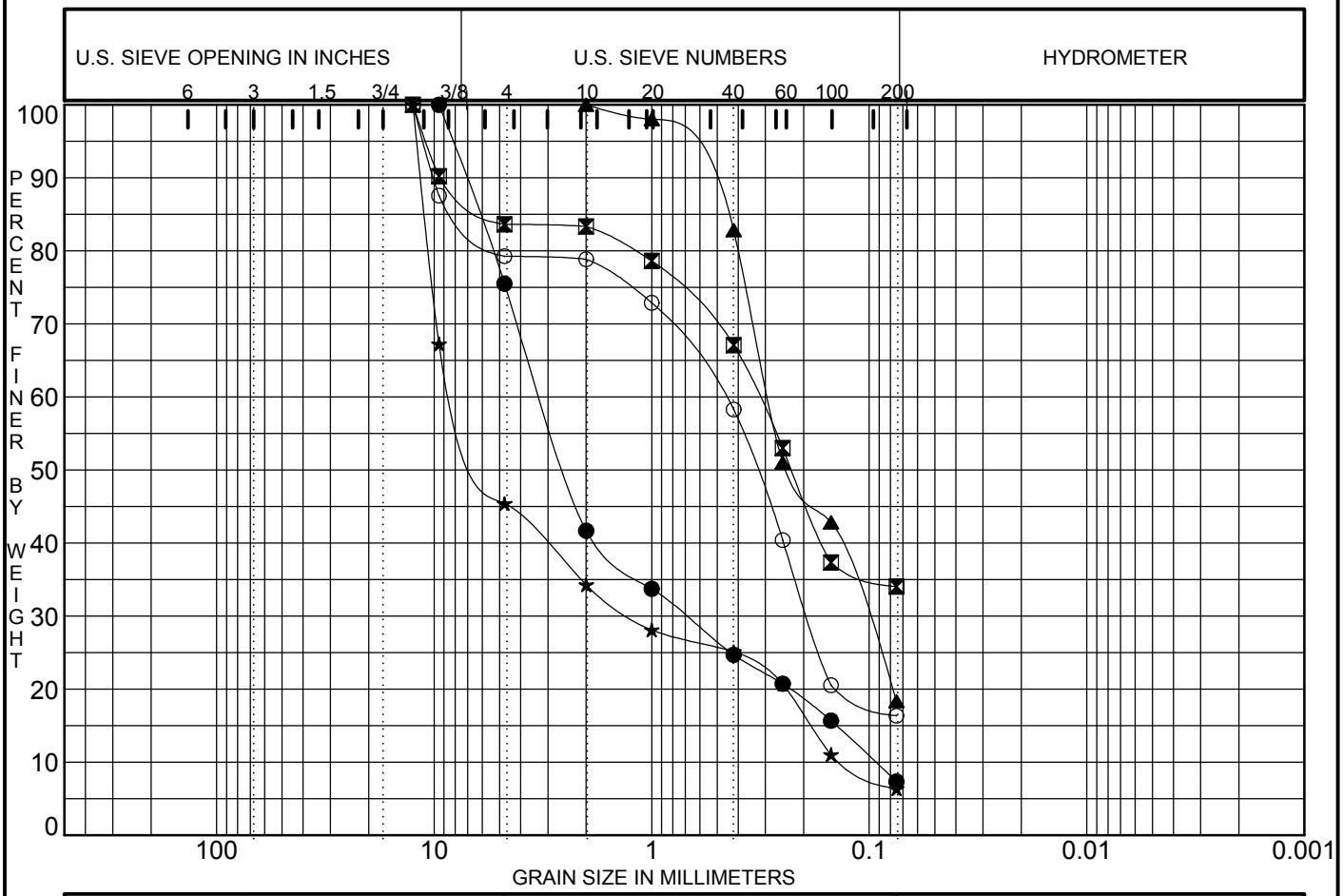
RIYADH GEOTECHNIQUE & FOUNDATIONS
AL-KHOBAR, SAUDI ARABIA

PROJECT # **MANSOURAH MASSARAH GOLD PROJECT**

CLIENT: **L & T CONSTRUCTION**

BOREHOLE NO. **TP-12**
LOCATION:

SIEVE ANALYSIS
ASTM D 422



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Sample No.	Depth(m)	Classification	MC%	LL	PL	PI	Cc	Cu
●	TP-09	0.50	WELL-GRADED SAND with SILT and GRAVEL (SW-SM)	1	NP	NP	NP	1.63	34.2
⊠	TP-10	0.00	CLAYEY SAND with GRAVEL (SC)	3	30	20	10		
▲	TP-11	0.00	SILTY SAND (SM)	0	NP	NP	NP		
★	TP-11	0.30	WELL-GRADED GRAVEL with SILT and SAND (GW-GM)	2	NP	NP	NP	1.57	58.7
⊙	TP-12	0.00	SILTY SAND with GRAVEL (SM)	1	NP	NP	NP		

	Sample No.	Depth(m)	D100	D50	D30	D10	%Gravel	%Sand	%Silt	%Clay
●	TP-09	0.50	9.50	2.47	0.698	0.0935	24.5	68.2	7.4	
⊠	TP-10	0.00	12.50	0.23			16.3	49.6	34.0	
▲	TP-11	0.00	2.00	0.23	0.104		0.0	81.6	18.4	
★	TP-11	0.30	12.50	5.49	1.236	0.1285	54.6	39.1	6.4	
⊙	TP-12	0.00	12.50	0.33	0.191		20.7	62.9	16.4	

RIYADH GEOTECHNIQUE & FOUNDATIONS

AL-KHOBAR, SAUDI ARABIA

PROJECT # **MANSOURAH MASSARAH GOLD PROJECT**

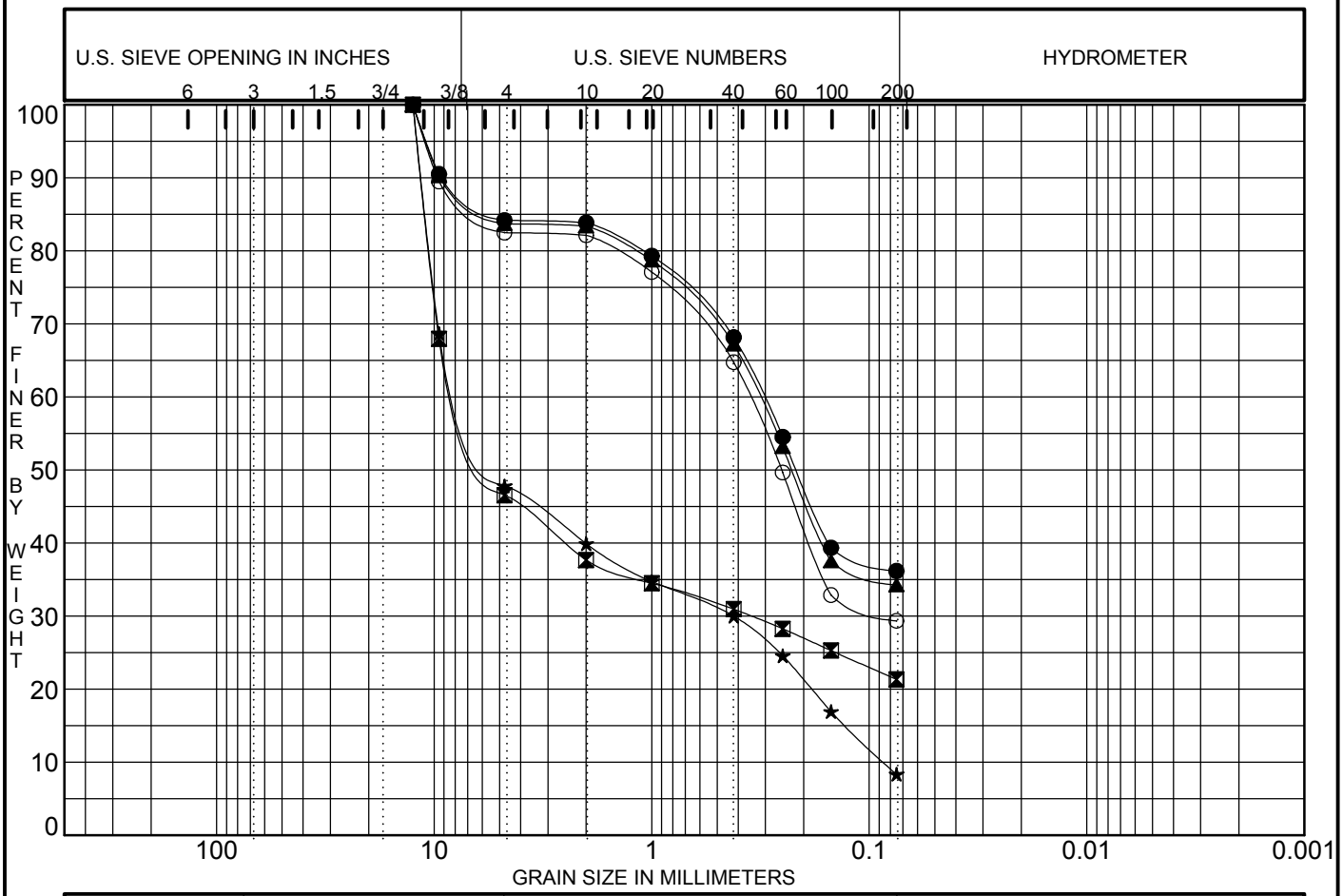
CLIENT: **L & T CONSTRUCTION**

BOREHOLE NO. **TP-15**

LOCATION:

SIEVE ANALYSIS

ASTM D 422



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Sample No.	Depth(m)	Classification	MC%	LL	PL	PI	Cc	Cu
●	TP-13	0.00	CLAYEY SAND with GRAVEL (SC)	2	28	19	9		
⊠	TP-13	0.70	SILTY GRAVEL with SAND (GM)	2	NP	NP	NP		
▲	TP-14	0.00	CLAYEY SAND with GRAVEL (SC)	2	29	21	8		
★	TP-14	0.50	POORLY GRADED GRAVEL with SILT and SAND (GP-GM)	1	NP	NP	NP	0.29	83.1
⊙	TP-15	0.00	CLAYEY SAND with GRAVEL (SC)	3	31	22	9		

	Sample No.	Depth(m)	D100	D50	D30	D10	%Gravel	%Sand	%Silt	%Clay
●	TP-13	0.00	12.50	0.21			15.8	48.0	36.2	
⊠	TP-13	0.70	12.50	5.31	0.349		53.4	25.2	21.4	
▲	TP-14	0.00	12.50	0.23			16.3	49.5	34.3	
★	TP-14	0.50	12.50	5.11	0.420	0.0856	52.2	39.5	8.4	
⊙	TP-15	0.00	12.50	0.25	0.085		17.5	53.1	29.4	

RIYADH GEOTECHNIQUE & FOUNDATIONS

AL-KHOBAR, SAUDI ARABIA

PROJECT # **MANSOURAH MASSARAH GOLD PROJECT**

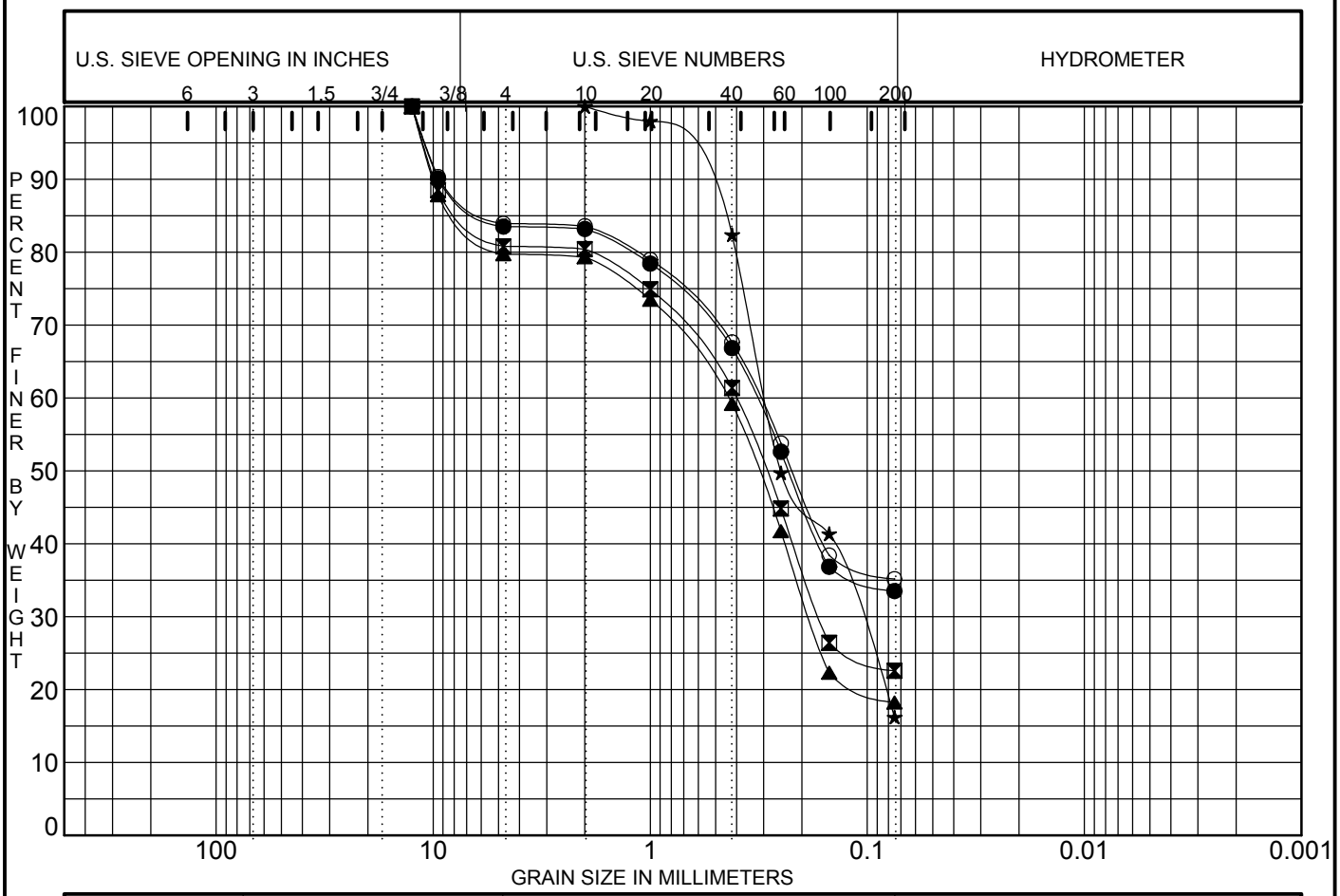
CLIENT: **L & T CONSTRUCTION**

BOREHOLE NO. **TP-19**

LOCATION:

SIEVE ANALYSIS

ASTM D 422



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Sample No.	Depth(m)	Classification	MC%	LL	PL	PI	Cc	Cu
●	TP-16	0.00	CLAYEY SAND with GRAVEL (SC)	3	29	21	8		
⊠	TP-17	0.00	SILTY SAND with GRAVEL (SM)	1	NP	NP	NP		
▲	TP-18	0.00	SILTY SAND with GRAVEL (SM)	2	NP	NP	NP		
★	TP-19	0.00	SILTY SAND (SM)	1	NP	NP	NP		
⊙	TP-19	0.20	CLAYEY SAND with GRAVEL (SC)	2	29	20	9		

	Sample No.	Depth(m)	D100	D50	D30	D10	%Gravel	%Sand	%Silt	%Clay
●	TP-16	0.00	12.50	0.23			16.5	50.0	33.5	
⊠	TP-17	0.00	12.50	0.29	0.166		19.2	58.2	22.6	
▲	TP-18	0.00	12.50	0.32	0.184		20.3	61.5	18.2	
★	TP-19	0.00	2.00	0.25	0.110		0.0	83.8	16.2	
⊙	TP-19	0.20	12.50	0.22			16.1	48.8	35.2	

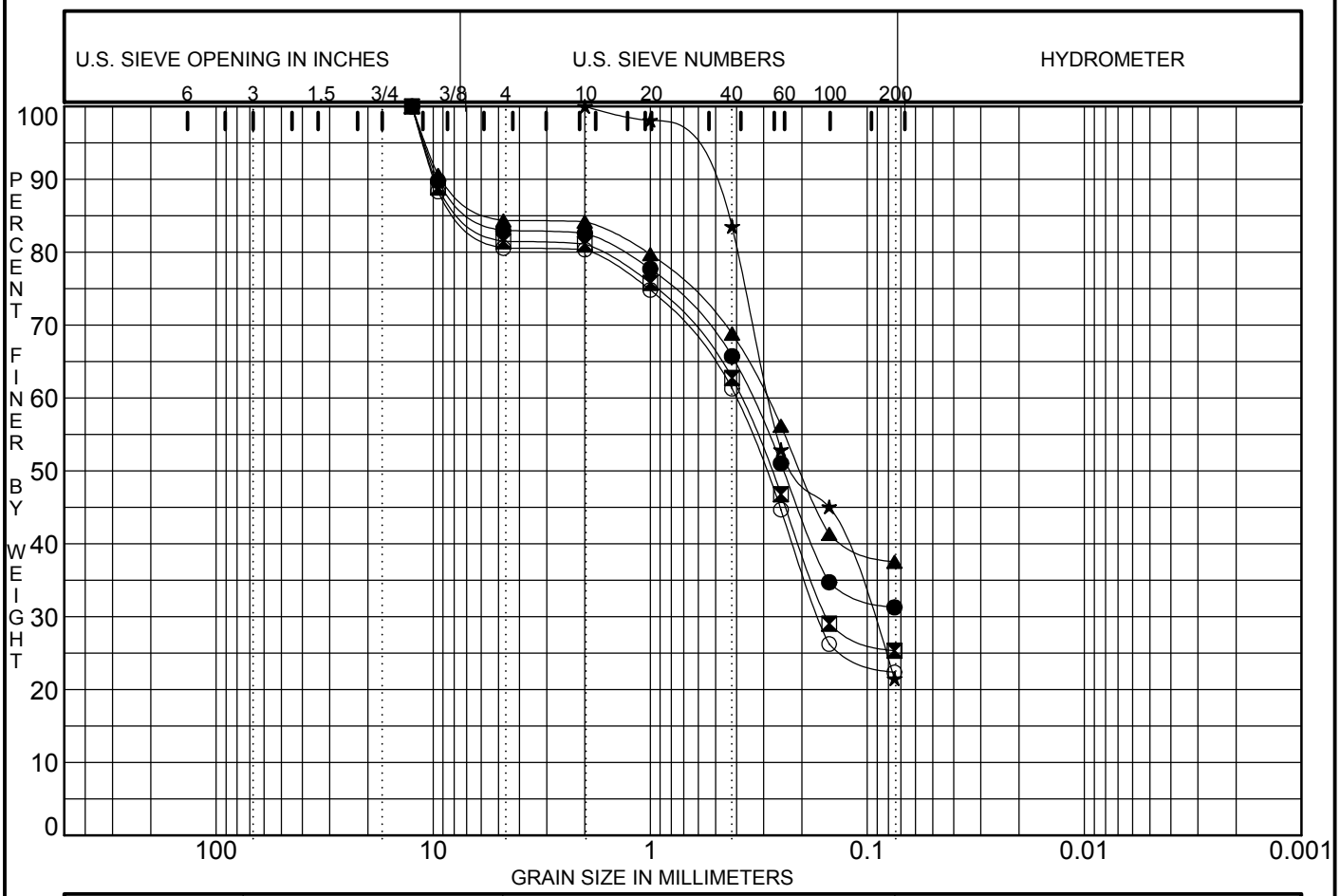
RIYADH GEOTECHNIQUE & FOUNDATIONS
AL-KHOBAR, SAUDI ARABIA

PROJECT # **MANSOURAH MASSARAH GOLD PROJECT**

CLIENT: **L & T CONSTRUCTION**

BOREHOLE NO. **TP-22**
LOCATION:

SIEVE ANALYSIS
ASTM D 422



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Sample No.	Depth(m)	Classification	MC%	LL	PL	PI	Cc	Cu
●	TP-20	0.00	CLAYEY SAND with GRAVEL (SC)	2	28	20	8		
⊠	TP-21	0.00	SILTY SAND with GRAVEL (SM)	1	NP	NP	NP		
▲	TP-21	0.40	CLAYEY SAND with GRAVEL (SC)	3	30	20	10		
★	TP-22	0.00	SILTY SAND (SM)	1	NP	NP	NP		
⊙	TP-22	0.50	SILTY SAND with GRAVEL (SM)	1	NP	NP	NP		

	Sample No.	Depth(m)	D100	D50	D30	D10	%Gravel	%Sand	%Silt	%Clay
●	TP-20	0.00	12.50	0.24			17.0	51.7	31.3	
⊠	TP-21	0.00	12.50	0.28	0.154		18.5	56.2	25.3	
▲	TP-21	0.40	12.50	0.20			15.6	46.8	37.5	
★	TP-22	0.00	2.00	0.21	0.096		0.0	78.5	21.5	
⊙	TP-22	0.50	12.50	0.30	0.166		19.4	58.2	22.4	

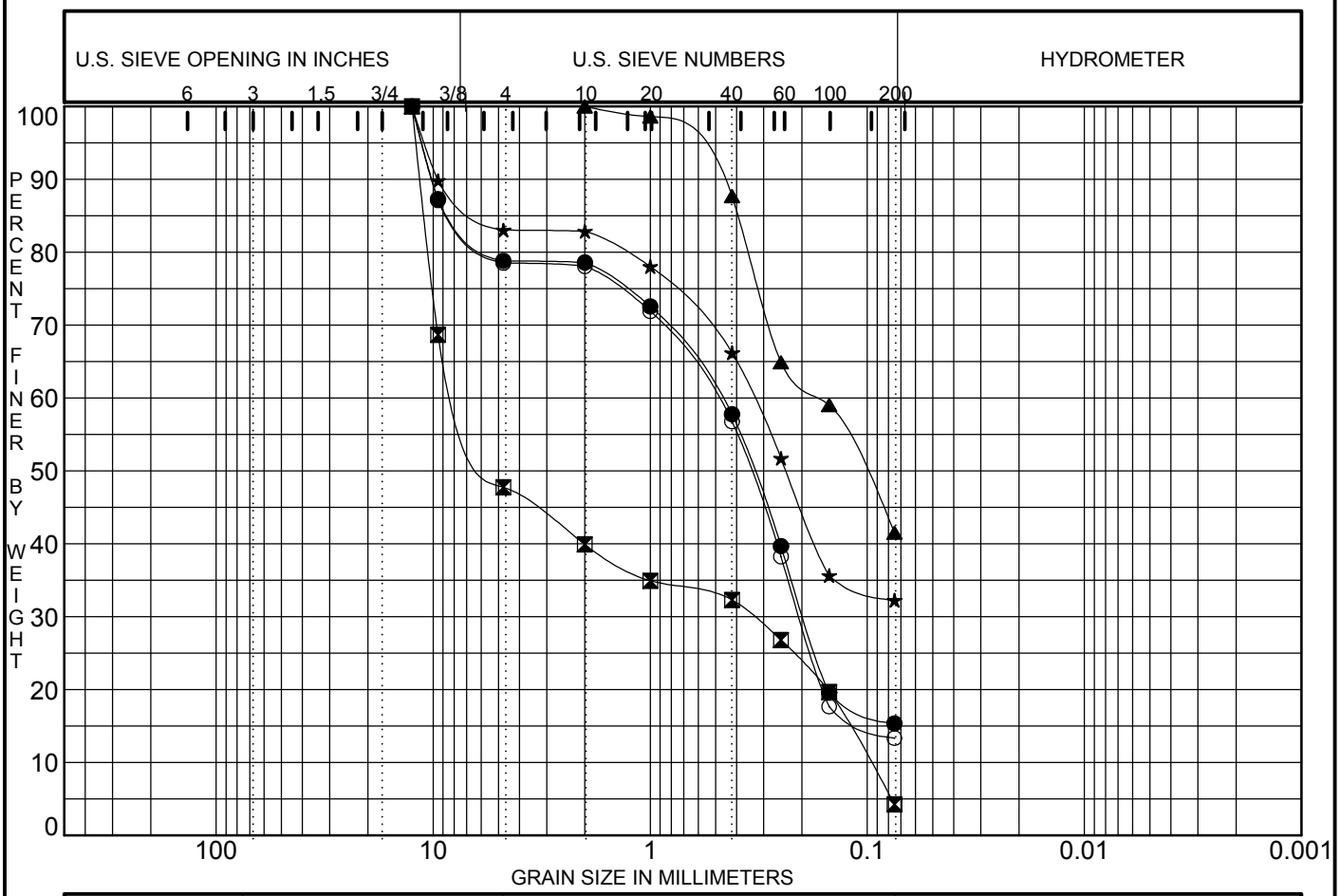
RIYADH GEOTECHNIQUE & FOUNDATIONS
AL-KHOBAR, SAUDI ARABIA

PROJECT # **MANSOURAH MASSARAH GOLD PROJECT**

CLIENT: **L & T CONSTRUCTION**

BOREHOLE NO. **TP-25**
LOCATION:

SIEVE ANALYSIS
ASTM D 422



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Sample No.	Depth(m)	Classification	MC%	LL	PL	PI	Cc	Cu
●	TP-23	0.00	SILTY SAND with GRAVEL (SM)	1	NP	NP	NP		
⊠	TP-23	0.30	POORLY GRADED GRAVEL with SAND (GP)	1	NP	NP	NP	0.16	73.4
▲	TP-24	0.00	SILTY SAND (SM)	2	NP	NP	NP		
★	TP-24	0.55	CLAYEY SAND with GRAVEL (SC)	3	29	21	8		
⊙	TP-25	0.00	SILTY SAND with GRAVEL (SM)	1	NP	NP	NP		

	Sample No.	Depth(m)	D100	D50	D30	D10	%Gravel	%Sand	%Silt	%Clay
●	TP-23	0.00	12.50	0.34	0.195		21.2	63.5	15.4	
⊠	TP-23	0.30	12.50	5.12	0.338	0.0971	52.2	43.5	4.3	
▲	TP-24	0.00	2.00	0.10			0.0	58.5	41.5	
★	TP-24	0.55	12.50	0.24			17.0	50.8	32.3	
⊙	TP-25	0.00	12.50	0.35	0.204		21.5	65.2	13.4	

RIYADH GEOTECHNIQUE & FOUNDATIONS

AL-KHOBAR, SAUDI ARABIA

PROJECT # **MANSOURAH MASSARAH GOLD PROJECT**

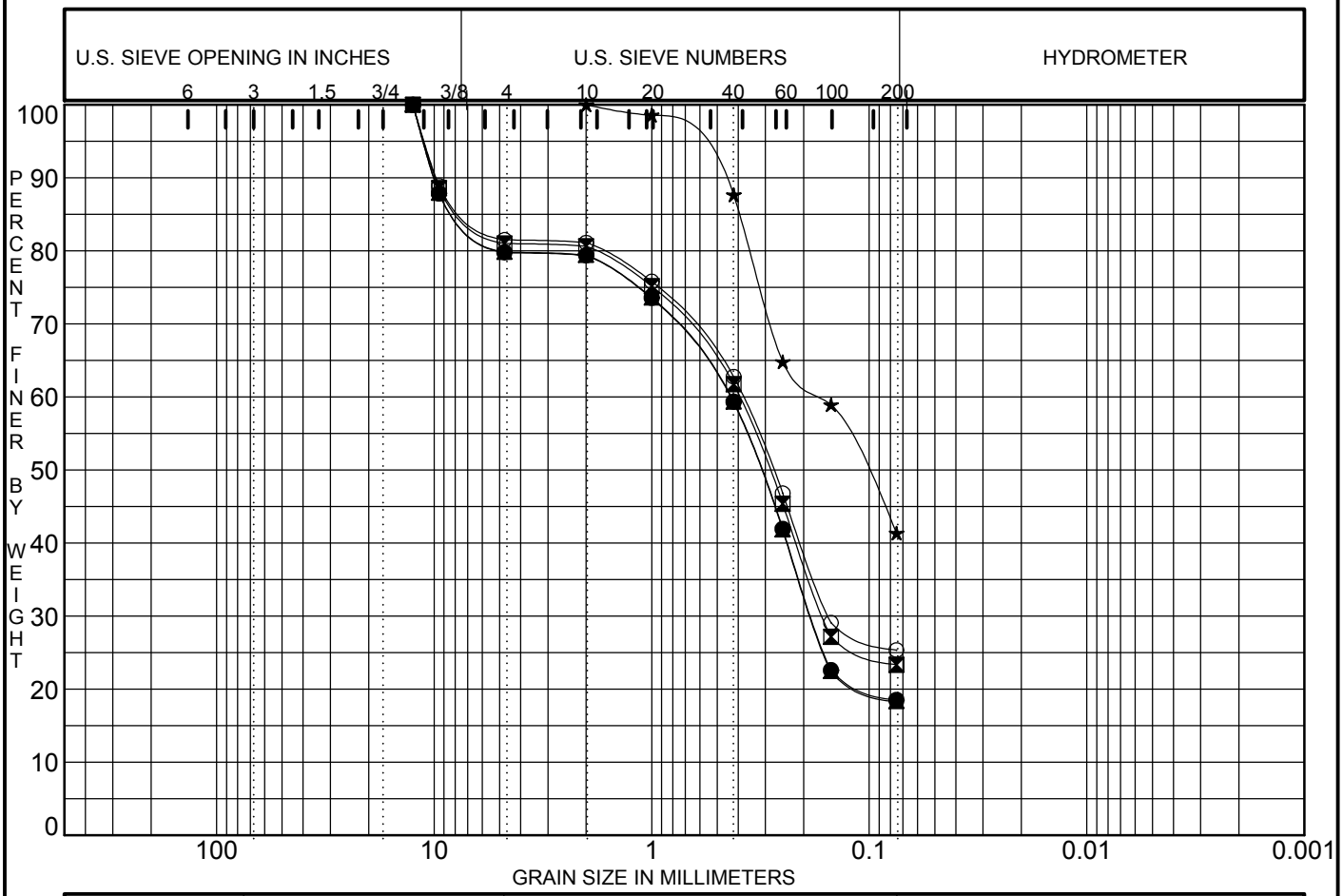
CLIENT: **L & T CONSTRUCTION**

BOREHOLE NO. **TP-28**

LOCATION:

SIEVE ANALYSIS

ASTM D 422



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Sample No.	Depth(m)	Classification	MC%	LL	PL	PI	Cc	Cu
●	TP-26	0.00	SILTY SAND with GRAVEL (SM)	1	NP	NP	NP		
⊠	TP-27	0.00	SILTY SAND with GRAVEL (SM)	0	NP	NP	NP		
▲	TP-27	0.50	SILTY SAND with GRAVEL (SM)	1	NP	NP	NP		
★	TP-28	0.00	CLAYEY SAND (SC)	2	29	20	9		
⊙	TP-28	0.70	SILTY SAND with GRAVEL (SM)	2	NP	NP	NP		

	Sample No.	Depth(m)	D100	D50	D30	D10	%Gravel	%Sand	%Silt	%Clay
●	TP-26	0.00	12.50	0.32	0.182		20.2	61.3	18.5	
⊠	TP-27	0.00	12.50	0.29	0.162		19.0	57.6	23.4	
▲	TP-27	0.50	12.50	0.32	0.183		20.3	61.5	18.3	
★	TP-28	0.00	2.00	0.11			0.0	58.6	41.4	
⊙	TP-28	0.70	12.50	0.28	0.154		18.5	56.1	25.4	

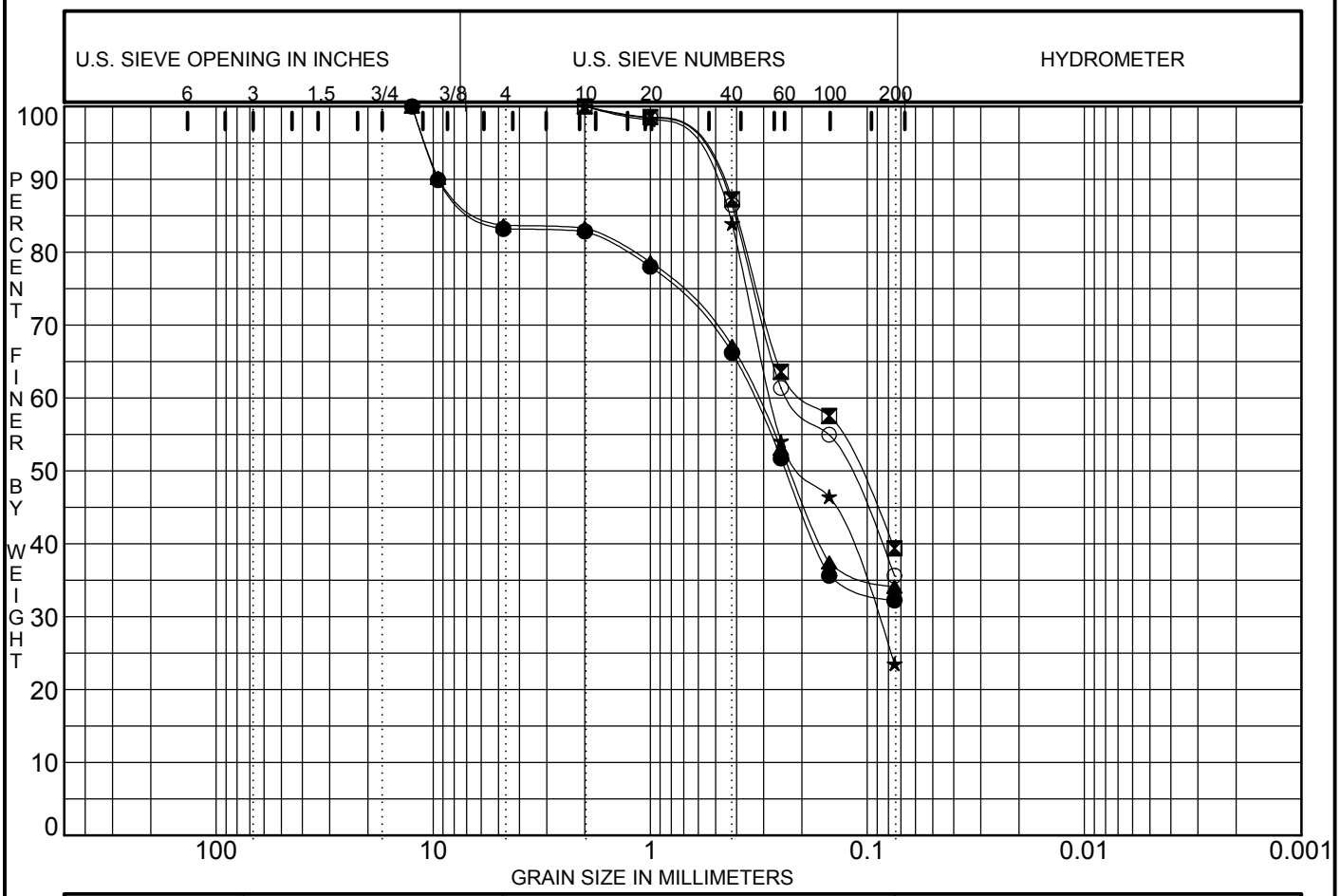
RIYADH GEOTECHNIQUE & FOUNDATIONS
AL-KHOBAR, SAUDI ARABIA

PROJECT # **MANSOURAH MASSARAH GOLD PROJECT**

CLIENT: **L & T CONSTRUCTION**

BOREHOLE NO. **TP-31**
LOCATION:

SIEVE ANALYSIS
ASTM D 422



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Sample No.	Depth(m)	Classification	MC%	LL	PL	PI	Cc	Cu
●	TP-29	0.00	SILTY SAND with GRAVEL (SM)	2	31	23	8		
⊠	TP-30	0.00	CLAYEY SAND (SC)	2	32	22	10		
▲	TP-30	0.40	CLAYEY SAND with GRAVEL (SC)	3	28	19	9		
★	TP-31	0.00	SILTY SAND (SM)	1	NP	NP	NP		
⊙	TP-31	0.40	CLAYEY SAND (SC)	2	33	23	10		

	Sample No.	Depth(m)	D100	D50	D30	D10	%Gravel	%Sand	%Silt	%Clay
●	TP-29	0.00	12.50	0.24			16.8	51.0	32.3	
⊠	TP-30	0.00	2.00	0.11			0.0	60.6	39.4	
▲	TP-30	0.40	12.50	0.23			16.3	49.5	34.1	
★	TP-31	0.00	2.00	0.19	0.091		0.0	76.5	23.5	
⊙	TP-31	0.40	2.00	0.13			0.0	64.4	35.6	

RIYADH GEOTECHNIQUE & FOUNDATIONS

AL-KHOBAR, SAUDI ARABIA

PROJECT # **MANSOURAH MASSARAH GOLD PROJECT**

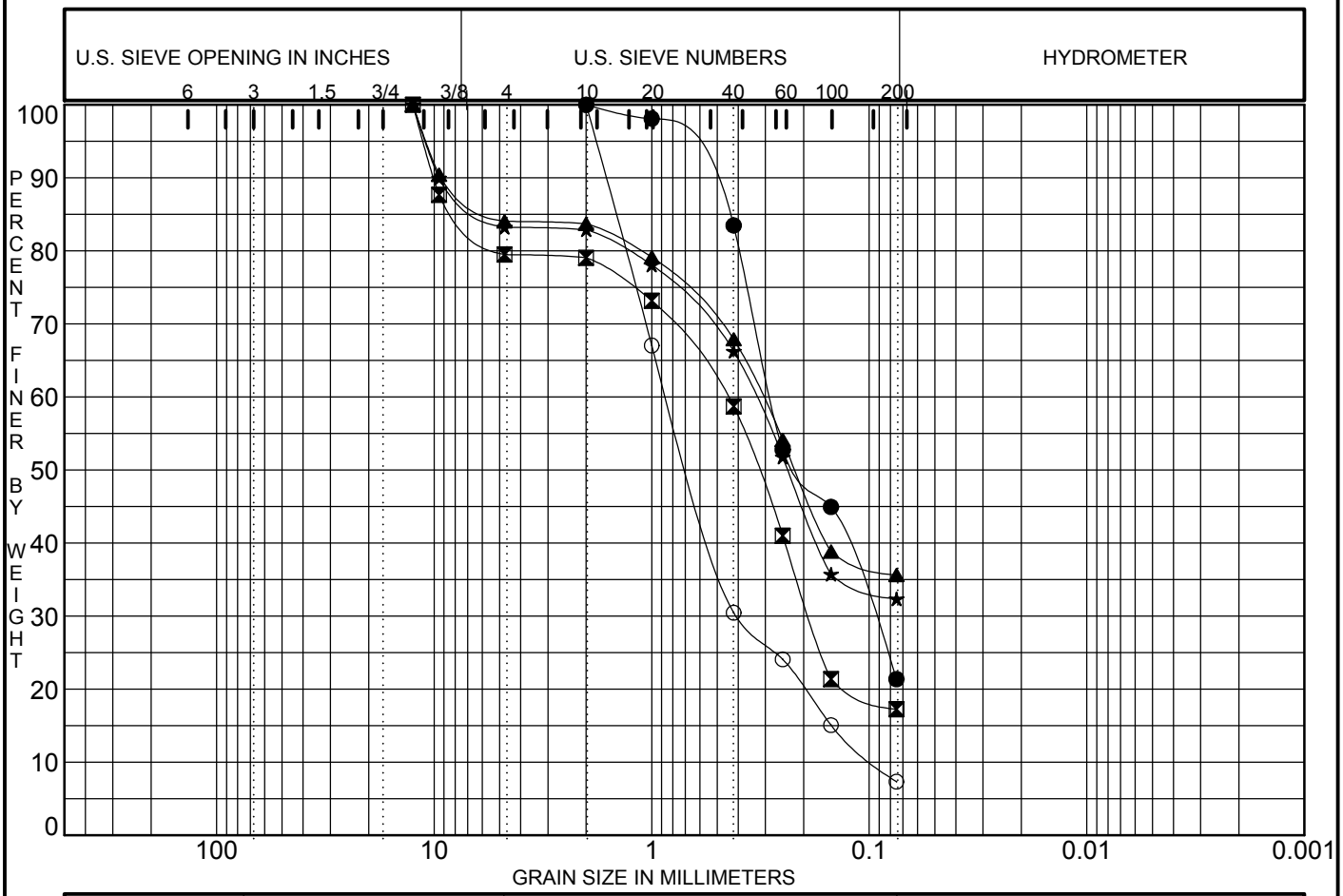
CLIENT: **L & T CONSTRUCTION**

BOREHOLE NO. **TP-34**

LOCATION:

SIEVE ANALYSIS

ASTM D 422



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Sample No.	Depth(m)	Classification	MC%	LL	PL	PI	Cc	Cu
●	TP-32	0.00	SILTY SAND (SM)	1	NP	NP	NP		
⊠	TP-33	0.00	SILTY SAND with GRAVEL (SM)	1	NP	NP	NP		
▲	TP-33	0.40	CLAYEY SAND with GRAVEL (SC)	2	30	20	10		
★	TP-34	0.00	CLAYEY SAND with GRAVEL (SC)	2	28	20	8		
⊙	TP-34	0.40	WELL-GRADED SAND with SILT (SW-SM)	1	NP	NP	NP	2.03	8.9

	Sample No.	Depth(m)	D100	D50	D30	D10	%Gravel	%Sand	%Silt	%Clay
●	TP-32	0.00	2.00	0.21	0.097		0.0	78.6	21.4	
⊠	TP-33	0.00	12.50	0.33	0.188		20.5	62.3	17.2	
▲	TP-33	0.40	12.50	0.22			16.0	48.4	35.6	
★	TP-34	0.00	12.50	0.24			16.8	50.9	32.4	
⊙	TP-34	0.40	2.00	0.67	0.404	0.0952	0.0	92.7	7.3	

RIYADH GEOTECHNIQUE & FOUNDATIONS

AL-KHOBAR, SAUDI ARABIA

PROJECT # **MANSOURAH MASSARAH GOLD PROJECT**

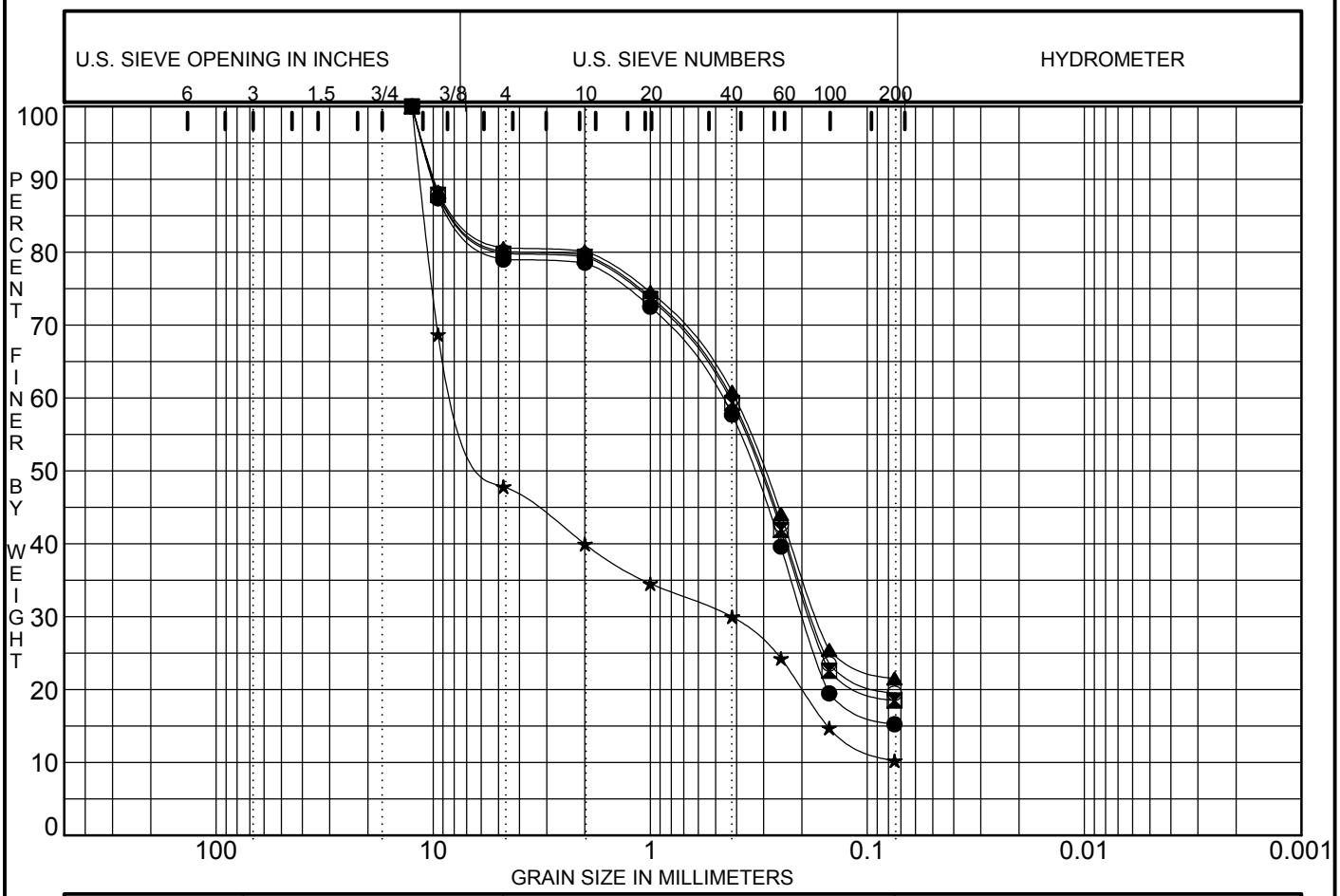
CLIENT: **L & T CONSTRUCTION**

BOREHOLE NO. **TP-42**

LOCATION:

SIEVE ANALYSIS

ASTM D 422



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Sample No.	Depth(m)	Classification	MC%	LL	PL	PI	Cc	Cu
●	TP-35	0.00	SILTY SAND with GRAVEL (SM)	1	NP	NP	NP		
⊠	TP-36	0.00	SILTY SAND with GRAVEL (SM)	1	NP	NP	NP		
▲	TP-41	0.00	SILTY SAND with GRAVEL (SM)	1	NP	NP	NP		
★	TP-41	0.50	POORLY GRADED GRAVEL with SILT and SAND (GP-GM)	2	NP	NP	NP	0.34	98.4
⊙	TP-42	0.00	SILTY SAND with GRAVEL (SM)	1	NP	NP	NP		

	Sample No.	Depth(m)	D100	D50	D30	D10	%Gravel	%Sand	%Silt	%Clay
●	TP-35	0.00	12.50	0.34	0.196		21.0	63.8	15.3	
⊠	TP-36	0.00	12.50	0.32	0.182		20.2	61.3	18.5	
▲	TP-41	0.00	12.50	0.30	0.170		19.4	59.0	21.5	
★	TP-41	0.50	12.50	5.10	0.420		52.2	37.6	10.2	
⊙	TP-42	0.00	12.50	0.31	0.178		19.9	60.6	19.5	

RIYADH GEOTECHNIQUE & FOUNDATIONS

AL-KHOBAR, SAUDI ARABIA

PROJECT # **MANSOURAH MASSARAH GOLD PROJECT**

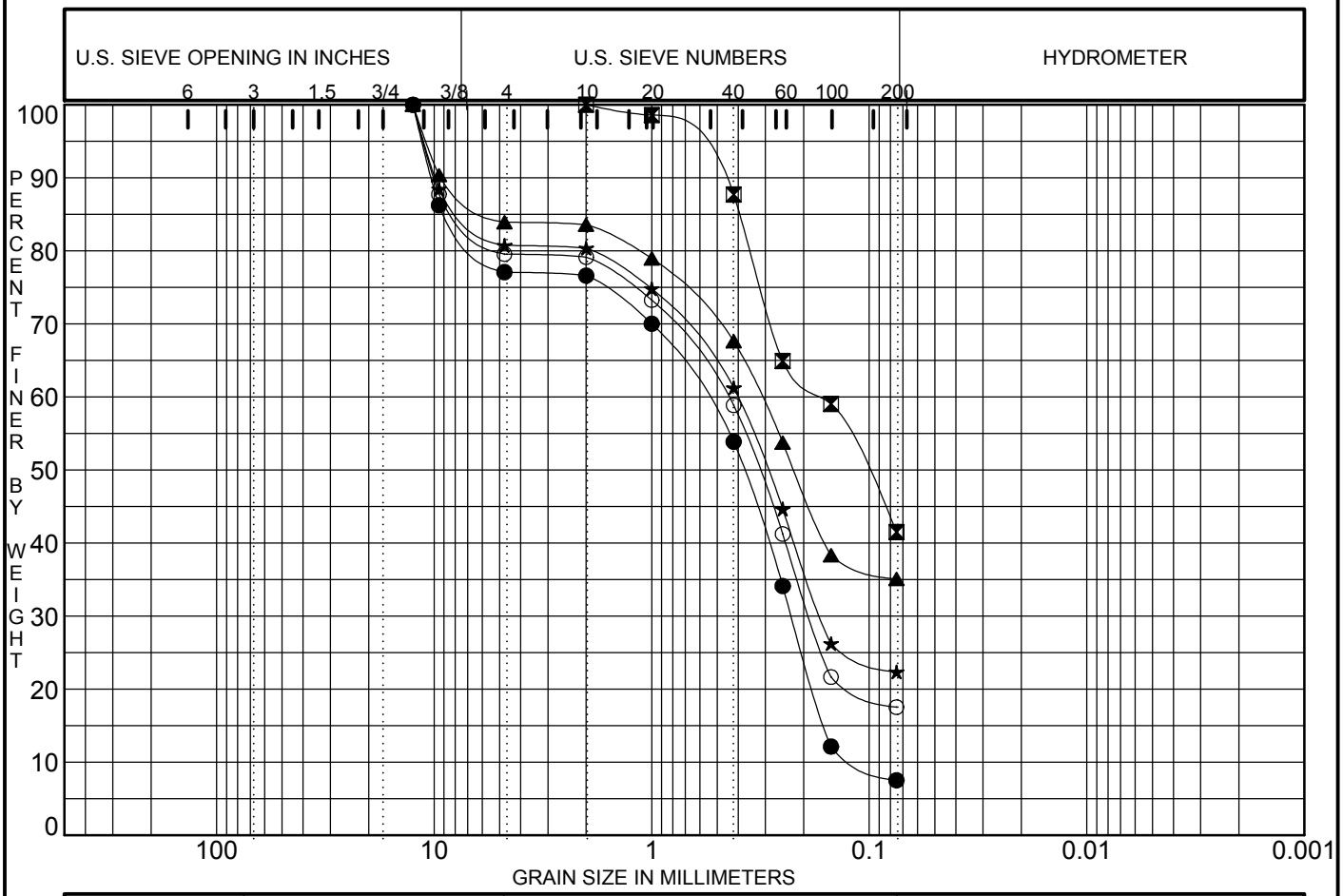
CLIENT: **L & T CONSTRUCTION**

BOREHOLE NO. **TP-45**

LOCATION:

SIEVE ANALYSIS

ASTM D 422



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Sample No.	Depth(m)	Classification	MC%	LL	PL	PI	Cc	Cu
●	TP-42	0.60	POORLY GRADED SAND with SILT and GRAVEL (SP-SM)	1	NP	NP	NP	0.81	5.4
⊠	TP-43	0.00	CLAYEY SAND (SC)	2	29	19	10		
▲	TP-43	0.20	CLAYEY SAND with GRAVEL (SC)	3	28	20	8		
★	TP-44	0.00	SILTY SAND with GRAVEL (SM)	1	NP	NP	NP		
⊙	TP-45	0.00	SILTY SAND with GRAVEL (SM)	1	NP	NP	NP		

	Sample No.	Depth(m)	D100	D50	D30	D10	%Gravel	%Sand	%Silt	%Clay
●	TP-42	0.60	12.50	0.38	0.227	0.1088	22.9	69.6	7.5	
⊠	TP-43	0.00	2.00	0.10			0.0	58.5	41.5	
▲	TP-43	0.20	12.50	0.22			16.1	48.8	35.1	
★	TP-44	0.00	12.50	0.30	0.166		19.2	58.4	22.4	
⊙	TP-45	0.00	12.50	0.32	0.186		20.4	62.0	17.5	

RIYADH GEOTECHNIQUE & FOUNDATIONS

AL-KHOBAR, SAUDI ARABIA

PROJECT # **MANSOURAH MASSARAH GOLD PROJECT**

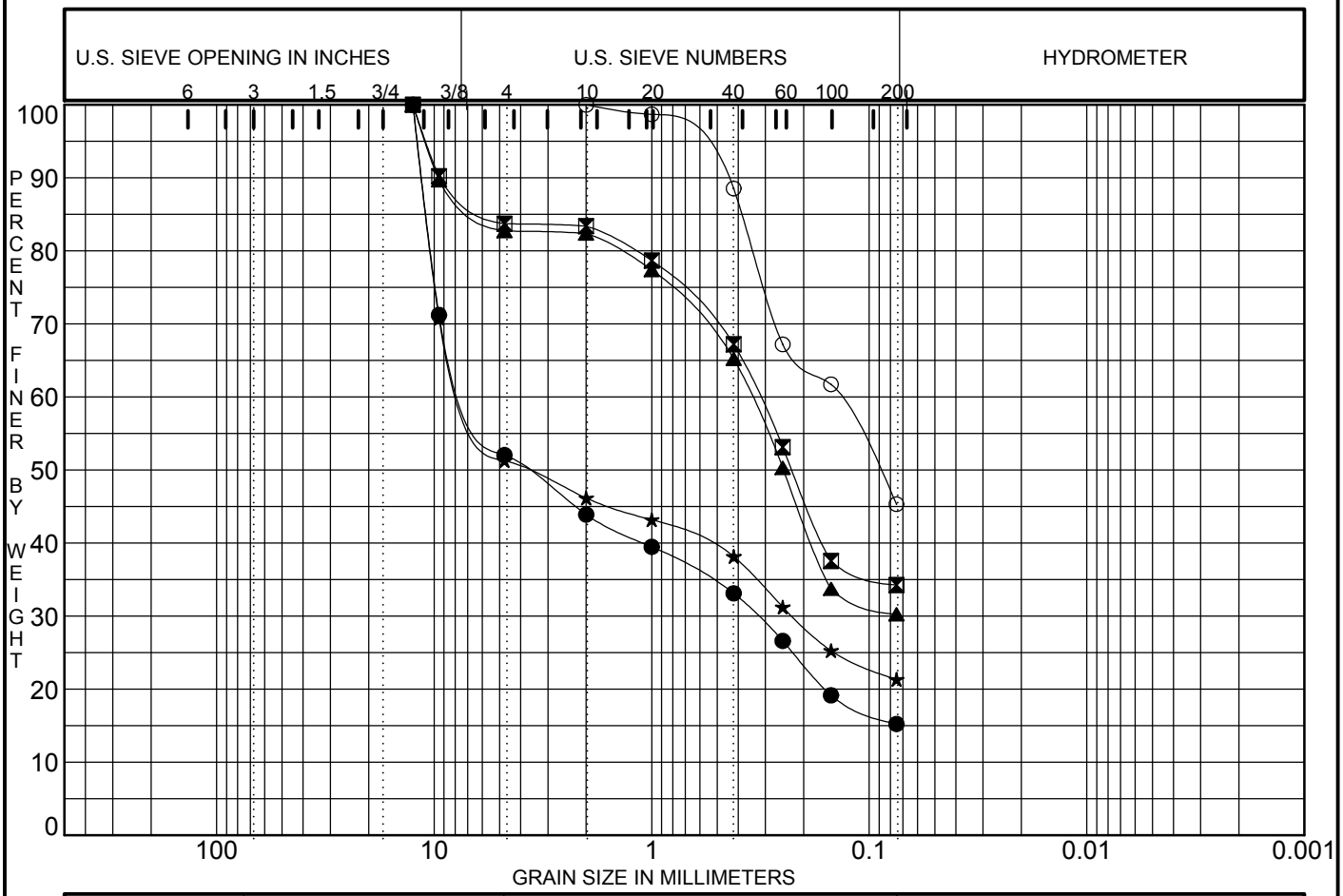
CLIENT: **L & T CONSTRUCTION**

BOREHOLE NO. **TP-48**

LOCATION:

SIEVE ANALYSIS

ASTM D 422



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Sample No.	Depth(m)	Classification	MC%	LL	PL	PI	Cc	Cu
●	TP-45	0.30	SILTY GRAVEL with SAND (GM)	1	NP	NP	NP		
⊠	TP-46	0.00	SILTY SAND with GRAVEL (SM)	2	NP	NP	NP		
▲	TP-47	0.00	CLAYEY SAND with GRAVEL (SC)	2	29	20	9		
★	TP-47	0.45	SILTY GRAVEL with SAND (GM)	2	NP	NP	NP		
⊙	TP-48	0.00	CLAYEY SAND (SC)	1	31	21	10		

	Sample No.	Depth(m)	D100	D50	D30	D10	%Gravel	%Sand	%Silt	%Clay
●	TP-45	0.30	12.50	3.82	0.327		48.0	36.8	15.2	
⊠	TP-46	0.00	12.50	0.23			16.3	49.5	34.2	
▲	TP-47	0.00	12.50	0.25			17.3	52.5	30.3	
★	TP-47	0.45	12.50	3.82	0.225		48.7	29.9	21.3	
⊙	TP-48	0.00	2.00	0.09			0.0	54.7	45.3	

RIYADH GEOTECHNIQUE & FOUNDATIONS

AL-KHOBAR, SAUDI ARABIA

PROJECT # **MANSOURAH MASSARAH GOLD PROJECT**

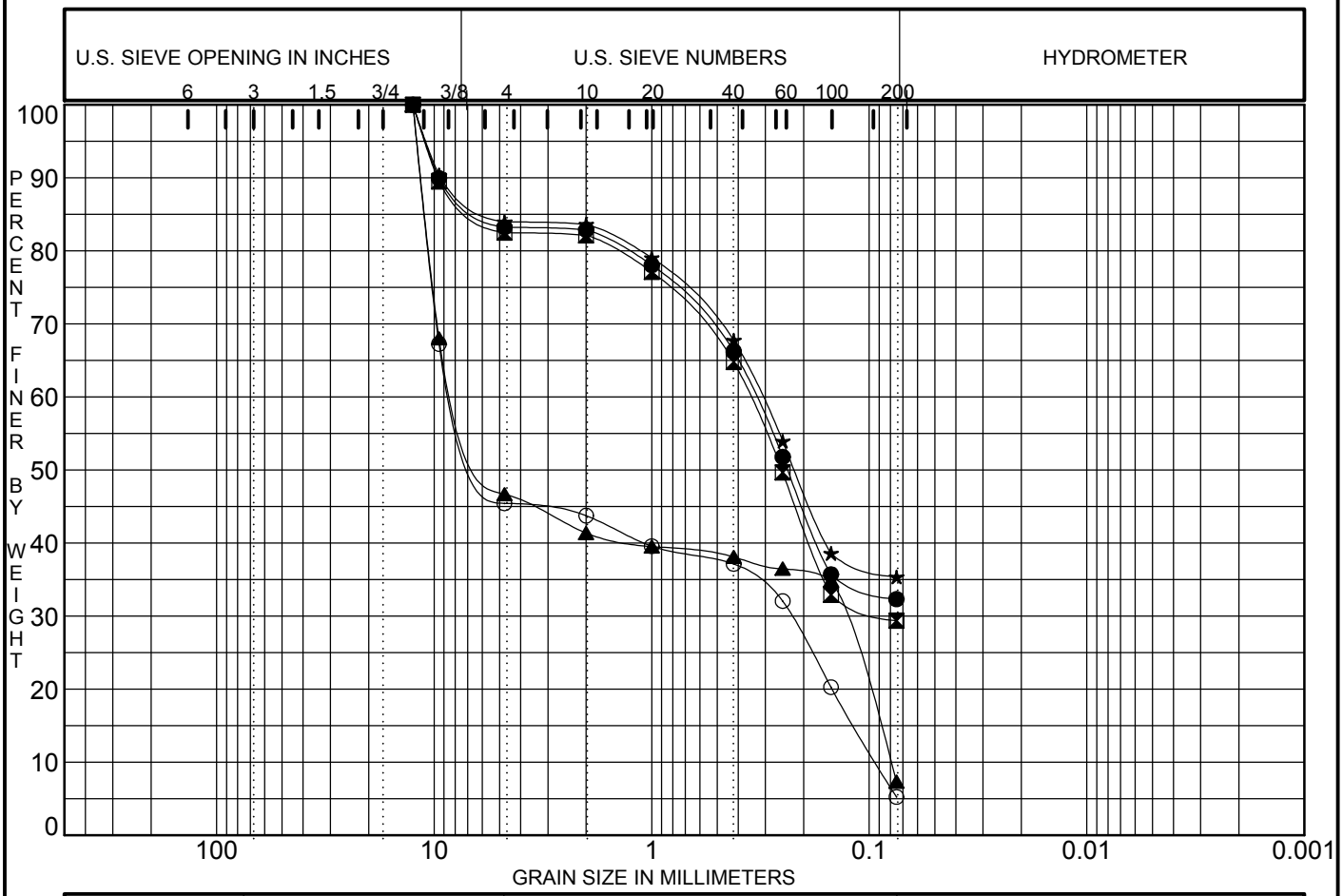
CLIENT: **L & T CONSTRUCTION**

BOREHOLE NO. **TP-50**

LOCATION:

SIEVE ANALYSIS

ASTM D 422



COBBLES	GRAVEL		SAND			SILT OR CLAY
	coarse	fine	coarse	medium	fine	

	Sample No.	Depth(m)	Classification	MC%	LL	PL	PI	Cc	Cu
●	TP-48	0.50	CLAYEY SAND with GRAVEL (SC)	2	29	20	9		
⊠	TP-49	0.00	CLAYEY SAND with GRAVEL (SC)	2	26	18	8		
▲	TP-49	0.20	POORLY GRADED GRAVEL with SILT and SAND (GP-GM)	2	NP	NP	NP	0.03	91.3
★	TP-50	0.00	CLAYEY SAND with GRAVEL (SC)	2	28	20	8		
⊙	TP-50	0.35	POORLY GRADED GRAVEL with SILT and SAND (GP-GM)	1	NP	NP	NP	0.07	80.8

	Sample No.	Depth(m)	D100	D50	D30	D10	%Gravel	%Sand	%Silt	%Clay
●	TP-48	0.50	12.50	0.24			16.8	50.9	32.3	
⊠	TP-49	0.00	12.50	0.25	0.085		17.5	53.1	29.4	
▲	TP-49	0.20	12.50	5.29	0.133	0.0802	53.3	39.4	7.3	
★	TP-50	0.00	12.50	0.22			16.0	48.6	35.4	
⊙	TP-50	0.35	12.50	5.49	0.229	0.0934	54.6	40.2	5.3	

CHEMICAL ANALYSIS RESULTS

الرياض للجيوتقنية والأساسات ذ.م.م
RIYADH GEOTECHNIQUE & FOUNDATIONS LLC
P. O. Box # 2870, Al-Khobar : 31952, Saudi Arabia, Tel : (03) 859 8503 / 04 /06 Fax : (03) 859 8507

CHEMICAL ANALYSIS OF SOIL/ROCK
TEST REPORT

Client:	**Larsen & Toubro Limited**	**Date:**	**03 September 2019**
Project:	**Mansourah Massarah Gold Project, Geotechnical Investigations**	**Ref. #:**	**CHK-2105/19**
Project #:	**19-K-3054**		

Borehole No.	Sample Depth (m)	pH BS 1377: Part 3 :1990	Water Soluble Chloride as Cl (%) AASHTO T 291-94	Class of Exposure Based on Chloride Content*	Water Soluble Sulfate as SO_4 (%) ASTM C 1580	Class of Exposure Based on Sulfate Content**	Total Carbonate as $CaCO_3$ (%) ASTM D 4373
BH – 02	150 – 3.00	7.9	0.06	C0	0.01	S0	50
BH – 09	1.50	7.5	0.29	C1	0.23	S2	21
BH – 18	2.25	7.8	0.04	C0	0.02	S2	16

Evaluation of class of exposure based on Chloride & Sulfate content as per Saudi Aramco Eng. Standard SAES-Q-001 (2018) Table 1.C* & 1.A**

Tested By : **SF/NA**

Checked/Approved By : _____
Laboratory Incharge

This report relates to the sample tested and may not be reproduced other than in full, except with prior written permission from RGF testing laboratory.

الرياض للجيوتقنية والأساسات ذ.م.م
RIYADH GEOTECHNIQUE & FOUNDATIONS LLC
P. O. Box # 2870, Al-Khobar : 31952, Saudi Arabia, Tel : (03) 859 8503 / 04 /06 Fax : (03) 859 8507

CHEMICAL ANALYSIS OF SOIL
TEST REPORT

Client:	**Larsen & Toubro Limited**	**Date:**	**10 September 2019**
Project:	**Mansourah Massarah Gold Project, Geotechnical Investigations**	**Ref. #:**	**CHK-2110/19**
Project #:	**19-K-3054**		

Borehole No.	Sample Depth (m)	pH BS 1377: Part 3 :1990	Water Soluble Chloride as Cl (%) AASHTO T 291-94	Class of Exposure Based on Chloride Content*	Water Soluble Sulfate as SO_4 (%) ASTM C 1580	Class of Exposure Based on Sulfate Content**	Total Carbonate as $CaCO_3$ (%) ASTM D 4373
BH – 05	3.00	7.9	0.11	C1	0.16	S1	2
BH – 12	1.50	7.6	0.17	C1	0.04	S0	72
BH – 17	2.25	8.0	0.02	C0	0.07	S0	10
BH – 25	2.25	7.3	0.06	C0	0.15	S1	4

Evaluation of class of exposure based on Chloride & Sulfate content as per Saudi Aramco Eng. Standard SAES-Q-001 (2018) Table 1.C* & 1.A**

Tested By : **SF/NA**

Checked/Approved By : _____
Laboratory Incharge

This report relates to the sample tested and may not be reproduced other than in full, except with prior written permission from RGF testing laboratory.

MODIFIED PROCTOR

MODIFIED PROCTOR TEST

ASTM D 1557 (METHOD " A ")

CLIENT	L & T CONSTRUCTION	REPORT DATE	02-Sep-19
PROJECT/LOCATION	MANSOURAH MASSARAH GOLD PROJECT	SAMPLING DATE	30-Aug-19
METHOD USED	"A"	TEST START DATE	31-Aug-19
SAMPLE SOURCE	ON SITE	TEST FINISH DATE	01-Sep-19
METHOD VARIATION	NONE	GEOTECHNICAL FILE #	-
SAMPLED BY	WAHAB	PROJECT #	19-K-3054

SOURCE / SAMPLE NO : B.S-1 (0.0-0.60m) TP-01 **MOULD NO :** 56

WEIGHT OF MOULD: 3334 **gms.** **VOLUME OF MOULD:** 942.415 **cc.**

TRIAL NO.		1	2	3	4	5
WATER ADDED	%	5%	7%	9%	11%	13%
WT. SOIL + MOULD (WET)	gm	5185	5269	5347	5351	5320
WEIGHT OF WET SOIL	gm	1851	1935	2013	2017	1986
WET DENSITY	**gm/cc**	1.964	2.053	2.136	2.140	2.107

CAN NO.		N#2	N#8	N#15	N#7	N#6
WT. SOIL + CAN (WET)	gm	734.7	732.7	752.5	782.2	777.2
WT. SOIL + CAN (DRY)	gm	706.3	694.1	703.9	720.6	704.8
WT. OF WATER	gm	28.4	38.6	48.6	61.6	72.4
WT. OF CAN	gm	170.8	157.6	164.3	181.1	172.6
WT. OF DRY SOIL	gm	535.5	536.5	539.6	539.5	532.2
WATER CONTENT	**%**	**5.3**	**7.2**	**9.0**	**11.4**	**13.6**
DRY DENSITY	**gm/cc**	**1.865**	**1.915**	**1.960**	**1.921**	**1.855**

Soil Description: ***SILTY SAND with GRAVEL***

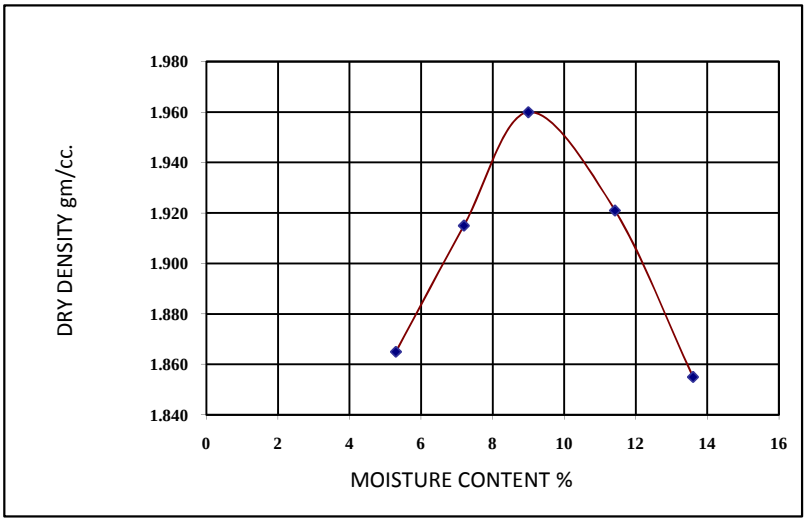
MAX. DRY DENSITY gm/cc : **1.960**

OPT. MOISTURE CONTENT % **9.0**

CORRECTED FOR OVER SIZE

MAX. DRY DENSITY gm/cc : **2.032**

OPT. MOISTURE CONTENT % **7.86**



REMARKS :

TESTED BY: REVIEWED BY:

MIN/RH

(TECHNICIAN) (SUPERVISOR) (LAB. INCHARGE)

MODIFIED PROCTOR TEST

ASTM D 1557 (METHOD " A ")

CLIENT	L & T CONSTRUCTION	REPORT DATE	02-Sep-19
PROJECT/LOCATION	MANSOURAH MASSARAH GOLD PROJECT	SAMPLING DATE	30-Aug-19
METHOD USED	"A"	TEST START DATE	31-Aug-19
SAMPLE SOURCE	ON SITE	TEST FINISH DATE	01-Sep-19
METHOD VARIATION	NONE	GEOTECHNICAL FILE #	-
SAMPLED BY	WAHAB	PROJECT #	19-K-3054

SOURCE / SAMPLE NO : B.S-2 (0.7-1.0m) TP-02 **MOULD NO :** 56

WEIGHT OF MOULD: 3334 **gms.** **VOLUME OF MOULD:** 942.415 **cc.**

TRIAL NO.		1	2	3	4	5
WATER ADDED	%	5%	7%	9%	11%	13%
WT. SOIL + MOULD (WET)	gm	5216	5305	5395	5382	5363
WEIGHT OF WET SOIL	gm	1882	1971	2061	2048	2029
WET DENSITY	**gm/cc**	1.997	2.091	2.187	2.173	2.153

CAN NO.		N#2	N#8	N#15	N#7	N#6
WT. SOIL + CAN (WET)	gm	710.7	709.9	730.1	758.4	762.9
WT. SOIL + CAN (DRY)	gm	683.0	672.1	680.5	699.7	692.1
WT. OF WATER	gm	27.7	37.8	49.6	58.7	70.8
WT. OF CAN	gm	170.8	157.6	164.3	181.1	172.6
WT. OF DRY SOIL	gm	512.2	514.5	516.2	518.6	519.5
WATER CONTENT	**%**	**5.4**	**7.4**	**9.6**	**11.3**	**13.6**
DRY DENSITY	**gm/cc**	**1.895**	**1.948**	**1.995**	**1.952**	**1.895**

Soil Description: ***CLAYEY SAND with GRAVELS***

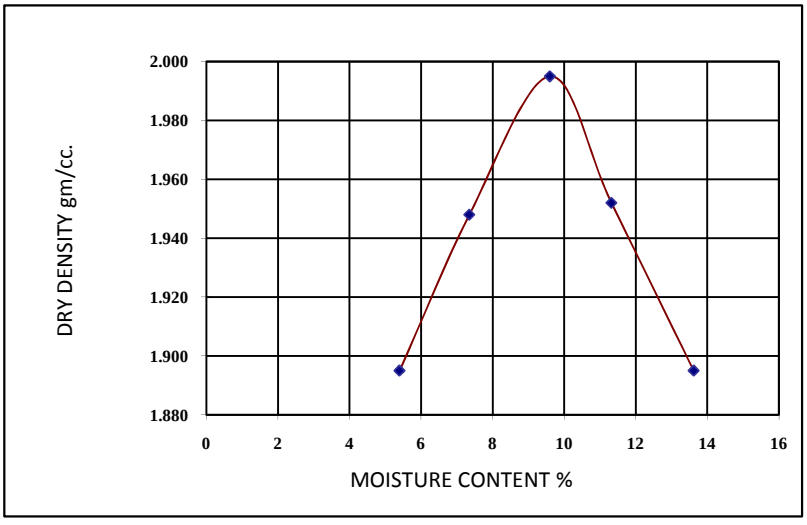
MAX. DRY DENSITY gm/cc : **1.995**

OPT. MOISTURE CONTENT % **9.6**

CORRECTED FOR OVER SIZE

MAX. DRY DENSITY gm/cc : **2.052**

OPT. MOISTURE CONTENT % **8.53**



REMARKS :

TESTED BY: REVIEWED BY:

MIN/RH

(TECHNICIAN) (SUPERVISOR) (LAB. INCHARGE)

MODIFIED PROCTOR TEST

ASTM D 1557 (METHOD " A ")

CLIENT	L & T CONSTRUCTION	REPORT DATE	02-Sep-19
PROJECT/LOCATION	MANSOURAH MASSARAH GOLD PROJECT	SAMPLING DATE	30-Aug-19
METHOD USED	"A"	TEST START DATE	31-Aug-19
SAMPLE SOURCE	ON SITE	TEST FINISH DATE	01-Sep-19
METHOD VARIATION	NONE	GEOTECHNICAL FILE #	-
SAMPLED BY	WAHAB	PROJECT #	19-K-3054

SOURCE / SAMPLE NO : B.S-1 (0.00-0.80m) TP-03 **MOULD NO :** 56

WEIGHT OF MOULD: 3334 **gms.** **VOLUME OF MOULD:** 942.415 **cc.**

TRIAL NO.		1	2	3	4	5
WATER ADDED	%	5%	7%	9%	11%	13%
WT. SOIL + MOULD (WET)	gm	5175	5264	5353	5339	5307
WEIGHT OF WET SOIL	gm	1841	1930	2019	2005	1973
WET DENSITY	**gm/cc**	1.954	2.048	2.143	2.128	2.094

CAN NO.		N#2	N#8	N#15	N#7	N#6
WT. SOIL + CAN (WET)	gm	705.9	704.7	724.4	753.6	757.4
WT. SOIL + CAN (DRY)	gm	684.3	671.8	680.5	700.6	693.8
WT. OF WATER	gm	21.6	32.9	43.9	53.0	63.6
WT. OF CAN	gm	170.8	157.6	164.3	181.1	172.6
WT. OF DRY SOIL	gm	513.5	514.2	516.2	519.5	521.2
WATER CONTENT	**%**	**4.2**	**6.4**	**8.5**	**10.2**	**12.2**
DRY DENSITY	**gm/cc**	**1.875**	**1.925**	**1.975**	**1.931**	**1.866**

Soil Description: ***SILTY SAND with GRAVEL***

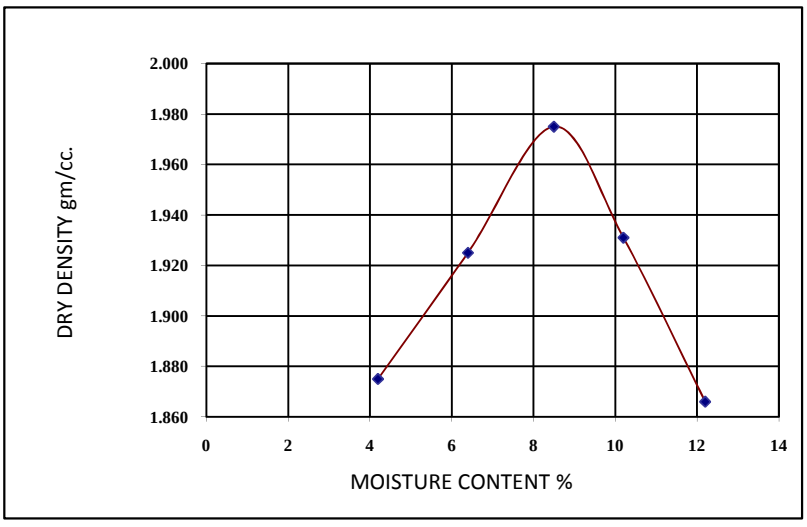
MAX. DRY DENSITY gm/cc : **1.975**

OPT. MOISTURE CONTENT % **8.5**

CORRECTED FOR OVER SIZE

MAX. DRY DENSITY gm/cc : **2.042**

OPT. MOISTURE CONTENT % **7.49**



REMARKS :

TESTED BY: REVIEWED BY:

MIN/RH

(TECHNICIAN) (SUPERVISOR) (LAB. INCHARGE)

MODIFIED PROCTOR TEST

ASTM D 1557 (METHOD " C ")

CLIENT	L & T CONSTRUCTION	REPORT DATE	03-Sep-19
PROJECT/LOCATION	MANSOURAH MASSARAH GOLD PROJECT	SAMPLING DATE	31-Aug-19
METHOD USED	"C"	TEST START DATE	01-Sep-19
SAMPLE SOURCE	ON SITE	TEST FINISH DATE	02-Sep-19
METHOD VARIATION	NONE	GEOTECHNICAL FILE #	-
SAMPLED BY	WAHAB	PROJECT #	19-K-3054

SOURCE / SAMPLE NO : B.S-1 (0.3-0.90m) TP-04 **MOULD NO :** 31

WEIGHT OF MOULD: 6577 **gms.** **VOLUME OF MOULD:** 2012.33 **cc.**

TRIAL NO.		1	2	3	4	5
WATER ADDED	%	5%	7%	9%	11%	13%
WT. SOIL + MOULD (WET)	gm	10713	10915	11110	11075	11041
WEIGHT OF WET SOIL	gm	4136	4338	4533	4498	4464
WET DENSITY	**gm/cc**	2.055	2.156	2.252	2.235	2.218

CAN NO.		N#2	N#8	N#15	N#7	N#6
WT. SOIL + CAN (WET)	gm	696.3	696.6	716.5	743.6	748.1
WT. SOIL + CAN (DRY)	gm	685.0	673.9	682.8	701.0	694.8
WT. OF WATER	gm	11.3	22.7	33.7	42.6	53.3
WT. OF CAN	gm	170.8	157.6	164.3	181.1	172.6
WT. OF DRY SOIL	gm	514.2	516.3	518.5	519.9	522.2
WATER CONTENT	**%**	**2.2**	**4.4**	**6.5**	**8.2**	**10.2**
DRY DENSITY	**gm/cc**	**2.011**	**2.065**	**2.115**	**2.066**	**2.013**

Soil Description: ***SILTY GRAVEL with SAND***

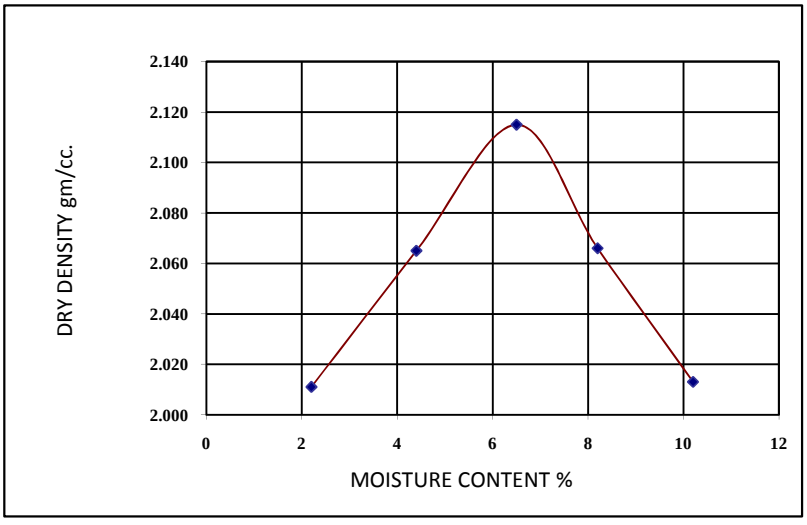
MAX. DRY DENSITY gm/cc : **2.115**

OPT. MOISTURE CONTENT % **6.5**

CORRECTED FOR OVER SIZE

MAX. DRY DENSITY gm/cc : **2.115**

OPT. MOISTURE CONTENT % **6.50**



REMARKS :

TESTED BY: REVIEWED BY:

MIN/RH

(TECHNICIAN) (SUPERVISOR) (LAB. INCHARGE)

MODIFIED PROCTOR TEST

ASTM D 1557 (METHOD " A ")

CLIENT	L & T CONSTRUCTION	REPORT DATE	03-Sep-19
PROJECT/LOCATION	MANSOURAH MASSARAH GOLD PROJECT	SAMPLING DATE	31-Aug-19
METHOD USED	"A"	TEST START DATE	01-Sep-19
SAMPLE SOURCE	ON SITE	TEST FINISH DATE	02-Sep-19
METHOD VARIATION	NONE	GEOTECHNICAL FILE #	-
SAMPLED BY	WAHAB	PROJECT #	19-K-3054

SOURCE / SAMPLE NO : B.S-1 (0.30-1.20m) TP-05 **MOULD NO :** 42

WEIGHT OF MOULD: 3352 **gms.** **VOLUME OF MOULD:** 943.42 **cc.**

TRIAL NO.		1	2	3	4	5
WATER ADDED	%	5%	7%	9%	11%	13%
WT. SOIL + MOULD (WET)	gm	5213	5304	5393	5381	5374
WEIGHT OF WET SOIL	gm	1861	1952	2041	2029	2022
WET DENSITY	**gm/cc**	1.973	2.069	2.164	2.151	2.143

CAN NO.		N#2	N#8	N#15	N#7	N#6
WT. SOIL + CAN (WET)	gm	707.7	706.5	799.0	783.1	749.5
WT. SOIL + CAN (DRY)	gm	686.0	674.5	749.8	727.4	684.8
WT. OF WATER	gm	21.6	32.0	49.2	55.7	64.7
WT. OF CAN	gm	170.8	157.6	164.3	181.1	172.6
WT. OF DRY SOIL	gm	515.2	516.9	585.5	546.3	512.2
WATER CONTENT	**%**	**4.2**	**6.2**	**8.4**	**10.2**	**12.6**
DRY DENSITY	**gm/cc**	**1.893**	**1.948**	**1.996**	**1.952**	**1.903**

Soil Description: ***CLAYEY SAND with GRAVELS***

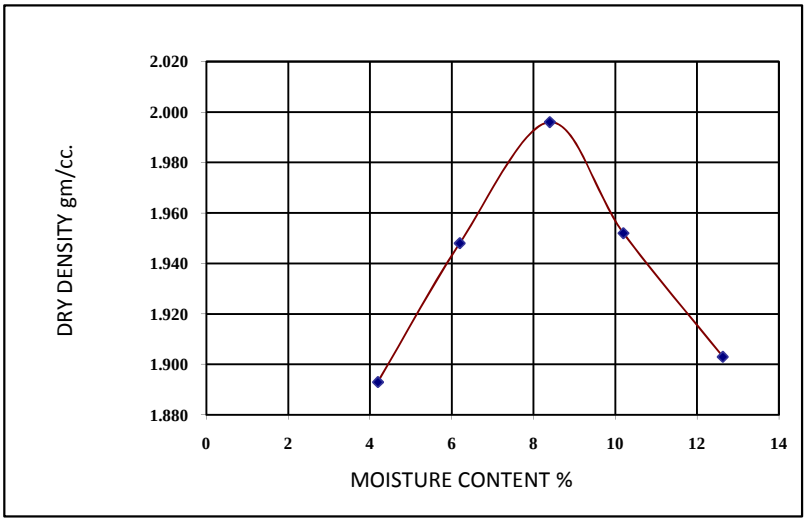
MAX. DRY DENSITY gm/cc : **1.996**

OPT. MOISTURE CONTENT % **8.4**

CORRECTED FOR OVER SIZE

MAX. DRY DENSITY gm/cc : **2.052**

OPT. MOISTURE CONTENT % **7.53**



REMARKS :

TESTED BY: REVIEWED BY:

MIN/RH

(TECHNICIAN) (SUPERVISOR) (LAB. INCHARGE)

MODIFIED PROCTOR TEST

ASTM D 1557 (METHOD " A ")

CLIENT	L & T CONSTRUCTION	REPORT DATE	03-Sep-19
PROJECT/LOCATION	MANSOURAH MASSARAH GOLD PROJECT	SAMPLING DATE	31-Aug-19
METHOD USED	"A"	TEST START DATE	01-Sep-19
SAMPLE SOURCE	ON SITE	TEST FINISH DATE	02-Sep-19
METHOD VARIATION	NONE	GEOTECHNICAL FILE #	-
SAMPLED BY	WAHAB	PROJECT #	19-K-3054

SOURCE / SAMPLE NO : B.S-1 (0.0-0.55m) TP-06 **MOULD NO :** 42

WEIGHT OF MOULD: 3352 **gms.** **VOLUME OF MOULD:** 943.42 **cc.**

TRIAL NO.		1	2	3	4	5
WATER ADDED	%	5%	7%	9%	11%	13%
WT. SOIL + MOULD (WET)	gm	5195	5278	5374	5360	5338
WEIGHT OF WET SOIL	gm	1843	1926	2022	2008	1986
WET DENSITY	**gm/cc**	1.954	2.041	2.144	2.129	2.105

CAN NO.		N#2	N#8	N#15	N#7	N#6
WT. SOIL + CAN (WET)	gm	708.7	708.2	729.1	756.7	765.3
WT. SOIL + CAN (DRY)	gm	687.0	676.1	683.9	703.3	698.9
WT. OF WATER	gm	21.7	32.1	45.2	53.4	66.4
WT. OF CAN	gm	170.8	157.6	164.3	181.1	172.6
WT. OF DRY SOIL	gm	516.2	518.5	519.6	522.2	526.3
WATER CONTENT	**%**	**4.2**	**6.2**	**8.7**	**10.2**	**12.6**
DRY DENSITY	**gm/cc**	**1.875**	**1.922**	**1.972**	**1.931**	**1.869**

Soil Description: ***CLAYEY SAND with GRAVELS***

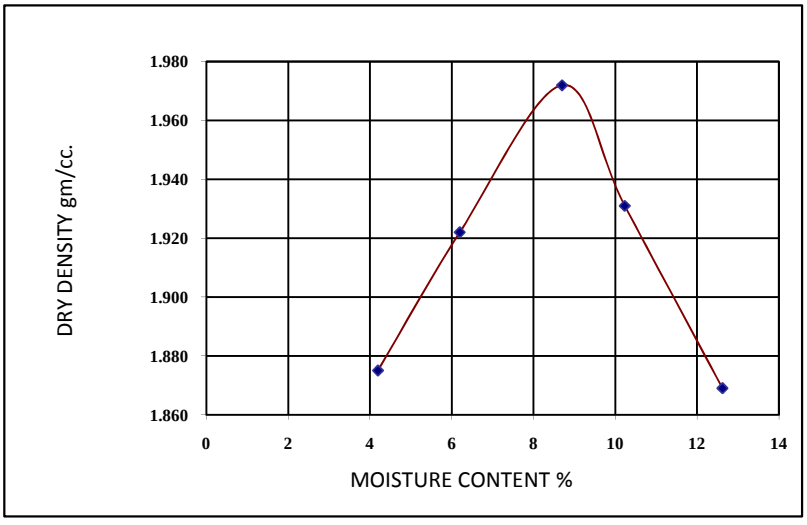
MAX. DRY DENSITY gm/cc : **1.972**

OPT. MOISTURE CONTENT % **8.7**

CORRECTED FOR OVER SIZE

MAX. DRY DENSITY gm/cc : **2.035**

OPT. MOISTURE CONTENT % **7.72**



REMARKS :

TESTED BY: REVIEWED BY:

MIN/RH

(TECHNICIAN) (SUPERVISOR) (LAB. INCHARGE)

MODIFIED PROCTOR TEST

ASTM D 1557 (METHOD " A ")

CLIENT	L & T CONSTRUCTION	REPORT DATE	04-Sep-19
PROJECT/LOCATION	MANSOURAH MASSARAH GOLD PROJECT	SAMPLING DATE	01-Sep-19
METHOD USED	"A"	TEST START DATE	02-Sep-19
SAMPLE SOURCE	ON SITE	TEST FINISH DATE	03-Sep-19
METHOD VARIATION	NONE	GEOTECHNICAL FILE #	-
SAMPLED BY	WAHAB	PROJECT #	19-K-3054

SOURCE / SAMPLE NO : B.S-1 (0.00-0.60m) TP-07 **MOULD NO :** 53

WEIGHT OF MOULD: 3345 **gms.** **VOLUME OF MOULD:** 944.5 **cc.**

TRIAL NO.		1	2	3	4	5
WATER ADDED	%	5%	7%	9%	11%	13%
WT. SOIL + MOULD (WET)	gm	5194	5289	5383	5353	5328
WEIGHT OF WET SOIL	gm	1849	1944	2038	2008	1983
WET DENSITY	**gm/cc**	1.958	2.059	2.158	2.126	2.100

CAN NO.		N#2	N#8	N#15	N#7	N#6
WT. SOIL + CAN (WET)	gm	704.6	705.5	725.5	754.2	764.9
WT. SOIL + CAN (DRY)	gm	682.0	672.1	680.6	700.0	699.0
WT. OF WATER	gm	22.6	33.4	44.9	54.2	65.9
WT. OF CAN	gm	170.8	157.6	164.3	181.1	172.6
WT. OF DRY SOIL	gm	511.2	514.5	516.3	518.9	526.4
WATER CONTENT	**%**	**4.4**	**6.5**	**8.7**	**10.5**	**12.5**
DRY DENSITY	**gm/cc**	**1.875**	**1.933**	**1.985**	**1.925**	**1.866**

Soil Description: ***SILTY SAND with GRAVEL***

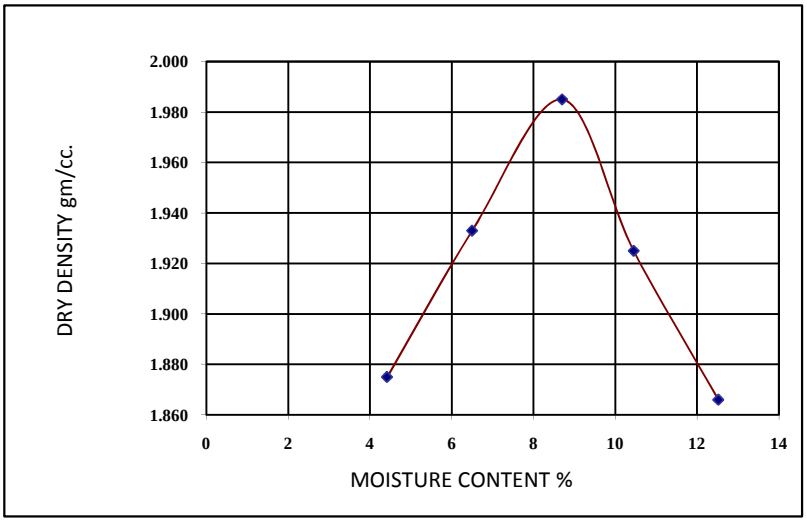
MAX. DRY DENSITY gm/cc : **1.985**

OPT. MOISTURE CONTENT % **8.7**

CORRECTED FOR OVER SIZE

MAX. DRY DENSITY gm/cc : **2.054**

OPT. MOISTURE CONTENT % **7.61**



REMARKS :

TESTED BY: REVIEWED BY:

MIN/RH

(TECHNICIAN) (SUPERVISOR) (LAB. INCHARGE)

MODIFIED PROCTOR TEST

ASTM D 1557 (METHOD " A ")

CLIENT	L & T CONSTRUCTION	REPORT DATE	04-Sep-19
PROJECT/LOCATION	MANSOURAH MASSARAH GOLD PROJECT	SAMPLING DATE	01-Sep-19
METHOD USED	"A"	TEST START DATE	02-Sep-19
SAMPLE SOURCE	ON SITE	TEST FINISH DATE	03-Sep-19
METHOD VARIATION	NONE	GEOTECHNICAL FILE #	-
SAMPLED BY	WAHAB	PROJECT #	19-K-3054

SOURCE / SAMPLE NO : B.S-1 (0.00-0.50m) TP-08 **MOULD NO :** 53

WEIGHT OF MOULD: 3345 **gms.** **VOLUME OF MOULD:** 944.5 **cc.**

TRIAL NO.		1	2	3	4	5
WATER ADDED	%	5%	7%	9%	11%	13%
WT. SOIL + MOULD (WET)	gm	5189	5276	5377	5357	5327
WEIGHT OF WET SOIL	gm	1844	1931	2032	2012	1982
WET DENSITY	**gm/cc**	1.953	2.044	2.151	2.131	2.099

CAN NO.		N#2	N#8	N#15	N#7	N#6
WT. SOIL + CAN (WET)	gm	708.1	755.8	738.8	778.9	754.1
WT. SOIL + CAN (DRY)	gm	686.5	720.9	692.8	723.4	690.4
WT. OF WATER	gm	21.7	34.9	46.0	55.5	63.7
WT. OF CAN	gm	170.8	157.6	164.3	181.1	172.6
WT. OF DRY SOIL	gm	515.6	563.3	528.5	542.3	517.8
WATER CONTENT	**%**	**4.2**	**6.2**	**8.7**	**10.2**	**12.3**
DRY DENSITY	**gm/cc**	**1.874**	**1.925**	**1.979**	**1.933**	**1.869**

Soil Description: ***SILTY SAND with GRAVEL***

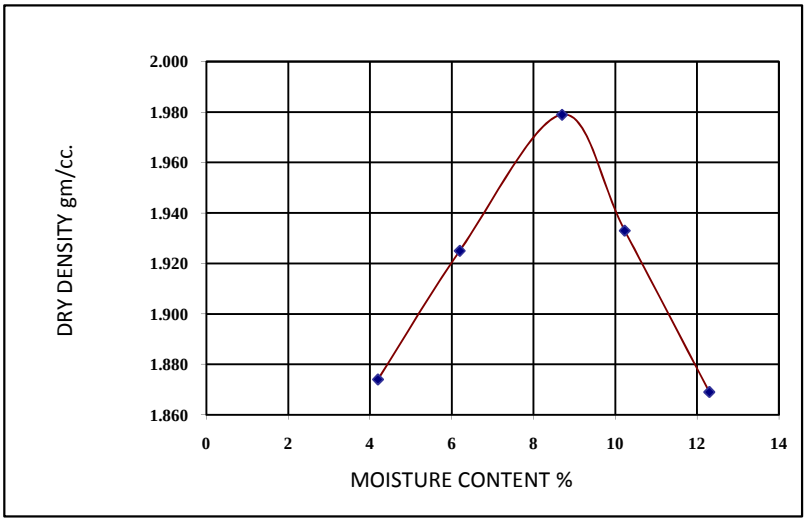
MAX. DRY DENSITY gm/cc : **1.979**

OPT. MOISTURE CONTENT % **8.7**

CORRECTED FOR OVER SIZE

MAX. DRY DENSITY gm/cc : **2.038**

OPT. MOISTURE CONTENT % **7.77**



REMARKS :

TESTED BY: REVIEWED BY:

MIN/RH

(TECHNICIAN) (SUPERVISOR) (LAB. INCHARGE)

MODIFIED PROCTOR TEST

ASTM D 1557 (METHOD " C ")

CLIENT	L & T CONSTRUCTION	REPORT DATE	05-Sep-19
PROJECT/LOCATION	MANSOURAH MASSARAH GOLD PROJECT	SAMPLING DATE	02-Sep-19
METHOD USED	"C"	TEST START DATE	03-Sep-19
SAMPLE SOURCE	ON SITE	TEST FINISH DATE	04-Sep-19
METHOD VARIATION	NONE	GEOTECHNICAL FILE #	-
SAMPLED BY	WAHAB	PROJECT #	19-K-3054

SOURCE / SAMPLE NO : B.S-1 (0.0-0.50m) TP-09 **MOULD NO :** 31

WEIGHT OF MOULD: 6577 **gms.** **VOLUME OF MOULD:** 2012.33 **cc.**

TRIAL NO.		1	2	3	4	5
WATER ADDED	%	5%	7%	9%	11%	13%
WT. SOIL + MOULD (WET)	gm	10744	10940	11127	11082	11054
WEIGHT OF WET SOIL	gm	4167	4363	4550	4505	4477
WET DENSITY	**gm/cc**	2.071	2.168	2.261	2.239	2.225

CAN NO.		N#2	N#8	N#15	N#7	N#6
WT. SOIL + CAN (WET)	gm	709.0	707.9	728.6	758.0	752.9
WT. SOIL + CAN (DRY)	gm	696.3	684.2	694.2	714.3	699.2
WT. OF WATER	gm	12.6	23.7	34.4	43.7	53.7
WT. OF CAN	gm	170.8	157.6	164.3	181.1	172.6
WT. OF DRY SOIL	gm	525.5	526.6	529.9	533.2	526.6
WATER CONTENT	**%**	**2.4**	**4.5**	**6.5**	**8.2**	**10.2**
DRY DENSITY	**gm/cc**	**2.022**	**2.075**	**2.123**	**2.069**	**2.019**

Soil Description: ***well graded SAND with SILT and GRAVELS***

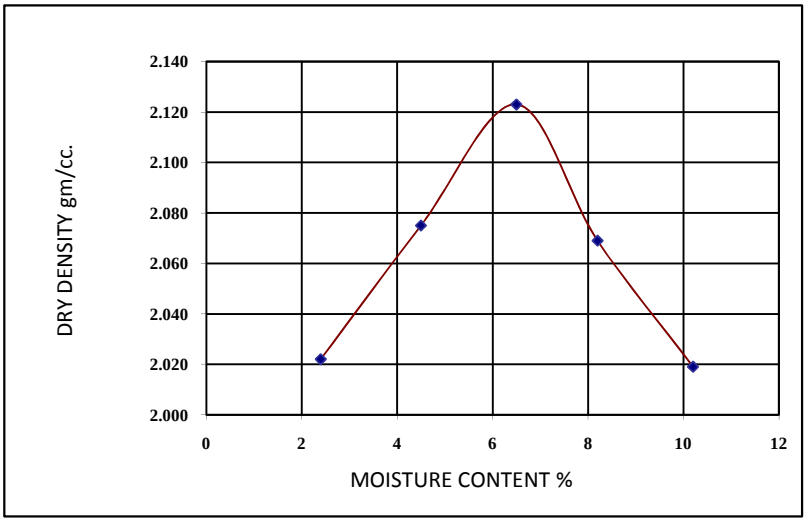
MAX. DRY DENSITY gm/cc : **2.123**

OPT. MOISTURE CONTENT % **6.5**

CORRECTED FOR OVER SIZE

MAX. DRY DENSITY gm/cc : **2.123**

OPT. MOISTURE CONTENT % **6.50**



REMARKS :

TESTED BY: REVIEWED BY:

MIN/RH

(TECHNICIAN) (SUPERVISOR) (LAB. INCHARGE)

MODIFIED PROCTOR TEST

ASTM D 1557 (METHOD " A ")

CLIENT	L & T CONSTRUCTION	REPORT DATE	05-Sep-19
PROJECT/LOCATION	MANSOURAH MASSARAH GOLD PROJECT	SAMPLING DATE	02-Sep-19
METHOD USED	"A"	TEST START DATE	03-Sep-19
SAMPLE SOURCE	ON SITE	TEST FINISH DATE	04-Sep-19
METHOD VARIATION	NONE	GEOTECHNICAL FILE #	-
SAMPLED BY	WAHAB	PROJECT #	19-K-3054

SOURCE / SAMPLE NO : B.S-1 (0.0-1.30m) TP-10 **MOULD NO :** 56

WEIGHT OF MOULD: 3354 **gms.** **VOLUME OF MOULD:** 942.415 **cc.**

TRIAL NO.		1	2	3	4	5
WATER ADDED	%	5%	7%	9%	11%	13%
WT. SOIL + MOULD (WET)	gm	5222	5311	5403	5395	5393
WEIGHT OF WET SOIL	gm	1868	1957	2049	2041	2039
WET DENSITY	**gm/cc**	1.982	2.076	2.174	2.166	2.163

CAN NO.		N#2	N#8	N#15	N#7	N#6
WT. SOIL + CAN (WET)	gm	726.4	716.5	751.2	753.7	823.8
WT. SOIL + CAN (DRY)	gm	704.0	683.9	705.5	700.6	750.8
WT. OF WATER	gm	22.4	32.6	45.7	53.1	73.0
WT. OF CAN	gm	170.8	157.6	164.3	181.1	172.6
WT. OF DRY SOIL	gm	533.2	526.3	541.2	519.5	578.2
WATER CONTENT	**%**	**4.2**	**6.2**	**8.5**	**10.2**	**12.6**
DRY DENSITY	**gm/cc**	**1.902**	**1.955**	**2.005**	**1.965**	**1.921**

Soil Description: ***CLAYEY SAND with GRAVELS***

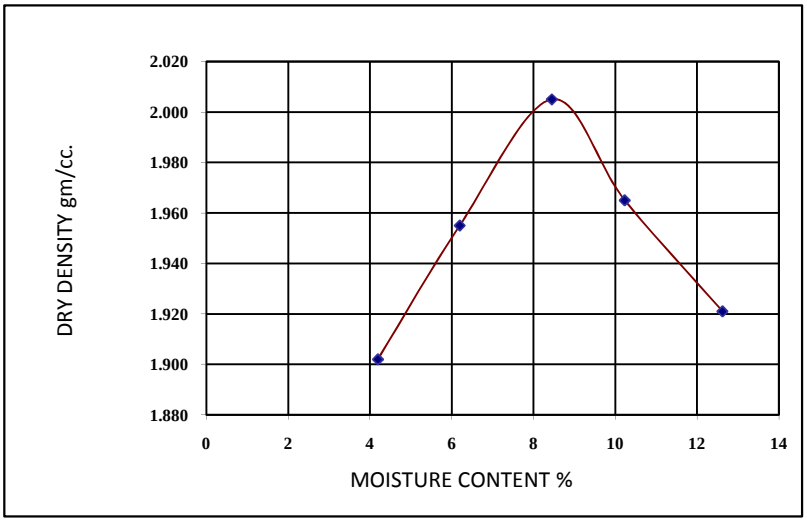
MAX. DRY DENSITY gm/cc : **2.005**

OPT. MOISTURE CONTENT % **8.5**

CORRECTED FOR OVER SIZE

MAX. DRY DENSITY gm/cc : **2.064**

OPT. MOISTURE CONTENT % **7.52**



REMARKS :

TESTED BY: REVIEWED BY:

MIN/RH

(TECHNICIAN) (SUPERVISOR) (LAB. INCHARGE)

MODIFIED PROCTOR TEST

ASTM D 1557 (METHOD " A ")

CLIENT	L & T CONSTRUCTION	REPORT DATE	06-Sep-19
PROJECT/LOCATION	MANSOURAH MASSARAH GOLD PROJECT	SAMPLING DATE	03-Sep-19
METHOD USED	"A"	TEST START DATE	04-Sep-19
SAMPLE SOURCE	ON SITE	TEST FINISH DATE	05-Sep-19
METHOD VARIATION	NONE	GEOTECHNICAL FILE #	-
SAMPLED BY	WAHAB	PROJECT #	19-K-3054

SOURCE / SAMPLE NO : B.S-1 (0.00-0.30m) TP-11 **MOULD NO :** 56

WEIGHT OF MOULD: 3354 **gms.** **VOLUME OF MOULD:** 942.415 **cc.**

TRIAL NO.		1	2	3	4	5
WATER ADDED	%	5%	7%	9%	11%	13%
WT. SOIL + MOULD (WET)	gm	5113	5195	5287	5275	5245
WEIGHT OF WET SOIL	gm	1759	1841	1933	1921	1891
WET DENSITY	**gm/cc**	1.866	1.953	2.051	2.038	2.006

CAN NO.		N#2	N#8	N#15	N#7	N#6
WT. SOIL + CAN (WET)	gm	731.7	721.8	756.9	759.3	828.3
WT. SOIL + CAN (DRY)	gm	704.0	683.9	705.5	700.6	750.8
WT. OF WATER	gm	27.7	37.9	51.4	58.7	77.5
WT. OF CAN	gm	170.8	157.6	164.3	181.1	172.6
WT. OF DRY SOIL	gm	533.2	526.3	541.2	519.5	578.2
WATER CONTENT	**%**	**5.2**	**7.2**	**9.5**	**11.3**	**13.4**
DRY DENSITY	**gm/cc**	**1.774**	**1.822**	**1.873**	**1.831**	**1.769**

Soil Description: ***SILTY SAND***

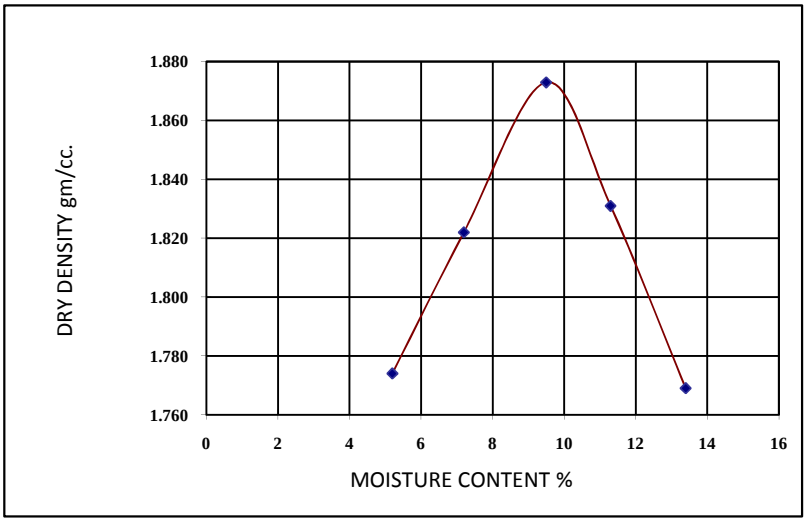
MAX. DRY DENSITY gm/cc : **1.873**

OPT. MOISTURE CONTENT % **9.5**

CORRECTED FOR OVER SIZE

MAX. DRY DENSITY gm/cc : **1.873**

OPT. MOISTURE CONTENT % **9.50**



REMARKS :

TESTED BY: REVIEWED BY:

MIN/RH

(TECHNICIAN) (SUPERVISOR) (LAB. INCHARGE)

MODIFIED PROCTOR TEST

ASTM D 1557 (METHOD " A ")

CLIENT	L & T CONSTRUCTION	REPORT DATE	06-Sep-19
PROJECT/LOCATION	MANSOURAH MASSARAH GOLD PROJECT	SAMPLING DATE	03-Sep-19
METHOD USED	"A"	TEST START DATE	04-Sep-19
SAMPLE SOURCE	ON SITE	TEST FINISH DATE	05-Sep-19
METHOD VARIATION	NONE	GEOTECHNICAL FILE #	-
SAMPLED BY	WAHAB	PROJECT #	19-K-3054

SOURCE / SAMPLE NO : B.S-1 (0.0-0.75m) TP-12 **MOULD NO :** 56

WEIGHT OF MOULD: 3354 **gms.** **VOLUME OF MOULD:** 942.415 **cc.**

TRIAL NO.		1	2	3	4	5
WATER ADDED	%	5%	7%	9%	11%	13%
WT. SOIL + MOULD (WET)	gm	5202	5286	5375	5370	5332
WEIGHT OF WET SOIL	gm	1848	1932	2021	2016	1978
WET DENSITY	**gm/cc**	1.961	2.050	2.144	2.140	2.099

CAN NO.		N#2	N#8	N#15	N#7	N#6
WT. SOIL + CAN (WET)	gm	706.6	734.4	740.7	786.4	771.1
WT. SOIL + CAN (DRY)	gm	683.0	699.2	694.1	727.4	704.6
WT. OF WATER	gm	23.6	35.2	46.6	59.0	66.5
WT. OF CAN	gm	170.8	157.6	164.3	181.1	172.6
WT. OF DRY SOIL	gm	512.2	541.6	529.8	546.3	532.0
WATER CONTENT	**%**	**4.6**	**6.5**	**8.8**	**10.8**	**12.5**
DRY DENSITY	**gm/cc**	**1.875**	**1.925**	**1.971**	**1.931**	**1.866**

Soil Description: ***SILTY SAND with GRAVEL***

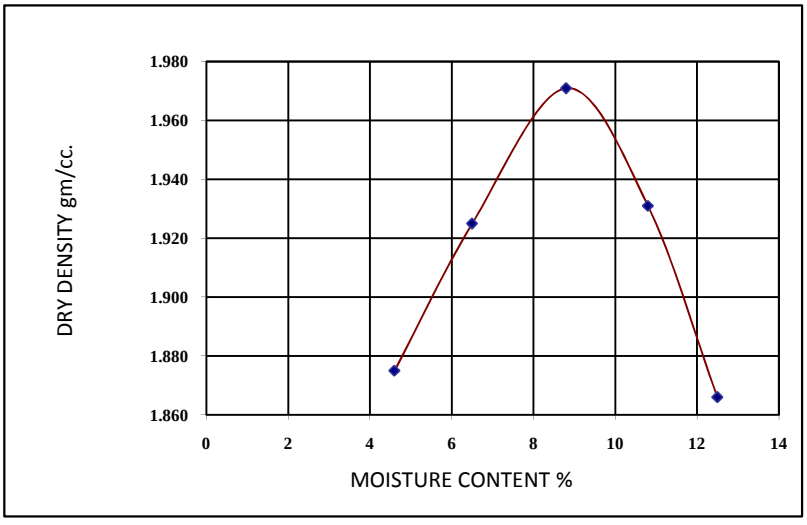
MAX. DRY DENSITY gm/cc : **1.971**

OPT. MOISTURE CONTENT % **8.8**

CORRECTED FOR OVER SIZE

MAX. DRY DENSITY gm/cc : **2.051**

OPT. MOISTURE CONTENT % **7.55**



REMARKS :

TESTED BY: REVIEWED BY:

MIN/RH

(TECHNICIAN) (SUPERVISOR) (LAB. INCHARGE)

MODIFIED PROCTOR TEST

ASTM D 1557 (METHOD " A ")

CLIENT	L & T CONSTRUCTION	REPORT DATE	07-Sep-19
PROJECT/LOCATION	MANSOURAH MASSARAH GOLD PROJECT	SAMPLING DATE	04-Sep-19
METHOD USED	"A"	TEST START DATE	05-Sep-19
SAMPLE SOURCE	ON SITE	TEST FINISH DATE	06-Sep-19
METHOD VARIATION	NONE	GEOTECHNICAL FILE #	-
SAMPLED BY	WAHAB	PROJECT #	19-K-3054

SOURCE / SAMPLE NO : B.S-1 (0.00-0.70m) TP-13 **MOULD NO :** 42

WEIGHT OF MOULD: 3352 **gms.** **VOLUME OF MOULD:** 943.42 **cc.**

TRIAL NO.		1	2	3	4	5
WATER ADDED	%	5%	7%	9%	11%	13%
WT. SOIL + MOULD (WET)	gm	5185	5271	5361	5351	5322
WEIGHT OF WET SOIL	gm	1833	1919	2009	1999	1970
WET DENSITY	**gm/cc**	1.943	2.034	2.129	2.119	2.088

CAN NO.		N#2	N#8	N#15	N#7	N#6
WT. SOIL + CAN (WET)	gm	706.6	709.4	723.5	751.6	756.2
WT. SOIL + CAN (DRY)	gm	685.0	677.2	680.6	698.6	691.1
WT. OF WATER	gm	21.6	32.2	42.9	52.9	65.1
WT. OF CAN	gm	170.8	157.6	164.3	181.1	172.6
WT. OF DRY SOIL	gm	514.2	519.6	516.3	517.5	518.5
WATER CONTENT	**%**	**4.2**	**6.2**	**8.3**	**10.2**	**12.6**
DRY DENSITY	**gm/cc**	**1.865**	**1.915**	**1.966**	**1.922**	**1.855**

Soil Description: ***CLAYEY SAND with GRAVELS***

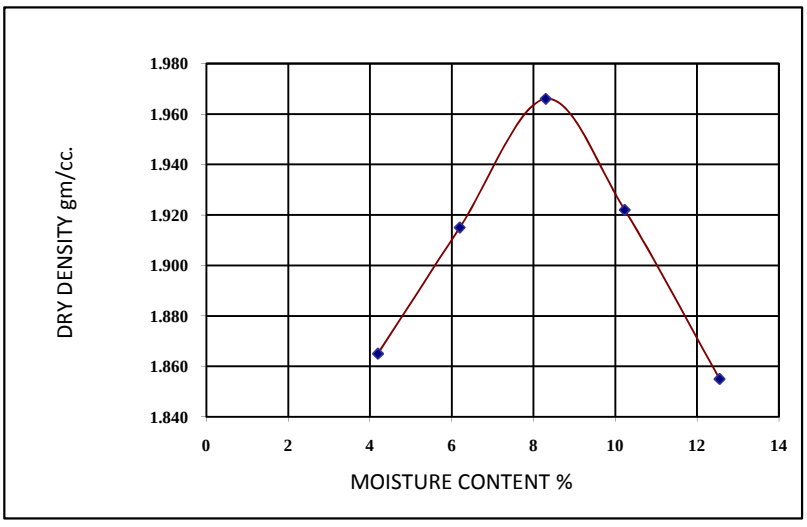
MAX. DRY DENSITY gm/cc : **1.966**

OPT. MOISTURE CONTENT % **8.3**

CORRECTED FOR OVER SIZE

MAX. DRY DENSITY gm/cc : **2.029**

OPT. MOISTURE CONTENT % **7.40**



REMARKS :

TESTED BY: REVIEWED BY:

MIN/RH

(TECHNICIAN) (SUPERVISOR) (LAB. INCHARGE)

MODIFIED PROCTOR TEST

ASTM D 1557 (METHOD " A ")

CLIENT	L & T CONSTRUCTION	REPORT DATE	07-Sep-19
PROJECT/LOCATION	MANSOURAH MASSARAH GOLD PROJECT	SAMPLING DATE	04-Sep-19
METHOD USED	"A"	TEST START DATE	05-Sep-19
SAMPLE SOURCE	ON SITE	TEST FINISH DATE	06-Sep-19
METHOD VARIATION	NONE	GEOTECHNICAL FILE #	-
SAMPLED BY	WAHAB	PROJECT #	19-K-3054

SOURCE / SAMPLE NO : B.S-1 (0.00-0.50m) TP-14 **MOULD NO :** 42

WEIGHT OF MOULD: 3352 **gms.** **VOLUME OF MOULD:** 943.42 **cc.**

TRIAL NO.		1	2	3	4	5
WATER ADDED	%	5%	7%	9%	11%	13%
WT. SOIL + MOULD (WET)	gm	5215	5304	5385	5385	5354
WEIGHT OF WET SOIL	gm	1863	1952	2033	2033	2002
WET DENSITY	**gm/cc**	1.975	2.069	2.155	2.155	2.122

CAN NO.		N#2	N#8	N#15	N#7	N#6
WT. SOIL + CAN (WET)	gm	724.3	712.6	777.0	768.6	789.2
WT. SOIL + CAN (DRY)	gm	702.0	680.2	730.6	713.2	721.1
WT. OF WATER	gm	22.3	32.4	46.4	55.4	68.1
WT. OF CAN	gm	170.8	157.6	164.3	181.1	172.6
WT. OF DRY SOIL	gm	531.2	522.6	566.3	532.1	548.5
WATER CONTENT	**%**	**4.2**	**6.2**	**8.2**	**10.4**	**12.4**
DRY DENSITY	**gm/cc**	**1.895**	**1.948**	**1.992**	**1.952**	**1.888**

Soil Description: ***CLAYEY SAND with GRAVELS***

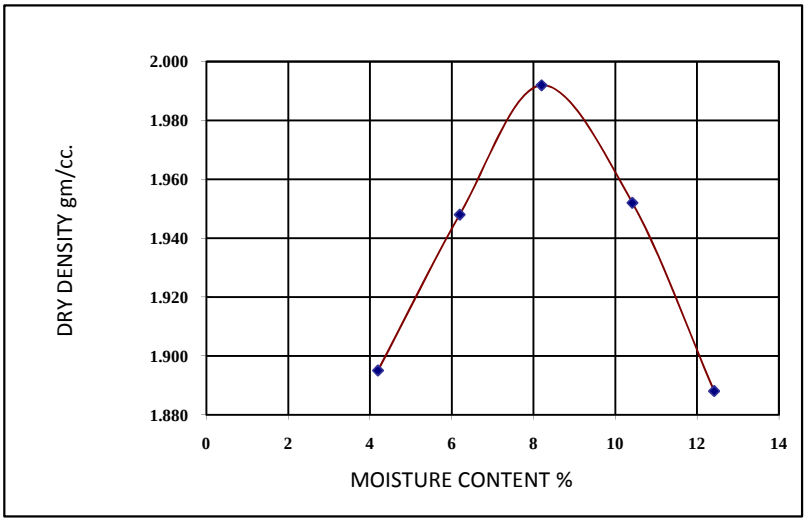
MAX. DRY DENSITY gm/cc : **1.992**

OPT. MOISTURE CONTENT % **8.2**

CORRECTED FOR OVER SIZE

MAX. DRY DENSITY gm/cc : **2.052**

OPT. MOISTURE CONTENT % **7.32**



REMARKS :

TESTED BY: REVIEWED BY:

MIN/RH

(TECHNICIAN) (SUPERVISOR) (LAB. INCHARGE)

MODIFIED PROCTOR TEST

ASTM D 1557 (METHOD " A ")

CLIENT	L & T CONSTRUCTION	REPORT DATE	08-Sep-19
PROJECT/LOCATION	MANSOURAH MASSARAH GOLD PROJECT	SAMPLING DATE	05-Sep-19
METHOD USED	"A"	TEST START DATE	06-Sep-19
SAMPLE SOURCE	ON SITE	TEST FINISH DATE	07-Sep-19
METHOD VARIATION	NONE	GEOTECHNICAL FILE #	-
SAMPLED BY	WAHAB	PROJECT #	19-K-3054

SOURCE / SAMPLE NO : B.S-1 (0.0-0.50m) TP-15 **MOULD NO :** 56

WEIGHT OF MOULD: 3354 **gms.** **VOLUME OF MOULD:** 942.415 **cc.**

TRIAL NO.		1	2	3	4	5
WATER ADDED	%	5%	7%	9%	11%	13%
WT. SOIL + MOULD (WET)	gm	5185	5269	5357	5350	5323
WEIGHT OF WET SOIL	gm	1831	1915	2003	1996	1969
WET DENSITY	**gm/cc**	1.942	2.032	2.125	2.118	2.089

CAN NO.		N#2	N#8	N#15	N#7	N#6
WT. SOIL + CAN (WET)	gm	724.1	716.3	720.0	747.3	753.5
WT. SOIL + CAN (DRY)	gm	707.4	689.7	683.2	700.1	695.0
WT. OF WATER	gm	16.6	26.6	36.8	47.2	58.5
WT. OF CAN	gm	170.8	157.6	164.3	181.1	172.6
WT. OF DRY SOIL	gm	536.6	532.1	518.9	519.0	522.4
WATER CONTENT	**%**	**3.1**	**5.0**	**7.1**	**9.1**	**11.2**
DRY DENSITY	**gm/cc**	**1.884**	**1.935**	**1.984**	**1.941**	**1.879**

Soil Description: ***CLAYEY SAND with GRAVELS***

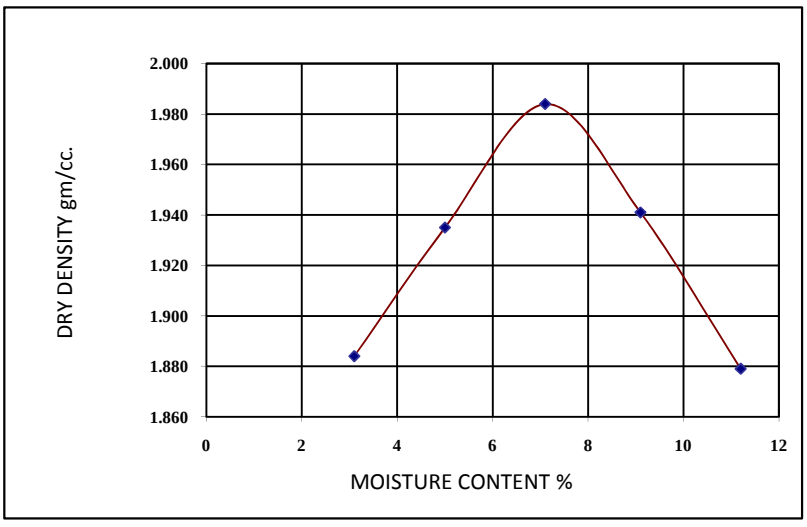
MAX. DRY DENSITY gm/cc : **1.984**

OPT. MOISTURE CONTENT % **7.1**

CORRECTED FOR OVER SIZE

MAX. DRY DENSITY gm/cc : **2.049**

OPT. MOISTURE CONTENT % **6.34**



REMARKS :

TESTED BY: REVIEWED BY:

MIN/RH

(TECHNICIAN) (SUPERVISOR) (LAB. INCHARGE)

MODIFIED PROCTOR TEST

ASTM D 1557 (METHOD " A ")

CLIENT	L & T CONSTRUCTION	REPORT DATE	09-Sep-19
PROJECT/LOCATION	MANSOURAH MASSARAH GOLD PROJECT	SAMPLING DATE	06-Sep-19
METHOD USED	"A"	TEST START DATE	07-Sep-19
SAMPLE SOURCE	ON SITE	TEST FINISH DATE	08-Sep-19
METHOD VARIATION	NONE	GEOTECHNICAL FILE #	-
SAMPLED BY	WAHAB	PROJECT #	19-K-3054

SOURCE / SAMPLE NO : B.S-1 (0.00-1.0m) TP-16 **MOULD NO :** 53

WEIGHT OF MOULD: 3345 **gms.** **VOLUME OF MOULD:** 944.5 **cc.**

TRIAL NO.		1	2	3	4	5
WATER ADDED	%	5%	7%	9%	11%	13%
WT. SOIL + MOULD (WET)	gm	5165	5260	5350	5335	5303
WEIGHT OF WET SOIL	gm	1820	1915	2005	1990	1958
WET DENSITY	**gm/cc**	1.926	2.028	2.123	2.107	2.073

CAN NO.		N#2	N#8	N#15	N#7	N#6
WT. SOIL + CAN (WET)	gm	742.0	751.4	748.2	779.5	788.0
WT. SOIL + CAN (DRY)	gm	726.4	724.2	710.5	731.1	727.6
WT. OF WATER	gm	15.6	27.2	37.7	48.4	60.5
WT. OF CAN	gm	170.8	157.6	164.3	181.1	172.6
WT. OF DRY SOIL	gm	555.6	566.6	546.2	550.0	555.0
WATER CONTENT	**%**	**2.8**	**4.8**	**6.9**	**8.8**	**10.9**
DRY DENSITY	**gm/cc**	**1.874**	**1.935**	**1.986**	**1.937**	**1.869**

Soil Description: ***CLAYEY SAND with GRAVELS***

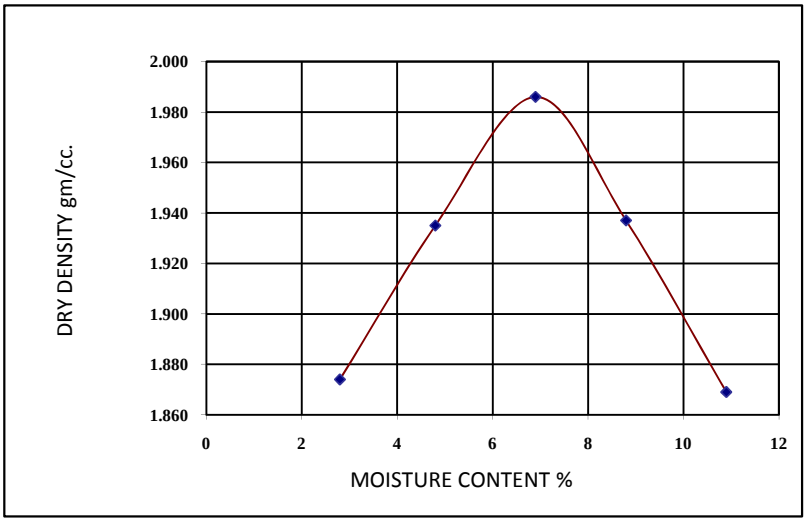
MAX. DRY DENSITY gm/cc : **1.986**

OPT. MOISTURE CONTENT % **6.9**

CORRECTED FOR OVER SIZE

MAX. DRY DENSITY gm/cc : **2.047**

OPT. MOISTURE CONTENT % **6.22**



REMARKS :

TESTED BY: REVIEWED BY:

MIN/RH

(TECHNICIAN) (SUPERVISOR) (LAB. INCHARGE)

MODIFIED PROCTOR TEST

ASTM D 1557 (METHOD " A ")

CLIENT	L & T CONSTRUCTION	REPORT DATE	09-Sep-19
PROJECT/LOCATION	MANSOURAH MASSARAH GOLD PROJECT	SAMPLING DATE	06-Sep-19
METHOD USED	"A"	TEST START DATE	07-Sep-19
SAMPLE SOURCE	ON SITE	TEST FINISH DATE	08-Sep-19
METHOD VARIATION	NONE	GEOTECHNICAL FILE #	-
SAMPLED BY	WAHAB	PROJECT #	19-K-3054

SOURCE / SAMPLE NO : B.S-1 (0.00-0.90m) TP-17 **MOULD NO :** 56

WEIGHT OF MOULD: 3354 **gms.** **VOLUME OF MOULD:** 942.415 **cc.**

TRIAL NO.		1	2	3	4	5
WATER ADDED	%	5%	7%	9%	11%	13%
WT. SOIL + MOULD (WET)	gm	5194	5283	5372	5364	5335
WEIGHT OF WET SOIL	gm	1840	1929	2018	2010	1981
WET DENSITY	**gm/cc**	1.953	2.047	2.141	2.133	2.102

CAN NO.		N#2	N#8	N#15	N#7	N#6
WT. SOIL + CAN (WET)	gm	743.6	753.1	749.8	781.8	790.4
WT. SOIL + CAN (DRY)	gm	726.4	724.2	710.5	731.1	727.6
WT. OF WATER	gm	17.2	28.9	39.3	50.8	62.8
WT. OF CAN	gm	170.8	157.6	164.3	181.1	172.6
WT. OF DRY SOIL	gm	555.6	566.6	546.2	550.0	555.0
WATER CONTENT	**%**	**3.1**	**5.1**	**7.2**	**9.2**	**11.3**
DRY DENSITY	**gm/cc**	**1.894**	**1.948**	**1.997**	**1.953**	**1.888**

Soil Description: ***SILTY SAND with GRAVEL***

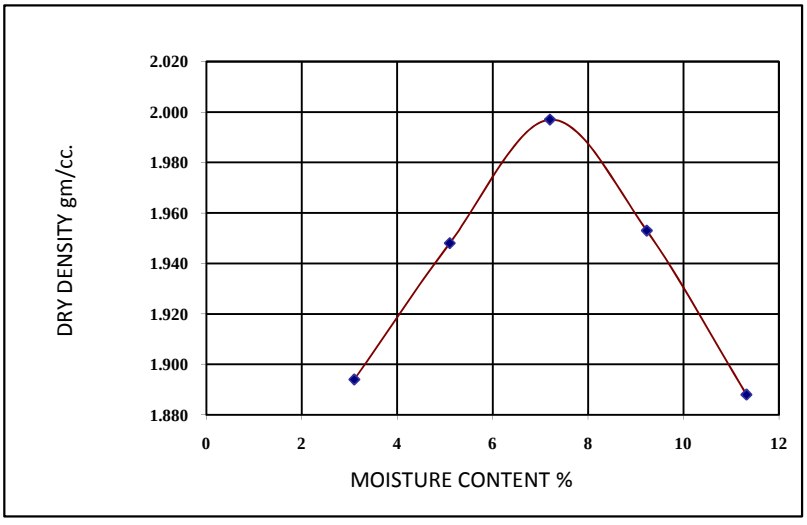
MAX. DRY DENSITY gm/cc : **1.997**

OPT. MOISTURE CONTENT % **7.2**

CORRECTED FOR OVER SIZE

MAX. DRY DENSITY gm/cc : **2.063**

OPT. MOISTURE CONTENT % **6.40**



REMARKS :

TESTED BY: REVIEWED BY:

MIN/RH

(TECHNICIAN) (SUPERVISOR) (LAB. INCHARGE)

MODIFIED PROCTOR TEST

ASTM D 1557 (METHOD " A ")

CLIENT	L & T CONSTRUCTION	REPORT DATE	10-Sep-19
PROJECT/LOCATION	MANSOURAH MASSARAH GOLD PROJECT	SAMPLING DATE	07-Sep-19
METHOD USED	"A"	TEST START DATE	08-Sep-19
SAMPLE SOURCE	ON SITE	TEST FINISH DATE	09-Sep-19
METHOD VARIATION	NONE	GEOTECHNICAL FILE #	-
SAMPLED BY	WAHAB	PROJECT #	19-K-3054

SOURCE / SAMPLE NO : B.S-1 (0.00-0.50m) TP-18 **MOULD NO :** 53

WEIGHT OF MOULD: 3345 **gms.** **VOLUME OF MOULD:** 944.5 **cc.**

TRIAL NO.		1	2	3	4	5
WATER ADDED	%	5%	7%	9%	11%	13%
WT. SOIL + MOULD (WET)	gm	5198	5286	5371	5354	5339
WEIGHT OF WET SOIL	gm	1853	1941	2026	2009	1994
WET DENSITY	**gm/cc**	1.962	2.055	2.145	2.127	2.111

CAN NO.		N#2	N#8	N#15	N#7	N#6
WT. SOIL + CAN (WET)	gm	755.6	734.1	800.2	746.3	777.8
WT. SOIL + CAN (DRY)	gm	737.4	706.1	757.5	698.7	715.8
WT. OF WATER	gm	18.1	28.0	42.7	47.6	62.0
WT. OF CAN	gm	170.8	157.6	164.3	181.1	172.6
WT. OF DRY SOIL	gm	566.6	548.5	593.2	517.6	543.2
WATER CONTENT	**%**	**3.2**	**5.1**	**7.2**	**9.2**	**11.4**
DRY DENSITY	**gm/cc**	**1.901**	**1.955**	**2.001**	**1.948**	**1.895**

Soil Description: ***SILTY SAND with GRAVEL***

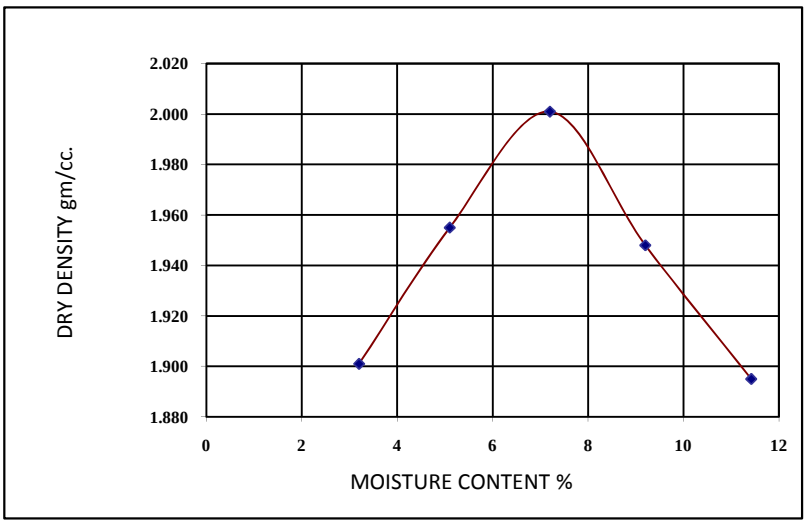
MAX. DRY DENSITY gm/cc : **2.001**

OPT. MOISTURE CONTENT % **7.2**

CORRECTED FOR OVER SIZE

MAX. DRY DENSITY gm/cc : **2.075**

OPT. MOISTURE CONTENT % **6.30**



REMARKS :

TESTED BY: REVIEWED BY:

MIN/RH

(TECHNICIAN) (SUPERVISOR) (LAB. INCHARGE)

MODIFIED PROCTOR TEST

ASTM D 1557 (METHOD " A ")

CLIENT	L & T CONSTRUCTION	REPORT DATE	10-Sep-19
PROJECT/LOCATION	MANSOURAH MASSARAH GOLD PROJECT	SAMPLING DATE	07-Sep-19
METHOD USED	"A"	TEST START DATE	08-Sep-19
SAMPLE SOURCE	ON SITE	TEST FINISH DATE	09-Sep-19
METHOD VARIATION	NONE	GEOTECHNICAL FILE #	-
SAMPLED BY	WAHAB	PROJECT #	19-K-3054

SOURCE / SAMPLE NO : B.S-1 (0.2-0.50m) TP-19 **MOULD NO :** 56

WEIGHT OF MOULD: 3354 **gms.** **VOLUME OF MOULD:** 942.415 **cc.**

TRIAL NO.		1	2	3	4	5
WATER ADDED	%	5%	7%	9%	11%	13%
WT. SOIL + MOULD (WET)	gm	5178	5265	5357	5348	5323
WEIGHT OF WET SOIL	gm	1824	1911	2003	1994	1969
WET DENSITY	**gm/cc**	1.935	2.027	2.126	2.116	2.089

CAN NO.		N#2	N#8	N#15	N#7	N#6
WT. SOIL + CAN (WET)	gm	748.5	748.6	766.7	774.7	778.1
WT. SOIL + CAN (DRY)	gm	731.1	719.9	726.2	724.7	717.1
WT. OF WATER	gm	17.4	28.7	40.5	50.0	61.0
WT. OF CAN	gm	170.8	157.6	164.3	181.1	172.6
WT. OF DRY SOIL	gm	560.3	562.3	561.9	543.6	544.5
WATER CONTENT	**%**	**3.1**	**5.1**	**7.2**	**9.2**	**11.2**
DRY DENSITY	**gm/cc**	**1.877**	**1.929**	**1.983**	**1.938**	**1.879**

Soil Description: ***CLAYEY SAND with GRAVELS***

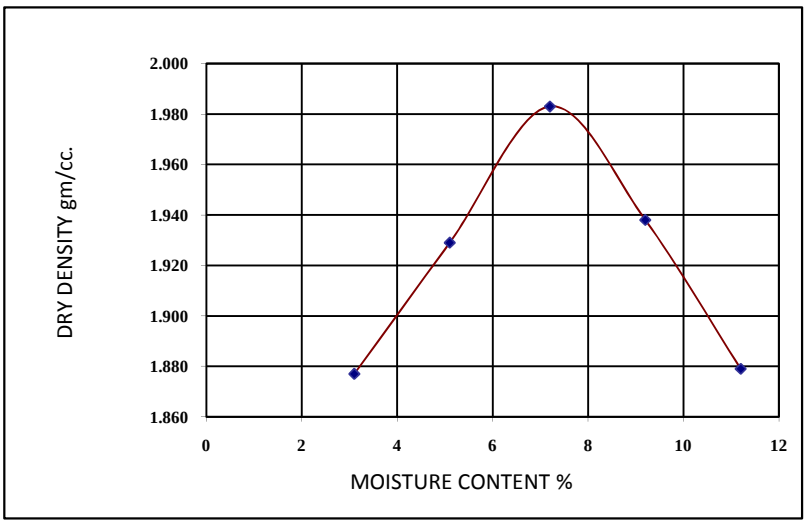
MAX. DRY DENSITY gm/cc : **1.983**

OPT. MOISTURE CONTENT % **7.2**

CORRECTED FOR OVER SIZE

MAX. DRY DENSITY gm/cc : **2.043**

OPT. MOISTURE CONTENT % **6.49**



REMARKS :

TESTED BY: REVIEWED BY:

MIN/RH

(TECHNICIAN) (SUPERVISOR) (LAB. INCHARGE)

MODIFIED PROCTOR TEST

ASTM D 1557 (METHOD " A ")

CLIENT	L & T CONSTRUCTION	REPORT DATE	11-Sep-19
PROJECT/LOCATION	MANSOURAH MASSARAH GOLD PROJECT	SAMPLING DATE	08-Sep-19
METHOD USED	"A"	TEST START DATE	09-Sep-19
SAMPLE SOURCE	ON SITE	TEST FINISH DATE	10-Sep-19
METHOD VARIATION	NONE	GEOTECHNICAL FILE #	-
SAMPLED BY	WAHAB	PROJECT #	19-K-3054

SOURCE / SAMPLE NO : B.S-1 (0.00-0.50m) TP-20 **MOULD NO :** 56

WEIGHT OF MOULD: 3354 **gms.** **VOLUME OF MOULD:** 942.415 **cc.**

TRIAL NO.		1	2	3	4	5
WATER ADDED	%	5%	7%	9%	11%	13%
WT. SOIL + MOULD (WET)	gm	5179	5275	5365	5352	5310
WEIGHT OF WET SOIL	gm	1825	1921	2011	1998	1956
WET DENSITY	**gm/cc**	1.936	2.039	2.134	2.120	2.075

CAN NO.		N#2	N#8	N#15	N#7	N#6
WT. SOIL + CAN (WET)	gm	711.1	711.6	732.8	799.5	775.5
WT. SOIL + CAN (DRY)	gm	694.3	684.2	694.2	747.4	714.8
WT. OF WATER	gm	16.8	27.4	38.7	52.1	60.7
WT. OF CAN	gm	170.8	157.6	164.3	181.1	172.6
WT. OF DRY SOIL	gm	523.5	526.6	529.9	566.3	542.2
WATER CONTENT	**%**	**3.2**	**5.2**	**7.3**	**9.2**	**11.2**
DRY DENSITY	**gm/cc**	**1.876**	**1.938**	**1.989**	**1.941**	**1.866**

Soil Description: ***CLAYEY SAND with GRAVELS***

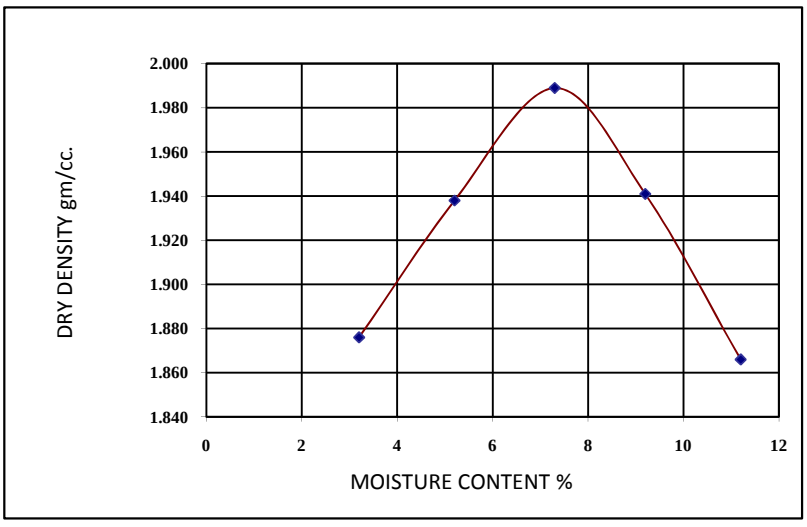
MAX. DRY DENSITY gm/cc : **1.989**

OPT. MOISTURE CONTENT % **7.3**

CORRECTED FOR OVER SIZE

MAX. DRY DENSITY gm/cc : **2.052**

OPT. MOISTURE CONTENT % **6.53**



REMARKS :

TESTED BY: REVIEWED BY:

MIN/RH

(TECHNICIAN) (SUPERVISOR) (LAB. INCHARGE)

MODIFIED PROCTOR TEST

ASTM D 1557 (METHOD " A ")

CLIENT	L & T CONSTRUCTION	REPORT DATE	11-Sep-19
PROJECT/LOCATION	MANSOURAH MASSARAH GOLD PROJECT	SAMPLING DATE	08-Sep-19
METHOD USED	"A"	TEST START DATE	09-Sep-19
SAMPLE SOURCE	ON SITE	TEST FINISH DATE	10-Sep-19
METHOD VARIATION	NONE	GEOTECHNICAL FILE #	-
SAMPLED BY	WAHAB	PROJECT #	19-K-3054

SOURCE / SAMPLE NO : B.S-1 (0.4-1.30m) TP-21 **MOULD NO :** 53

WEIGHT OF MOULD: 3345 **gms.** **VOLUME OF MOULD:** 944.5 **cc.**

TRIAL NO.		1	2	3	4	5
WATER ADDED	%	5%	7%	9%	11%	13%
WT. SOIL + MOULD (WET)	gm	5191	5277	5367	5359	5332
WEIGHT OF WET SOIL	gm	1846	1932	2022	2014	1987
WET DENSITY	**gm/cc**	1.954	2.045	2.140	2.133	2.103

CAN NO.		N#2	N#8	N#15	N#7	N#6
WT. SOIL + CAN (WET)	gm	713.9	728.1	733.3	796.4	790.9
WT. SOIL + CAN (DRY)	gm	697.5	699.9	693.6	744.3	727.9
WT. OF WATER	gm	16.4	28.2	39.7	52.1	63.0
WT. OF CAN	gm	170.8	157.6	164.3	181.1	172.6
WT. OF DRY SOIL	gm	526.6	542.3	529.3	563.2	555.3
WATER CONTENT	**%**	**3.1**	**5.2**	**7.5**	**9.3**	**11.4**
DRY DENSITY	**gm/cc**	**1.895**	**1.944**	**1.991**	**1.952**	**1.889**

Soil Description: ***CLAYEY SAND with GRAVELS***

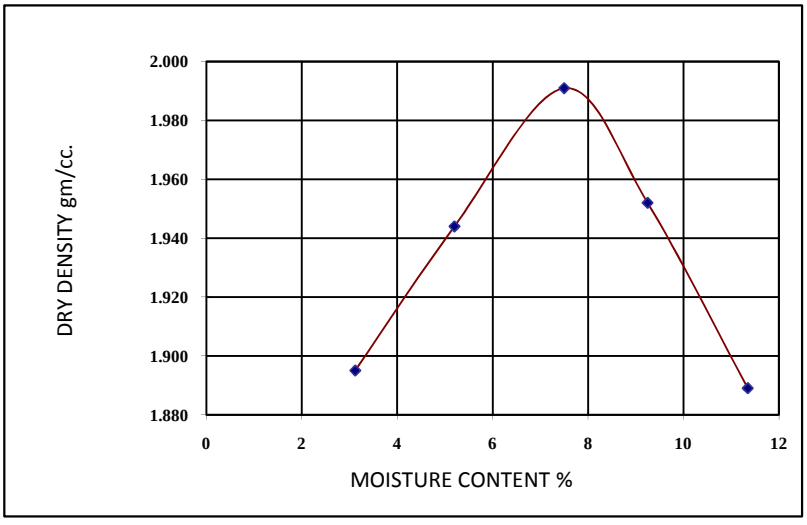
MAX. DRY DENSITY gm/cc : **1.991**

OPT. MOISTURE CONTENT % **7.5**

CORRECTED FOR OVER SIZE

MAX. DRY DENSITY gm/cc : **2.049**

OPT. MOISTURE CONTENT % **6.75**



REMARKS :

TESTED BY: MIN/RH (TECHNICIAN)

REVIEWED BY: (SUPERVISOR)

(LAB. INCHARGE)

MODIFIED PROCTOR TEST

ASTM D 1557 (METHOD " A ")

CLIENT	L & T CONSTRUCTION	REPORT DATE	12-Sep-19
PROJECT/LOCATION	MANSOURAH MASSARAH GOLD PROJECT	SAMPLING DATE	09-Sep-19
METHOD USED	"A"	TEST START DATE	10-Sep-19
SAMPLE SOURCE	ON SITE	TEST FINISH DATE	11-Sep-19
METHOD VARIATION	NONE	GEOTECHNICAL FILE #	-
SAMPLED BY	WAHAB	PROJECT #	19-K-3054

SOURCE / SAMPLE NO : B.S-1 (0.0-0.50m) TP-22 **MOULD NO :** 53

WEIGHT OF MOULD: 3345 **gms.** **VOLUME OF MOULD:** 944.5 **cc.**

TRIAL NO.		1	2	3	4	5
WATER ADDED	%	5%	7%	9%	11%	13%
WT. SOIL + MOULD (WET)	gm	5118	5201	5292	5279	5245
WEIGHT OF WET SOIL	gm	1773	1856	1947	1934	1900
WET DENSITY	**gm/cc**	1.877	1.965	2.062	2.047	2.012

CAN NO.		N#2	N#8	N#15	N#7	N#6
WT. SOIL + CAN (WET)	gm	707.6	706.2	729.5	753.2	768.2
WT. SOIL + CAN (DRY)	gm	681.0	669.8	680.5	695.6	699.2
WT. OF WATER	gm	26.5	36.4	49.0	57.6	69.0
WT. OF CAN	gm	170.8	157.6	164.3	181.1	172.6
WT. OF DRY SOIL	gm	510.2	512.2	516.2	514.5	526.6
WATER CONTENT	**%**	**5.2**	**7.1**	**9.5**	**11.2**	**13.1**
DRY DENSITY	**gm/cc**	**1.784**	**1.835**	**1.883**	**1.841**	**1.779**

Soil Description: ***SILTY SAND***

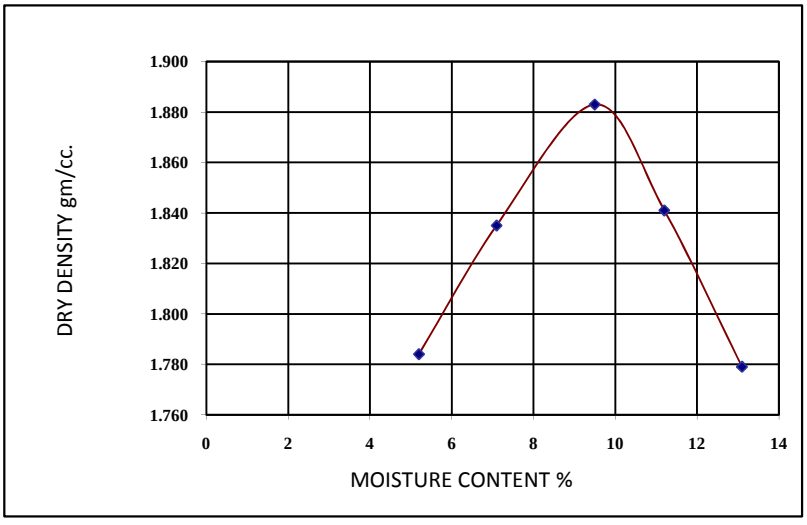
MAX. DRY DENSITY gm/cc : **1.883**

OPT. MOISTURE CONTENT % **9.5**

CORRECTED FOR OVER SIZE

MAX. DRY DENSITY gm/cc : **1.883**

OPT. MOISTURE CONTENT % **9.50**



REMARKS :

TESTED BY: REVIEWED BY:

MIN/RH

(TECHNICIAN) (SUPERVISOR) (LAB. INCHARGE)

MODIFIED PROCTOR TEST

ASTM D 1557 (METHOD " C ")

CLIENT	L & T CONSTRUCTION	REPORT DATE	12-Sep-19
PROJECT/LOCATION	MANSOURAH MASSARAH GOLD PROJECT	SAMPLING DATE	09-Sep-19
METHOD USED	"C"	TEST START DATE	10-Sep-19
SAMPLE SOURCE	ON SITE	TEST FINISH DATE	11-Sep-19
METHOD VARIATION	NONE	GEOTECHNICAL FILE #	-
SAMPLED BY	WAHAB	PROJECT #	19-K-3054

SOURCE / SAMPLE NO : B.S-1 (0.30-0.55m) TP-23 **MOULD NO :** 31

WEIGHT OF MOULD: 6577 **gms.** **VOLUME OF MOULD:** 2012.333 **cc.**

TRIAL NO.		1	2	3	4	5
WATER ADDED	%	5%	7%	9%	11%	13%
WT. SOIL + MOULD (WET)	gm	10766	10949	11153	11145	11087
WEIGHT OF WET SOIL	gm	4189	4372	4576	4568	4510
WET DENSITY	**gm/cc**	2.082	2.173	2.274	2.270	2.241

CAN NO.		N#2	N#8	N#15	N#7	N#6
WT. SOIL + CAN (WET)	gm	692.8	691.3	714.1	738.8	753.1
WT. SOIL + CAN (DRY)	gm	681.0	669.8	680.5	695.6	699.2
WT. OF WATER	gm	11.7	21.5	33.6	43.2	53.9
WT. OF CAN	gm	170.8	157.6	164.3	181.1	172.6
WT. OF DRY SOIL	gm	510.2	512.2	516.2	514.5	526.6
WATER CONTENT	**%**	**2.3**	**4.2**	**6.5**	**8.4**	**10.2**
DRY DENSITY	**gm/cc**	**2.035**	**2.085**	**2.135**	**2.094**	**2.033**

Soil Description: ***POORLY GRADED GRAVEL with SAND***

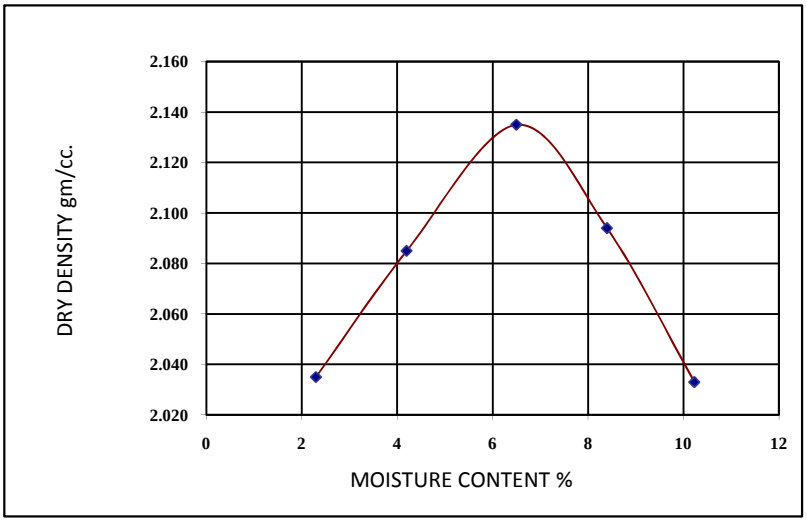
MAX. DRY DENSITY gm/cc : **2.135**

OPT. MOISTURE CONTENT % **6.5**

CORRECTED FOR OVER SIZE

MAX. DRY DENSITY gm/cc : **2.135**

OPT. MOISTURE CONTENT % **6.50**



REMARKS :

TESTED BY: REVIEWED BY:

MIN/RH

(TECHNICIAN) (SUPERVISOR) (LAB. INCHARGE)

MODIFIED PROCTOR TEST

ASTM D 1557 (METHOD " A ")

CLIENT	L & T CONSTRUCTION	REPORT DATE	13-Sep-19
PROJECT/LOCATION	MANSOURAH MASSARAH GOLD PROJECT	SAMPLING DATE	10-Sep-19
METHOD USED	"A"	TEST START DATE	11-Sep-19
SAMPLE SOURCE	ON SITE	TEST FINISH DATE	12-Sep-19
METHOD VARIATION	NONE	GEOTECHNICAL FILE #	-
SAMPLED BY	WAHAB	PROJECT #	19-K-3054

SOURCE / SAMPLE NO : B.S-1 (0.0-0.55m) TP-24 **MOULD NO :** 53

WEIGHT OF MOULD: 3345 **gms.** **VOLUME OF MOULD:** 944.5 **cc.**

TRIAL NO.		1	2	3	4	5
WATER ADDED	%	5%	7%	9%	11%	13%
WT. SOIL + MOULD (WET)	gm	5083	5168	5253	5238	5190
WEIGHT OF WET SOIL	gm	1738	1823	1908	1893	1845
WET DENSITY	**gm/cc**	1.840	1.930	2.020	2.005	1.954

CAN NO.		N#2	N#8	N#15	N#7	N#6
WT. SOIL + CAN (WET)	gm	709.1	708.2	729.5	753.7	769.0
WT. SOIL + CAN (DRY)	gm	681.0	669.8	680.5	695.6	699.2
WT. OF WATER	gm	28.1	38.4	49.0	58.1	69.8
WT. OF CAN	gm	170.8	157.6	164.3	181.1	172.6
WT. OF DRY SOIL	gm	510.2	512.2	516.2	514.5	526.6
WATER CONTENT	**%**	**5.5**	**7.5**	**9.5**	**11.3**	**13.3**
DRY DENSITY	**gm/cc**	**1.744**	**1.795**	**1.845**	**1.801**	**1.725**

Soil Description: ***SILTY SAND***

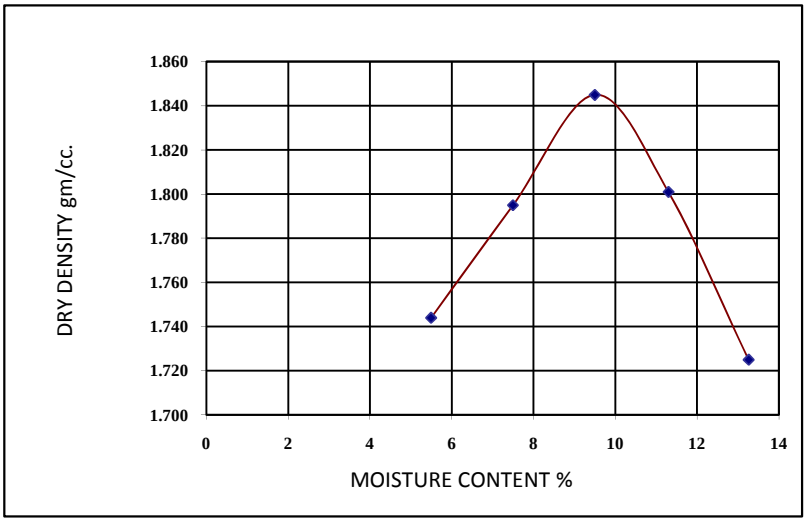
MAX. DRY DENSITY gm/cc : **1.845**

OPT. MOISTURE CONTENT % **9.5**

CORRECTED FOR OVER SIZE

MAX. DRY DENSITY gm/cc : **1.845**

OPT. MOISTURE CONTENT % **9.50**



REMARKS :

TESTED BY: REVIEWED BY:

MIN/RH

(TECHNICIAN) (SUPERVISOR) (LAB. INCHARGE)

MODIFIED PROCTOR TEST

ASTM D 1557 (METHOD " A ")

CLIENT	L & T CONSTRUCTION	REPORT DATE	13-Sep-19
PROJECT/LOCATION	MANSOURAH MASSARAH GOLD PROJECT	SAMPLING DATE	10-Sep-19
METHOD USED	"A"	TEST START DATE	11-Sep-19
SAMPLE SOURCE	ON SITE	TEST FINISH DATE	12-Sep-19
METHOD VARIATION	NONE	GEOTECHNICAL FILE #	-
SAMPLED BY	WAHAB	PROJECT #	19-K-3054

SOURCE / SAMPLE NO : B.S-1 (0.00-0.50m) TP-25 **MOULD NO :** 53

WEIGHT OF MOULD: 3345 **gms.** **VOLUME OF MOULD:** 944.5 **cc.**

TRIAL NO.		1	2	3	4	5
WATER ADDED	%	5%	7%	9%	11%	13%
WT. SOIL + MOULD (WET)	gm	5171	5255	5347	5334	5304
WEIGHT OF WET SOIL	gm	1826	1910	2002	1989	1959
WET DENSITY	**gm/cc**	1.933	2.022	2.119	2.106	2.075

CAN NO.		N#2	N#8	N#15	N#7	N#6
WT. SOIL + CAN (WET)	gm	702.4	702.1	723.9	748.1	763.4
WT. SOIL + CAN (DRY)	gm	681.0	669.8	680.5	695.6	699.2
WT. OF WATER	gm	21.4	32.3	43.4	52.5	64.2
WT. OF CAN	gm	170.8	157.6	164.3	181.1	172.6
WT. OF DRY SOIL	gm	510.2	512.2	516.2	514.5	526.6
WATER CONTENT	**%**	**4.2**	**6.3**	**8.4**	**10.2**	**12.2**
DRY DENSITY	**gm/cc**	**1.855**	**1.902**	**1.955**	**1.911**	**1.849**

Soil Description: ***SILTY SAND with GRAVEL***

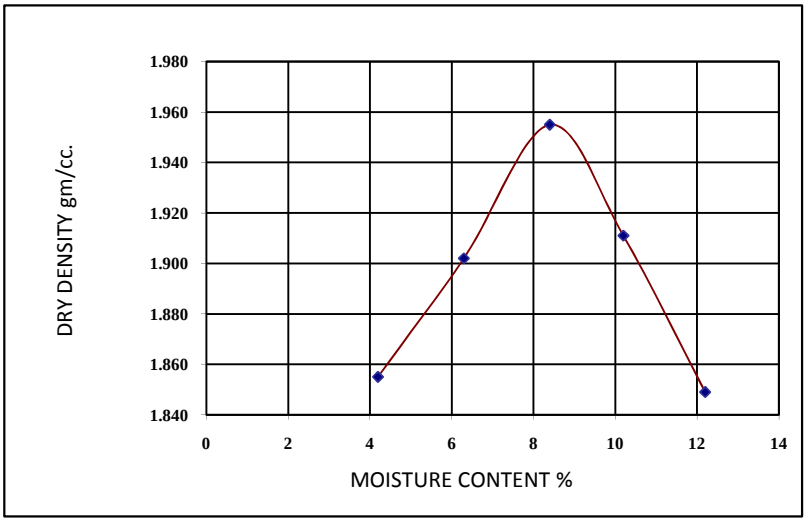
MAX. DRY DENSITY gm/cc : **1.955**

OPT. MOISTURE CONTENT % **8.4**

CORRECTED FOR OVER SIZE

MAX. DRY DENSITY gm/cc : **2.040**

OPT. MOISTURE CONTENT % **7.19**



REMARKS :

TESTED BY: REVIEWED BY:

MIN/RH

(TECHNICIAN) (SUPERVISOR) (LAB. INCHARGE)

MODIFIED PROCTOR TEST

ASTM D 1557 (METHOD " A ")

CLIENT	L & T CONSTRUCTION	REPORT DATE	14-Sep-19
PROJECT/LOCATION	MANSOURAH MASSARAH GOLD PROJECT	SAMPLING DATE	11-Sep-19
METHOD USED	"A"	TEST START DATE	12-Sep-19
SAMPLE SOURCE	ON SITE	TEST FINISH DATE	13-Sep-19
METHOD VARIATION	NONE	GEOTECHNICAL FILE #	-
SAMPLED BY	WAHAB	PROJECT #	19-K-3054

SOURCE / SAMPLE NO : B.S-1 (0.00-0.50m) TP-26 **MOULD NO :** 53

WEIGHT OF MOULD: 3345 **gms.** **VOLUME OF MOULD:** 944.5 **cc.**

TRIAL NO.		1	2	3	4	5
WATER ADDED	%	5%	7%	9%	11%	13%
WT. SOIL + MOULD (WET)	gm	5157	5242	5340	5317	5272
WEIGHT OF WET SOIL	gm	1812	1897	1995	1972	1927
WET DENSITY	**gm/cc**	1.919	2.008	2.112	2.088	2.041

CAN NO.		N#2	N#8	N#15	N#7	N#6
WT. SOIL + CAN (WET)	gm	696.8	695.9	720.2	742.9	758.2
WT. SOIL + CAN (DRY)	gm	681.0	669.8	680.5	695.6	699.2
WT. OF WATER	gm	15.8	26.1	39.7	47.3	59.0
WT. OF CAN	gm	170.8	157.6	164.3	181.1	172.6
WT. OF DRY SOIL	gm	510.2	512.2	516.2	514.5	526.6
WATER CONTENT	**%**	**3.1**	**5.1**	**7.7**	**9.2**	**11.2**
DRY DENSITY	**gm/cc**	**1.861**	**1.911**	**1.961**	**1.912**	**1.835**

Soil Description: ***SILTY SAND with GRAVEL***

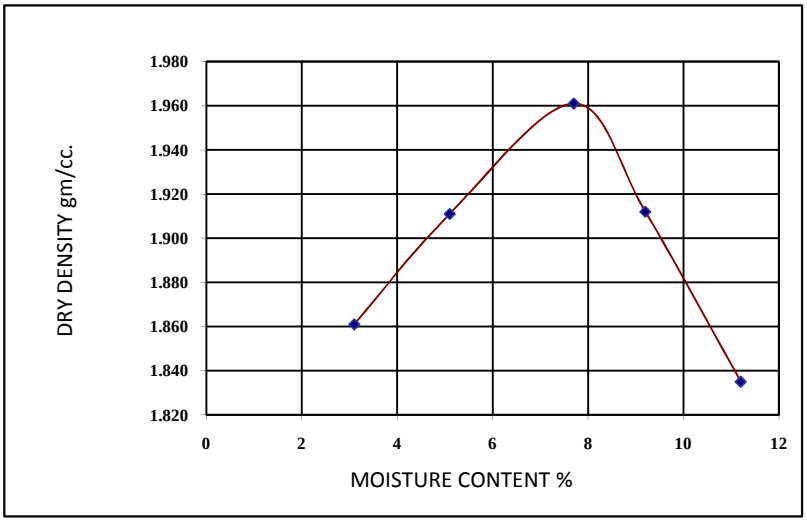
MAX. DRY DENSITY gm/cc : **1.961**

OPT. MOISTURE CONTENT % **7.7**

CORRECTED FOR OVER SIZE

MAX. DRY DENSITY gm/cc : **2.039**

OPT. MOISTURE CONTENT % **6.72**



REMARKS :

TESTED BY: REVIEWED BY:

MIN/RH

(TECHNICIAN) (SUPERVISOR) (LAB. INCHARGE)

MODIFIED PROCTOR TEST

ASTM D 1557 (METHOD " A ")

CLIENT	L & T CONSTRUCTION	REPORT DATE	14-Sep-19
PROJECT/LOCATION	MANSOURAH MASSARAH GOLD PROJECT	SAMPLING DATE	11-Sep-19
METHOD USED	"A"	TEST START DATE	12-Sep-19
SAMPLE SOURCE	ON SITE	TEST FINISH DATE	13-Sep-19
METHOD VARIATION	NONE	GEOTECHNICAL FILE #	-
SAMPLED BY	WAHAB	PROJECT #	19-K-3054

SOURCE / SAMPLE NO : B.S-1 (0.0-0.50m) TP-27 **MOULD NO :** 53

WEIGHT OF MOULD: 3345 **gms.** **VOLUME OF MOULD:** 944.5 **cc.**

TRIAL NO.		1	2	3	4	5
WATER ADDED	%	5%	7%	9%	11%	13%
WT. SOIL + MOULD (WET)	gm	5175	5258	5350	5344	5305
WEIGHT OF WET SOIL	gm	1830	1913	2005	1999	1960
WET DENSITY	**gm/cc**	1.937	2.025	2.123	2.117	2.076

CAN NO.		N#2	N#8	N#15	N#7	N#6
WT. SOIL + CAN (WET)	gm	697.3	696.4	719.2	744.5	758.3
WT. SOIL + CAN (DRY)	gm	681.0	669.8	680.5	695.6	699.2
WT. OF WATER	gm	16.3	26.6	38.7	48.9	59.1
WT. OF CAN	gm	170.8	157.6	164.3	181.1	172.6
WT. OF DRY SOIL	gm	510.2	512.2	516.2	514.5	526.6
WATER CONTENT	**%**	**3.2**	**5.2**	**7.5**	**9.5**	**11.2**
DRY DENSITY	**gm/cc**	**1.877**	**1.925**	**1.975**	**1.933**	**1.866**

Soil Description: ***SILTY SAND with GRAVEL***

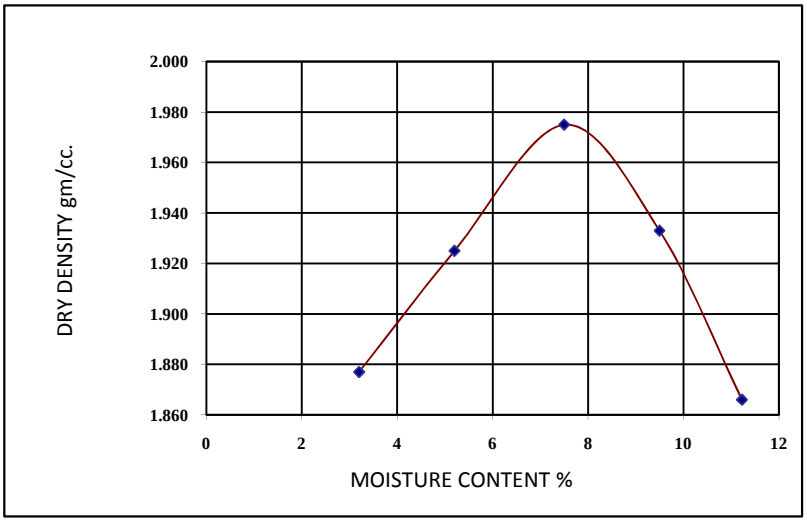
MAX. DRY DENSITY gm/cc : **1.975**

OPT. MOISTURE CONTENT % **7.5**

CORRECTED FOR OVER SIZE

MAX. DRY DENSITY gm/cc : **2.047**

OPT. MOISTURE CONTENT % **6.60**



REMARKS :

TESTED BY: REVIEWED BY:

MIN/RH

(TECHNICIAN) (SUPERVISOR) (LAB. INCHARGE)

MODIFIED PROCTOR TEST

ASTM D 1557 (METHOD " A ")

CLIENT	L & T CONSTRUCTION	REPORT DATE	15-Sep-19
PROJECT/LOCATION	MANSOURAH MASSARAH GOLD PROJECT	SAMPLING DATE	12-Sep-19
METHOD USED	"A"	TEST START DATE	13-Sep-19
SAMPLE SOURCE	ON SITE	TEST FINISH DATE	14-Sep-19
METHOD VARIATION	NONE	GEOTECHNICAL FILE #	-
SAMPLED BY	WAHAB	PROJECT #	19-K-3054

SOURCE / SAMPLE NO : B.S-1 (0.00-0.70m) TP-28 **MOULD NO :** 42

WEIGHT OF MOULD: 3352 **gms.** **VOLUME OF MOULD:** 943.42 **cc.**

TRIAL NO.		1	2	3	4	5
WATER ADDED	%	5%	7%	9%	11%	13%
WT. SOIL + MOULD (WET)	gm	4965	5046	5129	5098	5075
WEIGHT OF WET SOIL	gm	1613	1694	1777	1746	1723
WET DENSITY	**gm/cc**	1.709	1.796	1.884	1.851	1.827

CAN NO.		N#2	N#8	N#15	N#7	N#6
WT. SOIL + CAN (WET)	gm	719.5	718.4	811.3	794.2	757.7
WT. SOIL + CAN (DRY)	gm	686.0	674.5	749.8	727.4	684.8
WT. OF WATER	gm	33.5	43.9	61.5	66.8	72.9
WT. OF CAN	gm	170.8	157.6	164.3	181.1	172.6
WT. OF DRY SOIL	gm	515.2	516.9	585.5	546.3	512.2
WATER CONTENT	**%**	**6.5**	**8.5**	**10.5**	**12.2**	**14.2**
DRY DENSITY	**gm/cc**	**1.605**	**1.655**	**1.705**	**1.649**	**1.599**

Soil Description: ***CLAYEY SAND***

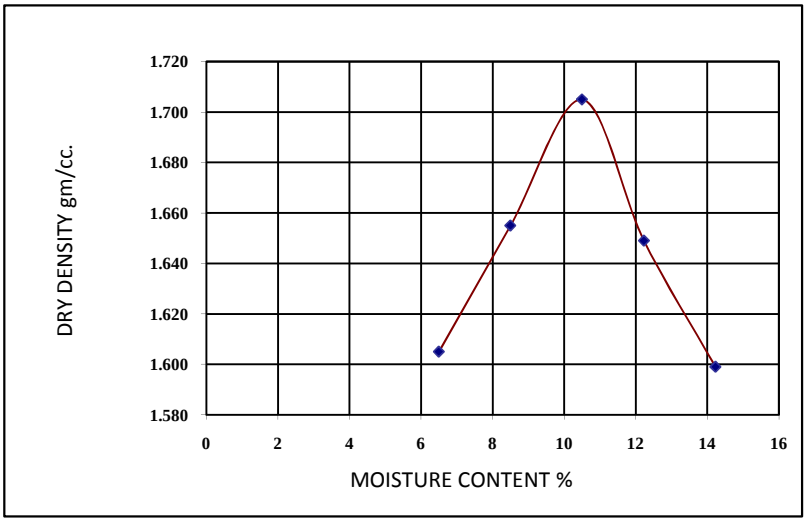
MAX. DRY DENSITY gm/cc : **1.705**

OPT. MOISTURE CONTENT % **10.5**

CORRECTED FOR OVER SIZE

MAX. DRY DENSITY gm/cc : **1.705**

OPT. MOISTURE CONTENT % **10.50**



REMARKS :

TESTED BY: REVIEWED BY:

MIN/RH

(TECHNICIAN) (SUPERVISOR) (LAB. INCHARGE)

MODIFIED PROCTOR TEST

ASTM D 1557 (METHOD " A ")

CLIENT	L & T CONSTRUCTION	REPORT DATE	15-Sep-19
PROJECT/LOCATION	MANSOURAH MASSARAH GOLD PROJECT	SAMPLING DATE	12-Sep-19
METHOD USED	"A"	TEST START DATE	13-Sep-19
SAMPLE SOURCE	ON SITE	TEST FINISH DATE	14-Sep-19
METHOD VARIATION	NONE	GEOTECHNICAL FILE #	-
SAMPLED BY	WAHAB	PROJECT #	19-K-3054

SOURCE / SAMPLE NO : B.S-1 (0.0-1.00m) TP-29 **MOULD NO :** 42

WEIGHT OF MOULD: 3352 **gms.** **VOLUME OF MOULD:** 943.42 **cc.**

TRIAL NO.		1	2	3	4	5
WATER ADDED	%	5%	7%	9%	11%	13%
WT. SOIL + MOULD (WET)	gm	5167	5251	5349	5332	5305
WEIGHT OF WET SOIL	gm	1815	1899	1997	1980	1953
WET DENSITY	**gm/cc**	1.924	2.012	2.117	2.099	2.070

CAN NO.		N#2	N#8	N#15	N#7	N#6
WT. SOIL + CAN (WET)	gm	703.5	703.1	722.9	752.6	759.5
WT. SOIL + CAN (DRY)	gm	687.0	676.1	683.9	703.3	698.9
WT. OF WATER	gm	16.5	27.0	39.0	49.3	60.6
WT. OF CAN	gm	170.8	157.6	164.3	181.1	172.6
WT. OF DRY SOIL	gm	516.2	518.5	519.6	522.2	526.3
WATER CONTENT	**%**	**3.2**	**5.2**	**7.5**	**9.5**	**11.5**
DRY DENSITY	**gm/cc**	**1.864**	**1.913**	**1.969**	**1.918**	**1.856**

Soil Description: ***SILTY SAND with GRAVEL***

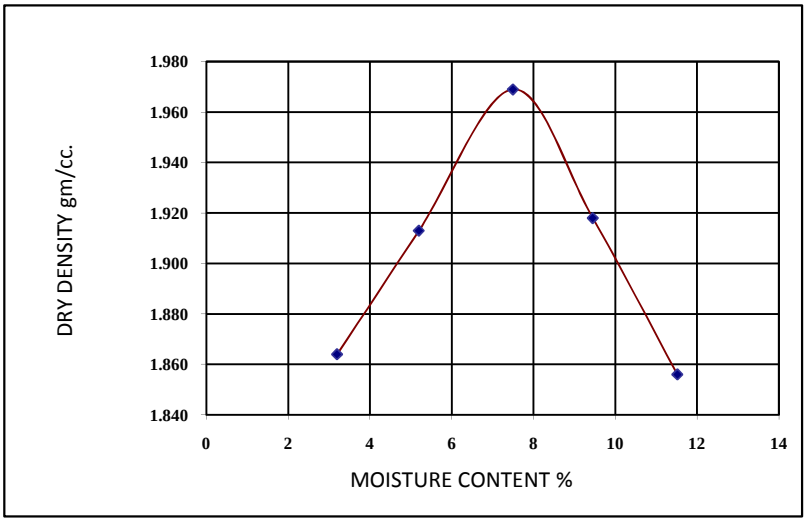
MAX. DRY DENSITY gm/cc : **1.969**

OPT. MOISTURE CONTENT % **7.5**

CORRECTED FOR OVER SIZE

MAX. DRY DENSITY gm/cc : **2.033**

OPT. MOISTURE CONTENT % **6.71**



REMARKS :

TESTED BY: REVIEWED BY:

MIN/RH

(TECHNICIAN) (SUPERVISOR) (LAB. INCHARGE)

MODIFIED PROCTOR TEST

ASTM D 1557 (METHOD " A ")

CLIENT	L & T CONSTRUCTION	REPORT DATE	16-Sep-19
PROJECT/LOCATION	MANSOURAH MASSARAH GOLD PROJECT	SAMPLING DATE	13-Sep-19
METHOD USED	"A"	TEST START DATE	14-Sep-19
SAMPLE SOURCE	ON SITE	TEST FINISH DATE	15-Sep-19
METHOD VARIATION	NONE	GEOTECHNICAL FILE #	-
SAMPLED BY	WAHAB	PROJECT #	19-K-3054

SOURCE / SAMPLE NO : B.S-1 (0.00-0.40m) TP-30 **MOULD NO :** 53

WEIGHT OF MOULD: 3345 **gms.** **VOLUME OF MOULD:** 944.5 **cc.**

TRIAL NO.		1	2	3	4	5
WATER ADDED	%	5%	7%	9%	11%	13%
WT. SOIL + MOULD (WET)	gm	5187	5274	5364	5358	5326
WEIGHT OF WET SOIL	gm	1842	1929	2019	2013	1981
WET DENSITY	**gm/cc**	1.950	2.043	2.137	2.131	2.098

CAN NO.		N#2	N#8	N#15	N#7	N#6
WT. SOIL + CAN (WET)	gm	709.6	710.2	729.6	759.1	769.8
WT. SOIL + CAN (DRY)	gm	682.0	672.1	680.6	700.0	699.0
WT. OF WATER	gm	27.6	38.1	49.0	59.1	70.8
WT. OF CAN	gm	170.8	157.6	164.3	181.1	172.6
WT. OF DRY SOIL	gm	511.2	514.5	516.3	518.9	526.4
WATER CONTENT	**%**	**5.4**	**7.4**	**9.5**	**11.4**	**13.5**
DRY DENSITY	**gm/cc**	**1.850**	**1.902**	**1.952**	**1.913**	**1.849**

Soil Description: ***CLAYEY SAND***

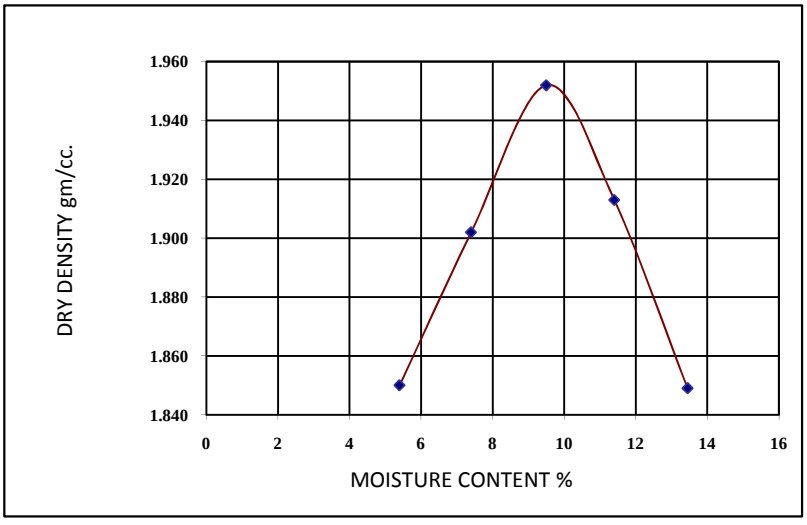
MAX. DRY DENSITY gm/cc : **1.952**

OPT. MOISTURE CONTENT % **9.5**

CORRECTED FOR OVER SIZE

MAX. DRY DENSITY gm/cc : **1.952**

OPT. MOISTURE CONTENT % **9.50**



REMARKS :

TESTED BY: REVIEWED BY:

MIN/RH

(TECHNICIAN) (SUPERVISOR) (LAB. INCHARGE)

MODIFIED PROCTOR TEST

ASTM D 1557 (METHOD " A ")

CLIENT	L & T CONSTRUCTION	REPORT DATE	16-Sep-19
PROJECT/LOCATION	MANSOURAH MASSARAH GOLD PROJECT	SAMPLING DATE	13-Sep-19
METHOD USED	"A"	TEST START DATE	14-Sep-19
SAMPLE SOURCE	ON SITE	TEST FINISH DATE	15-Sep-19
METHOD VARIATION	NONE	GEOTECHNICAL FILE #	-
SAMPLED BY	WAHAB	PROJECT #	19-K-3054

SOURCE / SAMPLE NO : B.S-1 (0.0-0.41m) TP-31 **MOULD NO :** 53

WEIGHT OF MOULD: 3345 **gms.** **VOLUME OF MOULD:** 944.5 **cc.**

TRIAL NO.		1	2	3	4	5
WATER ADDED	%	5%	7%	9%	11%	13%
WT. SOIL + MOULD (WET)	gm	5158	5245	5330	5310	5280
WEIGHT OF WET SOIL	gm	1813	1900	1985	1965	1935
WET DENSITY	**gm/cc**	1.920	2.012	2.102	2.080	2.048

CAN NO.		N#2	N#8	N#15	N#7	N#6
WT. SOIL + CAN (WET)	gm	713.3	761.5	741.4	784.7	758.2
WT. SOIL + CAN (DRY)	gm	686.5	720.9	692.8	723.4	690.4
WT. OF WATER	gm	26.8	40.6	48.6	61.3	67.8
WT. OF CAN	gm	170.8	157.6	164.3	181.1	172.6
WT. OF DRY SOIL	gm	515.6	563.3	528.5	542.3	517.8
WATER CONTENT	**%**	**5.2**	**7.2**	**9.2**	**11.3**	**13.1**
DRY DENSITY	**gm/cc**	**1.825**	**1.877**	**1.925**	**1.869**	**1.811**

Soil Description: ***SILTY SAND***

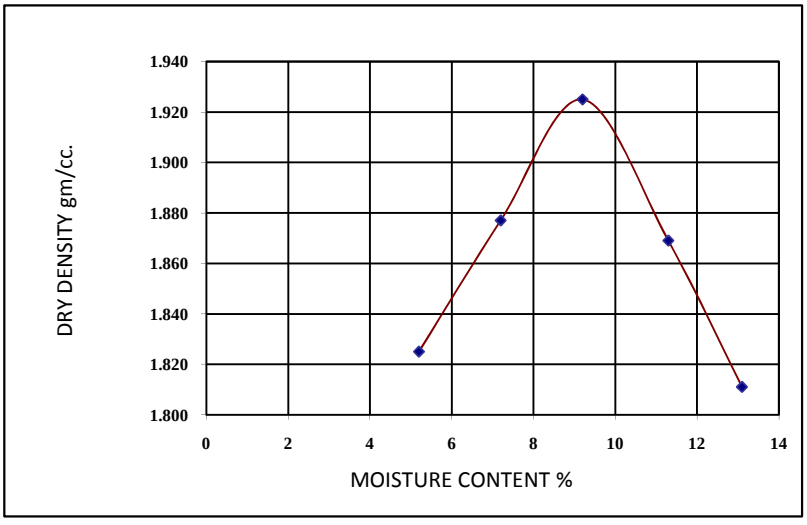
MAX. DRY DENSITY gm/cc : **1.925**

OPT. MOISTURE CONTENT % **9.2**

CORRECTED FOR OVER SIZE

MAX. DRY DENSITY gm/cc : **1.925**

OPT. MOISTURE CONTENT % **9.20**



REMARKS :

TESTED BY: REVIEWED BY:

MIN/RH

(TECHNICIAN) (SUPERVISOR) (LAB. INCHARGE)

MODIFIED PROCTOR TEST

ASTM D 1557 (METHOD " A ")

CLIENT	L & T CONSTRUCTION	REPORT DATE	17-Sep-19
PROJECT/LOCATION	MANSOURAH MASSARAH GOLD PROJECT	SAMPLING DATE	14-Sep-19
METHOD USED	"A"	TEST START DATE	15-Sep-19
SAMPLE SOURCE	ON SITE	TEST FINISH DATE	16-Sep-19
METHOD VARIATION	NONE	GEOTECHNICAL FILE #	-
SAMPLED BY	WAHAB	PROJECT #	19-K-3054

SOURCE / SAMPLE NO : B.S-1 (0.00-0.40m) TP-32 **MOULD NO :** 56

WEIGHT OF MOULD: 3354 **gms.** **VOLUME OF MOULD:** 942.415 **cc.**

TRIAL NO.		1	2	3	4	5
WATER ADDED	%	5%	7%	9%	11%	13%
WT. SOIL + MOULD (WET)	gm	5134	5217	5307	5289	5265
WEIGHT OF WET SOIL	gm	1780	1863	1953	1935	1911
WET DENSITY	**gm/cc**	1.889	1.977	2.073	2.053	2.028

CAN NO.		N#2	N#8	N#15	N#7	N#6
WT. SOIL + CAN (WET)	gm	724.2	722.1	742.9	774.6	769.8
WT. SOIL + CAN (DRY)	gm	696.3	684.2	694.2	714.3	699.2
WT. OF WATER	gm	27.9	37.9	48.7	60.3	70.6
WT. OF CAN	gm	170.8	157.6	164.3	181.1	172.6
WT. OF DRY SOIL	gm	525.5	526.6	529.9	533.2	526.6
WATER CONTENT	**%**	**5.3**	**7.2**	**9.2**	**11.3**	**13.4**
DRY DENSITY	**gm/cc**	**1.794**	**1.844**	**1.898**	**1.845**	**1.788**

Soil Description: ***SILTY SAND***

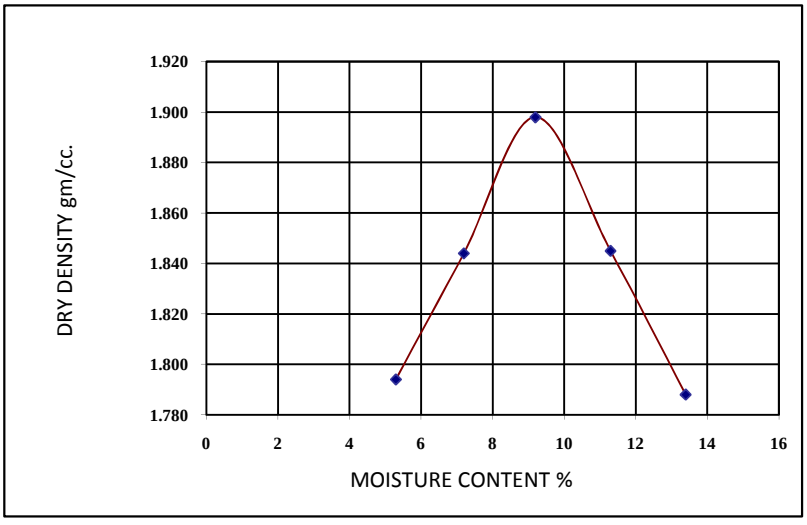
MAX. DRY DENSITY gm/cc : **1.898**

OPT. MOISTURE CONTENT % **9.2**

CORRECTED FOR OVER SIZE

MAX. DRY DENSITY gm/cc : **1.898**

OPT. MOISTURE CONTENT % **9.20**



REMARKS :

TESTED BY: REVIEWED BY:

MIN/RH

(TECHNICIAN) (SUPERVISOR) (LAB. INCHARGE)

MODIFIED PROCTOR TEST

ASTM D 1557 (METHOD " A ")

CLIENT	L & T CONSTRUCTION	REPORT DATE	17-Sep-19
PROJECT/LOCATION	MANSOURAH MASSARAH GOLD PROJECT	SAMPLING DATE	14-Sep-19
METHOD USED	"A"	TEST START DATE	15-Sep-19
SAMPLE SOURCE	ON SITE	TEST FINISH DATE	16-Sep-19
METHOD VARIATION	NONE	GEOTECHNICAL FILE #	-
SAMPLED BY	WAHAB	PROJECT #	19-K-3054

SOURCE / SAMPLE NO : B.S-1 (0.00-0.40m) TP-33 **MOULD NO :** 56

WEIGHT OF MOULD: 3354 **gms.** **VOLUME OF MOULD:** 942.415 **cc.**

TRIAL NO.		1	2	3	4	5
WATER ADDED	%	5%	7%	9%	11%	13%
WT. SOIL + MOULD (WET)	gm	5200	5285	5376	5366	5336
WEIGHT OF WET SOIL	gm	1846	1931	2022	2012	1982
WET DENSITY	**gm/cc**	1.958	2.049	2.145	2.135	2.103

CAN NO.		N#2	N#8	N#15	N#7	N#6
WT. SOIL + CAN (WET)	gm	728.0	718.1	751.5	754.9	823.2
WT. SOIL + CAN (DRY)	gm	704.0	683.9	705.5	700.6	750.8
WT. OF WATER	gm	24.0	34.2	46.0	54.3	72.4
WT. OF CAN	gm	170.8	157.6	164.3	181.1	172.6
WT. OF DRY SOIL	gm	533.2	526.3	541.2	519.5	578.2
WATER CONTENT	**%**	**4.5**	**6.5**	**8.5**	**10.5**	**12.5**
DRY DENSITY	**gm/cc**	**1.874**	**1.924**	**1.977**	**1.933**	**1.869**

Soil Description: ***SILTY SAND with GRAVEL***

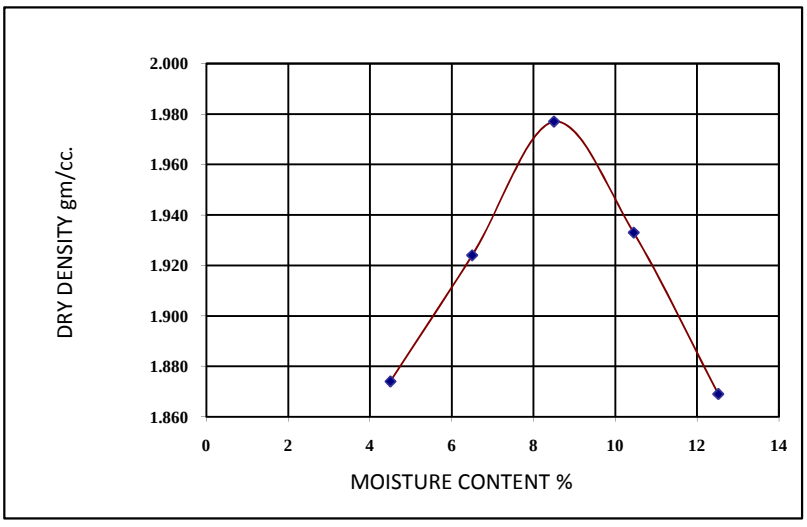
MAX. DRY DENSITY gm/cc : **1.977**

OPT. MOISTURE CONTENT % **8.5**

CORRECTED FOR OVER SIZE

MAX. DRY DENSITY gm/cc : **2.055**

OPT. MOISTURE CONTENT % **7.33**



REMARKS :

TESTED BY: REVIEWED BY:

MIN/RH

(TECHNICIAN) (SUPERVISOR) (LAB. INCHARGE)

MODIFIED PROCTOR TEST

ASTM D 1557 (METHOD " A ")

CLIENT	L & T CONSTRUCTION	REPORT DATE	18-Sep-19
PROJECT/LOCATION	MANSOURAH MASSARAH GOLD PROJECT	SAMPLING DATE	15-Sep-19
METHOD USED	"A"	TEST START DATE	16-Sep-19
SAMPLE SOURCE	ON SITE	TEST FINISH DATE	17-Sep-19
METHOD VARIATION	NONE	GEOTECHNICAL FILE #	-
SAMPLED BY	WAHAB	PROJECT #	19-K-3054

SOURCE / SAMPLE NO : B.S-1 (0.40-0.70m) TP-34 **MOULD NO :** 56

WEIGHT OF MOULD: 3354 **gms.** **VOLUME OF MOULD:** 942.415 **cc.**

TRIAL NO.		1	2	3	4	5
WATER ADDED	%	5%	7%	9%	11%	13%
WT. SOIL + MOULD (WET)	gm	5177	5266	5353	5347	5315
WEIGHT OF WET SOIL	gm	1823	1912	1999	1993	1961
WET DENSITY	**gm/cc**	1.934	2.029	2.121	2.115	2.080

CAN NO.		N#2	N#8	N#15	N#7	N#6
WT. SOIL + CAN (WET)	gm	728.0	718.1	751.5	754.9	823.2
WT. SOIL + CAN (DRY)	gm	704.0	683.9	705.5	700.6	750.8
WT. OF WATER	gm	24.0	34.2	46.0	54.3	72.4
WT. OF CAN	gm	170.8	157.6	164.3	181.1	172.6
WT. OF DRY SOIL	gm	533.2	526.3	541.2	519.5	578.2
WATER CONTENT	**%**	**4.5**	**6.5**	**8.5**	**10.5**	**12.5**
DRY DENSITY	**gm/cc**	**1.851**	**1.905**	**1.955**	**1.915**	**1.849**

Soil Description: ***WELL GRADED SAND with SILT***

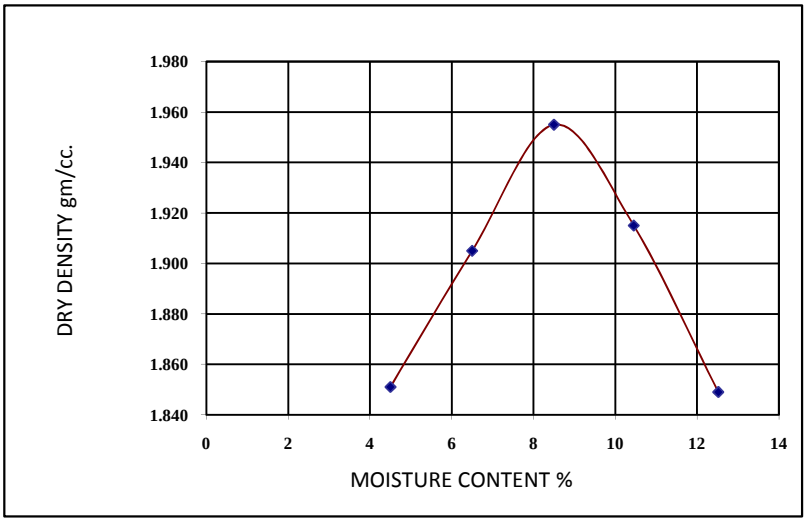
MAX. DRY DENSITY gm/cc : **1.955**

OPT. MOISTURE CONTENT % **8.5**

CORRECTED FOR OVER SIZE

MAX. DRY DENSITY gm/cc : **1.955**

OPT. MOISTURE CONTENT % **8.50**



REMARKS :

TESTED BY: REVIEWED BY:

MIN/RH

(TECHNICIAN) (SUPERVISOR) (LAB. INCHARGE)

MODIFIED PROCTOR TEST

ASTM D 1557 (METHOD " A ")

CLIENT	L & T CONSTRUCTION	REPORT DATE	18-Sep-19
PROJECT/LOCATION	MANSOURAH MASSARAH GOLD PROJECT	SAMPLING DATE	15-Sep-19
METHOD USED	"A"	TEST START DATE	16-Sep-19
SAMPLE SOURCE	ON SITE	TEST FINISH DATE	17-Sep-19
METHOD VARIATION	NONE	GEOTECHNICAL FILE #	-
SAMPLED BY	WAHAB	PROJECT #	19-K-3054

SOURCE / SAMPLE NO : B.S-1 (0.0-0.60m) TP-35 **MOULD NO :** 56

WEIGHT OF MOULD: 3354 **gms.** **VOLUME OF MOULD:** 942.415 **cc.**

TRIAL NO.		1	2	3	4	5
WATER ADDED	%	5%	7%	9%	11%	13%
WT. SOIL + MOULD (WET)	gm	5186	5272	5364	5351	5319
WEIGHT OF WET SOIL	gm	1832	1918	2010	1997	1965
WET DENSITY	**gm/cc**	1.944	2.036	2.133	2.119	2.085

CAN NO.		N#2	N#8	N#15	N#7	N#6
WT. SOIL + CAN (WET)	gm	705.0	733.3	738.6	783.3	770.7
WT. SOIL + CAN (DRY)	gm	683.0	699.2	694.1	727.4	704.6
WT. OF WATER	gm	22.0	34.1	44.5	55.9	66.1
WT. OF CAN	gm	170.8	157.6	164.3	181.1	172.6
WT. OF DRY SOIL	gm	512.2	541.6	529.8	546.3	532.0
WATER CONTENT	**%**	**4.3**	**6.3**	**8.4**	**10.2**	**12.4**
DRY DENSITY	**gm/cc**	**1.864**	**1.915**	**1.968**	**1.922**	**1.855**

Soil Description: ***SILTY SAND with GRAVEL***

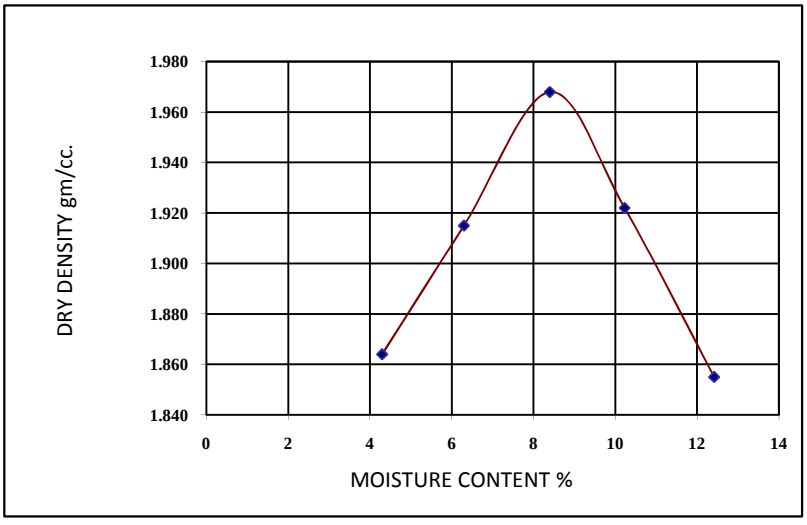
MAX. DRY DENSITY gm/cc : **1.968**

OPT. MOISTURE CONTENT % **8.4**

CORRECTED FOR OVER SIZE

MAX. DRY DENSITY gm/cc : **2.049**

OPT. MOISTURE CONTENT % **7.22**



REMARKS :

TESTED BY: REVIEWED BY:

MIN/RH

(TECHNICIAN) (SUPERVISOR) (LAB. INCHARGE)

MODIFIED PROCTOR TEST

ASTM D 1557 (METHOD " A ")

CLIENT	L & T CONSTRUCTION	REPORT DATE	19-Sep-19
PROJECT/LOCATION	MANSOURAH MASSARAH GOLD PROJECT	SAMPLING DATE	16-Sep-19
METHOD USED	"A"	TEST START DATE	17-Sep-19
SAMPLE SOURCE	ON SITE	TEST FINISH DATE	18-Sep-19
METHOD VARIATION	NONE	GEOTECHNICAL FILE #	-
SAMPLED BY	WAHAB	PROJECT #	19-K-3054

SOURCE / SAMPLE NO : B.S-1 (0.0-0.45m) TP-36 **MOULD NO :** 42

WEIGHT OF MOULD: 3352 **gms.** **VOLUME OF MOULD:** 943.42 **cc.**

TRIAL NO.		1	2	3	4	5
WATER ADDED	%	5%	7%	9%	11%	13%
WT. SOIL + MOULD (WET)	gm	5200	5285	5373	5354	5335
WEIGHT OF WET SOIL	gm	1848	1933	2021	2002	1983
WET DENSITY	**gm/cc**	1.959	2.049	2.142	2.122	2.101

CAN NO.		N#2	N#8	N#15	N#7	N#6
WT. SOIL + CAN (WET)	gm	708.3	711.0	724.5	751.6	756.6
WT. SOIL + CAN (DRY)	gm	685.0	677.2	680.6	698.6	691.1
WT. OF WATER	gm	23.2	33.8	43.9	52.9	65.4
WT. OF CAN	gm	170.8	157.6	164.3	181.1	172.6
WT. OF DRY SOIL	gm	514.2	519.6	516.3	517.5	518.5
WATER CONTENT	**%**	**4.5**	**6.5**	**8.5**	**10.2**	**12.6**
DRY DENSITY	**gm/cc**	**1.874**	**1.924**	**1.974**	**1.925**	**1.866**

Soil Description: ***SILTY SAND with GRAVEL***

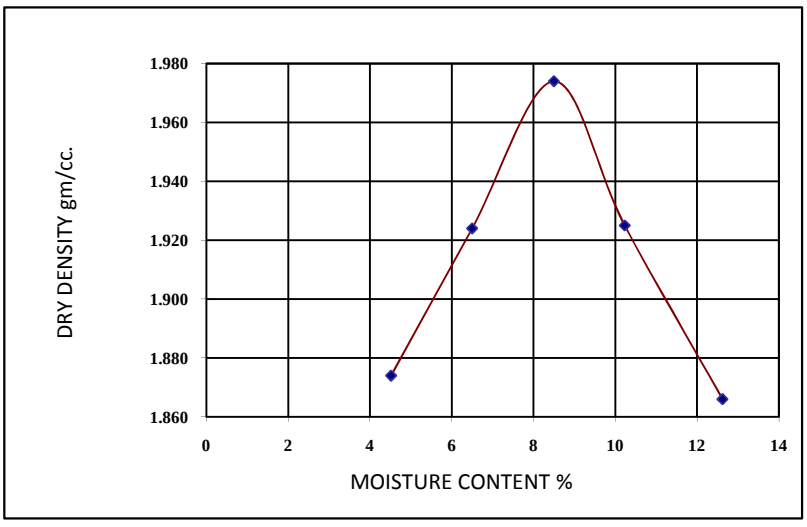
MAX. DRY DENSITY gm/cc : **1.974**

OPT. MOISTURE CONTENT % **8.5**

CORRECTED FOR OVER SIZE

MAX. DRY DENSITY gm/cc : **2.051**

OPT. MOISTURE CONTENT % **7.34**



REMARKS :

TESTED BY: REVIEWED BY:

MIN/RH

(TECHNICIAN) (SUPERVISOR) (LAB. INCHARGE)

MODIFIED PROCTOR TEST

ASTM D 1557 (METHOD " A ")

CLIENT	L & T CONSTRUCTION	REPORT DATE	19-Sep-19
PROJECT/LOCATION	MANSOURAH MASSARAH GOLD PROJECT	SAMPLING DATE	16-Sep-19
METHOD USED	"A"	TEST START DATE	17-Sep-19
SAMPLE SOURCE	ON SITE	TEST FINISH DATE	18-Sep-19
METHOD VARIATION	NONE	GEOTECHNICAL FILE #	-
SAMPLED BY	WAHAB	PROJECT #	19-K-3054

SOURCE / SAMPLE NO : B.S-1 (0.0-0.51m) TP-41 **MOULD NO :** 42

WEIGHT OF MOULD: 3352 **gms.** **VOLUME OF MOULD:** 943.42 **cc.**

TRIAL NO.		1	2	3	4	5
WATER ADDED	%	5%	7%	9%	11%	13%
WT. SOIL + MOULD (WET)	gm	5200	5289	5374	5365	5333
WEIGHT OF WET SOIL	gm	1848	1937	2022	2013	1981
WET DENSITY	**gm/cc**	1.958	2.053	2.144	2.133	2.099

CAN NO.		N#2	N#8	N#15	N#7	N#6
WT. SOIL + CAN (WET)	gm	725.9	714.2	779.3	767.5	789.4
WT. SOIL + CAN (DRY)	gm	702.0	680.2	730.6	713.2	721.1
WT. OF WATER	gm	23.9	34.0	48.7	54.3	68.3
WT. OF CAN	gm	170.8	157.6	164.3	181.1	172.6
WT. OF DRY SOIL	gm	531.2	522.6	566.3	532.1	548.5
WATER CONTENT	**%**	**4.5**	**6.5**	**8.6**	**10.2**	**12.5**
DRY DENSITY	**gm/cc**	**1.874**	**1.928**	**1.974**	**1.936**	**1.867**

Soil Description: ***SILTY SAND with GRAVEL***

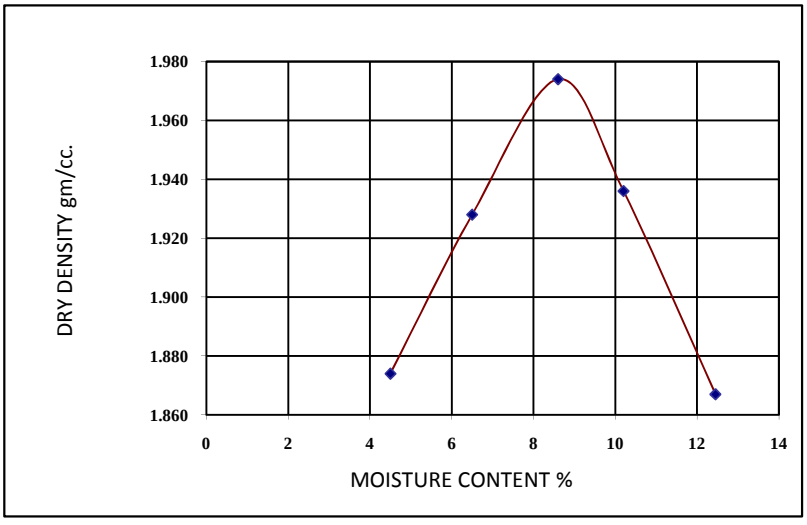
MAX. DRY DENSITY gm/cc : **1.974**

OPT. MOISTURE CONTENT % **8.6**

CORRECTED FOR OVER SIZE

MAX. DRY DENSITY gm/cc : **2.048**

OPT. MOISTURE CONTENT % **7.47**



REMARKS :

TESTED BY: REVIEWED BY:

MIN/RH

(TECHNICIAN) (SUPERVISOR) (LAB. INCHARGE)

MODIFIED PROCTOR TEST

ASTM D 1557 (METHOD " A ")

CLIENT	L & T CONSTRUCTION	REPORT DATE	20-Sep-19
PROJECT/LOCATION	MANSOURAH MASSARAH GOLD PROJECT	SAMPLING DATE	17-Sep-19
METHOD USED	"A"	TEST START DATE	18-Sep-19
SAMPLE SOURCE	ON SITE	TEST FINISH DATE	19-Sep-19
METHOD VARIATION	NONE	GEOTECHNICAL FILE #	-
SAMPLED BY	WAHAB	PROJECT #	19-K-3054

SOURCE / SAMPLE NO : B.S-1 (0.0-0.60m) TP-42 **MOULD NO :** 56

WEIGHT OF MOULD: 3354 **gms.** **VOLUME OF MOULD:** 942.415 **cc.**

TRIAL NO.		1	2	3	4	5
WATER ADDED	%	5%	7%	9%	11%	13%
WT. SOIL + MOULD (WET)	gm	5197	5294	5379	5370	5365
WEIGHT OF WET SOIL	gm	1843	1940	2025	2016	2011
WET DENSITY	**gm/cc**	1.956	2.058	2.149	2.140	2.134

CAN NO.		N#2	N#8	N#15	N#7	N#6
WT. SOIL + CAN (WET)	gm	730.0	722.7	725.7	753.1	759.8
WT. SOIL + CAN (DRY)	gm	707.4	689.7	683.2	700.1	695.0
WT. OF WATER	gm	22.5	33.0	42.5	53.1	64.9
WT. OF CAN	gm	170.8	157.6	164.3	181.1	172.6
WT. OF DRY SOIL	gm	536.6	532.1	518.9	519.0	522.4
WATER CONTENT	**%**	**4.2**	**6.2**	**8.2**	**10.2**	**12.4**
DRY DENSITY	**gm/cc**	**1.877**	**1.938**	**1.986**	**1.941**	**1.898**

Soil Description: ***SILTY SAND with GRAVEL***

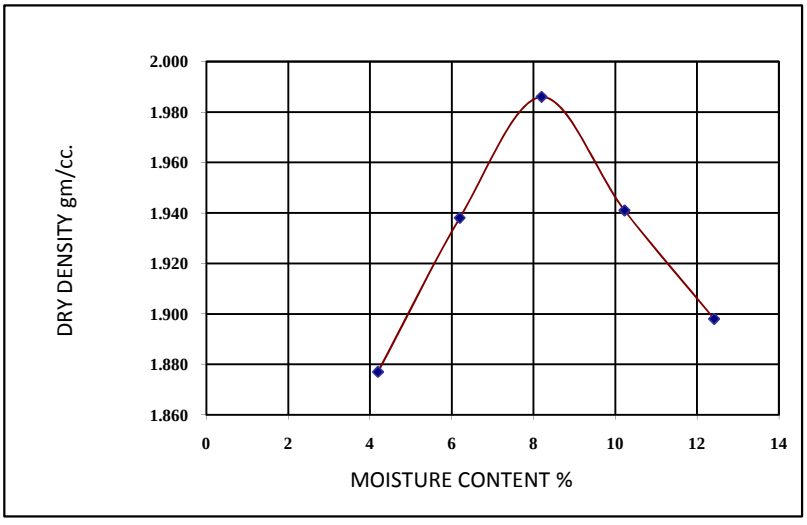
MAX. DRY DENSITY gm/cc : **1.986**

OPT. MOISTURE CONTENT % **8.2**

CORRECTED FOR OVER SIZE

MAX. DRY DENSITY gm/cc : **2.061**

OPT. MOISTURE CONTENT % **7.12**



REMARKS :

TESTED BY: REVIEWED BY:

MIN/RH

(TECHNICIAN) (SUPERVISOR) (LAB. INCHARGE)

MODIFIED PROCTOR TEST

ASTM D 1557 (METHOD " A ")

CLIENT	L & T CONSTRUCTION	REPORT DATE	20-Sep-19
PROJECT/LOCATION	MANSOURAH MASSARAH GOLD PROJECT	SAMPLING DATE	17-Sep-19
METHOD USED	"A"	TEST START DATE	18-Sep-19
SAMPLE SOURCE	ON SITE	TEST FINISH DATE	19-Sep-19
METHOD VARIATION	NONE	GEOTECHNICAL FILE #	-
SAMPLED BY	WAHAB	PROJECT #	19-K-3054

SOURCE / SAMPLE NO : B.S-1 (0.2-0.75m) TP-43 **MOULD NO :** 53

WEIGHT OF MOULD: 3345 **gms.** **VOLUME OF MOULD:** 944.5 **cc.**

TRIAL NO.		1	2	3	4	5
WATER ADDED	%	5%	7%	9%	11%	13%
WT. SOIL + MOULD (WET)	gm	5195	5279	5366	5349	5331
WEIGHT OF WET SOIL	gm	1850	1934	2021	2004	1986
WET DENSITY	**gm/cc**	1.959	2.047	2.140	2.122	2.103

CAN NO.		N#2	N#8	N#15	N#7	N#6
WT. SOIL + CAN (WET)	gm	751.5	760.5	756.4	787.3	797.6
WT. SOIL + CAN (DRY)	gm	726.4	724.2	710.5	731.1	727.6
WT. OF WATER	gm	25.1	36.3	45.9	56.3	70.0
WT. OF CAN	gm	170.8	157.6	164.3	181.1	172.6
WT. OF DRY SOIL	gm	555.6	566.6	546.2	550.0	555.0
WATER CONTENT	**%**	**4.5**	**6.4**	**8.4**	**10.2**	**12.6**
DRY DENSITY	**gm/cc**	**1.874**	**1.924**	**1.974**	**1.925**	**1.867**

Soil Description: ***CLAYEY SAND with GRAVELS***

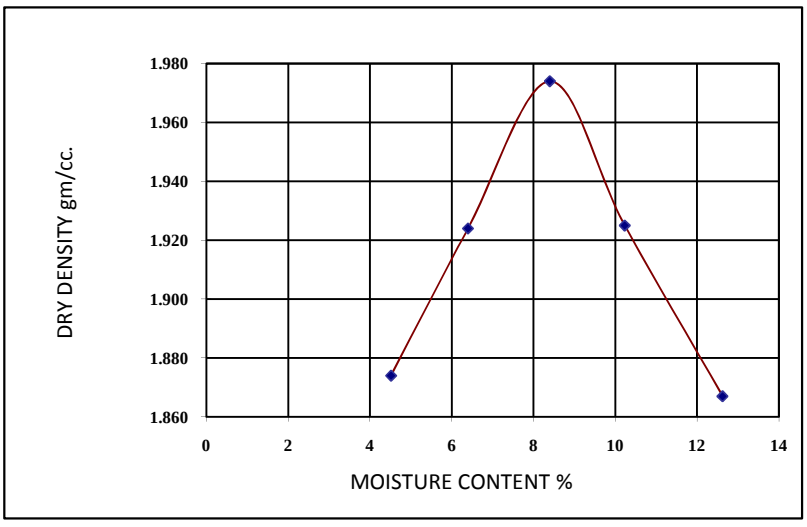
MAX. DRY DENSITY gm/cc : **1.974**

OPT. MOISTURE CONTENT % **8.4**

CORRECTED FOR OVER SIZE

MAX. DRY DENSITY gm/cc : **2.035**

OPT. MOISTURE CONTENT % **7.50**



REMARKS :

TESTED BY: REVIEWED BY:

MIN/RH

(TECHNICIAN) (SUPERVISOR) (LAB. INCHARGE)

MODIFIED PROCTOR TEST

ASTM D 1557 (METHOD " A ")

CLIENT	L & T CONSTRUCTION	REPORT DATE	21-Sep-19
PROJECT/LOCATION	MANSOURAH MASSARAH GOLD PROJECT	SAMPLING DATE	18-Sep-19
METHOD USED	"A"	TEST START DATE	19-Sep-19
SAMPLE SOURCE	ON SITE	TEST FINISH DATE	20-Sep-19
METHOD VARIATION	NONE	GEOTECHNICAL FILE #	-
SAMPLED BY	WAHAB	PROJECT #	19-K-3054

SOURCE / SAMPLE NO : B.S-1 (0.0-0.60m) TP-44 **MOULD NO :** 56

WEIGHT OF MOULD: 3354 **gms.** **VOLUME OF MOULD:** 942.415 **cc.**

TRIAL NO.		1	2	3	4	5
WATER ADDED	%	5%	7%	9%	11%	13%
WT. SOIL + MOULD (WET)	gm	5190	5276	5368	5357	5334
WEIGHT OF WET SOIL	gm	1836	1922	2014	2003	1980
WET DENSITY	**gm/cc**	1.948	2.039	2.137	2.126	2.101

CAN NO.		N#2	N#8	N#15	N#7	N#6
WT. SOIL + CAN (WET)	gm	751.4	761.0	758.0	789.4	797.6
WT. SOIL + CAN (DRY)	gm	726.4	724.2	710.5	731.1	727.6
WT. OF WATER	gm	25.0	36.8	47.5	58.3	70.0
WT. OF CAN	gm	170.8	157.6	164.3	181.1	172.6
WT. OF DRY SOIL	gm	555.6	566.6	546.2	550.0	555.0
WATER CONTENT	**%**	**4.5**	**6.5**	**8.7**	**10.6**	**12.6**
DRY DENSITY	**gm/cc**	**1.864**	**1.915**	**1.966**	**1.922**	**1.866**

Soil Description: ***SILTY SAND with GRAVEL***

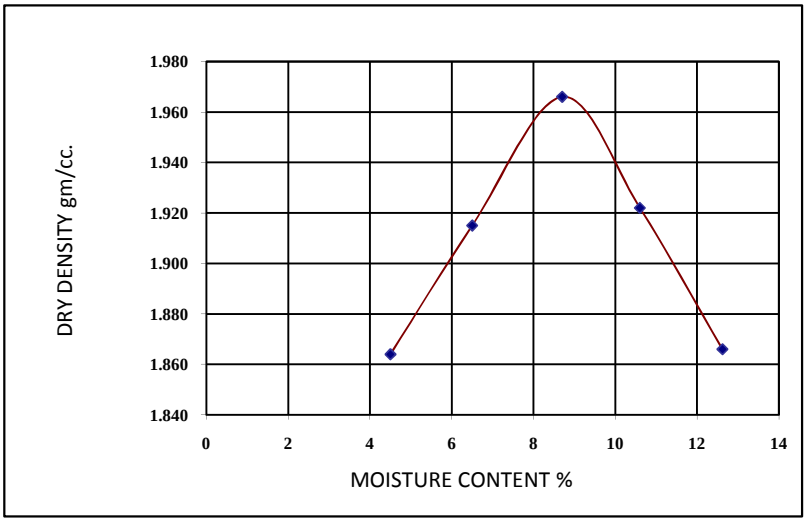
MAX. DRY DENSITY gm/cc : **1.966**

OPT. MOISTURE CONTENT % **8.7**

CORRECTED FOR OVER SIZE

MAX. DRY DENSITY gm/cc : **2.043**

OPT. MOISTURE CONTENT % **7.52**



REMARKS :

TESTED BY: REVIEWED BY:

MIN/RH

(TECHNICIAN) (SUPERVISOR) (LAB. INCHARGE)

MODIFIED PROCTOR TEST

ASTM D 1557 (METHOD " C ")

CLIENT	L & T CONSTRUCTION	REPORT DATE	21-Sep-19
PROJECT/LOCATION	MANSOURAH MASSARAH GOLD PROJECT	SAMPLING DATE	18-Sep-19
METHOD USED	"C"	TEST START DATE	19-Sep-19
SAMPLE SOURCE	ON SITE	TEST FINISH DATE	20-Sep-19
METHOD VARIATION	NONE	GEOTECHNICAL FILE #	-
SAMPLED BY	WAHAB	PROJECT #	19-K-3054

SOURCE / SAMPLE NO : B.S-1 (0.3-0.65m) TP-45 **MOULD NO :** 31

WEIGHT OF MOULD: 6577 **gms.** **VOLUME OF MOULD:** 2012.333 **cc.**

TRIAL NO.		1	2	3	4	5
WATER ADDED	%	5%	7%	9%	11%	13%
WT. SOIL + MOULD (WET)	gm	10735	10928	11131	11103	11071
WEIGHT OF WET SOIL	gm	4158	4351	4554	4526	4494
WET DENSITY	**gm/cc**	2.066	2.162	2.263	2.249	2.233

CAN NO.		N#2	N#8	N#15	N#7	N#6
WT. SOIL + CAN (WET)	gm	749.9	729.1	796.1	741.3	771.4
WT. SOIL + CAN (DRY)	gm	737.4	706.1	757.5	698.7	715.8
WT. OF WATER	gm	12.5	23.0	38.6	42.6	55.6
WT. OF CAN	gm	170.8	157.6	164.3	181.1	172.6
WT. OF DRY SOIL	gm	566.6	548.5	593.2	517.6	543.2
WATER CONTENT	**%**	**2.2**	**4.2**	**6.5**	**8.2**	**10.2**
DRY DENSITY	**gm/cc**	**2.022**	**2.075**	**2.125**	**2.078**	**2.026**

Soil Description: ***SILTY GRAVEL with SAND***

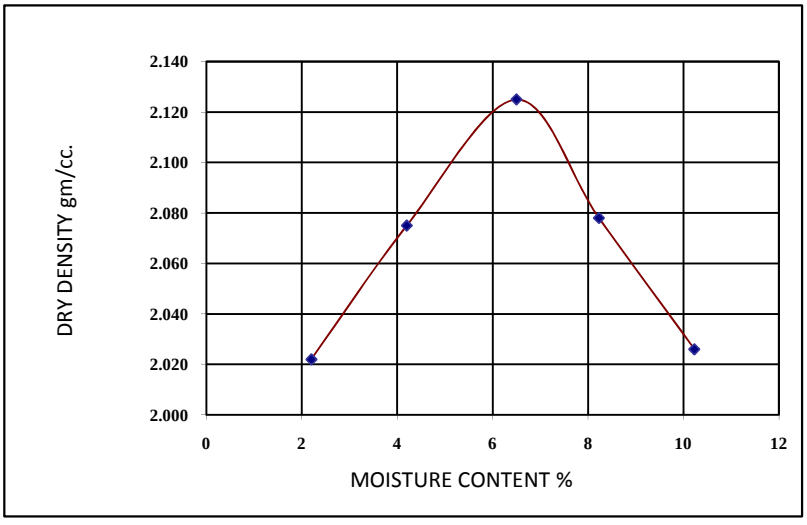
MAX. DRY DENSITY gm/cc : **2.125**

OPT. MOISTURE CONTENT % **6.5**

CORRECTED FOR OVER SIZE

MAX. DRY DENSITY gm/cc : **2.125**

OPT. MOISTURE CONTENT % **6.50**



REMARKS :

TESTED BY: REVIEWED BY:

MIN/RH

(TECHNICIAN) (SUPERVISOR) (LAB. INCHARGE)

MODIFIED PROCTOR TEST

ASTM D 1557 (METHOD " A ")

CLIENT	L & T CONSTRUCTION	REPORT DATE	22-Sep-19
PROJECT/LOCATION	MANSOURAH MASSARAH GOLD PROJECT	SAMPLING DATE	19-Sep-19
METHOD USED	"A"	TEST START DATE	20-Sep-19
SAMPLE SOURCE	ON SITE	TEST FINISH DATE	21-Sep-19
METHOD VARIATION	NONE	GEOTECHNICAL FILE #	-
SAMPLED BY	WAHAB	PROJECT #	19-K-3054

SOURCE / SAMPLE NO : B.S-1 (0.0-0.60m) TP-46 **MOULD NO :** 56

WEIGHT OF MOULD: 3354 **gms.** **VOLUME OF MOULD:** 942.415 **cc.**

TRIAL NO.		1	2	3	4	5
WATER ADDED	%	5%	7%	9%	11%	13%
WT. SOIL + MOULD (WET)	gm	5198	5283	5373	5362	5336
WEIGHT OF WET SOIL	gm	1844	1929	2019	2008	1982
WET DENSITY	**gm/cc**	1.956	2.046	2.142	2.131	2.103

CAN NO.		N#2	N#8	N#15	N#7	N#6	
WT. SOIL + CAN (WET)	gm	754.8	754.8	773.4	781.5	785.3	
WT. SOIL + CAN (DRY)	gm	731.1	719.9	726.2	724.7	717.1	
WT. OF WATER	gm	23.7	34.9	47.2	56.8	68.2	
WT. OF CAN	gm	170.8	157.6	164.3	181.1	172.6	
WT. OF DRY SOIL	gm	560.3	562.3	561.9	543.6	544.5	
WATER CONTENT	**%**	**4.2**	**6.2**	**8.4**	**10.5**	**12.5**	
DRY DENSITY	**gm/cc**	**1.877**	**1.927**	**1.976**	**1.929**	**1.869**	###

Soil Description: ***SILTY SAND with GRAVEL***

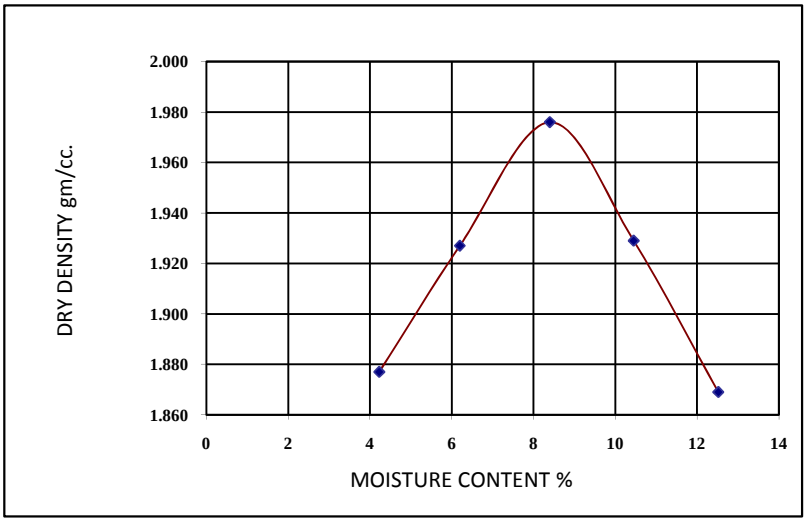
MAX. DRY DENSITY gm/cc : **1.976**

OPT. MOISTURE CONTENT % **8.4**

CORRECTED FOR OVER SIZE

MAX. DRY DENSITY gm/cc : **2.038**

OPT. MOISTURE CONTENT % **7.48**



REMARKS :

TESTED BY: REVIEWED BY:

MIN/RH

(TECHNICIAN) (SUPERVISOR) (LAB. INCHARGE)

MODIFIED PROCTOR TEST

ASTM D 1557 (METHOD " A ")

CLIENT	L & T CONSTRUCTION	REPORT DATE	22-Sep-19
PROJECT/LOCATION	MANSOURAH MASSARAH GOLD PROJECT	SAMPLING DATE	19-Sep-19
METHOD USED	"A"	TEST START DATE	20-Sep-19
SAMPLE SOURCE	ON SITE	TEST FINISH DATE	21-Sep-19
METHOD VARIATION	NONE	GEOTECHNICAL FILE #	-
SAMPLED BY	WAHAB	PROJECT #	19-K-3054

SOURCE / SAMPLE NO : B.S-1 (0.0-0.45m) TP-47 **MOULD NO :** 56

WEIGHT OF MOULD: 3354 **gms.** **VOLUME OF MOULD:** 942.415 **cc.**

TRIAL NO.		1	2	3	4	5
WATER ADDED	%	5%	7%	9%	11%	13%
WT. SOIL + MOULD (WET)	gm	5226	5315	5401	5392	5361
WEIGHT OF WET SOIL	gm	1872	1961	2047	2038	2007
WET DENSITY	**gm/cc**	1.986	2.080	2.173	2.163	2.130

CAN NO.		N#2	N#8	N#15	N#7	N#6
WT. SOIL + CAN (WET)	gm	719.4	720.0	741.3	808.6	784.2
WT. SOIL + CAN (DRY)	gm	694.3	684.2	694.2	747.4	714.8
WT. OF WATER	gm	25.1	35.8	47.2	61.2	69.4
WT. OF CAN	gm	170.8	157.6	164.3	181.1	172.6
WT. OF DRY SOIL	gm	523.5	526.6	529.9	566.3	542.2
WATER CONTENT	**%**	**4.8**	**6.8**	**8.9**	**10.8**	**12.8**
DRY DENSITY	**gm/cc**	**1.895**	**1.948**	**1.995**	**1.952**	**1.888**

Soil Description: ***CLAYEY SAND with GRAVELS***

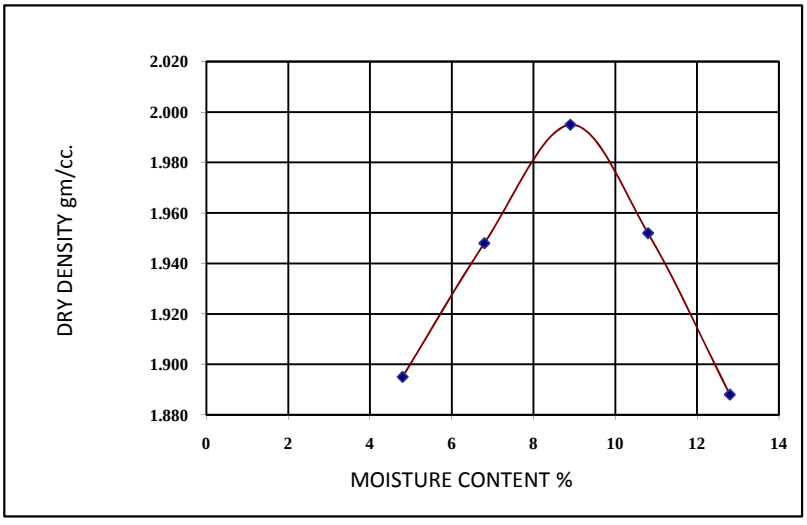
MAX. DRY DENSITY gm/cc : **1.995**

OPT. MOISTURE CONTENT % **8.9**

CORRECTED FOR OVER SIZE

MAX. DRY DENSITY gm/cc : **2.058**

OPT. MOISTURE CONTENT % **7.84**



REMARKS :

TESTED BY: REVIEWED BY:

MIN/RH

(TECHNICIAN) (SUPERVISOR) (LAB. INCHARGE)

RIYADH GEOTECHNIQUE & FOUNDATIONS LLC

MODIFIED PROCTOR TEST

ASTM D 1557 (METHOD " A ")

CLIENT	L & T CONSTRUCTION	REPORT DATE	23-Sep-19
PROJECT/LOCATION	MANSOURAH MASSARAH GOLD PROJECT	SAMPLING DATE	20-Sep-19
METHOD USED	"A"	TEST START DATE	21-Sep-19
SAMPLE SOURCE	ON SITE	TEST FINISH DATE	22-Sep-19
METHOD VARIATION	NONE	GEOTECHNICAL FILE #	-
SAMPLED BY	WAHAB	PROJECT #	19-K-3054

SOURCE / SAMPLE NO : B.S-1 (0.0-0.50m) TP-48 **MOULD NO :** 53

WEIGHT OF MOULD: 3345 **gms.** **VOLUME OF MOULD:** 944.5 **cc.**

TRIAL NO.		1	2	3	4	5
WATER ADDED	%	5%	7%	9%	11%	13%
WT. SOIL + MOULD (WET)	gm	4965	5046	5127	5117	5091
WEIGHT OF WET SOIL	gm	1620	1701	1782	1772	1746
WET DENSITY	**gm/cc**	1.715	1.800	1.887	1.876	1.849

CAN NO.		N#2	N#8	N#15	N#7	N#6
WT. SOIL + CAN (WET)	gm	730.1	744.4	747.6	815.4	806.8
WT. SOIL + CAN (DRY)	gm	697.5	699.9	693.6	744.3	727.9
WT. OF WATER	gm	32.7	44.5	54.0	71.1	78.9
WT. OF CAN	gm	170.8	157.6	164.3	181.1	172.6
WT. OF DRY SOIL	gm	526.6	542.3	529.3	563.2	555.3
WATER CONTENT	**%**	**6.2**	**8.2**	**10.2**	**12.6**	**14.2**
DRY DENSITY	**gm/cc**	**1.615**	**1.664**	**1.712**	**1.666**	**1.619**

Soil Description: ***CLAYEY SAND***

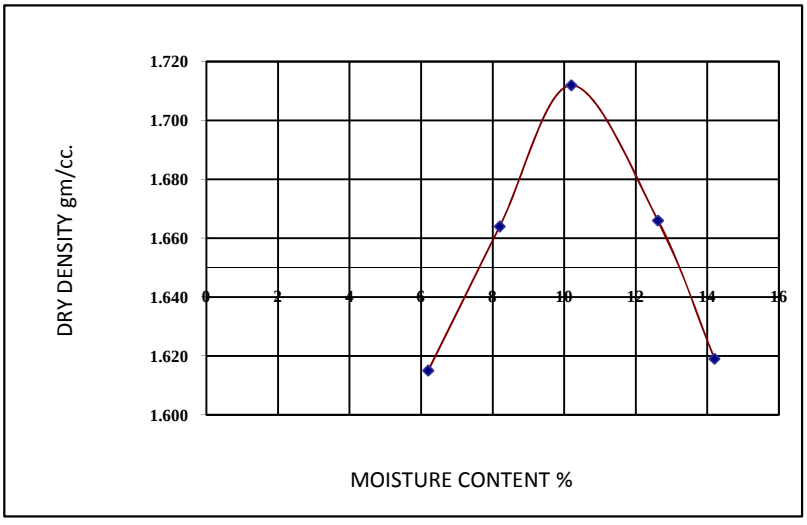
MAX. DRY DENSITY gm/cc : **1.712**

OPT. MOISTURE CONTENT % **10.2**

CORRECTED FOR OVER SIZE

MAX. DRY DENSITY gm/cc : **1.712**

OPT. MOISTURE CONTENT % **10.20**



REMARKS :

TESTED BY: REVIEWED BY:

MIN/RH

(TECHNICIAN) (SUPERVISOR) (LAB. INCHARGE)

MODIFIED PROCTOR TEST

ASTM D 1557 (METHOD " C ")

CLIENT	L & T CONSTRUCTION	REPORT DATE	23-Sep-19
PROJECT/LOCATION	MANSOURAH MASSARAH GOLD PROJECT	SAMPLING DATE	20-Sep-19
METHOD USED	"C"	TEST START DATE	21-Sep-19
SAMPLE SOURCE	ON SITE	TEST FINISH DATE	22-Sep-19
METHOD VARIATION	NONE	GEOTECHNICAL FILE #	-
SAMPLED BY	WAHAB	PROJECT #	19-K-3054

SOURCE / SAMPLE NO : B.S-1 (0.2-.60m) TP-49 **MOULD NO :** 53

WEIGHT OF MOULD: 3345 **gms.** **VOLUME OF MOULD:** 944.5 **cc.**

TRIAL NO.		1	2	3	4	5
WATER ADDED	%	5%	7%	9%	11%	13%
WT. SOIL + MOULD (WET)	gm	5318	5407	5503	5480	5462
WEIGHT OF WET SOIL	gm	1973	2062	2158	2135	2117
WET DENSITY	**gm/cc**	2.089	2.183	2.284	2.260	2.241

CAN NO.		N#2	N#8	N#15	N#7	N#6
WT. SOIL + CAN (WET)	gm	692.2	691.3	714.1	737.8	753.1
WT. SOIL + CAN (DRY)	gm	681.0	669.8	680.5	695.6	699.2
WT. OF WATER	gm	11.2	21.5	33.6	42.2	53.9
WT. OF CAN	gm	170.8	157.6	164.3	181.1	172.6
WT. OF DRY SOIL	gm	510.2	512.2	516.2	514.5	526.6
WATER CONTENT	**%**	**2.2**	**4.2**	**6.5**	**8.2**	**10.2**
DRY DENSITY	**gm/cc**	**2.044**	**2.095**	**2.145**	**2.089**	**2.033**

Soil Description: ***POORLY GRADED GRAVEL with SILT and SAND***

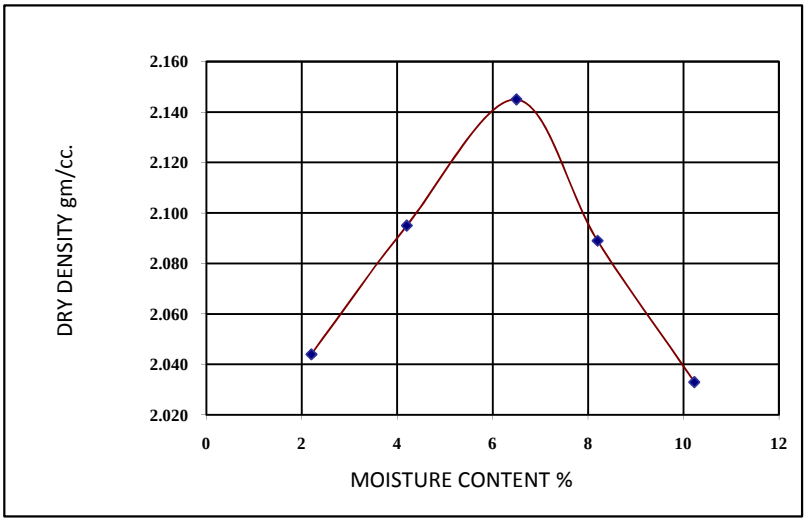
MAX. DRY DENSITY gm/cc : **2.145**

OPT. MOISTURE CONTENT % **6.5**

CORRECTED FOR OVER SIZE

MAX. DRY DENSITY gm/cc : **2.145**

OPT. MOISTURE CONTENT % **6.50**



REMARKS :

TESTED BY: REVIEWED BY:

MIN/RH

(TECHNICIAN) (SUPERVISOR) (LAB. INCHARGE)

MODIFIED PROCTOR TEST

ASTM D 1557 (METHOD " C ")

CLIENT	L & T CONSTRUCTION	REPORT DATE	24-Sep-19
PROJECT/LOCATION	MANSOURAH MASSARAH GOLD PROJECT	SAMPLING DATE	21-Sep-19
METHOD USED	"C"	TEST START DATE	22-Sep-19
SAMPLE SOURCE	ON SITE	TEST FINISH DATE	23-Sep-19
METHOD VARIATION	NONE	GEOTECHNICAL FILE #	-
SAMPLED BY	WAHAB	PROJECT #	19-K-3054

SOURCE / SAMPLE NO : B.S-1 (0.35-0.75m) TP-50 **MOULD NO :** 53

WEIGHT OF MOULD: 3345 **gms.** **VOLUME OF MOULD:** 944.5 **cc.**

TRIAL NO.		1	2	3	4	5
WATER ADDED	%	5%	7%	9%	11%	13%
WT. SOIL + MOULD (WET)	gm	5306	5396	5491	5479	5457
WEIGHT OF WET SOIL	gm	1961	2051	2146	2134	2112
WET DENSITY	**gm/cc**	2.077	2.172	2.272	2.260	2.237

CAN NO.		N#2	N#8	N#15	N#7	N#6
WT. SOIL + CAN (WET)	gm	692.2	691.3	712.5	737.9	753.1
WT. SOIL + CAN (DRY)	gm	681.0	669.8	680.5	695.6	699.2
WT. OF WATER	gm	11.2	21.5	32.0	42.3	53.9
WT. OF CAN	gm	170.8	157.6	164.3	181.1	172.6
WT. OF DRY SOIL	gm	510.2	512.2	516.2	514.5	526.6
WATER CONTENT	**%**	**2.2**	**4.2**	**6.2**	**8.2**	**10.2**
DRY DENSITY	**gm/cc**	**2.032**	**2.084**	**2.139**	**2.088**	**2.029**

Soil Description: ***POORLY GRADED GRAVEL with SILT and SAND***

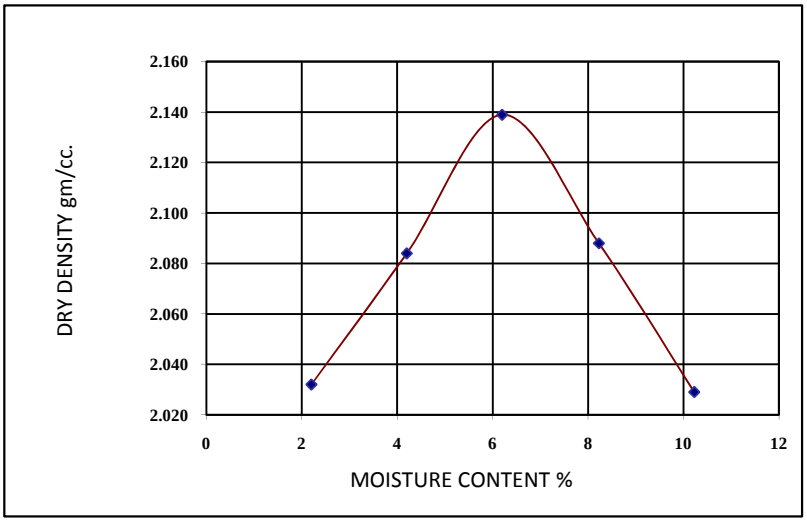
MAX. DRY DENSITY gm/cc : **2.139**

OPT. MOISTURE CONTENT % **6.2**

CORRECTED FOR OVER SIZE

MAX. DRY DENSITY gm/cc : **2.139**

OPT. MOISTURE CONTENT % **6.20**



REMARKS :

TESTED BY: REVIEWED BY:

MIN/RH

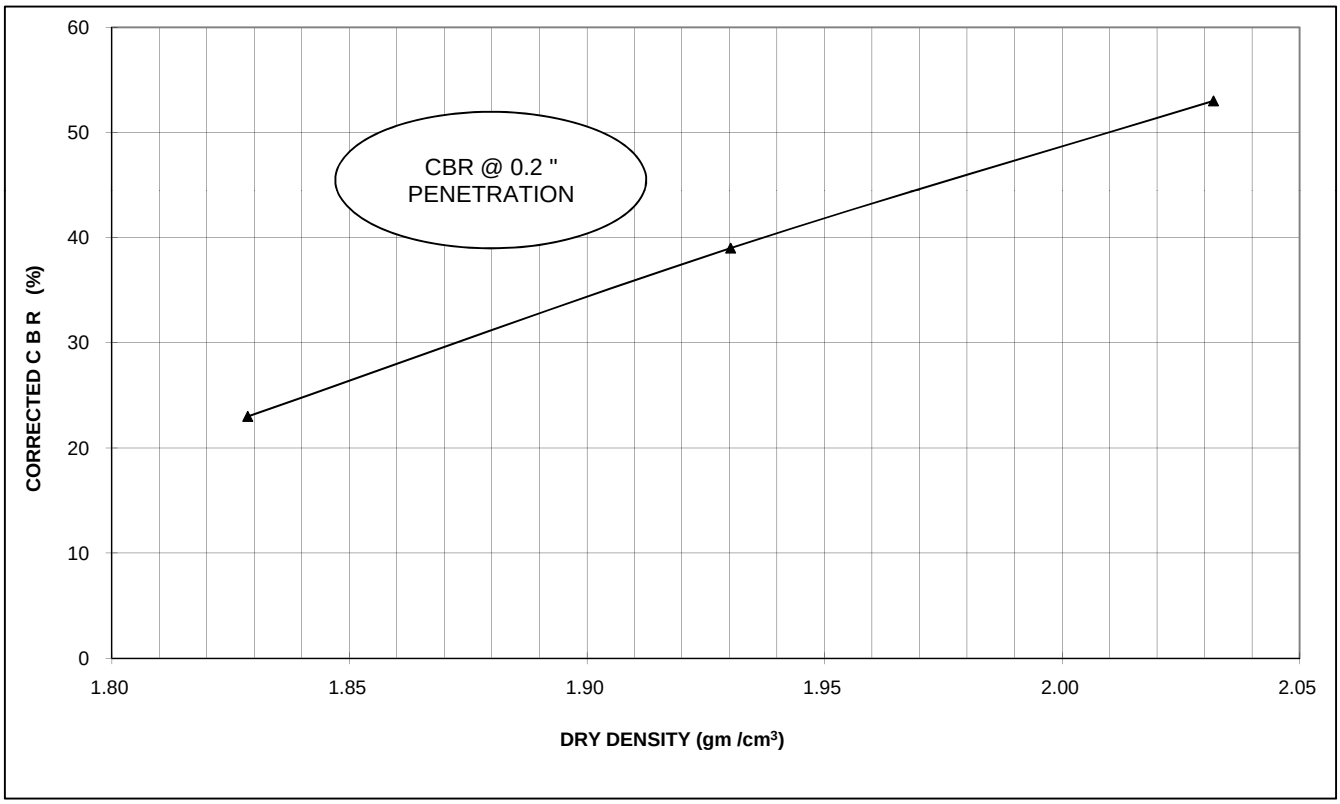
(TECHNICIAN) (SUPERVISOR) (LAB. INCHARGE)

CALIFORNIA BEARING RATIO (CBR)

CBR - DENSITY RELATIONSHIP
ASTM D 1883

CLIENT :	L & T CONSTRUCTION	REPORT DATE:	7-Sep-19
PROJECT :	MANSOURAH MASSARAH GOLD PROJECT GEOTECHNICAL INVESTIGATION	PROJECT #:	19-K-3054
		TRIAL PIT:	TP-01
		SAMPLE :	B.S-1
SAMPLE:	BULK SAMPLE	DEPTH-(m):	0.0-0.60
		SAMPLE DATE:	2-Sep-19
TP NO.:	TP-01	TEST DATE:	6-Sep-19

COMPACTION EFFORT	10 Blows	25 Blows	56 Blows	100%MDD
DRY DENSITY (gm/cm^3)	1.829	1.930	2.032	2.032
BEARING @ 0.2" PENETRATION (CBR)	23	39	53	53
SWELL (%)	nil	nil	nil	-



MAX. DRY DENSITY (γ_d): **2.032** g /cm^3
OPTIMUM M.C. (OMC): **7.86** %

Remarks:
Nill

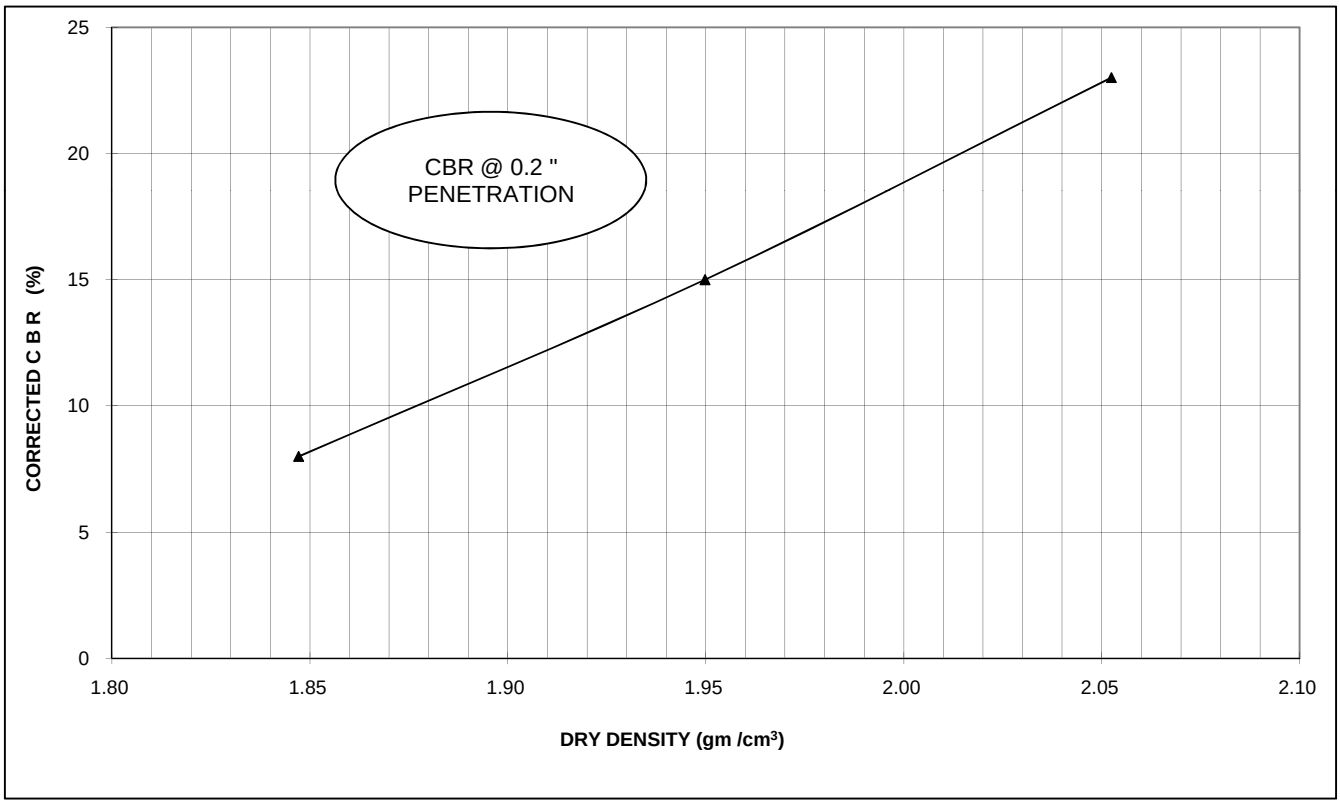
TESTED BY
WAHAB

LAB. SUPERVISOR
ANJAM

CBR - DENSITY RELATIONSHIP
ASTM D 1883

CLIENT :	L & T CONSTRUCTION	REPORT DATE:	7-Sep-19
PROJECT :	MANSOURAH MASSARAH GOLD PROJECT GEOTECHNICAL INVESTIGATION	PROJECT #:	19-K-3054
		TRIAL PIT:	TP-02
		SAMPLE :	B.S-1
SAMPLE:	BULK SAMPLE	DEPTH-(m):	0.7-1.0
		SAMPLE DATE:	2-Sep-19
TP NO.:	TP-02	TEST DATE:	6-Sep-19

COMPACTION EFFORT	10 Blows	25 Blows	56 Blows	100%MDD
DRY DENSITY (gm/cm^3)	1.847	1.950	2.052	2.052
BEARING @ 0.2" PENETRATION (CBR)	8	15	23	23
SWELL (%)	0.2	0.5	1.1	-



MAX. DRY DENSITY (γ_d): **2.052** g /cm^3
OPTIMUM M.C. (OMC): **8.53** %

Remarks:
Nil

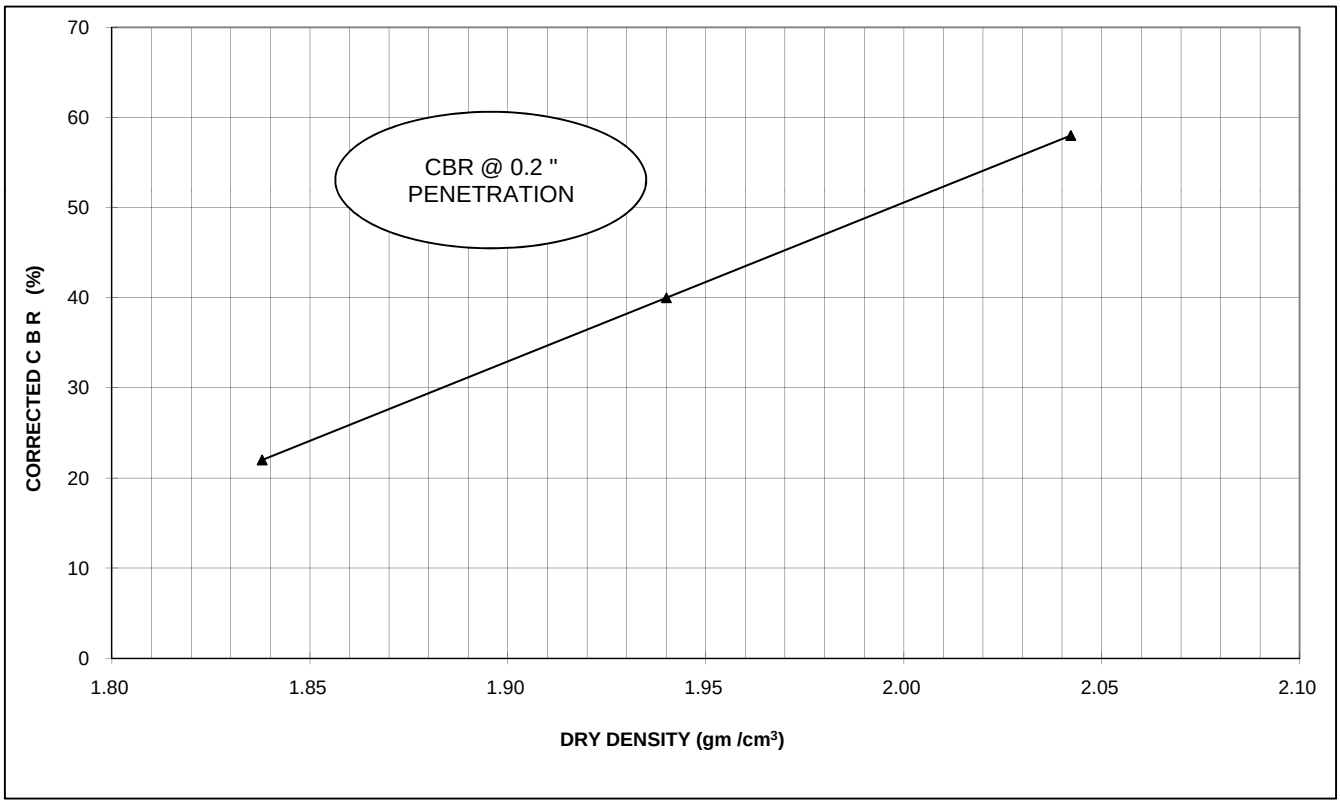
TESTED BY
WAHAB

LAB. SUPERVISOR
ANJAM

CBR - DENSITY RELATIONSHIP
ASTM D 1883

CLIENT :	L & T CONSTRUCTION	REPORT DATE:	7-Sep-19
PROJECT :	MANSOURAH MASSARAH GOLD PROJECT GEOTECHNICAL INVESTIGATION	PROJECT #:	19-K-3054
		TRIAL PIT:	TP-03
		SAMPLE :	B.S-1
SAMPLE:	BULK SAMPLE	DEPTH-(m):	0.00-0.80
		SAMPLE DATE:	2-Sep-19
TP NO.:	TP-03	TEST DATE:	6-Sep-19

COMPACTION EFFORT	10 Blows	25 Blows	56 Blows	100%MDD
DRY DENSITY (gm/cm^3)	1.838	1.940	2.042	2.042
BEARING @ 0.2" PENETRATION (CBR)	22	40	58	58
SWELL (%)	nil	nil	nil	-



MAX. DRY DENSITY (γ_d): **2.042** g /cm^3
OPTIMUM M.C. (OMC): **7.49** %

Remarks:
Nill

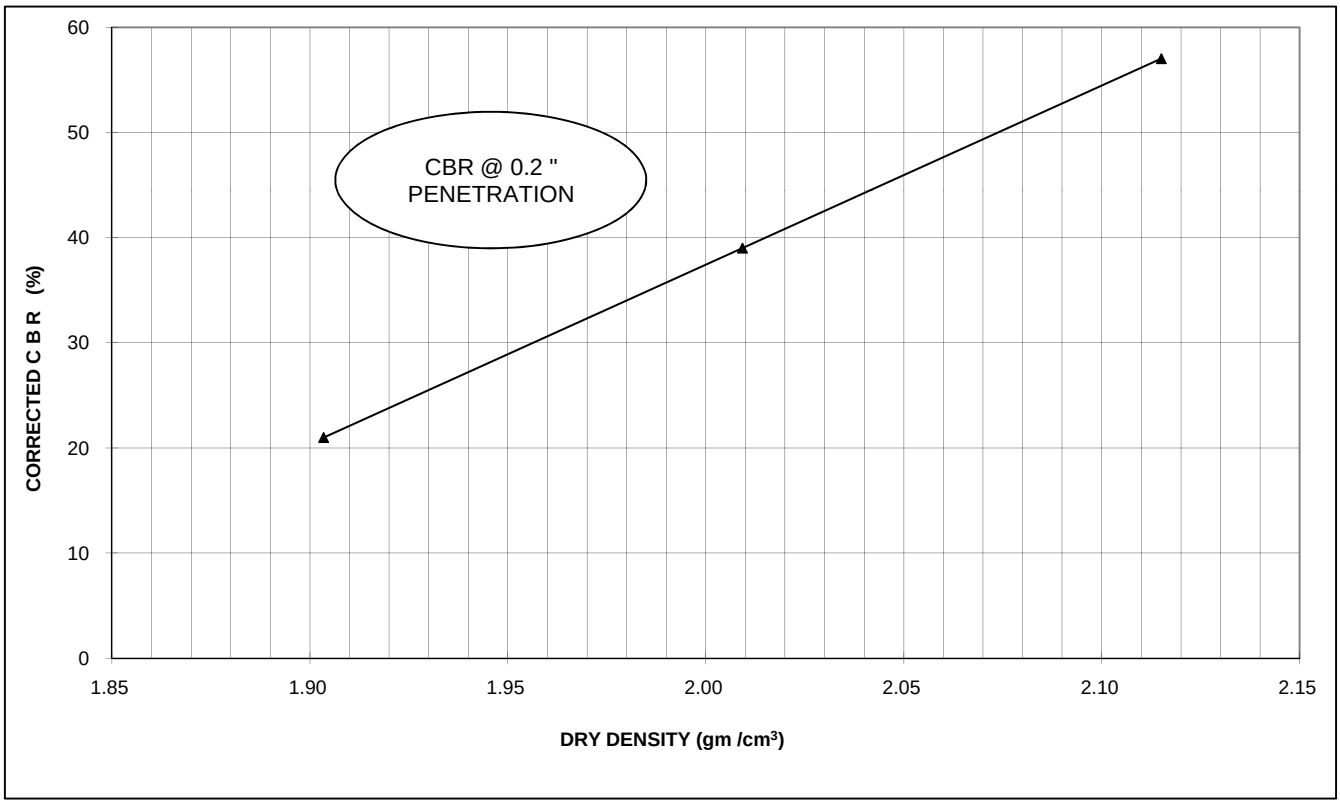
TESTED BY
WAHAB

LAB. SUPERVISOR
ANJAM

CBR - DENSITY RELATIONSHIP
ASTM D 1883

CLIENT :	L & T CONSTRUCTION	REPORT DATE:	8-Sep-19
PROJECT :	MANSOURAH MASSARAH GOLD PROJECT GEOTECHNICAL INVESTIGATION	PROJECT #:	19-K-3054
		TRIAL PIT:	TP-04
		SAMPLE :	B.S-1
SAMPLE:	BULK SAMPLE	DEPTH-(m):	0.3-0.90
		SAMPLE DATE:	3-Sep-19
TP NO.:	TP-04	TEST DATE:	7-Sep-19

COMPACTION EFFORT	10 Blows	25 Blows	56 Blows	100%MDD
DRY DENSITY (gm/cm^3)	1.904	2.009	2.115	2.115
BEARING @ 0.2" PENETRATION (CBR)	21	39	57	57
SWELL (%)	nil	nil	nil	-



MAX. DRY DENSITY (γ_d): **2.115** g /cm^3
OPTIMUM M.C. (OMC): **6.50** %

Remarks:
Nill

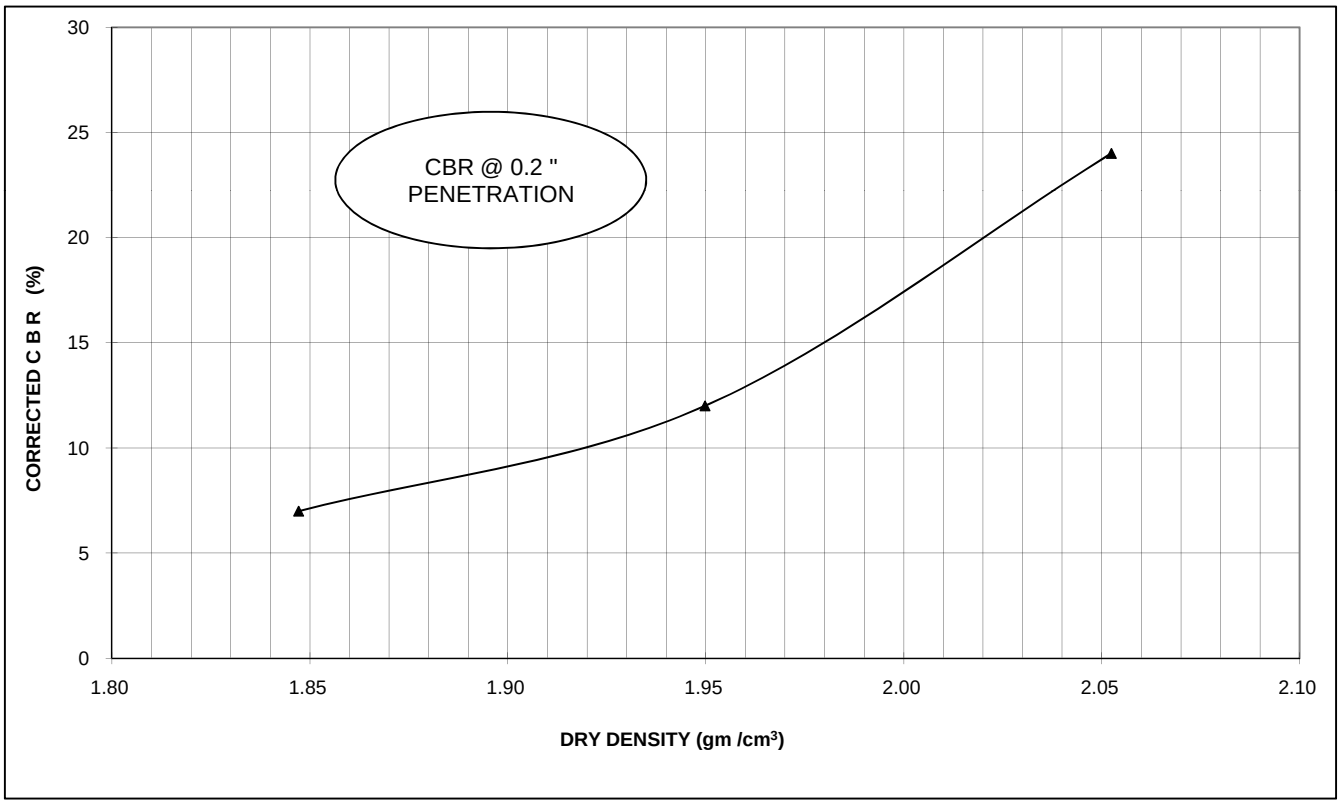
TESTED BY
WAHAB

LAB. SUPERVISOR
ANJAM

CBR - DENSITY RELATIONSHIP
ASTM D 1883

CLIENT :	L & T CONSTRUCTION	REPORT DATE:	8-Sep-19
PROJECT :	MANSOURAH MASSARAH GOLD PROJECT GEOTECHNICAL INVESTIGATION	PROJECT #:	19-K-3054
		TRIAL PIT:	TP-05
		SAMPLE :	B.S-1
SAMPLE:	BULK SAMPLE	DEPTH-(m):	0.30-1.20
		SAMPLE DATE:	3-Sep-19
TP NO.:	TP-05	TEST DATE:	7-Sep-19

COMPACTION EFFORT	10 Blows	25 Blows	56 Blows	100%MDD
DRY DENSITY (gm/cm^3)	1.847	1.950	2.052	2.052
BEARING @ 0.2" PENETRATION (CBR)	7	12	24	24
SWELL (%)	0.5	1.2	1.56	-



MAX. DRY DENSITY (γ_d): **2.052** g /cm^3
OPTIMUM M.C. (OMC): **7.53** %

Remarks:
Nil

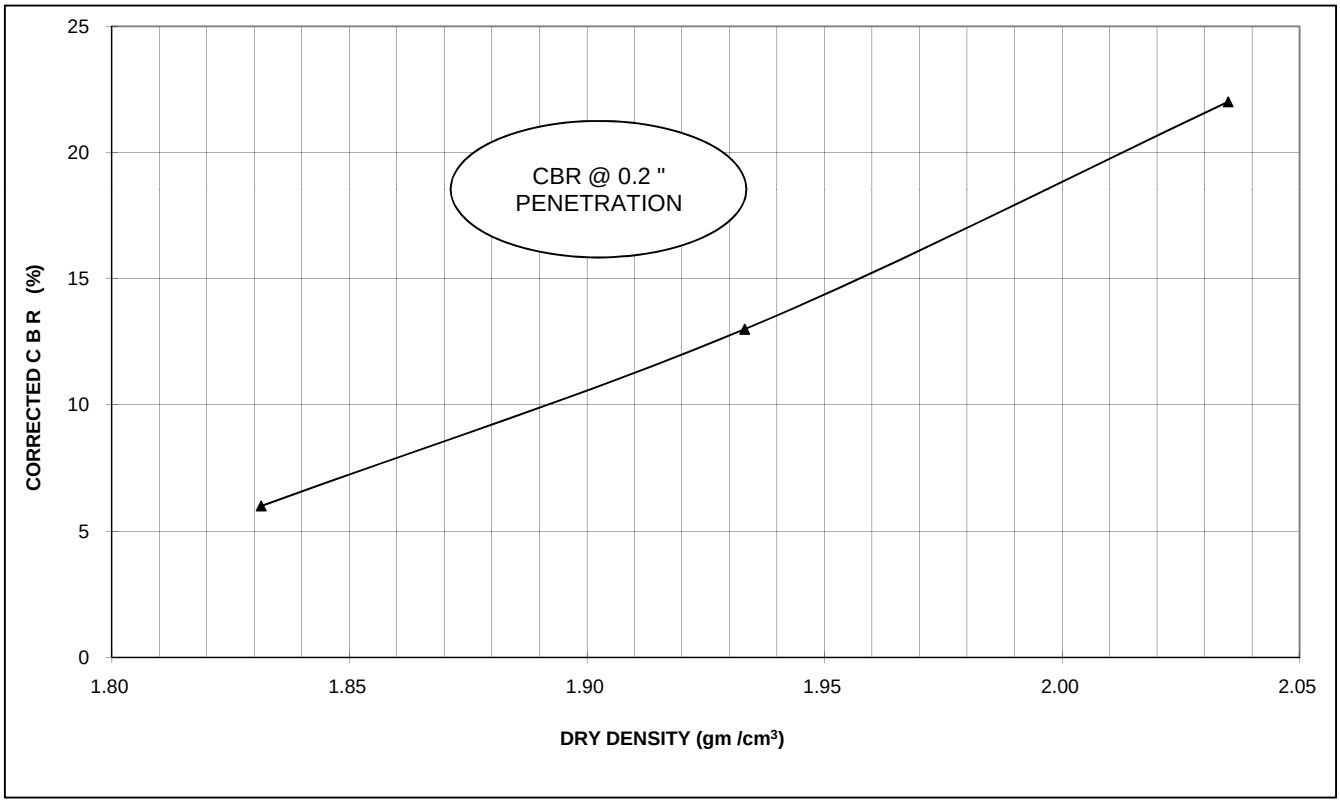
TESTED BY
WAHAB

LAB. SUPERVISOR
ANJAM

CBR - DENSITY RELATIONSHIP
ASTM D 1883

CLIENT :	L & T CONSTRUCTION	REPORT DATE:	8-Sep-19
PROJECT :	MANSOURAH MASSARAH GOLD PROJECT GEOTECHNICAL INVESTIGATION	PROJECT #:	19-K-3054
		TRIAL PIT:	TP-06
		SAMPLE :	B.S-1
SAMPLE:	BULK SAMPLE	DEPTH-(m):	0.0-0.55
		SAMPLE DATE:	3-Sep-19
TP NO.:	TP-06	TEST DATE:	7-Sep-19

COMPACTION EFFORT	10 Blows	25 Blows	56 Blows	100%MDD
DRY DENSITY (gm/cm^3)	1.831	1.933	2.035	2.035
BEARING @ 0.2" PENETRATION (CBR)	6	13	22	22
SWELL (%)	0.4	0.75	1.2	-



MAX. DRY DENSITY (γ_d): **2.035** g /cm^3
OPTIMUM M.C. (OMC): **7.72** %

Remarks:
Nil

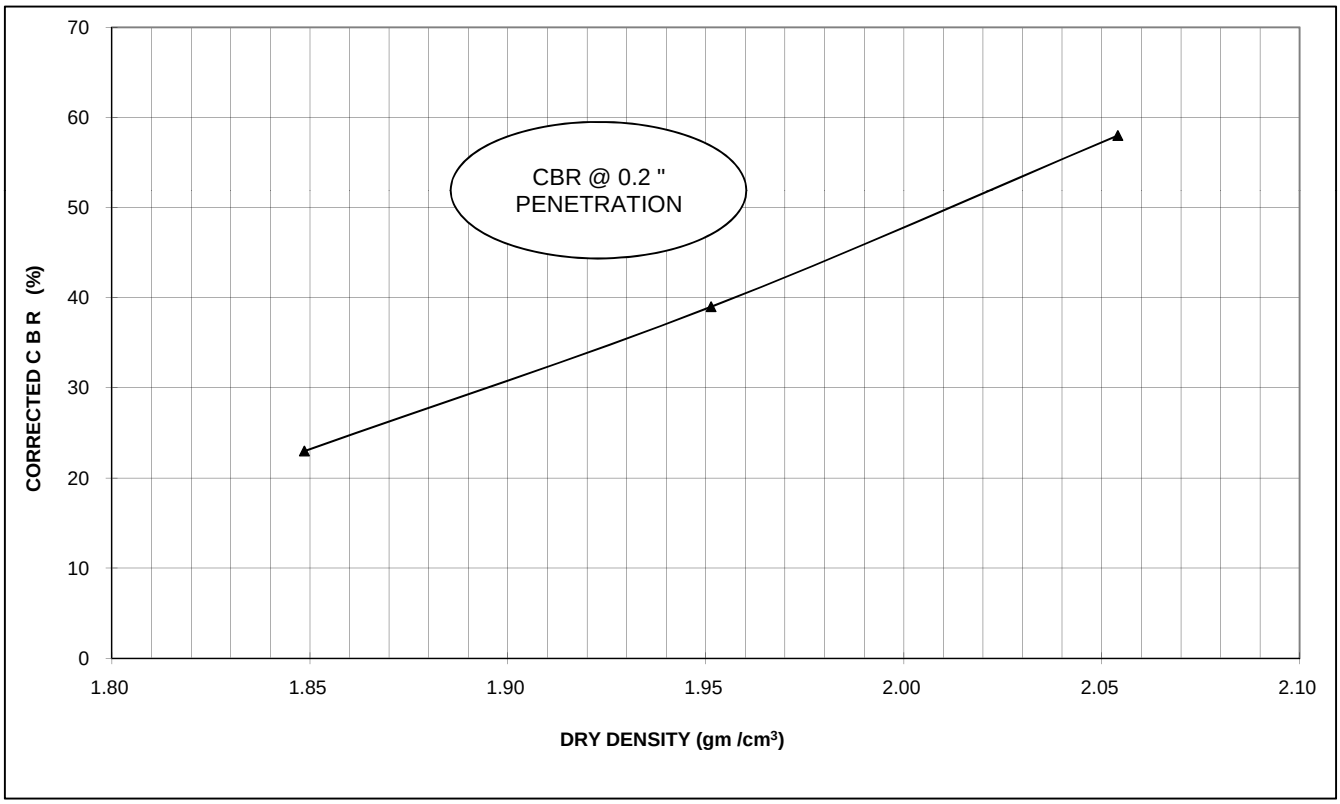
TESTED BY
WAHAB

LAB. SUPERVISOR
ANJAM

CBR - DENSITY RELATIONSHIP
ASTM D 1883

CLIENT :	L & T CONSTRUCTION	REPORT DATE:	9-Sep-19
PROJECT :	MANSOURAH MASSARAH GOLD PROJECT GEOTECHNICAL INVESTIGATION	PROJECT #:	19-K-3054
		TRIAL PIT:	TP-07
		SAMPLE :	B.S-1
SAMPLE:	BULK SAMPLE	DEPTH-(m):	0.00-0.60
		SAMPLE DATE:	4-Sep-19
TP NO.:	TP-07	TEST DATE:	8-Sep-19

COMPACTION EFFORT	10 Blows	25 Blows	56 Blows	100%MDD
DRY DENSITY (gm/cm^3)	1.849	1.951	2.054	2.054
BEARING @ 0.2" PENETRATION (CBR)	23	39	58	58
SWELL (%)	nil	nil	nil	-



MAX. DRY DENSITY (γ_d): **2.054** g /cm^3
OPTIMUM M.C. (OMC): **7.61** %

Remarks:
Nill

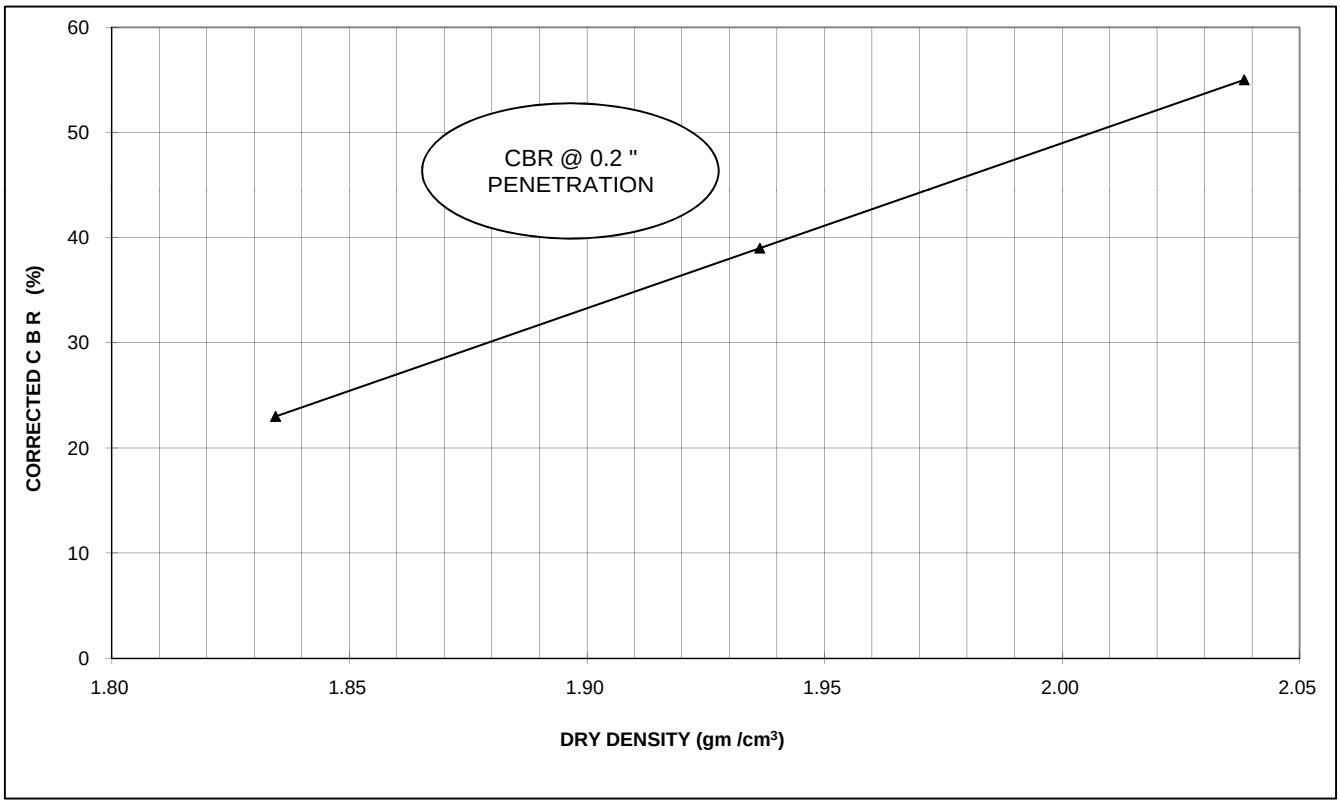
TESTED BY
WAHAB

LAB. SUPERVISOR
ANJAM

CBR - DENSITY RELATIONSHIP
ASTM D 1883

CLIENT :	L & T CONSTRUCTION	REPORT DATE:	9-Sep-19
PROJECT :	MANSOURAH MASSARAH GOLD PROJECT GEOTECHNICAL INVESTIGATION	PROJECT #:	19-K-3054
		TRIAL PIT:	TP-08
		SAMPLE :	B.S-1
SAMPLE:	BULK SAMPLE	DEPTH-(m):	0.00-0.50
		SAMPLE DATE:	4-Sep-19
TP NO.:	TP-08	TEST DATE:	8-Sep-19

COMPACTION EFFORT	10 Blows	25 Blows	56 Blows	100%MDD
DRY DENSITY (gm/cm^3)	1.834	1.936	2.038	2.038
BEARING @ 0.2" PENETRATION (CBR)	23	39	55	55
SWELL (%)	nil	nil	nil	-



MAX. DRY DENSITY (γ_d): **2.038** g /cm^3
OPTIMUM M.C. (OMC): **7.77** %

Remarks:
Nill

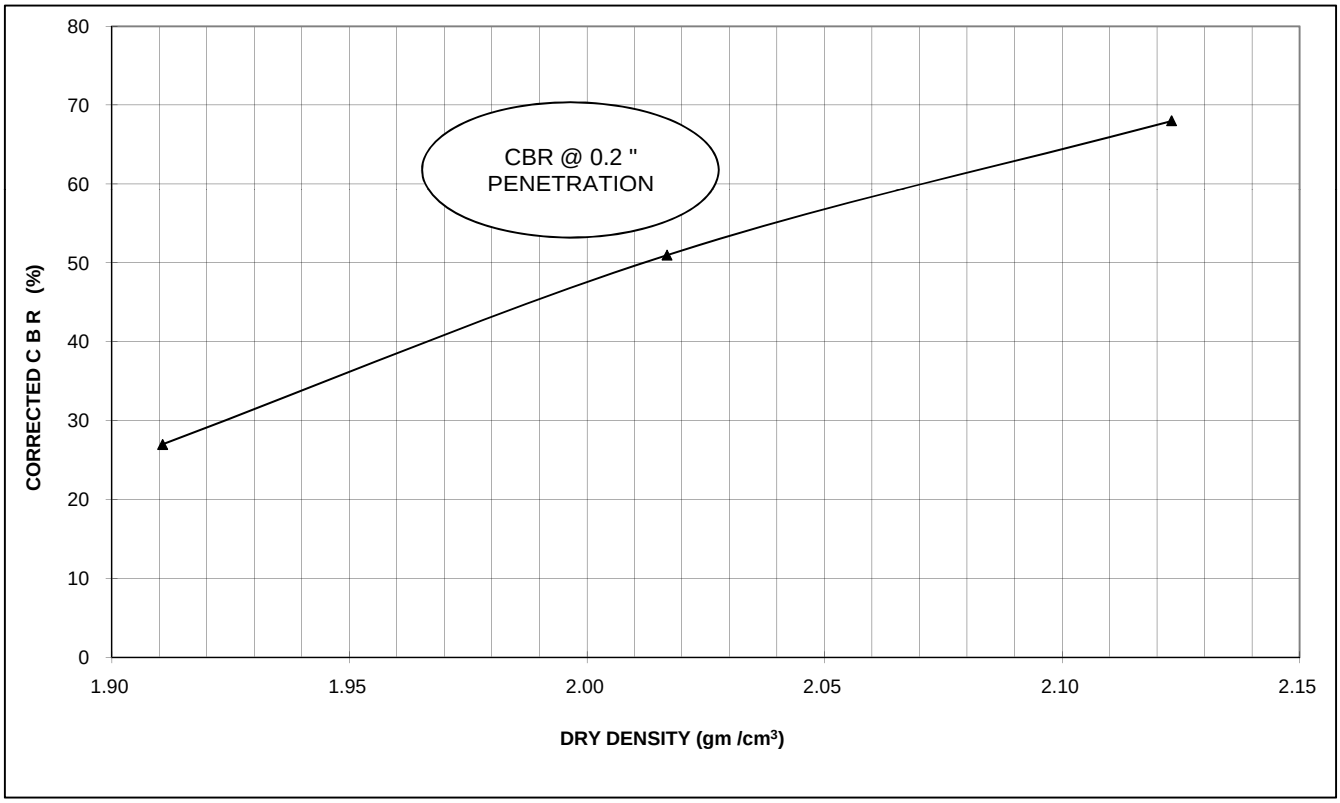
TESTED BY
WAHAB

LAB. SUPERVISOR
ANJAM

CBR - DENSITY RELATIONSHIP
ASTM D 1883

CLIENT :	L & T CONSTRUCTION	REPORT DATE:	10-Sep-19
PROJECT :	MANSOURAH MASSARAH GOLD PROJECT GEOTECHNICAL INVESTIGATION	PROJECT #:	19-K-3054
		TRIAL PIT:	TP-09
		SAMPLE :	B.S-1
SAMPLE:	BULK SAMPLE	DEPTH-(m):	0.0-0.50
		SAMPLE DATE:	5-Sep-19
TP NO.:	TP-09	TEST DATE:	9-Sep-19

COMPACTION EFFORT	10 Blows	25 Blows	56 Blows	100%MDD
DRY DENSITY (gm/cm^3)	1.911	2.017	2.123	2.123
BEARING @ 0.2" PENETRATION (CBR)	27	51	68	68
SWELL (%)	nil	nil	nil	-



MAX. DRY DENSITY (γ_d): **2.123** g /cm^3
OPTIMUM M.C. (OMC): **6.50** %

Remarks:
Nil

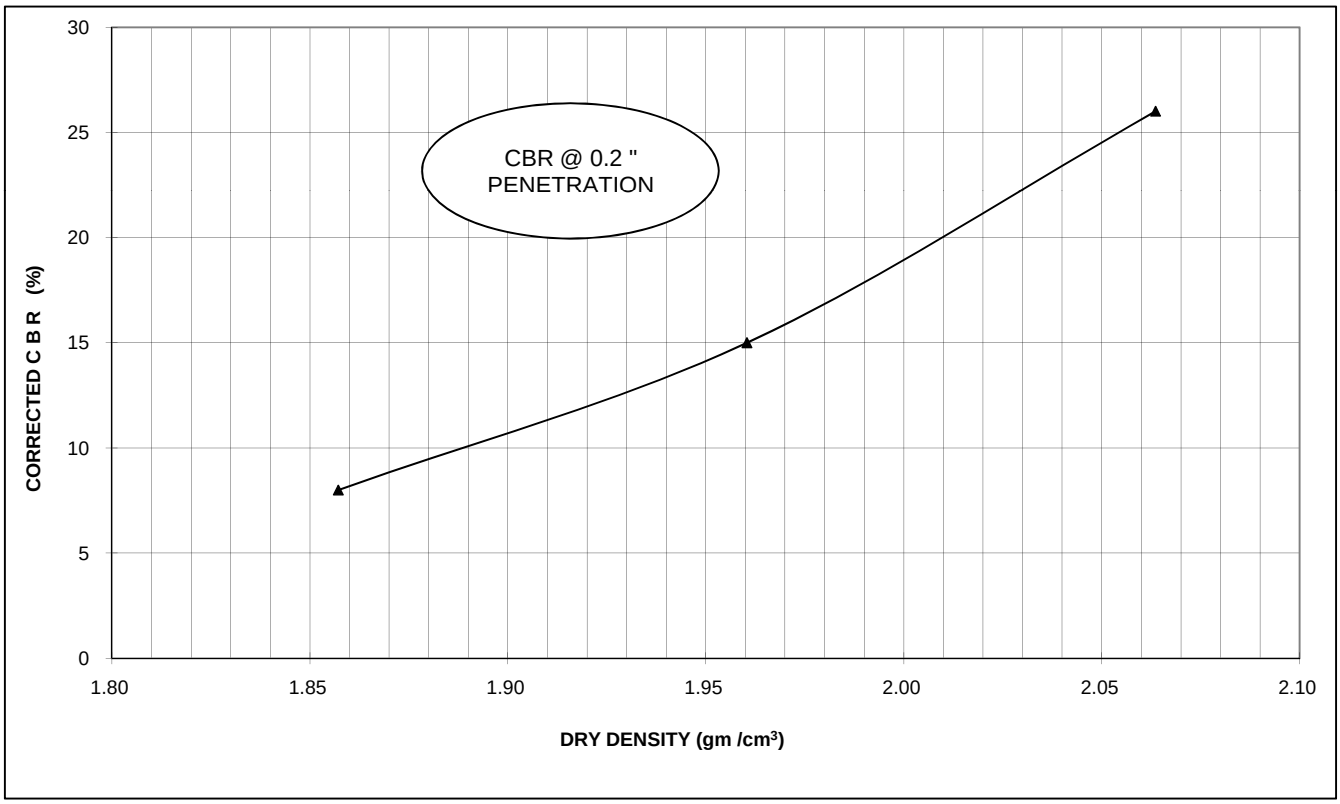
TESTED BY
WAHAB

LAB. SUPERVISOR
ANJAM

CBR - DENSITY RELATIONSHIP
ASTM D 1883

CLIENT :	L & T CONSTRUCTION	REPORT DATE:	10-Sep-19
PROJECT :	MANSOURAH MASSARAH GOLD PROJECT GEOTECHNICAL INVESTIGATION	PROJECT #:	19-K-3054
		TRIAL PIT:	TP-10
		SAMPLE :	B.S-1
SAMPLE:	BULK SAMPLE	DEPTH-(m):	0.0-1.30
		SAMPLE DATE:	5-Sep-19
TP NO.:	TP-10	TEST DATE:	9-Sep-19

COMPACTION EFFORT	10 Blows	25 Blows	56 Blows	100%MDD
DRY DENSITY (gm/cm^3)	1.857	1.960	2.064	2.064
BEARING @ 0.2" PENETRATION (CBR)	8	15	26	26
SWELL (%)	0.3	0.75	1.1	-



MAX. DRY DENSITY (γ_d): **2.064** g /cm^3
OPTIMUM M.C. (OMC): **7.52** %

Remarks:
Nil

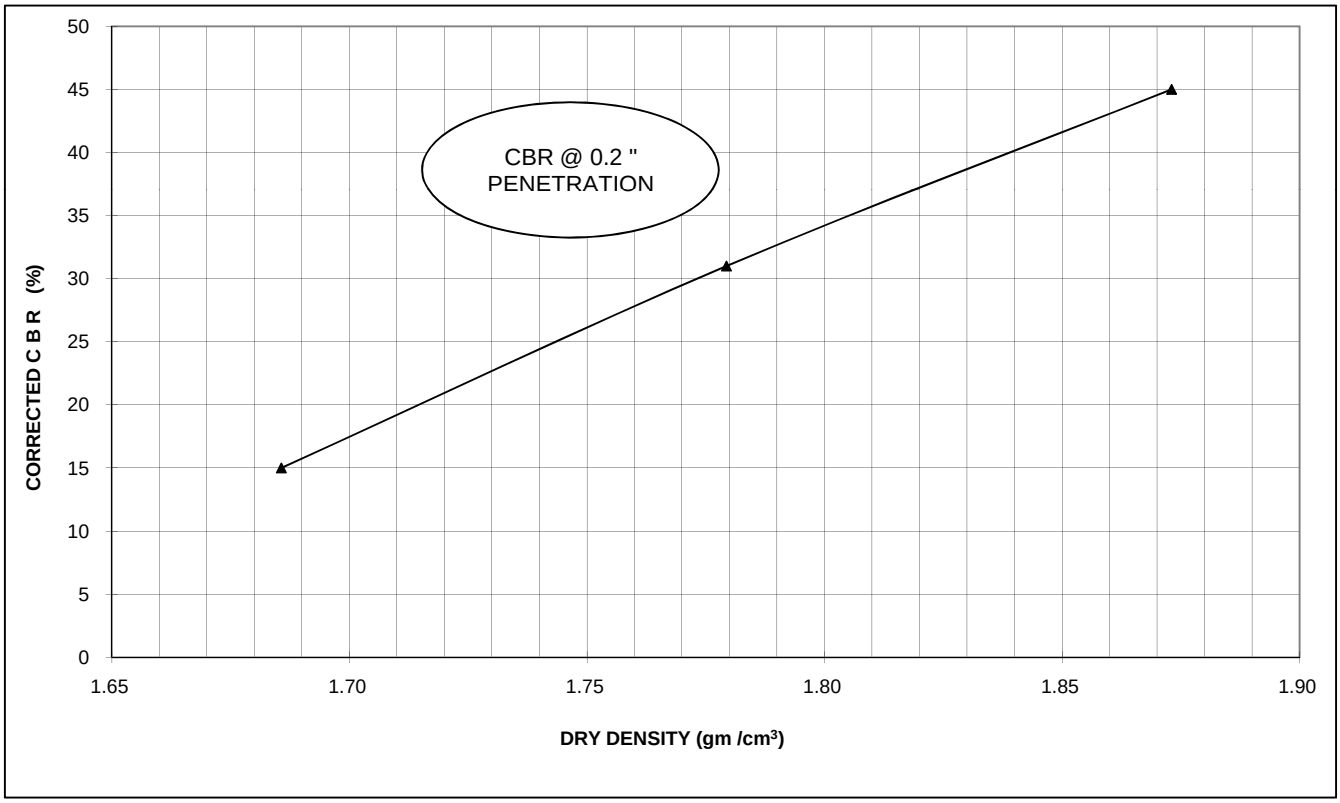
TESTED BY
WAHAB

LAB. SUPERVISOR
ANJAM

CBR - DENSITY RELATIONSHIP
ASTM D 1883

CLIENT :	L & T CONSTRUCTION	REPORT DATE:	11-Sep-19
PROJECT :	MANSOURAH MASSARAH GOLD PROJECT GEOTECHNICAL INVESTIGATION	PROJECT #:	19-K-3054
		TRIAL PIT:	TP-11
		SAMPLE :	B.S-1
SAMPLE:	BULK SAMPLE	DEPTH-(m):	0.00-0.30
		SAMPLE DATE:	6-Sep-19
TP NO.:	TP-11	TEST DATE:	10-Sep-19

COMPACTION EFFORT	10 Blows	25 Blows	56 Blows	100%MDD
DRY DENSITY (gm/cm^3)	1.686	1.779	1.873	1.873
BEARING @ 0.2" PENETRATION (CBR)	15	31	45	45
SWELL (%)	nil	nil	nil	-



MAX. DRY DENSITY (γ_d): **1.873** g /cm^3
OPTIMUM M.C. (OMC): **9.50** %

Remarks:
Nil

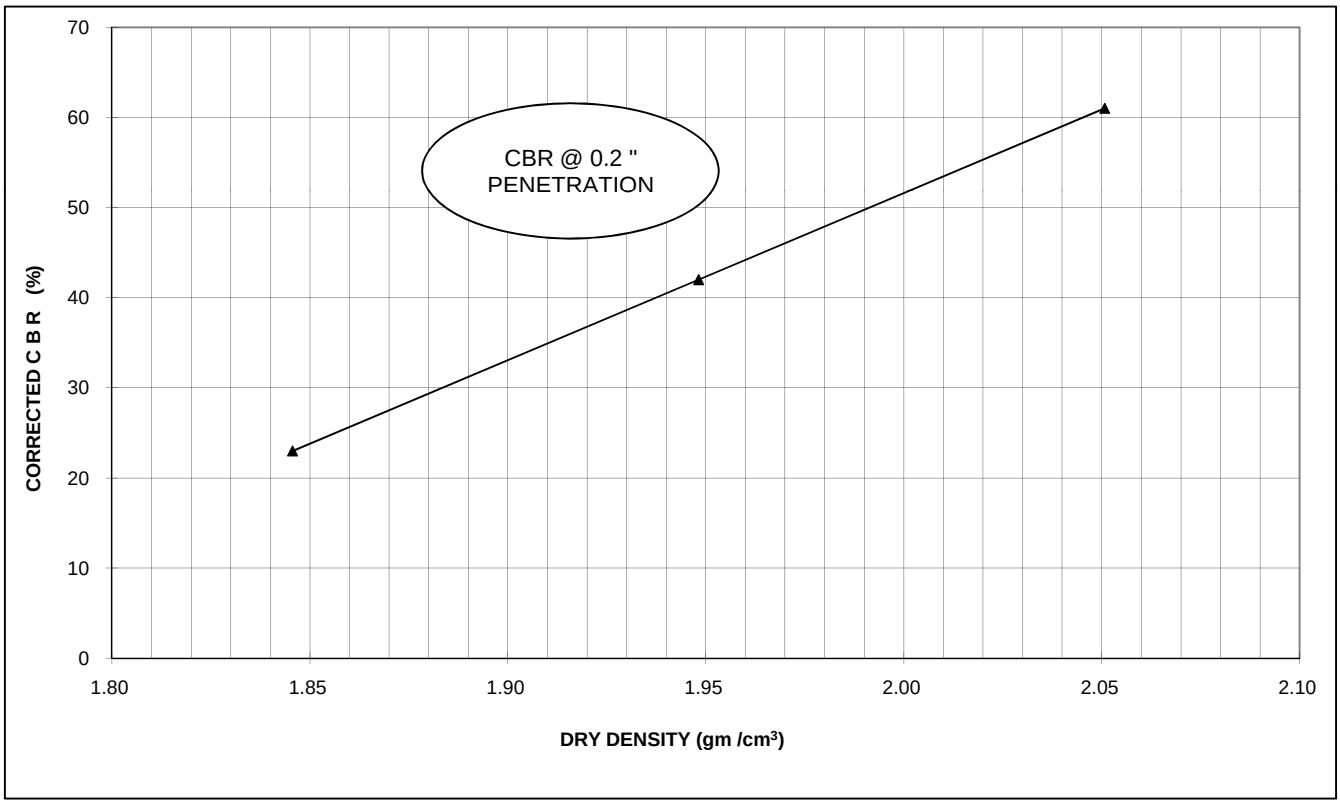
TESTED BY
WAHAB

LAB. SUPERVISOR
ANJAM

CBR - DENSITY RELATIONSHIP
ASTM D 1883

CLIENT :	L & T CONSTRUCTION	REPORT DATE:	11-Sep-19
PROJECT :	MANSOURAH MASSARAH GOLD PROJECT GEOTECHNICAL INVESTIGATION	PROJECT #:	19-K-3054
		TRIAL PIT:	TP-12
		SAMPLE :	B.S-1
SAMPLE:	BULK SAMPLE	DEPTH-(m):	0.0-0.75
		SAMPLE DATE:	6-Sep-19
TP NO.:	TP-12	TEST DATE:	10-Sep-19

COMPACTION EFFORT	10 Blows	25 Blows	56 Blows	100%MDD
DRY DENSITY (gm/cm^3)	1.846	1.948	2.051	2.051
BEARING @ 0.2" PENETRATION (CBR)	23	42	61	61
SWELL (%)	nil	nil	nil	-



MAX. DRY DENSITY (γ_d): **2.051** g /cm^3
OPTIMUM M.C. (OMC): **7.55** %

Remarks:
Nill

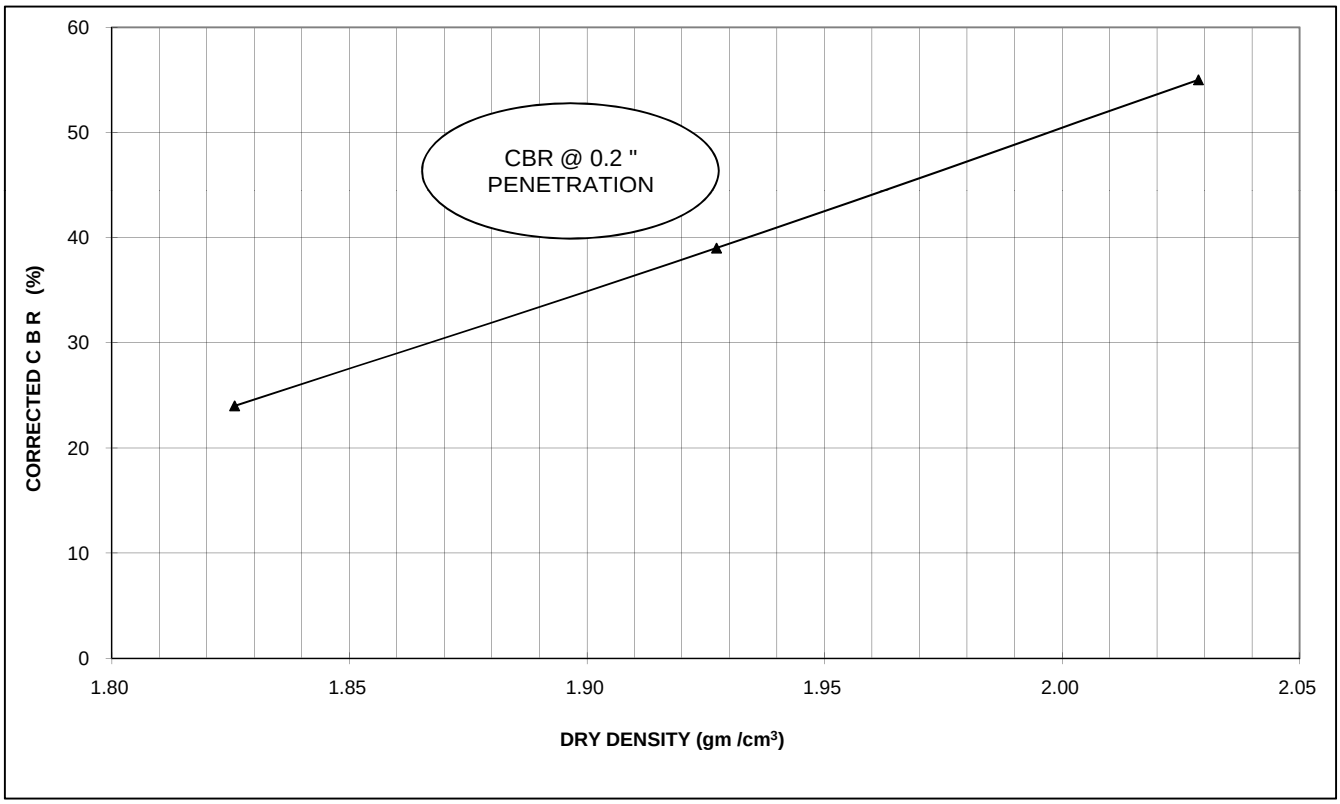
TESTED BY
WAHAB

LAB. SUPERVISOR
ANJAM

CBR - DENSITY RELATIONSHIP
ASTM D 1883

CLIENT :	L & T CONSTRUCTION	REPORT DATE:	12-Sep-19
PROJECT :	MANSOURAH MASSARAH GOLD PROJECT GEOTECHNICAL INVESTIGATION	PROJECT #:	19-K-3054
		TRIAL PIT:	TP-13
		SAMPLE :	B.S-1
SAMPLE:	BULK SAMPLE	DEPTH-(m):	0.00-0.70
		SAMPLE DATE:	7-Sep-19
TP NO.:	TP-13	TEST DATE:	11-Sep-19

COMPACTION EFFORT	10 Blows	25 Blows	56 Blows	100%MDD
DRY DENSITY (gm/cm^3)	1.826	1.927	2.029	2.029
BEARING @ 0.2" PENETRATION (CBR)	24	39	55	55
SWELL (%)	nil	nil	nil	-



MAX. DRY DENSITY (γ_d): **2.029** g /cm^3
OPTIMUM M.C. (OMC): **7.40** %

Remarks:
Nill

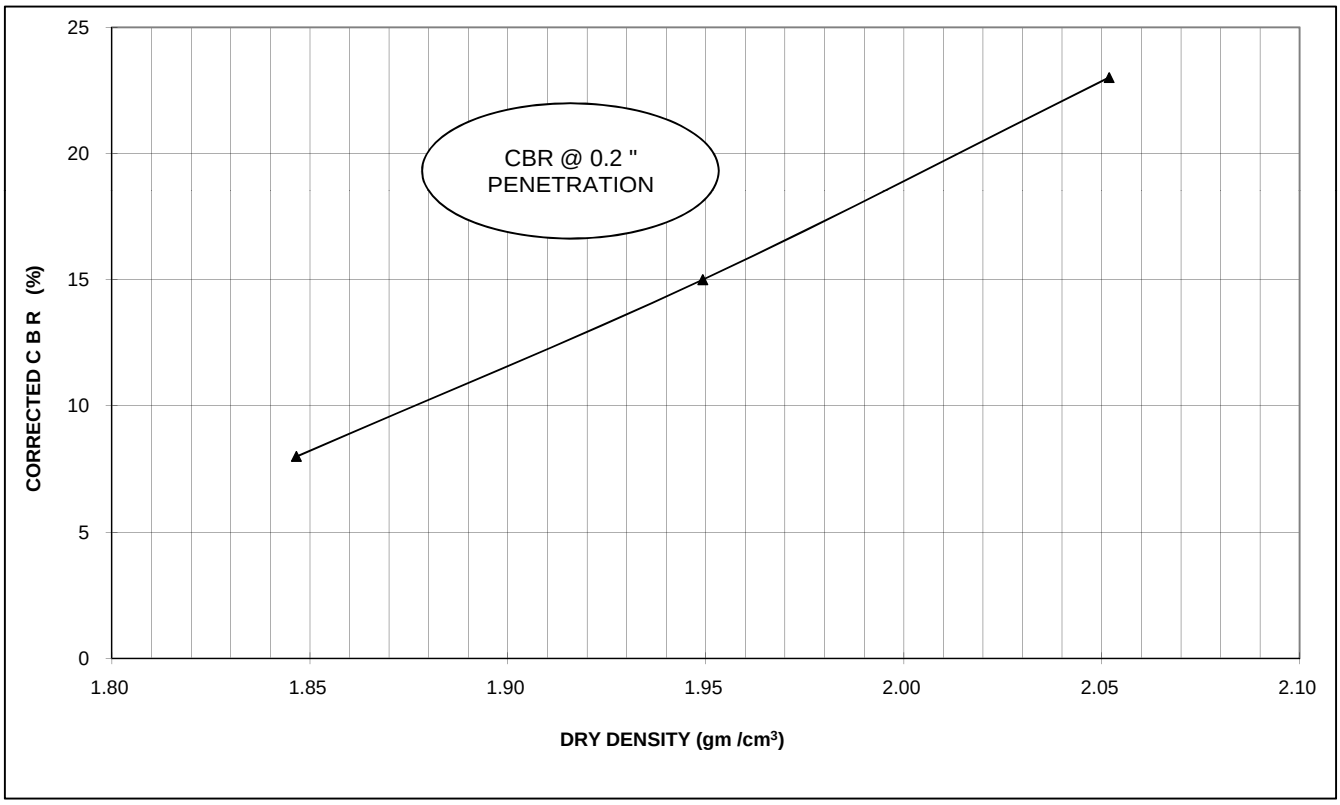
TESTED BY
WAHAB

LAB. SUPERVISOR
ANJAM

CBR - DENSITY RELATIONSHIP
ASTM D 1883

CLIENT :	L & T CONSTRUCTION	REPORT DATE:	12-Sep-19
PROJECT :	MANSOURAH MASSARAH GOLD PROJECT GEOTECHNICAL INVESTIGATION	PROJECT #:	19-K-3054
		TRIAL PIT:	TP-14
		SAMPLE :	B.S-1
SAMPLE:	BULK SAMPLE	DEPTH-(m):	0.00-0.50
		SAMPLE DATE:	7-Sep-19
TP NO.:	TP-14	TEST DATE:	11-Sep-19

COMPACTION EFFORT	10 Blows	25 Blows	56 Blows	100%MDD
DRY DENSITY (gm/cm^3)	1.847	1.949	2.052	2.052
BEARING @ 0.2" PENETRATION (CBR)	8	15	23	23
SWELL (%)	0.5	0.8	1.1	-



MAX. DRY DENSITY (γ_d): **2.052** g /cm^3
OPTIMUM M.C. (OMC): **7.32** %

Remarks:
Nil

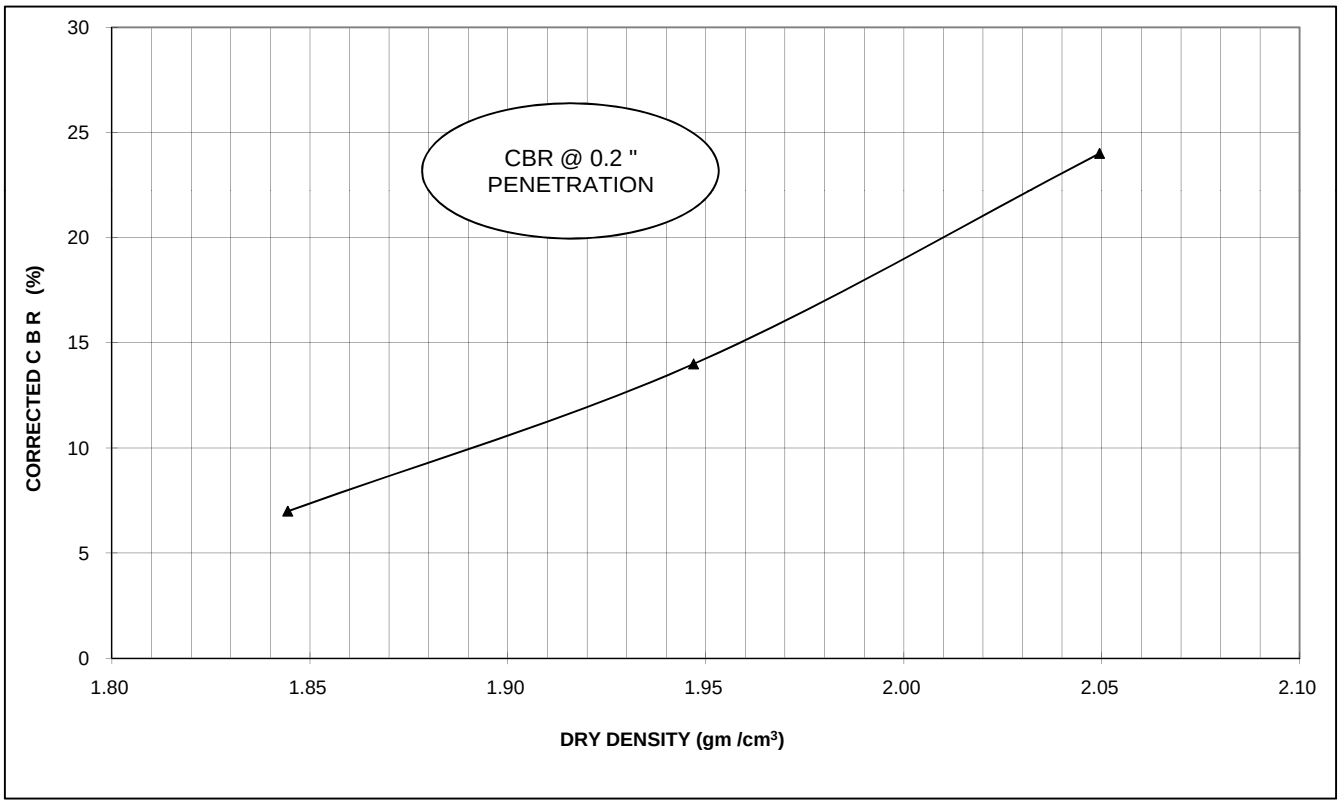
TESTED BY
WAHAB

LAB. SUPERVISOR
ANJAM

CBR - DENSITY RELATIONSHIP
ASTM D 1883

CLIENT :	L & T CONSTRUCTION	REPORT DATE:	13-Sep-19
PROJECT :	MANSOURAH MASSARAH GOLD PROJECT GEOTECHNICAL INVESTIGATION	PROJECT #:	19-K-3054
		TRIAL PIT:	TP-15
		SAMPLE :	B.S-1
SAMPLE:	BULK SAMPLE	DEPTH-(m):	0.0-0.50
		SAMPLE DATE:	8-Sep-19
TP NO.:	TP-15	TEST DATE:	12-Sep-19

COMPACTION EFFORT	10 Blows	25 Blows	56 Blows	100%MDD
DRY DENSITY (gm/cm^3)	1.844	1.947	2.049	2.049
BEARING @ 0.2" PENETRATION (CBR)	7	14	24	24
SWELL (%)	0.7	1.1	1.5	-



MAX. DRY DENSITY (γ_d): **2.049** g /cm^3
OPTIMUM M.C. (OMC): **6.34** %

Remarks:
Nil

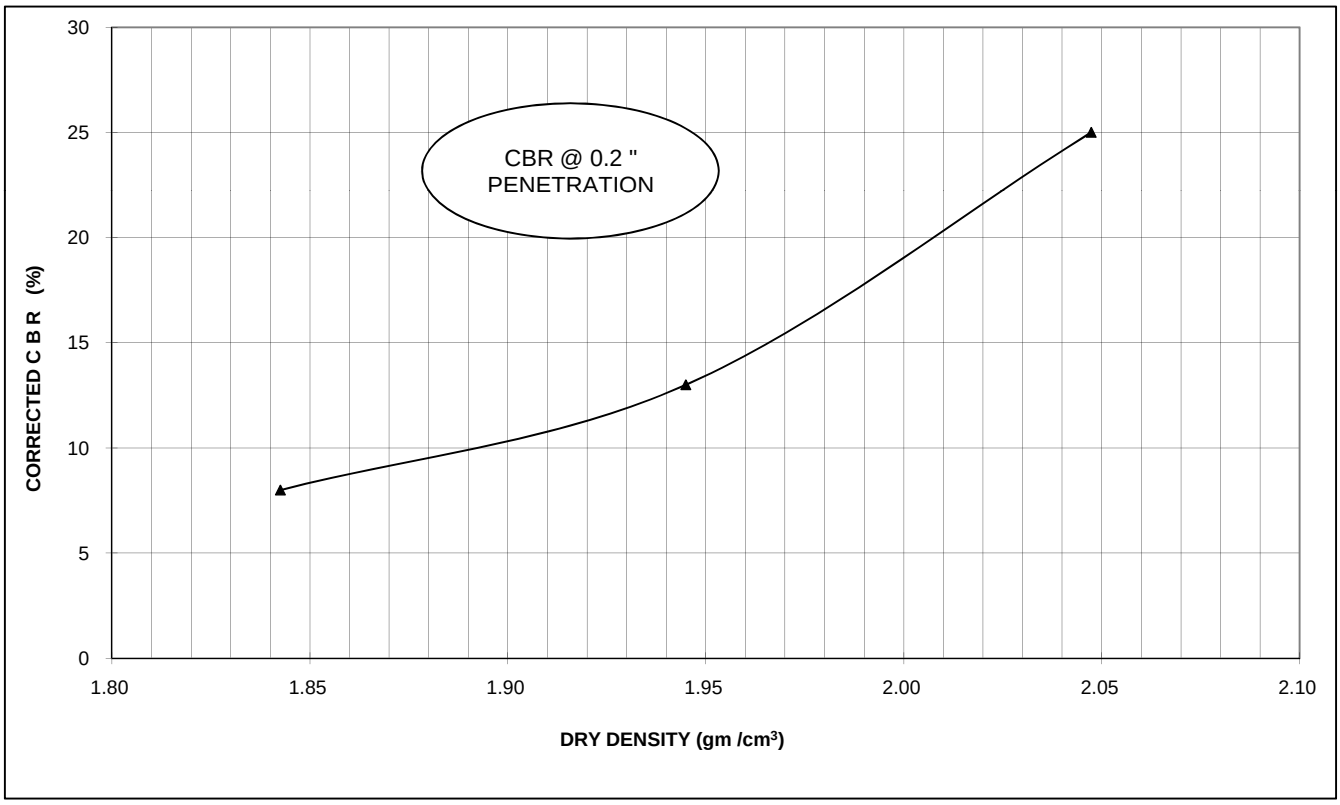
TESTED BY
WAHAB

LAB. SUPERVISOR
ANJAM

CBR - DENSITY RELATIONSHIP
ASTM D 1883

CLIENT :	L & T CONSTRUCTION	REPORT DATE:	14-Sep-19
PROJECT :	MANSOURAH MASSARAH GOLD PROJECT GEOTECHNICAL INVESTIGATION	PROJECT #:	19-K-3054
		TRIAL PIT:	TP-16
		SAMPLE :	B.S-1
SAMPLE:	BULK SAMPLE	DEPTH-(m):	0.00-1.0
		SAMPLE DATE:	9-Sep-19
TP NO.:	TP-16	TEST DATE:	13-Sep-19

COMPACTION EFFORT	10 Blows	25 Blows	56 Blows	100%MDD
DRY DENSITY (gm/cm^3)	1.843	1.945	2.047	2.047
BEARING @ 0.2" PENETRATION (CBR)	8	13	25	25
SWELL (%)	0.6	0.85	1.1	-



MAX. DRY DENSITY (γ_d): **2.047** g /cm^3
OPTIMUM M.C. (OMC): **6.22** %

Remarks:
Nil

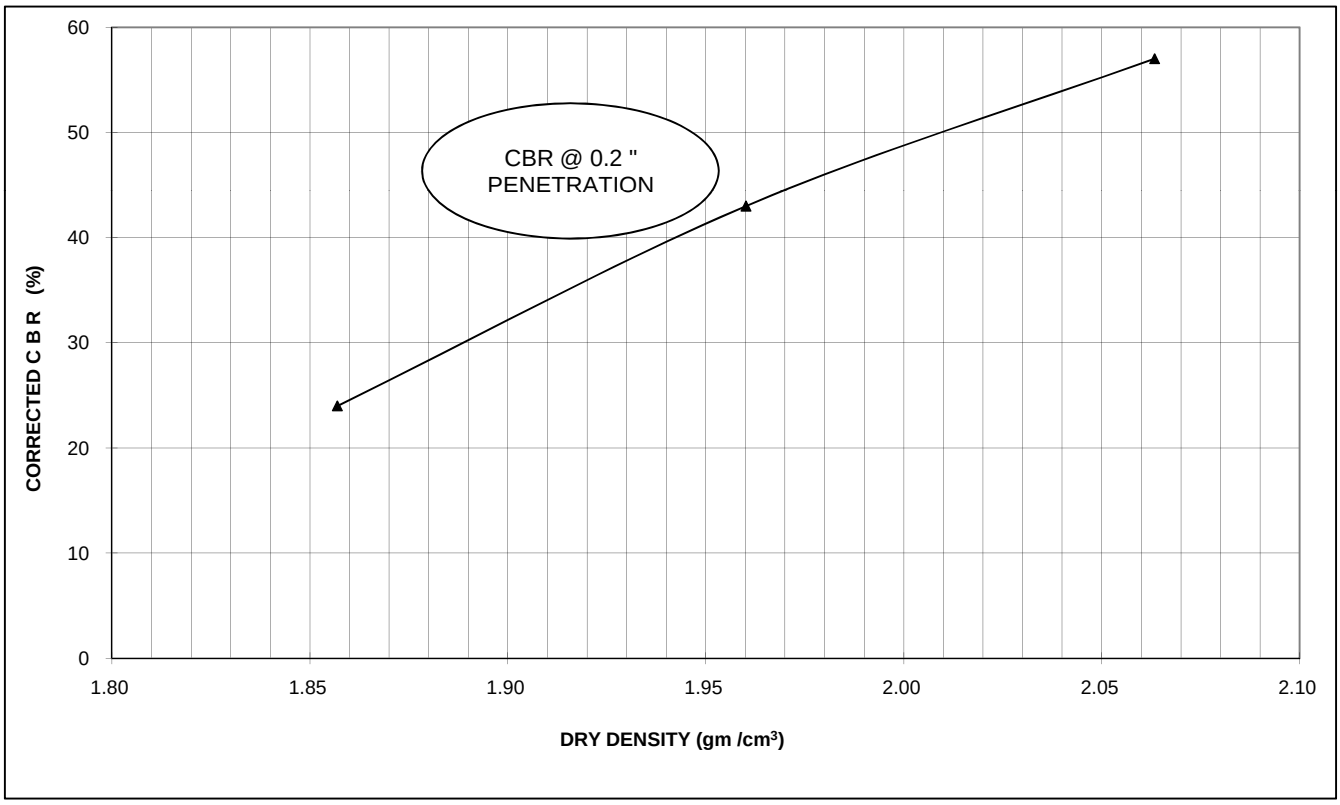
TESTED BY
WAHAB

LAB. SUPERVISOR
ANJAM

CBR - DENSITY RELATIONSHIP
ASTM D 1883

CLIENT :	L & T CONSTRUCTION	REPORT DATE:	14-Sep-19
PROJECT :	MANSOURAH MASSARAH GOLD PROJECT GEOTECHNICAL INVESTIGATION	PROJECT #:	19-K-3054
		TRIAL PIT:	TP-17
		SAMPLE :	B.S-1
SAMPLE:	BULK SAMPLE	DEPTH-(m):	0.00-0.90
		SAMPLE DATE:	9-Sep-19
TP NO.:	TP-17	TEST DATE:	13-Sep-19

COMPACTION EFFORT	10 Blows	25 Blows	56 Blows	100%MDD
DRY DENSITY (gm/cm^3)	1.857	1.960	2.063	2.063
BEARING @ 0.2" PENETRATION (CBR)	24	43	57	57
SWELL (%)	nil	nil	nil	-



MAX. DRY DENSITY (γ_d): **2.063** g /cm^3
OPTIMUM M.C. (OMC): **6.40** %

Remarks:
Nil

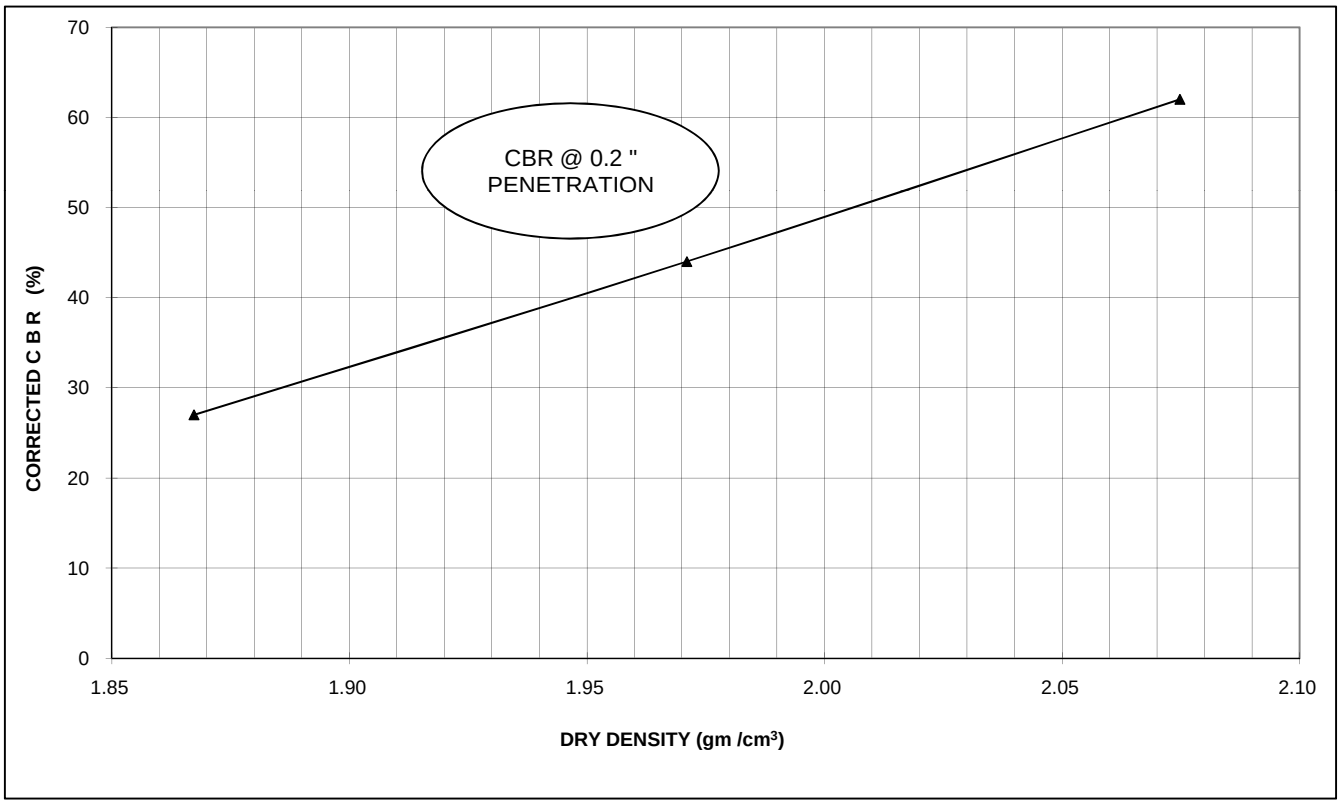
TESTED BY
WAHAB

LAB. SUPERVISOR
ANJAM

CBR - DENSITY RELATIONSHIP
ASTM D 1883

CLIENT :	L & T CONSTRUCTION	REPORT DATE:	15-Sep-19
PROJECT :	MANSOURAH MASSARAH GOLD PROJECT GEOTECHNICAL INVESTIGATION	PROJECT #:	19-K-3054
		TRIAL PIT:	TP-18
		SAMPLE :	B.S-1
SAMPLE:	BULK SAMPLE	DEPTH-(m):	0.00-0.50
		SAMPLE DATE:	10-Sep-19
TP NO.:	TP-18	TEST DATE:	14-Sep-19

COMPACTION EFFORT	10 Blows	25 Blows	56 Blows	100%MDD
DRY DENSITY (gm/cm^3)	1.867	1.971	2.075	2.075
BEARING @ 0.2" PENETRATION (CBR)	27	44	62	62
SWELL (%)	nil	nil	nil	-



MAX. DRY DENSITY (γ_d): **2.075** g /cm^3
OPTIMUM M.C. (OMC): **6.30** %

Remarks:
Nil

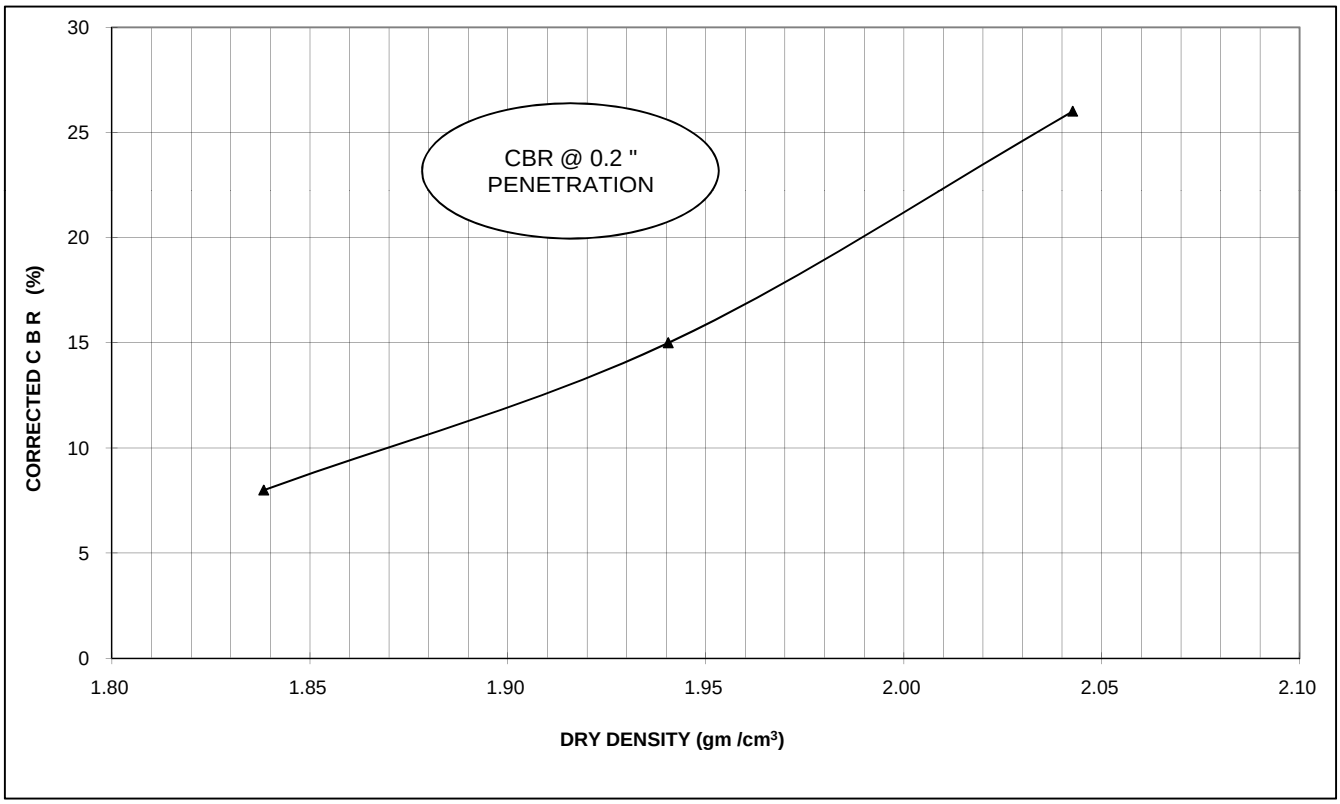
TESTED BY
WAHAB

LAB. SUPERVISOR
ANJAM

CBR - DENSITY RELATIONSHIP
ASTM D 1883

CLIENT :	L & T CONSTRUCTION	REPORT DATE:	15-Sep-19
PROJECT :	MANSOURAH MASSARAH GOLD PROJECT GEOTECHNICAL INVESTIGATION	PROJECT #:	19-K-3054
		TRIAL PIT:	TP-19
		SAMPLE :	B.S-1
SAMPLE:	BULK SAMPLE	DEPTH-(m):	0.2-0.50
		SAMPLE DATE:	10-Sep-19
TP NO.:	TP-19	TEST DATE:	14-Sep-19

COMPACTION EFFORT	10 Blows	25 Blows	56 Blows	100%MDD
DRY DENSITY (gm/cm^3)	1.838	1.941	2.043	2.043
BEARING @ 0.2" PENETRATION (CBR)	8	15	26	26
SWELL (%)	1.1	1.2	1.3	-



MAX. DRY DENSITY (γ_d): **2.043** g /cm^3
OPTIMUM M.C. (OMC): **6.49** %

Remarks:
Nil

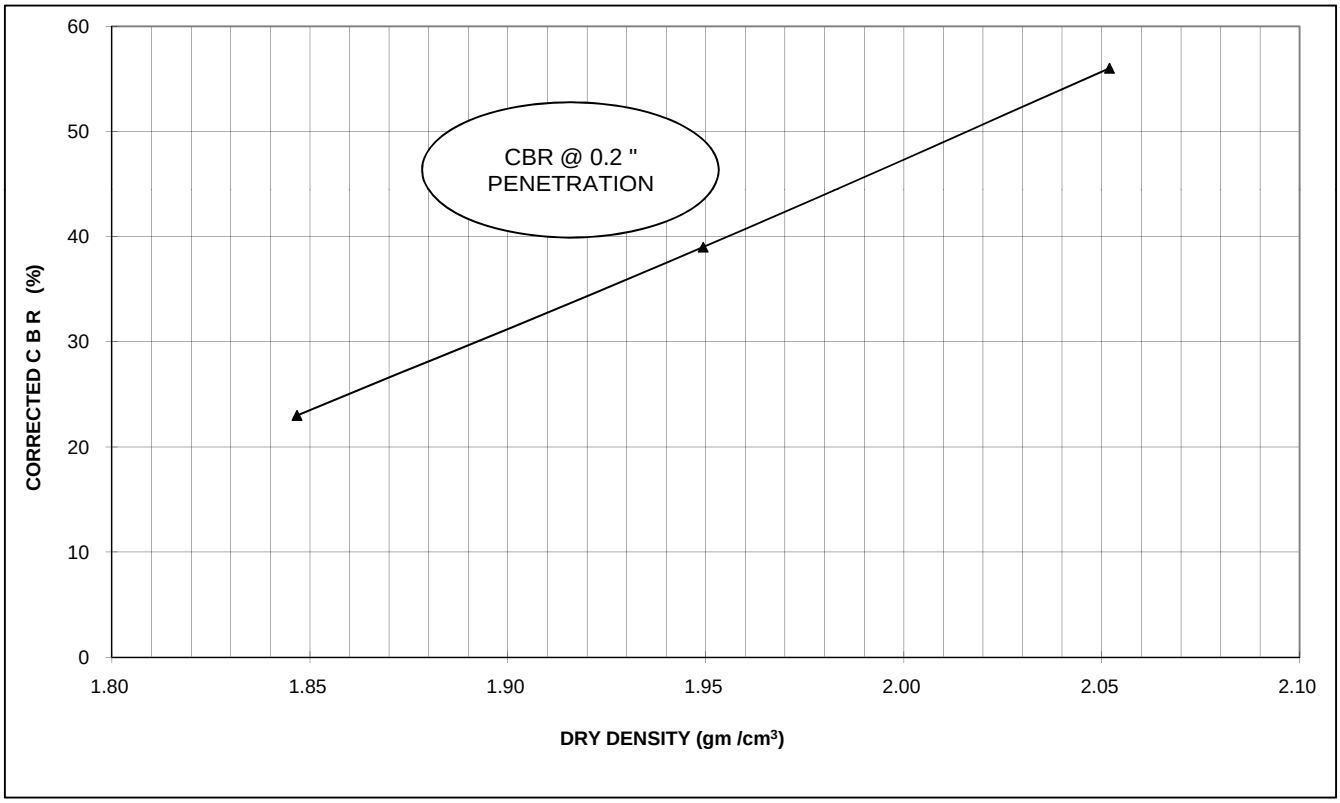
TESTED BY
WAHAB

LAB. SUPERVISOR
ANJAM

CBR - DENSITY RELATIONSHIP
ASTM D 1883

CLIENT :	L & T CONSTRUCTION	REPORT DATE:	16-Sep-19
PROJECT :	MANSOURAH MASSARAH GOLD PROJECT GEOTECHNICAL INVESTIGATION	PROJECT #:	19-K-3054
		TRIAL PIT:	TP-20
		SAMPLE :	B.S-1
SAMPLE:	BULK SAMPLE	DEPTH-(m):	0.00-0.50
		SAMPLE DATE:	11-Sep-19
TP NO.:	TP-20	TEST DATE:	15-Sep-19

COMPACTION EFFORT	10 Blows	25 Blows	56 Blows	100%MDD
DRY DENSITY (gm/cm^3)	1.847	1.949	2.052	2.052
BEARING @ 0.2" PENETRATION (CBR)	23	39	56	56
SWELL (%)	nil	nil	nil	-



MAX. DRY DENSITY (γ_d): **2.052** g /cm^3
OPTIMUM M.C. (OMC): **6.53** %

Remarks:
Nil

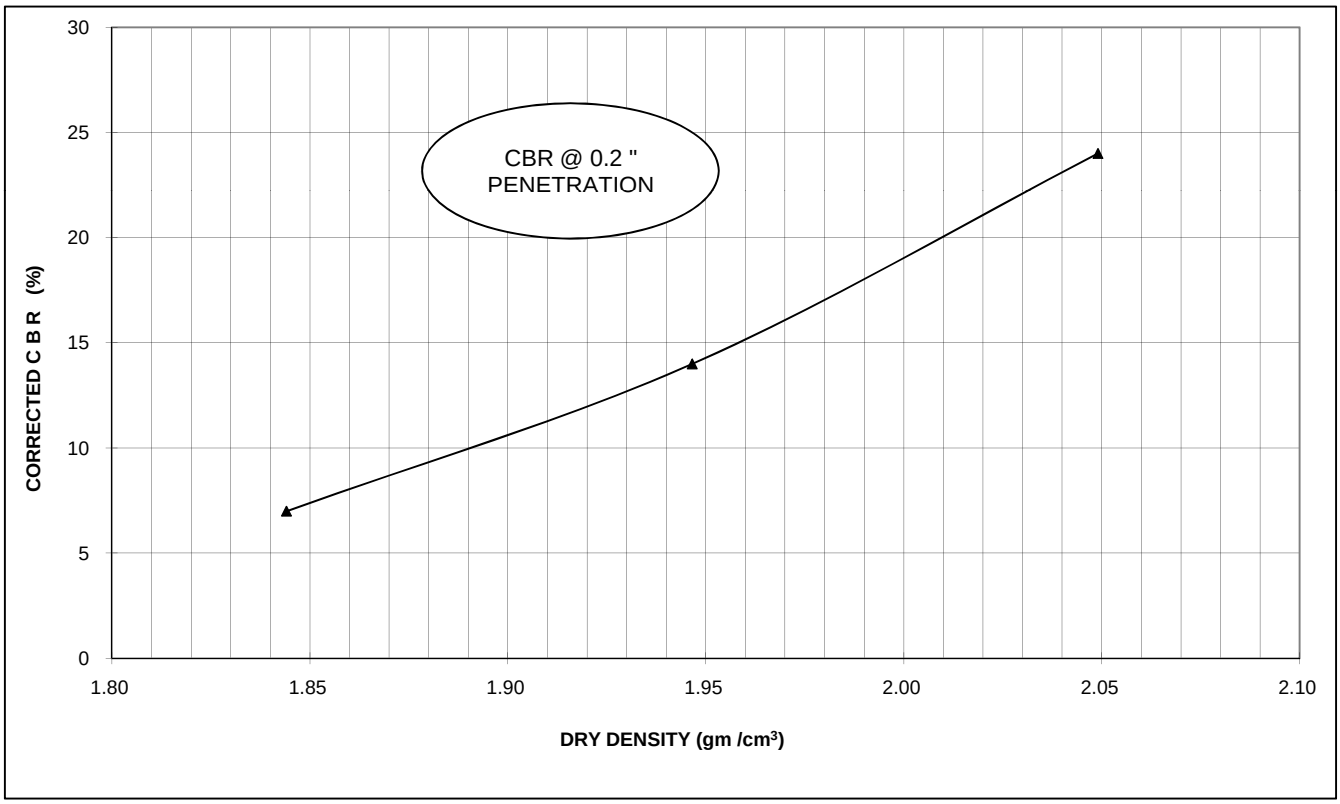
TESTED BY
WAHAB

LAB. SUPERVISOR
ANJAM

CBR - DENSITY RELATIONSHIP
ASTM D 1883

CLIENT :	L & T CONSTRUCTION	REPORT DATE:	16-Sep-19
PROJECT :	MANSOURAH MASSARAH GOLD PROJECT GEOTECHNICAL INVESTIGATION	PROJECT #:	19-K-3054
		TRIAL PIT:	TP-21
		SAMPLE :	B.S-1
SAMPLE:	BULK SAMPLE	DEPTH-(m):	0.4-1.30
		SAMPLE DATE:	11-Sep-19
TP NO.:	TP-21	TEST DATE:	15-Sep-19

COMPACTION EFFORT	10 Blows	25 Blows	56 Blows	100%MDD
DRY DENSITY (gm/cm^3)	1.844	1.947	2.049	2.049
BEARING @ 0.2" PENETRATION (CBR)	7	14	24	24
SWELL (%)	1.1	1.5	1.4	-



MAX. DRY DENSITY (γ_d): **2.049** g /cm^3
OPTIMUM M.C. (OMC): **6.75** %

Remarks:
Nil

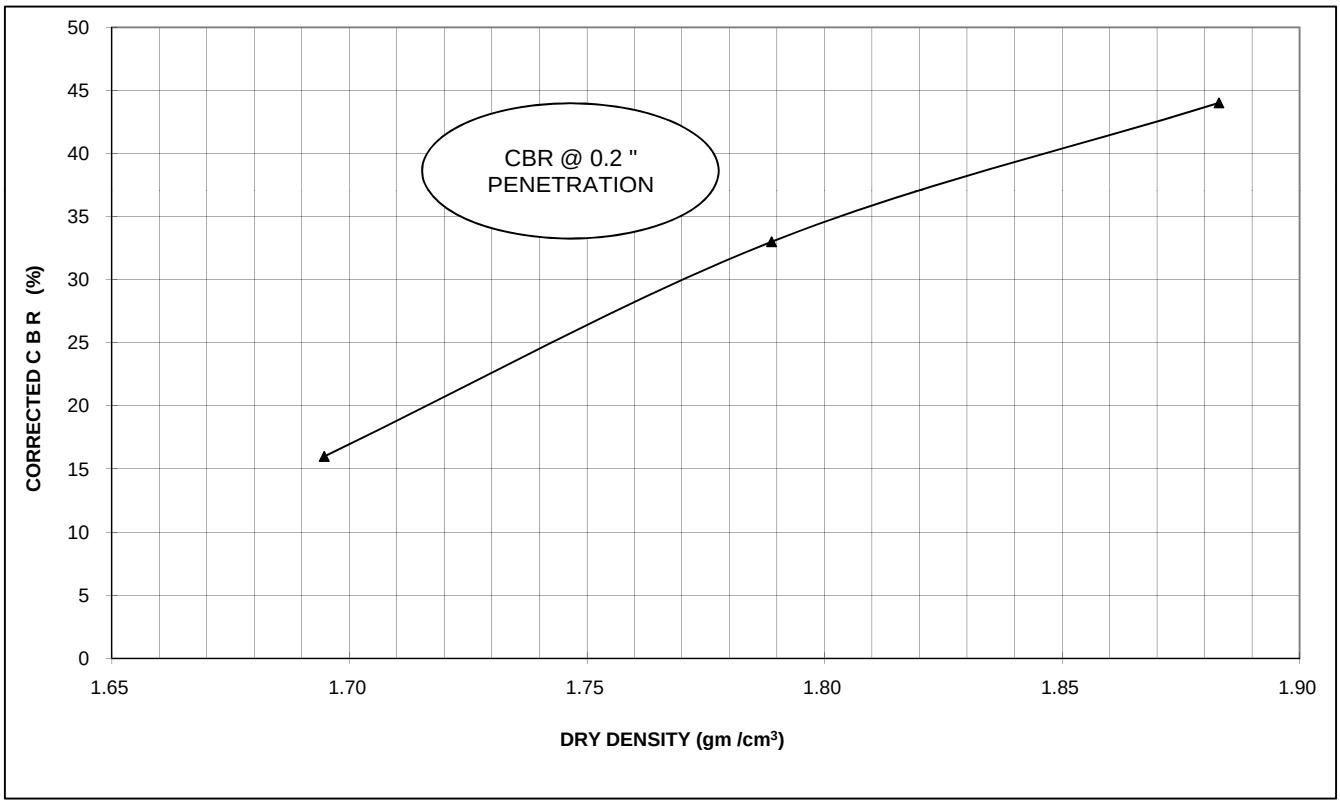
TESTED BY
WAHAB

LAB. SUPERVISOR
ANJAM

CBR - DENSITY RELATIONSHIP
ASTM D 1883

CLIENT :	L & T CONSTRUCTION	REPORT DATE:	17-Sep-19
PROJECT :	MANSOURAH MASSARAH GOLD PROJECT GEOTECHNICAL INVESTIGATION	PROJECT #:	19-K-3054
		TRIAL PIT:	TP-22
		SAMPLE :	B.S-1
SAMPLE:	BULK SAMPLE	DEPTH-(m):	0.0-0.50
		SAMPLE DATE:	12-Sep-19
TP NO.:	TP-22	TEST DATE:	16-Sep-19

COMPACTION EFFORT	10 Blows	25 Blows	56 Blows	100%MDD
DRY DENSITY (gm/cm^3)	1.695	1.789	1.883	1.883
BEARING @ 0.2" PENETRATION (CBR)	16	33	44	44
SWELL (%)	nil	nil	nil	-



MAX. DRY DENSITY (γ_d): **1.883** g /cm^3
OPTIMUM M.C. (OMC): **9.50** %

Remarks:
Nil

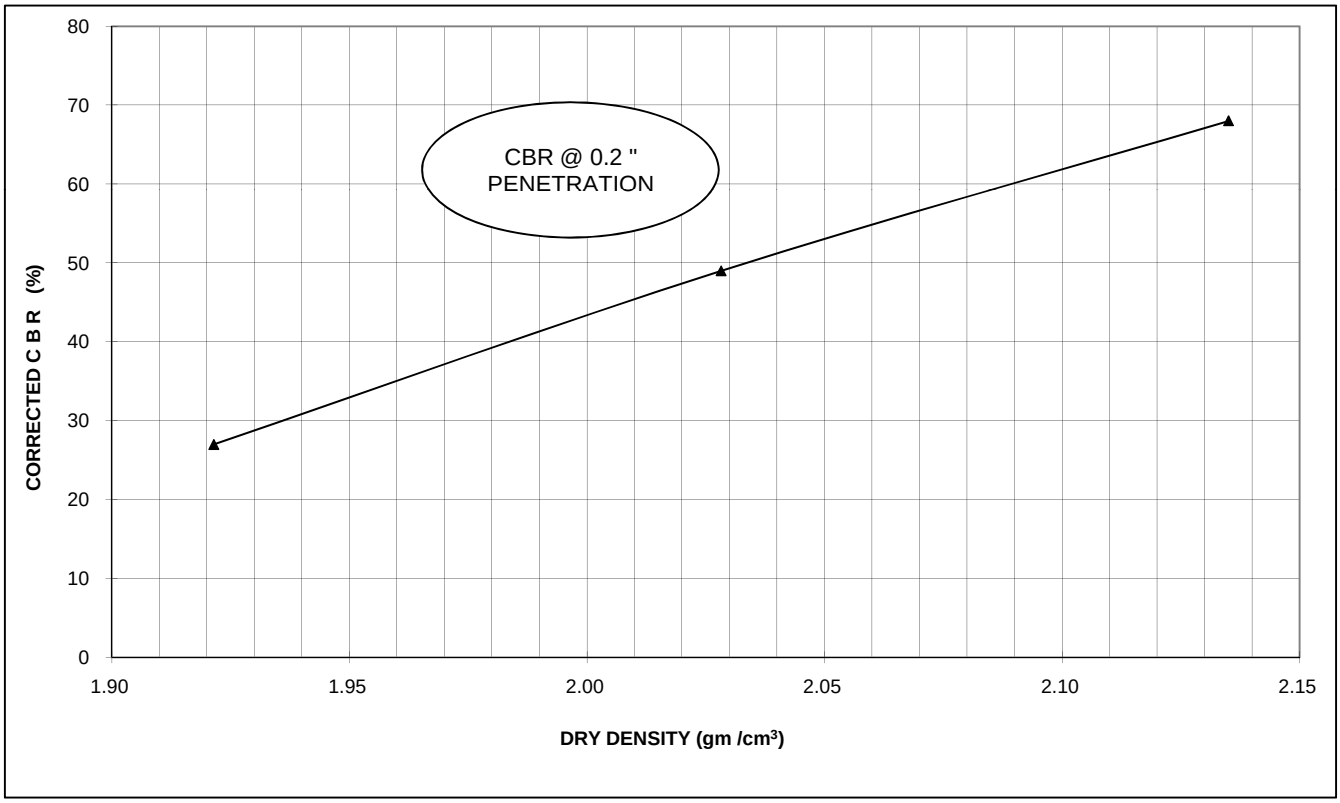
TESTED BY
WAHAB

LAB. SUPERVISOR
ANJAM

CBR - DENSITY RELATIONSHIP
ASTM D 1883

CLIENT :	L & T CONSTRUCTION	REPORT DATE:	17-Sep-19
PROJECT :	MANSOURAH MASSARAH GOLD PROJECT GEOTECHNICAL INVESTIGATION	PROJECT #:	19-K-3054
		TRIAL PIT:	TP-23
		SAMPLE :	B.S-1
SAMPLE:	BULK SAMPLE	DEPTH-(m):	0.30-0.55
		SAMPLE DATE:	12-Sep-19
TP NO.:	TP-23	TEST DATE:	16-Sep-19

COMPACTION EFFORT	10 Blows	25 Blows	56 Blows	100%MDD
DRY DENSITY (gm/cm^3)	1.922	2.028	2.135	2.135
BEARING @ 0.2" PENETRATION (CBR)	27	49	68	68
SWELL (%)	nil	nil	nil	-



MAX. DRY DENSITY (γ_d): **2.135** g /cm^3
OPTIMUM M.C. (OMC): **6.50** %

Remarks:
Nil

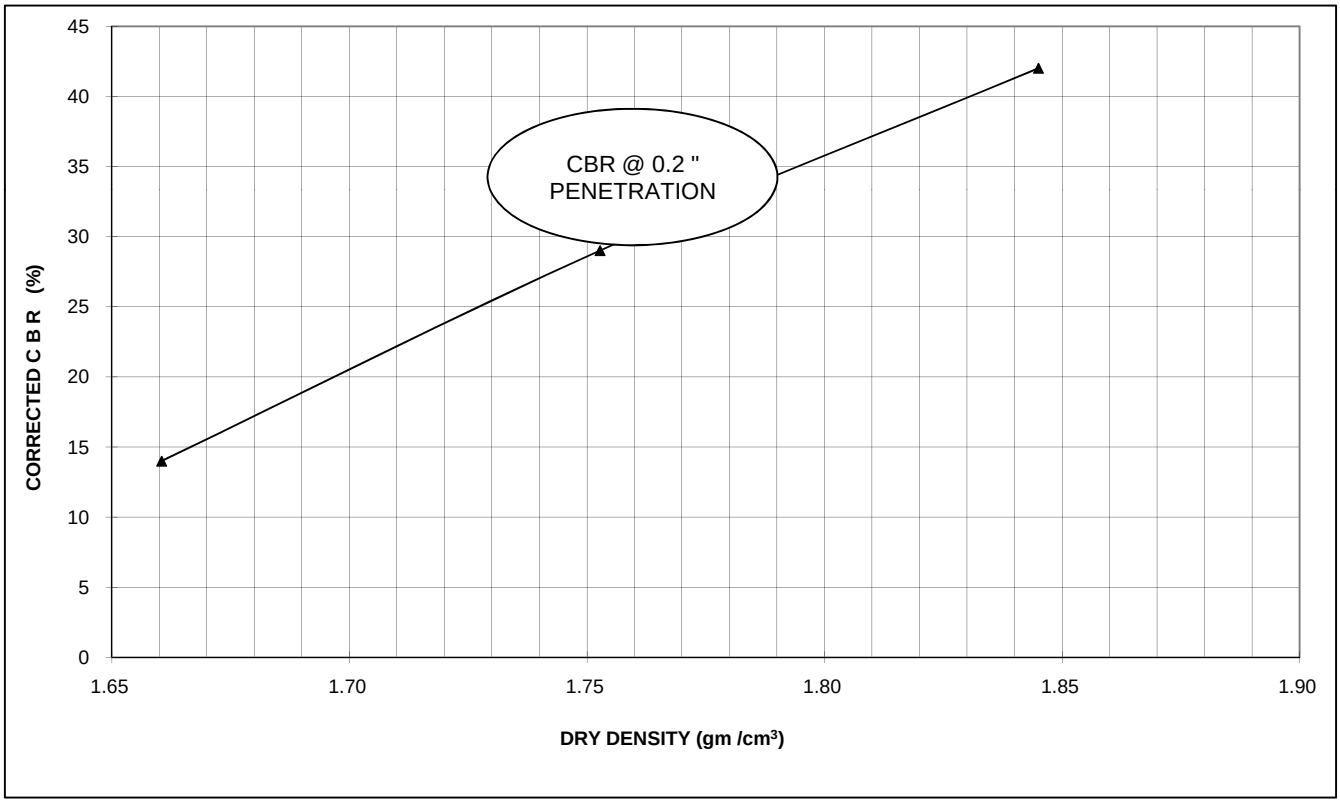
TESTED BY
WAHAB

LAB. SUPERVISOR
ANJAM

CBR - DENSITY RELATIONSHIP
ASTM D 1883

CLIENT :	L & T CONSTRUCTION	REPORT DATE:	18-Sep-19
PROJECT :	MANSOURAH MASSARAH GOLD PROJECT GEOTECHNICAL INVESTIGATION	PROJECT #:	19-K-3054
		TRIAL PIT:	TP-24
		SAMPLE :	B.S-1
SAMPLE:	BULK SAMPLE	DEPTH-(m):	0.0-0.55
		SAMPLE DATE:	13-Sep-19
TP NO.:	TP-24	TEST DATE:	17-Sep-19

COMPACTION EFFORT	10 Blows	25 Blows	56 Blows	100%MDD
DRY DENSITY (gm/cm^3)	1.661	1.753	1.845	1.845
BEARING @ 0.2" PENETRATION (CBR)	14	29	42	42
SWELL (%)	nil	nil	nil	-



MAX. DRY DENSITY (γ_d): **1.845** g /cm^3
OPTIMUM M.C. (OMC): **9.50** %

Remarks:
Nil

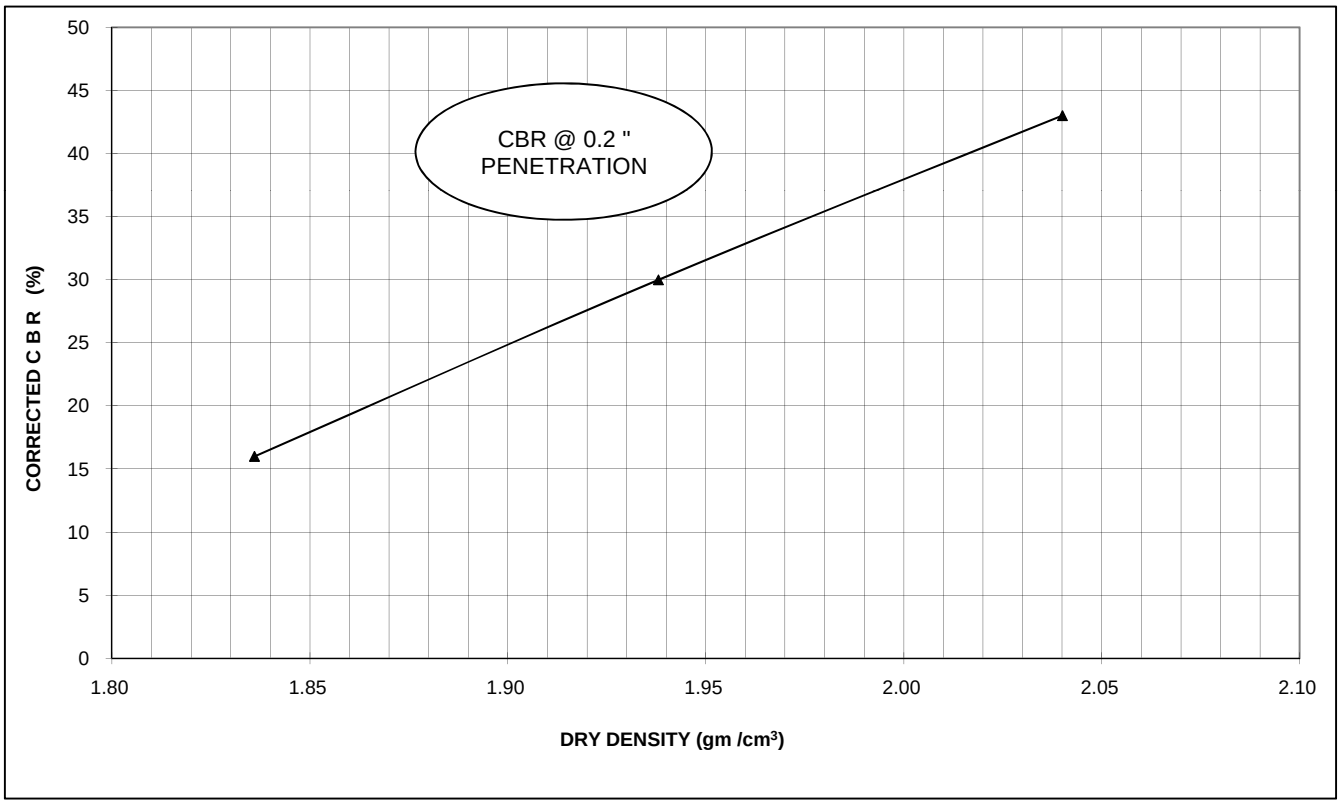
TESTED BY
WAHAB

LAB. SUPERVISOR
ANJAM

CBR - DENSITY RELATIONSHIP
ASTM D 1883

CLIENT :	L & T CONSTRUCTION	REPORT DATE:	18-Sep-19
PROJECT :	MANSOURAH MASSARAH GOLD PROJECT GEOTECHNICAL INVESTIGATION	PROJECT #:	19-K-3054
		TRIAL PIT:	TP-25
		SAMPLE :	B.S-1
SAMPLE:	BULK SAMPLE	DEPTH-(m):	0.00-0.50
		SAMPLE DATE:	13-Sep-19
TP NO.:	TP-25	TEST DATE:	17-Sep-19

COMPACTION EFFORT	10 Blows	25 Blows	56 Blows	100%MDD
DRY DENSITY (gm/cm^3)	1.836	1.938	2.040	2.040
BEARING @ 0.2" PENETRATION (CBR)	16	30	43	43
SWELL (%)	nil	nil	nil	-



MAX. DRY DENSITY (γ_d): **2.040** g /cm^3
OPTIMUM M.C. (OMC): **7.19** %

Remarks:
Nil

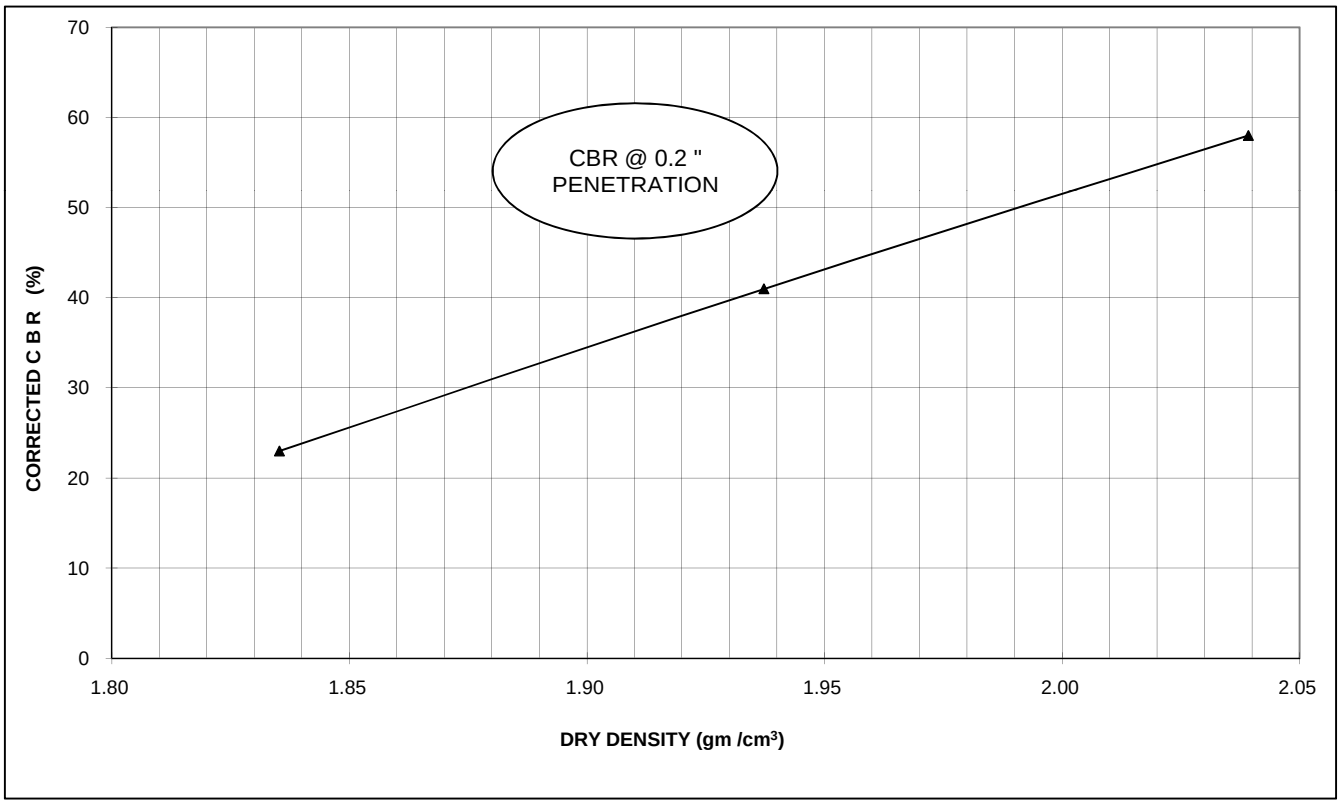
TESTED BY
WAHAB

LAB. SUPERVISOR
ANJAM

CBR - DENSITY RELATIONSHIP
ASTM D 1883

CLIENT :	L & T CONSTRUCTION	REPORT DATE:	19-Sep-19
PROJECT :	MANSOURAH MASSARAH GOLD PROJECT GEOTECHNICAL INVESTIGATION	PROJECT #:	19-K-3054
		TRIAL PIT:	TP-26
		SAMPLE :	B.S-1
SAMPLE:	BULK SAMPLE	DEPTH-(m):	0.00-0.50
		SAMPLE DATE:	14-Sep-19
TP NO.:	TP-26	TEST DATE:	18-Sep-19

COMPACTION EFFORT	10 Blows	25 Blows	56 Blows	100%MDD
DRY DENSITY (gm/cm^3)	1.835	1.937	2.039	2.039
BEARING @ 0.2" PENETRATION (CBR)	23	41	58	58
SWELL (%)	nil	nil	nil	-



MAX. DRY DENSITY (γ_d): **2.039** g /cm^3
OPTIMUM M.C. (OMC): **6.72** %

Remarks:
Nil

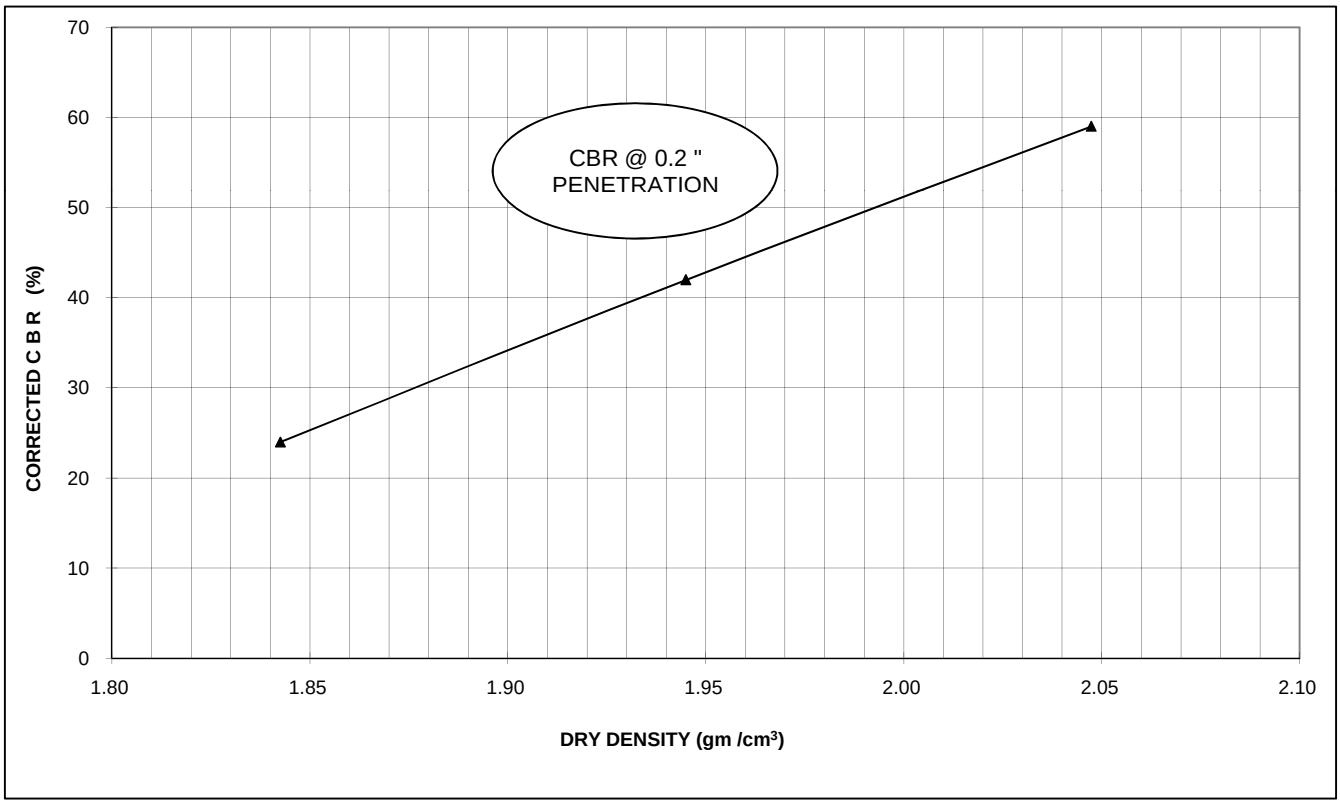
TESTED BY
WAHAB

LAB. SUPERVISOR
ANJAM

CBR - DENSITY RELATIONSHIP
ASTM D 1883

CLIENT :	L & T CONSTRUCTION	REPORT DATE:	19-Sep-19
PROJECT :	MANSOURAH MASSARAH GOLD PROJECT GEOTECHNICAL INVESTIGATION	PROJECT #:	19-K-3054
		TRIAL PIT:	TP-27
		SAMPLE :	B.S-1
SAMPLE:	BULK SAMPLE	DEPTH-(m):	0.0-0.50
		SAMPLE DATE:	14-Sep-19
TP NO.:	TP-27	TEST DATE:	18-Sep-19

COMPACTION EFFORT	10 Blows	25 Blows	56 Blows	100%MDD
DRY DENSITY (gm/cm^3)	1.843	1.945	2.047	2.047
BEARING @ 0.2" PENETRATION (CBR)	24	42	59	59
SWELL (%)	nil	nil	nil	-



MAX. DRY DENSITY (γ_d): **2.047** g /cm^3
OPTIMUM M.C. (OMC): **6.60** %

Remarks:
Nill

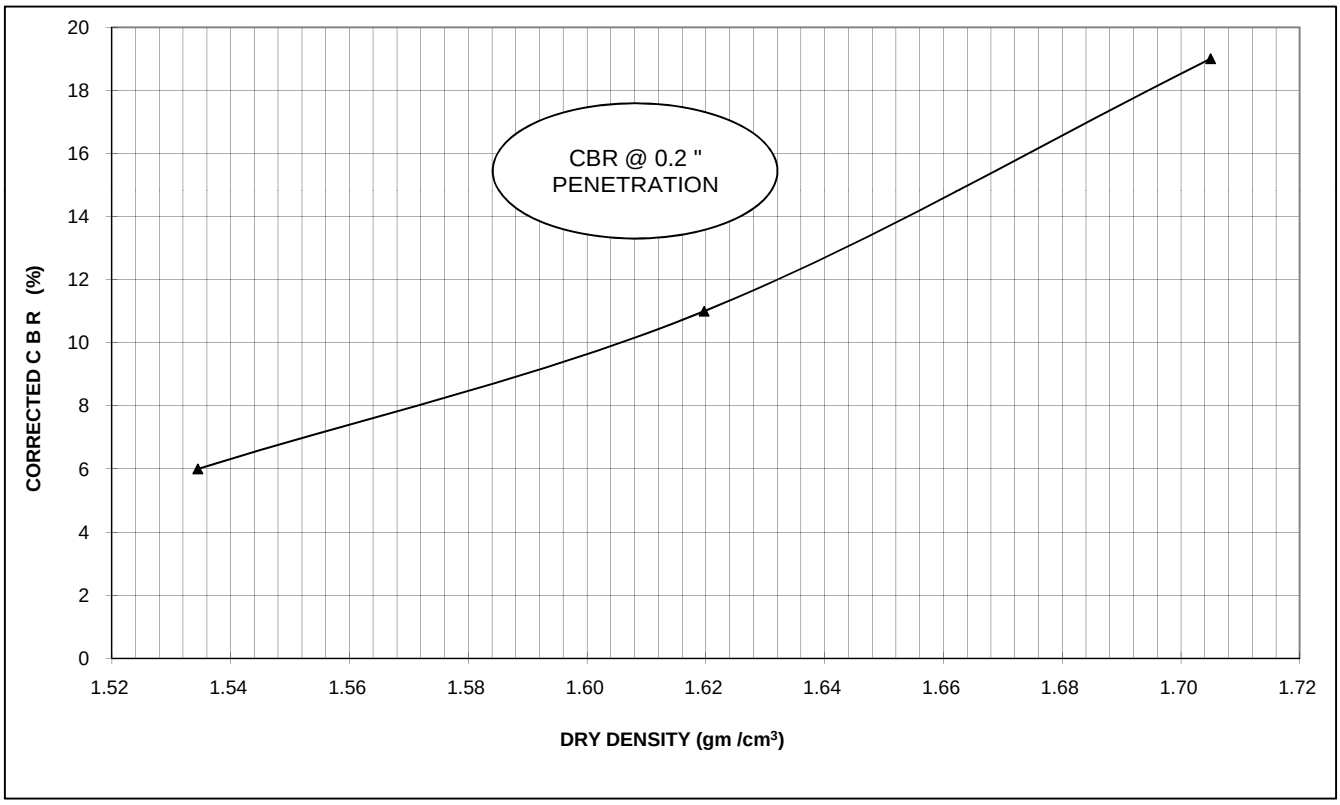
TESTED BY
WAHAB

LAB. SUPERVISOR
ANJAM

CBR - DENSITY RELATIONSHIP
ASTM D 1883

CLIENT :	L & T CONSTRUCTION	REPORT DATE:	20-Sep-19
PROJECT :	MANSOURAH MASSARAH GOLD PROJECT GEOTECHNICAL INVESTIGATION	PROJECT #:	19-K-3054
		TRIAL PIT:	TP-28
		SAMPLE :	B.S-1
SAMPLE:	BULK SAMPLE	DEPTH-(m):	0.00-0.70
		SAMPLE DATE:	15-Sep-19
TP NO.:	TP-28	TEST DATE:	19-Sep-19

COMPACTION EFFORT	10 Blows	25 Blows	56 Blows	100%MDD
DRY DENSITY (gm/cm^3)	1.535	1.620	1.705	1.705
BEARING @ 0.2" PENETRATION (CBR)	6	11	19	19
SWELL (%)	1.4	1.62	1.85	-



MAX. DRY DENSITY (γ_d): **1.705** g /cm^3
OPTIMUM M.C. (OMC): **10.50** %

Remarks:
Nil

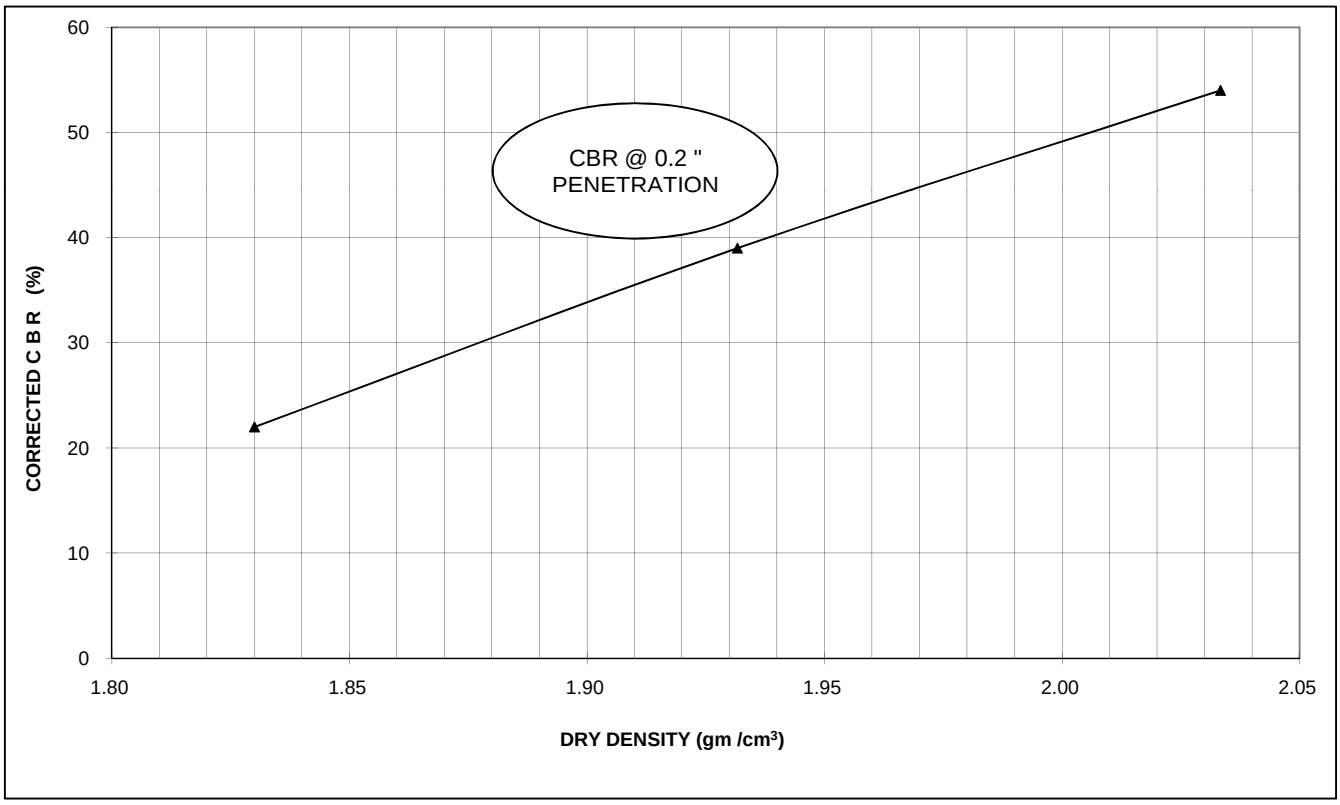
TESTED BY
WAHAB

LAB. SUPERVISOR
ANJAM

CBR - DENSITY RELATIONSHIP
ASTM D 1883

CLIENT :	L & T CONSTRUCTION	REPORT DATE:	20-Sep-19
PROJECT :	MANSOURAH MASSARAH GOLD PROJECT GEOTECHNICAL INVESTIGATION	PROJECT #:	19-K-3054
		TRIAL PIT:	TP-29
		SAMPLE :	B.S-1
SAMPLE:	BULK SAMPLE	DEPTH-(m):	0.0-1.00
		SAMPLE DATE:	15-Sep-19
TP NO.:	TP-29	TEST DATE:	19-Sep-19

COMPACTION EFFORT	10 Blows	25 Blows	56 Blows	100%MDD
DRY DENSITY (gm/cm^3)	1.830	1.932	2.033	2.033
BEARING @ 0.2" PENETRATION (CBR)	22	39	54	54
SWELL (%)	nil	nil	nil	-



MAX. DRY DENSITY (γ_d): **2.033** g /cm^3
OPTIMUM M.C. (OMC): **6.71** %

Remarks:
Nil

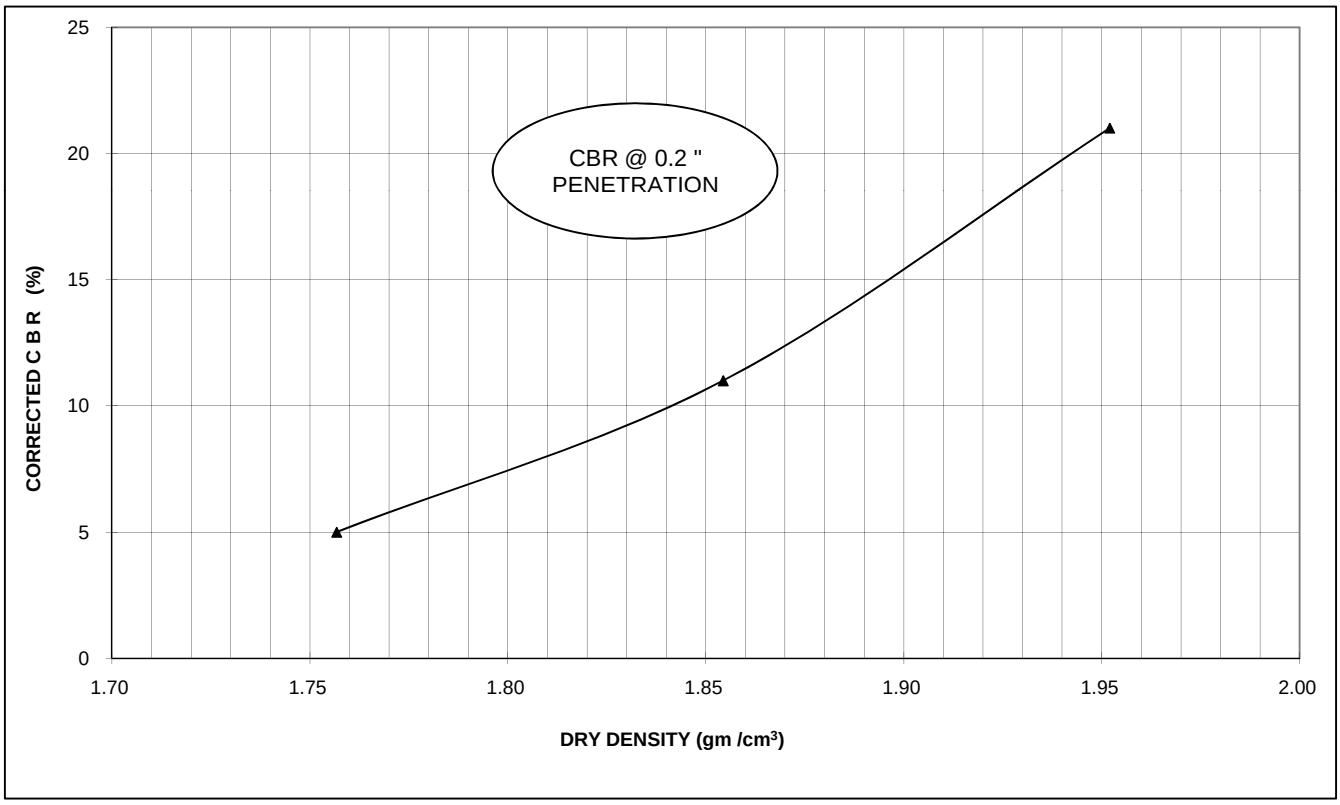
TESTED BY
WAHAB

LAB. SUPERVISOR
ANJAM

CBR - DENSITY RELATIONSHIP
ASTM D 1883

CLIENT :	L & T CONSTRUCTION	REPORT DATE:	21-Sep-19
PROJECT :	MANSOURAH MASSARAH GOLD PROJECT GEOTECHNICAL INVESTIGATION	PROJECT #:	19-K-3054
		TRIAL PIT:	TP-30
		SAMPLE :	B.S-1
SAMPLE:	BULK SAMPLE	DEPTH-(m):	0.00-0.40
		SAMPLE DATE:	16-Sep-19
TP NO.:	TP-30	TEST DATE:	20-Sep-19

COMPACTION EFFORT	10 Blows	25 Blows	56 Blows	100%MDD
DRY DENSITY (gm/cm^3)	1.757	1.854	1.952	1.952
BEARING @ 0.2" PENETRATION (CBR)	5	11	21	21
SWELL (%)	1.4	2.1	2.5	-



MAX. DRY DENSITY (γ_d): **1.952** g /cm^3
OPTIMUM M.C. (OMC): **9.50** %

Remarks:
Nil

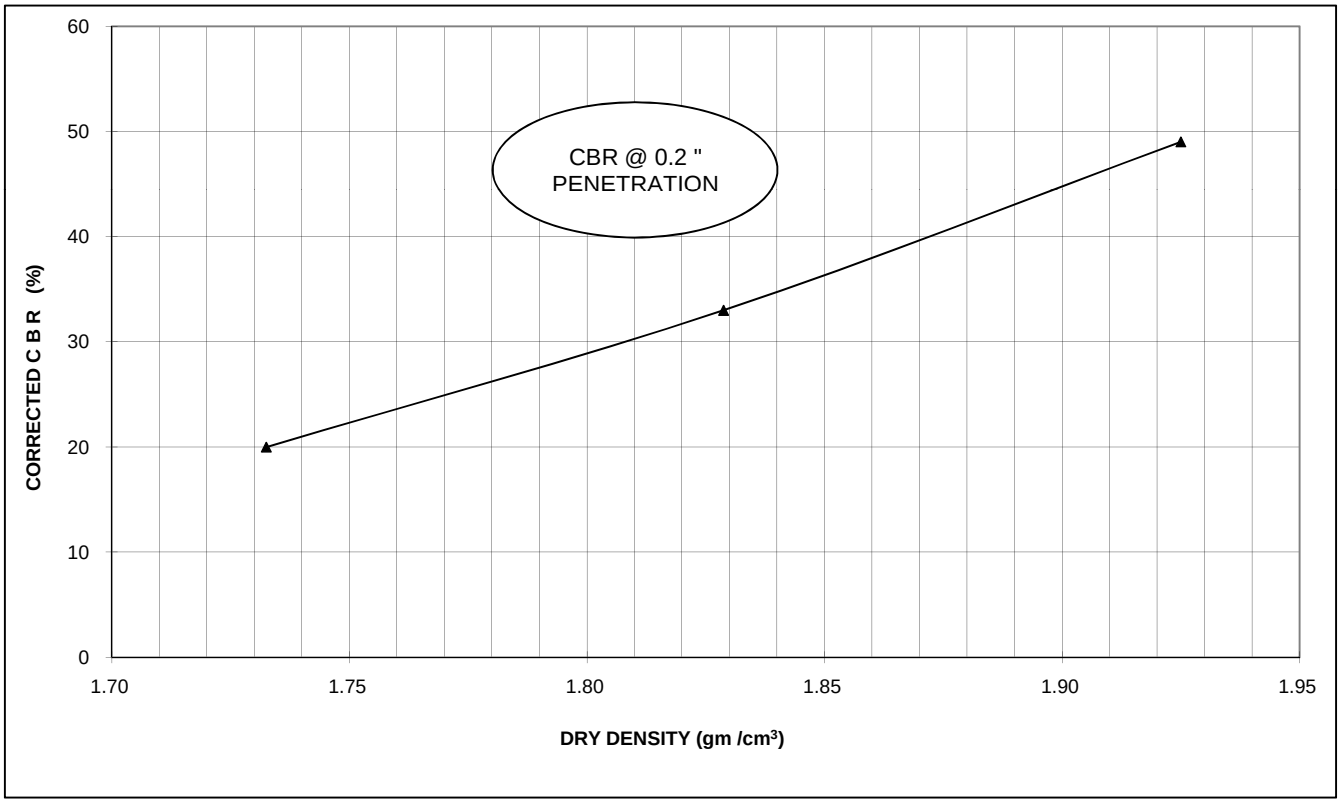
TESTED BY
WAHAB

LAB. SUPERVISOR
ANJAM

CBR - DENSITY RELATIONSHIP
ASTM D 1883

CLIENT :	L & T CONSTRUCTION	REPORT DATE:	21-Sep-19
PROJECT :	MANSOURAH MASSARAH GOLD PROJECT GEOTECHNICAL INVESTIGATION	PROJECT #:	19-K-3054
		TRIAL PIT:	TP-31
		SAMPLE :	B.S-1
SAMPLE:	BULK SAMPLE	DEPTH-(m):	0.0-0.41
		SAMPLE DATE:	16-Sep-19
TP NO.:	TP-31	TEST DATE:	20-Sep-19

COMPACTION EFFORT	10 Blows	25 Blows	56 Blows	100%MDD
DRY DENSITY (gm/cm^3)	1.733	1.829	1.925	1.925
BEARING @ 0.2" PENETRATION (CBR)	20	33	49	49
SWELL (%)	nil	nil	nil	-



MAX. DRY DENSITY (γ_d): **1.925** g /cm^3
OPTIMUM M.C. (OMC): **9.20** %

Remarks:
Nil

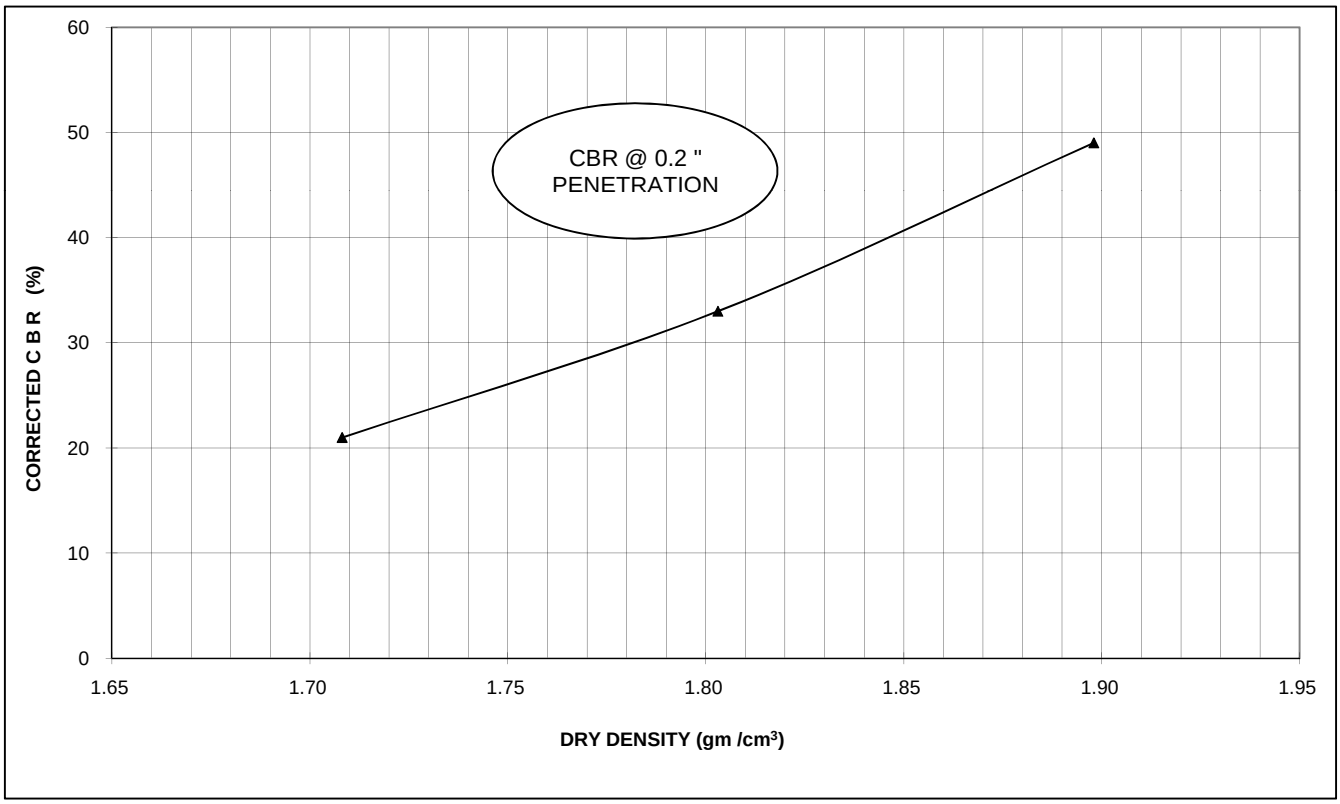
TESTED BY
WAHAB

LAB. SUPERVISOR
ANJAM

CBR - DENSITY RELATIONSHIP
ASTM D 1883

CLIENT :	L & T CONSTRUCTION	REPORT DATE:	22-Sep-19
PROJECT :	MANSOURAH MASSARAH GOLD PROJECT GEOTECHNICAL INVESTIGATION	PROJECT #:	19-K-3054
		TRIAL PIT:	TP-32
		SAMPLE :	B.S-1
SAMPLE:	BULK SAMPLE	DEPTH-(m):	0.00-0.40
		SAMPLE DATE:	17-Sep-19
TP NO.:	TP-32	TEST DATE:	21-Sep-19

COMPACTION EFFORT	10 Blows	25 Blows	56 Blows	100%MDD
DRY DENSITY (gm/cm^3)	1.708	1.803	1.898	1.898
BEARING @ 0.2" PENETRATION (CBR)	21	33	49	49
SWELL (%)	nil	nil	nil	-



MAX. DRY DENSITY (γ_d): **1.898** g /cm^3
OPTIMUM M.C. (OMC): **9.20** %

Remarks:
Nil

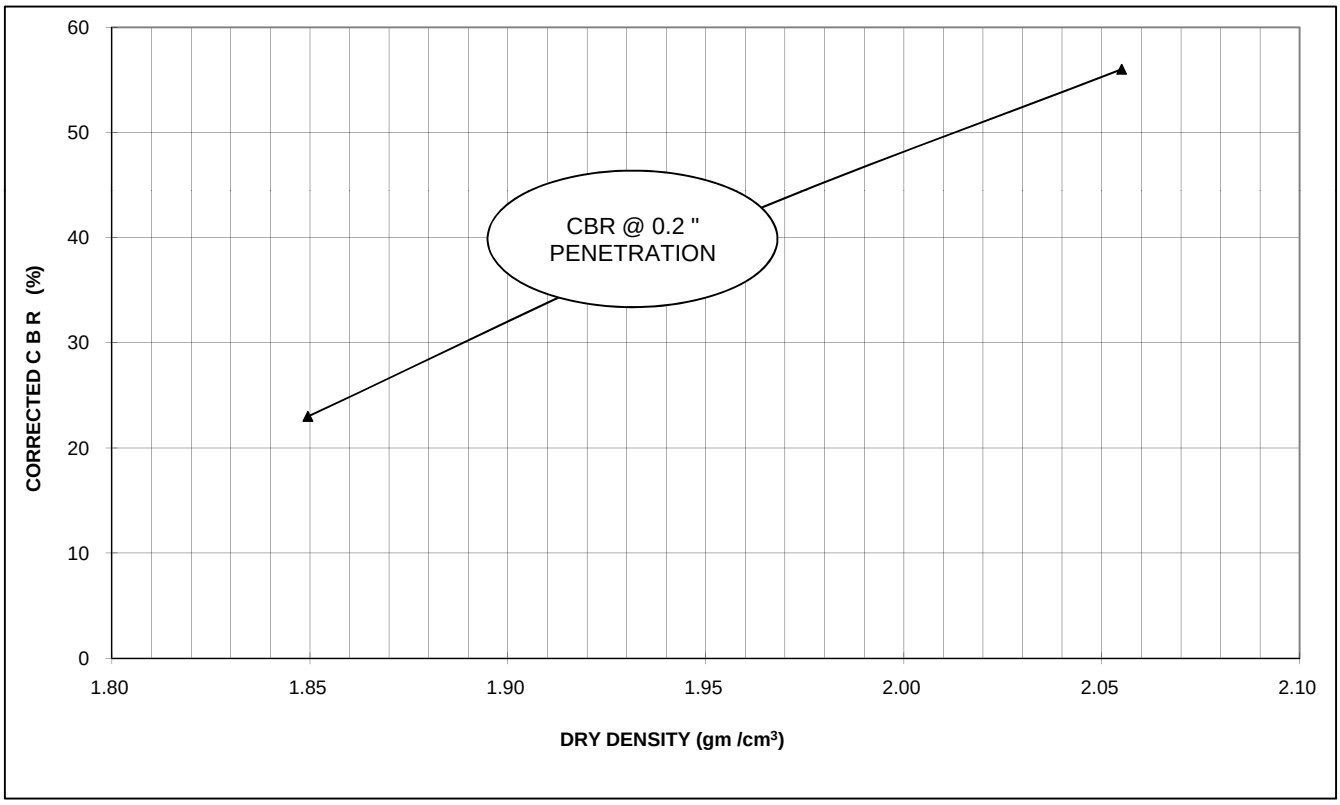
TESTED BY
WAHAB

LAB. SUPERVISOR
ANJAM

CBR - DENSITY RELATIONSHIP
ASTM D 1883

CLIENT :	L & T CONSTRUCTION	REPORT DATE:	22-Sep-19
PROJECT :	MANSOURAH MASSARAH GOLD PROJECT GEOTECHNICAL INVESTIGATION	PROJECT #:	19-K-3054
		TRIAL PIT:	TP-33
		SAMPLE :	B.S-1
SAMPLE:	BULK SAMPLE	DEPTH-(m):	0.00-0.40
		SAMPLE DATE:	17-Sep-19
TP NO.:	TP-33	TEST DATE:	21-Sep-19

COMPACTION EFFORT	10 Blows	25 Blows	56 Blows	100%MDD
DRY DENSITY (gm/cm^3)	1.850	1.952	2.055	2.055
BEARING @ 0.2" PENETRATION (CBR)	23	41	56	56
SWELL (%)	nil	nil	nil	-



MAX. DRY DENSITY (γ_d): **2.055** g /cm^3
OPTIMUM M.C. (OMC): **7.33** %

Remarks:
Nil

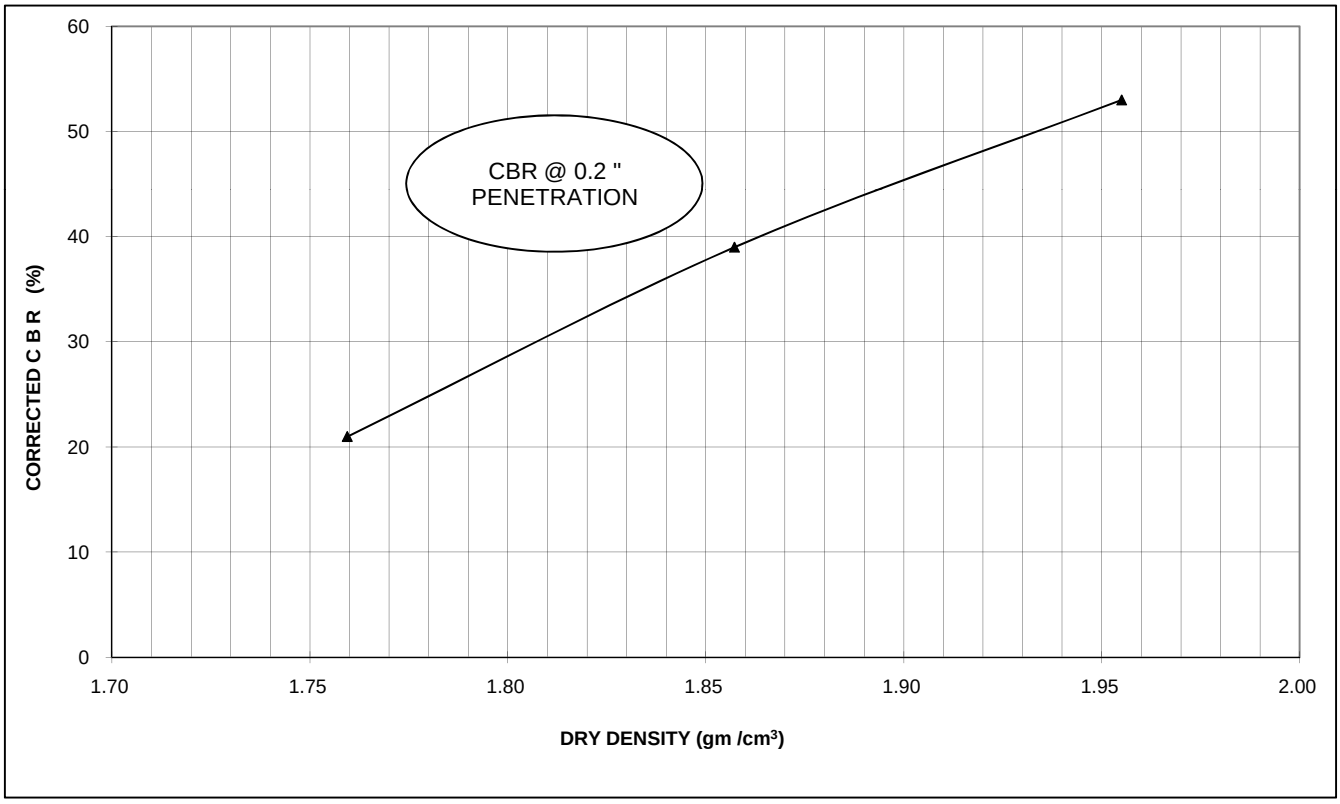
TESTED BY
WAHAB

LAB. SUPERVISOR
ANJAM

CBR - DENSITY RELATIONSHIP
ASTM D 1883

CLIENT :	L & T CONSTRUCTION	REPORT DATE:	23-Sep-19
PROJECT :	MANSOURAH MASSARAH GOLD PROJECT GEOTECHNICAL INVESTIGATION	PROJECT #:	19-K-3054
		TRIAL PIT:	TP-34
		SAMPLE :	B.S-1
SAMPLE:	BULK SAMPLE	DEPTH-(m):	0.40-0.70
		SAMPLE DATE:	18-Sep-19
TP NO.:	TP-34	TEST DATE:	22-Sep-19

COMPACTION EFFORT	10 Blows	25 Blows	56 Blows	100%MDD
DRY DENSITY (gm/cm^3)	1.760	1.857	1.955	1.955
BEARING @ 0.2" PENETRATION (CBR)	21	39	53	53
SWELL (%)	nil	nil	nil	-



MAX. DRY DENSITY (γ_d): **1.955** g /cm^3
OPTIMUM M.C. (OMC): **8.50** %

Remarks:
Nil

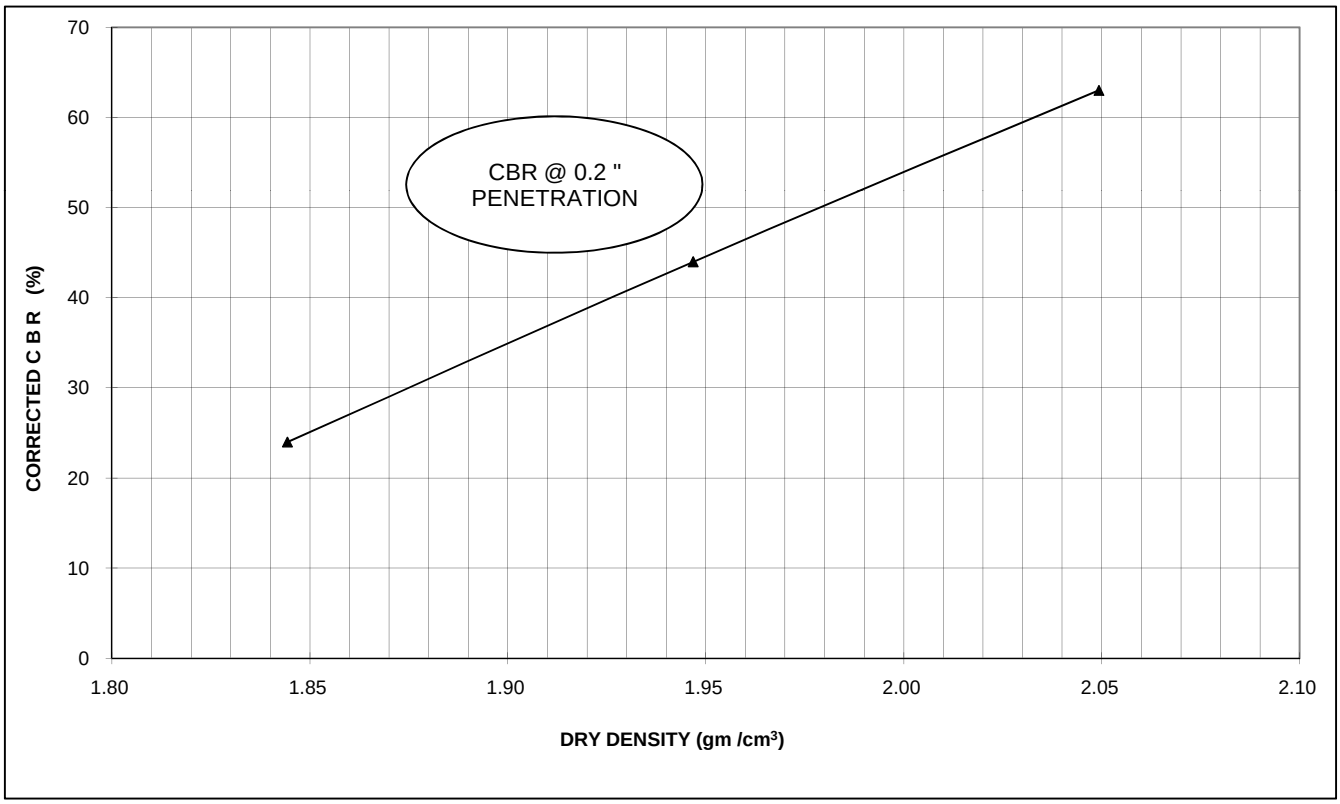
TESTED BY
WAHAB

LAB. SUPERVISOR
ANJAM

CBR - DENSITY RELATIONSHIP
ASTM D 1883

CLIENT :	L & T CONSTRUCTION	REPORT DATE:	23-Sep-19
PROJECT :	MANSOURAH MASSARAH GOLD PROJECT GEOTECHNICAL INVESTIGATION	PROJECT #:	19-K-3054
		TRIAL PIT:	TP-35
		SAMPLE :	B.S-1
SAMPLE:	BULK SAMPLE	DEPTH-(m):	0.0-0.60
		SAMPLE DATE:	18-Sep-19
TP NO.:	TP-35	TEST DATE:	22-Sep-19

COMPACTION EFFORT	10 Blows	25 Blows	56 Blows	100%MDD
DRY DENSITY (gm/cm^3)	1.844	1.947	2.049	2.049
BEARING @ 0.2" PENETRATION (CBR)	24	44	63	63
SWELL (%)	nil	35	nil	-



MAX. DRY DENSITY (γ_d): **2.049** g /cm^3
OPTIMUM M.C. (OMC): **7.22** %

Remarks:
Nil

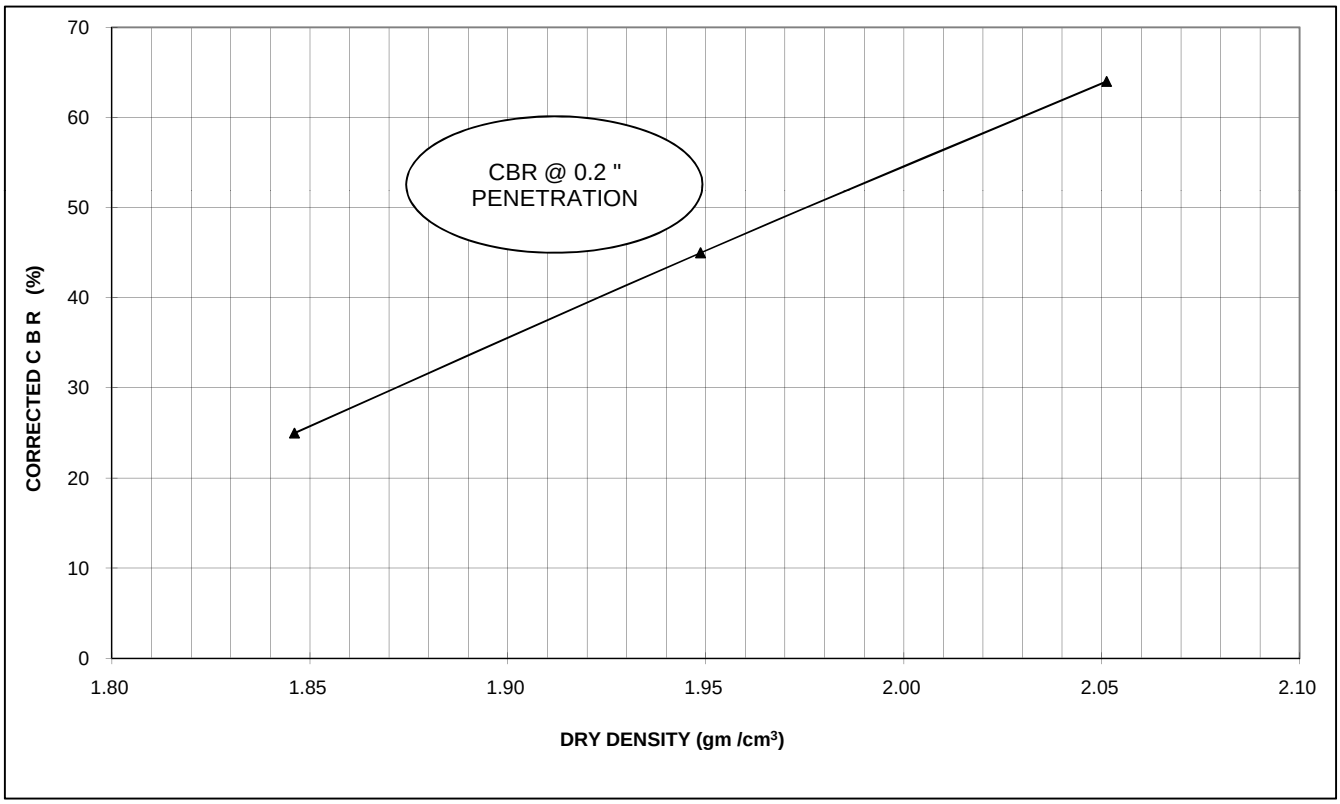
TESTED BY
WAHAB

LAB. SUPERVISOR
ANJAM

CBR - DENSITY RELATIONSHIP
ASTM D 1883

CLIENT :	L & T CONSTRUCTION	REPORT DATE:	24-Sep-19
PROJECT :	MANSOURAH MASSARAH GOLD PROJECT GEOTECHNICAL INVESTIGATION	PROJECT #:	19-K-3054
		TRIAL PIT:	TP-36
		SAMPLE :	B.S-1
SAMPLE:	BULK SAMPLE	DEPTH-(m):	0.0-0.45
		SAMPLE DATE:	19-Sep-19
TP NO.:	TP-36	TEST DATE:	23-Sep-19

COMPACTION EFFORT	10 Blows	25 Blows	56 Blows	100%MDD
DRY DENSITY (gm/cm^3)	1.846	1.949	2.051	2.051
BEARING @ 0.2" PENETRATION (CBR)	25	45	64	64
SWELL (%)	nil	35	nil	-



MAX. DRY DENSITY (γ_d): **2.051** g /cm^3
OPTIMUM M.C. (OMC): **7.34** %

Remarks:
Nil

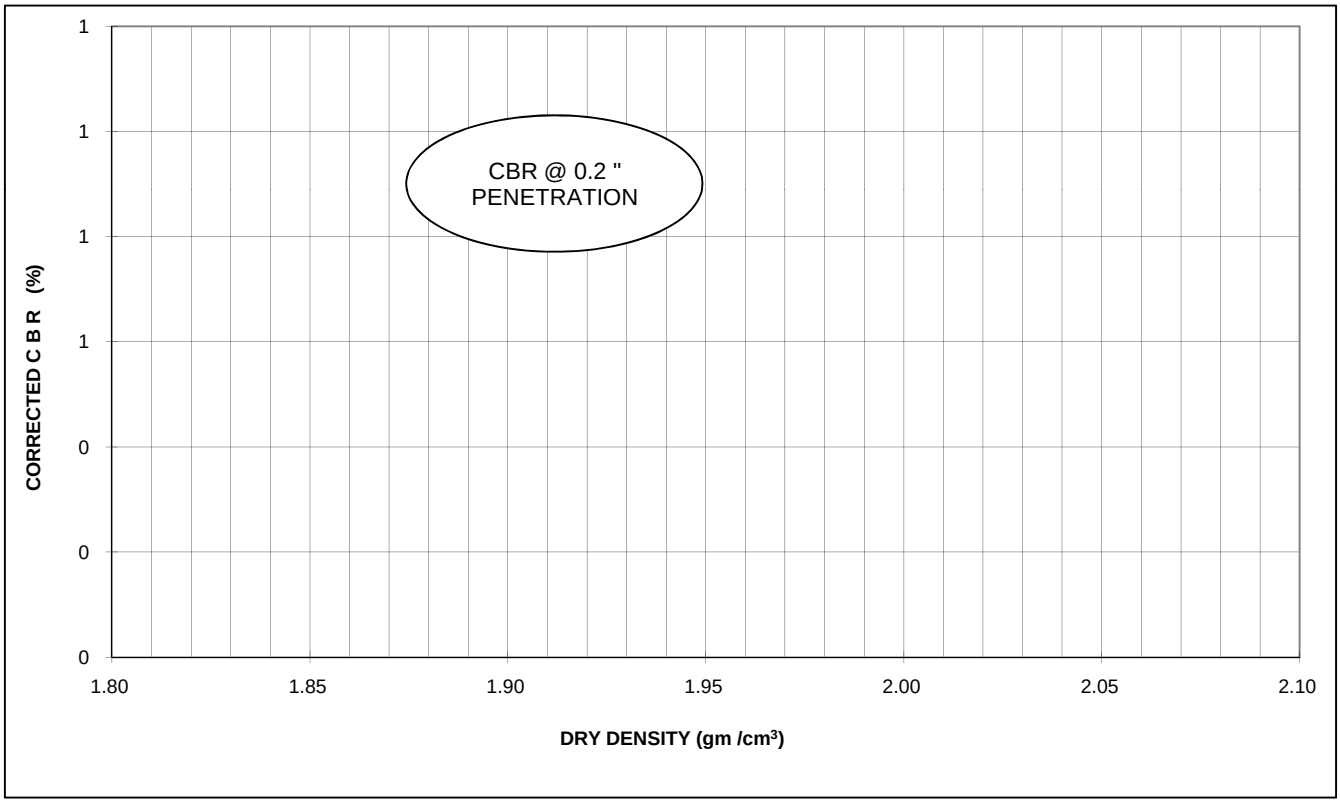
TESTED BY
WAHAB

LAB. SUPERVISOR
ANJAM

CBR - DENSITY RELATIONSHIP
ASTM D 1883

CLIENT :	L & T CONSTRUCTION	REPORT DATE:	24-Sep-19
PROJECT :	MANSOURAH MASSARAH GOLD PROJECT GEOTECHNICAL INVESTIGATION	PROJECT #:	19-K-3054
		TRIAL PIT:	TP-41
		SAMPLE :	B.S-1
SAMPLE:	BULK SAMPLE	DEPTH-(m):	0.0-0.51
		SAMPLE DATE:	19-Sep-19
TP NO.:	TP-41	TEST DATE:	23-Sep-19

COMPACTION EFFORT	10 Blows	25 Blows	56 Blows	100%MDD
DRY DENSITY (gm/cm^3)	1.843	1.946	2.048	2.048
BEARING @ 0.2" PENETRATION (CBR)				0
SWELL (%)	nil	35	nil	-



MAX. DRY DENSITY (γ_d): **2.048** g /cm^3
OPTIMUM M.C. (OMC): **7.47** %

Remarks:
Nil

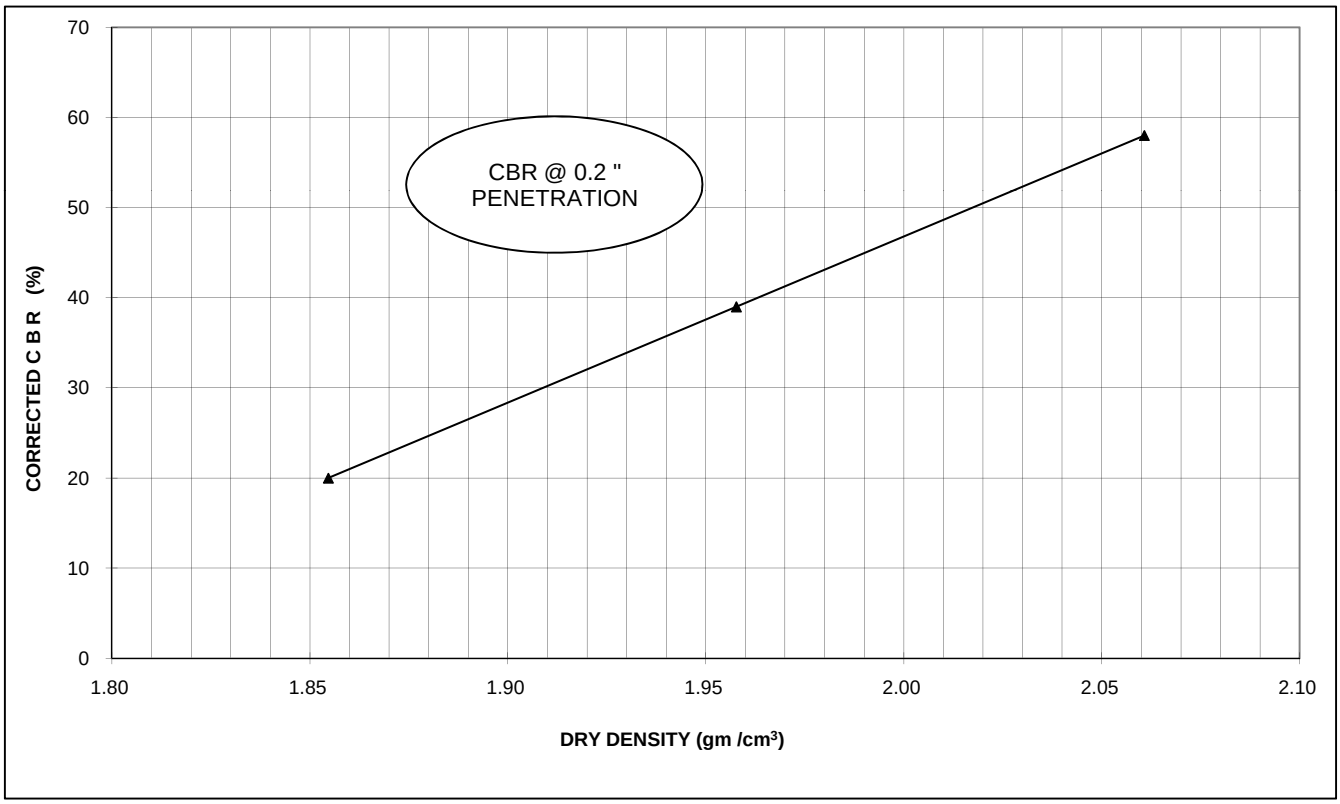
TESTED BY
WAHAB

LAB. SUPERVISOR
ANJAM

CBR - DENSITY RELATIONSHIP
ASTM D 1883

CLIENT :	L & T CONSTRUCTION	REPORT DATE:	25-Sep-19
PROJECT :	MANSOURAH MASSARAH GOLD PROJECT GEOTECHNICAL INVESTIGATION	PROJECT #:	19-K-3054
		TRIAL PIT:	TP-42
		SAMPLE :	B.S-1
SAMPLE:	BULK SAMPLE	DEPTH-(m):	0.0-0.60
		SAMPLE DATE:	20-Sep-19
TP NO.:	TP-42	TEST DATE:	24-Sep-19

COMPACTION EFFORT	10 Blows	25 Blows	56 Blows	100%MDD
DRY DENSITY (gm/cm^3)	1.855	1.958	2.061	2.061
BEARING @ 0.2" PENETRATION (CBR)	20	39	58	58
SWELL (%)	nil	35	nil	-



MAX. DRY DENSITY (γ_d): **2.061** g /cm^3
OPTIMUM M.C. (OMC): **7.12** %

Remarks:
Nil

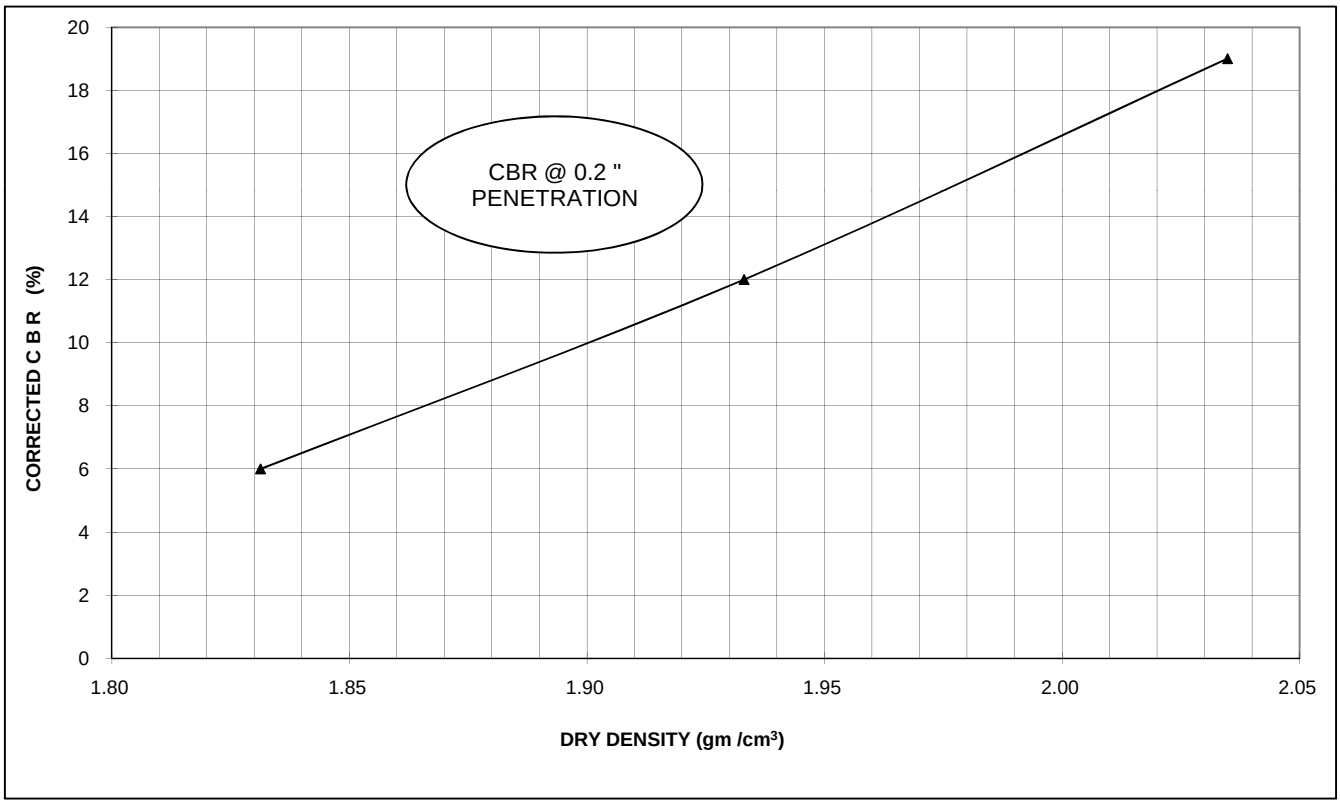
TESTED BY
WAHAB

LAB. SUPERVISOR
ANJAM

CBR - DENSITY RELATIONSHIP
ASTM D 1883

CLIENT :	L & T CONSTRUCTION	REPORT DATE:	25-Sep-19
PROJECT :	MANSOURAH MASSARAH GOLD PROJECT GEOTECHNICAL INVESTIGATION	PROJECT #:	19-K-3054
		TRIAL PIT:	TP-43
		SAMPLE :	B.S-1
SAMPLE:	BULK SAMPLE	DEPTH-(m):	0.2-0.75
		SAMPLE DATE:	20-Sep-19
TP NO.:	TP-43	TEST DATE:	24-Sep-19

COMPACTION EFFORT	10 Blows	25 Blows	56 Blows	100%MDD
DRY DENSITY (gm/cm^3)	1.831	1.933	2.035	2.035
BEARING @ 0.2" PENETRATION (CBR)	6	12	19	19
SWELL (%)	1.1	1.2	1.4	-



MAX. DRY DENSITY (γ_d): **2.035** g /cm^3
OPTIMUM M.C. (OMC): **7.50** %

Remarks:
Nil

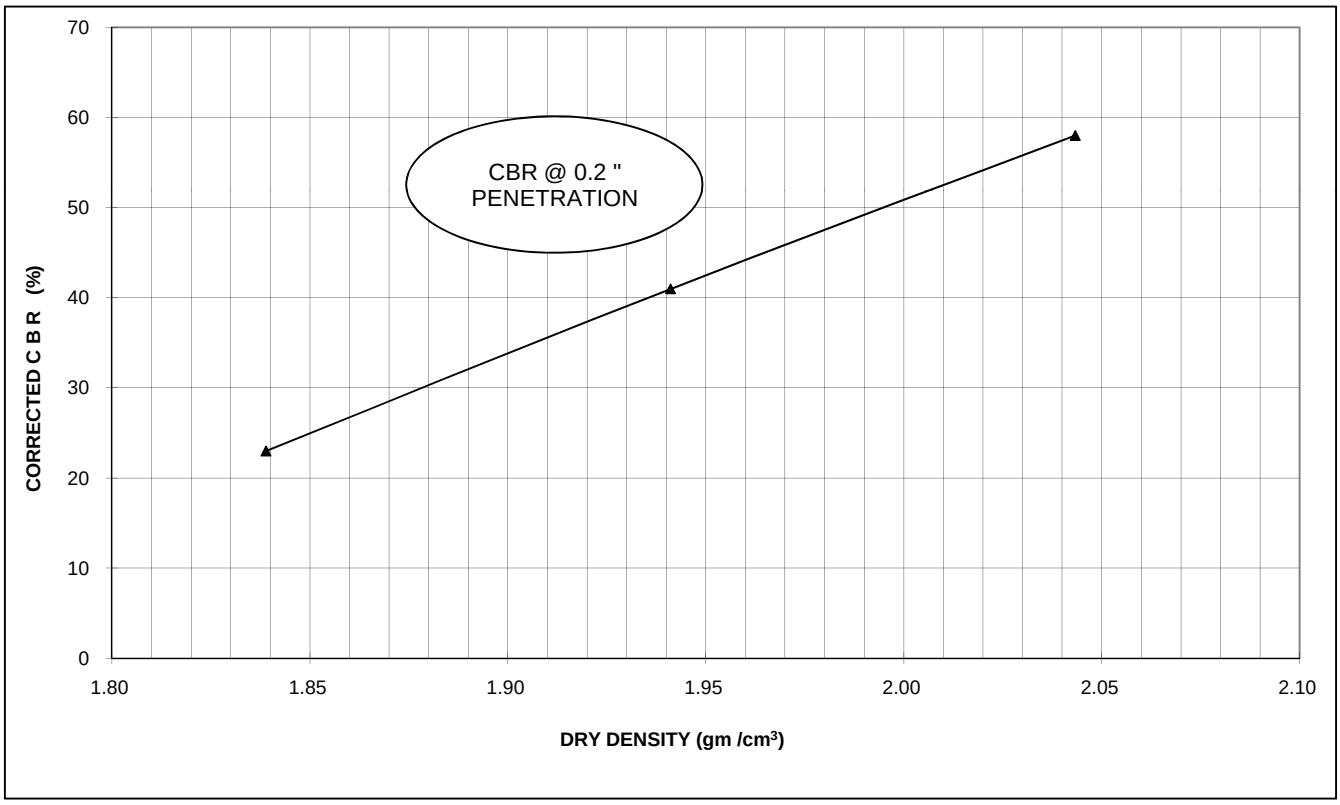
TESTED BY
WAHAB

LAB. SUPERVISOR
ANJAM

CBR - DENSITY RELATIONSHIP
ASTM D 1883

CLIENT :	L & T CONSTRUCTION	REPORT DATE:	26-Sep-19
PROJECT :	MANSOURAH MASSARAH GOLD PROJECT GEOTECHNICAL INVESTIGATION	PROJECT #:	19-K-3054
		TRIAL PIT:	TP-44
		SAMPLE :	B.S-1
SAMPLE:	BULK SAMPLE	DEPTH-(m):	0.0-0.60
		SAMPLE DATE:	21-Sep-19
TP NO.:	TP-44	TEST DATE:	25-Sep-19

COMPACTION EFFORT	10 Blows	25 Blows	56 Blows	100%MDD
DRY DENSITY (gm/cm^3)	1.839	1.941	2.043	2.043
BEARING @ 0.2" PENETRATION (CBR)	23	41	58	58
SWELL (%)	nil	35	nil	-



MAX. DRY DENSITY (γ_d): **2.043** g /cm^3
OPTIMUM M.C. (OMC): **7.52** %

Remarks:
Nil

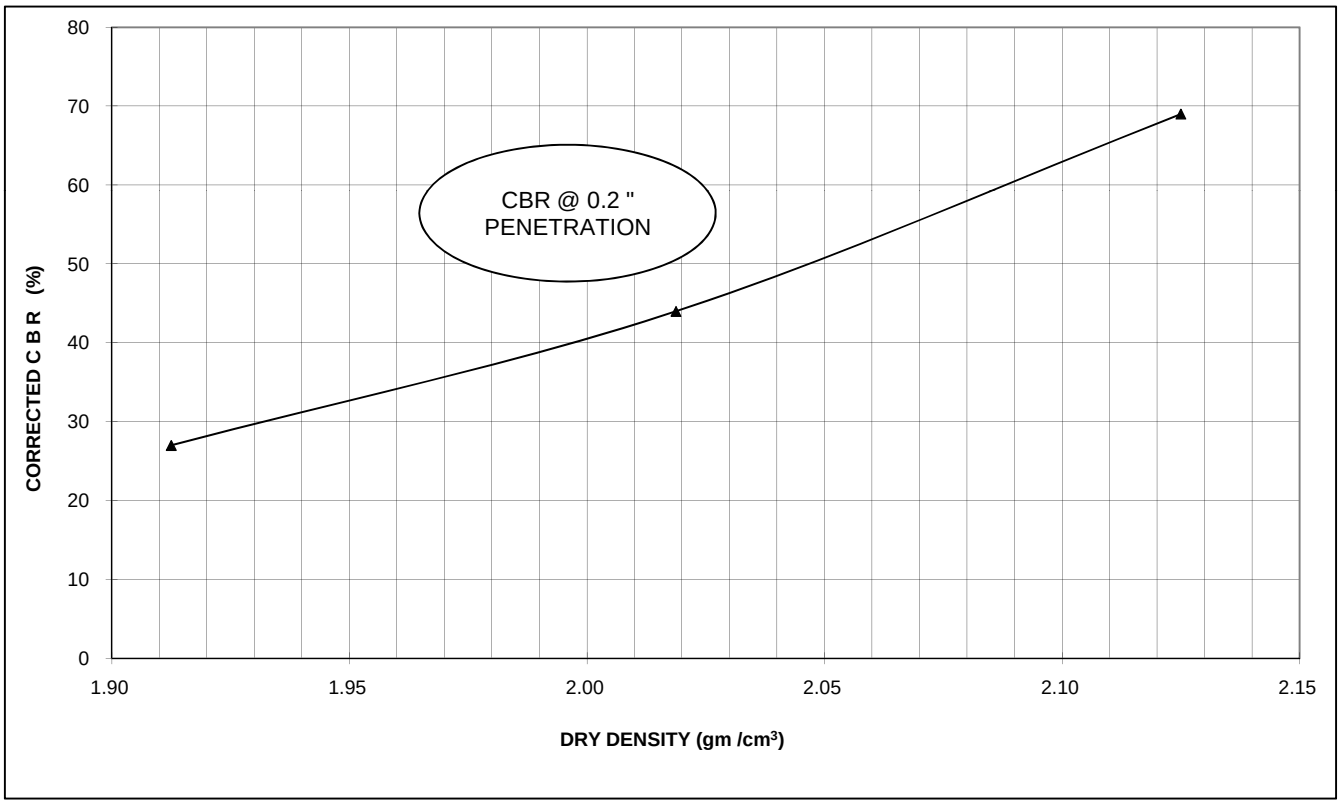
TESTED BY
WAHAB

LAB. SUPERVISOR
ANJAM

CBR - DENSITY RELATIONSHIP
ASTM D 1883

CLIENT :	L & T CONSTRUCTION	REPORT DATE:	26-Sep-19
PROJECT :	MANSOURAH MASSARAH GOLD PROJECT GEOTECHNICAL INVESTIGATION	PROJECT #:	19-K-3054
		TRIAL PIT:	TP-45
		SAMPLE :	B.S-1
SAMPLE:	BULK SAMPLE	DEPTH-(m):	0.3-0.65
		SAMPLE DATE:	21-Sep-19
TP NO.:	TP-45	TEST DATE:	25-Sep-19

COMPACTION EFFORT	10 Blows	25 Blows	56 Blows	100%MDD
DRY DENSITY (gm/cm^3)	1.913	2.019	2.125	2.125
BEARING @ 0.2" PENETRATION (CBR)	27	44	69	69
SWELL (%)	nil	35	nil	-



MAX. DRY DENSITY (γ_d): **2.125** g /cm^3
OPTIMUM M.C. (OMC): **6.50** %

Remarks:
Nill

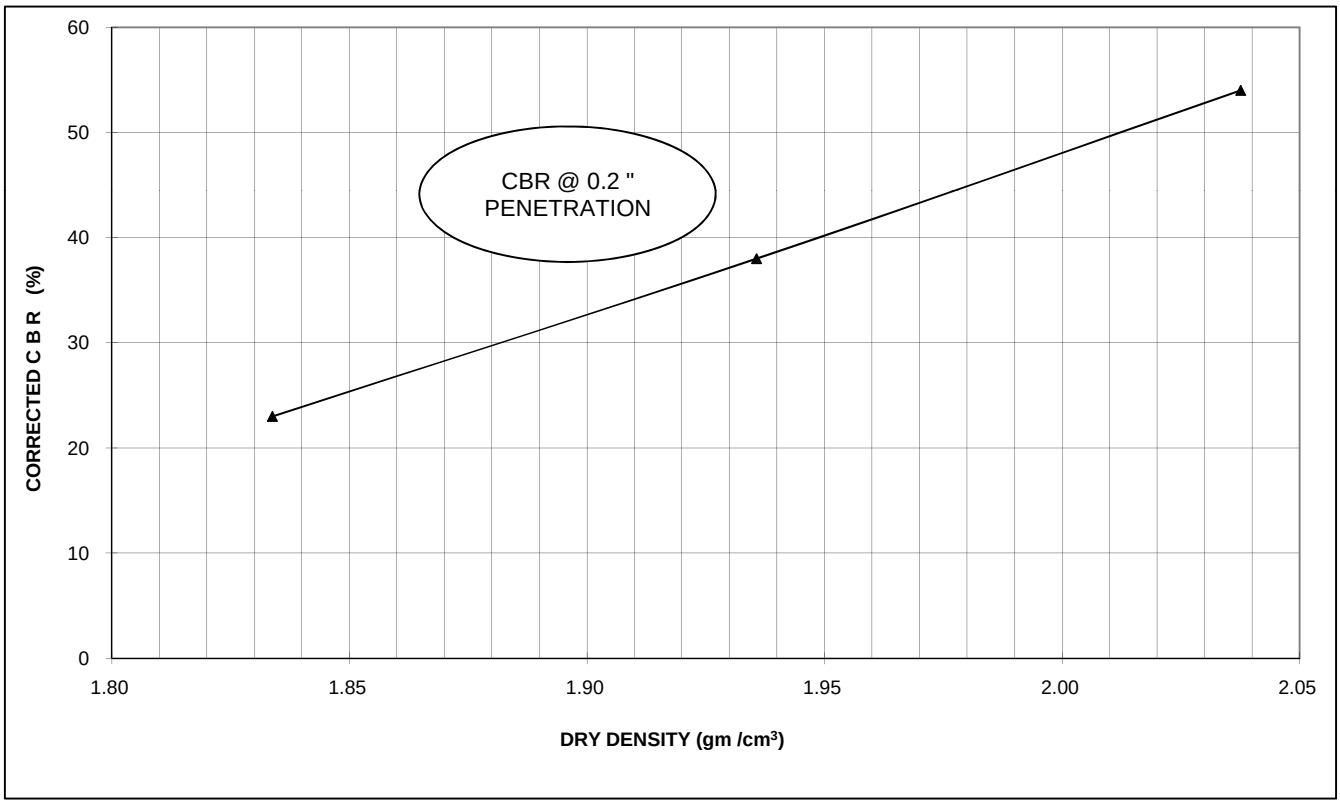
TESTED BY
WAHAB

LAB. SUPERVISOR
ANJAM

CBR - DENSITY RELATIONSHIP
ASTM D 1883

CLIENT :	L & T CONSTRUCTION	REPORT DATE:	27-Sep-19
PROJECT :	MANSOURAH MASSARAH GOLD PROJECT GEOTECHNICAL INVESTIGATION	PROJECT #:	19-K-3054
		TRIAL PIT:	TP-46
		SAMPLE :	B.S-1
SAMPLE:	BULK SAMPLE	DEPTH-(m):	0.0-0.60
		SAMPLE DATE:	22-Sep-19
TP NO.:	TP-46	TEST DATE:	26-Sep-19

COMPACTION EFFORT	10 Blows	25 Blows	56 Blows	100%MDD
DRY DENSITY (gm/cm^3)	1.834	1.936	2.038	2.038
BEARING @ 0.2" PENETRATION (CBR)	23	38	54	54
SWELL (%)	nil	35	nil	-



MAX. DRY DENSITY (γ_d): **2.038** g /cm^3
OPTIMUM M.C. (OMC): **7.48** %

Remarks:
Nil

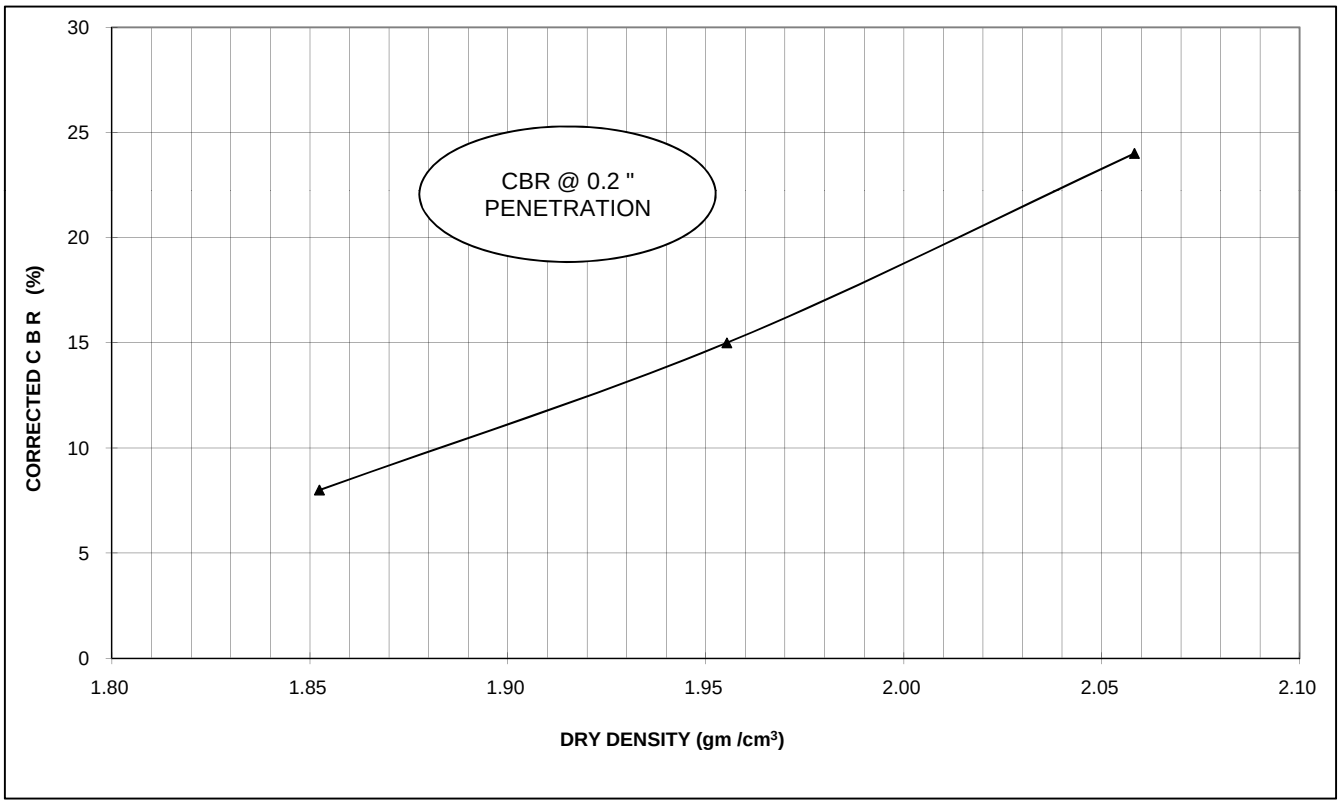
TESTED BY
WAHAB

LAB. SUPERVISOR
ANJAM

CBR - DENSITY RELATIONSHIP
ASTM D 1883

CLIENT :	L & T CONSTRUCTION	REPORT DATE:	27-Sep-19
PROJECT :	MANSOURAH MASSARAH GOLD PROJECT GEOTECHNICAL INVESTIGATION	PROJECT #:	19-K-3054
		TRIAL PIT:	TP-47
		SAMPLE :	B.S-1
SAMPLE:	BULK SAMPLE	DEPTH-(m):	0.0-0.45
		SAMPLE DATE:	22-Sep-19
TP NO.:	TP-47	TEST DATE:	26-Sep-19

COMPACTION EFFORT	10 Blows	25 Blows	56 Blows	100%MDD
DRY DENSITY (gm/cm^3)	1.852	1.955	2.058	2.058
BEARING @ 0.2" PENETRATION (CBR)	8	15	24	24
SWELL (%)	1.1	1.2	0.95	-



MAX. DRY DENSITY (γ_d): **2.058** g /cm^3
OPTIMUM M.C. (OMC): **7.84** %

Remarks:
Nil

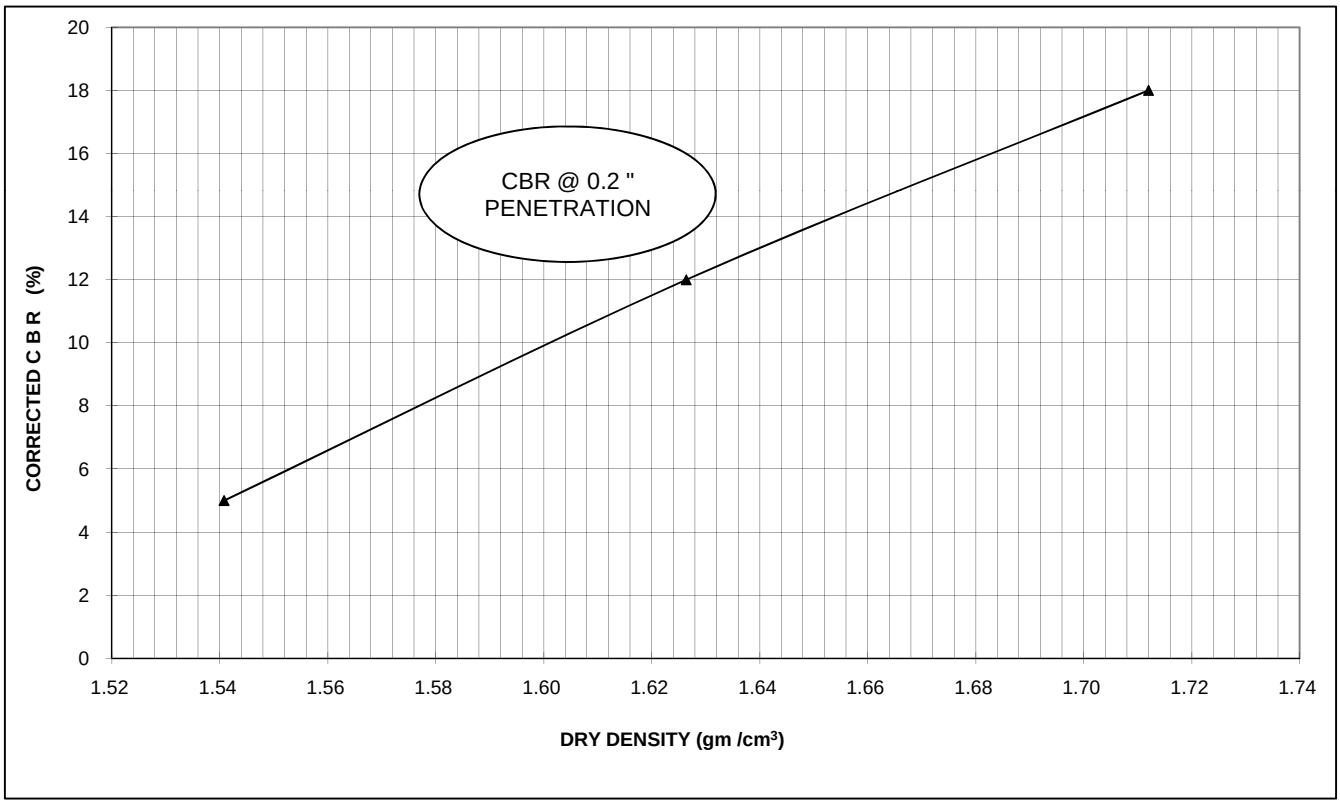
TESTED BY
WAHAB

LAB. SUPERVISOR
ANJAM

CBR - DENSITY RELATIONSHIP
ASTM D 1883

CLIENT :	L & T CONSTRUCTION	REPORT DATE:	28-Sep-19
PROJECT :	MANSOURAH MASSARAH GOLD PROJECT GEOTECHNICAL INVESTIGATION	PROJECT #:	19-K-3054
		TRIAL PIT:	TP-48
		SAMPLE :	B.S-1
SAMPLE:	BULK SAMPLE	DEPTH-(m):	0.0-0.50
		SAMPLE DATE:	23-Sep-19
TP NO.:	TP-48	TEST DATE:	27-Sep-19

COMPACTION EFFORT	10 Blows	25 Blows	56 Blows	100%MDD
DRY DENSITY (gm/cm^3)	1.541	1.626	1.712	1.712
BEARING @ 0.2" PENETRATION (CBR)	5	12	18	18
SWELL (%)	1.1	1.2	1.4	-



MAX. DRY DENSITY (γ_d): **1.712** g /cm^3
OPTIMUM M.C. (OMC): **10.20** %

Remarks:
Nil

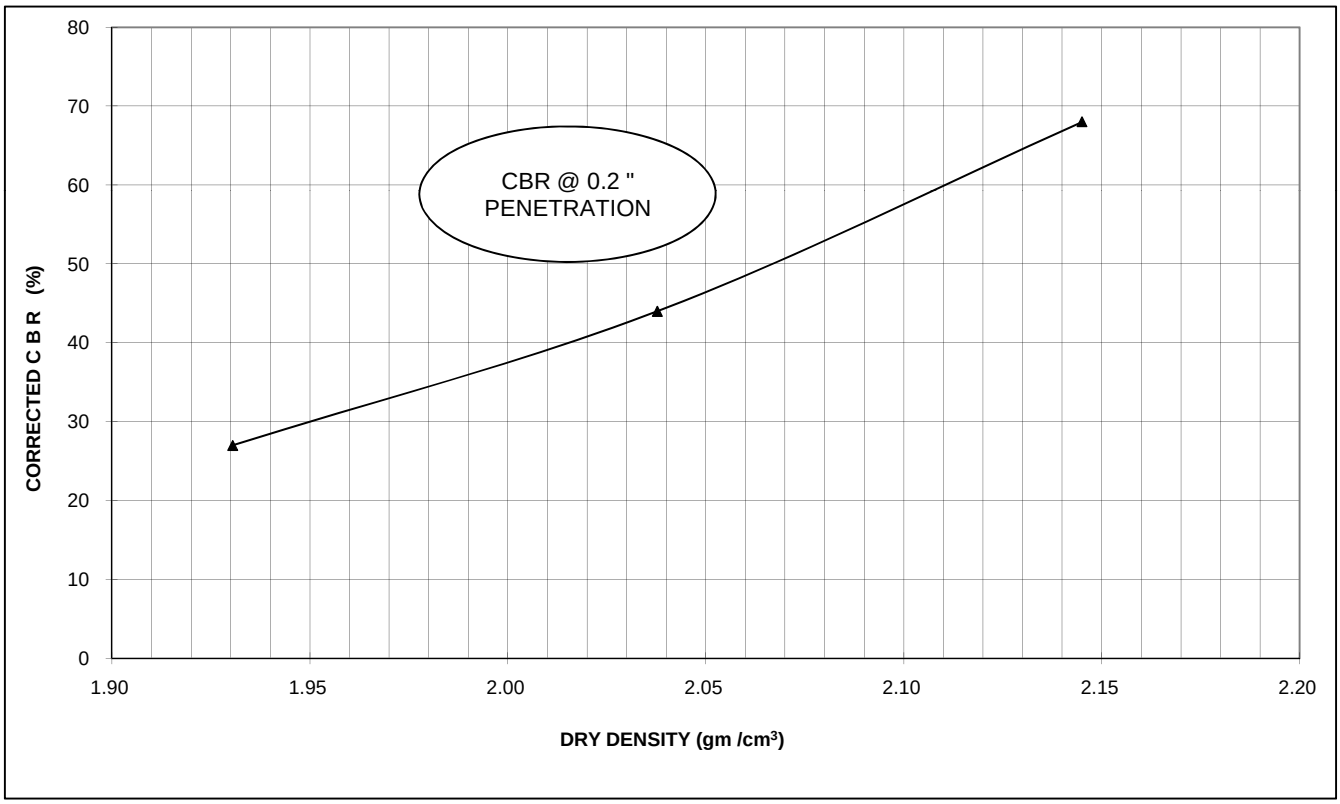
TESTED BY
WAHAB

LAB. SUPERVISOR
ANJAM

CBR - DENSITY RELATIONSHIP
ASTM D 1883

CLIENT :	L & T CONSTRUCTION	REPORT DATE:	28-Sep-19
PROJECT :	MANSOURAH MASSARAH GOLD PROJECT GEOTECHNICAL INVESTIGATION	PROJECT #:	19-K-3054
		TRIAL PIT:	TP-49
		SAMPLE :	B.S-1
SAMPLE:	BULK SAMPLE	DEPTH-(m):	0.2-.60
		SAMPLE DATE:	23-Sep-19
TP NO.:	TP-49	TEST DATE:	27-Sep-19

COMPACTION EFFORT	10 Blows	25 Blows	56 Blows	100%MDD
DRY DENSITY (gm/cm^3)	1.931	2.038	2.145	2.145
BEARING @ 0.2" PENETRATION (CBR)	27	44	68	68
SWELL (%)	nil	35	nil	-



MAX. DRY DENSITY (γ_d): **2.145** g /cm^3
OPTIMUM M.C. (OMC): **6.50** %

Remarks:
Nil

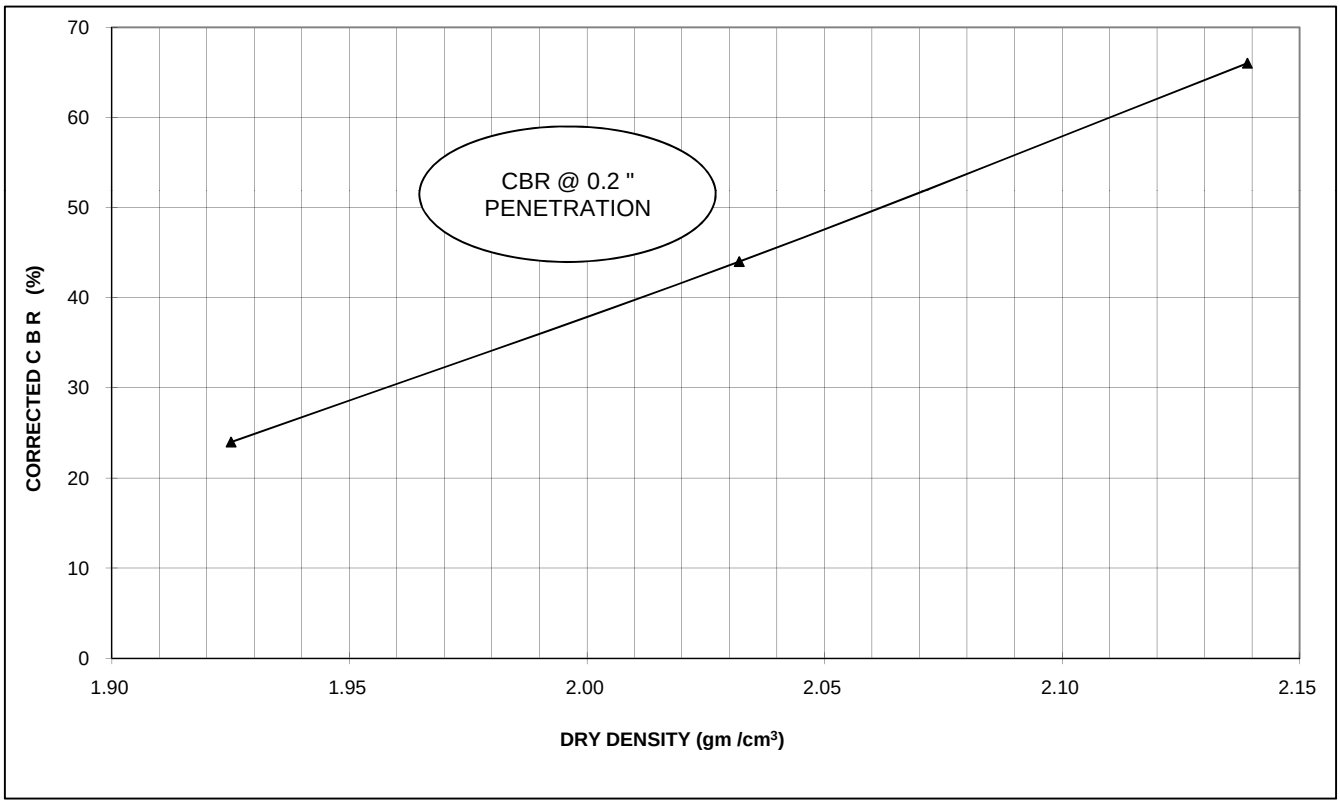
TESTED BY
WAHAB

LAB. SUPERVISOR
ANJAM

CBR - DENSITY RELATIONSHIP
ASTM D 1883

CLIENT :	L & T CONSTRUCTION	REPORT DATE:	29-Sep-19
PROJECT :	MANSOURAH MASSARAH GOLD PROJECT GEOTECHNICAL INVESTIGATION	PROJECT #:	19-K-3054
		TRIAL PIT:	TP-50
		SAMPLE :	B.S-1
SAMPLE:	BULK SAMPLE	DEPTH-(m):	0.35-0.75
		SAMPLE DATE:	24-Sep-19
TP NO.:	TP-50	TEST DATE:	28-Sep-19

COMPACTION EFFORT	10 Blows	25 Blows	56 Blows	100%MDD
DRY DENSITY (gm/cm^3)	1.925	2.032	2.139	2.139
BEARING @ 0.2" PENETRATION (CBR)	24	44	66	66
SWELL (%)	nil	35	nil	-



MAX. DRY DENSITY (γ_d): **2.139** g /cm^3
OPTIMUM M.C. (OMC): **6.20** %

Remarks:
Nil

TESTED BY
WAHAB

LAB. SUPERVISOR
ANJAM

UNCONFINED COMPRESSIVE STRENGTH (UCS)

RIYADH GEOTECHNIQUE & FOUNDATIONS LLC

UNCONFINED COMPRESSIVE STRENGTH OF INTACT ROCK CORE SPECIMENS

ASTM D 2938-95 (Reapproved 2002)

PROJECT	**MANSOURAH MASSARAH GOLD PROJECT**	**SAMPLING DATE**	**08-Sep-19**
CLIENT	**L & T CONSTRUCTION**	**DATE OF TESTING**	**09-Sep-19**
SAMPLE DESCRIPTION	**CORE**	**TEST FINISH DATE**	**11-Sep-19**
GEOTECHNICAL FILE #	-	**LAB. REF**	**KHO-P19-13**
SAMPLE BY	**WAHAB**	**PROJECT #**	**19-*K-3054**

SAMPLE #	Depth	Water Content	Length of Core before Capping	Length of Core after Capping	Diameter	Ratio (L/D)	Correction Factor	Area	Weight of Core	Bulk Density	Load	Load after Correction	Compressive Strength	
	(m)	(%)	(cm)	(cm)	(cm)			(cm^2)	(gm)	(gm/cm^3)	(kN)	(kN)	(N/mm^2)	(psi)
BH-01	12.00-13.50	0.10	10.6	10.6	6.1	1.74	0.98	29.228	837.60	2.70	106	104	35.5	5145
BH-12	9.00-10.50	0.20	10.7	10.7	6.1	1.75	0.98	29.228	919.80	2.94	132	129	44.2	6405
BH-13	4.50-6.00	0.10	10.9	10.9	6.1	1.79	1.00	29.228	724.20	2.27	69	69	23.5	3404
BH-15	3.0-4.50	0.30	10.6	10.6	8.3	1.28	0.93	54.113	1437.10	2.51	99	92	17.0	2465
BH-17	10.50-12.00	0.20	9.5	9.5	6.1	1.56	0.96	29.228	822.70	2.96	52	50	17.0	2468
BH-25	13.50-15.00	0.10	10.2	10.2	6.1	1.67	0.98	29.228	880.20	2.95	143	140	47.8	6935
BH-02	6.00-7.50	0.10	15.6	15.6	8.3	1.88	1.00	54.113	2244.20	2.66	178	178	32.9	4776
BH-09	13.50-15.00	0.20	11.7	11.7	6.1	1.92	1.00	29.228	887.80	2.60	73	73	24.8	3603
BH-18	13.50-15.00	0.10	9.5	9.5	6.1	1.56	0.96	29.228	936.20	3.37	59	57	19.3	2806

TESTED BY:
ARIF

REVIEWED BY
S.M.MINTU
(CTTA # 234)
SUPERVISOR

APPROVED BY
M.ISMAIL MASHWANI
(CTTA # 423)
LAB INCHARGE

This report relates to the samples tested and may not be reproduced other than in full, except with prior written permission from RGF laboratory

APPENDIX "C"

DESIGN AIDS

APPENDIX C

DESIGN AIDS

Annexure C-1

Shallow Foundations

Annexure C-2

Detailed Procedure of Settle 3D Software

Annexure C-3

RocLAB Theory

APPENDIX C

ANNEXURE C-1

SHALLOW FOUNDATIONS

1.0 SHALLOW FOUNDATIONS

1.1 SHEAR FAILURE ANALYSIS (Reference: Bowles, J.E, "Foundation Analysis and Design", 5th Edition)

Foundation on Sand

The net allowable bearing capacity of footings against shear failure of the load bearing strata was calculated using the following equation.

$$q_{A(net)} = \frac{1}{F}\left[q\, s_q N_q + \frac{\gamma'}{2} B s_\gamma N_\gamma \right] - \gamma' \mathrm{D}$$

$$N_q = \tan^2\left[45 + \frac{\phi}{2}\right] e^{\pi \tan\Phi}$$

$$N_\gamma = 2\left[N_q + 1\right]\tan\Phi$$

$$s_q = \left[1 + \frac{B}{L}\tan\Phi\right]$$

$$s_\gamma = \left[1 - 0.4\frac{B}{L}\right]$$

Where $q_{A(net)}$ = *Net Allowable Bearing Capacity (kN/m²), F.O.S. = 3.0*

γ' = *Effective unit weight of soil*

D = *Depth of foundation below ground level, m*

B = *Desired width of foundation, m*

$Nq, N\gamma$ = *Bearing capacity factors*

$s_q, s\gamma$ = *Shape factors*

φ = *Angle of internal friction*

Foundation on Clay

$$q_{A(net)} = \frac{1}{F}\left[N_c c_u\right] + \gamma D$$

Where $q_{A(net)}$ = *Net Allowable Bearing Capacity (kN/m²), F.O.S. = 3.0*

c = *Cohesion*

γ = *Unit weight of soil*

D = *Depth of foundation below ground level, m*

Nc = *5.14 for strip foundation and 6.17 for circular or square foundation*

1.2 SETTLEMENT ANALYSIS

1.2.1 Burland and Burbidge Method (1985)

For normally consolidated sand, the immediate average settlement (ρ_i) at the end of construction corresponding to the average effective foundation pressure (q') is given by:

$$\rho_i = q' \text{ x } B^{0.7} \text{ x } I_c$$

Where B is the width of foundation in mm and I_c is the compressibility index given by.

$$I_c = \frac{1.71}{\overline{N}^{1.4}}$$

Where $\overline{N}$ = Average SPT N value over the depth of influence (Z_1)

Depth of Influence (Z_1):

As per Burland and Burbidge, the depth of influence can be calculated as follows:

(i) When N is increasing or is constant, Z_1 can directly be read from *Fig. 3*. Alternative, following equation (**Ref. 3**) can be used instead of *Fig. 3* for the calculation of Z_1:

$$Z1 = 1.4 B_R \left(\frac{B}{B_R}\right)^{0.75}$$

Where B_R = reference width = 0.30m

(ii) When N is consistently decreasing, Z_1 is taken as **2B** or the bottom of soft layer whichever is lesser.

For an over consolidated sand or for loading at the base of an excavation for which the maximum previous effective overburden pressure is σ'_{vo} , the average settlement (ρ_i) at the end of construction is given by:

When $q' > \sigma'_{vo}$

$$\rho_i = \sigma'_{vo} \text{ x } B^{0.7} \text{ x } \frac{I_c}{3} + (q' - \sigma'_{vo}) \text{ x } B^{0.7} \text{ x } I_c$$

$$\rho_i = (q' - \frac{2}{3}\sigma'_{vo}) \text{ x } B^{0.7} \text{ x } I_c$$

When $q' < \sigma'_{vo}$

$$\rho_i = q' \text{ x } B^{0.7} \text{ x } \frac{I_c}{3}$$

Following factors are applied to the immediate average settlement (ρ_i) calculated above:

Shape factor:

$$f_S = \left[\frac{1.25 \text{ L/B}}{(\text{L/B}) + 0.25} \right]^2$$

Correction factor for depth of sand or gravel beneath the foundation (*Hs*):

$$f_1 = \frac{H_s}{z_1} \left(2 - \frac{H_s}{z_1} \right)$$

B_R = reference width = 0.30m

Time dependent settlement:

$$f_t = \frac{\rho_t}{\rho_i} = \left(1 + R_3 + R_t \log\frac{t}{3}\right)$$

Where f_t the correction factor is for time, t ≥ 3 Years, R_3 is the time dependent settlement (expressed as a proportion of ρ_i) that takes place during the first three years after construction and R_t is the time dependent settlement (expressed as a proportion of ρ_i) that takes place each log cycle of time after 3 Years.

For static loads conservative values of R_3 and R_t are 0.3 and 0.2 respectively. Thus at t = 30 years, f_t = 1.50.

For dynamic/fluctuating loads conservative values of R_3 and R_t are 0.7 and 0.8 respectively. Thus at t = 30 years, f_t = 2.50.

Variation of Standard Penetration Number with Depth

For gravel or sandy gravel:

$$N' = 1.25\, N_{60}$$

For fine sand or silty sand below groundwater table and $N_{60} > 15$:

$$N' = 15 + 0.50\,(N_{60} - 15)$$

APPENDIX C

ANNEXURE C-2

SETTLE3D-THEORY

1 STRESS

1.1 Effective Stress

Settlement depends on effective stress. The effective stress is the total stress due to gravity and external loads minus the pore water pressure. The vertical effective stress, σ' at any point is simply:

$$\sigma' = \sigma - u \qquad 1$$

Where σ is the total vertical stress and u is the pore water pressure. Compressive stresses are positive.

1.2 Initial stress and pore pressure

The initial total stress is just the stress due to gravity loading. At any point this stress is found by summing the weights of the above material layers. The weight of any given layer is the unit weight, γ, times the layer thickness. Therefore the initial total stress at any point is:

$$\sigma_i = \Sigma \gamma H \qquad 2$$

Where H is the layer thickness. The unit weight γ is set to either the saturated unit weight or the moist unit weight depending on whether the layer is below or above the piezometric line (water table).

The initial pore pressure at any point is the pressure due to the weight of overlying water. If the elevation of the point of interest is z and the water table is at elevation zwt, the initial pressure is:

$$u_i = (z - z_{wt})\ \gamma_{water} \qquad z > z_{wt}$$

$$u_i = 0 \qquad z > z_{wt} \qquad 3$$

Where γ_{water} is the unit weight of water.

The initial effective stress can then be calculated from equations 1, 2 and 3:

$$\sigma_i = \sigma_i - u_i \qquad 4$$

1.2 Stress change due to external load

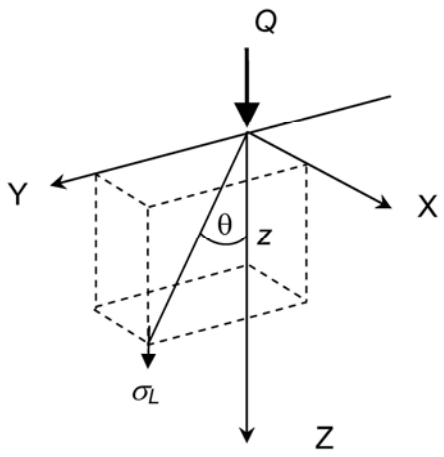
External loads such as embankments or fills can be applied in any stage. These external loads cause a change in total stress depending on the geometry and magnitude of the external load. The stress change can be calculated by four different methods.

1.3.1 Boussinesq

The Boussinesq method uses the theory of elasticity to calculate the vertical stress under a point load in a homogeneous, semi-infinite half space:

$$\sigma L = \frac{3Q}{2\pi z}{}_2 \cos^5 \theta \qquad 5$$

Where σ_L is the loading stress at any point and the meaning of the other symbols are as shown.



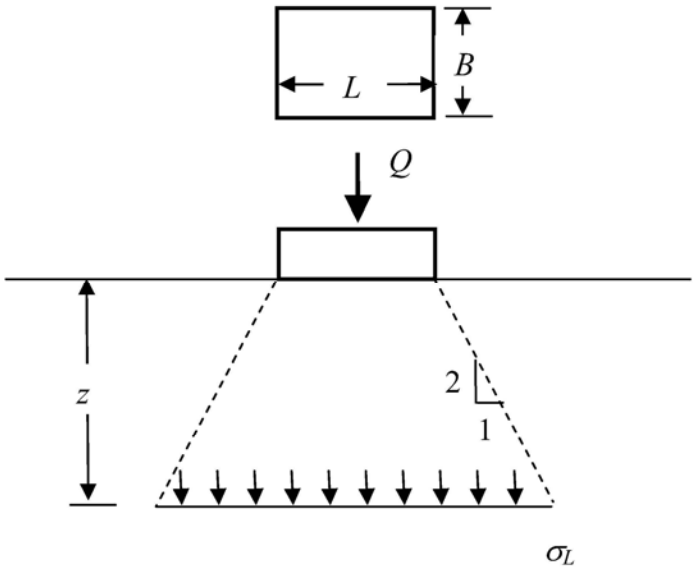
Useful solutions for stresses under different footing shapes can be obtained by integrating over the area of the footing.

1.3.2 2:1

The 2:1 method assumes that the zone of influence for an applied rectangular load has a slope of 2:1 as shown. The vertical loading stress at some depth is then calculated by:

$$\sigma_L = \frac{Q}{(B+z)(L+z)}$$

For a non-rectangular footing, the stress is calculated by computing the area of the load at the surface. With increasing depth, the area over which the load is applied increases at a 2:1 ratio and the magnitude of the loading stress decreases correspondingly.



1.3.3 Multi-layer solution

The complete elastic solution for an arbitrary shaped foundation resting on a layered elastic medium is computed by the integration of point load solution (Green's functions). Yue (1995, 1996) has recently provided a computational schema to efficiently compute the point load solutions through Hankel transforms. The elastic state of each homogeneous material body is governed by a set of partial differential equations and a set of boundary conditions. In this case of homogeneous elastic layers the boundary conditions are the set of matching stress and displacement conditions. Hankel transformation is used to convert the set of partial differential equations governing each material layer to a set of ordinary differential equations and the respective boundary conditions to algebraic expressions. Yue (1995, 1996) solved the systems of ordinary differential equations with the matching boundary conditions to a set of algebraic system of equations, unknowns being specially constructed functions in the Hankel transform domain. The stresses and displacements can be obtained through combinations of the inverse transforms of these functions. Due to extreme complexity of the functions in the Hankel transform domain, we need to resort to numerical methods which are computationally intensive and therefore very slow.

For our purpose, we need to perform two levels of numerical integration: the first level of computation evaluates the point load solutions, and the subsequent level of computation performs numerical quadrature to evaluate stresses and displacements due to the distributed loading over the given area. We have developed an extremely efficient computational method by combining the two computational levels in different order. We have also enhanced the numerical accuracy by converting the area integrals into boundary integrals by a method developed by Vijayakumar, Yacoub and Curran (2000). This boundary conversion method also circumvents the numerical difficulty associated with the singularity of the point load solutions. For any rapidly varying stress locations such as in the vicinity of boundaries, a form of adaptive special subdivision scheme is used.

1.3.4 Westergaard solution

The Boussinesq equation is one the most commonly used methods for obtaining the stress distribution induced by an applied vertical load. In developing his point load formula, Boussinesq assumed that a soil medium is elastic, isotropic and homogeneous. However, most soils are neither isotropic nor homogeneous. In fact, many soils encountered in practice exist in layers with distinctive characteristics. In particular, when the soil medium is a layered stratum of fine and coarse material, such as material underneath a road pavement or alternating layers of clay and sand, many experts believe that the Westergaard equations give a better estimate for the stress. Westergaard assumed that an elastic soil medium is laterally reinforced by numerous, closely spaced, horizontal sheets of negligible thickness but of infinite rigidity. These sheets prevent the mass as a whole from experiencing any lateral strains. Hence, only downward vertical deformations are allowed. These assumptions can better represent stratified soils, in which soft layers are reinforced by strong layers. The effect of which is a greater lateral stress distribution than what is usually obtained from methods that assume isotropic elastic solids. Unlike the Boussinesq solution, the Westergaard equations include the Poisson's ratio of the soil medium. For large lateral restraint, the Poisson's ratio may be taken to be zero. In addition, the main effect of Westergaard's assumptions is to reduce the stresses obtained directly below the center of the load as compared to Boussinesq. However, at a certain distance away from the center, stresses obtained from Westergaard begin to exceed those calculated from Boussinesq. For example, for radius r and depth z with a Poisson's ratio $\nu = 0.0$, Westergaard stresses are about 2 3 that of Boussinesq when $r\ z = 0$. At $r\ z = 1.5$, both methods give identical values, while for values where $r\ z > 1.5$, the Westergaard stresses are larger.

Similar to Boussinesq, closed form expressions for the stress profiles of: 1) the center of a uniformly loaded circular area and 2) the corner of a uniformly loaded rectangular area are both

available. Further, the contributions of adjacent loads can be superimposed to give the total stress at a given point.

In general, sedimentary soils like natural clay strata accentuate the non-isotropic condition of the soil medium. Hence, for these cases, the Westergaard equations serve as better models for reality. However, practicing geotechnical engineers often prefer Boussinesq primarily because this method gives more conservative results. In any case, the choice of analysis depends on how closely field conditions match a model's basic assumptions.

As described in Venkatramaiah (2006), for a soil medium with Poisson's ratio v, the vertical stress σ_z due to a point load Q as obtained by Westergaard is given by:

$$\sigma_z = \frac{Q}{z^2} \frac{\frac{1}{2\pi}\sqrt{\frac{1-2\nu}{2-2\nu}}}{\left[\left(\frac{1-2\nu}{2-2\nu}\right)+\left(\frac{r}{z}\right)^2\right]^{3/2}} \qquad 7$$

For large lateral restraint, ν may be taken as zero. The vertical stress below the center of a circular footing can be obtained analytically by integrating equation 7. The solution of which is given by:

$$\sigma_z = q\left[1-\frac{1}{\sqrt{1+\left(a/\eta z\right)^2}}\right] \qquad 8$$

where: $\eta = \sqrt{\frac{1-2\nu}{2-2\nu}}$

q is a uniform load.,and

a = radius

The vertical stress below the corner of a rectangular footing can be obtained analytically by integrating equation 7. The solution of which is given by:

$$\sigma_z = \frac{q}{2\pi}\cot^{-1}\sqrt{\left(\frac{1-2\nu}{2-2\nu}\right)\left(\frac{1}{m^2}+\frac{1}{n^2}\right)+\left(\frac{1-2\nu}{2-2\nu}\right)^2\left(\frac{1}{m^2 n^2}\right)} \qquad 9$$

where: $m = L/z$ $\quad n = W/z$

L and W are the respective lengths and widths of the rectangle

z is the depth
and q is a uniform load.

For the case of a square, L = B. Hence, m = n.

2 SETTLEMENT

The total settlement is the sum of three components:

- Immediate or initial settlement
- Settlement due to consolidation
- Secondary settlement (creep)

Each of these is calculated as follows.

2.1 *Immediate settlement*

2.1.1 Loading

The immediate settlement occurs instantly after load is applied and is assumed to be linear elastic. The strain for each element in a string can then be easily calculated from the 1D modulus and the effective stress. The 1D modulus (or constrained modulus) (*Es*) is input directly into *Settle3D*. The relationship between the 1D modulus and 3D Young's modulus (*E*) is:

$$E = E_s \frac{(1+\nu)(1-2\nu)}{1-\nu} \qquad 10$$

Where ν is the Poisson's Ratio.

The vertical strain in each sub layer is calculated by:

$$\varepsilon = \frac{\Delta\sigma}{E_s} \qquad 11$$

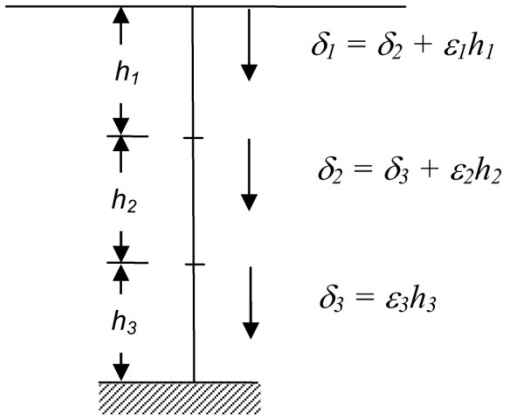
Where ere Δσ is the change in vertical total stress.

Initial settlement is then calculated from these strains. For each string, the bottom point is assumed to be fixed (non-moving). The vertical displacement of the point second from bottom is then:

$$\delta = \Delta z = \varepsilon h \qquad 12$$

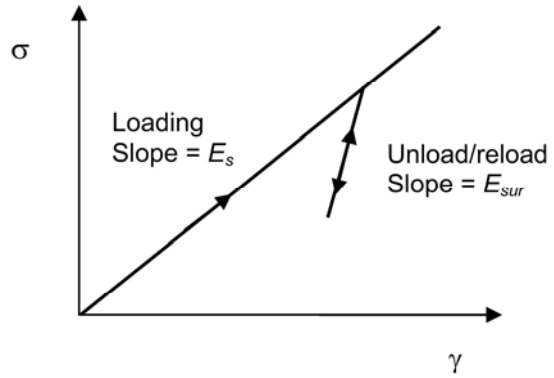
Where h is the original thickness of the bottom sublayer. The settlement of the i^{th} point is then the settlement of the point below (i+1) plus the settlement in sublayer i:

$$\delta_i = \delta_{i+1} + \varepsilon_i h_i \qquad 13$$



2.1.2 Unloading

In Settle3D, the user may supply an unload/reload modulus. If unloading occurs (due to decreasing the magnitude of an existing load, or adding an excavation) then the unload/reload modulus (*Esur*) is used in place of *Es* in equation 11. If the soil is then reloaded, *Esur* continues to be used until the soil sublayer reaches its previous stress state, at which time *Es* is again used in equation 11 to compute strain as shown.



2.1.3 Immediate settlement with mean stress

By default, settlement is calculated using only the vertical stress. A more accurate analysis can be performed by using the three-dimensional mean stress in the calculations. The mean stress at any point is the average of the volumetric stress components:

$$\sigma_M = \frac{1}{3}\left(\sigma_{xx} + \sigma_{yy} + \sigma_{zz}\right) \qquad or \qquad \sigma_M = \frac{1}{3}\left(\sigma_1 + \sigma_2 + \sigma_3\right) \tag{14}$$

The Immediate Settlement is then calculated by computing the strain in each sublayer:

$$\varepsilon = \frac{(1+\nu)(\Delta\sigma) - 3\nu(\Delta\sigma_M)}{E} \tag{15}$$

Where ν is the Poisson's Ratio, Δσ is the change in total vertical stress, M Δσ is the change in mean stress and E is the Young's modulus. When mean stress is being used, Young's modulus is input directly for each material (not the 1D modulus). The unload/reload modulus *Eur* may be used in place of E as described above.

The mean stress is not used in the calculation of Consolidation Settlement and Secondary Settlement because the relationship between strain and mean stress is not clearly defined for non-linear materials.

2.2 Settlement due to consolidation

Settlement due to consolidation progresses gradually as pore pressures dissipate and effective stresses increase. The settlements are calculated from the strains in each sublayer as in equation 13, however the calculation of the strains depends on the type of material as follows.

2.2.1 Linear

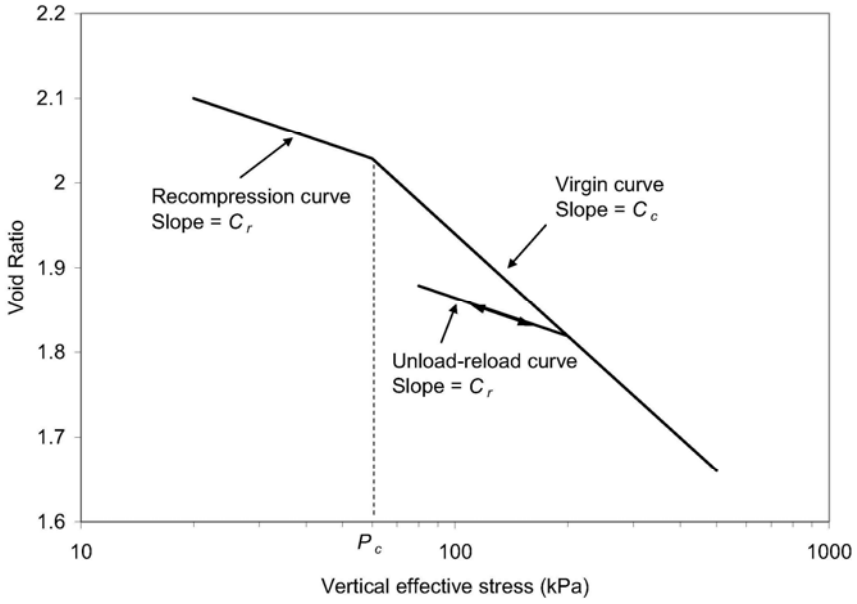
Linear material is assumed to be linear elastic. Therefore the change in vertical strain for any given element for a change in vertical effective stress $\Delta\sigma'$ is simply:

$$\Delta\varepsilon = m_v \Delta\sigma' \qquad 16$$

Where m_v is the one-dimensional compressibility. During unload/reload cycles, m_v is replaced with the unload/reload compressibility m_{vur}.

2.2.2 Non-linear

In non-linear material, the modulus is not constant but is a function of stress. This relationship is most commonly shown on a graph of void ratio versus the logarithm of effective stress as shown.



The void ratio in a soil is the ratio of the volume of voids to the volume of solids

$$e = \frac{V_v}{V_s} \tag{17}$$

The stress level *Pc* is the preconsolidation stress and represents the maximum effective stress experienced by the soil in the past. If a soil is experiencing an effective stress less than *Pc* it is *overconsolidated*. The relationship between the void ratio and the logarithm of the effective stress is given by the recompression index, *Cr*. A soil that has a stress greater than or equal to *Pc* is *normally* consolidated and its deformation is dictated by the compression index *Cc*. A soil that is unloading is always considered overconsolidated as shown.

For a stress change in an overconsolidated soil layer, the change in void ratio, Δe, can be calculated from the initial effective stress, σ_i' and the final effective stress, σ_f' by:

$$\Delta e = -C_r \log\left(\frac{\sigma_f'}{\sigma_i'}\right) \tag{18}$$

Vertical strain is related to void ratio by:

$$\varepsilon = -\frac{\Delta e}{1 + e_0} \tag{19}$$

Where *e0* is the initial void ratio.

Therefore,

$$\Delta\varepsilon = \frac{C_r}{1 + e_0} \log\left(\frac{\sigma_f'}{\sigma_i'}\right) \tag{20}$$

For a normally consolidated soil, the equation is the same with *Cr* replaced with *Cc*. It is also possible for a soil layer to start out overconsolidated and end up normally consolidated if the final stress is greater than *Pc*. In this case, the vertical strain can be calculated by:

$$\Delta\varepsilon = \frac{C_r}{1 + e_0} \log\left(\frac{P_c'}{\sigma_i'}\right) + \frac{C_c}{1 + e_0} \log\left(\frac{\sigma_f'}{P_c}\right) \tag{21}$$

Note that the initial stress does not necessarily have to refer to the initial in-situ stress due to gravitational loading. For a multi-stage analysis, the change in strain is computed for each stage by using the effective stress at the start of the stage and the effective stress at the end of the stage in equation 20 or 21.

Strains may also be calculated using strain based versions of the compression indices, $C_{c\varepsilon}$ and $C_{r\varepsilon}$. In this case, stress is related directly to strain, rather than void ratio so for a completely overconsolidated material equation 20 becomes:

$$\Delta\varepsilon = C_{r\varepsilon} \log\left(\frac{\sigma_f'}{\sigma_i'}\right)$$

Equations involving normally consolidated material are formed similarly.

3 PORE PRESSURES

3.1 *Initial pore pressure*

The initial pore pressure at any point is the pressure due to the weight of overlying water as given in equation 3.

3.2 *Excess pore pressure when load is applied*

By default, when a load is applied, the pore pressure at each node increases by an amount equal to the change in vertical stress at that node:

$$\Delta u = \Delta\sigma \qquad 22$$

The excess pore pressure is then simply the pore pressure minus the initial pore pressure due to gravity:

$$u_e = u - u_i \qquad 23$$

A more accurate approach is to set the change in pore pressure equal to the change in *undrained mean stress*. This can be accomplished by choosing the **Use mean 3D stress** option in the

$$\sigma_M^{undrained} = \frac{1}{3}\left(\sigma_{xx} + \sigma_{yy} + \sigma_{zz}\right)_{\nu=0.5} \quad or \quad \sigma_M^{undrained} = \frac{1}{3}\left(\sigma_1 + \sigma_2 + \sigma_3\right)_{\nu=0.5}$$

Project Settings (Advanced). The *undrained mean stress* is the average of the volumetric stress components for an undrained (incompressible) material:

An undrained material has a Poisson's ratio ν = 0.5.

Setting the initial pore pressure equal to the initial stress is only strictly true for a fluid of infinite stiffness in a fully saturated soil. For partially saturated material, the pore spaces are partly filled with air. In this case the pore fluid (air + water) cannot be considered infinitely stiff. Therefore some load is initially carried by the soil matrix. To account for this effect, the user can specify the Skempton pore pressure coefficient *B* such that

$$\Delta u = B\left(\Delta \sigma_M^{undrained}\right) \qquad 25$$

(Skempton, 1954).

The coefficient *B* ranges from 0 for dry soil to 1 for fully saturated soil. If only vertical stresses are being used, then the parameter *B* is set such that

$$\Delta u = \overline{B}\left(\Delta \sigma\right) \qquad 26$$

By default in Settle3D, *B* or *B* = 1.

3.2.1 Excess pore pressure above water table

By default, the soil is assumed to be dry above the water table, so an applied load does not generate excess pore pressure above the water table.

If you want the soil to behave as if it were saturated above the water table, then you can select the **Generate excess pore pressures above water table** checkbox under the Groundwater tab of the Project Settings dialog. In this case, applied loads will generate excess pore pressure above the water table, as if the material were saturated with *B* = 1.

3.3 *Permeability and Coefficient of Consolidation*

To solve for pore pressures in a multi-layered material, both the material permeability (*k*) and material coefficient of consolidation (*cv*) are required. However in Settle3D, the user only needs

to specify one or the other of these quantities. This is because these quantities are not independent and are related to each other through the material stiffness. The material stiffness depends on the material type. Therefore if the user enters *cv*, permeabilities are calculated as follows:

$k = c_v m_v \gamma_w$ linear material

$k = \frac{c_v C_c \gamma_w}{2.3(1+e_0)\sigma_{zi}'}$ non-linear material

$k = \frac{c_v \gamma_w}{m\sigma_r' \left(\frac{\sigma_{zi}'}{\sigma_r'}\right)^{1-a}}$ Janbu material

$k = \frac{c_v \gamma_w}{C_p \sigma_{zi}'}$ Koppejan material

Where σ_{zi}' is the initial effective stress and γ_w is the unit weight of pore water. For overconsolidated material, c_v is replaced by c_{vr}, C_c is replaced by C_r, m is replaced by m_r and C_p is replace by C'_p.

If the user enters *k* instead, then equation 50 can be rearranged to obtain c_v.

3.3.1 Variable permeability

It is often observed that increasing stress in a soil causes a permeability decrease as porosity (and void ratio) decreases. Several relationships have been proposed to mathematically account for changing permeabilities with changing stress. Settle3D currently implements two different relationships as described below.

The changing permeability is accounted for in Settle3D by computing stress (and/or void ratio) at the start of each stage and setting the permeability for the duration of the stage. Permeability does not change during a stage. If the material permeability is very sensitive to stress changes and the stress is changing rapidly, then the user should specify many stages of short duration to ensure accurate results.

Changing permeability with changing void ratio

Terzaghi et al. (1996) give the following relationship:

$$C_k = \frac{\Delta e}{\Delta \log k} \qquad 27$$

Where *e* is the void ratio, *k* is the permeability and *Ck* is a unitless parameter to be specified by the user. This equation cannot be used for Linear or Janbu material since no void ratio is specified for these material types.

It is up to the user to specify *Ck*, however Tavenas et al. (1983) propose the following empirical relationship between *Ck* and *e* for soft clay and silt deposits:

$$\frac{C_k}{e_0} = 0.50 \qquad 28$$

Permeability is a function of effective stress

Vaughan (1989) proposed that the permeability, *k*, varies exponentially with the mean effective stress, σ′:

$$k = k_0 \exp(-B\sigma') \qquad 29$$

Where *k0* is the permeability at zero mean effective stress and *B* is a user-specified material property with units of AREA/FORCE. This *B* should not be confused with the Skempton pore pressure coefficient – a completely different parameter. In Settle3D, vertical stress is used instead of mean stress in equation 29.

3.5 Degree of consolidation

The degree of consolidation is a number between 0 – 100% that indicates the stage of consolidation (0 for completely unconsolidated to 100% for totally consolidated). Lambe and Whitman (1969) give two definitions for degree of consolidation. The first is:

$$U = \frac{compression\ at\ time\ t}{compression\ at\ end\ of\ consolidation}$$

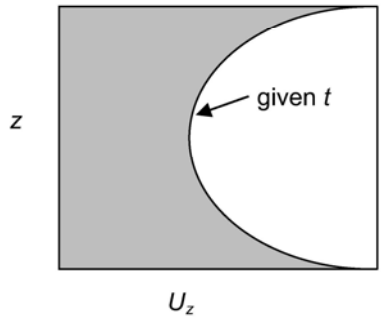
This is the equation used by Settle3D, where consolidation settlements are used in the calculation (not total settlements).

However, for some situations, namely when $B < 1$, then this may be unsatisfactory. When $B < 1$, or when the mean stress is used to calculate settlements, the applied load is not completely balanced by excess pore pressures so there is some instant settlement. In these cases you will see the degree of consolidation start from some non-zero value and progress to 100% as excess pore pressures dissipate. This may not be what you want, so Settle3D also provides another indicator for U called Average Degree of Consolidation. This follows the second method of calculation in Lambe and Whitman (1969) and the calculation is performed as follows:

First, at each point below the surface, the consolidation ratio is calculated:

$$U_z = 1 - \frac{u_e}{u_{peak}}$$

Where u_e is the excess pore pressure and u_{peak} is the maximum excess pore pressure in the past. These values are then summed over the thickness of a material layer as shown in the figure below. In this way, an average degree of consolidation can be calculated for each material layer.



$$U = \frac{shaded\ area}{total\ area}$$

For a linear material with $B = 1$, the values for U and U_{avg} are equal. In other cases they are not. It is up to the user to decide which indicator is most appropriate. Keep in mind however, that U reflects the consolidation of all soil below the point of interest, whereas U_{avg} reflects the consolidation of each material layer independently.

APPENDIX C

ANNEXURE C-3

ROCLAB THEORY

How Can I Use RocLab?

The following tasks can be accomplished with RocLab.

Determine Strength Parameters

Determine the Generalized Hoek-Brown strength parameters of a rock mass (**mb**, **s** and **a**), based on the following input data (see Ref 1):

- unconfined compressive strength of intact rock **sigci**
- the intact rock parameter **mi**
- the geological strength index **GSI**
- the disturbance factor **D**

Determine Rock Mass Deformation Modulus

Determine the rock mass deformation modulus based on the following input data (See **Generalized** Hoek-Diederichs equation in Ref 2):

- the intact rock deformation modulus **Ei**
- **Ei** can be optionally estimated using a modulus ratio **MR**

Plot Failure Envelopes

Plot the Hoek-Brown failure envelope in principal and / or shear-normal stress space.

- Interactively change **sigci**, **GSI**, **mi**, **D**, to see how the failure envelope changes with each parameter

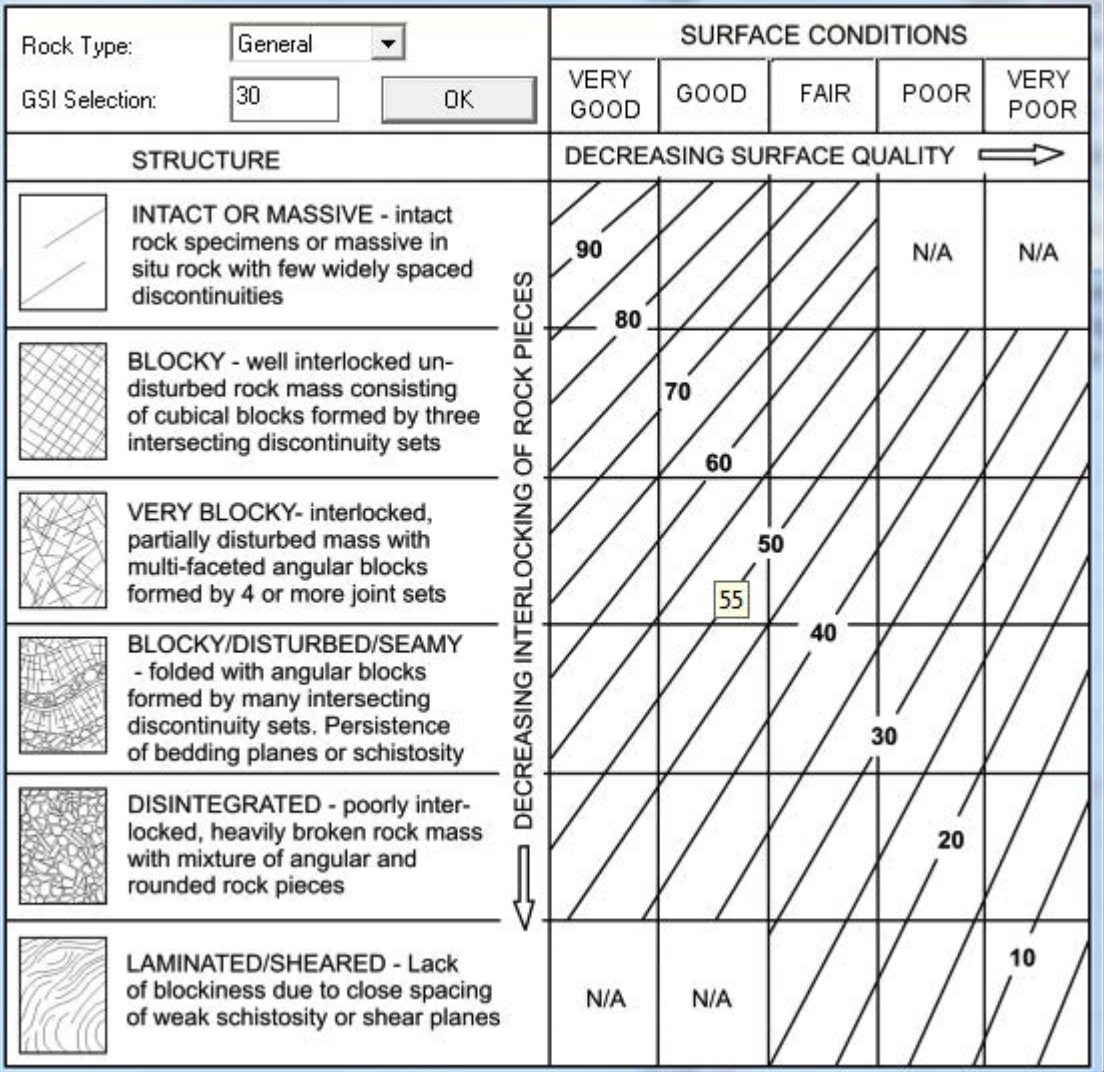
Estimation of Input Parameters

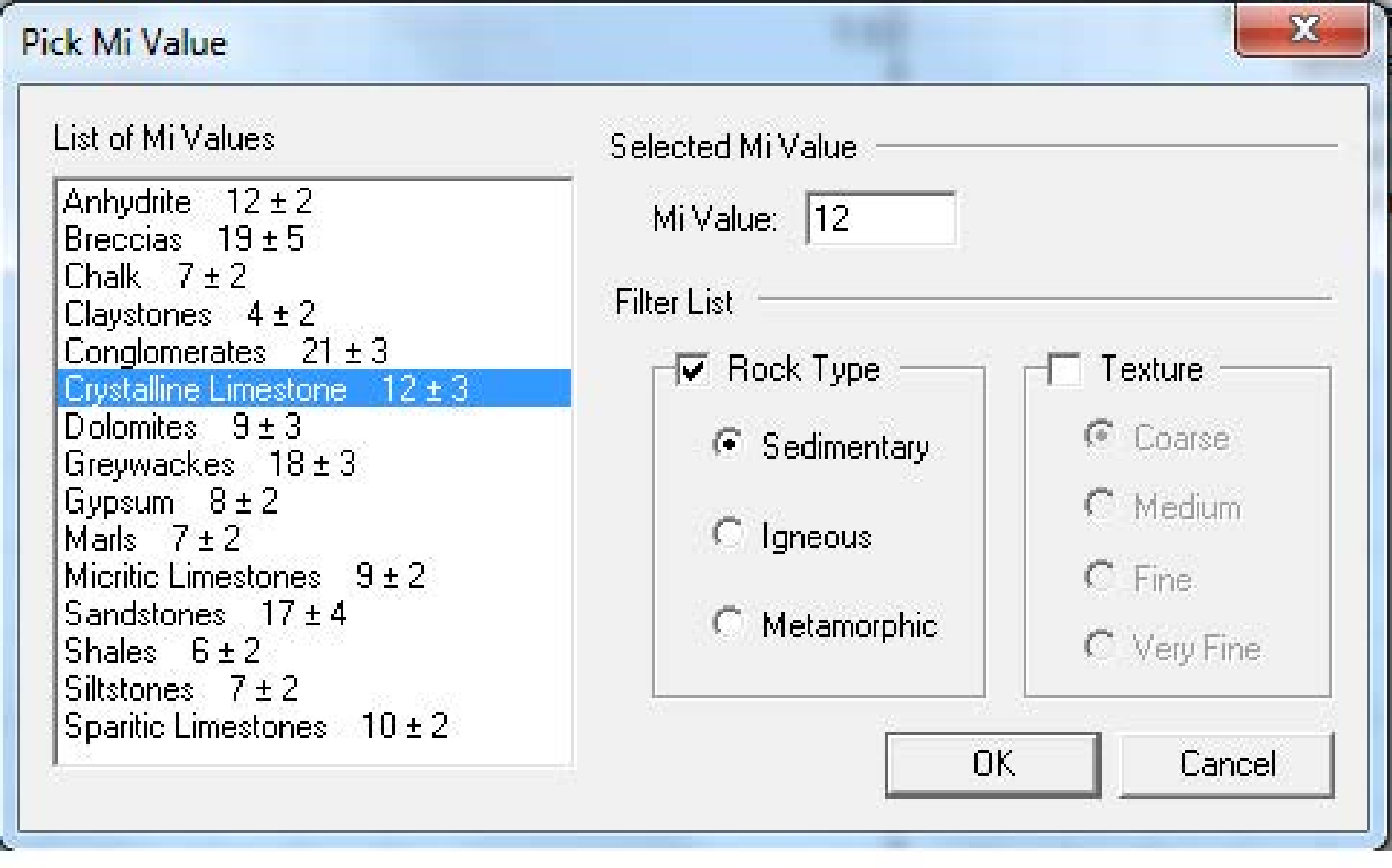
Each of the above 5 input parameters (**sigci**, **mi**, **GSI, D** and **Ei**), can be conveniently estimated from built-in charts and tables, based on rock type, geological conditions, etc.

Pick GSI Value

Rock Type: General

GSI Selection: 30 OK




Pick Mi Value
List of Mi Values
Anhydrite 12 ± 2
Breccias 19 ± 5
Chalk 7 ± 2
Claystones 4 ± 2
Conglomerates 21 ± 3
Crystalline Limestone 12 ± 3
Dolomites 9 ± 3
Greywackes 18 ± 3
Gypsum 8 ± 2
Marls 7 ± 2
Micritic Limestones 9 ± 2
Sandstones 17 ± 4
Shales 6 ± 2
Siltstones 7 ± 2
Sparitic Limestones 10 ± 2
Selected Mi Value
Mi Value: 12
Filter List
Rock Type
Sedimentary
Igneous
Metamorphic
Texture
Coarse
Medium
Fine
Very Fine
OK
Cancel

Estimate Ei using a Modulus Ratio (Ei=MR*sigci)

Using a modulus ratio (MR), it is possible to estimate the intact modulus (Ei) from:

Ei = MR*sigci

This relationship is useful when no direct values of the intact modulus are available or where completely undisturbed sampling for measurement of Ei is difficult.

Pick MR Value by Rock Type:

- Agglomerate 500 ± 100
- Amphibolites 450 ± 50
- Andesite 400 ± 100
- Anhydrite 350 ± 0
- Basalt 350 ± 100
- Breccia 500 ± 0
- Breccias 290 ± 60
- Chalk 1000 ± 0
- Claystones 250 ± 50
- Conglomerates 350 ± 50
- Crystalline Limestone 500 ± 100
- Dacite 400 ± 50
- Diabase 325 ± 25
- Diorite 325 ± 25
- Dolerite 350 ± 50
- Dolomites 425 ± 75
- Gabbro 450 ± 50
- Gneiss 525 ± 225

Estimated Intact Modulus (Ei)

Ei: 3000

MR Value: 500 sigci: 6

MR Filter List

Rock Type

- Sedimentary
- Igneous
- Metamorphic

Texture

- Coarse
- Medium
- Fine
- Very Fine

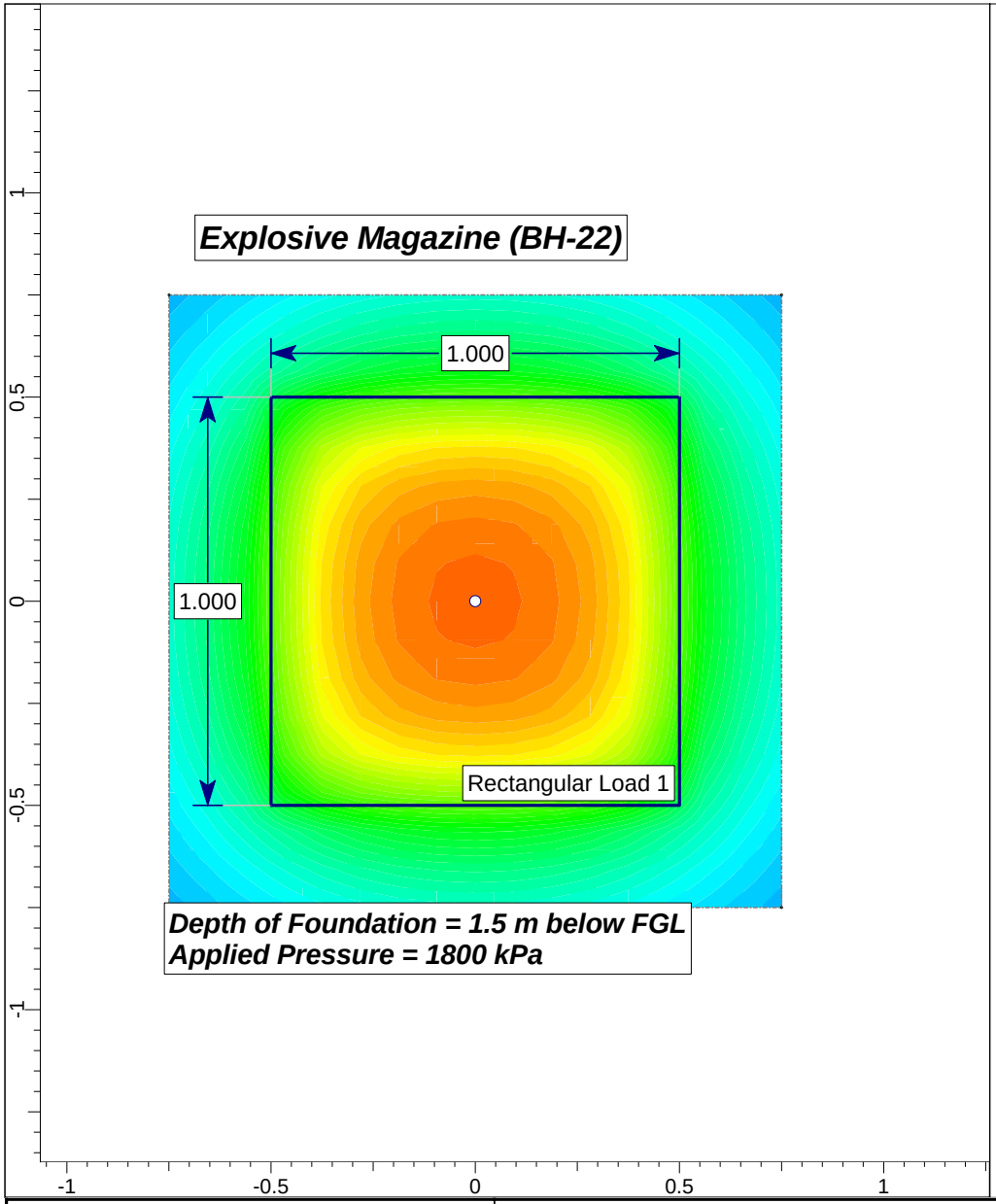
OK Cancel

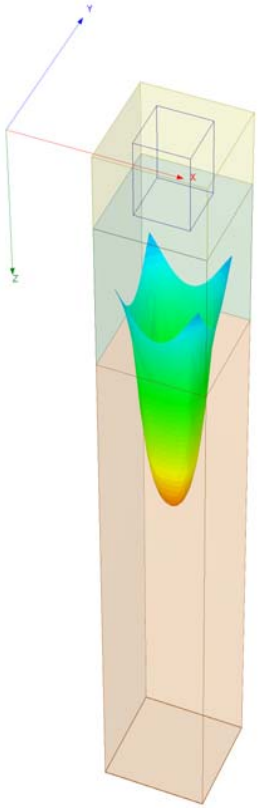
APPENDIX "D"

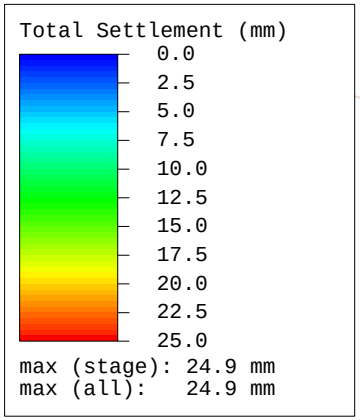
BEARING CAPACITY CALCULATION SHEET

Explosive Magazine (BH-22)

Depth of Foundation = 1.5 m below FGL
Applied Pressure = 1800 kPa

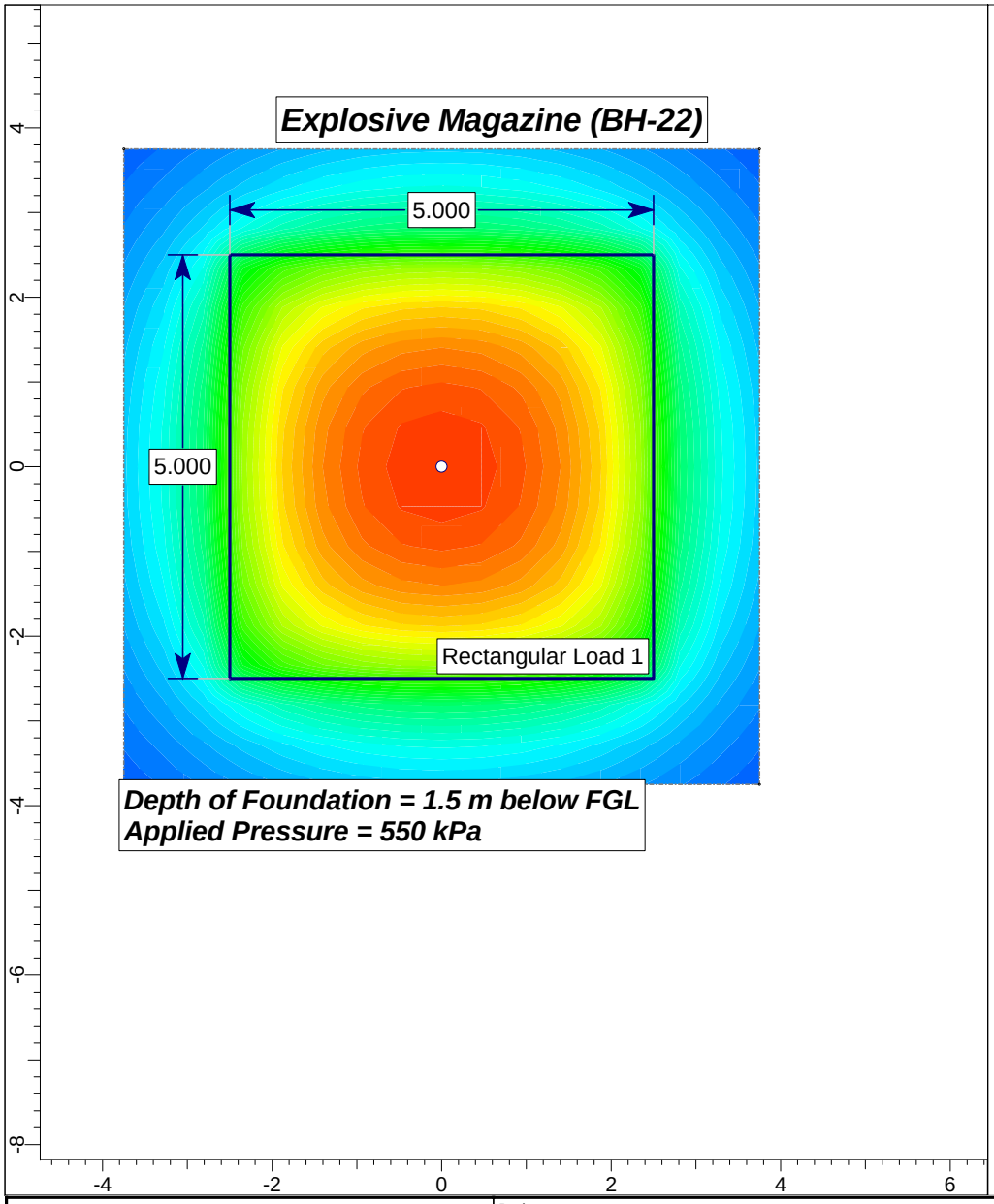


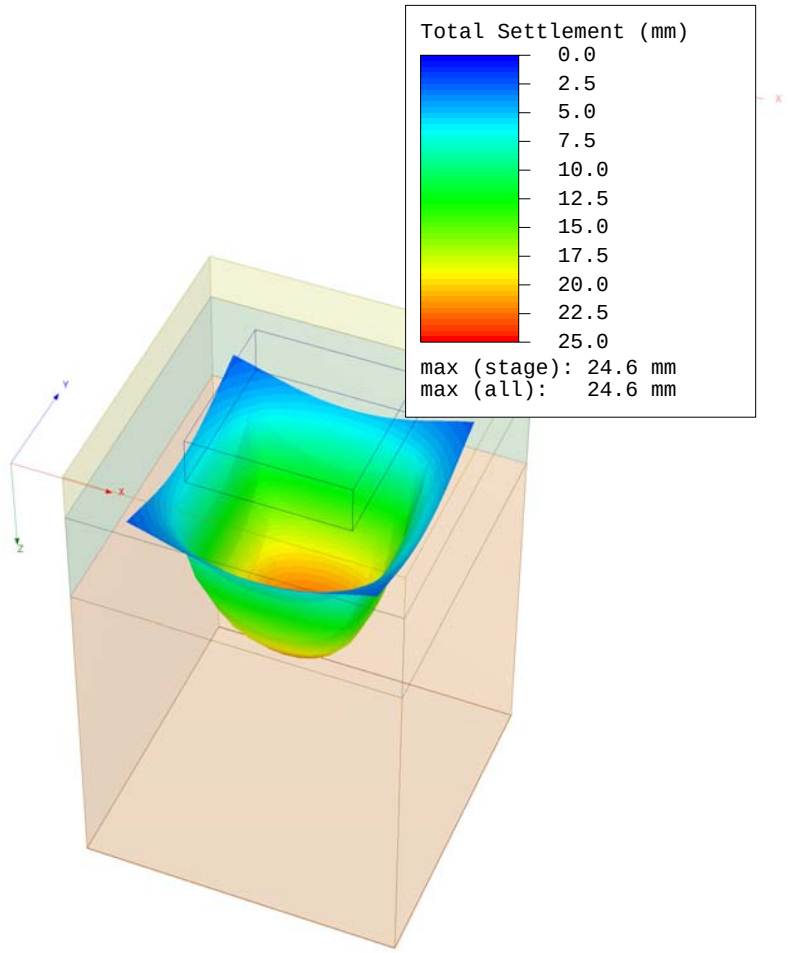




rocscience	*Project*	MANSOURAH - MASSARAH GOLD PROJECT	
	Analysis Description	Settlement Analysis	
	Drawn By	HUR/MF	*Company* RGF
	Date		*File Name* 8. Explosive Magazine (BH-22).s3z

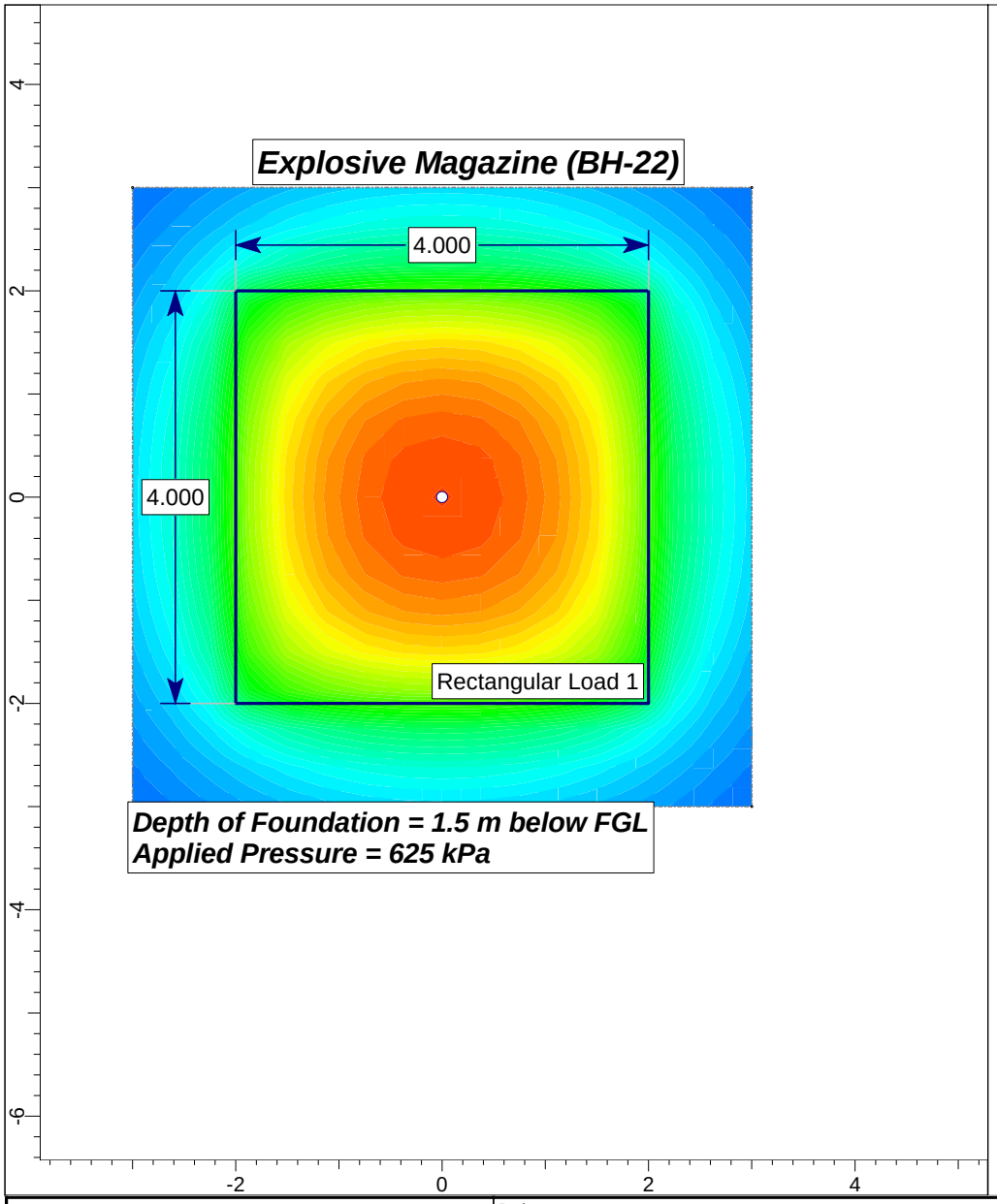
SETTLE3D 4.016

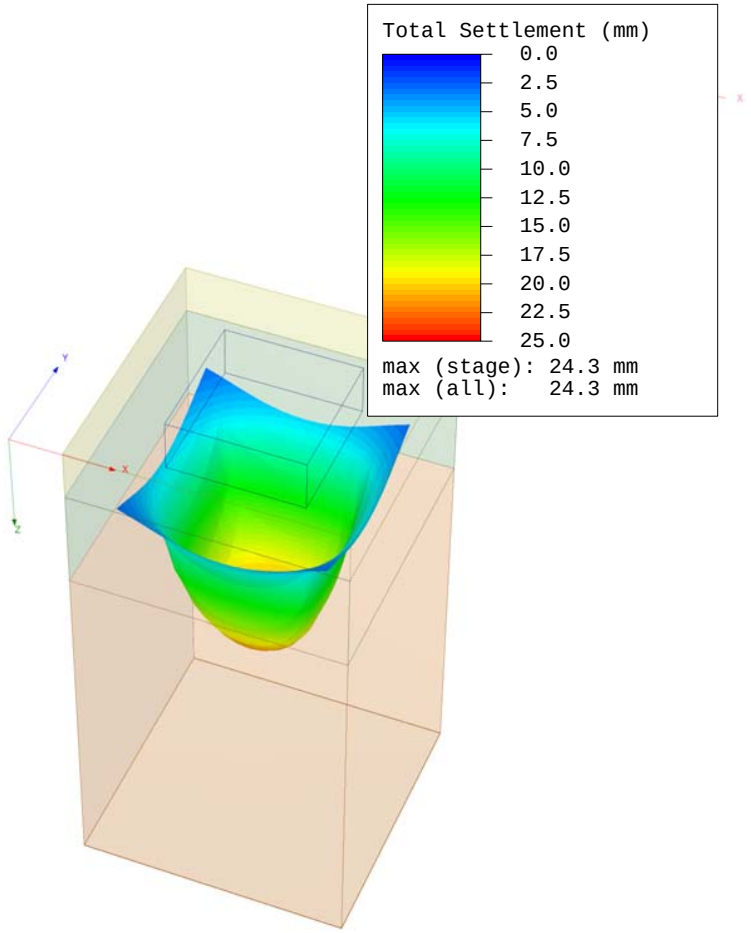




rocscience	Project	MANSOURAH - MASSARAH GOLD PROJECT	
	Analysis Description	Settlement Analysis	
	Drawn By	HUR/MF	Company RGF
	Date		File Name 8. Explosive Magazine (BH-22).s3z

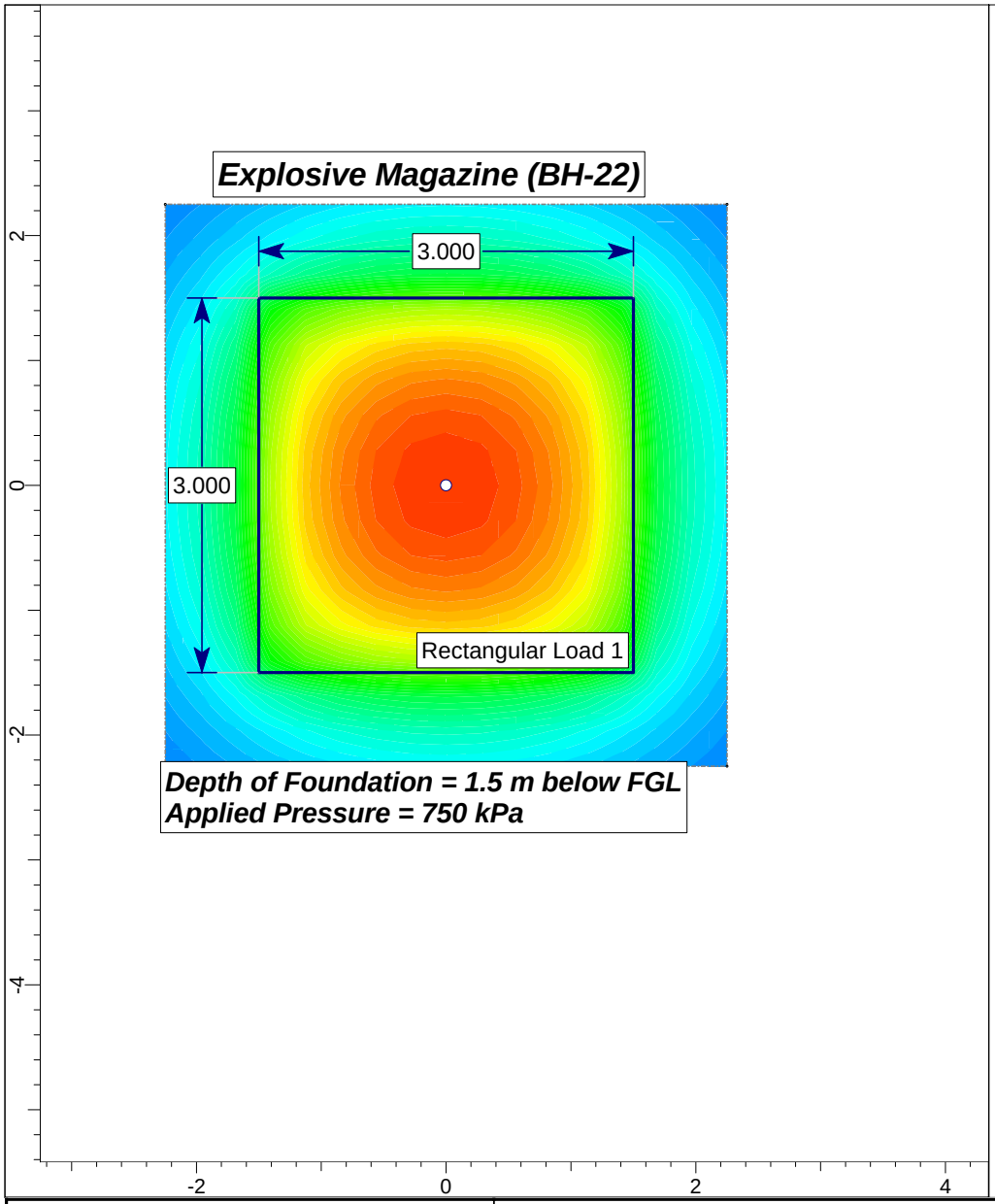
SETTLE3D 4.016

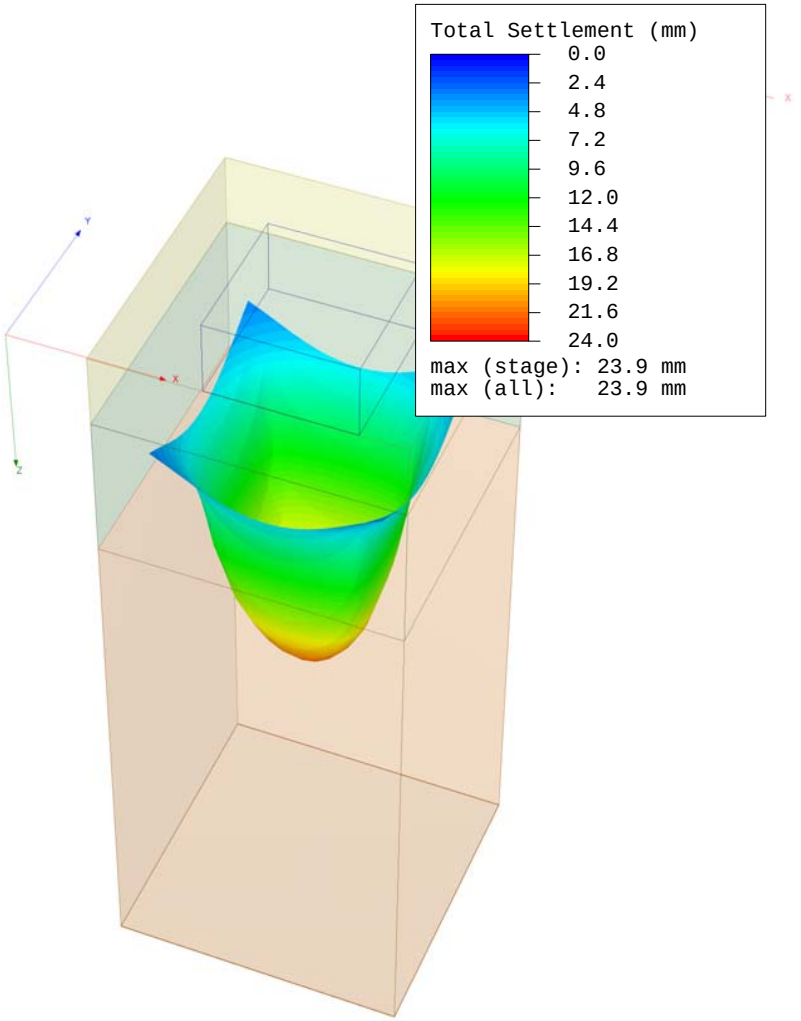




rocscience	Project	MANSOURAH - MASSARAH GOLD PROJECT	
	Analysis Description	Settlement Analysis	
	Drawn By	HUR/MF	Company RGF
	Date		File Name 8. Explosive Magazine (BH-22).s3z

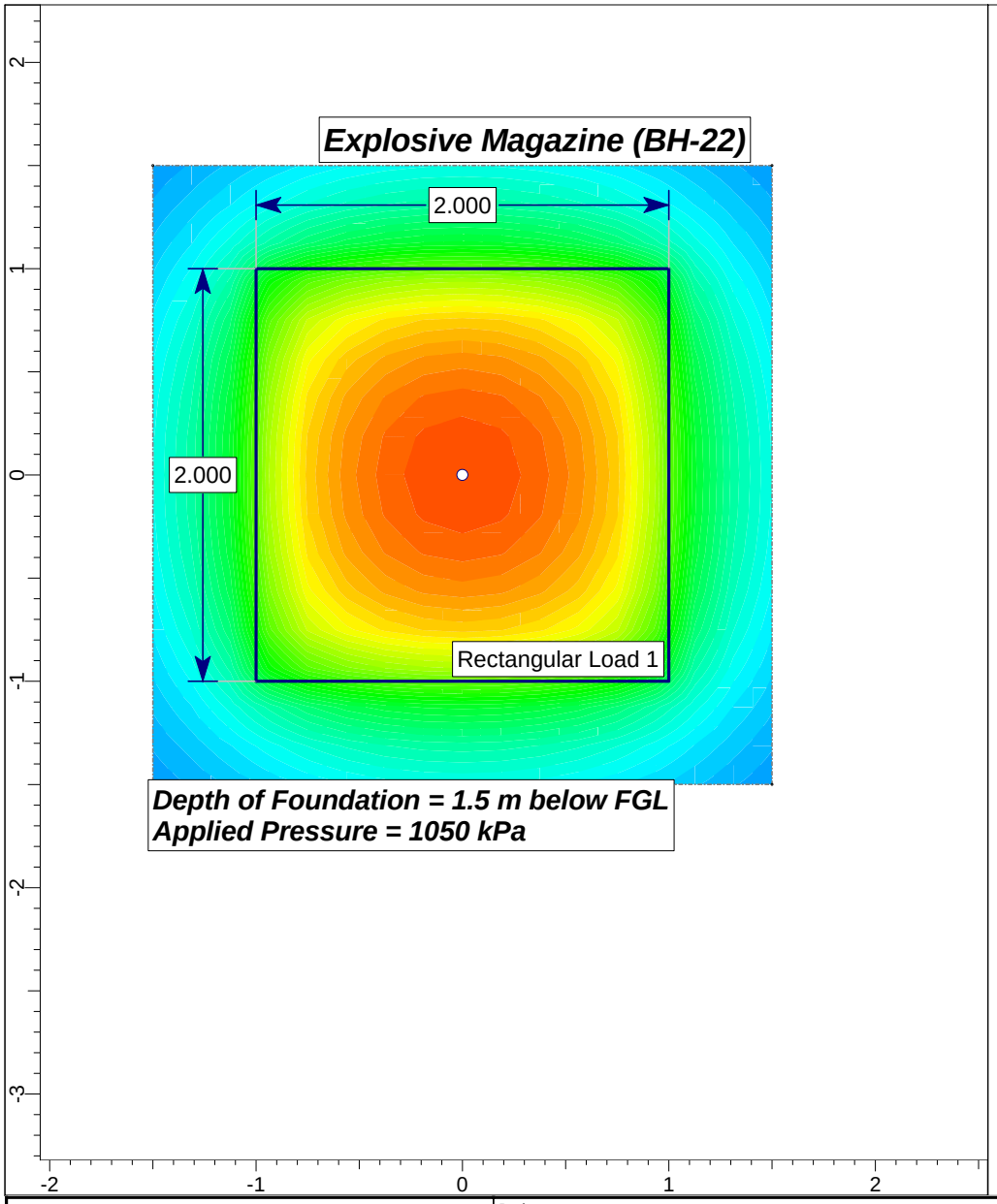
SETTLE3D 4.016

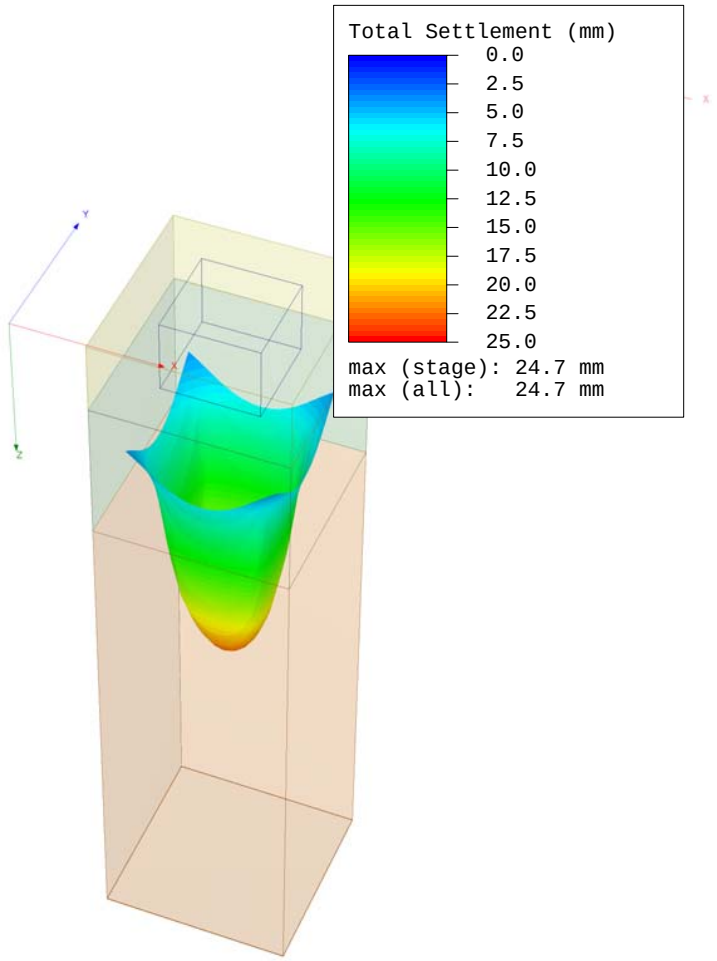




rocscience	Project	MANSOURAH - MASSARAH GOLD PROJECT	
	Analysis Description	Settlement Analysis	
	Drawn By	HUR/MF	Company RGF
	Date		File Name 8. Explosive Magazine (BH-22).s3z

SETTLE3D 4.016





rocscience	Project	MANSOURAH - MASSARAH GOLD PROJECT	
	Analysis Description	Settlement Analysis	
	Drawn By	HUR/MF	Company RGF
	Date		File Name 8. Explosive Magazine (BH-22).s3z

SETTLE3D 4.016

Settle3D Analysis Information
MANSOURAH - MASSARAH GOLD PROJECT

Project Settings

Document Name	8. Explosive Magazine (BH-22)
Project Title	MANSOURAH - MASSARAH GOLD PROJECT
Analysis	Settlement Analysis
Author	HUR/MF
Company	RGF
Stress Computation Method	Boussinesq

Calculate settlement with mean stress

Use properties from first layer to calculate layered stresses

Stage Settings

Stage #	Name
1	Stage 1

Results

Time taken to compute: 0.0529834 seconds

Stage: Stage 1

Data Type	Minimum	Maximum
Total Settlement [mm]	0	24.583
Total Consolidation Settlement [mm]	0	0
Virgin Consolidation Settlement [mm]	0	0
Recompression Consolidation Settlement [mm]	0	0
Immediate Settlement [mm]	0	24.583
Loading Stress ZZ [kPa]	0	550
Loading Stress XX [kPa]	-70.4895	468.994
Loading Stress YY [kPa]	-70.4895	468.994
Mean Stress [kPa]	0	476.667
Total Stress ZZ [kPa]	0	582.3
Total Stress XX [kPa]	-43.4895	495.994
Total Stress YY [kPa]	-43.4895	495.994
Modulus of Subgrade Reaction (Total) [kPa/m]	0	1.25187e+007
Modulus of Subgrade Reaction (Immediate) [kPa/m]	0	1.25187e+007
Modulus of Subgrade Reaction (Consolidation) [kPa/m]	0	0
Total Strain	-0.000259404	0.0054883
Degree of Consolidation [%]	0	0
Pre-consolidation Stress [kPa]	1.35	580.081
Over-consolidation Ratio	1	1
Void Ratio	0	0
Hydroconsolidation Settlement [mm]	0	0
Undrained Shear Strength	0	0

Loads

1. Rectangular Load: "Rectangular Load 1"

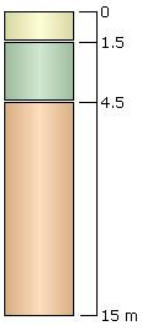
Length	5 m
Width	5 m
Rotation angle	0 degrees
Load Type	Flexible
Area of Load	25 m^2
Load	550 kPa
Depth	1.5 m
Installation Stage	Stage 1

Coordinates

X [m]	Y [m]
-2.5	-2.5
2.5	-2.5
2.5	2.5
-2.5	2.5

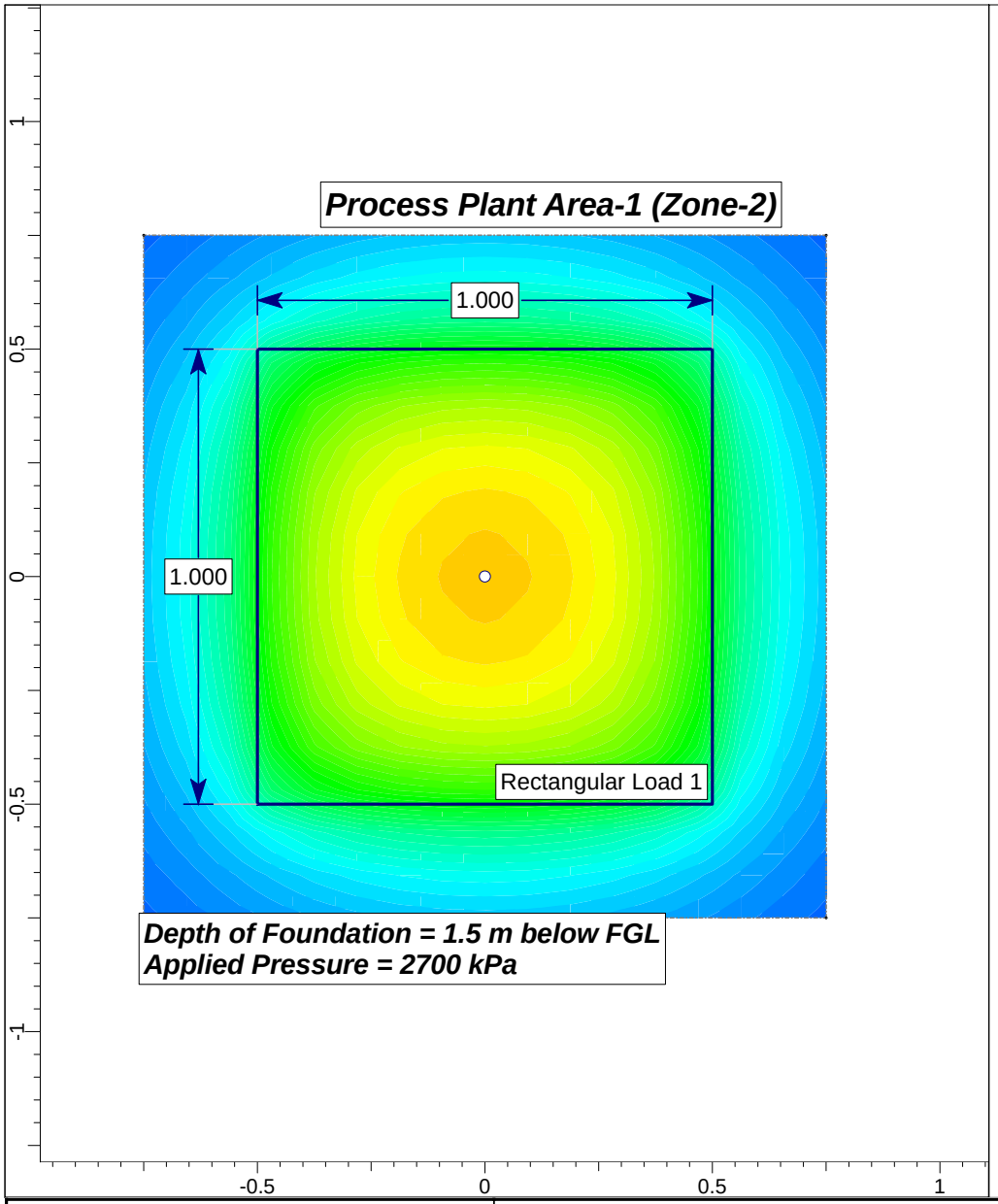
Soil Layers

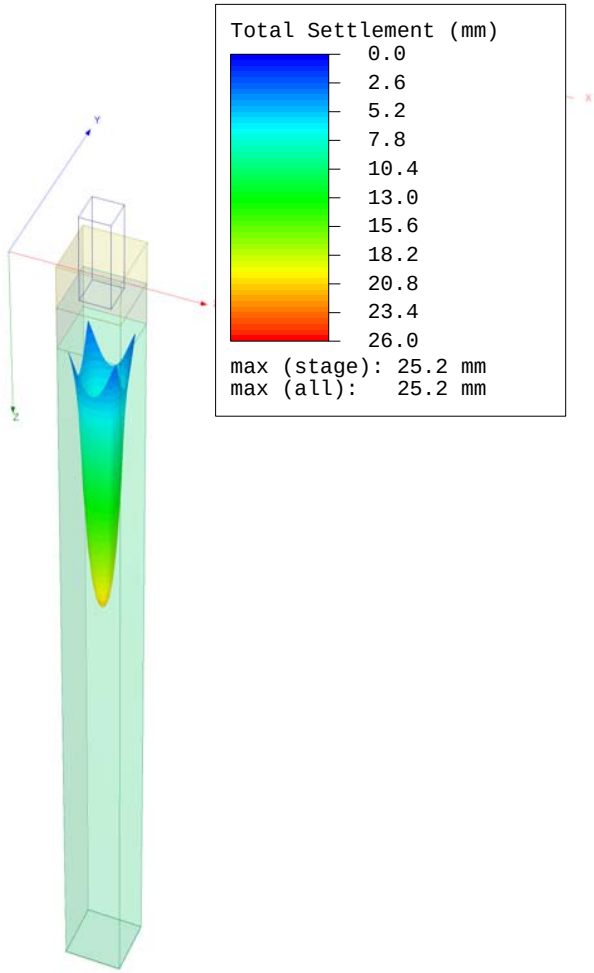
Layer #	Type	Thickness [m]	Depth [m]
1	Soil 1	1.5	0
2	Soil 2	3	1.5
3	Soil 3	10.5	4.5



Soil Properties

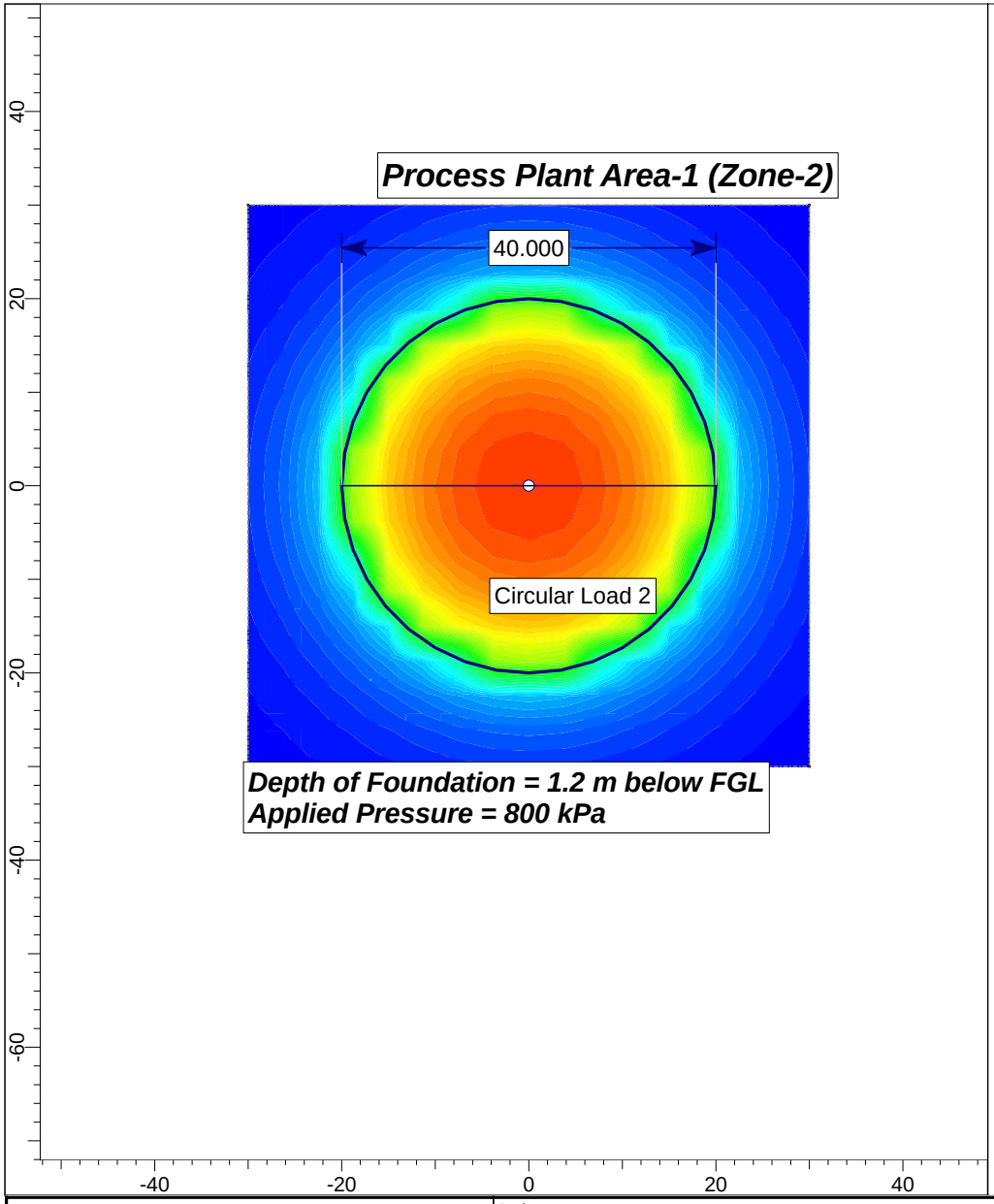
Property	Soil 1	Soil 2	Soil 3
Color			
Unit Weight [kN/m^3]	18	20	20
Poisson's Ratio	0.3	0.3	0.3
K0	1	1	1
Immediate Settlement	Enabled	Enabled	Enabled
E [kPa]	33000	74000	134000
Eur [kPa]	33000	74000	134000
Undrained Su A [kN/m2]	0	0	0
Undrained Su S	0	0	0
Undrained Su m	0	0	0

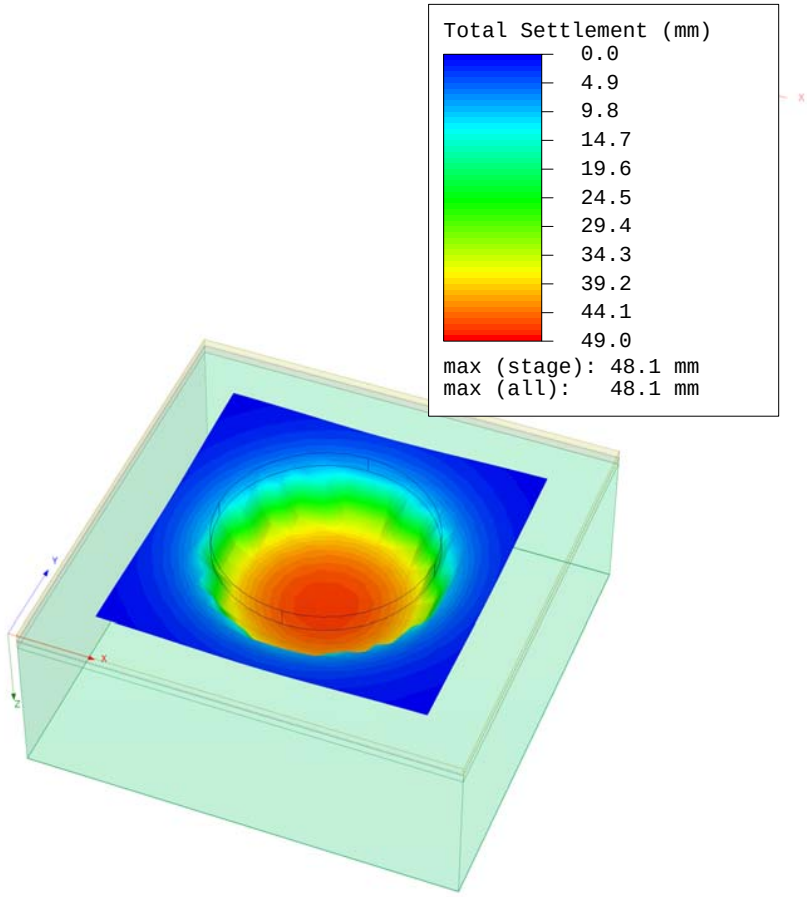




rocscience	Project	MANSOURAH - MASSARAH GOLD PROJECT	
	Analysis Description	Settlement Analysis	
	Drawn By	HUR/MF	Company RGF
	Date		File Name 3. Process Plant-1 (Z-2).s3z

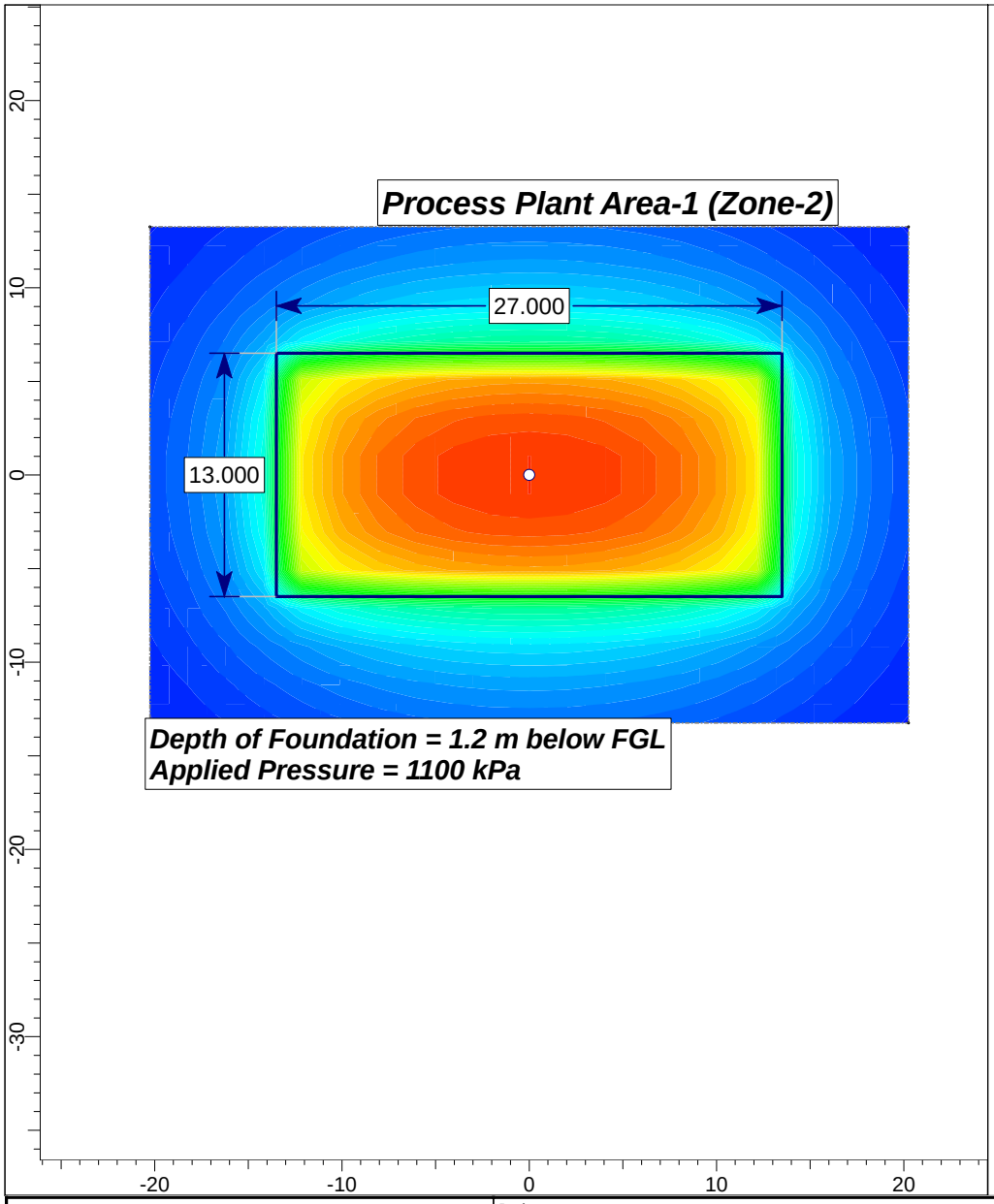
SETTLE3D 4.016

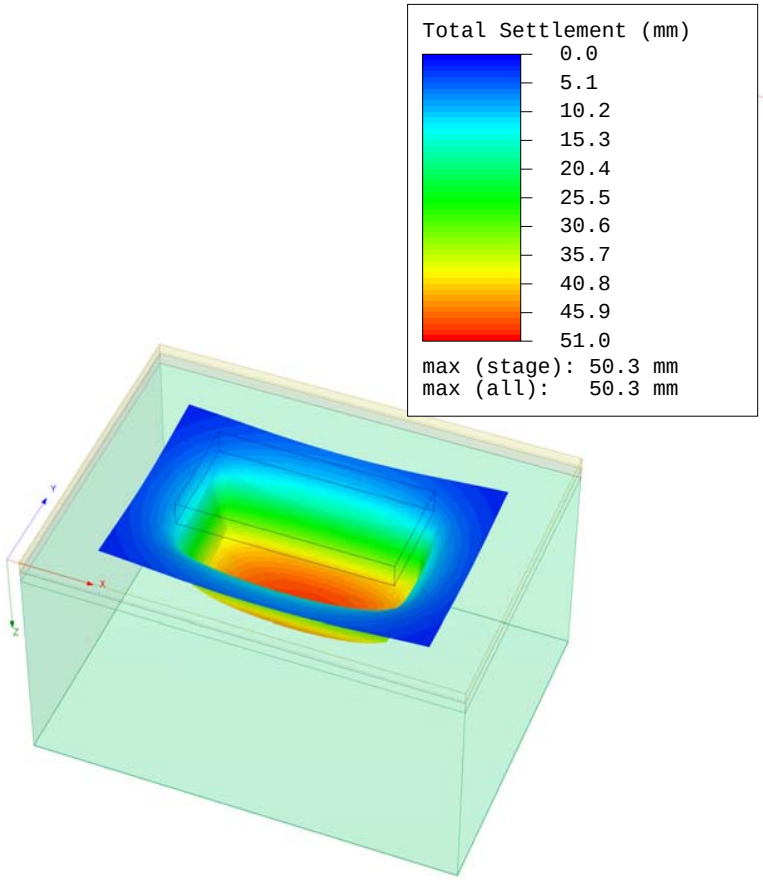




rocscience	Project	MANSOURAH - MASSARAH GOLD PROJECT	
	Analysis Description	Settlement Analysis	
	Drawn By	HUR/MF	Company RGF
	Date		File Name 3. Process Plant-1 (Z-2).s3z

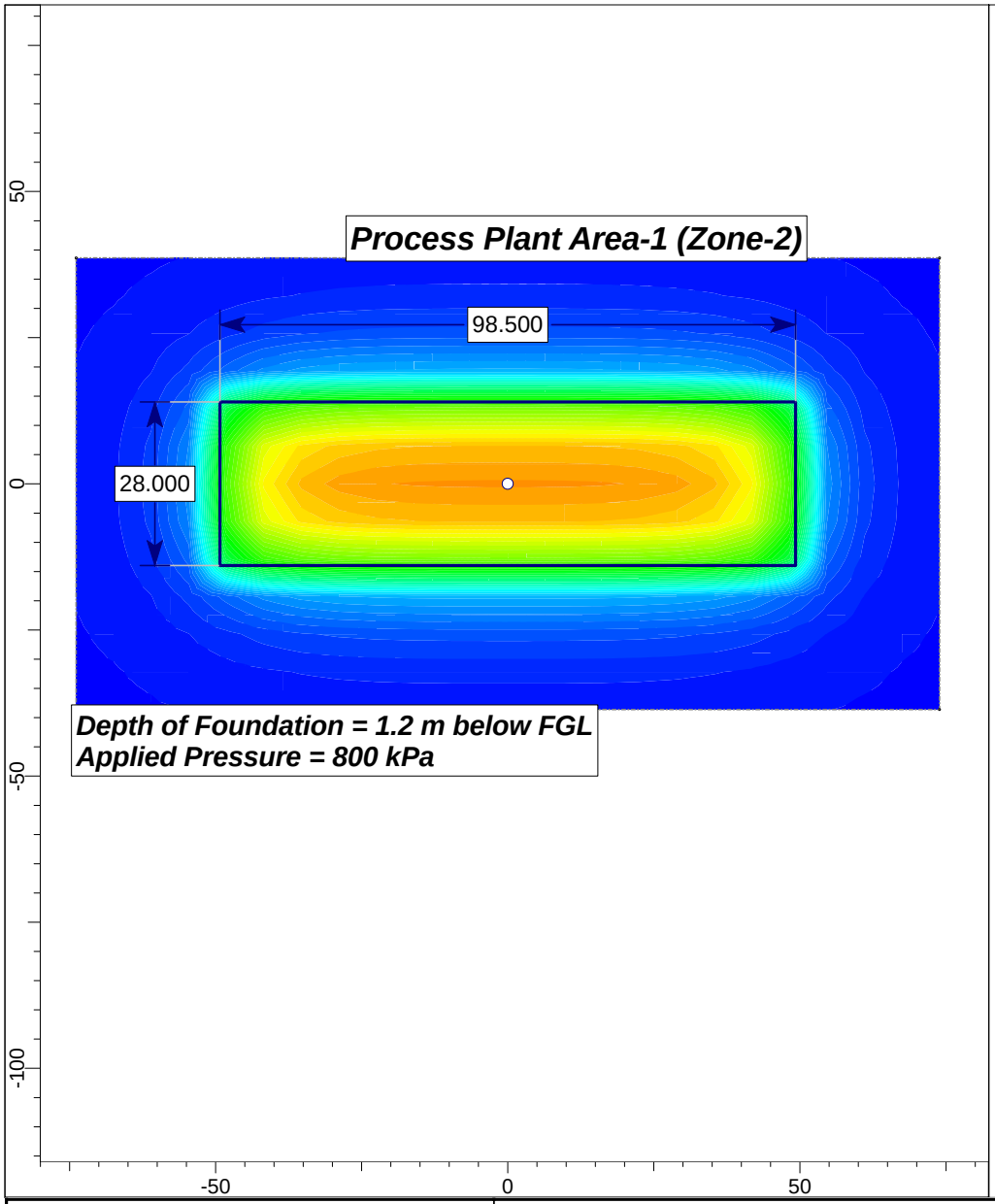
SETTLE3D 4.016

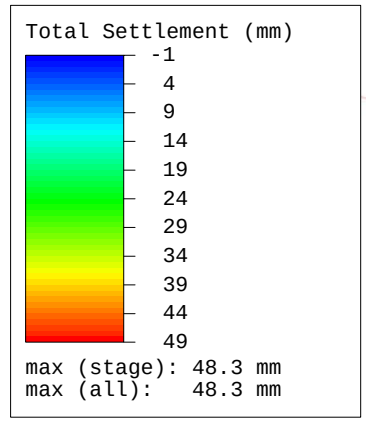


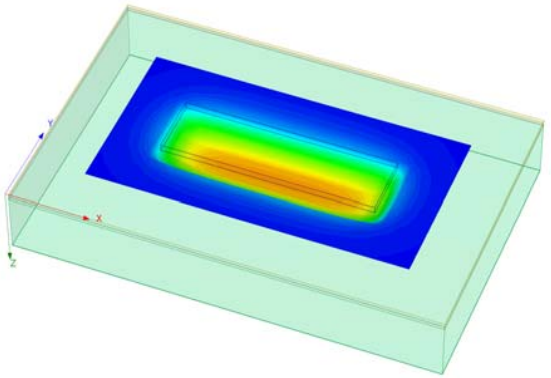


rocscience	Project	MANSOURAH - MASSARAH GOLD PROJECT	
	Analysis Description	Settlement Analysis	
	Drawn By	HUR/MF	Company RGF
	Date		File Name 3. Process Plant-1 (Z-2).s3z

SETTLE3D 4.016



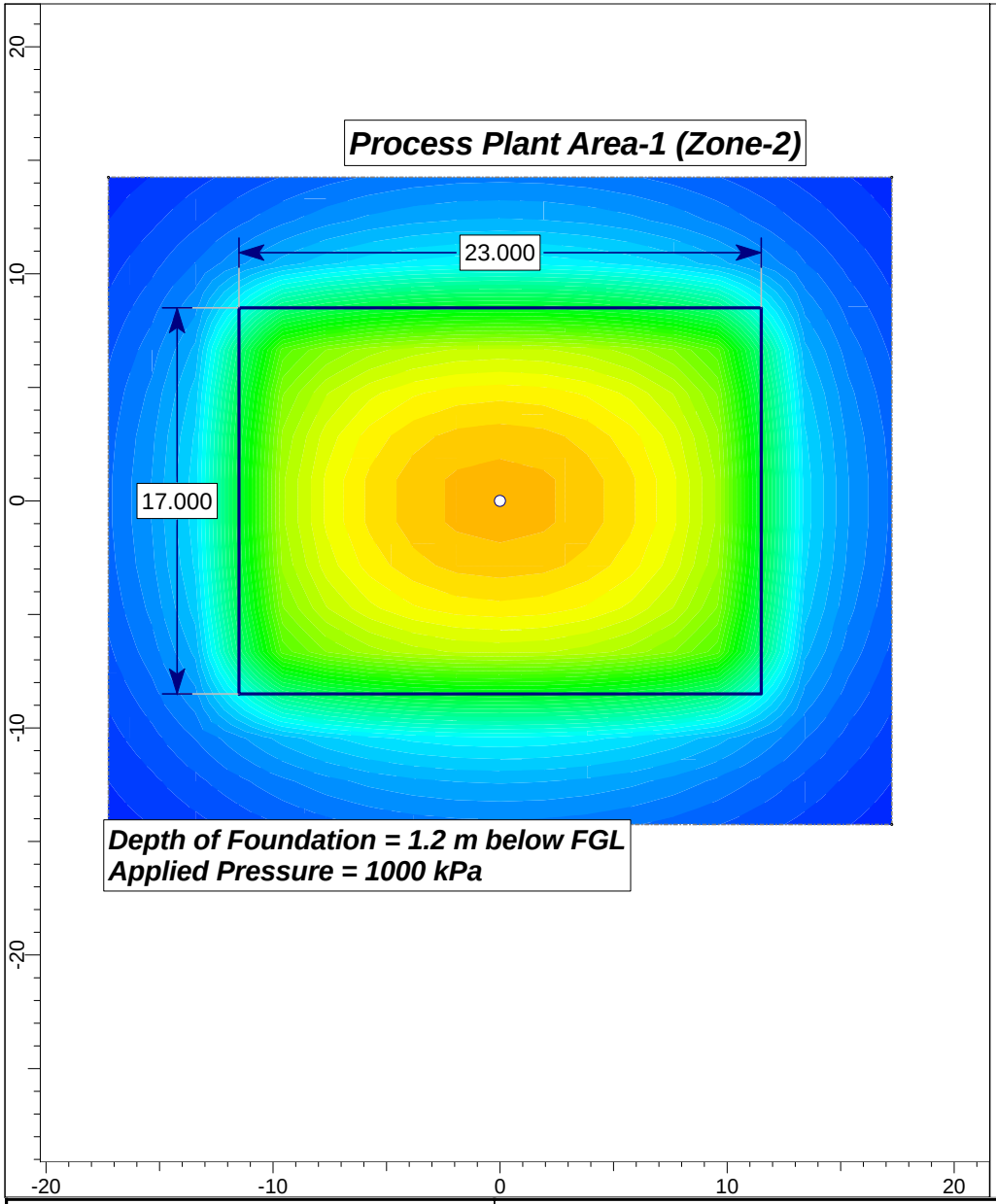




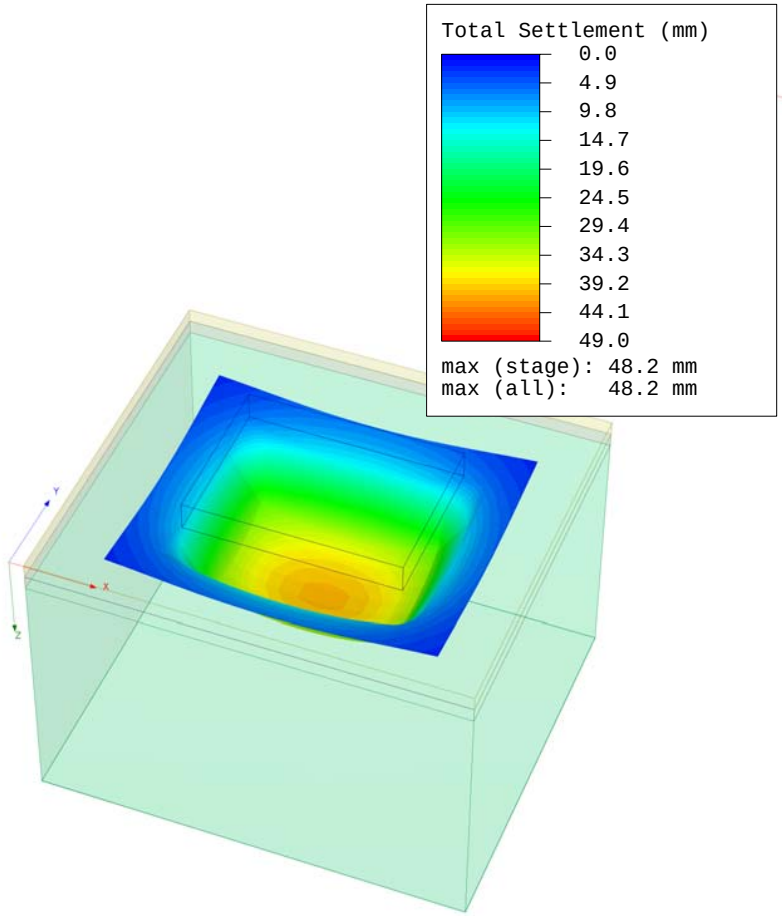
rocscience	Project	MANSOURAH - MASSARAH GOLD PROJECT	
	Analysis Description	Settlement Analysis	
	Drawn By	HUR/MF	Company RGF
	Date		File Name 3. Process Plant-1 (Z-2).s3z

SETTLE3D 4.016

Process Plant Area-1 (Zone-2)

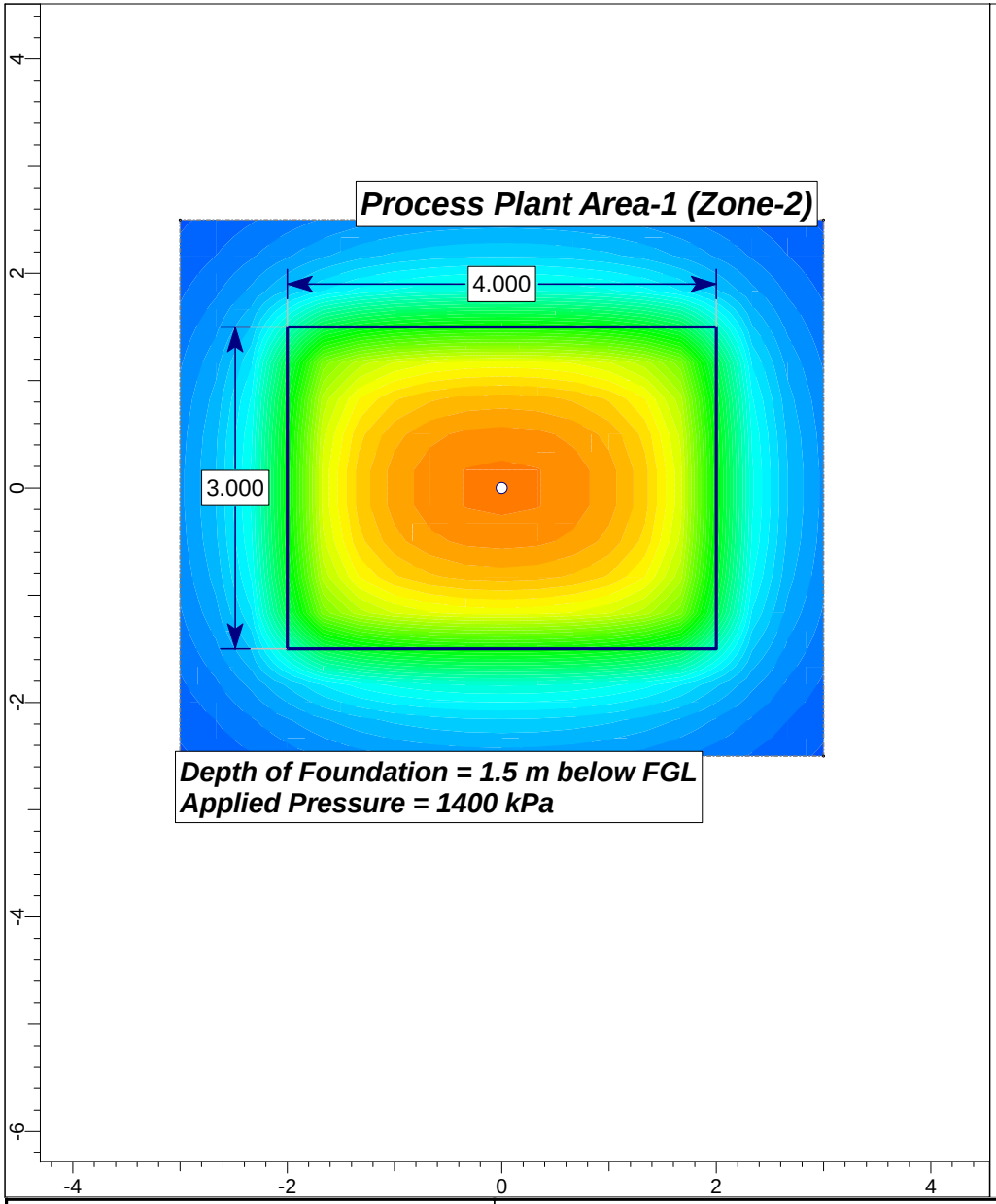


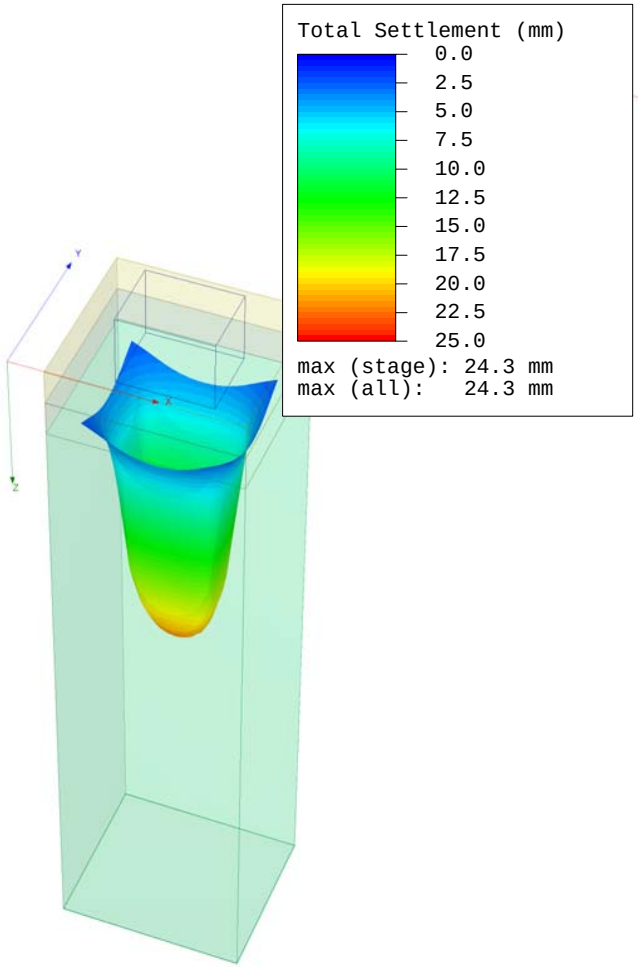
Depth of Foundation = 1.2 m below FGL
Applied Pressure = 1000 kPa



rocscience	Project	MANSOURAH - MASSARAH GOLD PROJECT	
	Analysis Description	Settlement Analysis	
	Drawn By	HUR/MF	Company RGF
	Date		File Name 3. Process Plant-1 (Z-2).s3z

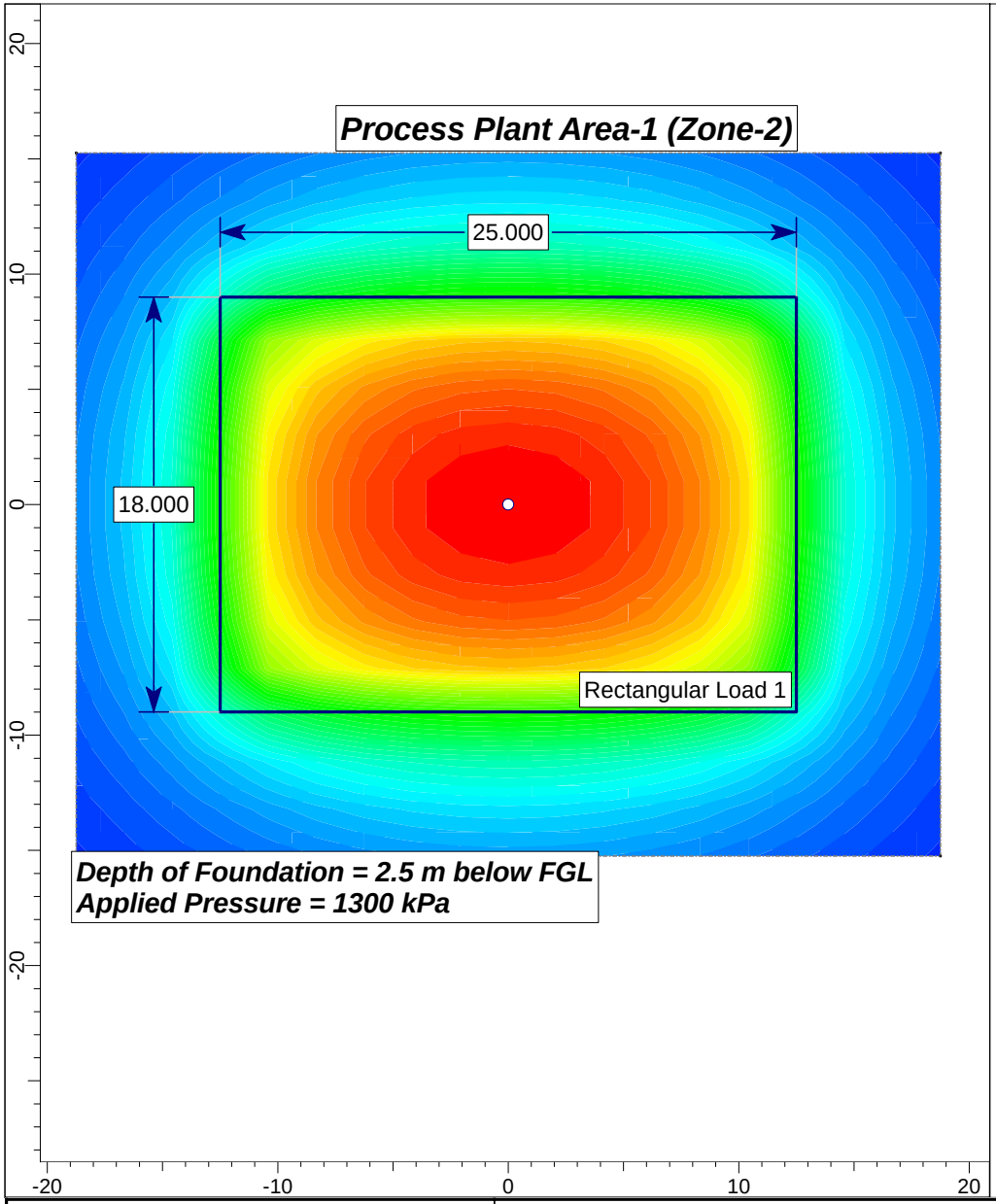
SETTLE3D 4.016

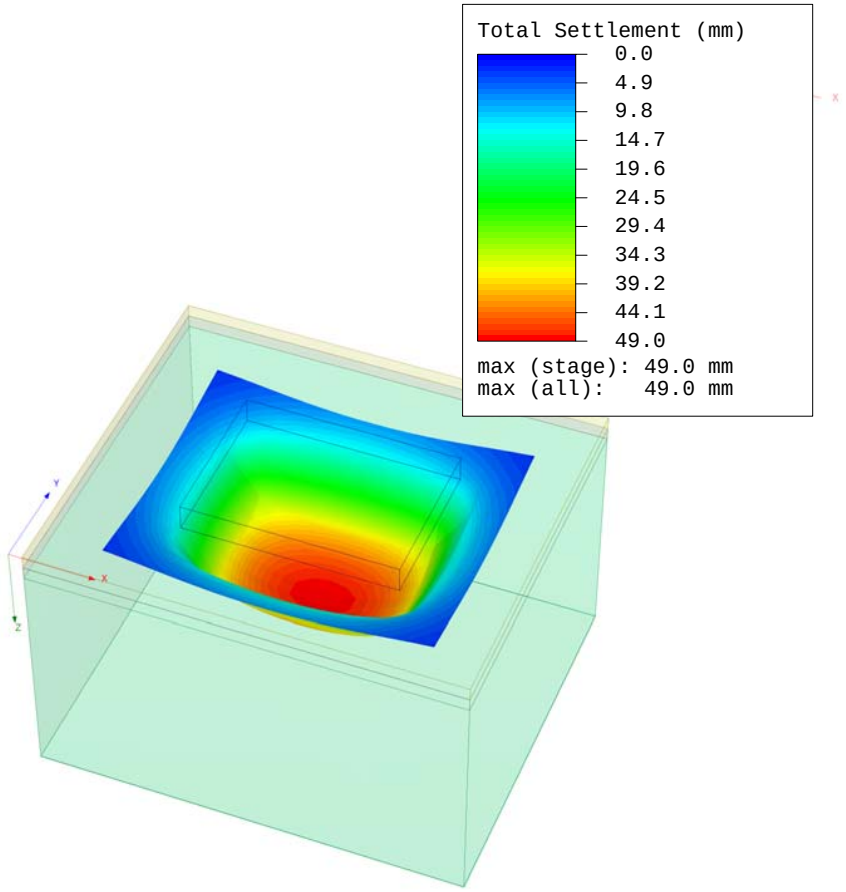




rocscience	Project	MANSOURAH - MASSARAH GOLD PROJECT	
	Analysis Description	Settlement Analysis	
	Drawn By	HUR/MF	Company RGF
	Date		File Name 3. Process Plant-1 (Z-2).s3z

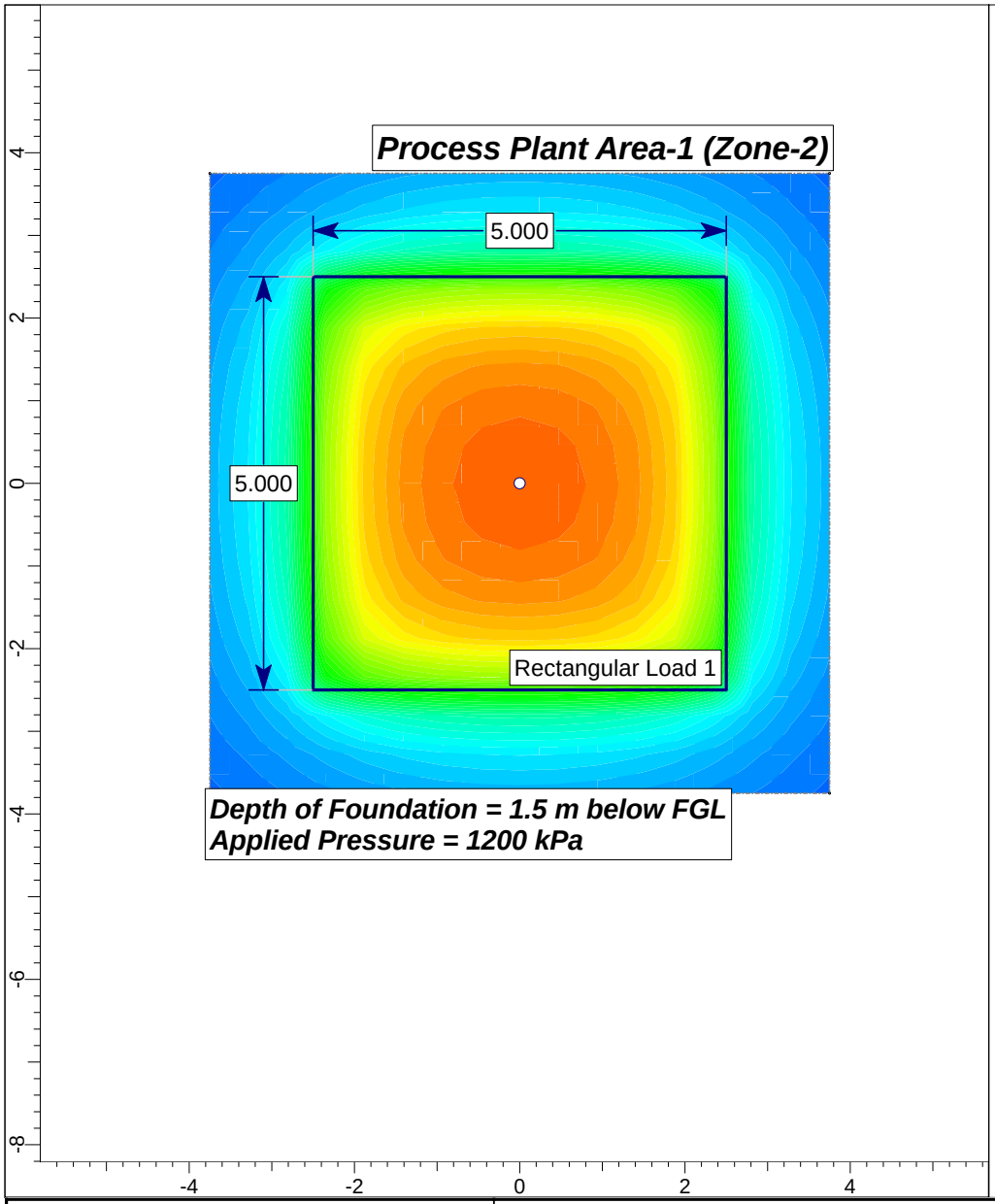
SETTLE3D 4.016

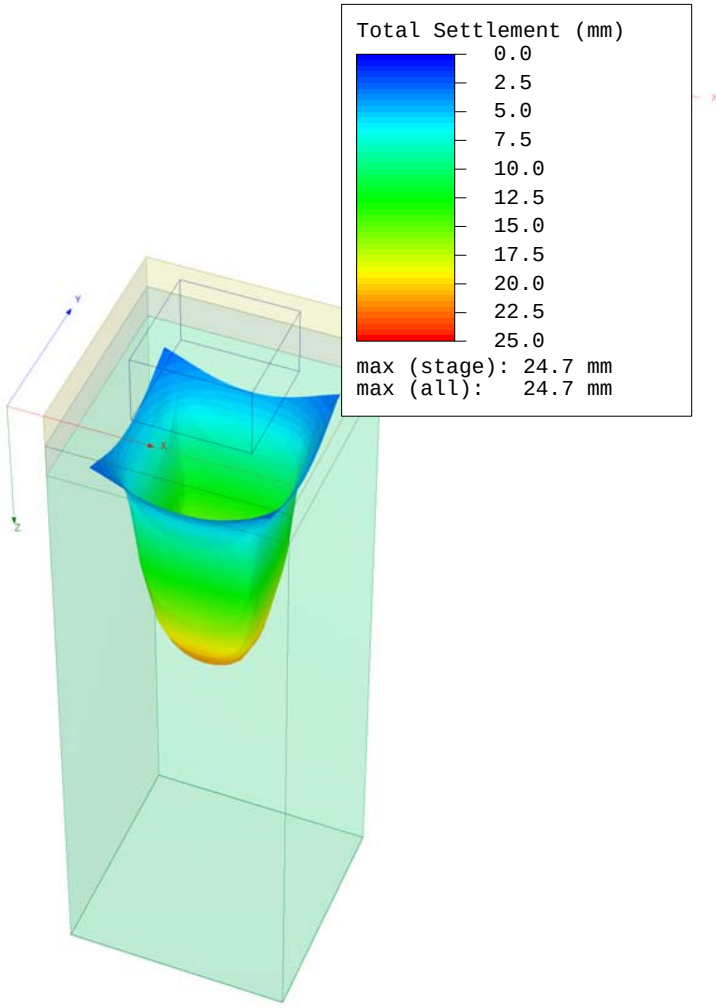




rocscience	Project	MANSOURAH - MASSARAH GOLD PROJECT	
	Analysis Description	Settlement Analysis	
	Drawn By	HUR/MF	Company RGF
	Date		File Name 3. Process Plant-1 (Z-2).s3z

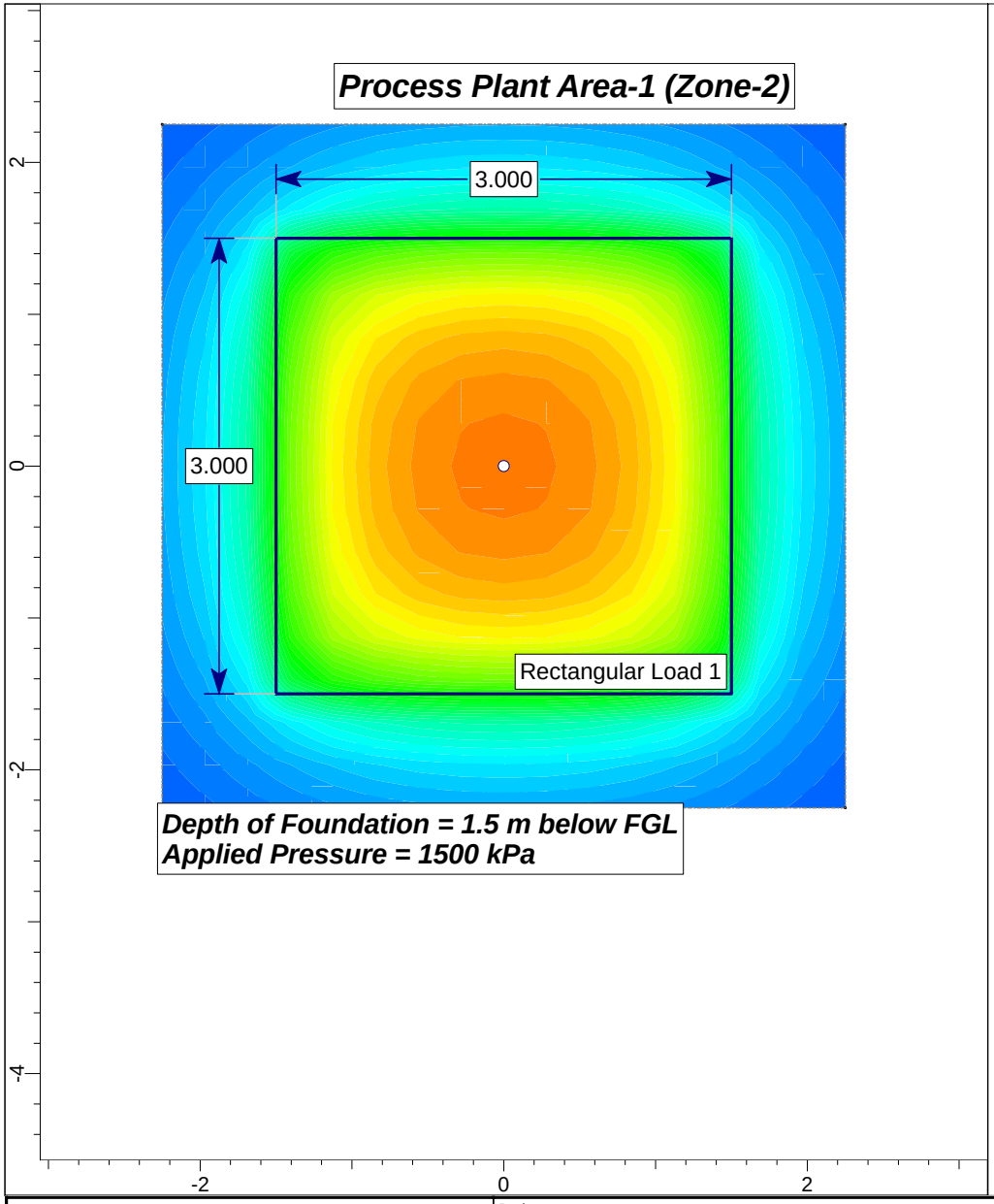
SETTLE3D 4.016

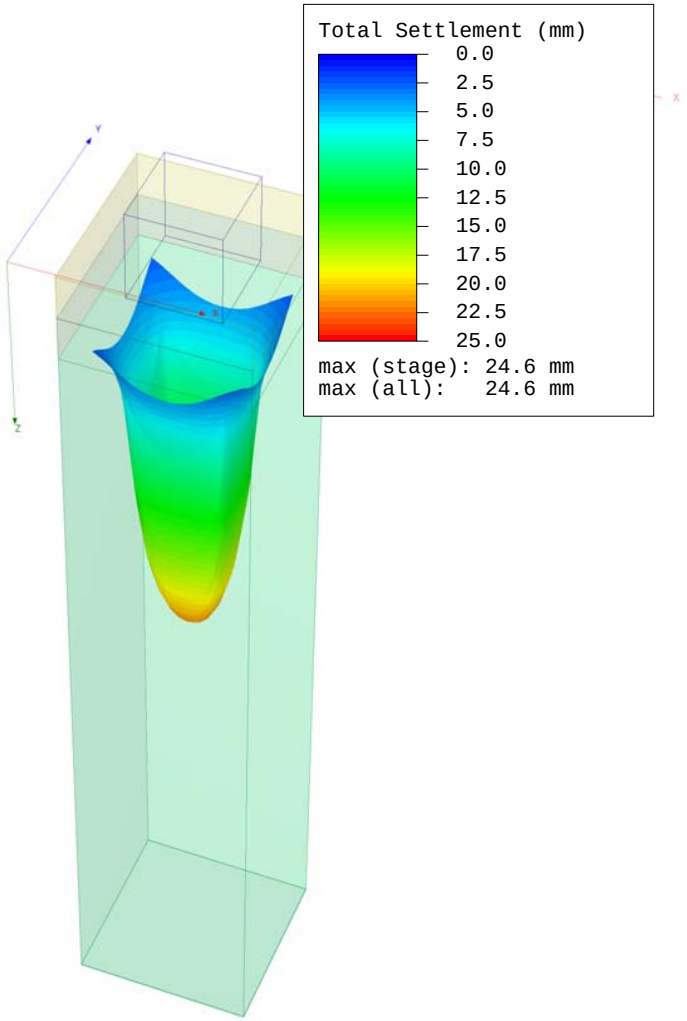




rocscience	Project	MANSOURAH - MASSARAH GOLD PROJECT	
	Analysis Description	Settlement Analysis	
	Drawn By	HUR/MF	Company RGF
	Date		File Name 3. Process Plant-1 (Z-2).s3z

SETTLE3D 4.016



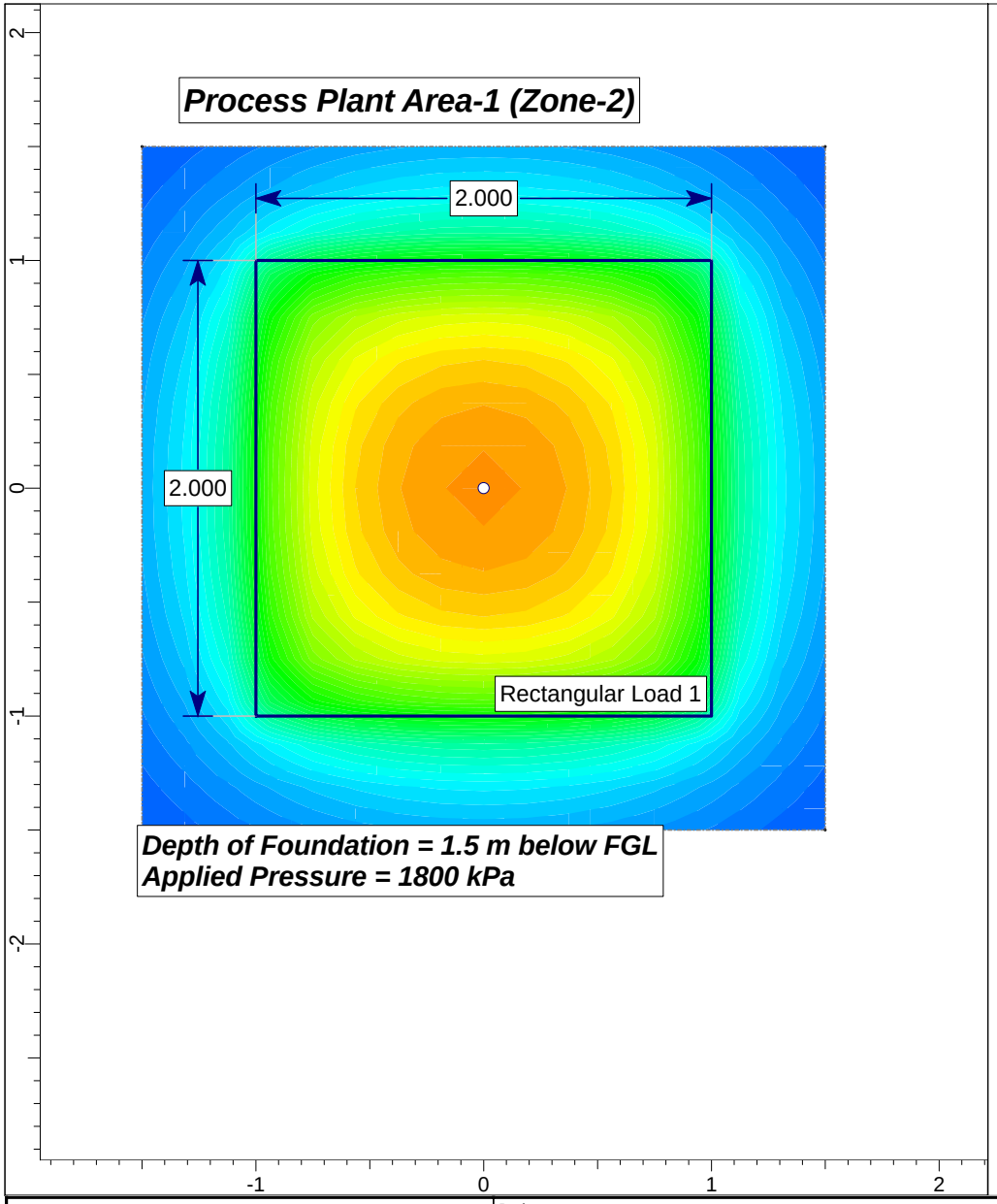


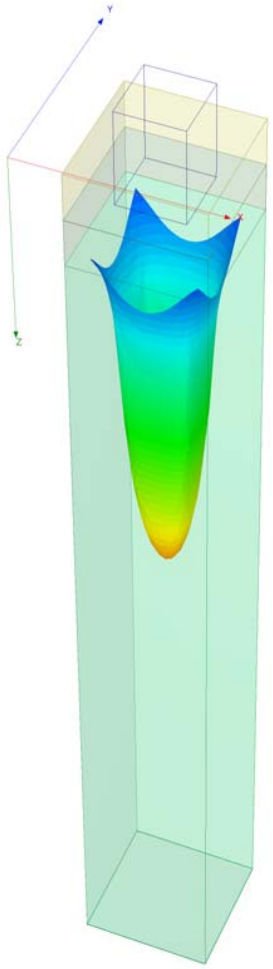
rocscience	Project	MANSOURAH - MASSARAH GOLD PROJECT	
	Analysis Description	Settlement Analysis	
	Drawn By	HUR/MF	Company RGF
	Date		File Name 3. Process Plant-1 (Z-2).s3z

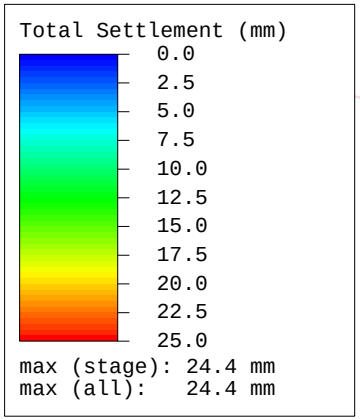
SETTLE3D 4.016

Process Plant Area-1 (Zone-2)

Depth of Foundation = 1.5 m below FGL
Applied Pressure = 1800 kPa

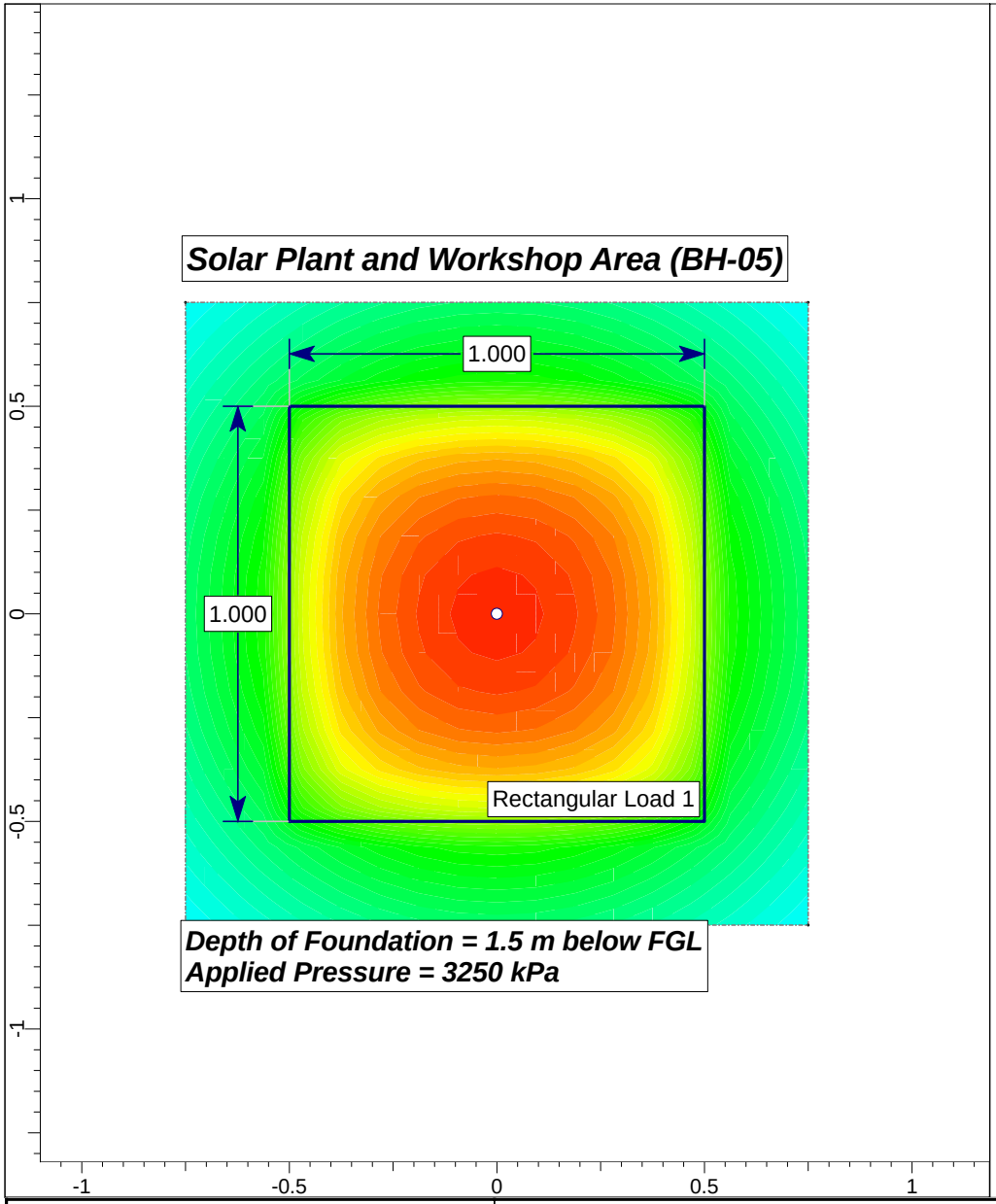


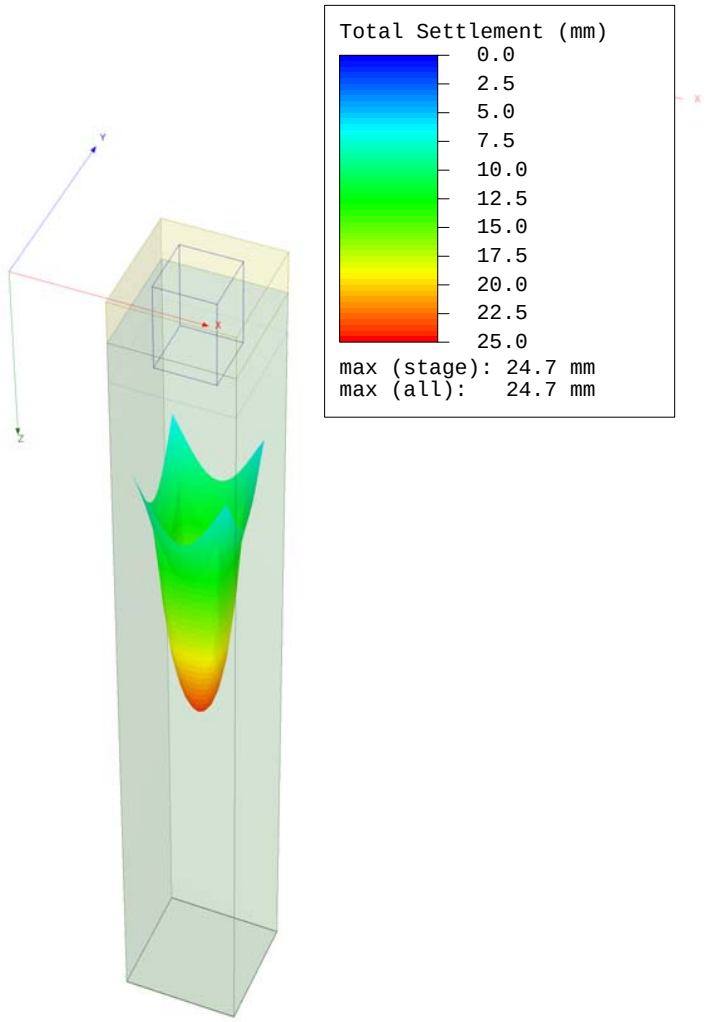




rocscience	*Project*	MANSOURAH - MASSARAH GOLD PROJECT	
	Analysis Description	Settlement Analysis	
	Drawn By	HUR/MF	*Company* RGF
	Date		*File Name* 3. Process Plant-1 (Z-2).s3z

SETTLE3D 4.016



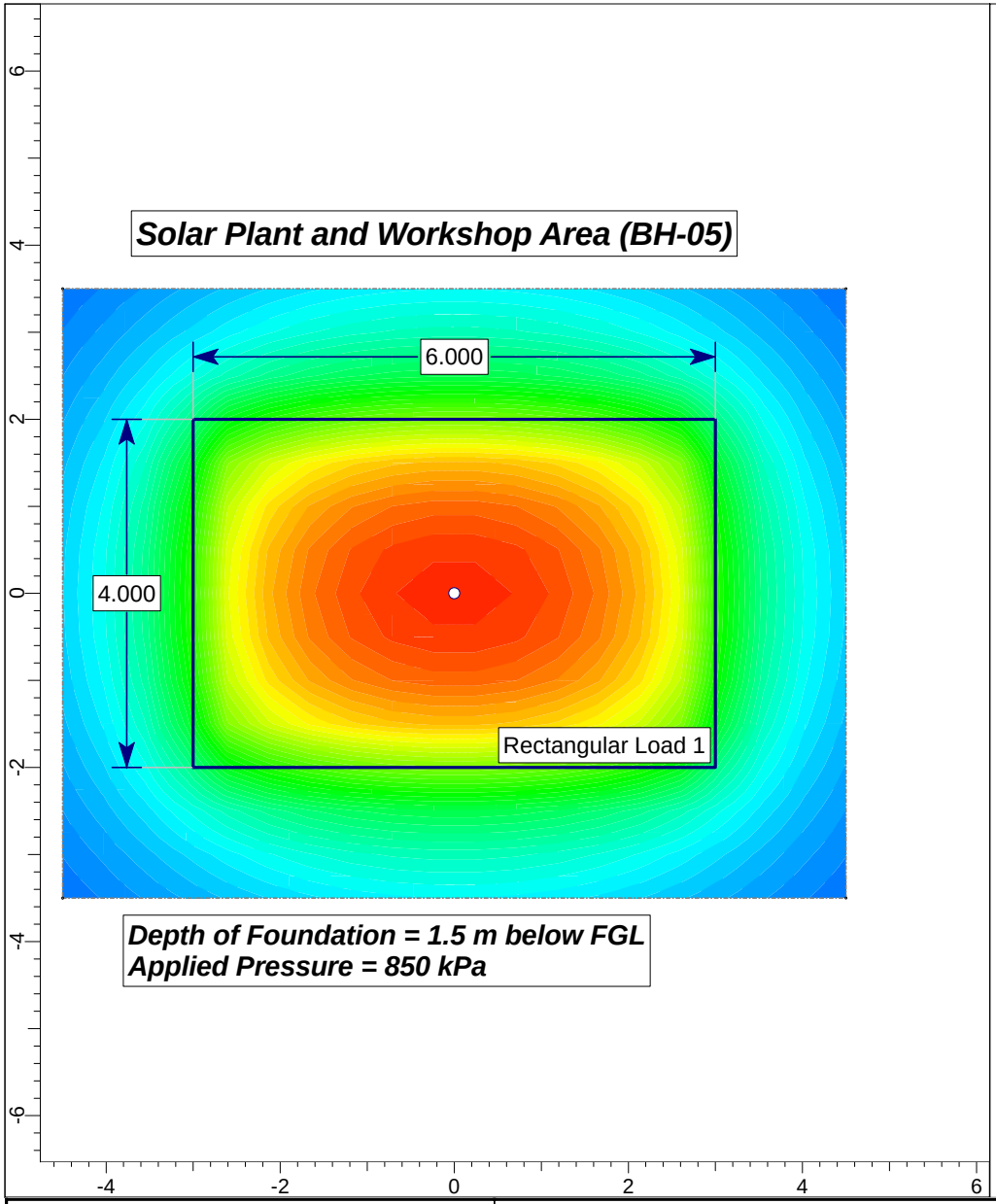


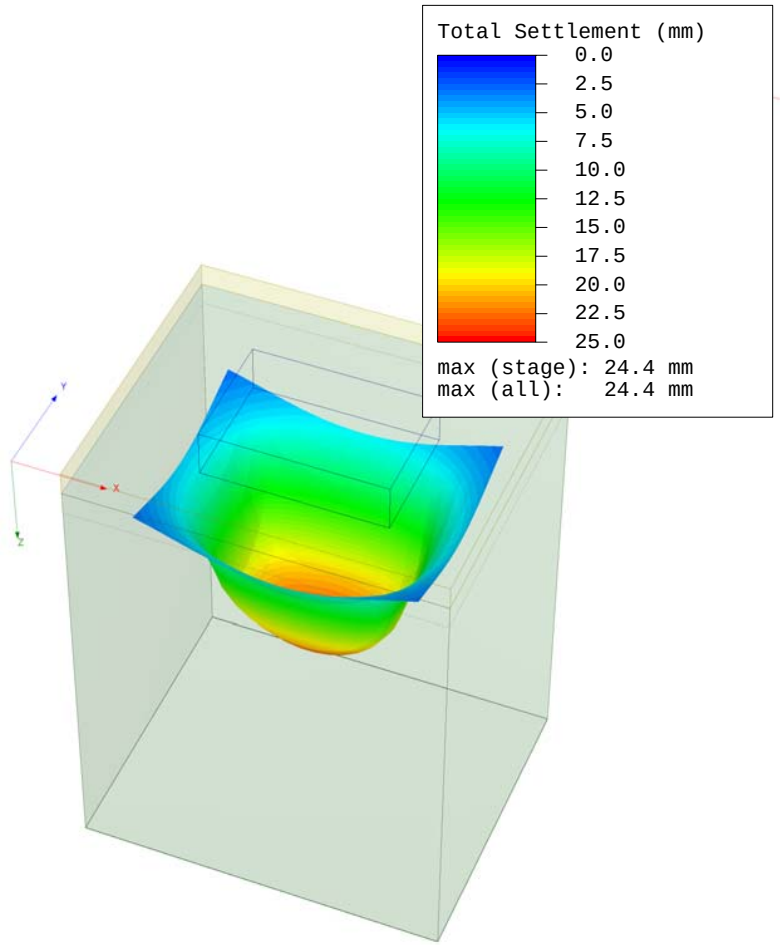
rocscience	Project	MANSOURAH - MASSARAH GOLD PROJECT	
	Analysis Description	Settlement Analysis	
	Drawn By	HUR/MF	Company RGF
	Date		File Name 6. Solar Plant & Workshop Area (BH-05).s3z

SETTLE3D 4.016

Solar Plant and Workshop Area (BH-05)

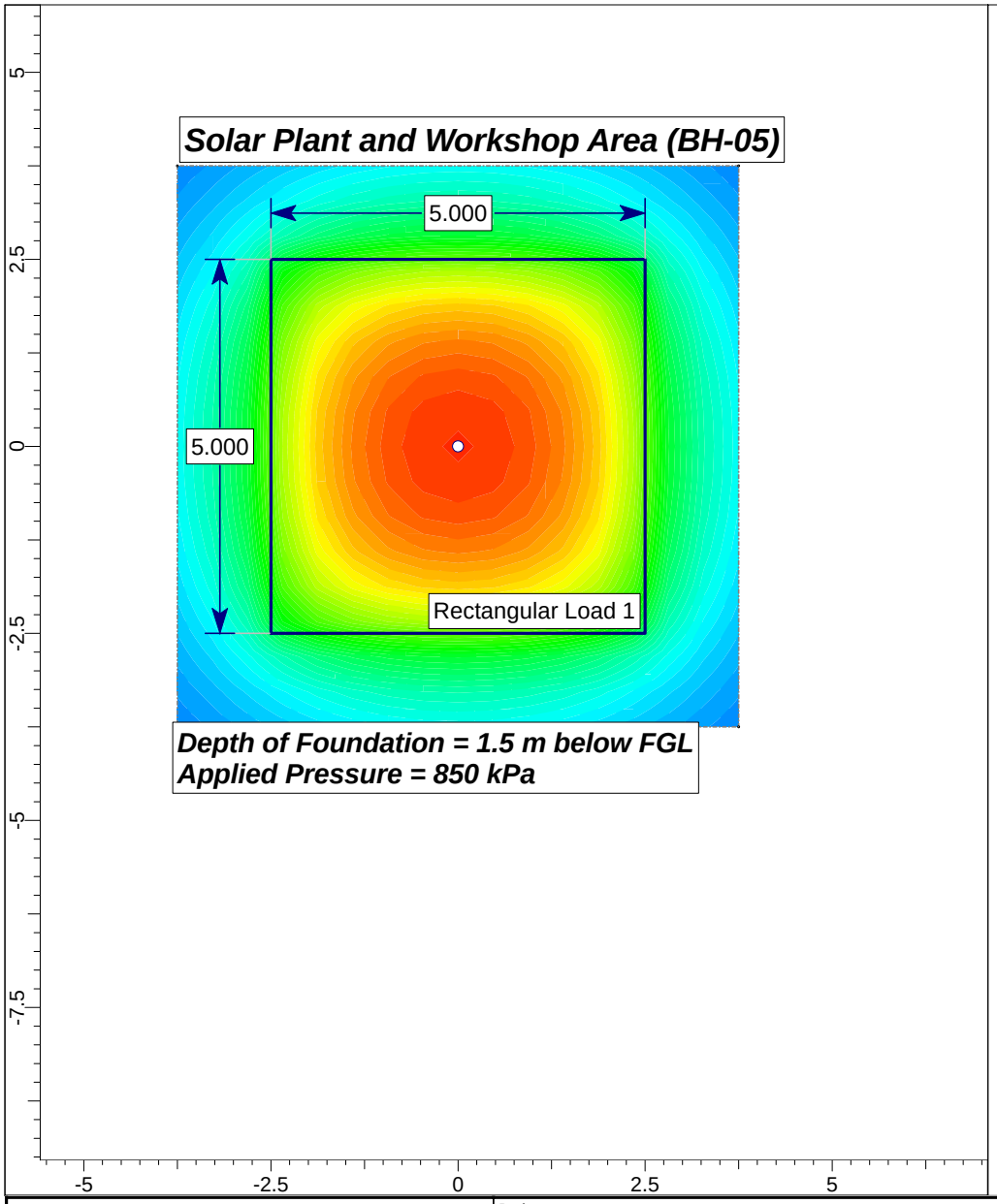
Depth of Foundation = 1.5 m below FGL
Applied Pressure = 850 kPa

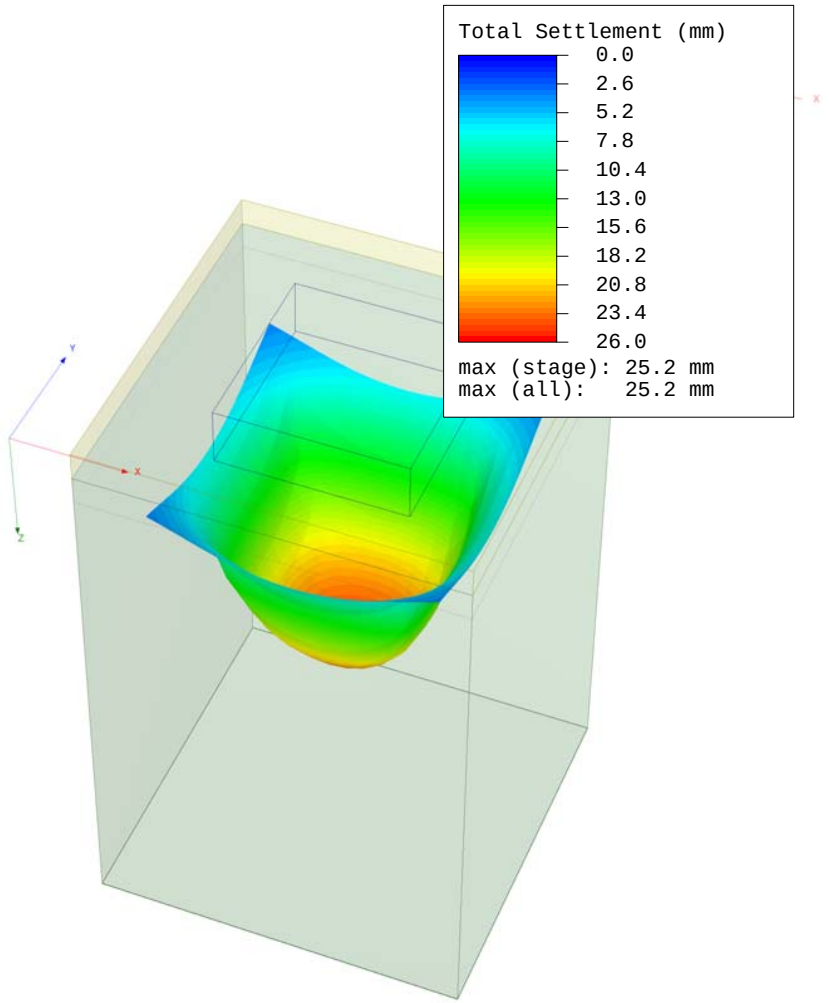




rocscience	*Project*	MANSOURAH - MASSARAH GOLD PROJECT	
	Analysis Description	Settlement Analysis	
	Drawn By	HUR/MF	*Company* RGF
	Date		*File Name* 6. Solar Plant & Workshop Area (BH-05).s3z

SETTLE3D 4.016

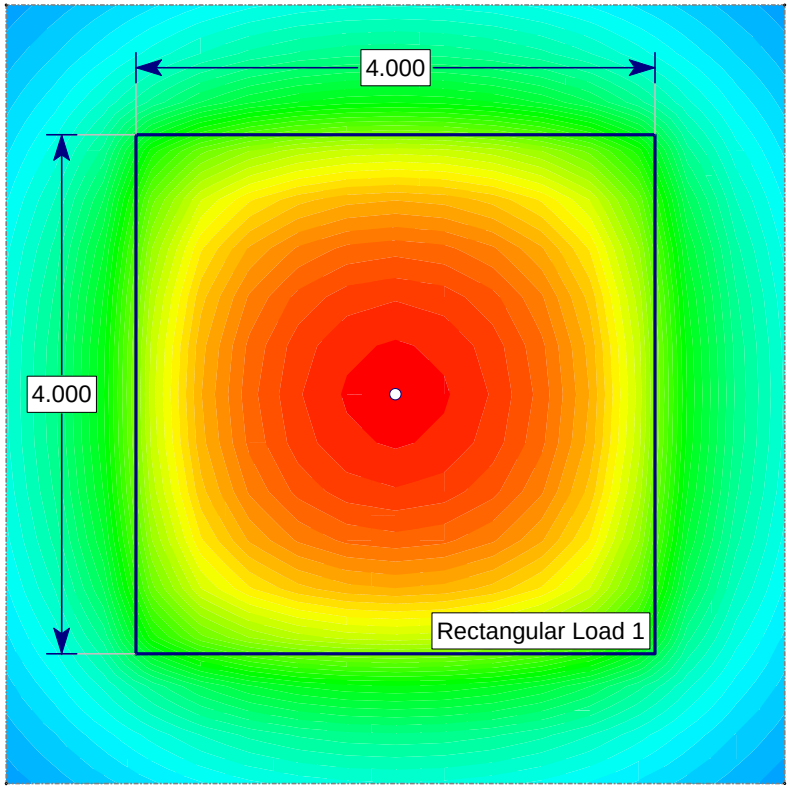




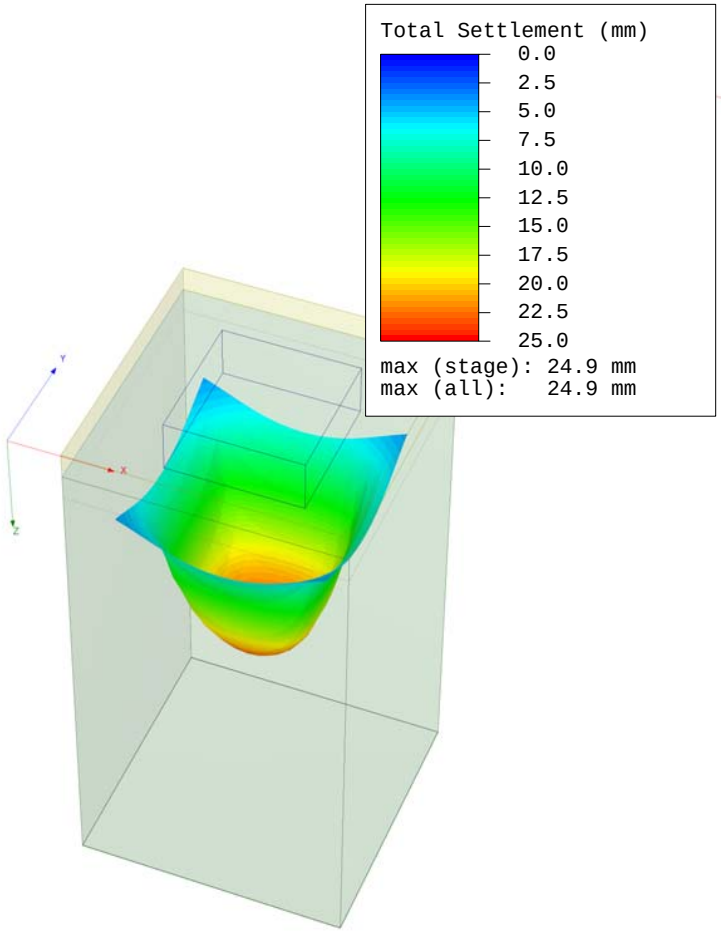
rocscience	Project	MANSOURAH - MASSARAH GOLD PROJECT	
	Analysis Description	Settlement Analysis	
	Drawn By	HUR/MF	Company RGF
	Date		File Name 6. Solar Plant & Workshop Area (BH-05).s3z

SETTLE3D 4.016

Solar Plant and Workshop Area (BH-05)

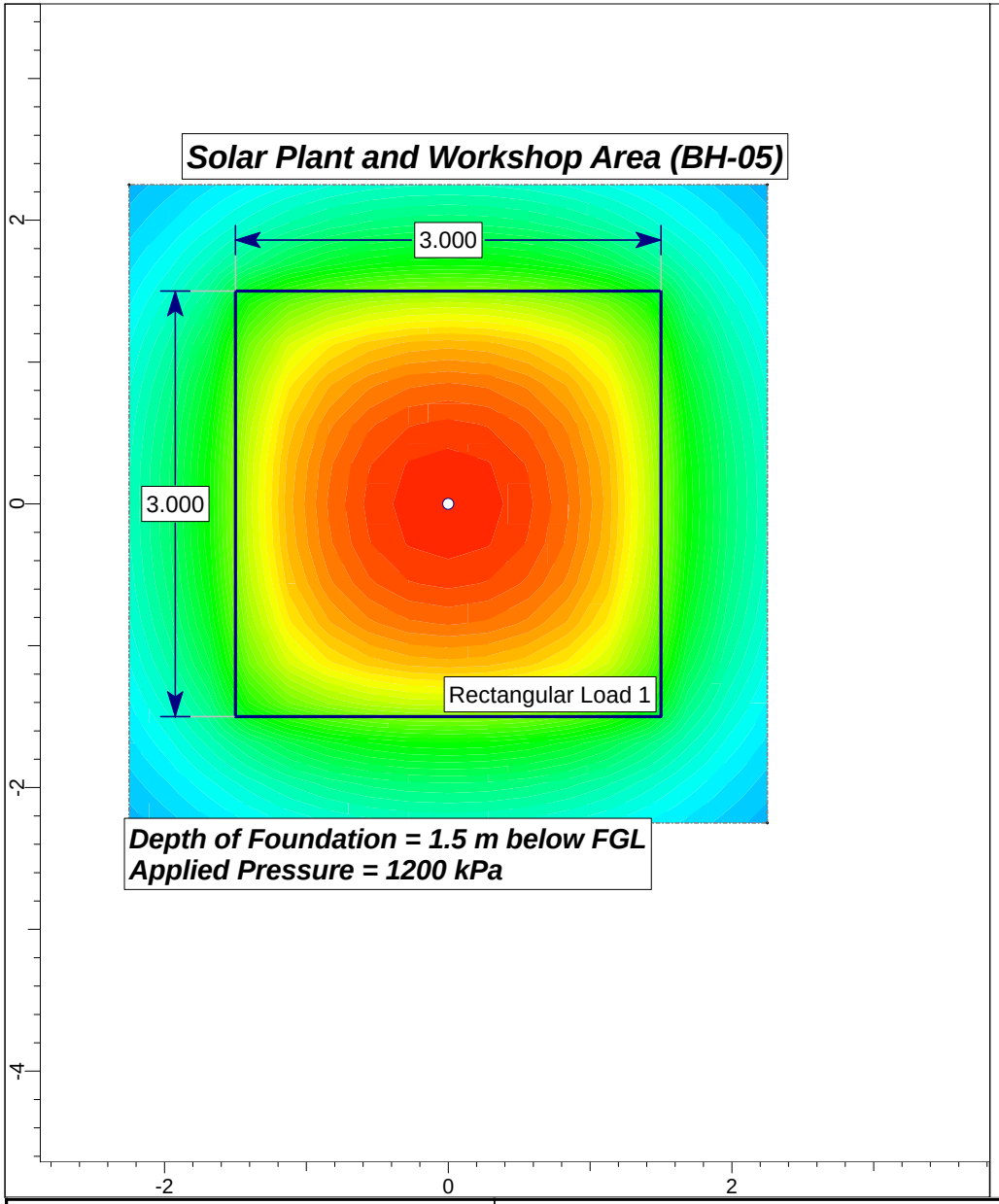


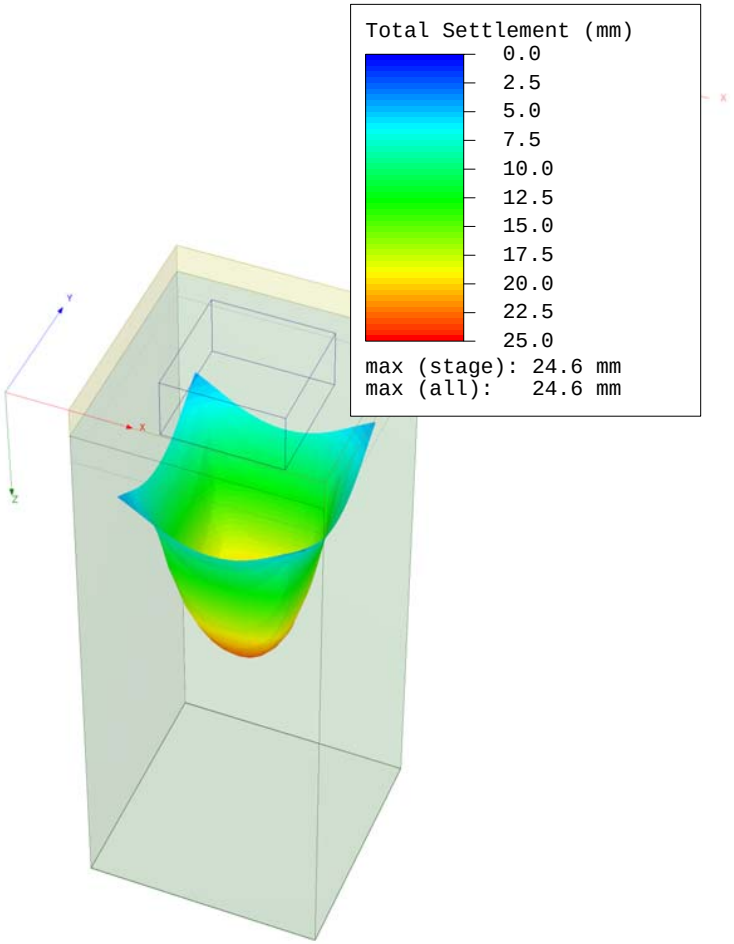
Depth of Foundation = 1.5 m below FGL
Applied Pressure = 950 kPa



rocscience	*Project*	MANSOURAH - MASSARAH GOLD PROJECT	
	Analysis Description	Settlement Analysis	
	Drawn By	HUR/MF	*Company* RGF
	Date		*File Name* 6. Solar Plant & Workshop Area (BH-05).s3z

SETTLE3D 4.016

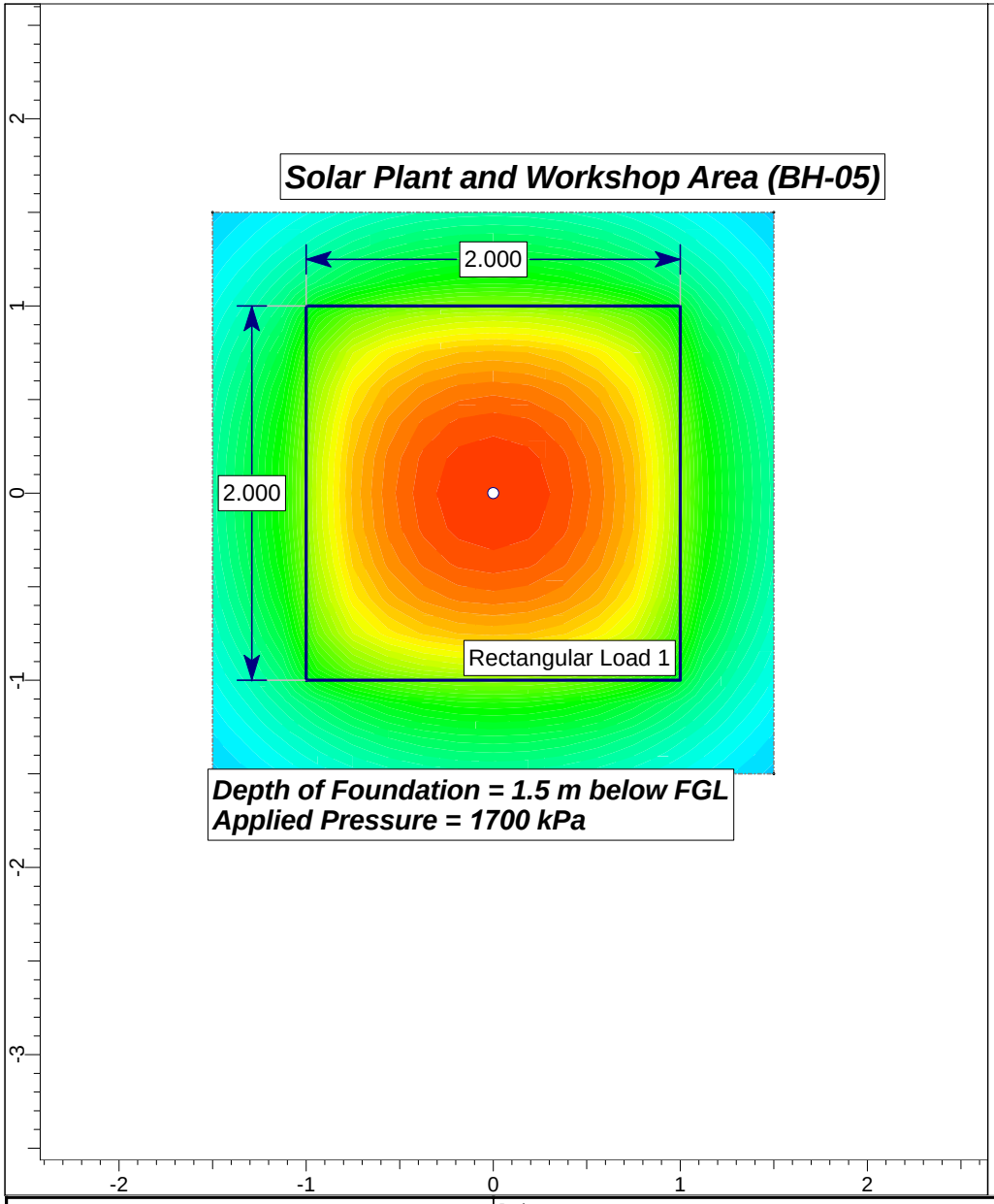




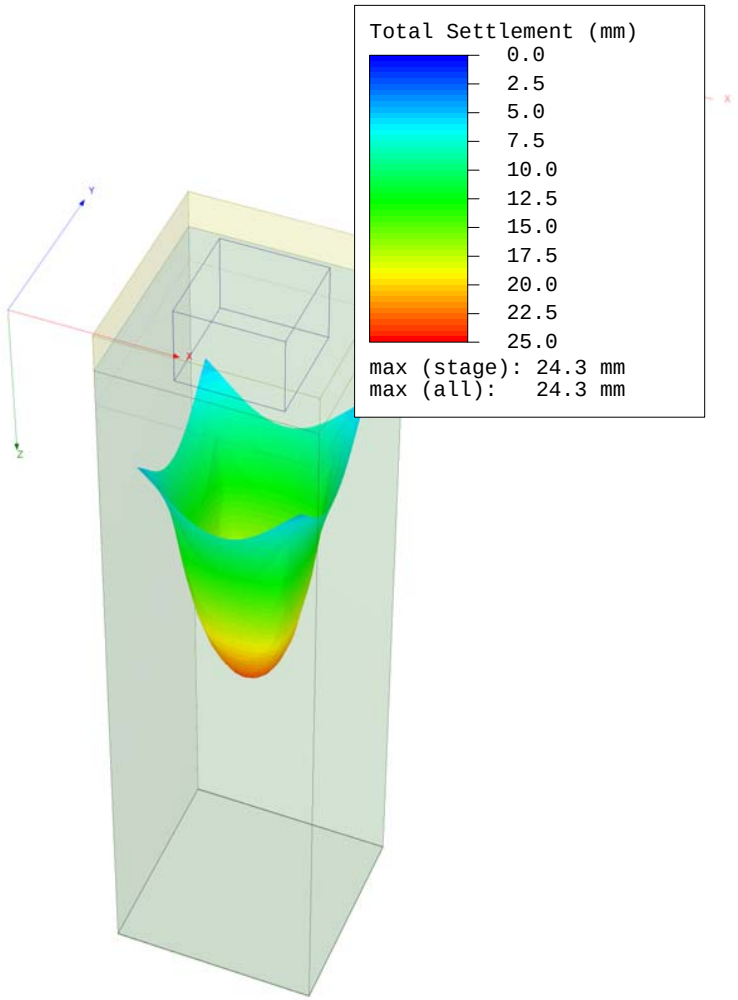
rocscience	Project	MANSOURAH - MASSARAH GOLD PROJECT	
	Analysis Description	Settlement Analysis	
	Drawn By	HUR/MF	Company RGF
	Date		File Name 6. Solar Plant & Workshop Area (BH-05).s3z

SETTLE3D 4.016

Solar Plant and Workshop Area (BH-05)



Depth of Foundation = 1.5 m below FGL
Applied Pressure = 1700 kPa



rocscience	*Project*	MANSOURAH - MASSARAH GOLD PROJECT	
	Analysis Description	Settlement Analysis	
	Drawn By	HUR/MF	*Company* RGF
	Date		*File Name* 6. Solar Plant & Workshop Area (BH-05).s3z

SETTLE3D 4.016

Settle3D Analysis Information
MANSOURAH - MASSARAH GOLD PROJECT

Project Settings

Document Name	6. Solar Plant & Workshop Area (BH-05)
Project Title	MANSOURAH - MASSARAH GOLD PROJECT
Analysis	Settlement Analysis
Author	HUR/MF
Company	RGF
Stress Computation Method	Boussinesq

Calculate settlement with mean stress

Use properties from first layer to calculate layered stresses

Stage Settings

Stage #	Name
1	Stage 1

Results

Time taken to compute: 0.0358325 seconds

Stage: Stage 1

Data Type	Minimum	Maximum
Total Settlement [mm]	0	25.1704
Total Consolidation Settlement [mm]	0	0
Virgin Consolidation Settlement [mm]	0	0
Recompression Consolidation Settlement [mm]	0	0
Immediate Settlement [mm]	0	25.1704
Loading Stress ZZ [kPa]	0	800
Loading Stress XX [kPa]	-5.26259	682.173
Loading Stress YY [kPa]	-5.26259	682.173
Mean Stress [kPa]	0	693.333
Total Stress ZZ [kPa]	0	833.905
Total Stress XX [kPa]	0	711.423
Total Stress YY [kPa]	0	711.423
Modulus of Subgrade Reaction (Total) [kPa/m]	0	9.7386e+006
Modulus of Subgrade Reaction (Immediate) [kPa/m]	0	9.7386e+006
Modulus of Subgrade Reaction (Consolidation) [kPa/m]	0	0
Total Strain	-8.10702e-005	0.0043261
Degree of Consolidation [%]	0	0
Pre-consolidation Stress [kPa]	0.7125	832.908
Over-consolidation Ratio	1	1
Void Ratio	0	0
Hydroconsolidation Settlement [mm]	0	0
Undrained Shear Strength	0	0

Loads

1. Rectangular Load: "Rectangular Load 1"

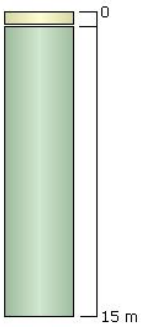
Length	5 m
Width	5 m
Rotation angle	0 degrees
Load Type	Flexible
Area of Load	25 m^2
Load	800 kPa
Depth	1.5 m
Installation Stage	Stage 1

Coordinates

X [m]	Y [m]
-2.5	-2.5
2.5	-2.5
2.5	2.5
-2.5	2.5

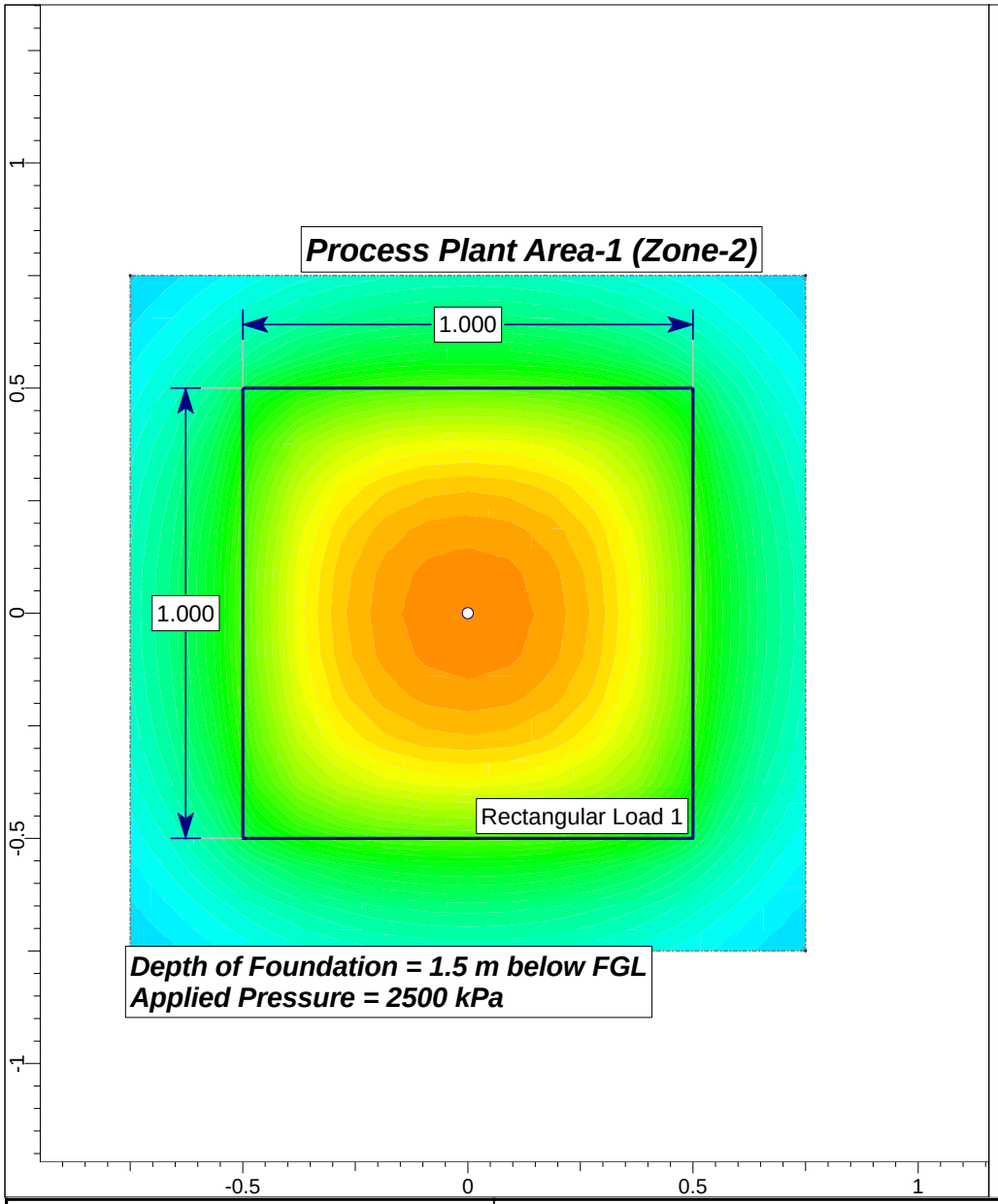
Soil Layers

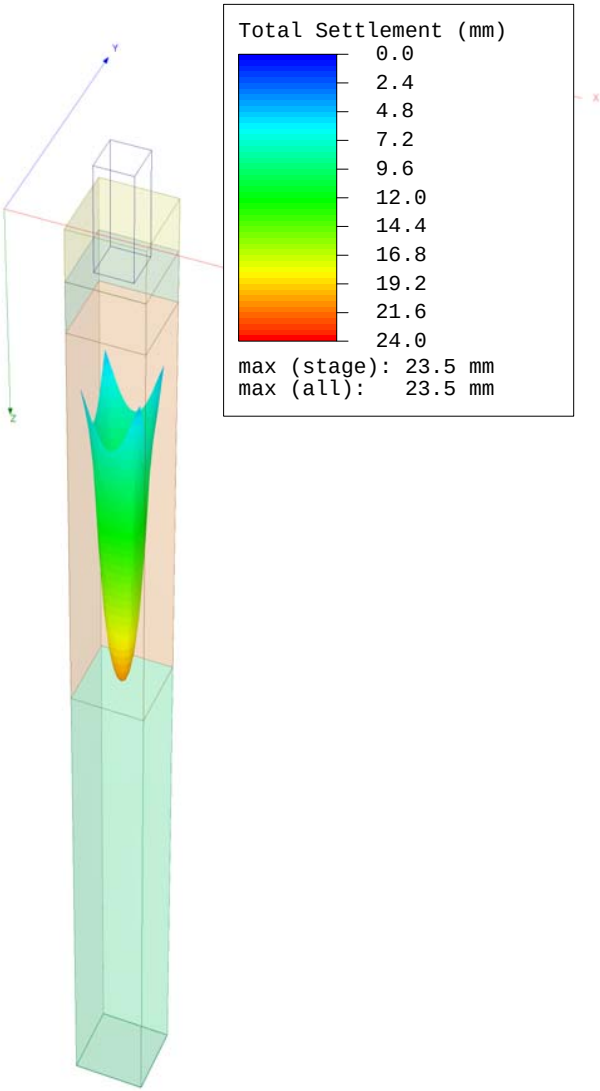
Layer #	Type	Thickness [m]	Depth [m]
1	Soil 1	0.75	0
2	Soil 2	14.25	0.75



Soil Properties

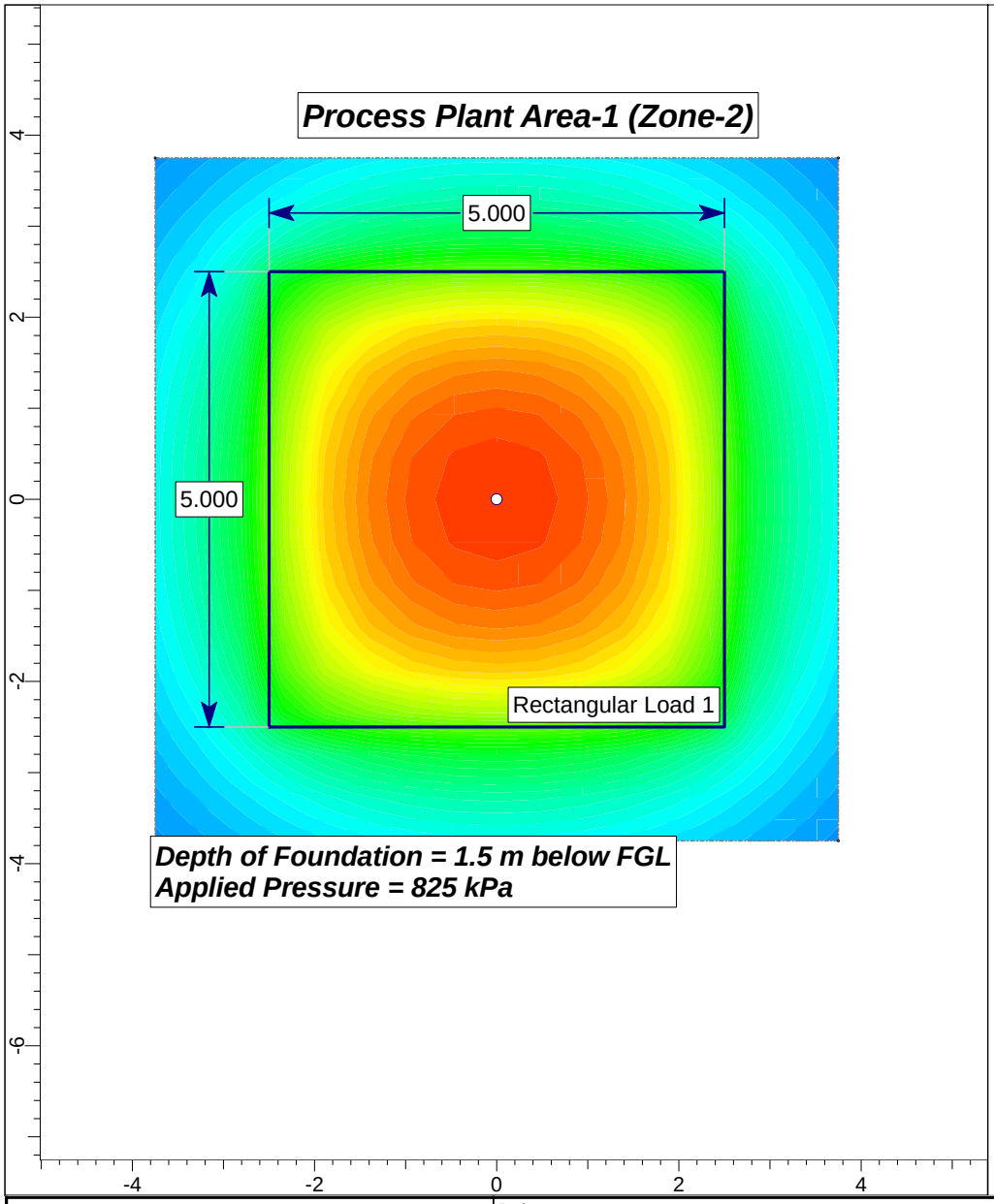
Property	Soil 1	Soil 2
Color		
Unit Weight [kN/m^3]	19	20
Poisson's Ratio	0.3	0.3
K0	1	1
Immediate Settlement	Enabled	Enabled
E [kPa]	40000	134000
Eur [kPa]	40000	134000
Undrained Su A [kN/m2]	0	0
Undrained Su S	0	0
Undrained Su m	0	0

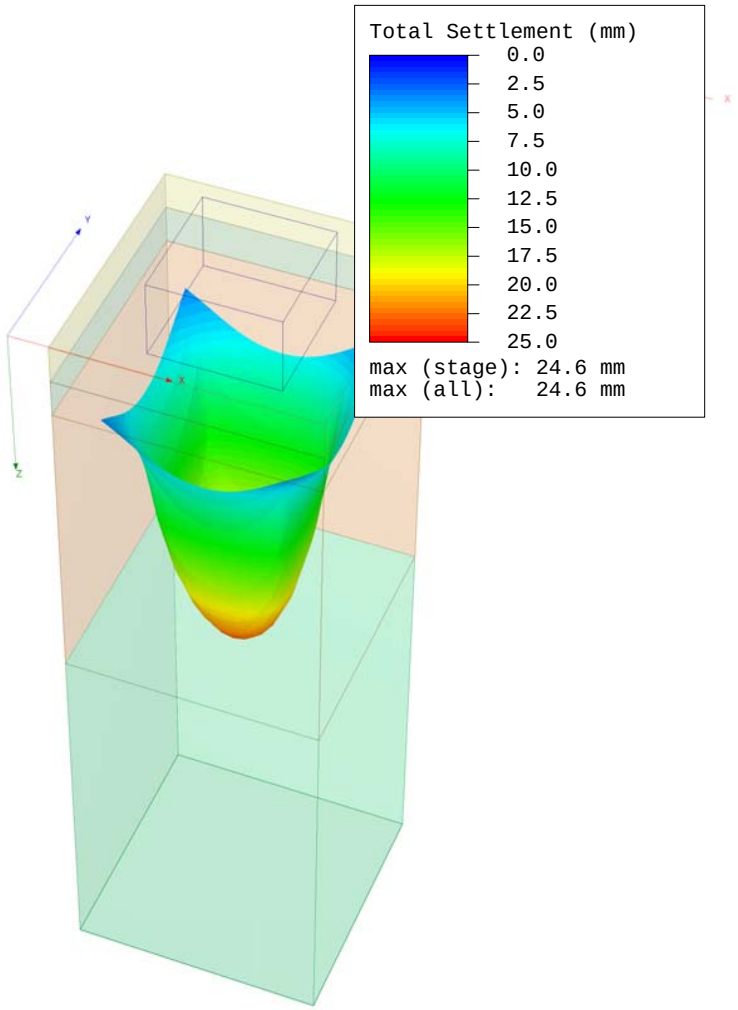




rocscience	Project	MANSOURAH - MASSARAH GOLD PROJECT	
	Analysis Description	Settlement Analysis	
	Drawn By	HUR/MF	Company RGF
	Date		File Name 9. TMT.s3z

SETTLE3D 4.016

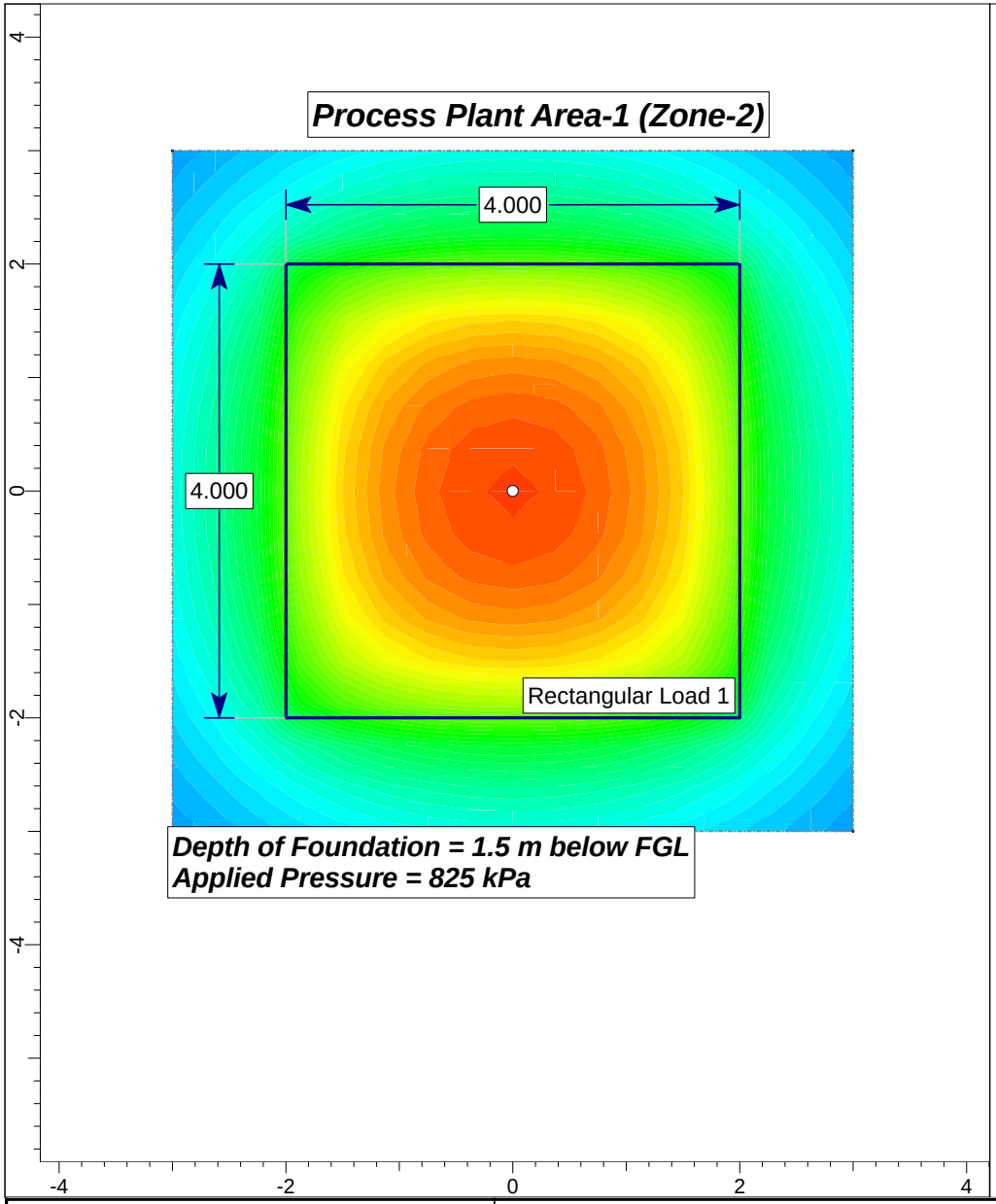




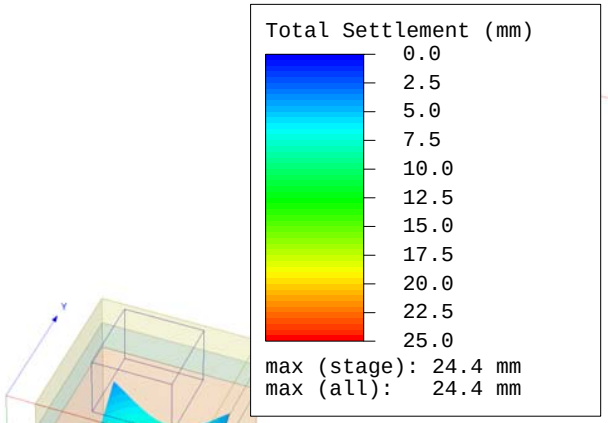
rocscience	Project	MANSOURAH - MASSARAH GOLD PROJECT	
	Analysis Description	Settlement Analysis	
	Drawn By	HUR/MF	Company RGF
	Date		File Name 9. TMT.s3z

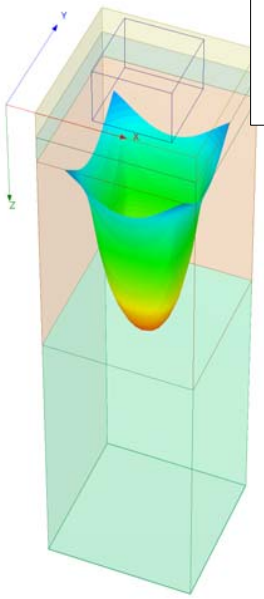
SETTLE3D 4.016

Process Plant Area-1 (Zone-2)



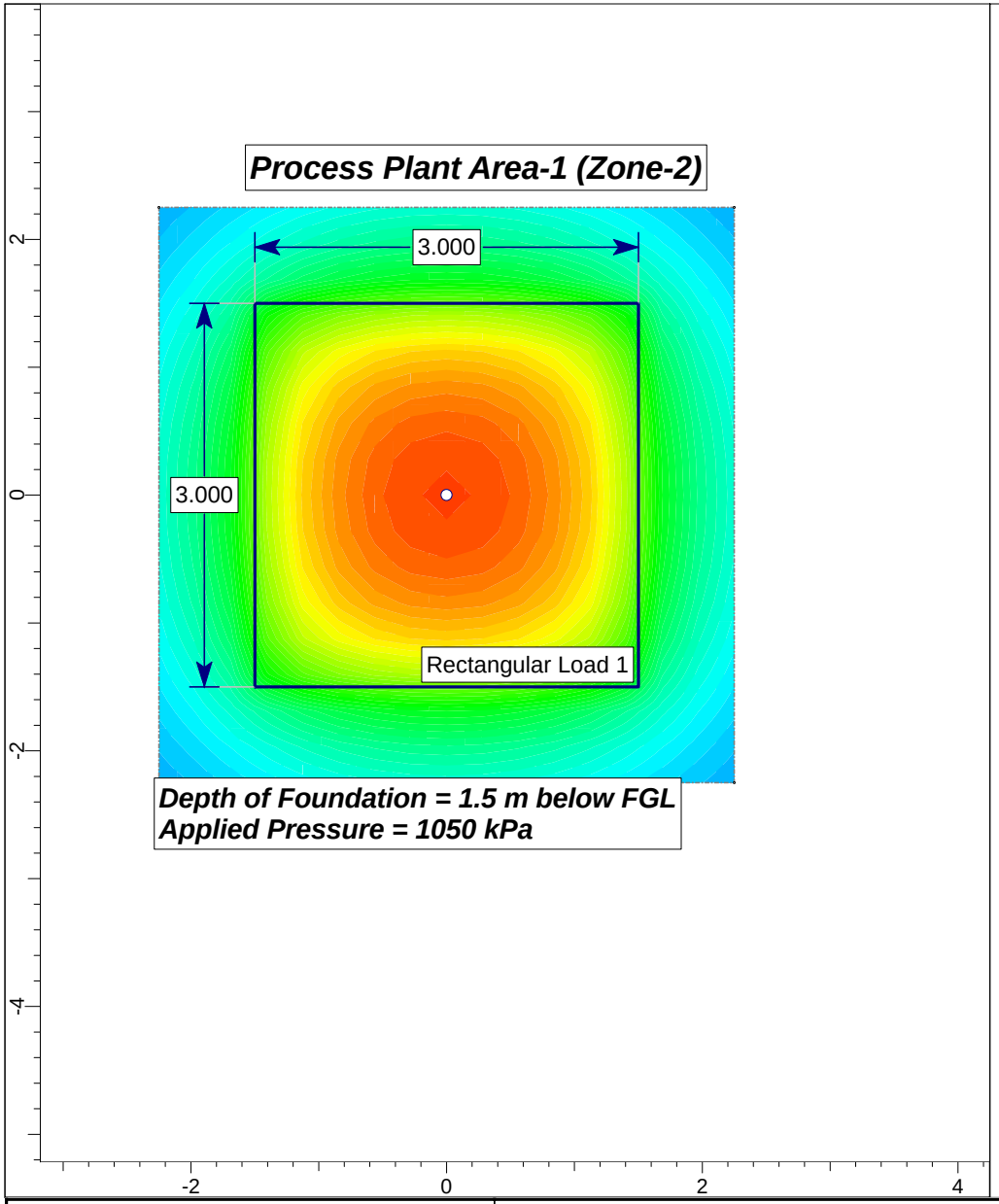
Depth of Foundation = 1.5 m below FGL
Applied Pressure = 825 kPa

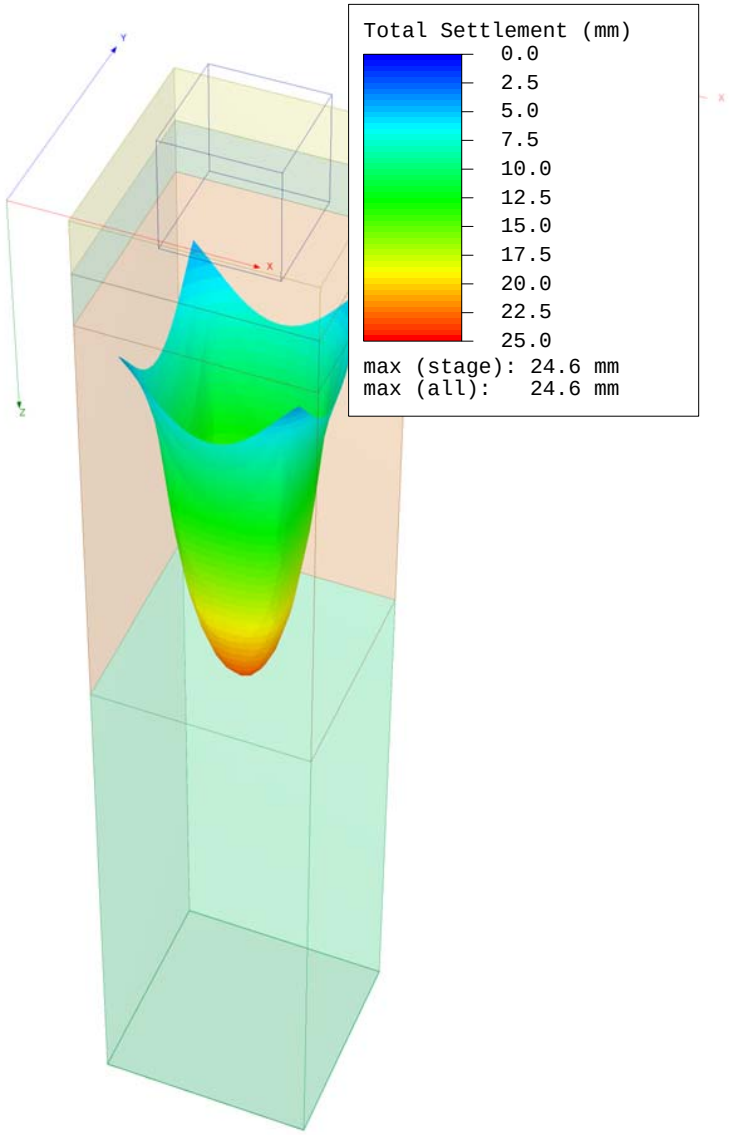




rocscience	*Project*	MANSOURAH - MASSARAH GOLD PROJECT	
	Analysis Description	Settlement Analysis	
	Drawn By HUR/MF	*Company*	RGF
	Date	*File Name*	9. TMT.s3z

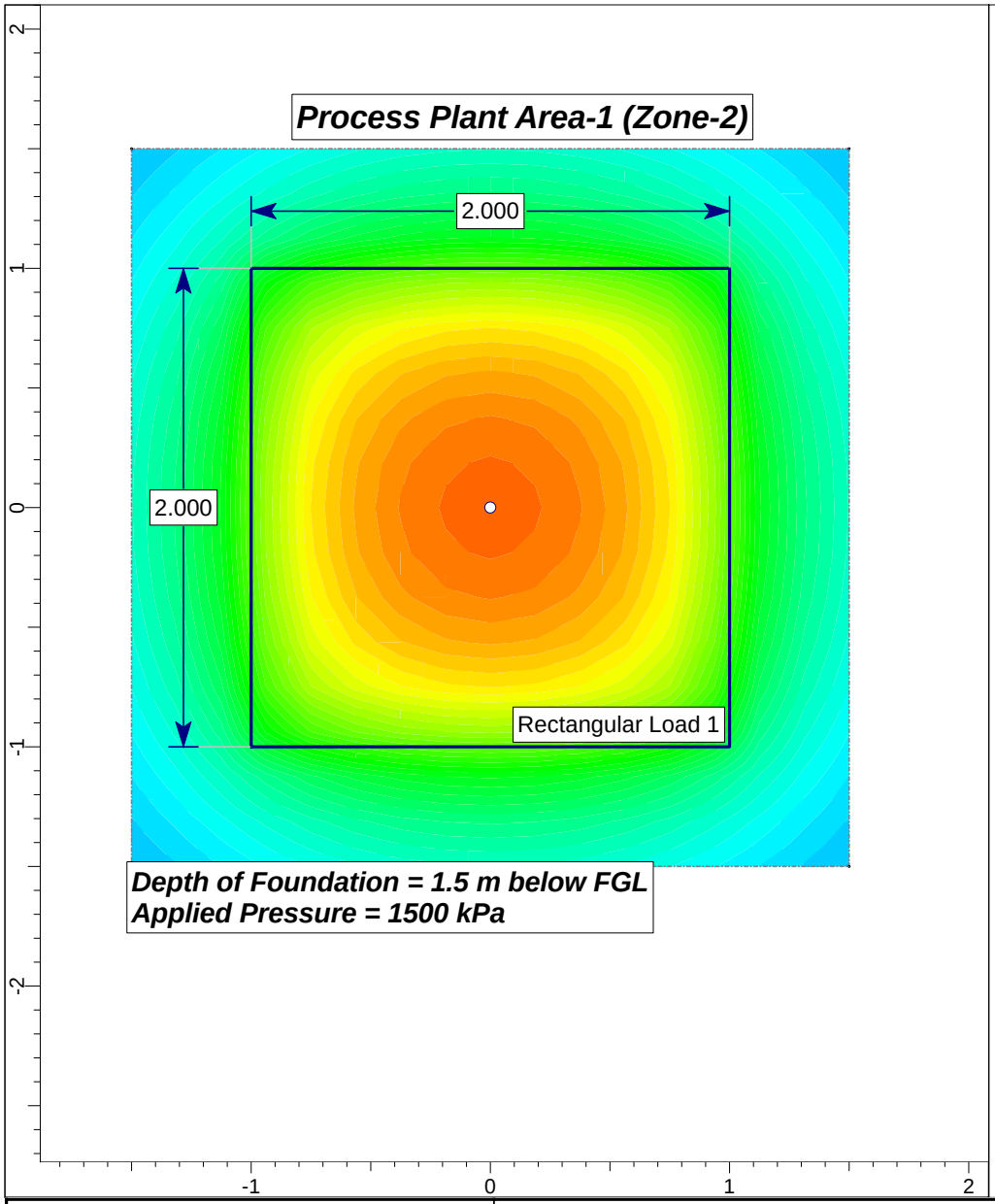
SETTLE3D 4.016

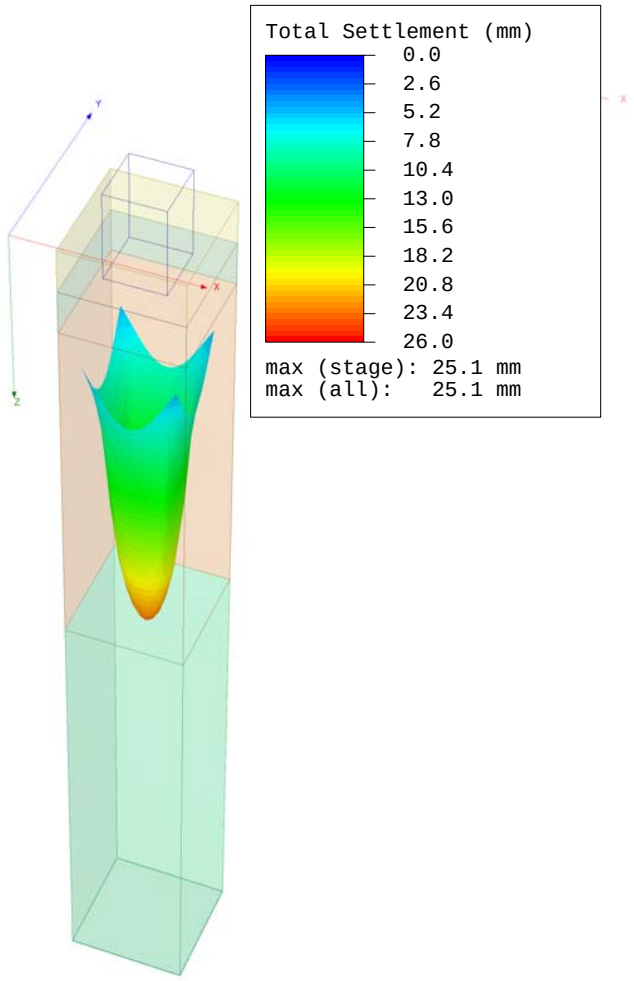




rocscience	Project	MANSOURAH - MASSARAH GOLD PROJECT	
	Analysis Description	Settlement Analysis	
	Drawn By	HUR/MF	Company RGF
	Date		File Name 9. TMT.s3z

SETTLE3D 4.016





SETTLE3D 4.016	Project	MANSOURAH - MASSARAH GOLD PROJECT	
	Analysis Description	Settlement Analysis	
	Drawn By	HUR/MF	Company RGF
	Date		File Name 9. TMT.s3z

Settle3D Analysis Information
MANSOURAH - MASSARAH GOLD PROJECT

Project Settings

Document Name	9. TMT
Project Title	MANSOURAH - MASSARAH GOLD PROJECT
Analysis	Settlement Analysis
Author	HUR/MF
Company	RGF
Stress Computation Method	Boussinesq

Calculate settlement with mean stress

Use properties from first layer to calculate layered stresses

Stage Settings

Stage #	Name
1	Stage 1

Results

Time taken to compute: 0.0284352 seconds

Stage: Stage 1

Data Type	Minimum	Maximum
Total Settlement [mm]	0	24.6443
Total Consolidation Settlement [mm]	0	0
Virgin Consolidation Settlement [mm]	0	0
Recompression Consolidation Settlement [mm]	0	0
Immediate Settlement [mm]	0	24.6443
Loading Stress ZZ [kPa]	0	700
Loading Stress XX [kPa]	-89.7139	596.901
Loading Stress YY [kPa]	-89.7139	596.901
Mean Stress [kPa]	0	606.667
Total Stress ZZ [kPa]	0	733.609
Total Stress XX [kPa]	-61.2139	658.778
Total Stress YY [kPa]	-61.2139	658.778
Modulus of Subgrade Reaction (Total) [kPa/m]	0	1.35294e+007
Modulus of Subgrade Reaction (Immediate) [kPa/m]	0	1.35294e+007
Modulus of Subgrade Reaction (Consolidation) [kPa/m]	0	0
Total Strain	-0.00029947	0.0046396
Degree of Consolidation [%]	0	0
Pre-consolidation Stress [kPa]	1.425	732.356
Over-consolidation Ratio	1	1
Void Ratio	0	0
Hydroconsolidation Settlement [mm]	0	0
Undrained Shear Strength	0	0

Loads

1. Rectangular Load: "Rectangular Load 1"

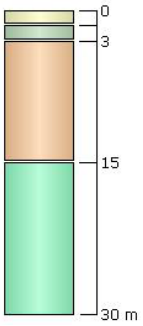
Length	5 m
Width	5 m
Rotation angle	0 degrees
Load Type	Flexible
Area of Load	25 m^2
Load	700 kPa
Depth	1.5 m
Installation Stage	Stage 1

Coordinates

X [m]	Y [m]
-2.5	-2.5
2.5	-2.5
2.5	2.5
-2.5	2.5

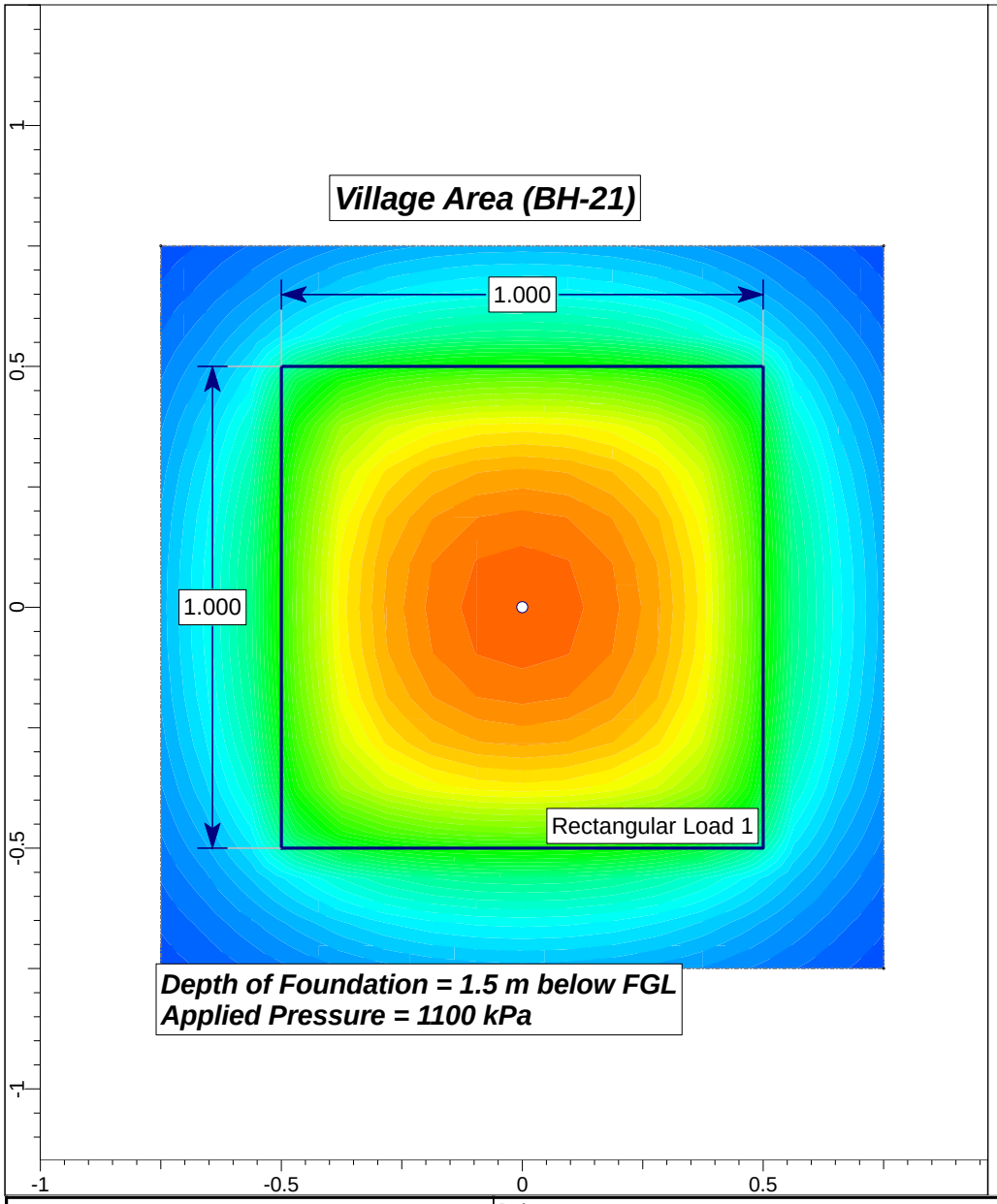
Soil Layers

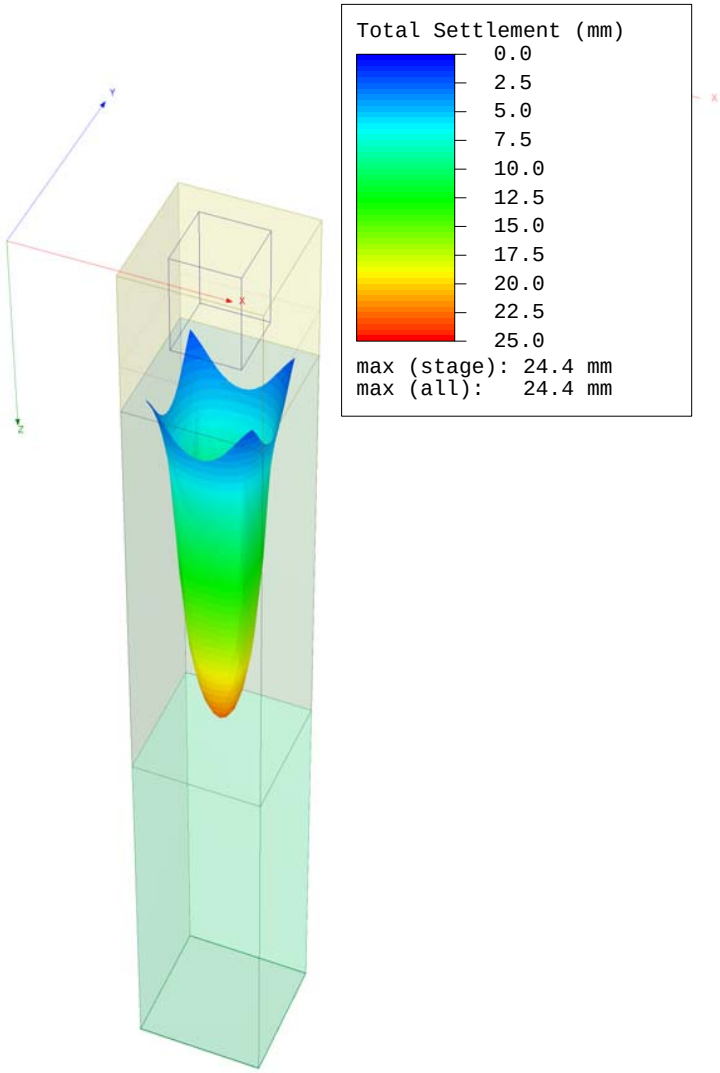
Layer #	Type	Thickness [m]	Depth [m]
1	Soil 1	1.5	0
2	Soil 2	1.5	1.5
3	Soil 3	12	3
4	Rock 1	15	15



Soil Properties

Property	Soil 1	Soil 2	Soil 3	Rock 1
Color				
Unit Weight [kN/m^3]	19	20	20	24
Poisson's Ratio	0.3	0.3	0.3	0.2
K0	1	1	1	1
Immediate Settlement	Enabled	Enabled	Enabled	Enabled
E [kPa]	40000	113000	134000	415000
Eur [kPa]	40000	113000	134000	415000
Undrained Su A [kN/m2]	0	0	0	0
Undrained Su S	0	0	0	0
Undrained Su m	0	0	0	0

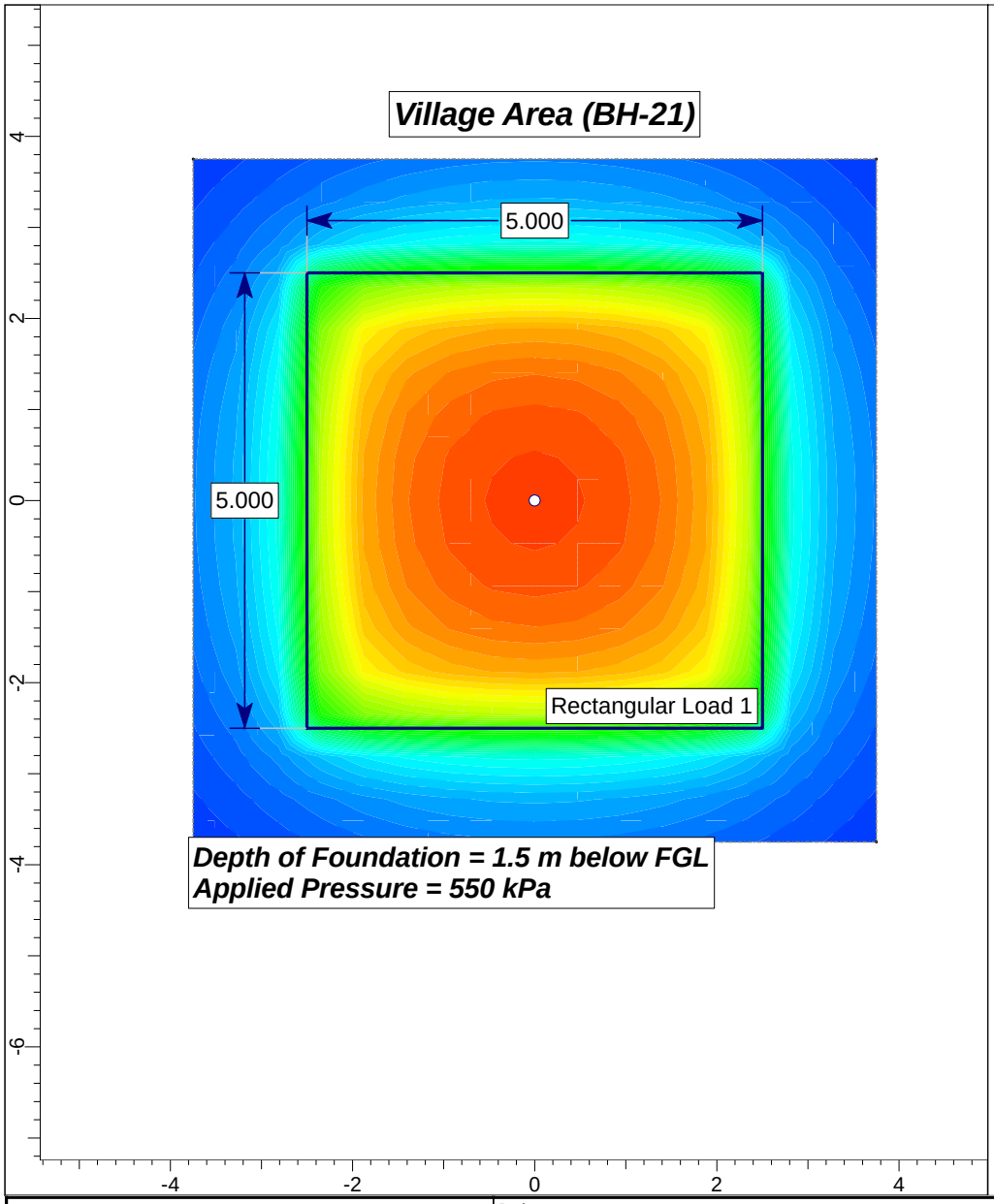




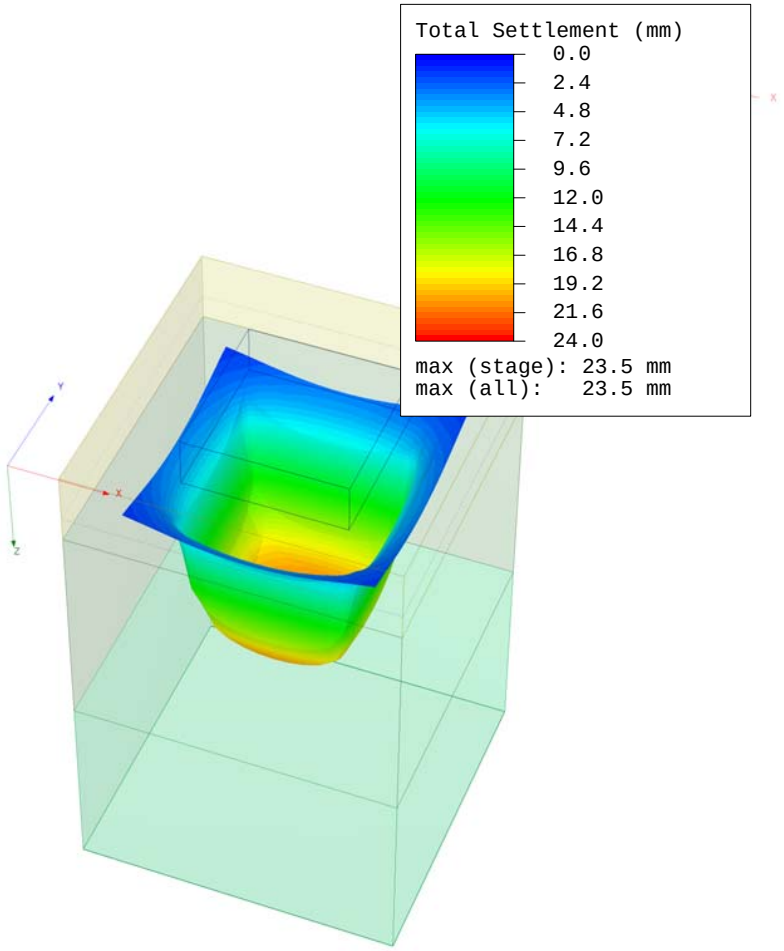
rocscience	Project	MANSOURAH - MASSARAH GOLD PROJECT	
	Analysis Description	Settlement Analysis	
	Drawn By	HUR/MF	Company RGF
	Date		File Name 7. Village Area (BH-21).s3z

SETTLE3D 4.016

Village Area (BH-21)

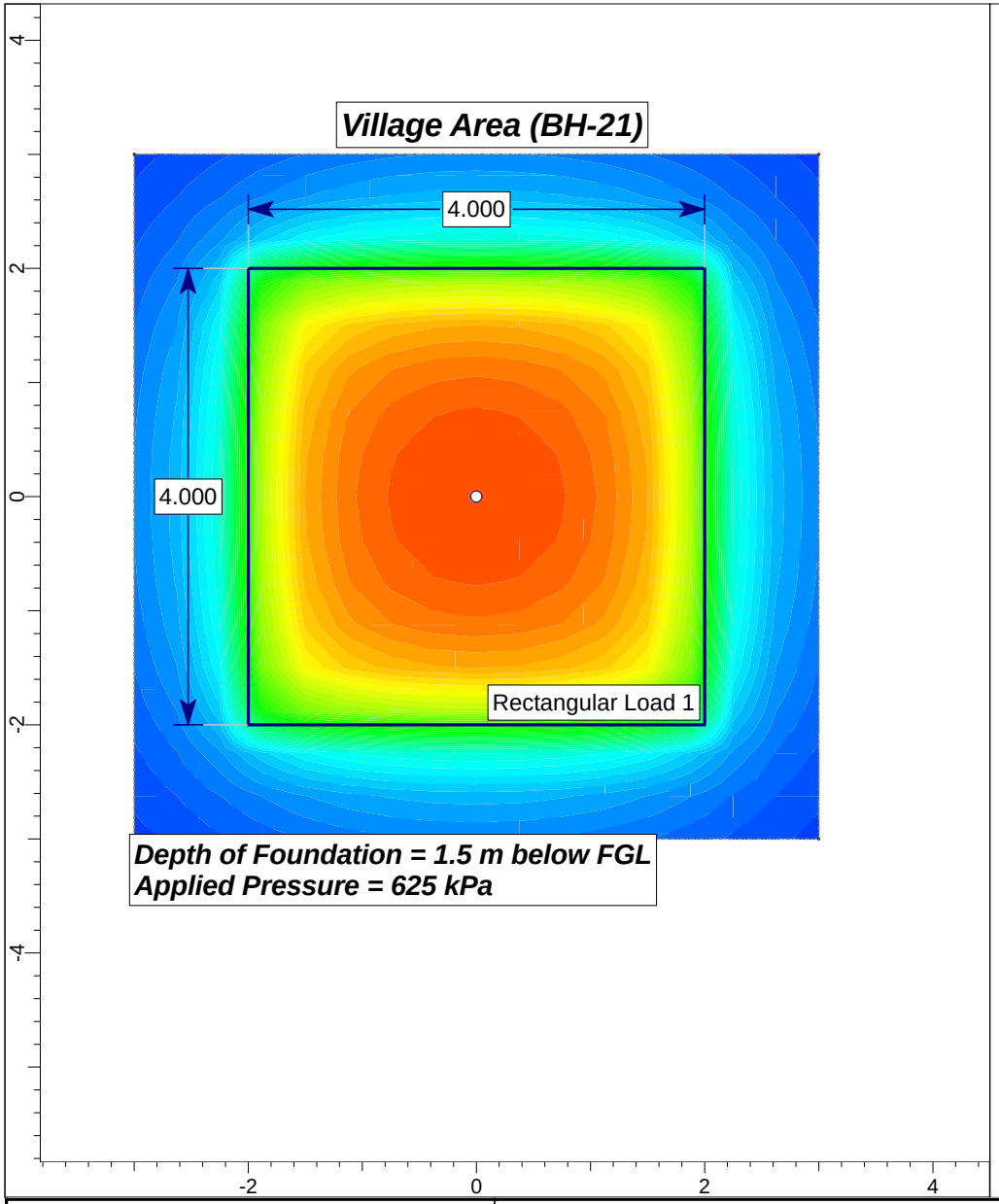


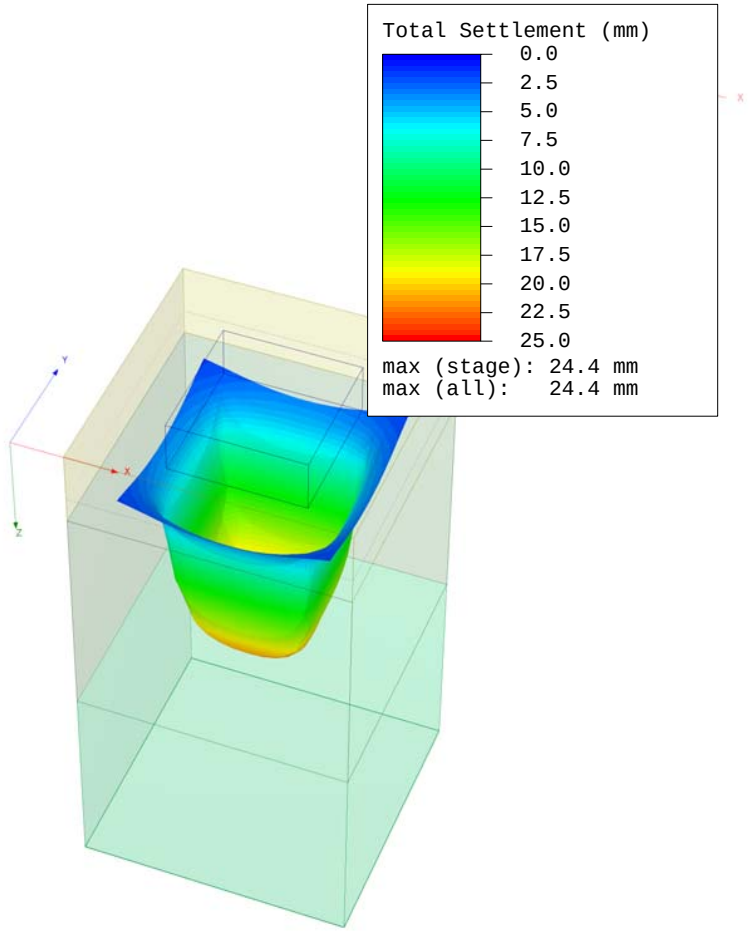
Depth of Foundation = 1.5 m below FGL
Applied Pressure = 550 kPa



rocscience	*Project*	MANSOURAH - MASSARAH GOLD PROJECT	
	Analysis Description	Settlement Analysis	
	Drawn By	HUR/MF	*Company* RGF
	Date		*File Name* 7. Village Area (BH-21).s3z

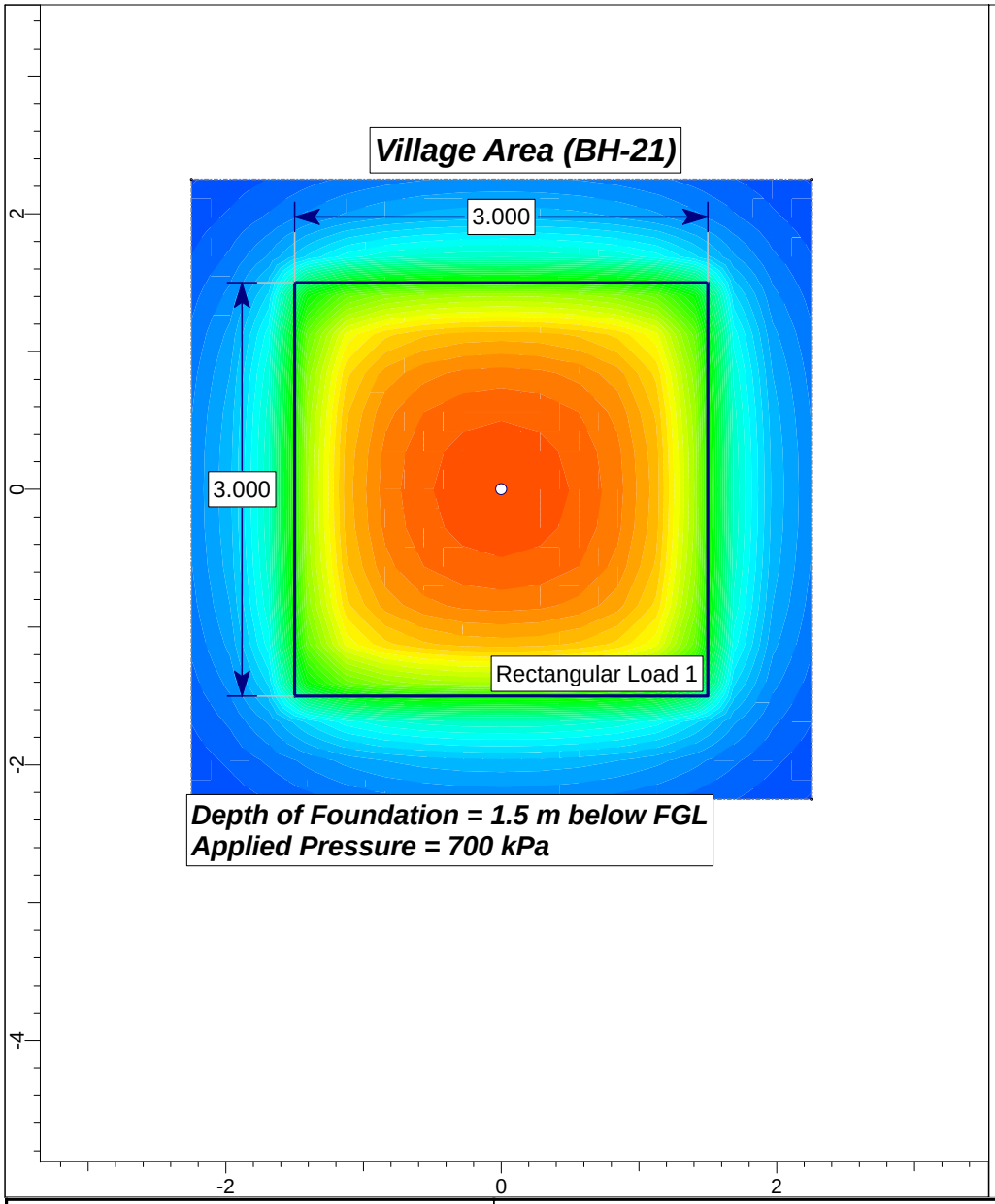
SETTLE3D 4.016

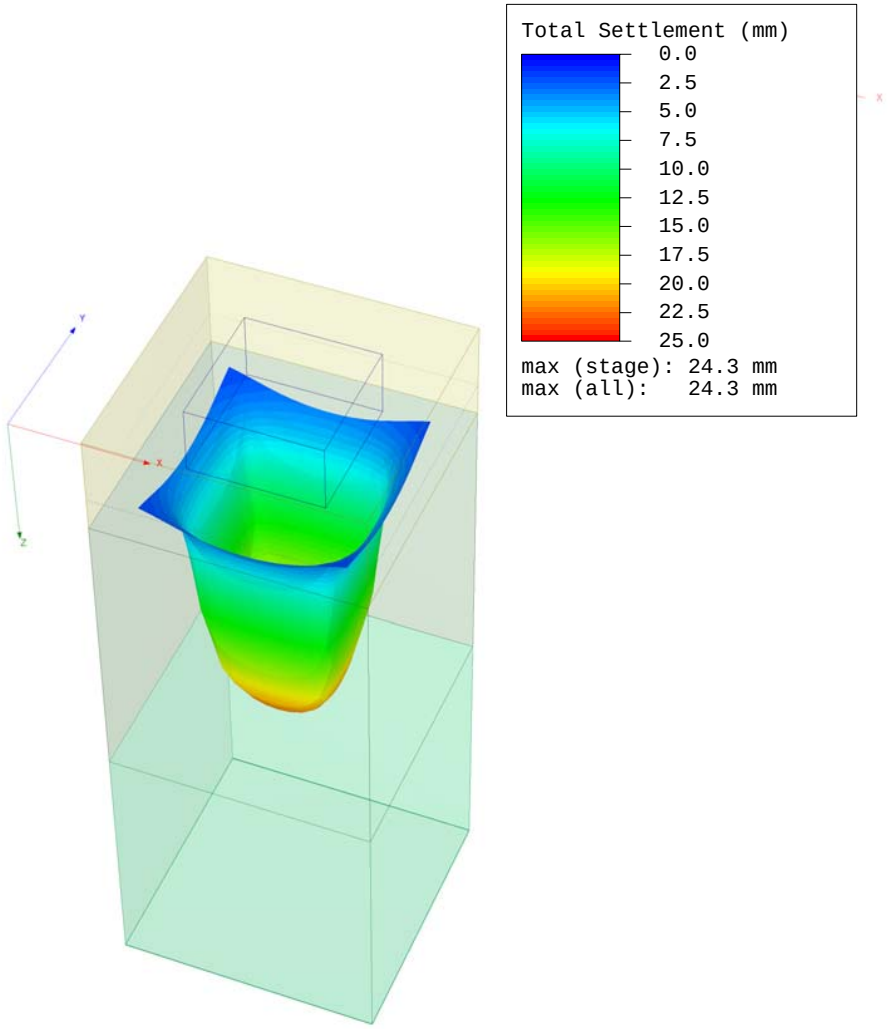




rocscience	Project	MANSOURAH - MASSARAH GOLD PROJECT	
	Analysis Description	Settlement Analysis	
	Drawn By	HUR/MF	Company RGF
	Date		File Name 7. Village Area (BH-21).s3z

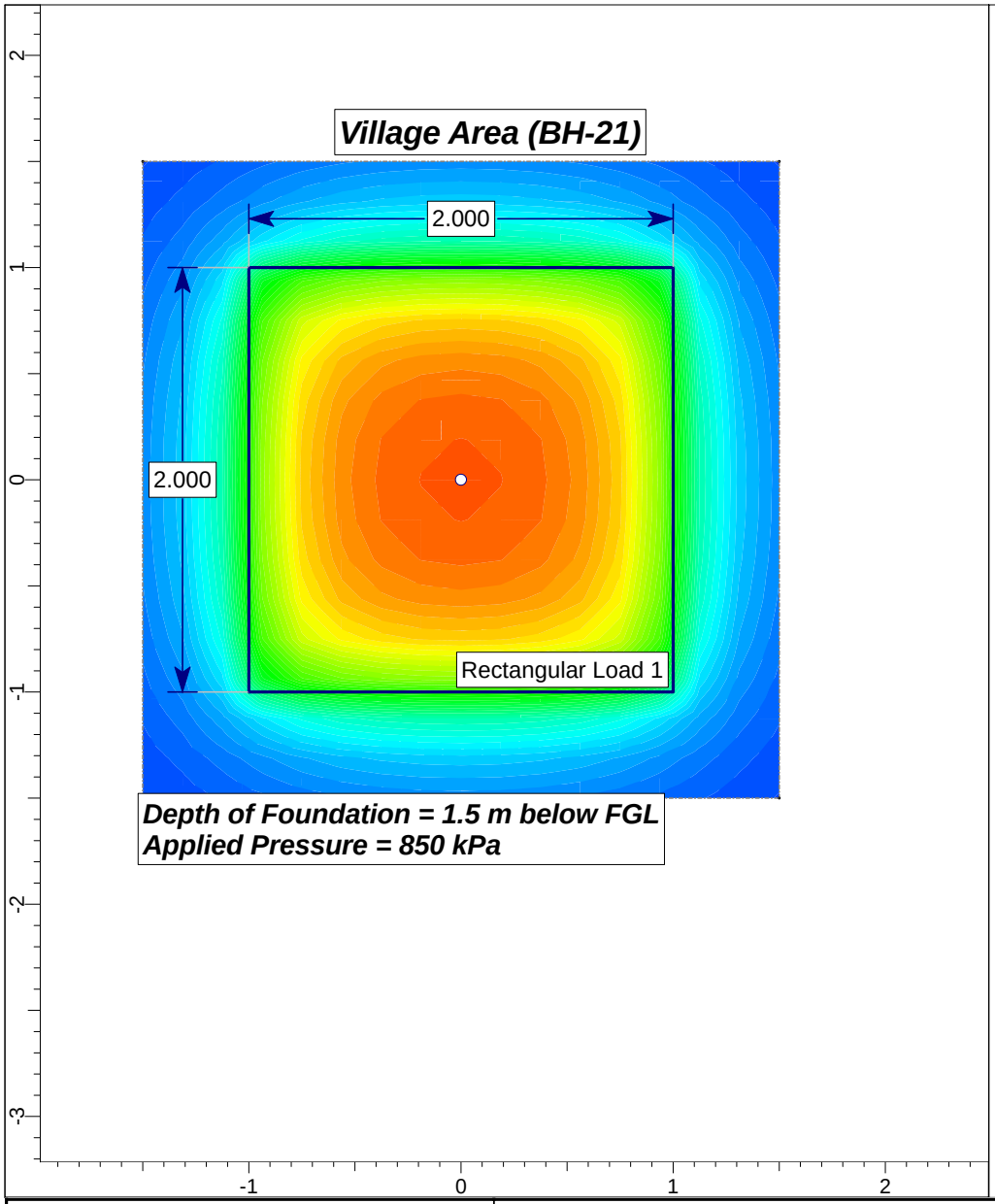
SETTLE3D 4.016

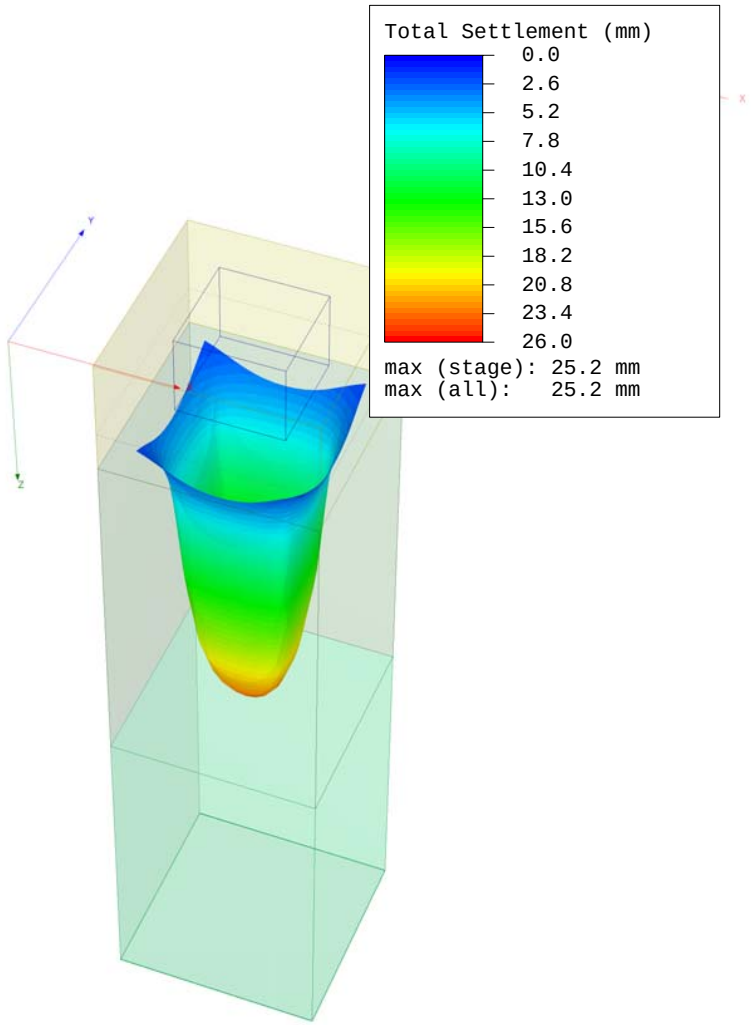




rocscience	Project	MANSOURAH - MASSARAH GOLD PROJECT	
	Analysis Description	Settlement Analysis	
	Drawn By	HUR/MF	Company RGF
	Date		File Name 7. Village Area (BH-21).s3z

SETTLE3D 4.016





rocscience	Project	MANSOURAH - MASSARAH GOLD PROJECT	
	Analysis Description	Settlement Analysis	
	Drawn By	HUR/MF	Company RGF
	Date		File Name 7. Village Area (BH-21).s3z

SETTLE3D 4.016

Settle3D Analysis Information
MANSOURAH - MASSARAH GOLD PROJECT

Project Settings

Document Name	7. Village Area (BH-21)
Project Title	MANSOURAH - MASSARAH GOLD PROJECT
Analysis	Settlement Analysis
Author	HUR/MF
Company	RGF
Stress Computation Method	Boussinesq

Calculate settlement with mean stress

Use properties from first layer to calculate layered stresses

Stage Settings

Stage #	Name
1	Stage 1

Results

Time taken to compute: 0.0331775 seconds

Stage: Stage 1

Data Type	Minimum	Maximum
Total Settlement [mm]	0	23.5061
Total Consolidation Settlement [mm]	0	0
Virgin Consolidation Settlement [mm]	0	0
Recompression Consolidation Settlement [mm]	0	0
Immediate Settlement [mm]	0	23.5061
Loading Stress ZZ [kPa]	0	550
Loading Stress XX [kPa]	-3.62035	468.994
Loading Stress YY [kPa]	-3.62035	468.994
Mean Stress [kPa]	0	476.667
Total Stress ZZ [kPa]	0	582.494
Total Stress XX [kPa]	0	495.994
Total Stress YY [kPa]	0	495.994
Modulus of Subgrade Reaction (Total) [kPa/m]	0	3.08599e+007
Modulus of Subgrade Reaction (Immediate) [kPa/m]	0	3.08599e+007
Modulus of Subgrade Reaction (Consolidation) [kPa/m]	0	0
Total Strain	-0.000790678	0.0138195
Degree of Consolidation [%]	0	0
Pre-consolidation Stress [kPa]	1.35	582.28
Over-consolidation Ratio	1	1
Void Ratio	0	0
Hydroconsolidation Settlement [mm]	0	0
Undrained Shear Strength	0	0

Loads

1. Rectangular Load: "Rectangular Load 1"

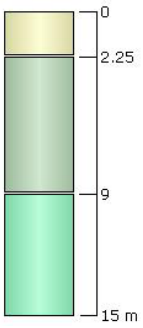
Length	5 m
Width	5 m
Rotation angle	0 degrees
Load Type	Flexible
Area of Load	25 m^2
Load	550 kPa
Depth	1.5 m
Installation Stage	Stage 1

Coordinates

X [m]	Y [m]
-2.5	-2.5
2.5	-2.5
2.5	2.5
-2.5	2.5

Soil Layers

Layer #	Type	Thickness [m]	Depth [m]
1	Soil 1	2.25	0
2	Soil 2	6.75	2.25
3	Rock 1	6	9



Soil Properties

Property	Soil 1	Soil 2	Rock 1
Color			
Unit Weight [kN/m^3]	18	20	23
Poisson's Ratio	0.3	0.3	0.2
K0	1	1	1
Immediate Settlement	Enabled	Enabled	Enabled
E [kPa]	28000	134000	375000
Eur [kPa]	28000	134000	375000
Undrained Su A [kN/m2]	0	0	0
Undrained Su S	0	0	0
Undrained Su m	0	0	0

BEARING CPAPACITY ANAL. FOR SHALLOW FOUNDATIONS
(Shear Failure Criteria - Vesic Method)

RGF-KHO-GEO-SF-03-01

Project: MANSOURAH - MASSARAH GOLD PROJECT

Location: Crushing Plant

DATA

FOOTING TYPE	***ISO***	Footing Width (B) =	1.00	m	ϕ_{des}	c_{des} (kPa)
		Footing Length (L) =	1.00	m		
		Footing Depth (D) =	1.50	m		
		Bulk Unit Weight =	20	kN/m^3	35	0

WATER TABLE EFFECT

Design Ground Water Level (m) =	50.0
Design Case =	Case 3

BEARING CAPACITY FACTORS

Nq	Nc	Nγ (v)
33.30	46.12	48.03

SHAPE FACTORS

Sq (v)	Sγ (v)	Sc (v)
1.70	0.60	1.72

d_c	d_q	d_γ
1.39	1.25	1.00

RESULTS

Ultimate Bearing Capacity	*Allowable Bearing Capacity**	*Net Allowable Bearing Capacity**
q_u (kPa)	q_a (kPa)	$q_{a(net)}$ (kPa)
1986	***662***	***632***

** FOS=3.0*

Note:

1. Depth factors have not been applied in the analysis.

BEARING CPAPACITY ANAL. FOR SHALLOW FOUNDATIONS
(Shear Failure Criteria - Vesic Method)

RGF-KHO-GEO-SF-03-01

Project: MANSOURAH - MASSARAH GOLD PROJECT

Location: Process Plant Area-1 (Zone-1)

DATA

FOOTING TYPE	***ISO***	Footing Width (B) =	3.00	m	ϕ_{des}	c_{des} (kPa)
		Footing Length (L) =	3.00	m		
		Footing Depth (D) =	2.00	m		
		Bulk Unit Weight =	19	kN/m^3	34	0

WATER TABLE EFFECT

Design Ground Water Level (m) =	50.0
Design Case =	Case 3

BEARING CAPACITY FACTORS

Nq	Nc	Nγ (v)
29.44	42.16	41.06

SHAPE FACTORS

Sq (v)	Sγ (v)	Sc (v)
1.67	0.60	1.70

d_c	d_q	d_γ
1.27	1.17	1.00

RESULTS

Ultimate Bearing Capacity	*Allowable Bearing Capacity**	*Net Allowable Bearing Capacity**
q_u (kPa)	q_a (kPa)	$q_{a(net)}$ (kPa)
2575	***858***	***820***

** FOS=3.0*

Note:

1. Depth factors have not been applied in the analysis.

BEARING CPAPACITY ANAL. FOR SHALLOW FOUNDATIONS
(Shear Failure Criteria - Vesic Method)

RGF-KHO-GEO-SF-03-01

Project: MANSOURAH - MASSARAH GOLD PROJECT

Location: Process Plant Area-1 (Zone-2)

DATA

FOOTING TYPE	***ISO***	Footing Width (B) =	6.00	m	ϕ_{des}	c_{des} (kPa)
		Footing Length (L) =	6.00	m		
		Footing Depth (D) =	1.50	m		
		Bulk Unit Weight =	20	kN/m^3	36	0

WATER TABLE EFFECT

Design Ground Water Level (m) =	50.0
Design Case =	Case 3

BEARING CAPACITY FACTORS

Nq	Nc	$N\gamma_{(v)}$
35.44	48.29	51.99

SHAPE FACTORS

$Sq_{(v)}$	$S\gamma_{(v)}$	$Sc_{(v)}$
1.71	0.60	1.73

d_c	d_q	d_γ
1.10	1.06	1.00

RESULTS

Ultimate Bearing Capacity	*Allowable Bearing Capacity**	*Net Allowable Bearing Capacity**
q_u (kPa)	q_a (kPa)	$q_{a(net)}$ (kPa)
3693	***1231***	***1201***

** FOS=3.0*

Note:

1. Depth factors have not been applied in the analysis.

BEARING CPAPACITY ANAL. FOR SHALLOW FOUNDATIONS
(Shear Failure Criteria - Vesic Method)

RGF-KHO-GEO-SF-03-01

Project: MANSOURAH - MASSARAH GOLD PROJECT

Location: Process Plant Area-1 (BH-28)

DATA

FOOTING TYPE	***ISO***	Footing Width (B) =	3.00	m	ϕ_{des}	c_{des} (kPa)
		Footing Length (L) =	3.00	m		
		Footing Depth (D) =	2.00	m		
		Bulk Unit Weight =	19	kN/m^3	34	0

WATER TABLE EFFECT

Design Ground Water Level (m) =	50.0
Design Case =	Case 3

BEARING CAPACITY FACTORS

Nq	Nc	Nγ (v)
29.44	42.16	41.06

SHAPE FACTORS

Sq (v)	Sγ (v)	Sc (v)
1.67	0.60	1.70

d_c	d_q	d_γ
1.27	1.17	1.00

RESULTS

Ultimate Bearing Capacity	*Allowable Bearing Capacity**	*Net Allowable Bearing Capacity**
q_u (kPa)	q_a (kPa)	$q_{a(net)}$ (kPa)
2575	***858***	***820***

** FOS=3.0*

Note:

1. Depth factors have not been applied in the analysis.

BEARING CPAPACITY ANAL. FOR SHALLOW FOUNDATIONS
(Shear Failure Criteria - Vesic Method)

RGF-KHO-GEO-SF-03-01

Project: MANSOURAH - MASSARAH GOLD PROJECT

Location: Process Plant Area-2 (BH-01)

DATA

FOOTING TYPE	***ISO***	Footing Width (B) =	5.00	m	ϕ_{des}	c_{des} (kPa)
		Footing Length (L) =	5.00	m		
		Footing Depth (D) =	1.50	m		
		Bulk Unit Weight =	20	kN/m^3	37	0

WATER TABLE EFFECT

Design Ground Water Level (m) =	50.0
Design Case =	Case 3

BEARING CAPACITY FACTORS

Nq	Nc	$N\gamma_{(v)}$
41.82	54.57	64.07

SHAPE FACTORS

$Sq_{(v)}$	$S\gamma_{(v)}$	$Sc_{(v)}$
1.75	0.60	1.77

d_c	d_q	d_γ
1.12	1.07	1.00

RESULTS

Ultimate Bearing Capacity	*Allowable Bearing Capacity**	*Net Allowable Bearing Capacity**
q_u (kPa)	q_a (kPa)	$q_{a(net)}$ (kPa)
4116	***1372***	***1342***

** FOS=3.0*

Note:

1. Depth factors have not been applied in the analysis.

BEARING CPAPACITY ANAL. FOR SHALLOW FOUNDATIONS
(Shear Failure Criteria - Vesic Method)

RGF-KHO-GEO-SF-03-01

Project: MANSOURAH - MASSARAH GOLD PROJECT

Location: Solar Plant & Workshop Area (BH-05)

DATA

FOOTING TYPE	***ISO***	Footing Width (B) =	4.00	m	ϕ_{des}	c_{des} (kPa)
		Footing Length (L) =	6.00	m		
		Footing Depth (D) =	1.50	m		
		Bulk Unit Weight =	20	kN/m^3	37	0

WATER TABLE EFFECT

Design Ground Water Level (m) =	50.0
Design Case =	Case 3

BEARING CAPACITY FACTORS

Nq	Nc	$N\gamma_{(v)}$
42.92	55.63	66.19

SHAPE FACTORS

$Sq_{(v)}$	$S\gamma_{(v)}$	$Sc_{(v)}$
1.50	0.73	1.51

d_c	d_q	d_γ
1.15	1.09	1.00

RESULTS

Ultimate Bearing Capacity	*Allowable Bearing Capacity**	*Net Allowable Bearing Capacity**
q_u (kPa)	q_a (kPa)	$q_{a(net)}$ (kPa)
3876	***1292***	***1262***

** FOS=3.0*

Note:

1. Depth factors have not been applied in the analysis.

BEARING CPAPACITY ANAL. FOR SHALLOW FOUNDATIONS
(Shear Failure Criteria - Vesic Method)

RGF-KHO-GEO-SF-03-01

Project: MANSOURAH - MASSARAH GOLD PROJECT

Location: Village Area

DATA

FOOTING TYPE	***ISO***	Footing Width (B) =	5.00	m	ϕ_{des}	c_{des} (kPa)
		Footing Length (L) =	5.00	m		
		Footing Depth (D) =	1.50	m		
		Bulk Unit Weight =	20	kN/m^3	36	0

WATER TABLE EFFECT

Design Ground Water Level (m) =	50.0
Design Case =	Case 3

BEARING CAPACITY FACTORS

Nq	Nc	$N\gamma_{(v)}$
39.73	52.53	60.05

SHAPE FACTORS

$Sq_{(v)}$	$S\gamma_{(v)}$	$Sc_{(v)}$
1.74	0.60	1.76

d_c	d_q	d_γ
1.12	1.07	1.00

RESULTS

Ultimate Bearing Capacity	*Allowable Bearing Capacity**	*Net Allowable Bearing Capacity**
q_u (kPa)	q_a (kPa)	$q_{a(net)}$ (kPa)
3814	***1271***	***1242***

** FOS=3.0*

Note:

1. Depth factors have not been applied in the analysis.

BEARING CPAPACITY ANAL. FOR SHALLOW FOUNDATIONS
(Shear Failure Criteria - Vesic Method)

RGF-KHO-GEO-SF-03-01

Project: MANSOURAH - MASSARAH GOLD PROJECT

Location: Explosive Magazine (BH-22)

DATA

FOOTING TYPE	***ISO***	Footing Width (B) =	5.00	m	ϕ_{des}	c_{des} (kPa)
		Footing Length (L) =	5.00	m		
		Footing Depth (D) =	1.50	m		
		Bulk Unit Weight =	19	kN/m^3	36	0

WATER TABLE EFFECT

Design Ground Water Level (m) =	50.0
Design Case =	Case 3

BEARING CAPACITY FACTORS

Nq	Nc	Nγ (v)
36.81	49.65	54.54

SHAPE FACTORS

Sq (v)	Sγ (v)	Sc (v)
1.72	0.60	1.74

d_c	d_q	d_γ
1.12	1.07	1.00

RESULTS

Ultimate Bearing Capacity	*Allowable Bearing Capacity**	*Net Allowable Bearing Capacity**
q_u (kPa)	q_a (kPa)	$q_{a(net)}$ (kPa)
3431	***1144***	***1114***

** FOS=3.0*

Note:

1. Depth factors have not been applied in the analysis.

BEARING CPAPACITY ANAL. FOR SHALLOW FOUNDATIONS
(Shear Failure Criteria - Vesic Method)

RGF-KHO-GEO-SF-03-01

Project: MANSOURAH - MASSARAH GOLD PROJECT

Location: TMT

DATA

FOOTING TYPE	***ISO***	Footing Width (B) =	5.00	m	ϕ_{des}	c_{des} (kPa)
		Footing Length (L) =	5.00	m		
		Footing Depth (D) =	1.50	m		
		Bulk Unit Weight =	20	kN/m^3	37	0

WATER TABLE EFFECT

Design Ground Water Level (m) =	50.0
Design Case =	Case 3

BEARING CAPACITY FACTORS

Nq	Nc	Nγ (v)
42.92	55.63	66.19

SHAPE FACTORS

Sq (v)	Sγ (v)	Sc (v)
1.75	0.60	1.77

d_c	d_q	d_γ
1.12	1.07	1.00

RESULTS

Ultimate Bearing Capacity	*Allowable Bearing Capacity**	*Net Allowable Bearing Capacity**
q_u (kPa)	q_a (kPa)	$q_{a(net)}$ (kPa)
4244	***1415***	***1385***

** FOS=3.0*

Note:

1. Depth factors have not been applied in the analysis.